

5G RAN

Network Impact Report

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1 About This Document

Change History

Issue	Date	Description
Draft B	2026-01-30	Product version: gNodeB V100R022C10SPC050 This issue introduces the following changes to gNodeB V100R022C10SPC010: - Added the following changes that were incorporated into V100R022C10SPC010: 5.2.4.1.20 Enhanced Dormancy Functions in Separate-MPT Base Stations Including GSM or UMTS# 5.2.4.2.18 Optimized the Effective MACsec Connection Type When MACsec Is Used for eCPRI# 5.3.3.25.3 Frequency Selection Policy for QCI-based Preallocation 5.3.4.1.45 PUSCH Scheduling Prohibition in Self-Contained Slots 5.3.4.1.46 Added the CA UE Data Split Stop Optimization Function - For the following changes in "Changes from V100R021C10SPC260 to V100R022C10SPC010": 5.2.4.1.19 Added Support for Reliability Protection During Dormancy in GSM and UMTS#: Renamed this change from "Added Support for Configuration Reliability Protection During Dormancy in GSM and UMTS", and revised the feature/function description and impact on the network. 5.3.4.1.59 Change in the Value Range of the Uplink Scheduling Priority

Issue	Date	Description
		Weight Factor Parameter in the gNBDUMacParamGroup MO: Renamed this change from "Change in the Value Range of Uplink Scheduling Priority Weight Factor". – 5.3.3.10.1 RFSP-based Flexible Configuration of Intra-RAT Handover Measurement Parameters for Voice and Data Service UEs: Renamed this change from "RFSP-based Flexible Configuration of Handover Measurement Parameters for Voice and Data Service UEs" and revised the feature/function description. – 5.3.2.1.2 Optimized Non-Overlapping Between Gap and NR Resources: Revised the descriptions of the impact on the network. – 5.3.2.9.1 Added Uplink Fast Intra-gNodeB CA: Revised the descriptions of the impact on the network. – 5.3.3.13.3 Added the SRS and PUSCH/PUCCH Concurrency Compatibility Function for FOFD-020205 Intra-Band CA and 5.3.3.21.3 Added the SRS and PUSCH/PUCCH Concurrency Compatibility Function for FOFD-031201 Intra-FR Inter-Band CA: Revised the descriptions of impacted features/functions. – 5.3.3.24.1 User-Rate-based Dynamic Network Slice Resource Management: Added the impact on hardware requirements and impacted features/functions. – 5.3.4.1.42 Added User-Rate-based Dynamic QCI Resource Management: Added the impact on hardware requirements and impacted features/functions. – TDD changes in FOFD-061203 Intelligent Channel Dynamic Shutdown: Moved these changes to the "Revised Features" node. – Changes in FBFD-050102 Basic Positioning, FBFD-091100 ANBR, and FOFD-050208 Service-Differentiated Low

Issue	Date	Description
		Latency: Moved these changes to the "Revised Features" node. – Changes in FOFD-010205 UL and DL Decoupling: Moved these changes to the "Enhanced Features" node.
Draft A	2025-12-31	Product version: gNodeB V100R022C10SPC010

Introduction

This document provides the following gNodeB product information.

Information	Organization
Version Requirements	Describes all information about this gNodeB product version, for example, software, hardware, documentation, and any other required products.
Quickly Obtaining Change Information	Describes how to quickly find out the changes in the features, functions, and operation and maintenance (O&M) of this version and how to quickly find major changes.
Change Summary	• Hardware: Describes any additions and modifications to the hardware in this version. • Features and Functions: Summarizes all feature changes, such as new, enhanced, and revised features; function changes; feature name changes; feature removal; and delayed-release changes^a.
Node	Describes logical functions of base station equipment and transmission.
gNodeB Function	Describes logical functions of NR.
New Features	Describes features that have been introduced in this version. This type of feature is labeled "New" in the feature list.
Enhanced Features	Describes features that deliver greater network performance benefits or can be used in a wider variety of scenarios. This type of feature is labeled "Enhanced" in the feature list.
Revised Features	Describes features that have undergone revisions, which are generally minor, in functions, man-machine interfaces and so on. This type of feature is labeled "Maintenance" in the feature list.
Function Changes	Describes function-related changes in this version, such as changes to the functions or man-machine interfaces.

Information	Organization
Operation and Maintenance	Describes changes to configuration management, performance management, fault management, inventory management, license management, and their management mechanisms and rules in this version.
a: Delayed-release changes are changes whose descriptions were omitted previously and are added in a later version. For each of these changes, the following statement is present: This change was first incorporated into <i>X</i> but is first described in the network impact report of <i>Y</i> . <i>X</i> is the earliest version in which the change is implemented, and <i>Y</i> is the earliest version in which the description of the change is added in the network impact report after the description omission is identified.	

Intended Audience

- This document is intended for:
- Network planning engineers
 - System engineers
 - Network operators

2 Version Requirements

2.1 Product Version

Product Name	gNodeB
Product Model	DBS3900_5G DBS5900_5G DBS5900A_5G DBS3900_LampSite_5G DBS5900_LampSite_5G
Product Version	V100R022C10SPC050
Protocol Version	Release 18 2025Q4
Description	To enable a multimode base station to support New Radio (NR), the existing multimode base station (which may support GSM, UMTS, and LTE) must be upgraded to V100R013C10 or a later version.

2.2 Related Product Versions

Related Product	Version	Remarks
MAE	MAE V100R026C10SPC010 or later MAE V100R026C10SPC010 is recommended.	For details about the changes, see <i>iMaster MAE V100R0XXCXXSPCX XX Documentation</i> .

Related Product	Version	Remarks
Offline Tool	Web LMT V100R026C10SPC500 or later	For the download link of the Offline Tool, see Support > Product Support > Wireless Network > SingleOSS-MBB > SingleOSS-MBB > MAE-M > WEB LMT .
Co-MPT Reconstruction Tool	Latest SPC version corresponding to V100R026C10 under iMaster MAE SiteMate TOOL	For the download link of the co-MPT Reconstruction Tool, see Support > Product Support > Wireless Network > SingleOSS-MBB > SingleOSS-MBB > M2000-Common > iMaster MAE SiteMate TOOL .

2.3 Related Documentation

Solution Version	Product Version	Documentation Version
5G RAN10.1	V100R022C10SPC050	(For Customer)5G RAN10.1 TDD Feature Documentation Draft B (HDX)-EN
5G RAN10.1	V100R022C10SPC010	(For Customer)5G RAN10.1 TDD Feature Documentation Draft A (HDX)-EN

2.4 Disclaimer for Trial Features

Trial features are features that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these features can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial features shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial features. Trial features are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial features may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial features but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial features are free, Huawei is not liable for any trial feature malfunctions or any losses incurred by using the trial features. Huawei does not promise that problems with trial features will be resolved in the current version. Huawei reserves the rights to convert trial features into commercial features in later R/C versions. If trial features are converted into commercial features in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial features. If a customer fails to purchase such a license, the trial feature(s) will be invalidated automatically when the product is upgraded.

3 Quickly Obtaining Change Information

3.1 Quickly Obtaining Feature and Function Change Summary

To briefly view the changes in features, functions, and O&M, see *Feature and Function Change Summary* provided with *Network Impact Report*.

3.2 Quickly Finding Major Changes

Before upgrading, carefully read about the major changes between versions, which include the following:

- All new features
- All enhanced features
- Revised features and function changes whose titles contain the # symbol
- All operation and maintenance changes

You can use one of the following methods to find major changes.

Major Change	Method 1: from <i>Network Impact Report</i>	Method 2: from <i>Feature and Function Change Summary</i>
All new features	See section "New Features" of each change path.	Filter the Major Change column to find changes with the value of Yes .
All enhanced features	See section "Enhanced Features" of each change path.	

Major Change	Method 1: from <i>Network Impact Report</i>	Method 2: from <i>Feature and Function Change Summary</i>
Revised features and function changes whose titles contain the # symbol	Find topics whose titles contain the # symbol in sections "Revised Features" and "Function Changes" of each change path.	
All operation and maintenance changes	See section "Operation and Maintenance" of each change path.	

4 Changes from V100R022C10SPC010 to V100R022C10SPC050

4.1 Change Summary

4.1.1 Hardware

See section "Hardware Changes" in "Library Information" of *3900 & 5900 Series Base Station Product Documentation*.

4.1.2 Features and Functions

For details, see the descriptions of new features, enhanced features, revised features, function changes, and operation and maintenance.

In addition to the new, enhanced, and revised features, the following changes have been incorporated.

Change Type	Change Description	RAT
Feature removal	Deleted FOFD-101202 Smooth Video Interaction and related descriptions because this feature is not recommended for commercial use in the current version.	Low-frequency NR TDD

4.2 Node

4.2.1 New Features

None

4.2.2 Enhanced Features

None

4.2.3 Revised Features

None

4.2.4 Function Changes

4.2.4.1 Radio

4.2.4.1.1 Added Support for Adaptive BBU Deep Dormancy When the Working Modes of the RF Modules Associated with the Baseband Processing Unit Include GSM

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

When the working modes of the RF modules associated with the baseband processing unit include GSM, adaptive BBU deep dormancy cannot take effect for the baseband processing unit before a base station upgrade but can take effect after the upgrade.

Implementation

For details, see *Multi-RAT Coordinated Energy Saving*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

No impact

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

Whether the change takes effect by default after an upgrade varies as follows:

- Yes if the working modes of the RF modules associated with the baseband processing unit include GSM and adaptive BBU deep dormancy has been enabled for the base station before the upgrade
 - No in other scenarios
- For details, see the corresponding "Operation and Maintenance" section in *Multi-RAT Coordinated Energy Saving*.

Requirement/Trouble Ticket Number

FeR-IR20260104002018-0010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS

4.2.4.1.2 Added Support for Dormancy Functions in RRU Single-Level Ring Networking

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Dormancy functions, such as RF module super dormancy and adaptive BBU deep dormancy, are now supported in RRU single-level ring networking.

In this networking, the **RRUCHAIN.TT (LTE eNodeB, 5G gNodeB)** parameter is set to **RING** and the RF module on the ring is only an RRU.

This change is irrelevant to LampSite base stations.

Implementation

For details, see *Multi-RAT Coordinated Energy Saving*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

For details about the RRUs that support RF module super dormancy or adaptive BBU deep dormancy, see the "Hardware" section related to the function.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

Whether the change takes effect by default after an upgrade varies as follows:

- Yes if the RRU single-level ring networking has been configured, RF module super dormancy or adaptive BBU deep dormancy has been enabled, and dormancy entry conditions are met before the upgrade
- No in other scenarios

For details, see the corresponding "Operation and Maintenance" section in *Multi-RAT Coordinated Energy Saving*.

Requirement/Trouble Ticket Number

FeR-IR20260104002018-0110

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

4.2.4.2 Transport

None

4.2.4.3 Equipment

None

4.2.5 Operation and Maintenance

4.2.5.1 General Management

There are no changes in general management.

4.2.5.2 Configuration Management

There are no changes in configuration management.

For details about changes in MML commands, MOs, and parameters, see *MML Command Changes* and *MO and Parameter Changes*, which are provided with the software version. You can obtain *MML Command Changes* and *MO and Parameter Changes* from the following directory: **BTS3900&BTS5900 VXXXRXXXCXXSPCXXX Release Documents-EN\01-Release Notes\Delta(MMI)**.

4.2.5.3 Performance Management

This section describes the changes in performance management.

For details about changes in performance counters and EBCs, see *Performance Counter Changes* and *EBC Changes*, which are provided with the software version. You can obtain *Performance Counter Changes* and *EBC Changes* from the following directory: **BTS3900&BTS5900 VXXXRXXXCXXSPCXXX Release Documents-EN\01-Release Notes\Delta(MMI)**.

4.2.5.3.1 Flexible Performance Counters Supporting 1-Minute Intervals

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

This function enables flexible performance measurement at 1-minute intervals. It allows real-time monitoring of user experience of VIP and non-VIP users across different service types in stadium scenarios or key event assurance.

It is recommended that this function be enabled in stadium scenarios or key event assurance. It is not recommended that this function be used in routine monitoring.

Implementation

Measurement of DU Cell Flexible Object Performance is introduced on the MAE. Specific counters can be selected for subscription. With this subscription, the corresponding NE reports the values of these counters every 1 minute. [Table 4-1](#) lists these counters.

Table 4-1 Performance counters supporting 1-minute reporting intervals

Indicator Type	Performance Counter
Number of active UEs	N.User.RRCCConn.Active.Flex.Max N.User.RRCCConn.Active.Flex.Avg
Traffic volume	N.ThpVol.UL.Flex N.ThpVol.DL.Flex
Average rate	N.ThpVol.DL.LastSlot.Flex N.ThpVol.DL.Flex N.ThpTime.DL.RmvLastSlot.Flex N.ThpVolForRate.UL.SmallPkt.Flex N.ThpVolForRate.UL.Flex N.ThpTimeForRate.UL.RmvSmallPkt.Flex
Rate distribution	N.Thp.UL.Flex.Index0 to N.Thp.UL.Flex.Index14 N.Thp.DL.Flex.Index0 to N.Thp.DL.Flex.Index14

Impact on the Network

After this function is enabled, the CPU usage of the main control board increases, which increases the impact of flexible-counter subscription on basic KPIs.

Impact on NEs

This change is implemented on the gNodeB and MAE.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

The related operations are as follows:

- Step 1** On the MAE, select the scenario to be observed.
- Step 2** Select the evaluation function option.
- Step 3** Set the period to 5 minutes. This setting corresponds to data collection at 1-minute intervals.
- Step 4** Select a service type.

----End

Requirement/Trouble Ticket Number

FeR-IR20260104002018-0130

RAT

NR FDD, NR TDD

4.2.5.4 Software Management

There are no changes in software management.

4.2.5.5 Fault Management

This section describes the changes in fault management.

For details about changes in alarms and events, see *Alarm Changes* and *Event Changes*, which are provided with the software version. You can obtain *Alarm Changes* and *Event Changes* from the following directory: **BTS3900&BTS5900 VXXRXXXCXXSPCXXX Release Documents-EN\01-Release Notes\Delta(MMI).**

4.2.5.5.1 Added an Upper Limit on the Number of NEs Supported by Short-Period KPI Analysis in All Scenario KPI Overview

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

An upper limit has been added on the number of NEs supported by KPI analysis with the KPI period being **P30**, **P15**, or **P5** in all scenario KPI overview.

Implementation

For details, see *FMA*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This change does not involve NE requirements and is implemented only on the MAE.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20260107000936

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD

4.2.5.6 Inventory Management

There are no changes in inventory management.

4.2.5.7 License Management

There are no changes in license management.

For details about changes in licenses, see *License Changes*, which is provided with the software version. You can obtain *License Changes* from the following directory:
BTS3900&BTS5900 VXXXRXXXCXXSPCXXX Release Documents-EN\01-Release Notes\Delta(MMI).

4.3 gNodeBFunction

4.3.1 New Features

None

4.3.2 Enhanced Features

None

4.3.3 Revised Features

4.3.3.1 FBFD-050103 Flexible User Steering

4.3.3.1.1 RFSP-based Flexible Configuration of Inter-RAT Handover Measurement Parameters for Voice and Data Service UEs

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Voice and data service UEs require different handover measurement parameters. The change in this feature now supports RFSP-based flexible configuration of inter-RAT handover measurement parameters for voice and data service UEs. In this context, voice service UEs refer to UEs performing VoNR services, while data service UEs refer to those not performing VoNR services. "Flexible configuration of handover measurement parameters for voice and data service UEs" applies only to SA networking.

"Flexible configuration of handover measurement parameters for voice and data service UEs" is implemented as follows:

For a specific RFSP index, the RFSP-specific handover measurement parameter group ID can be configured using the parameters in the **gNBRfspConfig** MO. The handover measurement parameters associated with the parameter group ID are the handover measurement parameters associated with the general handover measurement parameter group ID that has the same value as the RFSP-specific handover measurement parameter group ID.

If the RFSP-specific handover measurement parameter group ID is set to **255** or there is no general handover measurement parameter group ID with the same value in the serving cells of the UEs, the RFSP-specific handover measurement parameter group ID does not take effect.

Table 4-2 lists the parameters of inter-RAT handover measurement parameter group IDs for different types of UEs.

Table 4-2 Parameters of inter-RAT handover measurement parameter group IDs for different types of UEs

UE Type	RFSP-specific Handover Measurement Parameter Group ID Parameter	General Handover Measurement Parameter Group ID Parameter
Voice service UE	gNBRfspConfig.VoInterRatHoGroupId	NRCellInterRHoMeaGrp.InterRatHoMeasGroupId

UE Type	RFSP-specific Handover Measurement Parameter Group ID Parameter	General Handover Measurement Parameter Group ID Parameter
Voice service UE	gNB RfSP Config.VoInterRHoEutraMeasGrpId	NRCellHoEutranMeasGrp.InterRHoToEutranMeasGrpId
Data service UE	gNB RfSP Config.DataInterRatHoGroupId	NRCellInterRHoMeasGrp.InterRatHoMeasGroupId
Data service UE	gNB RfSP Config.DataInterRHoEutraMeasGrpId	NRCellHoEutranMeasGrp.InterRHoToEutranMeasGrpId
Note 1: For details about the general handover measurement parameter group ID and its involved handover measurement parameters in inter-RAT handover scenarios, see <i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i>. Note 2: For voice service UEs, the ranges of handover measurement parameters corresponding to the gNB RfSP Config.VoInterRatHoGroupId and gNB RfSP Config.VoInterRHoEutraMeasGrpId parameters are different. The differences are similar to those between the parameters corresponding to the general handover measurement parameter group ID. The same rules apply to data service UEs.		

Implementation

For details, see *Flexible User Steering*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature is not under license control.

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	<i>gNBRfspConfig.DataInterRatHoGroupId</i>	ADD GNBRFSPCO NFIG MOD GNBRFSPCO NFIG LST GNBRFSPCO NFIG	gNodeB	This parameter has been added to specify the ID of an inter-RAT handover parameter group for data services.
New	<i>gNBRfspConfig.VoInterRatHoGroupId</i>	ADD GNBRFSPCO NFIG MOD GNBRFSPCO NFIG LST GNBRFSPCO NFIG	gNodeB	This parameter has been added to specify the ID of an inter-RAT handover parameter group for voice services.
New	<i>gNBRfspConfig.DataInterRHoEutraMeasGrpId</i>	ADD GNBRFSPCO NFIG MOD GNBRFSPCO NFIG LST GNBRFSPCO NFIG	gNodeB	This parameter has been added to specify the ID of a measurement parameter group related to inter-RAT handover to E-UTRAN for data services.
New	<i>gNBRfspConfig.VoInterRHoEutraMeasGrpId</i>	ADD GNBRFSPCO NFIG MOD GNBRFSPCO NFIG LST GNBRFSPCO NFIG	gNodeB	This parameter has been added to specify the ID of a measurement parameter group related to inter-RAT handover to E-UTRAN for voice services.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

You can run the **MOD GNBFRSPCONFIG** command to configure the **gNBFRspConfig.DataInterRatHoGroupIid**, **gNBFRspConfig.VolInterRatHoGroupIid**, **gNBFRspConfig.DataInterRHoEutraMeasGrpIid**, and **gNBFRspConfig.VolInterRHoEutraMeasGrpIid** parameters to implement the flexible configuration of inter-RAT handover measurement parameters for voice and data service UEs.

Requirement/Trouble Ticket Number

FeR-IR20260104002018-0120

RAT

NR FDD, NR TDD

4.3.3.2 FOFD-050201 Network Slicing Introduction

4.3.3.2.1 Modified the User-Rate-based Dynamic Network Slice Resource Management Mechanism

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The user-rate-based dynamic network slice resource management mechanism has been modified as follows:

- Before the modification: When the **DYNAMIC_NETWORKSLICE_SW** option of the **NRDUCellCommonSw.ServiceDiffAlgoExtSwitch** parameter is selected and **NRDUCellRes.UlRbAverageRatio** or **NRDUCellRes.DlRbAverageRatio** is configured for NR DU cell resources, the gNodeB dynamically adjusts RB

allocation based on the minimum rate of non-GBR or GBR services of UEs in a network slice group.

- After the modification: When the **DYNAMIC_NETWORKSLICE_SW** option of the **NRDUCellCommonSw.ServiceDiffAlgoExtSwitch** parameter is selected and **NRDUCellRes.UlRbAverageRatio** or **NRDUCellRes.DlRbAverageRatio** is configured for NR DU cell resources, the gNodeB dynamically adjusts RB allocation based on **NRDUCellQciBearer.DynResUlTargetRate** (indicating the uplink target rate specific to dynamic resources) or **NRDUCellQciBearer.DynResDlTargetRate** (indicating the downlink target rate specific to dynamic resources) specified for UEs in a network slice group.

The change in this feature is supported only in SA networking.

LampSite base stations do not support the change in this feature.

The change in this feature is supported only in low frequency bands.

Implementation

For details, see *Network Slicing*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellQciBearer.DynResULTargetRate	ADD NRDUCELLQ CIBEARER MOD NRDUCELLQ CIBEARER LST NRDUCELLQ CIBEARER	gNodeB	This parameter has been added to specify the uplink target rate specific to dynamic resources.
New	NRDUCellQciBearer.DynResDLTargetRate	ADD NRDUCELLQ CIBEARER MOD NRDUCELLQ CIBEARER LST NRDUCELLQ CIBEARER	gNodeB	This parameter has been added to specify the downlink target rate specific to dynamic resources.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Network Slicing*.

Requirement/Trouble Ticket Number

IR20260104002018

DTS2026012908119

RAT

NR FDD, NR TDD

4.3.3.3 FOFD-051211 3CC CA

4.3.3.3.1 Added Downlink RLC Throughput Distribution Measurement for 3CC or Higher CA UEs

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

This feature change adds downlink RLC throughput distribution measurement for 3CC or higher CA UEs by setting the **NRDUCellOamParam.UeSpecStatisticsPolicy** parameter to **DL_NCC_UE**. After this function is enabled, the counters **N.Thp.DL.Ue.Spec.Samp.Index0** to **N.Thp.DL.Ue.Spec.Samp.Index14** can be used to measure the downlink RLC throughput distributions of 3CC or higher CA UEs. For details about the counters **N.Thp.DL.Ue.Spec.Samp.Index0** to **N.Thp.DL.Ue.Spec.Samp.Index14** and the settings of throughput distribution ranges, see the description of the corresponding counters.

The change in this feature is supported only in low-frequency bands.

Implementation

For details, see *CA Experience Improvement*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management

The change in this feature has no impact. It is controlled by the existing feature license.

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellOamParam.UeSpecStatisticsPolicy	MOD NRDUCELLOAMPARAM LST NRDUCELLOAMPARAM	gNodeB	The value DL_NCC_UE has been added. When the NRDUCellOamParam.UeSpecStatisticsPolicy parameter is set to DL_NCC_UE , N.Thp.DL.Ue.Spec.Samp.Index0 to N.Thp.DL.Ue.Spec.Samp.Index14 can be used to measure the downlink RLC throughput distributions of 3CC or higher CA UEs.

- Performance management

The change in this feature has no impact.

- Fault management

The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

The change in this feature has no impact.

- Mutually exclusive features/functions

The change in this feature has no impact.

- Impacted features/functions

The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *CA Experience Improvement*.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0197

RAT

NR FDD, NR TDD

4.3.3.4 FOFD-070316 RedCap Introduction

4.3.3.4.1 Modified the Mechanism of Mapping Between 5QI Bearers and Specific QCI in "Differentiated QoS Management for RedCap UEs"

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In differentiated QoS management for RedCap UEs, when the **gNBOP5qiConfig.RedCapQci** parameter is set to **65535**, the following are true:

- Before the change: The value of the **gNBOP5qiConfig.Qci** parameter is the QCI actually used by RedCap UEs.
- After the change: If the **QCI_MAP_POLICY_OPT_SW** option of the **gNodeBAlgo.QciAlgoOptSw** parameter is selected, the QCI actually used by RedCap UEs is determined based on the configuration of certain functions (The functions include SAI-based service experience improvement. For details about the involved functions, see "Function Impacts" of "Differentiated QoS Management for RedCap UEs"). For details about SAI-based service experience improvement, see *SAI-based Service Experience Improvement*.

The change in this feature is supported only in low-frequency SA networking.

Implementation

For details, see *RedCap*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	gNodeBAlgo.QciAlgoOptSw	MOD GNODEBAL GO LST GNODEBAL GO	gNodeB	The QCI_MAP_POLICY_OPT_SW has been added to specify whether to change the mechanism of mapping between 5QI bearers and specific QCIs in "differentiated QoS management for RedCap UEs".

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The following impacted functions have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SAI-based service experience improvement	gNBSchAdaptToTfcMdl.SchAdaptToTfcModelPol set to SAI	<i>SAI-based Service Experience Improvement</i>	If the QCI_MAP_POLICY_OPT_SW option of the gNodeBAlgo.QciAlgoOptSw parameter is deselected and the corresponding gNBOp5qiConfig MO is configured for a bearer, the equivalent QCI configured for the bearer for SAI-based service experience improvement (specified by gNBServExpGroup.EquivalentQci) and SAI-based service experience improvement do not take effect. If the QCI_MAP_POLICY_OPT_SW option of the gNodeBAlgo.QciAlgoOptSw parameter is selected and a 5QI-specific bearer has been configured with a QCI specified by gNBOp5qiConfig.RedCapQci for "differentiated QoS management for RedCap UEs", the following are true: If the gNBOp5qiConfig.RedCapQci parameter is set to 65535, SAI-based service experience improvement does not take effect on non-RedCap UEs.

RAT	Function Name	Function Switch	Reference	Description
				• If the gNBOP5qiConfig.RedCapQci parameter is set to a value other than 65535, the following are true: – If the gNBOP5qiConfig.Qci parameter is set to 65535, SAI-based service experience improvement does not take effect on RedCap UEs. – If the gNBOP5qiConfig.Qci parameter is set to a value other than 65535, SAI-based service experience improvement does not take effect on all UEs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Service identification-based assurance	gNBSchAdaptToTfcMdl.SchAdaptToTfcModelPol set to TAI	<i>Device-Pipe Synergy</i>	If the QCI_MAP_POLICY_OPT_SW option of the gNodeBAlgo.QciAlgoOptSw parameter is deselected and the corresponding gNBOp5qiConfig MO is configured for a bearer, the equivalent QCI configured for the bearer for service identification-based assurance (specified by gNBServExpGroup.EquivalentQci) and service identification-based assurance do not take effect. If the QCI_MAP_POLICY_OPT_SW option of the gNodeBAlgo.QciAlgoOptSw parameter is selected and a 5QI-specific bearer has been configured with a QCI specified by gNBOp5qiConfig.RedCapQci for "differentiated QoS management for RedCap UEs", the following are true: If the gNBOp5qiConfig.RedCapQci parameter is set to 65535, service identification-based assurance does not take effect on non-RedCap UEs.

RAT	Function Name	Function Switch	Reference	Description
				• If the gNBOP5qiConfig.RedCapQci parameter is set to a value other than 65535, the following are true: – If the gNBOP5qiConfig.Qci parameter is set to 65535, service identification-based assurance does not take effect on RedCap UEs. – If the gNBOP5qiConfig.Qci parameter is set to a value other than 65535, service identification-based assurance does not take effect on all UEs.

Related Operations

By default, the change does not take effect after an upgrade.

Run the **MOD GNODEBALGO** command with the **QCI_MAP_POLICY_OPT_SW** option of the parameter **gNodeBAlgo.QciAlgoOptSw** parameter selected to modify the mechanism of mapping between 5QI bearers and specific QCIs in "differentiated QoS management for RedCap UEs".

Requirement/Trouble Ticket Number

IR20260104002018

DTS2026011413547

RAT

NR FDD, NR TDD

4.3.3.5 FOFD-080012 Massive CA

4.3.3.5.1 Added Downlink RLC Throughput Distribution Measurement for 3CC or Higher CA UEs

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

This feature change adds downlink RLC throughput distribution measurement for 3CC or higher CA UEs by setting the **NRDUCellOamParam.UeSpecStatisticsPolicy** parameter to **DL_NCC_UE**. After this function is enabled, the counters **N.Thp.DL.Ue.Spec.Samp.Index0** to **N.Thp.DL.Ue.Spec.Samp.Index14** can be used to measure the downlink RLC throughput distributions of 3CC or higher CA UEs. For details about the counters **N.Thp.DL.Ue.Spec.Samp.Index0** to **N.Thp.DL.Ue.Spec.Samp.Index14** and the settings of throughput distribution ranges, see the description of the corresponding counters.

The change in this feature is supported only in low-frequency SA networking.

Implementation

For details, see *CA Experience Improvement*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management

The change in this feature has no impact. It is controlled by the existing feature license.

- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCelloamParam.UeSpecStatisticsPolicy	MOD NRDUCELLO AMPARAM LST NRDUCELLO AMPARAM	gNodeB	The value DL_NCC_UE has been added. When the NRDUCelloamParam.UeSpecStatisticsPolicy parameter is set to DL_NCC_UE , N.Thp.DL.Ue.Spec.Samp.Index0 to N.Thp.DL.Ue.Spec.Samp.Index14 can be used to measure the downlink RLC throughput distributions of 3CC or higher CA UEs.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *CA Experience Improvement*.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0197

RAT

NR FDD, NR TDD

4.3.3.6 FOFD-100210 Flexible Rate Grading (Trial)

4.3.3.6.1 Change of Flexible Rate Grading from a Commercial Feature into a Trial Feature

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

FOFD-100210 Flexible Rate Grading is now changed from a commercial feature into a trial feature.

Implementation

For details, see *Service-differentiated Experience Guarantee*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The change in this feature has no impact.
- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

- The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0201

RAT

NR TDD

4.3.3.6.2 Uplink and Downlink Decoupling of the Unlimited Rate Switch for Data Rate Limitation in Flexible Experience Differentiation

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The **Unlimited Rate Switch** of data rate limitation in flexible experience differentiation now supports uplink and downlink decoupling. Specifically, the **UNLIMITED_RATE_SW** option of the **gNBFlexExpDiffCfg.EquityLevelAlgoSw** parameter is added to the disuse list and replaced by the **UL_UNLIMITED_RATE_SW** and **DL_UNLIMITED_RATE_SW** options of this parameter. These new options are defined as follows:

- For the uplink, the **UL_UNLIMITED_RATE_SW** option of the **gNBFlexExpDiffCfg.EquityLevelAlgoSw** parameter specifies whether "data rate limitation in flexible experience differentiation" takes effect for users with a privilege level. This option is deselected by default. If this option is deselected, "data rate limitation in flexible experience differentiation" takes effect for users with the corresponding privilege level. This option can be selected if this function is not required for users with the corresponding privilege level.
- For the downlink, the **DL_UNLIMITED_RATE_SW** option of the **gNBFlexExpDiffCfg.EquityLevelAlgoSw** parameter specifies whether "data rate limitation in flexible experience differentiation" takes effect for users with a privilege level. This option is deselected by default. If this option is deselected, "data rate limitation in flexible experience differentiation" takes effect for users with the corresponding privilege level. This option can be selected if this function is not required for users with the corresponding privilege level.

The upgrade rules for the related parameter are as follows:

- **UL_UNLIMITED_RATE_SW** of **gNBFlexExpDiffCfg.EquityLevelAlgoSw**: After an upgrade from a version in the range of V100R022C00SPC050–V100R022C10SPC050 (not inclusive) to V100R022C10SPC050 or later, this option will inherit the value of **UNLIMITED_RATE_SW** of **gNBFlexExpDiffCfg.EquityLevelAlgoSw** before the upgrade.
- **DL_UNLIMITED_RATE_SW** of **gNBFlexExpDiffCfg.EquityLevelAlgoSw**: After an upgrade from a version in the range of V100R022C00SPC050–V100R022C10SPC050 (not inclusive) to V100R022C10SPC050 or later, this option will inherit the value of **UNLIMITED_RATE_SW** of **gNBFlexExpDiffCfg.EquityLevelAlgoSw** before the upgrade.

Implementation

For details, see *Service-differentiated Experience Guarantee*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameter has changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNBFlexExpDiffCf g.EquityLevelAlgoSw	ADD GNBFLEXEXP DIFFCFG MOD GNBFLEXEXP DIFFCFG LST GNBFLEXEXP DIFFCFG	gNodeB	The functions of the UNLIMITED_RATE_SW option of this parameter are now controlled by the new options UL_UNLIMITED_RATE_SW and DL_UNLIMITED_RATE_SW of this parameter. The UNLIMITED_RATE_SW option will be deleted in later versions. Although the configuration interface in this version supports synchronization and delivery of the setting of this option, the system does not use this option. Therefore, avoid using the UNLIMITED_RATE_SW option.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change takes effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Service-differentiated Experience Guarantee*.

Requirement/Trouble Ticket Number

DTS2025122727412

RAT

NR TDD

4.3.3.6.3 Flexible Rate Grading Taking Effect Only in SA Networking

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The constraints of functions related to Flexible Rate Grading take effect only for SA UEs.

Implementation

For details, see *Service-differentiated Experience Guarantee*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The change in this feature has no impact.
- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change takes effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Service-differentiated Experience Guarantee*.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0201

RAT

NR TDD

4.3.4 Function Changes

4.3.4.1 Radio

4.3.4.1.1 Added Measurement of Operator-level Numbers of CA-capable UEs

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The following counters have been added to measure the numbers of CA-capable UEs for a specific operator:

- **N.User.UL.CA.Capability.Avg.PLMN**: average number of uplink CA-capable UEs for a specific operator in a cell
- **N.User.3CC.CA.Capability.Avg.PLMN**: average number of 3CC CA-capable UEs for a specific operator in a cell

Implementation

The following counters have been added to measure the numbers of CA-capable UEs for a specific operator:

- **N.User.UL.CA.Capability.Avg.PLMN**: average number of uplink CA-capable UEs for a specific operator in a cell

- **N.User.3CC.CA.Capability.Avg.PLMN**: average number of 3CC CA-capable UEs for a specific operator in a cell

The measurement of **N.User.UL.CA.Capability.Avg.PLMN** and **N.User.3CC.CA.Capability.Avg.PLMN** is subject to the "optimized measurement of the numbers of CA-capable UEs in cells" function (controlled by the **CA_CAP_USER_STATS_POL_SW** option of the **gNodeBAlgo.NrSmartMultiCarrSw** parameter). For details about optimized measurement of the numbers of CA-capable UEs in cells, see *Carrier Aggregation*.

Impact on the Network

No impact

Impact on NEs

No impact

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management

The following counters have changed.

Change Type	Counter Name	Function Subset	NE	Description
New	N.User.UL.CA.Capability.Avg.PLMN	Cell User Quantity Measurement per Operator	gNode B	This counter has been added to measure the average number of uplink CA-capable UEs for a specific operator in a cell.
New	N.User.3CC.CA.Capability.Avg.PLMN	Cell User Quantity Measurement per Operator	gNode B	This counter has been added to measure the average number of 3CC CA-capable UEs for a specific operator in a cell.

- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0189

RAT

NR FDD, NR TDD

4.3.4.1.2 GP Symbol Shutdown in Special Slots

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Due to long-distance transmission of downlink signals, a victim base station may receive downlink signals from other base stations in GP symbols. If these interfering downlink signals are strong, the RF module's interference protection mechanism is triggered in the corresponding GP symbols, affecting the reception of subsequent uplink symbols. As a result, uplink user experience may deteriorate. For a wideband RF module (an RF module that supports multiple frequency bands), GP symbols are more susceptible to interference due to wideband uplink reception. To address this issue, the GP symbol shutdown in special slots function is introduced.

After GP symbol shutdown in special slots is enabled, if no data is transmitted in the first two GP symbols of a special slot, the gNodeB shuts down these GP symbols and postpones the timing of downlink-to-uplink switching, thereby reducing the impact of interference in the GP symbols on uplink performance. This function is controlled by the **SS_GAP_SYMBOL_SHUTDN_SW** option of the **NRDUCellTimeIntrf. TimeIntrfAvoidSw** parameter.

In LNR multi-mode scenarios (where the RF module also serves an LTE cell), the GP symbol shutdown in special subframes function must also be enabled for the

LTE cell. For details, see section "GP Symbol Shutdown in Special Subframes" of *Interference Detection and Suppression* in eRAN feature documentation.

Implementation

For details, see *Interference Avoidance*. The specific sections are listed in the change history of the document.

Impact on the Network

When the RF module's interference protection mechanism is triggered due to strong interference in GP symbols, enabling this function mitigates the impact of uplink interference on uplink user experience, decreases the value of **N.GAP.ThirdSymbolFromLast.Pwr**, and increases the **User Uplink Average Throughput (DU)**.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

RF modules: Only the AAU5685md supports this function.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCell TimeIntrf. TimeIntrf AvoidSw	MOD NRDUCELLS RSMEAS LST NRDUCELLS RSMEAS	gNodeB	The SS_GAP_SYMBOL_SHUTDN_SW option has been added to specify whether to shut down the first two GP symbols of a special slot and to postpone the timing of downlink-to-uplink switching. These measures prevent time-domain interference in GP symbols.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions

Function Name	Function Switch	Reference	Description
8_2_SS54 adaptation	8_2_SS54_ADAPT_SW option of the gNodeBAlgo.UISchSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	When the NRDUCell.SlotStructure parameter is set to SS54 and both GP symbol shutdown in special slots and 8_2_SS54 adaptation are enabled, 8_2_SS54 adaptation takes effect preferentially.
Uplink symbol power saving	UL_SYMBOL_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When both uplink symbol power saving and GP symbol shutdown in special slots are enabled, the value of VS.RADIOEQM.UL.SymbolShutdown.Dur increases due to GP symbol shutdown.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Interference Avoidance*.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0176

RAT

NR TDD

4.3.4.1.3 Added Support for AAU-level Baseband Processor Shutdown to Not Take Effect on the AAU5685md

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

For the baseband shutdown for energy saving function (controlled by the **BB_SHUTDOWN_POWERSAVING_SW** option of the **gNodeBParam.BaseStationDeepDormancySw** parameter), AAU-level baseband processor shutdown does not take effect on the AAU5685md.

This function is supported only in low frequency bands.

Implementation

For details, see *Energy Conservation and Emission Reduction*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

Except the AAU5685md, all NR-capable AAUs that comply with the eCPRI protocol, work in low frequency bands, and are configured with NR carriers support AAU-level baseband processor shutdown in the baseband shutdown for energy saving function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20260104002018-0060

RAT

NR TDD

4.3.4.1.4 Optimized the Equivalent QCI Mapping Mechanism for SAI-based Service Experience Improvement

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

If a bearer is configured with a QCI for differentiated QoS management for RedCap UEs (specified by **gNBOP5qiConfig.RedCapQci**) and requires service assurance using the equivalent QCI for SAI-based service experience improvement (specified by **gNBServExpGroup.EquivalentQci**), QCI mapping policy optimization must be enabled by selecting the **QCI_MAP_POLICY_OPT_SW** option of the **gNodeBAlgo.QciAlgoOptSw** parameter.

- If **gNBOP5qiConfig.RedCapQci** is set to **65535** and **gNBOP5qiConfig.Qci** is set to a value other than **65535**, the equivalent QCI for SAI-based service experience improvement can be used for bearers of RedCap UEs.
- If **gNBOP5qiConfig.RedCapQci** is set to a value other than **65535** and **gNBOP5qiConfig.Qci** is set to **65535**, the equivalent QCI for SAI-based service experience improvement can be used for bearers of non-RedCap UEs.

Implementation

For details, see *SAI-based Service Experience Improvement*.

Impact on the Network

No impact

Impact on NEs

No impact

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	gNodeBAlgo.QciAlgoOptSw	MOD GNODEBALG O LST GNODEBALG O	gNodeB	The gNodeBAlgo.QciAlgoOptSw parameter has been added to specify the QCI algorithm optimization switch, with the QCI_MAP_POLICY_OPT_SW option used to specify whether to enable QCI mapping policy optimization.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
The descriptions of the following impact relationships have been revised.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Differentiated QoS management for RedCap UEs	gNBOP5qiConfig.RedCapQci set to a value other than 65535	<i>RedCap</i>	- When the QCI_MAP_POLICY_OPT_SW option of the gNodeBAlgo.QciAlgoOptSw parameter is deselected and the corresponding gNBOP5qiConfig MO is configured for a bearer, the equivalent QCI for SAI-based service experience improvement (specified by gNBServExpGroup.EquivalentQci) does not apply to that bearer, and SAI-based service experience improvement does not take effect. - When the QCI_MAP_POLICY_OPT_SW option of the gNodeBAlgo.QciAlgoOptSw parameter is selected and a bearer is configured with a QCI for differentiated QoS management for RedCap UEs (specified by gNBOP5qiConfig.RedCapQci): - If gNBOP5qiConfig.RedCapQci is set to 65535, SAI-based

RAT	Function Name	Function Switch	Reference	Description
				service experience improvement does not take effect for non-RedCap UEs. - If gNBOP5qiConfig.RedCapQci is set to a value other than 65535: - With gNBOP5qiConfig.Qci set to 65535, SAI-based service experience improvement does not take effect for RedCap UEs. - With gNBOP5qiConfig.Qci set to a value other than 65535, SAI-based service experience improvement does not take effect for any UE.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *SAI-based Service Experience Improvement*.

Requirement/Trouble Ticket Number

IR20260104002018

DTS2026011413547

RAT

NR FDD, NR TDD

4.3.4.1.5 Optimized Equivalent QCI Mapping Mechanism for Service Identification-based Assurance

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

If a bearer is configured with a QCI for differentiated QoS management for RedCap UEs (specified by **gNBOP5qiConfig.RedCapQci**) and requires service assurance using the equivalent QCI for service identification-based assurance (specified by **gNBServExpGroup.EquivalentQci**), QCI mapping policy optimization must be enabled by selecting the **QCI_MAP_POLICY_OPT_SW** option of the **gNodeBAlgo.QciAlgoOptSw** parameter.

- If **gNBOP5qiConfig.RedCapQci** is set to **65535** and **gNBOP5qiConfig.Qci** is set to a value other than **65535**, the equivalent QCI for service identification-based assurance can be used for the bearers of RedCap UEs.
- If **gNBOP5qiConfig.RedCapQci** is set to a value other than **65535** and **gNBOP5qiConfig.Qci** is set to **65535**, the equivalent QCI for service identification-based assurance can be used for the bearers of non-RedCap UEs.

Implementation

For details, see *Device-Pipe Synergy*.

Impact on the Network

No impact

Impact on NEs

No impact

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameter has changed.

Change Type	Parameter ID	MML Command	NE	Description
New	gNodeBAlgo.QciAlgoOptSw	MOD GNODEBALG O LST GNODEBALG O	gNodeB	The gNodeBAlgo.QciAlgoOptSw parameter has been added to specify the QCI algorithm optimization switch, with the QCI_MAP_POLICY_OPT_SW option used to specify whether to enable QCI mapping policy optimization.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
The descriptions of the following impact relationships have been revised.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Differentiated QoS management for RedCap UEs	gNBOp5qiConfig.RedCapQci set to a value other than 65535	<i>RedCap</i>	- If the QCI_MAP_POLICY_OPT_SW option of the gNodeBAlgo.QciAlgoOptSw parameter is deselected and the corresponding gNBOp5qiConfig MO is configured for a bearer, the equivalent QCI for service identification-based assurance (specified by gNBServExpGroup.EquivalentQci) of the bearer and service identification-based assurance do not take effect. - If the QCI_MAP_POLICY_OPT_SW option of the gNodeBAlgo.QciAlgoOptSw parameter is selected and a QCI for differentiated QoS management for RedCap UEs (specified by gNBOp5qiConfig.RedCapQci) has been configured for a bearer: - If gNBOp5qiConfig.RedCapQci is set to 65535, service identification-

RAT	Function Name	Function Switch	Reference	Description
				based assurance does not take effect for non-RedCap UEs. - If gNBOp5qiConfig.RedCapQci is set to a value other than 65535: - If gNBOp5qiConfig.Qci is set to 65535, service identification-based assurance does not take effect for RedCap UEs. - If gNBOp5qiConfig.Qci is set to a value other than 65535, service identification-based assurance does not take effect for all UEs.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Device-Pipe Synergy*.

Requirement/Trouble Ticket Number

IR20260104002018

DTS2026011413547

RAT

NR FDD, NR TDD

4.3.4.1.6 Expanded the Application Scenarios of PRACH Fixed Delay Compensation

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In low-frequency scenarios, new slot configurations and preamble formats are added for the PRACH fixed delay compensation function (controlled by the **NRDUCellTrpAlgo.FixedDelayCompensation** parameter). [Table 4-3](#) lists the expanded slot configurations and their supported PRACH configuration indexes.

When the preamble format is **0**, the **NRDUCellTrpAlgo.FixedDelayCompensation** parameter can only be set to a value between **DELAY_5_9** and **DELAY_90_94**.

Table 4-3 Slot configurations and preamble formats supported by PRACH fixed delay compensation

Duplex Mode	Slot Configuration	PUSCH SCS	Cell Radius	PRACH Configuration Index	Preamble Format	PRACH SCS
CELL_TDD	4_1_DDDSU	30 kHz	≤ 9.65 km	200, 202, 210, 65535	C2	15 kHz
CELL_TDD	8_2_DDDDDDD SUU or 7_3_DDDSUDD SUU	30 kHz	> 14.5 km and ≤ 60 km	0 to 7, 12, 17, 65535	0	1.25 kHz
CELL_TDD	8_2_DDDDDDD SUU or 7_3_DDDSUDD SUU	30 kHz	≤ 14.5 km	0 to 7, 12, 17, 65535	0	1.25 kHz
CELL_TDD	8_2_DDDSUUD DDD	30 kHz	≤ 60 km	9, 14, 19, 65535	0	1.25 kHz

Implementation

For details, see *Random Access Control*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Random Access Control*.

Requirement/Trouble Ticket Number

FeR-IR20260109000916-0001

RAT

NR TDD

4.3.4.1.7 Modified the Mechanism of User-Rate-based Dynamic QCI Resource Management

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The mechanism of user-rate-based dynamic QCI resource management has been modified as follows:

- Before the modification: When the **DYNAMIC_QCI_GROUP_SW** option of the **NRDUCellFeatureSw.QciBasedDiffServExpSw** is selected and the **NRDUCellRes.UlRbAverageRatio** or **NRDUCellRes.DlRbAverageRatio** parameter is configured for a QCI-group-based resource group, the gNodeB dynamically adjusts RB allocation based on the minimum rate for non-GBR or GBR services of UEs in the QCI group.
- After the modification: When the **DYNAMIC_QCI_GROUP_SW** option of **NRDUCellFeatureSw.QciBasedDiffServExpSw** is selected and the **NRDUCellRes.UlRbAverageRatio** or **NRDUCellRes.DlRbAverageRatio** parameter is configured for a QCI-group-based resource group, the gNodeB dynamically adjusts RB allocation based on the uplink or downlink target rate specific to dynamic resources for UEs in the QCI group. The uplink and downlink target rates specific to dynamic resources are specified by the **NRDUCellQciBearer.DynResUlTargetRate** and **NRDUCellQciBearer.DynResDlTargetRate** parameters, respectively.

This change is supported only in SA networking.

LampSite base stations do not support this change.

This change is supported only in low frequency bands.

Implementation

For details, see *QCI-based Resource Control*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

Base station models: 3900 and 5900 series base stations (macro base stations).
3900 series base stations must be configured with the BBU3910.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellQciBearer.DynResULTargetRate	ADD NRDUCELLQ CIBEARER MOD NRDUCELLQ CIBEARER LST NRDUCELLQ CIBEARER	gNodeB	This parameter has been added to specify the uplink target rate specific to dynamic resources.
New	NRDUCellQciBearer.DynResDLTargetRate	ADD NRDUCELLQ CIBEARER MOD NRDUCELLQ CIBEARER LST NRDUCELLQ CIBEARER	gNodeB	This parameter has been added to specify the downlink target rate specific to dynamic resources.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *QCI-based Resource Control*.

Requirement/Trouble Ticket Number

IR20260104002018

DTS2026012908119

RAT

NR FDD, NR TDD

4.3.4.1.8 Change in the Value Ranges of the Downlink Scheduling Priority Weight Factor Parameters

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The downlink scheduling priority weighting factors include **gNBDUMacParamGroup.DLSchPriWeightFactor** and **gNBArp.DLSchPriWeightFactor**, which are applicable to different scenarios. Larger values of these parameters result in more scheduling opportunities for services on the corresponding radio bearers and higher user-perceived data rates when multiple services are scheduled concurrently in the cell. For details about **gNBDUMacParamGroup.DLSchPriWeightFactor** and **gNBArp.DLSchPriWeightFactor**, see the "ARP- and QCI-based Scheduling Priority Control" section in *QoS Management*.

Before the change in this function is introduced, the maximum GUI value of **gNBDUMacParamGroup.DLSchPriWeightFactor** and **gNBArp.DLSchPriWeightFactor** is **1000**. In some scenarios, services on radio bearers require more scheduling opportunities. Against this backdrop, the value ranges of the **gNBDUMacParamGroup.DLSchPriWeightFactor** and **gNBArp.DLSchPriWeightFactor** parameters are expanded.

Implementation

- The GUI value range of **gNBDUMacParamGroup.DLSchPriWeightFactor** has been changed from range "**1–1000**" to range "**1–10000**". Considering the step of 0.001, the actual value range has been changed from range "**0.001–1**" to range "**0.001–10**".
- The GUI value range of **gNBArp.DLSchPriWeightFactor** has been changed from range "**1–1000**" to range "**1–10000**". Considering the step of 0.001, the actual value range has been changed from range "**0.001–1**" to range "**0.001–10**".

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNBDUMacParamGroup.DLSchPriWeightFactor	ADD GNBUMAC PARAMGROUP MOD GNBUMAC PARAMGROUP LST GNBUMAC PARAMGROUP	gNodeB	This parameter now supports the values ranging from 1001 to 10000 . The value range has been changed from range " 1-1000 " to range " 1-10000 ". Considering the step of 0.001, the actual value range has been changed from range " 0.001-1 " to range " 0.001-10 ".
Modified	gNBArp.DLSchPriWeightFactor	ADD GNBARP MOD GNBARP LST GNBARP	gNodeB	This parameter now supports the values ranging from 1001 to 10000 . The value range has been changed from range " 1-1000 " to range " 1-10000 ". Considering the step of 0.001, the actual value range has been changed from range " 0.001-1 " to range " 0.001-10 ".

- Performance management

No impact

- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions

The following impact relationship has been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Downlink inter-gNodeB CA	INTER_GNODEB_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>CA Experience Improvement</i>	Consider a downlink inter-gNodeB CA scenario where the version of the gNodeB serving the PCell is V100R022C10SPC050 or later and the version of the gNodeB serving the SCell is V100R022C10SPC010 or earlier. If the gNBDUMacParamGroup.DLSchPriWeightFactor or gNBArp.DLSchPriWeightFactor parameter (Downlink Scheduling Priority Weight Factor) is greater than 1000 , the value of $\gamma(QCI) \times \text{Coeff}_{RFSP}$ (product of the GUI values) may be greater than 65535000 . In this case, due to the version restriction of the gNodeB serving the SCell, the SCell's effective downlink scheduling priority weighting factor (which should use the PCell configuration in a downlink inter-gNodeB CA scenario) is actually less than the PCell's configured value. When it comes to concurrent scheduling of multiple services, the SCell delivers lower data rates to CA UEs than expected based on the PCell's configured scheduling priority weighting factor. For details about $\gamma(QCI) \times \text{Coeff}_{RFSP}$, see the "QoS Guarantee for Downlink Scheduling" section in <i>QoS Management</i> .

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0185

RAT

NR FDD, NR TDD

4.3.4.1.9 Change in the Value Range of the Uplink Scheduling Priority Weight Factor Parameter in the gNBArp MO

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

gNBArp.UlSchPriWeightFactor specifies the weighting factor used for calculating the uplink scheduling priority specific to an ARP value. A larger value of this parameter results in more scheduling opportunities for services on the corresponding radio bearer and a higher user-perceived data rate when multiple services are scheduled concurrently in the cell.

Before the change in this function is introduced, the maximum GUI value of the **gNBArp.UlSchPriWeightFactor** parameter is **1000**. In some scenarios, services on radio bearers require more scheduling opportunities. Against this backdrop, the value range of the **gNBArp.UlSchPriWeightFactor** parameter is expanded.

Implementation

The GUI value range of **gNBArp.UlSchPriWeightFactor** has been changed from range "**1-1000**" to range "**1-10000**". Considering the step of 0.001, the actual value range has been changed from range "**0.001-1**" to range "**0.001-10**".

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameter has changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNBArp.UlSc hPriWeightFactor	ADD GNBARP MOD GNBARP LST GNBARP	gNodeB	This parameter now supports the values ranging from 1001 to 10000 . The value range has been changed from range "1-1000" to range "1-10000". Considering the step of 0.001, the actual value range has been changed from range "0.001-1" to range "0.001-10".

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0185

RAT

NR FDD, NR TDD

4.3.4.2 Transport

4.3.4.2.1 eXn Interface Transmission Capability Improvement

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

This function improves the transmission capability of the eXn interface, thereby increasing the probability that inter-base-station coordination services take effect in heavy-load scenarios.

Implementation

This function improves the transmission capability of the eXn interface, thereby increasing the probability that inter-base-station coordination services take effect in heavy-load scenarios.

Impact on the Network

For inter-base-station coordination services in heavy-load scenarios:

- The values of the counters related to the following function subsets may increase: Cell CA Measurement (CA.Cell), Cell SUL Measurement (SUL.Cell), gNodeB Transmission Interface Measurement (Traffic.TRPIP.gNodeB), Cell User Quantity Measurement (User.Cell), NR DU Cell Scheduled User Quantity Measurement (ScheUser.DuCell), DU Cell SUL Measurement (SUL.DuCell), SCTP Link Measurement (Transport.SCTPLNK), BBU Board Load Measurement (Load.BBUBoard), and gNodeB Board User-Plane CPU Measurement (UserPlane.CPU.NRBoard).
- The values of the counters related to the following function subsets may change: Throughput and Data Volume Measurement (Thp.DuCell), UE Throughput Distribution Measurement (Thp.PDF.DuCell), and DU Cell CA Measurement (CA.DuCell).
- The values of the following counters may change:

Cell Downlink Average Throughput (DU)

User Downlink Average Throughput (DU)

Cell Uplink Average Throughput (DU)

User Uplink Average Throughput (DU)

Average CQI of a cell = $\text{SUM}(k \times \text{N.ChMeas.CQI.SingleCW}.k) / \text{SUM}(\text{N.ChMeas.CQI.SingleCW}.k)$ ($k = 0$ to 15)

N.PRB.DL.DrbUsed.Avg

N.PRB.DL.Used.Avg

Contention-based RA success rate ($\text{N.RA.Contention.Resolution.Succ} / \text{N.RA.Contention.Att} \times 100\%$) and non-contention-based RA success rate ($\text{N.RA.Dedicated.Msg3} / \text{N.RA.Dedicated.Att} \times 100\%$)

- After the data channel power decreases, the overall interference on the network changes. As a result, some KPIs, such as the RRC setup success rate, handover success rate, and service drop rate, may slightly fluctuate.
- The values of the **N.IntergNB.UL.CoMP.SchUser.Sum** and **N.IntergNB.DL.JT.SchUser.Sum** counters may increase.
- The values of the **N.User.IntergNB.ULEnhTimeFreq.Avg** and **N.User.IntergNB.SUL.UL.Avg** counters may increase.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0195

RAT

NR FDD, NR TDD

4.3.4.3 Equipment

None

4.3.5 Operation and Maintenance

4.3.5.1 General Management

There are no changes in general management.

4.3.5.2 Configuration Management

There are no changes in configuration management.

For details about changes in MML commands, MOs, and parameters, see *MML Command Changes* and *MO and Parameter Changes*, which are provided with the software version. You can obtain *MML Command Changes* and *MO and Parameter Changes* from the following directory: **BTS3900&BTS5900 VXXXRXXXCXXSPCXXX Release Documents-EN\01-Release Notes\Delta(MMI)**.

4.3.5.3 Performance Management

This section describes the changes in performance management.

For details about changes in performance counters, see *Performance Counter Changes* delivered with the software version. You can obtain *Performance Counter Changes* from the following directory: **BTS3900&BTS5900 VXXXRXXXCXXSPCXXX Release Documents-EN\01-Release Notes\Delta(MMI)**.

4.3.5.3.1 Optimized 5QI-specific Measurement of Abnormal Releases

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Optimized 5QI-specific measurement of abnormal releases is now supported. It is controlled by the **AMF_QOS_FLOW_REL_STAT_OPT_SW** option of the **gNBOperator.OperatorAlgoSwitch** parameter. If this option is selected, this function is enabled, and an abnormal UE release that occurs in the following scenario is counted as a normal QoS flow release: The gNodeB receives a PDU SESSION RESOURCE MODIFY REQUEST message that requests a QoS flow release, and then the gNodeB sends an RRC Reconfiguration message to the UE. If this option is deselected, this function is disabled, and such a release is counted as an abnormal QoS flow release.

This change is supported only in SA networking.

Implementation

Optimized 5QI-specific measurement of abnormal releases is now supported. When this function is enabled, abnormal releases in the scenario described above now count toward the counters measuring normal releases in the following counter sets, instead of counters measuring abnormal releases:

- Number of QoS Flow Releases in SA Networking
- Number of QoS Flow Releases for a Specific 5QI
- Operator-specific QoS Flow Releases
- Operator-specific VoNR QoS Flow Releases in SA Networking
- QoS Flow Releases for a Network Slice
- Number of QoS Flow Releases with a Specific 5QI per Operator
- QoS Flow Releases for a Specific Flexible Object

Impact on the Network

After an upgrade, if the preceding option is selected in the scenario described above, the values of the counters measuring normal releases increase, and the values of the counters measuring abnormal releases decrease in the following counter sets:

- Number of QoS Flow Releases in SA Networking
- Number of QoS Flow Releases for a Specific 5QI
- Operator-specific QoS Flow Releases
- Operator-specific VoNR QoS Flow Releases in SA Networking
- QoS Flow Releases for a Network Slice
- Number of QoS Flow Releases with a Specific 5QI per Operator
- QoS Flow Releases for a Specific Flexible Object

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNBOperator.OperatorAlgoSwitch	ADD GNBOPERATOR MOD GNBOPERATOR LST GNBOPERATOR	gNodeB	The AMF_QOS_FLOW_REL_STAT_OPT_SW option has been added to specify whether to enable optimized 5QI-specific measurement of abnormal releases.

- Performance management
The following counters have changed.

Change Type	Counter Name	Function Subset	NE	Description
Modified	All counters in the "Number of QoS Flow Releases in SA Networking" counter set	QoS Flow Release Measurement	gNodeB	After this function is enabled, an abnormal UE release that occurs in the following scenario is counted as a normal QoS flow release: The gNodeB receives a PDU SESSION RESOURCE MODIFY REQUEST message that requests a QoS flow release, and then the gNodeB sends an RRC Reconfiguration message to the UE.
Modified	All counters in the "Number of QoS Flow Releases for a Specific 5QI" counter set	5QI QoS Flow Release Measurement	gNodeB	
Modified	All counters in the "Operator-specific QoS Flow Releases" counter set	QoS Flow Measurement per Operator	gNodeB	
Modified	All counters in the "Operator-specific VoNR QoS Flow Releases in SA Networking" counter set	QoS Flow Measurement per Operator	gNodeB	
Modified	All counters in the "QoS Flow Releases for a Network Slice" counter set	Cell Network Slice QoS Flow Measurement	gNodeB	

Change Type	Counter Name	Function Subset	NE	Description
Modified	All counters in the "Number of QoS Flow Releases with a Specific 5QI per Operator" counter set	5QI QoS Flow Measurement per Operator	gNodeB	
Modified	All counters in the "QoS Flow Releases for a Specific Flexible Object" counter set	Cell Flexible Object QoS Flow Release Measurement	gNodeB	

- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

To enable this function, run the **ADD GNBOPERATOR** or **MOD GNBOPERATOR** command with the **AMF_QOS_FLOW_REL_STAT_OPT_SW** option of the **gNBOperator.OperatorAlgoSwitch** parameter selected.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0194

RAT

NR FDD, NR TDD

4.3.5.3.2 Optimized Measurement of the Number of Active UEs

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Optimized measurement of the number of active UEs is now supported. It is controlled by the **ACTIVE_USR_STAT_OPT_SW** option of the **gNBOamParam.StatisticsStrategySwitch** parameter. This function is enabled only if this option is selected. With this function, the number of active UEs is defined as the number of UEs with buffered data at the sampling time.

Implementation

Optimized measurement of the number of active UEs is now supported. With this function, the following counters measure only the number of UEs with buffered data at the sampling time:

- **N.User.RRCCConn.Active.UL.Max**
- **N.User.RRCCConn.Active.UL.Avg**
- **N.User.RRCCConn.Active.DL.Max**
- **N.User.RRCCConn.Active.DL.Avg**
- **N.User.RRCCConn.Active.Max**
- **N.User.RRCCConn.Active.Avg**
- **N.User.RRCCConn.Active.B2H.Avg**
- **N.DRB.Active.DL.Avg**
- **N.DRB.Active.DL.Max**
- **N.User.RRCCConn.Active.UL.Edge.Avg**
- **N.User.RRCCConn.Active.RedCap.Avg**
- **N.User.RRCCConn.Active.RedCap.Max**
- **N.User.RRCCConn.Active.QCI.Max**
- **N.User.RRCCConn.Active.QCI.Avg**
- **N.User.RRCCConn.Active.DuCell5QI.PLMN.Avg**
- **N.User.RRCCConn.Active.DuCell5QI.PLMN.Max**
- **N.User.RRCCConn.Active.DuCell5QI.Max**
- **N.User.RRCCConn.Active.DuCell5QI.Avg**
- **N.DRB.Active.DL.Max.PLMN**
- **N.DRB.Active.DL.Avg.PLMN**
- **N.User.RRCCConn.Active.Max.PLMN**
- **N.User.RRCCConn.Active.Avg.PLMN**
- **N.User.RRCCConn.Active.UL.Edge.Avg.PLMN**
- **N.User.RRCCConn.Active.UL.Avg.PLMN**
- **N.User.RRCCConn.Active.DL.Avg.PLMN**
- **N.User.RRCCConn.Active.RedCap.Avg.PLMN**
- **N.User.RRCCConn.Active.RedCap.Max.PLMN**
- **N.User.RRCCConn.Active.Flex.Max**
- **N.User.RRCCConn.Active.Flex.Avg**
- **N.User.RRCCConn.Active.Max.Ns**

- **N.User.RRCCConn.Active.Avg.Ns**

Impact on the Network

After an upgrade, if the preceding option is selected, the preceding counters will return lower values because they measure only the number of UEs with buffered data at the sampling time. In addition, if the measurement period is short (less than or equal to 15 minutes), the values of the preceding counters are susceptible to service bursts and will therefore vary greatly at different measurement time points.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNBOamParam.StatisticsStrategy-Switch	MOD GNBOAMPARAM LST GNBOAMPARAM	gNodeB	The ACTIVE_USR_STAT_OPT_SW option has been added to specify whether to enable optimized measurement of the number of active UEs.

- Performance management

The following counters have changed.

Change Type	Counter Name	Function Subset	NE	Description
Modified	- N.User.RRCConn.Active.UL.Max - N.User.RRCConn.Active.UL.Avg - N.User.RRCConn.Active.DL.Max - N.User.RRCConn.Active.DL.Avg - N.User.RRCConn.Active.Max - N.User.RRCConn.Active.Avg - N.User.RRCConn.Active.B2H.Avg - N.DRB.Active.DL.Avg - N.DRB.Active.DL.Max - N.User.RRCConn.Active.UL.Edge.Avg - N.User.RRCConn.Active.RedCap.Avg - N.User.RRCConn.Active.RedCap.Max	NR DU Cell Scheduled User Quantity Measurement	gNodeB	After this function is enabled, these counters measure the number of UEs with buffered data at the sampling time.
Modified	- N.User.RRCConn.Active.QCI.Max - N.User.RRCConn.Active.QCI.Avg	DU Cell QCI User Number Measurement	gNodeB	
Modified	- N.User.RRCConn.Active.DuCell5QI.PLMN.Avg - N.User.RRCConn.Active.DuCell5QI.PLMN.Max	DU Cell Operator 5QI Traffic Measurement	gNodeB	

Change Type	Counter Name	Function Subset	NE	Description
Modified	- N.User.RRCConn.Active.DuCell5QI.Max - N.User.RRCConn.Active.DuCell5QI.Avg	DU Cell 5QI Traffic Measurement	gNodeB	
Modified	- N.DRB.Active.DL.Max.PLMN - N.DRB.Active.DL.Avg.PLMN - N.User.RRCConn.Active.Max.PLMN - N.User.RRCConn.Active.Avg.PLMN - N.User.RRCConn.Active.UL.Edge.Avg.PLMN - N.User.RRCConn.Active.UL.Avg.PLMN - N.User.RRCConn.Active.DL.Avg.PLMN - N.User.RRCConn.Active.RedCap.Avg.PLMN - N.User.RRCConn.Active.RedCap.Max.PLMN	NR DU Cell Scheduled User Quantity Measurement per Operator	gNodeB	
Modified	- N.User.RRCConn.Active.Flex.Max - N.User.RRCConn.Active.Flex.Avg	Flexible Object Traffic Volume or Throughput Measurement	gNodeB	
Modified	- N.User.RRCConn.Active.Max.Ns - N.User.RRCConn.Active.Avg.Ns	Network Slice Scheduled UE Number Measurement in a DU Cell	gNodeB	

- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

To enable this function, run the **MOD GNBOAMPARAM** command with the **ACTIVE_USR_STAT_OPT_SW** option of the **gNBAmParam.StatisticsStrategySwitch** parameter selected.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0190

RAT

NR FDD, NR TDD

4.3.5.3.3 Performance Measurement for Flexible Objects with QCI and Operators Specified

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

This function allows for performance measurement for flexible objects with QCI and operators specified in NSA mode. It is recommended that this function be enabled on MOCN networks where performance counter measurement needs to be differentiated by QCI and operator.

This change is supported only in NSA networking.

Implementation

When **gNBFlexCounterObjCfg.FlexCounterObjType** is set to **NRCELL_FLEX_AND_NRDUCCELL_FLEX** or **NRDUCCELL_FLEX**, the following mechanism is applied: If **gNBFlexCounterObjCfg.QciValidPolicy** is set to **NSA**, **gNBFlexCounterObjCfg.Qci** and **gNBFlexCounterObjCfg.OperatorId** can be configured. In this case, the values of **gNBFlexCounterObjCfg.RfspIndex**, **gNBFlexCounterObjCfg.Nr5qi**, and **gNBFlexCounterObjCfg.Arp** are invalid.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

You can run the **MOD GNBFLXCOUNTEROBJCFG** command with the **gNBFlexCounterObjCfg.Qci** and **gNBFlexCounterObjCfg.OperatorId** parameters set.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0180

RAT

NR FDD, NR TDD

4.3.5.4 Software Management

There are no changes in software management.

4.3.5.5 Fault Management

There are no changes in fault management.

4.3.5.6 Inventory Management

There are no changes in inventory management.

4.3.5.7 License Management

There are no changes in license management.

For details about changes in licenses, see *License Changes*, which is provided with the software version. You can obtain *License Changes* from the following directory: **BTS3900&BTS5900 VXXXRXXCXXSPCXXX Release Documents-EN\01-Release Notes\Delta(MMI)**.

5 Changes from V100R021C10SPC260 to V100R022C10SPC010

5.1 Change Summary

5.1.1 Hardware

See section "Hardware Changes" in "Library Information" of *3900 & 5900 Series Base Station Product Documentation*.

5.1.2 Features and Functions

For details, see the descriptions of new features, enhanced features, revised features, function changes, and operation and maintenance.

The changes in the following features are different between high frequency bands and low frequency bands.

Feature ID	Feature Name	Change in Low-Frequency TDD	Change in High-Frequency TDD	Change Difference Between High Frequency Bands and Low Frequency Bands
FOFD-050201	Network Slicing Introduction	Revised	New	- Low-frequency TDD This is a revised feature. - High-frequency TDD This is a new feature.

5.2 Node

5.2.1 New Features

5.2.1.1 MRFD-221264 iLiBattScan (NR)

5.2.1.1.1 iLiBattScan

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

iLiBattScan is now supported by the APM, MTS, and third-party power systems. In Huawei voltage-boosting lithium-ion battery scenarios, polling-based voltage adjustment is performed on the configured battery packs (the voltage of the battery under test must be greater than both the PSU voltage and the voltage of each battery in the queue). This enables the batteries to discharge sequentially and proactively, helping update their state of health (SOH).

iLiBattScan works as follows: The BBU delivers the function switch status and related parameters. Using this information, the power monitoring system raises the voltage of the configured Huawei voltage-boosting lithium-ion battery packs above both the PSU voltage and the voltage of each battery in the queue in polling mode. This ensures that the batteries discharge sequentially, enabling SOH updates.

iLiBattScan can be configured as follows:

Step 1 Verify function support. Specifically, run the **DSP PMU** command to check whether intelligent lithium battery capacity verification tests are supported.

Step 2 Turn on the function switch and configure the required parameters.

- Set **BATCTPA.ILIBATTSCANSW (LTE eNodeB, 5G gNodeB)** to **ON**.
- Set **BATCTPA.ILIBATTSCTM (LTE eNodeB, 5G gNodeB)** to **120**.

----End

LampSite base stations do not support this feature.

Implementation

For details, see *Power Supply Management*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

- Boards

- Main control boards: Only the UMPT supports this function.
- Baseband processing units: N/A
- RF modules: N/A
- Cabinets
 - APM cabinets: Only APM5930 (excluding those using the PMU.22 hardware version) and APM5950 series power cabinets are supported.
 - MTS cabinets: Only the SMU02C power monitoring module is supported.
 - In all power supply scenarios, the configured batteries must be Huawei voltage-boosting lithium-ion batteries, including the ESM-48100C1, ESM-48100B1, ESM-48150B1, ESM-48100A7, ESM-48200B1, ESM-48150A3, ESM-48200A1, and ESM-48100B5.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

This feature is controlled by the following new license control items.

Feature ID	Feature Name	Model	Sales Unit
MRFD-221264	iLiBattScan (NR)	NR0SILIBAS00	per gNodeB
MRFD-221224	iLiBattScan (LTE FDD)	LT1SILIBAS00	per eNodeB
MRFD-221234	iLiBattScan (LTE TDD)	LT4SILIBAS00	per eNodeB
MRFD-221214	iLiBattScan (UMTS)	LQW9ILBS01	per NodeB
MRFD-221204	iLiBattScan (GSM)	LGB3LDHR	per BTS

- Configuration management

The following MML commands have changed.

Change Type	MML Command	NE	Description
Modified	DSP PMU	eGBTS NodeB eNodeB gNodeB	This MML command now supports querying the functions supported by the PMU.

Change Type	MML Command	NE	Description
Modified	DSP BATTR	eGBTS NodeB eNodeB gNodeB	This MML command now supports querying intelligent lithium battery capacity verification test results.

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	BATCTPA.I/LIB ATTSCANSW (LTE eNodeB, 5G gNodeB)	MOD BATCTPA LST BATCTPA	eGBTS NodeB eNodeB gNodeB	This parameter has been added to specify whether to enable intelligent lithium battery capacity verification tests for a board. If this parameter is set to OFF , such a test cannot be performed. If this parameter is set to ON , such a test can be performed.
New	BATCTPA.I/LIB ATTSCTM (LTE eNodeB, 5G gNodeB)	MOD BATCTPA LST BATCTPA	eGBTS NodeB eNodeB gNodeB	This parameter has been added to specify the interval at which intelligent lithium battery capacity verification tests are performed.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions

RAT	Function Name	Function Switch	Reference
NR FDD Low- freque ncy NR TDD High- freque ncy NR TDD LTE FDD LTE TDD UMTS GSM	PSU intelligent coordinated energy saving	IPWRM.PSUICESS set to ON	<i>Green Site</i>
NR FDD Low- freque ncy NR TDD High- freque ncy NR TDD LTE FDD LTE TDD UMTS GSM	PSU current limiting protection in case of AC circuit breaker capability insufficiency	PMU.ACCLF set to ENABLE	<i>Power Supply Management</i>

RAT	Function Name	Function Switch	Reference
NR FDD Low- freque ncy NR TDD High- freque ncy NR TDD LTE FDD LTE TDD UMTS GSM	Staggering electricity usage	EQUIPMENT.STAGGER EDPOWERSWITCH set to ON	<i>Green Site</i>
NR FDD Low- freque ncy NR TDD High- freque ncy NR TDD LTE FDD LTE TDD UMTS GSM	PSU intelligent shutdown	PSUIS.PSUISS set to ENABLE	<i>Green Site</i>

RAT	Function Name	Function Switch	Reference
NR FDD Low- freque ncy NR TDD High- freque ncy NR TDD LTE FDD LTE TDD UMTS GSM	Automatic PSU voltage optimization/ Automatic voltage optimization for intelligent lithium batteries	PMU.BVR set to ON	<i>Green Site</i>
NR FDD Low- freque ncy NR TDD High- freque ncy NR TDD LTE FDD LTE TDD UMTS GSM	SOC level-based low power consumption mode	IPWRM.SOCLVLADJS W set to ON	<i>Green Site</i>

- Impacted features/functions

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low- freque ncy NR TDD High- freque ncy NR TDD LTE FDD LTE TDD UMTS GSM	Power Backup Perception	None	<i>Green Site</i>	iLiBattScan involves charging and discharging the battery. If the power supply is disconnected during the discharge phase, the backup power duration is reduced, causing an alarm related to Power Backup Perception (backup power exhaustion warning) to be reported earlier than expected. This may affect normal base station operation.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Power Supply Management*.

Requirement/Trouble Ticket Number

FeR-IR20250117000486-0020

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.2 Enhanced Features

5.2.2.1 MRFD-191261 Intelligent Coordinated PSU Energy Saving (NR)

5.2.2.1.1 Added Support for PSU Intelligent Coordinated Energy Saving by MTS Series Power Systems

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

MTS series power systems now support PSU intelligent coordinated energy saving.

This function can be enabled by setting the **IPWRM.PSUICESS (5G gNodeB, LTE eNodeB)** parameter to **ON**.

If MTS series power systems are used, the **IPWRM.LRPSW (5G gNodeB, LTE eNodeB)** parameter can be used to specify whether to enable this function based on the load rate.

- If **IPWRM.LRPSW** is set to **ON**, this function is enabled based on the load rate. The load rate range is [**IPWRM.MINLR**, **IPWRM.MAXLR**].
 - If the PSU load rate is within the load rate range, no adjustment is performed.
 - If the PSU load rate is beyond the load rate range, the base station dynamically adjusts the number of working PSUs so that the load rate of each working PSU is within the load rate range.
- If **IPWRM.LRPSW** is set to **OFF**, the base station dynamically adjusts the number of working PSUs by enabling PSU intelligent coordinated energy saving not based on the load rate, so that the conversion efficiency of each working PSU is optimal.

This change is irrelevant to LampSite base stations.

Implementation

For details, see *Green Site*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

If an MTS series power system (whose power monitoring module is an SMU02C) is used, the following requirements must be met:

- The base station is AC powered and runs a version of BTS3900 V100R022C00 or later. And the MTS series power system in use is the one whose power monitoring module is an SMU02C running a version of iSitePower-A V100R024C10 or later.
- The base station is configured with batteries. The circuit breakers between the batteries and the cabinet are turned on, and no battery-related alarm is reported.
- At least three PSUs are installed in the MTS series cabinet.
- No temperature alarm or PMU-/PSU-related alarm is reported in the cabinet.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management

The change in this feature has no impact. It is controlled by the existing feature license.

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	IPWRM.LRP SW (5G gNodeB, LTE eNodeB)	SET IPWRM LST IPWRM	eGBTS NodeB eNodeB gNodeB	This parameter has been added to specify whether to enable PSU intelligent coordinated energy saving based on the load rate.
New	IPWRM.MIN LR (5G gNodeB, LTE eNodeB)	SET IPWRM LST IPWRM	eGBTS NodeB eNodeB gNodeB	This parameter has been added to specify the minimum load rate.
New	IPWRM.MAX LR (5G gNodeB, LTE eNodeB)	SET IPWRM LST IPWRM	eGBTS NodeB eNodeB gNodeB	This parameter has been added to specify the maximum load rate.

- Performance management

The change in this feature has no impact.

- Fault management

The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

The change in this feature has no impact.

- Mutually exclusive features/functions

The change in this feature has no impact.

- Impacted features/functions

The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Green Site*.

Requirement/Trouble Ticket Number

FeR-IR20240808004257-0010

RAT

NR FDD, NR TDD

5.2.3 Revised Features

None

5.2.4 Function Changes

5.2.4.1 Radio

5.2.4.1.1 Modified GNSS and 1588V2 Difference Backhaul According to G.8275.1

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

GNSS and 1588V2 difference backhaul now complies with the G.8275.1 protocol. Specifically, the reported clock phase deviation is the average value within a period.

Implementation

When the **IPCLKLNK.PROFILETYPE (LTE eNodeB, 5G gNodeB)** parameter is set to **G.8275.1** and the **IPCLKLNK.AUTOCMPSTSW (LTE eNodeB, 5G gNodeB)** parameter is set to **ON**, the base station reports the clock phase deviation to the transport network according to the G.8275.1 protocol. The reported value is the average clock phase deviation within a period.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

To enable this function, run the **ADD IPCLKLINK** command with the **IPCLKLNK.PROFILETYPE (LTE eNodeB, 5G gNodeB)** parameter set to **G.8275.1**, and run the **ADD IPCLKLINK** or **MOD IPCLKLINK** command with the **IPCLKLNK.AUTOCMPSTSW (LTE eNodeB, 5G gNodeB)** parameter set to **ON**.

Requirement/Trouble Ticket Number

FeR-IR20250121001496-0020

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.1.2 Optimized the Value of the GNSS Card State Field When the Satellite Card Type Cannot Be Identified

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The value of the **GNSS Card State** field in the **DSP GNSS** command output has been optimized to address the issue that the base station cannot identify the type of the satellite card.

Implementation

In the scenario where the base station cannot identify the type of the satellite card, the cause of the identification failure is displayed in the **GNSS Card State** field in the **DSP GNSS** command output.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact

- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20250319000813

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.1.3 Added Automatic Switchback Between the IEEE 1588V2 and 1PPS+TOD Clock Sources, and Between the IEEE 1588V2 and Peer Clock Sources

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

When the **TASM.SWITCHPOLICY** parameter is set to **IPCLK+TOD**, automatic switchback between the IEEE 1588V2 and 1PPS+TOD clock sources is supported.

- In dual-reference-clock mode of IEEE 1588V2+"1PPS+TOD" in time synchronization:
 - With 1PPS+TOD being used as the current synchronization source, base stations automatically switch back to the IEEE 1588V2 clock if it is available, regardless of whether the 1PPS+TOD clock is normal.
 - With IEEE 1588V2 being used as the current synchronization source: (1) Base stations automatically switch to the 1PPS+TOD clock if the IEEE 1588V2 clock becomes faulty; (2) Base stations automatically switch back to the IEEE 1588V2 clock if it recovers, regardless of whether the 1PPS+TOD clock is normal.
- In other scenarios, automatic switchback to the reference clock source is not supported.

When the **TASM.SWITCHPOLICY** parameter is set to **IPCLK+PEERCLK**, automatic switchback between the IEEE 1588V2 and peer clock sources is supported.

- In dual-reference-clock mode of IEEE 1588V2+peer clock in time synchronization:
 - With the peer clock being used as the current synchronization source, base stations automatically switch back to the IEEE 1588V2 clock if it is available, regardless of whether the peer clock is normal.
 - With IEEE 1588V2 being used as the current synchronization source: (1) Base stations automatically switch to the peer clock if the IEEE 1588V2

clock becomes faulty; (2) Base stations automatically switch back to the IEEE 1588V2 clock if it recovers, regardless of whether the peer clock is normal.

- In other scenarios, automatic switchback to the reference clock source is not supported.

Implementation

For details, see *Synchronization*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameter has changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	TASM.SWITCHPOLICY	SET CLKSWITCHPOLICY LST CLKSWITCHPOLICY	eGBTS NodeB eNodeB gNodeB	- The value IPCLK+TOD has been added to control automatic switchback between the IEEE 1588V2 and 1PPS+TOD clock sources. - The value IPCLK+PEERCLK has been added to control automatic switchback between the IEEE 1588V2 and peer clock sources.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.

Function Name	Function Switch	Reference	Description
Time synchronization	TASM.CLKSYNCMODE (LTE eNodeB, 5G gNodeB) set to TIME	<i>Synchronization</i>	Automatic switchback between the IEEE 1588V2 and 1PPS+TOD clock sources and automatic switchback between the IEEE 1588V2 and peer clock sources require time synchronization and automatic clock selection.
Automatic clock selection	TASM.MODE (LTE eNodeB, 5G gNodeB) set to AUTO	<i>Synchronization</i>	

- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Synchronization*.

Requirement/Trouble Ticket Number

FeR-IR20250806001609-0190

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.1.4 Optimized Maintenance and Test Related to the Duration of IP Clock Unavailability

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The **VS.IPCLK.Status.Unavail.Dur (LTE eNodeB, 5G gNodeB)** counter has been added to measure the duration of IP clock source unavailability. This counter can be used when the base station is configured with the 1588V2 clock source and the IP Clock Link Failure alarm is reported with the specific problem being "Link Negotiation Abnormal", "Clock Quality Unavailable", or others for this clock source.

Implementation

The **VS.IPCLK.Status.Unavail.Dur (LTE eNodeB, 5G gNodeB)** counter is measured every 5s. If the IP Clock Link Failure alarm is reported with the specific problem being "Link Negotiation Abnormal", "Clock Quality Unavailable", or others for the IP clock source, this counter is incremented by 5s. The accumulated value of this counter is reported at the end of a measurement period.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
The following counter has changed.

Change Type	Counter Name	Function Subset	NE	Description
Modified	VS.IPCLK.Status.Unavail.Dur	IP Clock Status Measurement	eGBTS NodeB eNodeB gNodeB	The VS.IPCLK.Status.Unavail.Dur (LTE eNodeB, 5G gNodeB) counter has been added to measure the duration of IP clock source unavailability.

- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250806001609-0090

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.1.5 Synchronization Source Selection Policy After the Maximum Holdover Duration of the Base Station Elapses

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In the case of synchronization source switching, after the maximum holdover duration of the base station elapses, the synchronization source is selected as follows:

- If the priorities of the configured synchronization sources are different, the synchronization sources are selected in descending order of priority.
- If the priorities of the configured synchronization sources are the same, the synchronization source that was last locked is preferentially selected.

Implementation

For details, see *Synchronization*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250806001609-0400

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.1.6 Enhanced Optimization for Scenarios with an Exception on Board Startup in the BBU

NEs Involved

5900 series base stations

Feature/Function Description

When the backplane of a board in the BBU is in poor contact, the base station detects the in-position status change of the board, sets the board status to starting, and waits for the board to restart. However, the board does not restart because it is not removed and inserted. As a result, the board cannot start up to work and remains in the starting state. Optimization has been made for the scenarios with an exception on board startup in the BBU.

Implementation

When the backplane of a board in the BBU is in poor contact, the base station detects the in-position status change of the board and periodically checks the startup status of boards. If a board is in the starting state for a long time, the base station checks the startup flag of the board to determine whether the board has restarted. If the board has not restarted, the base station resets the board and triggers the startup procedure again for the board.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20240902002286

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.1.7 Enhanced Synchronization Reliability for a Clock Source Involving GLONASS When Certain Satellite Cards Are Configured

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

A mechanism has been added to ensure synchronization reliability when a specific satellite card switches back to a GLONASS-only clock source or a multimode clock source involving GLONASS after working in non-GLONASS mode for a long time.

Implementation

In the preceding scenario, this new mechanism ensures that the clock source involving GLONASS can provide timing services properly.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the eGBTS, NodeB, eNodeB, and gNodeB.

Impact on Hardware Requirements

Satellite card type: Specific satellite cards are required. For details, contact Huawei engineers.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20240902002286

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.1.8 Optimization for Scenarios with an Exception on Board Startup in the BBU

NEs Involved

5900 series base stations

Feature/Function Description

Optimization has been made for the following scenarios with an exception on board startup in the BBU:

- Scenario 1: When the backplane of a board in the BBU is in poor contact, the base station detects the power-on/off status change of the board, sets the board status to starting, and waits for the board to restart. However, the board does not restart because it is not powered on or off. As a result, the board cannot start up to work and remains in the starting state.
- Scenario 2: When a board in the BBU starts up and message congestion occurs, the board receives retransmitted startup identification messages after starting up because it does not preserve the sequence of these messages. As a result, the board enters the starting state again after starting up to work properly.

Implementation

The preceding scenarios are optimized as follows:

- Scenario 1: The base station periodically checks the startup status of boards. If a board is in the starting state for a long time, the base station checks the startup flag of the board to determine whether the board has restarted. If the board has not restarted, the base station resets the board and triggers the startup procedure again for the board.
- Scenario 2: After determining that a board has started up, the base station uses board link setup messages with the reliable sequence-preserving retransmission mechanism as board startup identification messages.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

SR-IR20240902002296-0410

SR-IR20240902002296-0500

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.1.9 Added Support for Querying the Historical Status of Grandmaster Quality Class of an IP Clock Server

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

ClockClass is a parameter defined in IP clock protocols (such as IEEE 1588V2 and ITU-T G.8275.1). It indicates the traceability of the time or frequency precision distributed by the grandmaster clock. This change enables query of the historical status of Grandmaster Quality Class by running the **DSP IPCLKCLASSREC** command, facilitating fault locating and improving O&M efficiency. When the IP clock link is normal and the quality level of the IP clock server changes, the main control board records this change. The board can store the latest 100 historical records.

Implementation

For details, see *Synchronization*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management

The following MML commands have changed.

Change Type	Command	NE	Description
New	DSP IPCLKCLASS REC	eGBTS NodeB eNodeB gNodeB	This MML command has been added to query the grandmaster quality level records of an IP clock server.

NOTE

The parameters in these new or deleted MML commands are no longer listed separately.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

Run the **DSP IPCLKCLASSREC** command to query the historical quality levels.

Requirement/Trouble Ticket Number

FeR-IR20250114003265-0010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.1.10 Added an MML Command to Directly Modify the Configuration of a Specified IP Clock Link

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Currently, only adding or removing an **IPCLKLNK** MO is supported on base stations. The parameters of an existing IP clock link cannot be directly modified, and the parameter modification efficiency is low. Therefore, the **MOD IPCLKLINK** command is added to modify the parameters of an IP clock link specified by the link number.

Implementation

For details, see *Synchronization*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following MML commands have changed.

Change Type	Command	NE	Description
New	MOD IPCLKLINK	eGBTS NodeB eNodeB gNodeB	This MML command has been added to directly modify parameters in an IPCLKLNK MO.

NOTE

The parameters in these new or deleted MML commands are no longer listed separately.

- Performance management

No impact

- Fault management

No impact

Impact on Features/Functions

- Prerequisite features/functions

No impact

- Mutually exclusive features/functions

No impact

- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Synchronization*.

Requirement/Trouble Ticket Number

FeR-IR20240822000244-0010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.1.11 PTP Trace Packet Optimization

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

If the **IPCLKLNK.DELAYTYPE (LTE eNodeB, 5G gNodeB)** parameter is set to **P2P**, Pdelay-related packets can be displayed in the **PTP Trace** output.

Implementation

If the **IPCLKLNK.DELAYTYPE (LTE eNodeB, 5G gNodeB)** parameter is set to **P2P** and **Trace Type** is set to **All** on the **PTP Trace** page, the restriction on the **IPCLKLNK.PREHOPIP (LTE eNodeB, 5G gNodeB)** parameter is removed. In this way, Pdelay-related packets, such as Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_up, can be displayed on the **PTP Trace** page.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the eGBTS, NodeB, eNodeB, and gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

SR-IR20240902002286-0500

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.1.12 Optimized the Display of the IP Clock Link Status

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The **DSP IPCLKLINK** command can be executed on the base station to query the running status of an IP clock link. This change optimizes the display of **Parent Port ID** in the command output.

Implementation

The display of **Parent Port ID** in the **DSP IPCLKLINK** command output has been optimized. It now includes the **ClockIdentity** and **SourcePortID** fields.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following MML commands have changed.

Change Type	Command	NE	Description
Modified	DSP IPCLKLINK	eGBTS NodeB eNodeB gNodeB	The display of Parent Port ID in the DSP IPCLKLINK command output has been optimized. It now includes the ClockIdentity and SourcePortID fields.

- Performance management

No impact

- Fault management

No impact

Impact on Features/Functions

- Prerequisite features/functions

No impact

- Mutually exclusive features/functions

No impact

- Impacted features/functions

No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20240902002286

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.1.13 Enhanced Reliability of Synchronization with GLONASS with Certain Satellite Cards Configured

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

A mechanism has been added to ensure the reliability in the scenario where a specific satellite card switches back to a single GLONASS clock source after working in non-GLONASS mode for a long time.

Implementation

In the preceding scenario, this new mechanism ensures that the single GLONASS clock source can provide timing services properly.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the eGBTS, NodeB, eNodeB, and gNodeB.

Impact on Hardware Requirements

Satellite card type: Specific satellite cards are required. For details, contact Huawei engineers.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management

- No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20240902002286

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.1.14 Optimized the DSP GNSSSNR Command Output for ST Satellite Cards Working in BeiDou Mode

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The **DSP GNSSSNR** command output has been optimized for ST satellite cards working in BeiDou mode. Specifically, unavailable BeiDou satellites are no longer displayed.

Implementation

The **DSP GNSSSNR** command output has been optimized for ST satellite cards. For some BeiDou satellites, the availability is determined based on their health status and orbit parameters in the almanac. If the related conditions are not met, these satellites are no longer displayed in the command output.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20250908001612

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.1.15 Removed the Requirement on the Temperature Control Mode of the BBU Fan When Adaptive BBU Deep Dormancy Is Enabled

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The requirements on the temperature control mode of the BBU fan when adaptive BBU deep dormancy is enabled have been removed, allowing adaptive BBU deep dormancy to work independently of the BBU fan's temperature control mode settings.

Implementation

For details, see *Multi-RAT Coordinated Energy Saving*. The specific sections are listed in the change history of the document.

LampSite base stations do not support this feature.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the gNodeB and eNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
The following mutually exclusive relationships have been removed.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD NR FDD Low-frequency NR TDD High-frequency NR TDD	Temperature Control Mode	BBUFAN.TC MODE (5G gNodeB, LTE eNodeB)	None	If the Temperature Control Mode parameter is not set to 2 , BBP deep dormancy cannot be enabled. In separate-MPT scenarios, it is recommended that this parameter be set to 2 for both the local and peer main control boards.

- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250806001609-0420

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD

5.2.4.1.16 Added RHUB Intelligent Shutdown

NEs Involved

5900 series base stations

Feature/Function Description

An RHUB contains two sets of independent data processing units. In scenarios without RHUB-based split, only one set is required to meet service requirements, making the other set redundant and causing energy waste. To address this, the RHUB intelligent shutdown function is introduced to reduce device energy consumption by disabling the redundant data processing unit. For more details on RHUB-based split, see *LampSite Pro Solution Description in 3900 & 5900 Series Base Station Product Documentation*.

This function is controlled by the **RHUB_SHUTDOWN_SW** option of the **EQUIPMENT.EQMDOROPTFEATSW** parameter.

Macro base stations do not support this function.

Implementation

For details, see *Multi-RAT Coordinated Energy Saving*. The specific sections are listed in the change history of the document.

Impact on the Network

It is recommended that RHUB intelligent shutdown be enabled in scenarios without RHUB-based split to reduce base station energy consumption.

Impact on NEs

This function is implemented on the NodeB, gNodeB, and eNodeB.

Impact on Hardware Requirements

Only the RHUB5775 supports this function.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following MML commands have changed.

Change Type	MML Command	NE	Description
Modified	DSP RHUB	NodeB eNodeB gNodeB	The Shutdown Status and Not Shutdown Cause fields have been added to the MML command output.

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	EQUIPMENT .EQMDORO PTFEATSW (LTE eNodeB, 5G gNodeB)	SET EQUIPMENT LST EQUIPMENT	NodeB eNodeB gNodeB	The RHUB_SHUTDOWN_SW option has been added to specify whether to enable RHUB intelligent shutdown.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
The following impact relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
UMTS LTE FDD LTE TDD NR FDD Low-frequency NR TDD	RHUB hardware testing	Commands such as STR HWTST and STR HWOLTST	None	After an RHUB enters the shutdown state, commands such as STR HWTST and STR HWOLTST cannot be used to initiate any RHUB hardware status check.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Multi-RAT Coordinated Energy Saving*.

Requirement/Trouble Ticket Number

FeR-IR20250822000747-001

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS

5.2.4.1.17 Added Main Control Board Dynamic Energy Saving

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

When the transmission rates over the main control board ports are low, the traffic volume is also low. Configuring the main control board with full-load resources in such conditions will lead to energy waste.

To address this, main control board dynamic energy saving is introduced. When the transmission rates over the main control board ports are low, this function can be enabled to dynamically adjust the port transmission rates, reducing the base station's energy consumption.

This function is controlled by the **MPT_DYN_ENG_SAV_FORBID_SW** option of the **EQUIPMENT.EQMDOROPTFEATSW** parameter.

- If this option is selected, main control board dynamic energy saving is forbidden from taking effect.
- If this option is deselected, main control board dynamic energy saving is allowed to take effect.

By default, this option is deselected, allowing main control board dynamic energy saving to take effect.

The BTS5929R does not support this function.

Implementation

For details, see *Multi-RAT Coordinated Energy Saving*. The specific sections are listed in the change history of the document.

Impact on the Network

- Main control board dynamic energy saving can dynamically adjust the transmission rates over the main control board ports, reducing the base station's energy consumption.
- When main control board dynamic energy saving is in effect, the maximum and average board CPU usages (**VS.BBUBoard.CPULoad.Max** and **VS.BBUBoard.CPULoad.Mean**) of the main control board may increase.
- When main control board dynamic energy saving is in effect, the maximum and average user-plane CPU usage of a board in a gNodeB (**VS.NRBoard.UPlane.CPULoad.Max** and **VS.NRBoard.UPlane.CPULoad.Avg**) may increase.

Impact on NEs

This function is implemented on the Node.

Impact on Hardware Requirements

Only the UMPThb series boards support this function.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	EQUIPMENT .EQMDORO PTFEATSW (LTE eNodeB, 5G gNodeB)	SET EQUIPME NT LST EQUIPME NT	gNodeB eNodeB NodeB eGBTS	The MPT_DYN_ENG_SAV_FOR BID_SW option has been added to specify whether to enable main control board dynamic energy saving. By default, this option is deselected, allowing main control board dynamic energy saving to take effect.

- Performance management

No impact

- Fault management

No impact

Impact on Features/Functions

- Prerequisite features/functions

No impact

- Mutually exclusive features/functions

No impact

- Impacted features/functions

The following impact relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TD High-frequency NR TD	Primary/Backup function in intra-frequency dual-network redundancy networking	PRIMARY_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	The primary/backup function in intra-frequency dual-network redundancy networking and main control board dynamic energy saving cannot take effect together.

Related Operations

By default, the change takes effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Multi-RAT Coordinated Energy Saving*.

Requirement/Trouble Ticket Number

FeR-IR20250212002019-0010

IR20250318001011

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.1.18 Optimized the DSP GNSSSNR Command Output for the UBLOX Satellite Card

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The **DSP GNSSSNR** command output for the UBLOX satellite card has been optimized. Specifically, information about satellites with SNR being 0 is no longer displayed in the command output.

Implementation

The **DSP GNSSSNR** command output for the UBLOX satellite card has been optimized. Specifically, the satellites' SNRs in the satellite information captured by the satellite card are checked. If the SNR of a satellite is 0, information about this satellite is no longer displayed in the command output.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20251204000006

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.1.19 Added Support for Reliability Protection During Dormancy in GSM and UMTS#

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Reliability protection during dormancy now applies to GSM and UMTS. It involves license reliability protection, configuration reliability protection, reliability protection against faults, and O&M reliability protection.

LampSite base stations do not support GSM, so this change affects only LampSite base stations in other RATs.

Implementation

For details, see *Multi-RAT Coordinated Energy Saving*. The specific sections are listed in the change history of the document.

Impact on the Network

If any of the following operations is performed, dormancy exit or pending will be triggered for the entire base station. This reduces the dormancy duration of all RF modules (including those working in LTE and NR) with the dormancy function enabled, thereby decreasing energy saving gains.

- License installation is performed on the GSM or UMTS base station.
- An XML configuration file or a batch processing script containing a GBTSFunction or NodeBFunction MML command that affects cell setup or a Node MML command is executed.
- The base station experiences a GSM- or UMTS-related service fault, such as a configuration or resource conflict, which impacts dormancy reliability.
- Software management operations, such as software upgrades, are performed on the GSM or UMTS base station.

Impact on NEs

This change is implemented on the NodeB and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20251204000006-0260

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.1.20 Enhanced Dormancy Functions in Separate-MPT Base Stations Including GSM or UMTS#

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

This change involves RF module deep dormancy and RF module super dormancy.

In a separate-MPT base station where the local RAT is limited to GSM or UMTS, the peer RAT includes LTE or NR, and RF modules are configured on both sides with the working RAT only including LTE or NR, RF module deep or super dormancy is not allowed to take effect before an upgrade but is allowed after the upgrade.

LampSite base stations do not support GSM, so this change affects only LampSite base stations in other RATs.

NOTE

This change was first incorporated into V100R022C10SPC010 but is first described in the network impact report of V100R022C10SPC050.

Implementation

For details, see *Multi-RAT Coordinated Energy Saving*. The specific sections are listed in the change history of the document.

Impact on the Network

In a separate-MPT base station where the local RAT is limited to GSM or UMTS, the peer RAT includes LTE or NR, and RF modules are configured on both sides with the working RAT only including LTE or NR, RF module deep or super dormancy is not allowed to take effect before an upgrade but is allowed after the upgrade. This improves energy saving gains.

Impact on NEs

This change is implemented on the NodeB and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250417002185-0010

DTS2026011015248

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.2 Transport

5.2.4.2.1 Added Function of Using ESNs Defined in the ESP Protocol

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The base station supports the extended sequence numbers (ESNs) defined in the ESP protocol for IPsec SAs to cope with re-negotiation caused by sequence number (SN) reversion upon high-speed traffic. The SN in IPsec traffic is 32 bits. The local end can negotiate with the peer end to support 64-bit SN, reducing the number of IPsec SA re-negotiations before the life cycle expires.

The function of using ESNs defined in the ESP protocol is controlled by the **IPSECPROPOSAL.ESPESNSW** parameter.

Implementation

For details, see *IPsec*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the eGBTS, NodeB, eNodeB, and gNodeB.

Impact on Hardware Requirements

Only the UMPThb series boards support this function.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions

Function Name	Function Switch	Reference	Description
IKE version	IKEPEER.IKEVER SION (LTE eNodeB, 5G gNodeB)	<i>IPsec</i>	If the IKEPEER.IKEVER SION (LTE eNodeB, 5G gNodeB) parameter is set to IKE_V1 , the function of using ESNs defined in the ESP protocol cannot be enabled.

Function Name	Function Switch	Reference	Description
IKE version	SECURITYTEMPLATE.IKEVERSION (LTE eNodeB, 5G gNodeB)	None	If the SECURITYTEMPLATE.IKEVERSION (LTE eNodeB, 5G gNodeB) parameter is set to IKE_V1 , the function of using ESNs defined in the ESP protocol cannot be enabled.
Anti-Replay Window Size	IPSECPOLICY.REPLAYWND (LTE eNodeB, 5G gNodeB)	<i>IPsec</i>	If the IPSECPOLICY.REPLAYWND (LTE eNodeB, 5G gNodeB) parameter is set to WND_DISABLE , the function of using ESNs defined in the ESP protocol cannot be enabled.
Anti-Replay Window Size	SECURITYTEMPLATE.REPLAYWND (LTE eNodeB, 5G gNodeB)	None	If the SECURITYTEMPLATE.REPLAYWND (LTE eNodeB, 5G gNodeB) parameter is set to WND_DISABLE , the function of using ESNs defined in the ESP protocol cannot be enabled.

Function Name	Function Switch	Reference	Description
ESP Authentication Algorithm	IPSECPROPOSAL.ESPAUTHALG (5G gNodeB, LTE eNodeB)	None	If the IPSECPROPOSAL.ESPAUTHALG (5G gNodeB, LTE eNodeB) parameter is set to NULL , the function of using ESNs defined in the ESP protocol cannot be enabled.

- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *IPsec*.

Requirement/Trouble Ticket Number

FeR-IR20250206002234-0150

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.2.2 Optimized SPI Management Mechanism for IKE Negotiation

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The SPI management mechanism for IPsec IKE negotiation has been optimized.

Implementation

The SPI management mechanism for IPsec IKE negotiation has been optimized. Specifically, the internal mechanism for delayed SA release has been optimized.

Impact on the Network

When operations such as unbinding IPsec policies, removing ACL rules, or deleting Ethernet/tunnel interfaces are performed on the base station, this function may

prevent IPsec from processing tail packet data streams, resulting in an increased loss rate of packets received by the IPsec module.

Impact on NEs

This function is implemented on the eGBTS, NodeB, eNodeB, and gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20250319000835

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.2.3 Enhanced Reliability for the IKE Negotiation Failure Alarm

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Reliability has been enhanced for ALM-25891 IKE Negotiation Failure. Specifically, the base station periodically checks the IKE link status when ALM-25891 IKE Negotiation Failure is reported with the **Specific Problem** being **IKE Negotiation Parameter Inconsistency**, **IKE Negotiation Timeout**, or **Other Cause**. This alarm is cleared once the IKE link returns to normal.

Implementation

The base station periodically checks the IKE link status when ALM-25891 IKE Negotiation Failure is reported with the **Specific Problem** being any of the following. This alarm is cleared once the IKE link returns to normal.

- IPsec Negotiation Parameter Inconsistency
- IKE Negotiation Timeout
- Other Cause

Impact on the Network

No impact

Impact on NEs

This function is implemented on the eGBTS, NodeB, eNodeB, and gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
The following alarm has changed.

Change Type	Name	NE	Description
Modified	ALM-25891 IKE Negotiation Failure	eGBTS NodeB eNodeB gNodeB	Reliability has been enhanced for ALM-25891 IKE Negotiation Failure. Specifically, the base station periodically checks the IKE link status when ALM-25891 IKE Negotiation Failure is reported with the Specific Problem being IKE Negotiation Parameter Inconsistency, IKE Negotiation Timeout, or Other Cause . This alarm is cleared once the IKE link returns to normal.

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20250319000835

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.2.4 Added the Support Forcible Execution Parameter to Commands That Modify ACL Rules for IPsec Tunnel Services

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The **Support Forcible Execution** parameter has been added to commands that modify ACL rules for IPsec tunnel services. Specifically, **ACLRULE.FORCEEXECUTE**

(5G gNodeB, LTE eNodeB) has been introduced in **MOD ACLRULE**, and **ACLRULE6.FORCEEXECUTE (5G gNodeB, LTE eNodeB)** has been introduced in **MOD ACLRULE6**. Source IP address verification for ACL rules is supported for the **MOD ACLRULE**, **MOD ACLRULE6**, and **MOD IPSECBINDITF** commands. When the preceding MML commands are executed, the **Support Forcible Execution** parameter is handled as follows:

- If this parameter is set to **NO**, the source IP address in an ACL rule referenced by IPsec is verified upon MML command delivery.
 - If the source IP address does not match any IP address configured on the base station, the system returns a failure message with a description of the cause.
 - If the source IP address matches an IP address configured on the base station, the system proceeds with the configuration change.
- If this parameter is set to **YES**, the configuration change is executed directly upon MML command delivery without assessing service impact.

This function extends the execution time of a single MML command by up to 2s.

Implementation

For details, see *IPsec*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the eGBTS, NodeB, eNodeB, and gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following MML commands have changed.

Change Type	MML Command	NE	Description
Modified	MOD ACLRULE	eGBTS NodeB eNode B gNode B	The ACLRULE.FORCEEXECUTE (5G gNodeB, LTE eNodeB) parameter has been added to this command to specify whether an ACL rule configuration is forcibly adjusted.
Modified	MOD ACLRULE6	eGBTS NodeB eNode B gNode B	The ACLRULE6.FORCEEXECUTE (5G gNodeB, LTE eNodeB) parameter has been added to this command to specify whether an IPv6 ACL rule configuration is forcibly adjusted.

No impact

- Performance management

No impact

- Fault management

No impact

Impact on Features/Functions

- Prerequisite features/functions

No impact

- Mutually exclusive features/functions

No impact

- Impacted features/functions

No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250806001609-0150

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.2.5 Expanded Transmission Capacity of the UMPThb

NEs Involved

5900 series base stations

Feature/Function Description

The UMPThb has been added to expand the transmission capability of the UMPTh.

- The transmission and forwarding rate of the UMPThb can now reach 50 Gbit/s.
- The number of IPsec SAs supported by the UMPThb has been increased from 512 to 1024.
- The minimum effective value of the **IPv4 Maximum Transmission Unit** parameter is **256**. If this parameter is set to a value ranging from **46** to **256**, the value **256** takes effect and ALM-26245 Configuration Data Inconsistency is reported.

This function involves the modification of the following parameters:

ETHPORT.SPEED (LTE eNodeB, 5G gNodeB), **RSCGRP.TXBW (LTE eNodeB, 5G gNodeB)**, **RSCGRP.RXBW (LTE eNodeB, 5G gNodeB)**, **RSCGRP.TXCBS (LTE eNodeB, 5G gNodeB)**, **RSCGRP.TXEBS (LTE eNodeB, 5G gNodeB)**, **RSCGRP.TXCIR (LTE eNodeB, 5G gNodeB)**, **RSCGRP.RXCIR (LTE eNodeB, 5G gNodeB)**, **RSCGRP.TXPIR (LTE eNodeB, 5G gNodeB)**, **RSCGRP.RXPIR (LTE eNodeB, 5G gNodeB)**, **RSCGRP.TXPBS (LTE eNodeB, 5G gNodeB)**, **RSCGRP.PALG.TXRSVBW (LTE eNodeB, 5G gNodeB)**, **RSCGRP.PALG.TXBWAMIN (LTE eNodeB, 5G gNodeB)**, **RSCGRP.PALG.RXBWAMIN (LTE eNodeB, 5G gNodeB)**, and **RSCGRP.PALG.RXRSVBW (LTE eNodeB, 5G gNodeB)**.

Implementation

For details, see *IPv4 Transmission*, *IPv6 Transmission*, or *IP Active Performance Measurement*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the eNodeB, gNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

The UMPThb is required.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	ETHPORT.SPEED (LTE eNodeB, 5G gNodeB)	ADD ETHPORT MOD ETHPORT	eGBTS NodeB eNodeB gNodeB	The enumerated value 50G has been added to this parameter. 50G optical modules support only RS-FEC and FEC cannot be disabled. Therefore, when this parameter is set to 50G , FECMODE is unavailable.
Modified	RSCGRP.TXBW (LTE eNodeB, 5G gNodeB)	ADD RSCGRP MOD RSCGRP	eGBTS NodeB eNodeB gNodeB	The GUI value range of this parameter has been changed to 32 to 50000000 .
Modified	RSCGRP.RXBW (LTE eNodeB, 5G gNodeB)	ADD RSCGRP MOD RSCGRP	eGBTS NodeB eNodeB gNodeB	The GUI value range of this parameter has been changed to 32 to 50000000 .
Modified	RSCGRP.TXCBS (LTE eNodeB, 5G gNodeB)	ADD RSCGRP MOD RSCGRP	eGBTS NodeB eNodeB gNodeB	The GUI value range of this parameter has been changed to 64 to 50000000 .
Modified	RSCGRP.TXEBS (LTE eNodeB, 5G gNodeB)	ADD RSCGRP MOD RSCGRP	eGBTS NodeB eNodeB gNodeB	The GUI value range of this parameter has been changed to 64 to 50000000 .
Modified	RSCGRP.TXCIR (LTE eNodeB, 5G gNodeB)	ADD RSCGRP MOD RSCGRP	eGBTS NodeB eNodeB gNodeB	The GUI value range of this parameter has been changed to 64 to 50000000 .

Change Type	Parameter ID	MML Command	NE	Description
Modified	RSCGRP.RXCIR (LTE eNodeB, 5G gNodeB)	ADD RSCGRP MOD RSCGRP	eGBTS NodeB eNodeB gNodeB	The GUI value range of this parameter has been changed to 64 to 50000000 .
Modified	RSCGRP.TXPIR (LTE eNodeB, 5G gNodeB)	ADD RSCGRP MOD RSCGRP	eGBTS NodeB eNodeB gNodeB	The GUI value range of this parameter has been changed to 64 to 50000000 .
Modified	RSCGRP.RXPIR (LTE eNodeB, 5G gNodeB)	ADD RSCGRP MOD RSCGRP	eGBTS NodeB eNodeB gNodeB	The GUI value range of this parameter has been changed to 64 to 50000000 .
Modified	RSCGRP.TXPBS (LTE eNodeB, 5G gNodeB)	ADD RSCGRP MOD RSCGRP	eGBTS NodeB eNodeB gNodeB	The GUI value range of this parameter has been changed to 64 to 50000000 .
Modified	RSCGRPALG.TXRSVBW (LTE eNodeB, 5G gNodeB)	SET RSCGRPALG	eGBTS NodeB eNodeB gNodeB	The GUI value range of this parameter has been changed to 0 to 50000000 .
Modified	RSCGRPALG.TXBWAMIN (LTE eNodeB, 5G gNodeB)	SET RSCGRPALG	eGBTS NodeB eNodeB gNodeB	The GUI value range of this parameter has been changed to 64 to 50000000 .
Modified	RSCGRPALG.RXBWAMIN (LTE eNodeB, 5G gNodeB)	SET RSCGRPALG	eGBTS NodeB eNodeB gNodeB	The GUI value range of this parameter has been changed to 64 to 50000000 .
Modified	RSCGRPALG.RXRSVBW (LTE eNodeB, 5G gNodeB)	SET RSCGRPALG	eGBTS NodeB eNodeB gNodeB	The GUI value range of this parameter has been changed to 0 to 50000000 .

- Performance management
No impact
- Fault management
The following alarms have changed.

Change Type	Alarm Name	NE	Description
Modified	ALM-25879 Ethernet Port Broadcast Packets Exceeding Alarm	eGBTS NodeB eNodeB gNodeB	The port rate 50 Gbit/s has been added to the alarm description.

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250206002234-0150

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.2.6 Optimized LiteCA Whitelist Mechanisms for Expired Initial Certificates and ESNs

NEs Involved

3900 series base stations, 5900 series base stations, USU3910

Feature/Function Description

This function includes the following optimizations:

- The whitelist function for expired initial certificates can be enabled on LiteCA when HTTPS authentication is not used.

- LiteCA can synchronize the whitelist of expired ESNs on the MAE.

Implementation

This function is implemented as follows:

- LiteCA relaxes its constraints, allowing the whitelist function for expired initial certificates to be enabled when HTTPS authentication is not used.
- After the whitelist function for expired initial certificates is enabled, LiteCA will synchronize the whitelist of expired ESNs on the MAE.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the eGBTS, NodeB, eNodeB, and gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250610000483-0020

IR20250701003000

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.2.7 Automatic Retrieval of an Operator-Issued Device Certificate for the Standby UMPT

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

An operator-issued device certificate can be automatically retrieved for the standby UMPT. If a base station is configured with two UMPT boards of the same model and the two boards are configured to work in active/standby mode, the base station requests operator-issued device certificates from the connected CA as follows before and after an upgrade:

- If the source version of a base station is earlier than V100R022C00 before an upgrade, an operator-issued device certificate is requested for the active UMPT during the deployment phase. The base station can automatically request an operator-issued device certificate for the original standby UMPT only if the active and standby boards are switched over. If the source version of a base station is between V100R022C00 (inclusive) and V100R022C10 (not inclusive) before an upgrade, users can request an operator-issued device certificate for the standby UMPT by running the **REQ DEVCERT** command on the base station with the cabinet, subrack, and slot numbers of the standby UMPT specified. When the active/standby relationship of the two boards is switched, the standby UMPT automatically requests an operator-issued device certificate that was unavailable before the switchover according to the configuration file.
- After the base station is upgraded to V100R022C10 or later, automatic management and manual update are supported for the standby operator-issued device certificate on the standby board.
 - Automatic management of the standby operator-issued device certificate on the standby board: The standby UMPT periodically checks the status of its standby certificate corresponding to the certificate used by the active UMPT. If the standby certificate is abnormal (expired, damaged, lost, revoked, or invalid), the base station automatically obtains a new certificate for the standby board. If the standby operator-issued device certificate is nearing expiration and **CERTCHKTSK.UPDATEMETHOD (5G gNodeB, LTE eNodeB)** is set to **CMP**, the certificate is automatically updated.
 - Manual update of the standby operator-issued device certificate on the standby board: If the standby operator-issued device certificate is nearing expiration and **CERTCHKTSK.UPDATEMETHOD (5G gNodeB, LTE**

eNodeB) is set to **MANUAL**, users can update the operator-issued device certificate on the standby UMPT by running the **UPD DEVCERT** command with the cabinet, subrack, and slot numbers of the standby UMPT specified.

This function has also been optimized as follows:

- For trust certificates with identical **CA.CANAME (LTE eNodeB, 5G gNodeB)** parameter values, the BBU distinguishes them by the specific identifier of each CA when applying for operator-issued device certificates on behalf of the AAU, instead of randomly selecting one.
- If the operator-issued certificate of the active board or the certificate in a single main control board scenario has expired, is not yet effective, or has been revoked, the base station automatically attempts to re-apply for a certificate to obtain a valid certificate file, ensuring that service setup is not affected.

Implementation

For details, see *PKI*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the eGBTS, NodeB, eNodeB, and gNodeB.

Impact on Hardware Requirements

UMPTb/UMPTe/UMPTg/UMPTga/UMPTHa series boards support this function.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following MML commands have changed.

Change Type	MML Command	NE	Description
Modified	UPD DEVCERT	eGBTS NodeB eNode B gNode B	The Cabinet No. , Subrack No. , and Slot No. parameters have been added to this command to specify the standby board so that an operator-issued device certificate can be updated for the standby board.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250604000235-0010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.2.8 Added the Support for Obtaining CRLs over HTTP

NEs Involved

3900 series base stations, 5900 series base stations, USU3910

Feature/Function Description

The base station complies with RFC 2585 and can obtain CRLs over HTTP. Specifically, it can obtain CRLs from a connected CRL server over HTTP or HTTPS. To support this, **CRLTSK.CRLGETMETHOD (LTE eNodeB, 5G gNodeB)** must be set

to **HTTP**. Other parameters should be configured as follows, based on whether HTTP or HTTPS is used.

- If HTTP is used to obtain CRLs, **CRLTSK.PORT (LTE eNodeB, 5G gNodeB)** must be configured to specify the port number used to find the desired HTTP server.
- If HTTPS is used to obtain CRLs, **CRLTSK.PORT (LTE eNodeB, 5G gNodeB)** must be configured to specify the port number used to find the desired HTTP server, **CRLTSK.CONNMODE (5G gNodeB, LTE eNodeB)** must be set, and **CRLTSK.AUTHPEER (5G gNodeB, LTE eNodeB)** must be set to **ENABLE**.

NOTE

When the base station uses HTTP to obtain CRLs, the HTTP URL for obtaining the CRL consists of the HTTP packet header, connection mode (**CRLTSK.CONNMODE (5G gNodeB, LTE eNodeB)**), colon (:) and slash (/), IP address (**CRLTSK.IP (5G gNodeB, LTE eNodeB)**) or IPv6 address (**CRLTSK.IP6 (LTE eNodeB, 5G gNodeB)**), port number (**CRLTSK.PORT (LTE eNodeB, 5G gNodeB)**), and CRL file name (**CRLTSK.FILENAME (LTE eNodeB, 5G gNodeB)**). For example, if the connection mode is SSL, the IP address is 192.168.23.112, the port number is 443, and the CRL file name is /test.crl, then the HTTP URL is HTTPS://192.168.23.112:443/test.crl.

When the base station uses HTTP to obtain CRLs, the certificate deployment position must remain at the default, and neither username nor password is required. The slave IP address (**CRLTSK.SLVIP (5G gNodeB, LTE eNodeB)**) or slave IPv6 address (**CRLTSK.SLVIP6 (LTE eNodeB, 5G gNodeB)**) cannot be set.

Implementation

For details, see *PKI, Equipment Security, SSL, and Transmission Security Overview*. The specific sections are listed in the change history of the documents.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the eGBTS, NodeB, eNodeB, and gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

An interface with RFC 2585 capability for retrieving CRLs over HTTP has been added.

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	CRLTSK.CRLGETMETHOD (LTE eNodeB, 5G gNodeB)	ADD CRLTSK LST CRLTSK	eGBTS NodeB eNodeB gNodeB	The enumerated value HTTP has been added to the CRLTSK.CRLGETMETHOD (LTE eNodeB, 5G gNodeB) parameter to enable periodic CRL retrieval over HTTP.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250912001539-0010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.2.9 Enhanced Automatic Certificate Management on the MAE

NEs Involved

3900 series base stations, 5900 series base stations, USU3910

Feature/Function Description

This function enhances automatic certificate management on the MAE. Specifically, after the link between the MAE and a disconnected NE is restored, the MAE can automatically replace the NE's initial certificate with an operator-issued certificate.

Implementation

After an automatic initial certificate replacement task is created for a disconnected NE on the MAE, the MAE automatically replaces the NE's initial certificate with an operator-issued certificate once the link between the NE and MAE is restored.

Impact on the Network

No impact

Impact on NEs

This change does not involve NE requirements and is implemented only on the MAE.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250213000210-0010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.2.10 Added Switches for Disabling Base Station OMCH Self-recovery

NEs Involved

3900 series base stations, 5900 series base stations, USU3910

Feature/Function Description

Disable switches have been added to the OMCH self-recovery function, which includes automatic remote OMCH establishment, automatic remote OMCH recovery by using running data, and transmission link fault rectification.

Implementation

Disable switches have been added to the OMCH self-recovery function, which includes automatic remote OMCH establishment, automatic remote OMCH recovery by using running data, and transmission link fault rectification. For details about the parameters, see [Table 5-1](#).

When a disable switch is turned on, the function is disabled. When the disable switch is turned off, the function is enabled. The disable switches are turned off by default. If the switches are turned on, the remote OMCH may fail to automatically recover after being interrupted, leading to continuous interruption of the remote OMCH. Therefore, do not turn on the disable switches unless necessary.

Table 5-1 Added switches for disabling the OMCH self-recovery function

Parameter	Switch Name	Function Description
AUTO_OMCH_ESTAB_POL1_DIS_SW option of the TRANSALGEXTPARA.OMCH MNGSW (LTE eNodeB, 5G gNodeB) parameter	AUTO_OMCH_ESTAB_POL1_DIS_SW	The automatic OMCH establishment function allows a base station to recover the remote OMCH through the DHCP Detect process. Policy 1 applies to OMCH recovery when services are normal. If this option is selected, the function is disabled. If this option is deselected, the function is enabled.

Parameter	Switch Name	Function Description
AUTO_OMCH_ESTAB_POL2_DIS_SW option of the TRANSALGEXTPARA.OMCH MNGSW (LTE eNodeB, 5G gNodeB) parameter	AUTO_OMCH_ESTAB_POL2_DIS_SW	The automatic OMCH establishment function allows a base station to recover the remote OMCH through the DHCP Detect process. Policy 2 applies to OMCH recovery when services are abnormal. If this option is selected, the function is disabled. If this option is deselected, the function is enabled.
AUTO_OMCH_RECOV_POL1_DIS_SW option of the TRANSALGEXTPARA.OMCH MNGSW (LTE eNodeB, 5G gNodeB) parameter	AUTO_OMCH_RECOV_POL1_DIS_SW	The automatic OMCH recovery function recovers the OMCH of a base station by using the running data of the OMCH during stable connection. Policy 1 applies to OMCH recovery when services are normal. If this option is selected, the function is disabled. If this option is deselected, the function is enabled.
AUTO_OMCH_RECOV_POL2_DIS_SW option of the TRANSALGEXTPARA.OMCH MNGSW (LTE eNodeB, 5G gNodeB) parameter	AUTO_OMCH_RECOV_POL2_DIS_SW	The automatic OMCH recovery function recovers the OMCH of a base station by using the running data of the OMCH during stable connection. Policy 2 applies to OMCH recovery when services are abnormal. If this option is selected, the function is disabled. If this option is deselected, the function is enabled.
TRANS_LNK_FAULT_RECT_DIS_SW option of the TRANSALGEXTPARA.OMCH MNGSW (LTE eNodeB, 5G gNodeB) parameter	TRANS_LNK_FAULT_RECT_DIS_SW	If this option is selected, the transmission link fault rectification function is disabled. If this option is deselected, the transmission link fault rectification function is enabled.

Parameter	Switch Name	Function Description
OMCH_MAINT_REPORT_DISABLE_SW option of the TRANSALGEXTPARA.OMCH MNGSW (LTE eNodeB, 5G gNodeB) parameter	OMCH_MAINT_REPORT_DISABLE_SW	If this option is selected, the OMCH maintenance information reporting function is disabled. If this option is deselected, the OMCH maintenance information reporting function is enabled.
PRECISE_OMCH_RECOVERY_DISABLE_SW option of the TRANSALGEXTPARA.OMCH MNGSW (LTE eNodeB, 5G gNodeB) parameter	PRECISE_OMCH_RECOVERY_DISABLE_SW	If this option is selected, the precise OMCH recovery function is disabled. If this option is deselected, the precise OMCH recovery function is enabled.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, eGBTS, and GBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	TRANSALGEXTPARA.OMCHMNGSW (LTE eNodeB, 5G gNodeB)	SET TRANSALGEXTPARA LST TRANSALGEXTPARA	eGBTS NodeB eNodeB gNodeB	The AUTO_OMCH_ESTAB_POLICY1 option has been added to specify whether to disable automatic OMCH establishment policy 1.
Modified	TRANSALGEXTPARA.OMCHMNGSW (LTE eNodeB, 5G gNodeB)	SET TRANSALGEXTPARA LST TRANSALGEXTPARA	eGBTS NodeB eNodeB gNodeB	The AUTO_OMCH_ESTAB_POLICY2 option has been added to specify whether to disable automatic OMCH establishment policy 2.
Modified	TRANSALGEXTPARA.OMCHMNGSW (LTE eNodeB, 5G gNodeB)	SET TRANSALGEXTPARA LST TRANSALGEXTPARA	eGBTS NodeB eNodeB gNodeB	The AUTO_OMCH_RECOVERY_POLICY1 option has been added to specify whether to disable automatic OMCH recovery policy 1.

Change Type	Parameter ID	MML Command	NE	Description
Modified	TRANSALGE XTPARA.OM CHMNGSW (LTE eNodeB, 5G gNodeB)	SET TRANSALGEX TPARA LST TRANSALGEX TPARA	eGBTS NodeB eNodeB gNodeB	The AUTO_OMC H_RECOV_P OL2_DIS_SW option has been added to specify whether to disable automatic OMCH recovery policy 2.
Modified	TRANSALGE XTPARA.OM CHMNGSW (LTE eNodeB, 5G gNodeB)	SET TRANSALGEX TPARA LST TRANSALGEX TPARA	eGBTS NodeB eNodeB gNodeB	The TRANS_LNK _FAULT_REC T_DIS_SW option has been added to specify whether to disable transmission link fault rectification.
Modified	TRANSALGE XTPARA.OM CHMNGSW (LTE eNodeB, 5G gNodeB)	SET TRANSALGEX TPARA LST TRANSALGEX TPARA	eGBTS NodeB eNodeB gNodeB	The OMCH_MAI NT_REPORT _DIS_SW option has been added to specify whether to disable OMCH maintenance information reporting.

Change Type	Parameter ID	MML Command	NE	Description
Modified	TRANSALGE XTPARA.OM CHMNGSW (LTE eNodeB, 5G gNodeB)	SET TRANSALGEX TPARA LST TRANSALGEX TPARA	eGBTS NodeB eNodeB gNodeB	The PRECISE_OMCH_RECOVERY_DIS_SW option has been added to specify whether to disable precise OMCH recovery.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20250319000835

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.2.11 Added Support for Precise Reporting of X2/eX2/gNBCUX2/gNBCUXn/gNBDUeXn User-Plane Fault Alarms

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

When static GTP-U echo is enabled, the mechanism of reporting X2/eX2/gNBCUX2/gNBCUXn/gNBDeXn user-plane fault alarms is optimized to improve the accuracy of user-plane fault alarms. If a user-plane path fault is detected and there are X2/eX2/gNBCUX2/gNBCUXn/gNBDeXn services triggering a bearer setup attempt, an X2/eX2/gNBCUX2/gNBCUXn/gNBDeXn service alarm will be reported, enabling precise reporting of user-plane fault alarms. This function is enabled when the **INTER_UP_ALM_PRECISE_SW** option of the **TRANSALGEXTPARA.TNLBEARERMNGALGSW (LTE eNodeB, 5G gNodeB)** parameter is selected.

If an X2/eX2/gNBCUX2/gNBCUXn/gNBDeXn user-plane fault alarm has been reported before the **INTER_UP_ALM_PRECISE_SW** option of the **TRANSALGEXTPARA.TNLBEARERMNGALGSW (LTE eNodeB, 5G gNodeB)** parameter is selected but there are no X2/eX2/gNBCUX2/gNBCUXn/gNBDeXn services triggering a bearer setup attempt, the alarm will be cleared. If there are services triggering a bearer setup attempt, a corresponding alarm will be reported.

If the **INTER_UP_ALM_PRECISE_SW** option of the **TRANSALGEXTPARA.TNLBEARERMNGALGSW (LTE eNodeB, 5G gNodeB)** parameter is selected and there are no X2/eX2/gNBCUX2/gNBCUXn/gNBDeXn services triggering a bearer setup attempt, any X2/eX2/gNBCUX2/gNBCUXn/gNBDeXn user-plane fault alarm reported before any of the following situations occurs will be cleared after any situations occurs. If there are services triggering a bearer setup attempt, a corresponding alarm will be reported.

- The base station is reset.
- The main control board is either reset or removed and re-added.
- Links are reestablished because an application is either reset or removed and re-added. The application type must be set to gNodeB or eNodeB.
- The parameters of the X2/eX2/gNBCUX2/gNBCUXn/gNBDeXn object are either modified or removed and re-added.

NOTE

- Precise reporting of eX2/gNBCUX2/gNBCUXn/gNBDeXn user-plane fault alarms only works in the endpoint configuration mode, while precise reporting of X2 user-plane fault alarms works in both the link and endpoint configuration modes.
- Precise reporting of eX2 user-plane fault alarms applies only to eX2 interfaces in non-ideal backhaul mode (with **EX2.BackhaulType** set to **NON_IDEAL**).

Implementation

For details, see *IPv4 Transmission* or *IPv6 Transmission*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the eNodeB and gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	TRANSALGEXTPARA.TNLBEARERMNGALGSW (LTE eNodeB, 5G gNodeB)	SET TRANSALGEXTPARA LST TRANSALGEXTPARA	eGBTS NodeB eNodeB gNodeB	The INTER_UP_ALM_PRECISE_SW option has been added to specify whether to enable precise reporting of X2/eX2/gNBCUX2/gNBCUXn/gNBDUeXn user-plane fault alarms.

- Performance management

No impact

- Fault management

No impact

Impact on Features/Functions

- Prerequisite features/functions

No impact

- Mutually exclusive features/functions

No impact

- Impacted features/functions

No impact

Related Operations

By default, the change does not take effect after an upgrade.

Select the **INTER_UP_ALM_PRECISE_SW** option of the **TRANSALGEXTPARA.TNLBEARERMNGALGSW (LTE eNodeB, 5G gNodeB)** parameter to enable precise reporting of X2/eX2/gNBCUX2/gNBCUXn/gNBDUeXn user-plane fault alarms.

Requirement/Trouble Ticket Number

FeR-IR20240909002853-0010

IR20241101000991

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD

5.2.4.2.12 Support for Filtering Fragmented Packets by Protocol in IPv6 Tracing

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Optimizations have been implemented to resolve the issue where fragmented passerby packets cannot be traced when IPv6 protocol tracing is enabled and the filtering protocol type is ICMPv6, TCP, UDP, SCTP, or GTPU.

The following constraints apply:

1. Only IPv6 packets containing basic IPv6 packet header, fragment extension header, and upper-layer extension header are supported.
2. The first, middle, and last fragments can be traced only when the protocol type is specified.
3. Only the first fragment can be traced if both the protocol type and port number are specified. (The middle and last fragments do not have port number information.)

Implementation

The ICMPv6/TCP/UDP/SCTP/GTPU protocol type and port number are parsed to match and report fragmented packets.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the eNodeB and gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20240902002286

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD

5.2.4.2.13 Support for TWAMP Defined in RFC 5938#

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

TWAMP defined in RFC 5938 is supported. This is, individual session control mode can be used. The individual session control function is controlled by the **TWAMPCLIENT.INDIVSESSCTRLSW (LTE eNodeB, 5G gNodeB)** parameter. This function takes effect only in full architecture when the Controller and Responder both comply with RFC 5938.

Implementation

For details, see *IP Active Performance Measurement*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following MML commands have changed.

Change Type	MML Command	NE	Description
Modified	DSP TWAMPCLIENT	eGBTS NodeB eNodeB gNodeB	The Individual Session Control Switch State field has been added to the output of this command.
Modified	DSP TWAMPRESPONDER	eGBTS NodeB eNodeB gNodeB	The Individual Session Control Switch State field has been added to the output of this command.

Change Type	MML Command	NE	Description
Modified	DSP TWAMPRES PONDERSTA	eGBTS NodeB eNodeB gNodeB	The following fields have been added to the statistics for TWAMP responders in the output of this command: • Number of Start-N-Sessions Packets Received(packet) • Number of Start-N-Sessions Rej Packets Sent(packet) • Number of Start-N-ACK Packets Sent(packet) • Number of Stop-N-Sessions Packets Received(packet) • Number of Stop-N-Sessions Rej Packets Sent(packet) • Number of Stop-N-ACK Packets Sent(packet)
Modified	LST TWAMPCLIENT	eGBTS NodeB eNodeB gNodeB	The Individual Session Control Switch field has been added to the output of this command.

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	TWAMPCLIE NT.INDIVSE SSCTRLSW (LTE eNodeB, 5G gNodeB)	ADD TWAMPCLIE NT MOD TWAMPCLIE NT LST TWAMPCLIE NT	eGBTS NodeB eNodeB gNodeB	This parameter has been added to specify the individual session control switch for a TWAMP client. If this parameter is set to ENABLE , the individual session control function is enabled. If this parameter is set to DISABLE , this function is disabled.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *IP Active Performance Measurement*.

Requirement/Trouble Ticket Number

FeR-IR20240819000922-0010

IR20241101001025

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.2.14 Base Stations Supporting Certificates with a Validity Period Longer Than 68 Years

NEs Involved

3900 series base stations, 5900 series base stations, USU3910

Feature/Function Description

The base station complies with RFC 5280 and can load device certificates, trust certificates, and cross-certificates with a validity period of more than 68 years.

Implementation

The base station can use device certificates, trust certificates, and cross-certificates with a validity period of more than 68 years on SSL, IPsec, and DTLS links.

The following description has been deleted from the notes for the commands **ADD CERTMK** (for adding a device certificate), **ADD TRUSTCERT** (for adding a trust certificate), and **ADD CROSSCERT** (for adding a cross-certificate): The difference between current system time and both the effective and expiration time of the *device/trust/cross* certificate to be loaded must be less than 68 years. In addition, a permanently valid certificate is not supported.

Before the upgrade, the base station will fail to load a device certificate, trust certificate, or cross-certificate with a validity period of more than 68 years and report ALM-26841 Certificate Invalid. After the upgrade, the base station can load such a certificate and clear the preceding alarm. Before the upgrade, if the IKE peer uses a device certificate, trust certificate, or cross-certificate with a validity period of more than 68 years, the base station will report ALM-25891 IKE Negotiation Failure. After the upgrade, the base station can verify such a certificate of the IKE peer and clear the preceding alarm.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the eGBTS, NodeB, eNodeB, and gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following MML commands have changed.

Change Type	MML Command	NE	Description
Modified	ADD CERTMK	eGBTS NodeB eNode B gNode B	The following description has been deleted from the notes: The difference between current system time and both the effective and expiration time of the device certificate to be loaded must be less than 68 years. In addition, a permanently valid certificate is not supported.
Modified	ADD TRUSTCERT	eGBTS NodeB eNode B gNode B	The following description has been deleted from the notes: The difference between current system time and both the effective and expiration time of the trust certificate to be loaded must be less than 68 years. In addition, a permanently valid certificate is not supported.
Modified	ADD CROSSCERT	eGBTS NodeB eNode B gNode B	The following description has been deleted from the notes: The difference between current system time and both the effective and expiration time of the cross certificate to be loaded must be less than 68 years. In addition, a permanently valid certificate is not supported.

- Performance management

No impact

- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20240819001235-0010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.2.15 Changed the Default and Recommended Values of the Certificate Validity Check Period

NEs Involved

3900 series base stations, 5900 series base stations, USU3910

Feature/Function Description

The default and recommended values of the certificate validity check period (specified by the **CERTCHKTSK.PERIOD (LTE eNodeB, 5G gNodeB)** parameter) have been changed from **7** to **1**. The certificate validity check frequency has been changed from once every seven days to once every day. This accelerates certificate self-checks and clearing of certificate-related alarms generated by the BBU when exceptions occur. Specifically:

- ALM-26841 Certificate Invalid is reported when a certificate is yet to take effect or has expired because the base station time is incorrect.
- ALM-26840 Imminent Certificate Expiry is reported when a certificate is about to expire because the base station time is incorrect.

The base station inherits the pre-upgrade value of **CERTCHKTSK.PERIOD (LTE eNodeB, 5G gNodeB)** after an upgrade.

Implementation

For details, see *PKI* and *Base Station Supporting Multi-operator PKI*. The specific sections are listed in the change history of the documents.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the eGBTS, NodeB, eNodeB, and gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	CERTCHKT SK.PERIOD (5G gNodeB, LTE eNodeB)	SET CERTCHKTS K LST CERTCHKTS K	eGBTS NodeB eNodeB gNodeB	The default and recommended values of this parameter have been changed from 7 to 1 .

- Performance management

No impact

- Fault management

No impact

Impact on Features/Functions

- Prerequisite features/functions

No impact

- Mutually exclusive features/functions

No impact

- Impacted features/functions

No impact

Related Operations

Whether the change takes effect by default after an upgrade varies as follows:

- New base station deployment scenarios: Yes
 - Non-new base station deployment scenarios: No
- For details, see the corresponding "Operation and Maintenance" section in *PKI*.

Requirement/Trouble Ticket Number

FeR-IR20241101001116-0010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.2.16 Optimized the Mechanism for Verifying the Validity Period of Initial Certificates

NEs Involved

3900 series base stations, 5900 series base stations, USU3910

Feature/Function Description

This function optimizes the mechanism for verifying the validity period of initial certificates. A validity period check, which determines whether an initial certificate is about to expire or has expired, is performed only when the certificate is being used. If the check result shows that the certificate is about to expire or has expired, a related alarm will be reported.

Implementation

For details, see *PKI*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the eGBTS, NodeB, eNodeB, and gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

- No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20240809000284-0030

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.2.17 Changed Configurations for Ethernet Link Aggregation

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The **ETHTRUNK.INTERBOARDFLAG (5G gNodeB, LTE eNodeB)** parameter has been added to improve the transmission reliability.

Implementation

The **ETHTRUNK.INTERBOARDFLAG (5G gNodeB, LTE eNodeB)** parameter has been added to support the combination of Ethernet links on two boards into one logical link. If one link is faulty, services on this link will be switched over to functional links, preventing service interruption and improving transmission reliability. This parameter has been added to specify whether the Ethernet trunk is configured across boards.

Impact on the Network

After Ethernet link aggregation is enabled, the CPU usage may increase.

Impact on NEs

This function is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	ETHTRUNK.INTERBOARDFLAG (5G gNodeB, LTE eNodeB)	ADD ETHTRUNK MOD ETHTRUNK DSP ETHTRUNK LST ETHTRUNK	eGBTS NodeB eNodeB gNodeB	This parameter has been added to specify whether the Ethernet trunk is configured across boards.

- Performance management

No impact

- Fault management

No impact

Impact on Features/Functions

- Prerequisite features/functions

No impact

- Mutually exclusive features/functions

No impact

- Impacted features/functions

No impact

Related Operations

By default, the change does not take effect after an upgrade.

You can run the **MOD ETHTRUNK** command to set the **ETHTRUNK.INTERBOARDFLAG (5G gNodeB, LTE eNodeB)** parameter.

Requirement/Trouble Ticket Number

IR20250908001644

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.2.18 Optimized the Effective MACsec Connection Type When MACsec Is Used for eCPRI#

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The effective MACsec connection type is now optimized when MACsec is used for eCPRI. The optimization applies when the following conditions are met: (1) The protocol type that takes effect on CPRI ports is eCPRI and these ports support the MACsec protocol. (2) The **CPRIPORT.MACSECCONNTYPE (LTE eNodeB, 5G gNodeB)** parameter is set to **MACSEC**. (3) The involved RF modules do not support the MACsec protocol.

NOTE

This change was first incorporated into V100R022C10SPC010 but is first described in the network impact report of V100R022C10SPC050.

Implementation

When MACsec is used for eCPRI, the **CPRIPORT.MACSECCONNTYPE (LTE eNodeB, 5G gNodeB)** parameter is set to **MACSEC**, and the involved RF modules do not support the MACsec protocol, then the effective MACsec connection type changes from plaintext at the MAC layer to the MACsec protocol. After the optimization, the effective MACsec connection type is consistent with the configured value.

Impact on the Network

For an upgrade from a version earlier than V100R022C10SPC010 to V100R022C10SPC010 or later, if MACsec is used for eCPRI, the **CPRIPORT.MACSECCONNTYPE (LTE eNodeB, 5G gNodeB)** parameter is set to **MACSEC**, and the involved RF modules do not support the MACsec protocol:

- Before the upgrade, the effective MACsec connection type is plaintext at the MAC layer and service data is normal.

- After the upgrade, the effective MACsec connection type is the MACsec protocol. As a result, service data flows fail to be transmitted, and the cell becomes unavailable.

Impact on NEs

No impact

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20250908001658

DTS2025082231771

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.3 Equipment

5.2.4.3.1 Optimized the Mechanism for Reporting ALM-26564 RF Unit DC Input Power Failure

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The mechanism for reporting ALM-26564 RF Unit DC Input Power Failure on the NE that uses CPRI MUX and is configured with three main control boards has been optimized when the RRU connected to all these main control boards is powered off. ALM-26564 RF Unit DC Input Power Failure may not be reported for one of the main control boards before the optimization and can be reported for all the main control boards after the optimization.

Implementation

After the RRU is powered off, the power-off packets sent to the BBU carry the link setup addresses of the three main control boards. After receiving the packets, the main control boards identify the NE-allocated link setup addresses carried by the packets and report the RF Unit DC Input Power Failure alarm.

Impact on the Network

No impact

Impact on NEs

No impact

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management

No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20250319000813

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.3.2 Support for Exporting Lithium Battery Logs from the MAE

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In scenarios where lithium batteries are connected to wireless power supplies and managed by the MAE, lithium battery logs can now be exported from the MAE. This enhancement improves O&M capabilities and supports remote fault diagnosis.

Implementation

Lithium battery logs can be exported from the MAE using any of the methods listed in [Table 5-2](#).

Table 5-2 Exporting lithium battery logs on the MAE

No.	Method of Exporting Lithium Battery Logs	Procedure
1	MML command	Run the ULD FILE command with Source File Type set to BATTERYLOG(Battery Log) and the following parameters specified: Cabinet No. , Subrack No. , Slot No. , Log No. , Destination File Name , IP Mode , FTP Server IP , User Name , and Password .
2	NE file transfer	On the MAE, choose Access > Software > NE File Transfer . On the displayed page, click From NE to OSS Client , select the target NE and the BATTERYLOG(Battery Log) file type, and enter the cabinet number, subrack number, slot number, and log number.
3	NIC-based collection	On the MAE, choose Access > Maintenance > Network Information Collection . Click Customized Collection on the displayed page and follow the system prompts to create a customized collection task. In this step, select the Obtaining Battery Logs collection item, and enter the cabinet number, subrack number, slot number, and log number.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

Device Model	PMU Hardware Version	Supported or Not
APM5930	PMU.19	Supported
	PMU.22	Not supported
APM5950	PMU.35	Supported
OPM200 series	PMU.29	Supported
PMU15A Ver.B	PMU.40	Supported

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	LN	ULD FILE	eGBTS NodeB eNode B gNode B	This parameter has been added to specify the battery log number.
Modified	SRCF	ULD FILE	eGBTS NodeB eNode B gNode B	The value BATTERYLOG (Battery Log) has been added.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

Method 1: Run the **ULD FILE** command with **Source File Type** set to **BATTERYLOG(Battery Log)** and the following parameters specified: **Cabinet No.**, **Subrack No.**, **Slot No.**, **Log No.**, **Destination File Name**, **IP Mode**, **FTP Server IP**, **User Name**, and **Password**.

Method 2: On the MAE, choose **Access > Software > NE File Transfer**. On the displayed page, click **From NE to OSS Client**, select the target NE and the **BATTERYLOG(Battery Log)** file type, and enter the cabinet number, subrack number, slot number, and log number.

Method 3: On the MAE, choose **Access > Maintenance > Network Information Collection**. Click **Customized Collection** on the displayed page and follow the system prompts to create a customized collection task. In this step, select the **Obtaining Battery Logs** collection item, and enter the cabinet number, subrack number, slot number, and log number.

Requirement/Trouble Ticket Number

FeR-IR20250217003316-0010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.3.3 Enhanced Basic MTS O&M on the MAE

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

This function enhances the MAE's basic O&M capabilities for the MTS power system.

O&M is enhanced as follows:

1. When the power system type is SMU for the PMU, the value range of the **PMU.DCVUTHD (LTE eNodeB, 5G gNodeB)** parameter is extended from 580–600 to 530–600.
2. When the power system type is SMU for the PMU, lithium battery software lock is now supported.
3. When the power system type is SMU for the PMU, the reference busbar voltage can now be set to 53.5 V or 57 V.
4. The limit of configuring up to four SMUs per BBU is removed.
5. For the SMU, power management reports are supported on the MAE. These reports include the Energy Efficiency Report, Mains Running Report, Load Running Report, Solar Power Running Report, Battery Running Report, and Power Availability Report.
6. When the power system type is SMU for the PMU, the following battery information cannot be queried: power outage counts, charging counts, and discharging counts. This information is displayed as **NULL** in the **DSP BATTERY** command output.
7. The **DSP BATTERYITEM** command is added to query detailed information about each battery, including the lithium battery rated capacity, remaining capacity, lithium battery efficiency, first power-on time, battery voltage, and battery current.

Implementation

For details, see *Power Supply Management*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

- SMU-related function optimization is only applicable to the SMU02C monitoring module (BOM code: 02312MML-009).
- The **DSP BATTERYITEM** command is only applicable to specific wireless power monitoring modules, including the APM5930 (excluding the PMU.22 hardware version), APM5950, OPM200, and PMU 15A (Ver.B).

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact

- Configuration management

The following MML commands have changed.

Change Type	MML Command	NE	Description
New	DSP BATTERYITEM	eGBTS NodeB eNodeB gNodeB	This command has been added to query dynamic information of batteries.

NOTE

The parameters in the new MML commands are not listed separately.

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	PMU.DCVUTH D (LTE eNodeB, 5G gNodeB)	ADD PMU LST PMU MOD PMU	eGBTS NodeB eNodeB gNodeB	The value range of this parameter has been changed to 530–600 to adapt to the SMU's actual capability.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250217003314-0010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.3.4 Added Voltage-based Port-Level Hierarchical Power-Off

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

For a PMU whose power system type is SMU (that is, a PMU for which **PMU.PTYPE (LTE eNodeB, 5G gNodeB)** is set to **SMU**), when the **PDPATTR.CUSLOADSHUTDOWNMODE (5G gNodeB, LTE eNodeB)** parameter is set to **ON_VOL** and the base station voltage drops to the threshold specified by the **PDPATTR.CUSLOADSHUTDOWNVOLTAGE (5G gNodeB, LTE eNodeB)** parameter after the mains supply is disconnected, power-off is triggered for the corresponding port.

This change is irrelevant to LampSite base stations.

Implementation

For details, see *Green Site*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

Power supply:

Only the SMU02C (BOM number: 02312MML-009) supports voltage-based port-level power-off.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	PDPATTR.CUS LOADSHUTD OWNMODE (5G gNodeB, LTE eNodeB)	MOD PDPATTR LST PDPATTR DSP PDPATTR	eGBTS NodeB eNode B gNode B	This parameter has been added to specify the optional customized power-off mode of a power distribution port when SMU voltage-based port-level power-off is enabled.
New	PDPATTR.CUS LOADSHUTD OWNVOLTAGE (5G gNodeB, LTE eNodeB)	MOD PDPATTR LST PDPATTR DSP PDPATTR	eGBTS NodeB eNode B gNode B	This parameter has been added to specify the customized load power-off voltage when SMU voltage-based port-level power-off is enabled.

- Performance management

No impact

- Fault management

No impact

Impact on Features/Functions

- Prerequisite features/functions

No impact

- Mutually exclusive features/functions

No impact

- Impacted features/functions

No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Green Site*.

Requirement/Trouble Ticket Number

FeR-IR20250217003314-0010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.3.5 Added Applicable Scenarios of SitePower Backup Policy Management

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The following changes are involved:

- More applicable scenarios of SitePower Backup Policy Management:
 - This feature now applies to scenarios where an MTS series power system is used with a DCDCU in an intelligent subrack.
 - In scenarios where an iSDU06D/iSDU03D interconnected with an MTS series power system using SNMP is used, the MTS series power system can be connected to the NetEco through BBU in-band networking.
- Design optimization:
 - RFUs are now compatible.
 - The **VS.PMUBoard.DCLoadEnergy (5G gNodeB, LTE eNodeB)** counter has been added to measure the DC energy of the EPMU.
 - Feature triggering optimization: This feature takes effect when the site meets the license, software, and hardware verification requirements. If the mains supply is disconnected, the base station proactively reports that mains supply unavailability. After the MAE recognizes the mains supply disconnection, SitePower Backup Policy Management takes effect.

LampSite base stations do not support this function.

Implementation

The task creation, service priority, lossless running duration, and target backup power use duration for SitePower Backup Policy Management are configured on the MAE-Access.

For details, see *Green Site*. The specific sections are listed in the change history of the document.

Impact on the Network

When the mains supply is disconnected and the backup power configuration cannot meet the target backup power use duration, SitePower Backup Policy Management shuts down low-priority services (at the cell level) at a proper time based on the service priorities defined by the operator, historical power consumption, service load trend, and remaining battery capacity. In this way, the service time of sites is extended and high-priority services can operate for the target backup power use duration with a small amount of or no additional backup power investment. After SitePower Backup Policy Management takes effect, the backup power use duration for high-priority services can be extended up to twice the standard length.

When the mains supply is disconnected, non-primary services may be shut down beyond the lossless running duration. As a result, the corresponding services are unavailable.

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, eGBTS, and MAE.

Impact on Hardware Requirements

Boards:

- Scenarios where an MTS series power system is used with a DCDU in an intelligent subrack: Only the main control boards that are upgraded to V100R022C10 or later are compatible.
- Scenarios where an iSDU06D/iSDU03D interconnected with an MTS series power system using SNMP is used (the MTS series power system can be connected to the NetEco through BBU in-band networking): Only the main control boards that are upgraded to V100R021C10 or later are compatible.

Power supply:

- Only Huawei lithium batteries support this function. The use of only lead-acid batteries and hybrid use of lead-acid and lithium batteries are both inapplicable. Power supply requirements have been added for scenarios where an MTS series power system is used with a DCDU in an intelligent subrack.
- Only centralized power supply is supported. In the centralized power supply scenario, the BBU and RRU use the same power supply and backup system.

Others:

- MAE: The MAE version must be iMaster MAE V100R026C10 or later. NEs in the same power supply and backup group must be managed by the same MAE. If NEs in the same group are managed by different MAEs, the calculated target backup power use duration will be incorrect.
- SitePower Backup Policy Management requires that all main equipment be provided by Huawei.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

The task creation, service priority, lossless running duration, and target backup power use duration for SitePower Backup Policy Management are configured on the MAE-Access instead of the base station side.

Requirement/Trouble Ticket Number

FeR-IR20250321000450-0010

FeR-IR20251204001679-0010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.3.6 Added Coordinated Power Backup of the Entire Site in Combination of the Base Station and Microwave

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

If the mains supply is disconnected and the base station enters the backup power saving state, the microwave device works with the base station so that the power saving status of the base station can be transmitted to the microwave device. In this way, the microwave device also enters the power saving state, increasing the backup power use duration of the entire site.

LampSite base stations do not support this function.

Implementation

For details, see *Green Site*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

Power supply:

The base station supports only the following backup power saving functions on the condition that the power supply requirements for them are met.

- Low power consumption mode (LTE)
- Low power consumption mode (NR)
- SOC level-based low power consumption mode (LTE)

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	SOC level-based low power consumption mode	• IPWRM.SOCLVLADJSW • SOCLVL_BASED_LOW_PWR_SW option of the CellLowPower.LowPwrConsumeModeExtSw parameter	<i>Green Site</i>	Coordinated power backup of the entire site in combination of the base station and microwave can be enabled only when low power consumption mode is in effect.
LTE FDD LTE TDD	LTE low power consumption mode	CellLowPower.LowPwrSwitch	<i>Green Site</i>	Coordinated power backup of the entire site in combination of the base station and microwave can be enabled only when low power consumption mode is in effect.
NR FDD Low-frequency NR TDD	NR low power consumption mode	LOW_PWR_MODE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Green Site</i>	Coordinated power backup of the entire site in combination of the base station and microwave can be enabled only when low power consumption mode is in effect.

- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Green Site*.

Requirement/Trouble Ticket Number

FeR-IR20250303001670-0010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.3.7 Optimized the Configuration for the Number of Ports on the CPRI Panel

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The number of ports on the CPRI panel (specified by **BBP.CPRIPANELPORTNUM (LTE eNodeB, 5G gNodeB)**) must be the same as the number of CPRI ports provided by the board. Otherwise, the board cannot be started. When the UBBP board provides nine ports on the fronthaul panel, this parameter must be set to **9**; when it provides only six ports on the fronthaul panel, the parameter must be set to **6**. You can change the number of ports on the CPRI panel using MML commands or the MAE-Deployment. The details are as follows:

- Changing the number of ports on the CPRI panel by running the **MOD BBP** command
 - If the number of ports on the CPRI panel is changed from **9** to **6**, **Port Usage** (specified by **CASCADEPORT.PORTUSAGE (LTE eNodeB, 5G gNodeB)**) of cascade port 6 will be changed to **INTERCONNECT** and **Port Mode** (specified by **CASCADEPORT.PM (LTE eNodeB, 5G gNodeB)**) will be changed to **AUTO**. In addition, cascade ports 7 and 8 will be deleted.
 - If the number of ports on the CPRI panel is changed from **6** to **9**, **Port Usage** of cascade port 6 will be changed to **FRONTHAUL** and **Port Mode** will be changed to **AUTO**. In addition, cascade ports 7 and 8 will be created, with **Port Usage** set to **FRONTHAUL** and **Port Mode** set to **AUTO**.
- If the number of ports on the CPRI panel is changed using the MAE-Deployment, **Port Usage** (specified by **CASCADEPORT.PORTUSAGE (LTE eNodeB, 5G gNodeB)**) of the cascade port corresponding to the board must be changed accordingly.
 - If the number of ports on the CPRI panel is changed from **9** to **6**, **Port Usage** of cascade port 6 must be set to **INTERCONNECT**.
 - If the number of ports on the CPRI panel is changed from **6** to **9**, **Port Usage** cascade port 6 must be set to **FRONTHAUL**.

Implementation

For details, see *Base Station Cabinets and Subracks (Including the BBU Subrack) Configuration*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the eNodeB, gNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

UBBP series

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	CASCADEPORT.PORTUSAGE (LTE eNodeB, 5G gNodeB)	SET PORTMODE LST PORTMODE	eNodeB gNodeB NodeB eGBTS	This parameter is added to specify the usage of the interconnection port.

- Performance management

No impact

- Fault management

No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

You can run the **MOD BBP** command or use the MAE-Deployment to change the number of ports on the CPRI panel.

Requirement/Trouble Ticket Number

FeR-IR20250806001609-0230

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.3.8 Optimized Descriptions of Specific Problems for a Digital-Antenna-related Alarm

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Digital Information Management Disabled has been added as a specific problem for ALM-26757 RET Antenna Running Data and Configuration Mismatch.

Implementation

For details, see *ALD Management*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

No impact

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
The specific problem **Digital Information Management Disabled** has been added for ALM-26757 RET Antenna Running Data and Configuration Mismatch.

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *ALD Management*.

Requirement/Trouble Ticket Number

AR-FuR-IR20250528001543-01-0010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD

5.2.4.3.9 Optimized License Functions for Digital Antennas

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In LTE/NR, certain authorization conditions must be met when the following functions are to be used for digital antennas:

- Horizontal azimuth and horizontal beamwidth adjustment: The ALD (RAS unit or RVD) must be configured with required licenses. Only licenses of the same type can be configured for one ALD, and one license cannot be used for angle adjustment on multiple ALDs at the same time.
- Interference diagnosis: The base station must be deployed with required licenses.

Implementation

For details, see *ALD Management*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

No impact

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management

The following MML commands have changed.

Change Type	MML Command	NE	Description
New	ADD DIGIANTINFO	eNodeB gNodeB	This command is used to add digital antenna information management configurations.
New	DSP DIGIANTINFO	eNodeB gNodeB	This command is used to display dynamic digital antenna management information.

Change Type	MML Command	NE	Description
New	MOD DIGIANTINFO	eNodeB gNodeB	This command is used to modify digital antenna information management configurations.
New	LST DIGIANTINFO	NodeB gNodeB	This command is used to list digital antenna information management configurations.

NOTE

The parameters in these new or deleted MML commands are no longer listed separately.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *ALD Management*.

Requirement/Trouble Ticket Number

AR-FuR-IR20250528001543-01-0020

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD

5.2.4.3.10 Optimized SitePower Backup Policy Management

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Optimized SitePower Backup Policy Management can now be enabled on the MAE-Access and no longer requires the iFaultCare app.

This change is irrelevant to LampSite base stations.

Implementation

The task creation, service priority, lossless running duration, and target backup power use duration for SitePower Backup Policy Management are configured on the MAE-Access.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, eGBTS, and MAE.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact

- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

The task creation, service priority, lossless running duration, and target backup power use duration for SitePower Backup Policy Management are configured on the MAE-Access instead of the base station side.

Requirement/Trouble Ticket Number

FeR-IR20240812001108-0060

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS

5.2.4.3.11 Optimized Recovery of the Automatically Recognized Topology in the Case of iSDU Power-On Reset

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The iSDU saves the recognized topology upon successful automatic topology recognition. After a power-on reset is performed for the iSDU, the topology remains unchanged.

This change is irrelevant to LampSite base stations.

Implementation

For details, see *Green Site*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250409002767-0050

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.3.12 Added Support for Base Station Hierarchical Power-Off and Intelligent Coordinated Voltage Boosting Power Distribution by the iSDU03T

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The three ports on the iSDU03T all support base station hierarchical power-off and intelligent coordinated voltage boosting power distribution. For details about these functions, see *Green Site*.

This change is irrelevant to LampSite base stations.

Implementation

For details, see *Green Site*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

No impact

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20241019000177-0010

FeR-IR20241019000177-0020

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.3.13 Added Automatic Voltage Optimization for Intelligent Lithium Batteries

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In scenarios where a third-party power system is used with Huawei intelligent lithium batteries, when the mains supply is disconnected, the base station used with the PMU 15A(Ver.B) in the power cabinet can keep the voltage of those batteries at a relatively high level through the automatic voltage optimization for intelligent lithium batteries function, thereby finding the optimal voltage and reducing power loss during cable transmission.

Automatic voltage optimization for intelligent lithium batteries is controlled by the **PMU.BVR** parameter. If this parameter is set to **AUTO**, this function is enabled.

LampSite base stations do not support this function.

Implementation

For details, see *Green Site*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

No impact

Impact on Hardware Requirements

Automatic voltage optimization for intelligent lithium batteries has the following power supply requirements:

- The base station must be configured with the PMU 15A(Ver.B). The PMU software version can be queried by running the **DSP BRDVER** command.
- The base station must be configured with Huawei intelligent lithium batteries. For details, contact Huawei engineers. (ESM-100B1, ESM-150B1, ESM-100A7, ESM-150A3, and ESM-200A1).
- If Huawei intelligent lithium batteries are used, they can be used together with lead-acid batteries and common lithium batteries without affecting the automatic voltage optimization for intelligent lithium batteries function.
- If Huawei intelligent lithium batteries and third-party lead-acid batteries are used together, it is recommended that battery temperature compensation be disabled for lead-acid batteries when automatic voltage optimization for intelligent lithium batteries is enabled.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

Whether the change takes effect by default after an upgrade varies as follows:

- It does not take effect if **PMU.BVR (5G gNodeB, LTE eNodeB)** is not set to **AUTO** before the upgrade.
- It takes effect if **PMU.BVR (5G gNodeB, LTE eNodeB)** is set to **AUTO** before the upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Green Site*.

Requirement/Trouble Ticket Number

FeR-IR20240812001108-0040

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.3.14 Added Descriptions of Whether an ICC Series Power System Supports Functions Related to a Green Site

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Whether an ICC series power system supports functions related to a green site is as follows:

Table 5-3 Whether an ICC series power system supports functions related to a green site

Function Name	Supported or Not	Description
PSU intelligent shutdown	Not supported	None
PSU intelligent coordinated energy saving	Supported	None
Automatic PSU voltage optimization	Not supported	None
Staggering electricity usage	Not supported	None
PV deployment for base stations	Supported	None
Third-party green power access measurement and O&M management	Not supported	None
Low power consumption mode (LTE)	Not supported	Low power consumption mode (LTE) is not supported when an ICC series power system is used in hybrid power supply scenarios.
Low power consumption mode (NR)	Not supported	Low power consumption mode (NR) is not supported when an ICC series power system is used in hybrid power supply scenarios.
SOC level-based low power consumption mode	Supported	None

Function Name	Supported or Not	Description
Base station hierarchical power-off	Partially supported	Time-based port-level power-off is supported. Voltage-based power-system-level power-off is supported. The customized port-level power-off time is supported only if a power subrack with intelligent circuit breakers is configured; and voltage-based subrack-level power-off is supported only if a power subrack without intelligent circuit breakers is configured.
Adaptive Power Backup	Not supported	Adaptive Power Backup is not supported when an ICC series power system is used in hybrid power supply scenarios.
SitePower Backup Policy Management	Not supported	SitePower Backup Policy Management is not supported when an ICC series power system is used in hybrid power supply scenarios.
iSDU06D support for connection with a third-party power system	N/A	None
Power Backup Perception (trial)	Not supported	None
PMU support for associated temperature control of battery cabinets	Not supported	None
iSDU03S intelligent coordinated voltage boosting power distribution	N/A	None

The power monitoring module of an ICC series power system is an SMU02C.

The ICC power cabinet in use must meet the capability requirements for its connection with the MAE for O&M management, which is supported from

SRAN22.0 onwards. For details about the ICC series power system model and version requirements, contact Huawei engineers.

Implementation

For details, see *Green Site*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

No impact

Impact on Hardware Requirements

The SMU02C is used as the power monitoring module of the MTS or ICC series power system, and the SMU version must be iSitePower-A V100R025C10 or later.

The ICC power cabinet in use must meet the capability requirements for its connection with the MAE for O&M management, which is supported from SRAN22.0 onwards. For details about the ICC series power system model and version requirements, contact Huawei engineers.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
The impact relationship between Adaptive Power Backup and Power Backup Perception has been removed.

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20240808004212-0010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.3.15 Support for Lithium Battery Management by the PMU 15A (Ver.B) When Third-Party Power Supply Is Used

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

When third-party power supply is used, configuration management and basic O&M of lithium batteries can be achieved by the PMU 15A (Ver.B) that is connected to the MAE.

Implementation

When third-party power supply is used, the PMU 15A (Ver.B) can be used to manage lithium batteries, including basic functions such as battery configuration, status query, and alarm reporting. The operation methods (such as MML-based configurations) are the same as those when Huawei power supply is used.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the eGBTS, NodeB, eNodeB, and gNodeB.

Impact on Hardware Requirements

- Power monitoring unit: PMU 15A (Ver.B)
- Third-party power cabinet: an MTS series one or a power cabinet of another vendor
- Lithium battery model: ESM-48100B1, ESM-48150B1, ESM-48100A7, ESM-48150A3, ESM-48200A1, and ESM-48100U2

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade. This function can take effect only when third-party power supply is used and lithium batteries are configured.

For details, see the "Operation and Maintenance" section of "Basic Power Supply Management Functions" in *Power Supply Management*.

Requirement/Trouble Ticket Number

IR20240812001108

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.3.16 Optimized Lithium Battery Software Lock When the PMU 15A Is Used

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The lithium battery software lock is now applicable to lithium batteries when third-party power supply is used. In addition, the lithium battery software lock timeout threshold is now configurable.

Implementation

- When third-party power supply is used, the PMU 15A (Ver.B) can be used to manage lithium batteries through the lithium battery software lock. The operations are the same as those when Huawei power supply is used.
- The **IPWRM.LIBATTLOCKTIMEOUT (LTE eNodeB, 5G gNodeB)** parameter has been added to set the timeout threshold for the lithium battery software lock. If the lock is enabled and the duration of disconnection between a lithium battery and the PMU 15A or PMU 15A (Ver.B) exceeds this threshold, the lithium battery disables the DC output and no longer provides backup power. In addition, a buzzer alarm is generated. The **Li Bat Software Lock Timeout Mgmt Capb** field has been added to the **DSP PMU** command output to indicate whether the lithium battery software lock timeout threshold can be configured.

Impact on the Network

No impact

Impact on NEs

No impact

Impact on Hardware Requirements

- When third-party power supply is used, the lithium battery software lock requires that the power monitoring unit be PMU 15A (Ver.B), the power cabinet be either an MTS series one or a power cabinet of another vendor, and the lithium battery model be one of the following: ESM-48100B1, ESM-48150B1, ESM-48100A7, ESM-48150A3, ESM-48200A1, and ESM-48100U2.
- Configuring the timeout threshold for the lithium battery software lock requires that the power monitoring unit be PMU 15A or PMU 15A (Ver.B) and the cabinet type be APM30H (Ver.D) or APM30H (Ver.E).

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	<i>IPWRM.LIBAT TLOCKTIMEOUT (LTE eNodeB, 5G gNodeB)</i>	SET IPWRM LST IPWRM	eGBTS NodeB eNodeB gNodeB	This parameter has been added to specify the lithium battery software lock timeout threshold. The value range is 10 to 43200 , in minutes. For the PMU 15A (Ver.B), the value range of this parameter is 10 to 43200 . For other PMUs, the value of this parameter is fixed to 8640 and cannot be configured or modified.

- Performance management
No impact
- Fault management
The following alarms have changed.

Change Type	Alarm Name	NE	Description
Modified	ALM-26276 Inter-System Site-Level Configuration Conflict	eGBTS NodeB eNodeB gNodeB	The fault scenario, a conflict in the configuration of lithium battery software lock timeout threshold, has been added to this alarm. This alarm is reported if the lithium battery software lock timeout thresholds configured for different NEs of a separate-MPT base station are different. It is recommended that the threshold be set to the same value for all the NEs.

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade. This function can take effect only when third-party power supply is used and lithium batteries are configured.

For details, see the "Operation and Maintenance" section of "Lithium Battery Software Lock Management" in *Power Supply Management*.

Requirement/Trouble Ticket Number

IR20240812001108

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.3.17 Support for ICC Power System Management on the MAE

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

This function implements basic O&M management of ICC power systems on the MAE, providing functions such as parameter configuration, status query, alarm and event reporting, performance counter reporting, electronic label query, and inventory management.

Implementation

The ICC power system can be connected to the MAE in either of the following ways:

- ICC -> BBU -> MAE: An SMU in an ICC power cabinet is connected to the UPEU in the BBU through an RS485 serial cable for communication. In this way, the ICC power system can be managed by the MAE.
- ICC -> MEB -> RRU -> BBU -> MAE: An SMU in an ICC power cabinet is connected to the MEB through an RS485 serial cable for communication; the MEB, which can convert RS485 signals into power loop signals, is connected to an RRU through a power cable; the RRU is connected to the BBU through a CPRI optical fiber. In this way, the ICC power system can be managed by the MAE.

After a link is set up, you can implement basic O&M management (such as status query, alarm and event reporting, performance counter reporting, electronic label query, and inventory management) of SMUs, power supply modules, and batteries in the ICC power cabinet on the MAE.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the eGBTS, NodeB, eNodeB, and gNodeB.

Impact on Hardware Requirements

- For the monitoring module, only the SMU02C (BOM number: 02312MML-009) supports this function.
- For the power modules, only the DCDU-150AN2, DCDU-400AN5, DCDU-600AN1, and DCDU-600AN2 support this function.
- For the power cabinets, only the ICC30-N1, ICC200-N1-C12, ICC330-D1, ICC330-H1-C6, ICC330-H1-C8, ICC330-HA1-C7, ICC330-HA1-C10, ICC330-HA1-C11, ICC330-HD1-C6, ICC350-H1-C5, ICC360-HA1-C1, ICC360-HA1-C2, ICC720-HA1-C2, and ICC800-A1-C2 support this function.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	DIESELGEN.DIESELGENMFR (5G gNodeB, LTE eNodeB)	ADD DIESELGEN LST DIESELGEN MOD DIESELGEN	eGBTS NodeB eNodeB gNodeB	This parameter has been added to specify the manufacturer of a diesel generator.
New	DIESELGEN.DIESELGENBRAND (5G gNodeB, LTE eNodeB)	ADD DIESELGEN LST DIESELGEN MOD DIESELGEN	eGBTS NodeB eNodeB gNodeB	This parameter has been added to specify the brand of a diesel generator.

Change Type	Parameter ID	MML Command	NE	Description
New	DIESELGEN.IN STDATE (5G gNodeB, LTE eNodeB)	ADD DIESELGE N LST DIESELGE N MOD DIESELGE N	eGBTS NodeB eNode B gNode B	This parameter has been added to specify the date when a diesel generator is installed.
New	DIESELGEN.O UTPUTACPT (5G gNodeB, LTE eNodeB)	ADD DIESELGE N LST DIESELGE N MOD DIESELGE N	eGBTS NodeB eNode B gNode B	This parameter has been added to specify the output AC phase type of a diesel generator.
New	DIESELGEN.R ATEDPHVOLT (5G gNodeB, LTE eNodeB)	ADD DIESELGE N LST DIESELGE N MOD DIESELGE N	eGBTS NodeB eNode B gNode B	This parameter has been added to specify the rated phase voltage of a diesel generator.
New	DIESELGEN.R ATEDFREQ (5G gNodeB, LTE eNodeB)	ADD DIESELGE N LST DIESELGE N MOD DIESELGE N	eGBTS NodeB eNode B gNode B	This parameter has been added to specify the rated frequency of a diesel generator.

Change Type	Parameter ID	MML Command	NE	Description
New	DIESELGEN.MAXOUTPUTPWRATIO (5G gNodeB, LTE eNodeB)	ADD DIESELGEN LST DIESELGEN MOD DIESELGEN	eGBTS NodeB eNodeB gNodeB	This parameter has been added to specify the maximum output power ratio of a diesel generator.

The following MML commands have changed.

Change Type	MML Command	NE	Description
New	ADD PMUEXTINFO	eGBTS NodeB eNodeB gNodeB	This command has been added for the ICC power system in diesel-grid hybrid power supply scenarios.
New	MOD PMUEXTINFO	eGBTS NodeB eNodeB gNodeB	This command has been added for the ICC power system in diesel-grid hybrid power supply scenarios.
New	RMV PMUEXTINFO	eGBTS NodeB eNodeB gNodeB	This command has been added for the ICC power system in diesel-grid hybrid power supply scenarios.
New	LST PMUEXTINFO	eGBTS NodeB eNodeB gNodeB	This command has been added for the ICC power system in diesel-grid hybrid power supply scenarios.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions

RAT	Function Name	Function Switch	Reference
NR LTE FDD LTE TDD UMT S GSM	Intelligent battery management	EQUIPMENT.BATIMS (5G gNodeB, LTE eNodeB)	<i>Power Supply Management</i>
NR LTE FDD LTE TDD UMT S GSM	Reporting of the Loss of Power Supply Redundancy alarm	EQUIPMENT.PAE (5G gNodeB, LTE eNodeB)	<i>Power Supply Management</i>
NR LTE FDD LTE TDD UMT S GSM	Hybrid use of lead-acid and lithium batteries	BATTERY.BTYPE (LTE eNodeB, 5G gNodeB)	<i>Power Supply Management</i>
NR LTE FDD LTE TDD UMT S GSM	Staggering electricity usage	EQUIPMENT.STAGGEREDPOWERSWITCH (LTE eNodeB, 5G gNodeB)	<i>Green Site</i>

RAT	Function Name	Function Switch	Reference
NR LTE FDD LTE TDD UMTS GSM	PSU intelligent shutdown	PSUIS.PSUISS (5G gNodeB, LTE eNodeB)	<i>Green Site</i>
NR LTE FDD LTE TDD UMTS GSM	Third-party green power access measurement and O&M management	PDU.PDUTYPE (LTE eNodeB, 5G gNodeB)	<i>Green Site</i>
NR	Precise backup power saving policies in low power consumption mode	NRDUCellPowerSaving.BackupPwrSavingPolicy set to PRECISE_BACKUP_POLICY1 , PRECISE_BACKUP_POLICY2 , or PRECISE_BACKUP_POLICY3	<i>Green Site</i>
LTE FDD LTE TDD	Precise backup power saving policies in low power consumption mode	CellLowPower.BackPwrSavPolicy set to PRECISE_BACKUP_POLICY1 , PRECISE_BACKUP_POLICY2 , or PRECISE_BACKUP_POLICY3	<i>Green Site</i>

- Impacted features/functions

RA T	Function Name	Function Switch	Referenc e	Description
NR LTE FD D LTE TD D UM TS GS M	Optimized access policy on power monitoring modules	IPWRM.PMUTYPE PECHSW (LTE eNodeB, 5G gNodeB)	<i>Power Supply Manage ment</i>	Whether the ICC power system can be managed by the MAE is affected by optimized access policy on power monitoring modules. • When the IPWRM.PMUTYPE CHKSW (LTE eNodeB, 5G gNodeB) parameter is set to OFF, the base station does not check the PMU type. As a result, management of ICC power cabinets on the MAE is not affected. • When the IPWRM.PMUTYPE CHKSW (LTE eNodeB, 5G gNodeB) parameter is set to ON, the base station checks the PMU type. Among ICC power systems, only the ICC power cabinets that are compatible with this function (listed in Impact on Hardware Requirements) can be managed by the MAE.

Related Operations

By default, the change does not take effect after an upgrade. An ICC power system needs to be configured to enable this function.

For details, see the "Operation and Maintenance" section of "Basic Power Supply Management Functions" in *Power Supply Management*.

Requirement/Trouble Ticket Number

IR20240808004212

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.3.18 Support for Upgrading the Monitoring Management Software of MTS Cabinets Using the MAE

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The monitoring management software of MTS cabinets can be upgraded using the MAE.

Implementation

The procedures for upgrading the monitoring management software of MTS cabinets using the MAE are as follows:

1. Obtain the monitoring management software upgrade package *iSitePower-A_V100R025C10SPC010_02X-ALL.7z* from the support website by choosing **Product Support > Wireless Network > Site Energy > Telecom Power > Hybrid Power > iSitePower-A**.
2. Load the software upgrade package on the SME of the MAE-Access. If an MTS cabinet is directly connected to the MAE through the BBU, the software package loading duration is 4 to 5 hours (when the transmission baud rate is 115200 symbol/s) or 17 to 18 hours (when the transmission baud rate is 9600 symbol/s). If an MTS cabinet is connected to the MAE through the "MTS->MEB->RRU->BBU->MAE" method, the loading duration is 17 to 18 hours.
3. Activate the software upgrade package to upgrade the MTS cabinets. A maximum of 1000 sites can be upgraded at the same time.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the eGBTS, NodeB, eNodeB, and gNodeB.

Impact on Hardware Requirements

- For the monitoring module, only the SMU02C (BOM number: 02312MML-009) supports this function.
- For embedded power modules, only the ETP48400-C3B1 (BOM number: 01074979-019) and ETP48400-C4B1 (BOM number: 01075865-002) support this function.

- For power cabinets, only the MTS9304A-HX10A1 (BOM number: 01075308) and MTS9514A-AM2003 (BOM number: 01075875) support this function.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
When the software upgrade package is loaded, the MTS power system cannot be configured.
- Performance management
When the software upgrade package is loaded, the performance counter data of the MTS power system cannot be obtained.
- Fault management
When the software upgrade package is loaded, fault alarms related to the MTS power system cannot be reported. When the package is activated, ALM-25600 Monitoring Device Maintenance Link Failure is reported. This alarm is automatically cleared after the activation is complete.

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
The following functions are affected when the software upgrade package is loaded, and they will be restored after the loading is complete.

RA T	Function Name	Function Switch	Reference	Description
NR LTE FDD LTE TDD UMTS GSM	MAE power management reports	None	None	When the software upgrade package is loaded, the performance counter data of the MTS power system cannot be obtained. As a result, empty data is displayed in power management reports.

RA T	Function Name	Function Switch	Reference	Description
NR LTE FDD LTE TDD	Low power consumption mode	- NR: LOW_PWR_MODE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter - LTE: CellLowPower.LowPwrSwitch	<i>Green Site</i>	Low power consumption mode depends on the alarm information reported by the PMU. However, when the software upgrade package is loaded, the alarm information reported by the PMU cannot be obtained. As a result, low power consumption mode cannot be triggered.
NR LTE FDD LTE TDD UMTS GSM	Adaptive Power Backup	- NR: LOW_PWR_MODE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter - LTE: CellLowPower.LowPwrSwitch - UMTS: NODEBALGP ARA.LOWPOWERCONSUMPTIONMODESW - GSM: GBTSENERGY MGTPARA.LOWPOWERCONSUMPTIONMODESW	<i>Green Site</i>	Adaptive Power Backup depends on the alarm information reported by the PMU. However, when the software upgrade package is loaded, the alarm information reported by the PMU cannot be obtained. As a result, Adaptive Power Backup cannot be triggered.

RAT	Function Name	Function Switch	Reference	Description
GSM UMTS LTE FDD LTE TDD NR	Base station hierarchical power-off	PDPATTR.CSDTS <i>(LTE eNodeB, 5G gNodeB)</i>	<i>Green Site</i>	Base station hierarchical power-off depends on the alarm information reported by the PMU. However, when the software upgrade package is loaded, the alarm information reported by the PMU cannot be obtained. As a result, base station hierarchical power-off cannot be triggered.

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20240808004257-0030

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.3.19 Optimized the Leap Second Compensation Mechanism for the Base Station Clock

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

This change optimizes the leap second compensation mechanism to improve the accuracy of leap second information obtained by a base station when the base station clock experiences external interference.

Implementation

The base station clock is obtained through leap second compensation based on the clock source. When external signal interference exists, the leap second compensation information may be inaccurate, causing base station clock jumps.

This change optimizes the leap second compensation mechanism for the base station clock, so that the base station can obtain accurate leap second information, which prevents base station clock jumps.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20241209002193-0063

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.3.20 Support for Tracing of MAC Packets by Port Number on the MAC Trace Page

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

MAC packets over an eCPRI port can now be traced using a **MAC Trace** task even if no HEI port is available. Also, MAC packets can be traced by port number.

This change is irrelevant to LampSite base stations.

Implementation

On the **MAC Trace** page of the MAE or LMT, when **Subboard Type** is set to **Eth Cover Board**, the **Port Filter** field is displayed. When the fronthaul port number is specified, only the MAC packets over this port are reported on the **MAC Trace** page.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the gNodeB, eNodeB, and MAE.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20240827002260-0010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD

5.2.4.3.21 Optimized DSP PMU Command Output

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In separate-MPT scenarios, **PSU Intel Coordinated Energy Saving Status = Enable** is displayed in the output of the **DSP PMU** command executed on the main control board with the function disabled if the following conditions are met: (1) Two main control boards are configured with one PMU. (2) PSU intelligent shutdown or PSU intelligent coordinated energy saving is enabled on one main control board but disabled on the other main control board. (3) The function has taken effect on the entire base station.

Implementation

For details, see *Green Site*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

No impact

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following MML commands have changed.

Change Type	MML Command	NE	Description
Modified	DSP PMU	eGBTS NodeB eNodeB gNodeB	In separate-MPT scenarios, PSU Intel Coordinated Energy Saving Status = Enable is displayed in the output of the DSP PMU command executed on the main control board with the function disabled if the following conditions are met: (1) Two main control boards are configured with one PMU. (2) PSU intelligent shutdown or PSU intelligent coordinated energy saving is enabled on one main control board but disabled on the other main control board. (3) The function has taken effect on the entire base station.

- Performance management

No impact

- Fault management

No impact

Impact on Features/Functions

- Prerequisite features/functions

No impact

- Mutually exclusive features/functions

No impact

- Impacted features/functions

No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

DTS2025051234122

FeR-IR20240808004257-0010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.3.22 Optimized Display of iSDU Automatic Topology Recognition Results

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The iSDU saves the recognized topology upon successful automatic topology recognition. If the conditions for triggering iSDU port power-off are met, RF modules are powered off upon iSDU port power-off. The value of **Equipment Serial Number(AUTO)** in the **DSP PDPATTR** command output still indicates the previous recognized topology.

This change is irrelevant to LampSite base stations.

NOTE

This change was first incorporated into V100R022C00SPC010 but is first described in the network impact report of V100R022C00SPC050.

Implementation

For details, see *Green Site*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250409002767-0050

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.3.23 Digital Antennas Supporting Log Upload in Batches

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

- This change supports the query of non-standard protocol capabilities of antenna devices: log upload.

Implementation

The **DSP RAEFUNCTION** and **DSP ALDFUNCTION** commands can now be used to query the non-standard protocol capabilities of antenna devices, that is, log upload.

Impact on the Network

No impact

Impact on NEs

No impact

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management

The following MML commands have changed.

Change Type	MML Command	NE	Description
Modified	DSP RAEFUNCTION	eGBTS NodeB eNodeB gNodeB	This command can now be used to query a non-standard protocol capability of antenna devices, that is, log upload.
Modified	DSP ALDFUNCTION	eGBTS NodeB eNodeB gNodeB	This command can now be used to query a non-standard protocol capability of antenna devices, that is, log upload.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

AR-HERT_26B-FuR- IR20250603002185-01-01

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.3.24 External and Internal AISU Products Supporting Roll Angle Queries and Digital Antennas Supporting Antenna Attitude Detection

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

- The roll angle can be queried.
- The antenna attitude detection function is supported.

Implementation

- The **DSP RAEDVICEDATA** and **DSP ALDDEVICEDATA** commands can be used to query the roll angle.
- The **DSP RAEFUNCTION** and **DSP ALDFUNCTION** commands can be used to query the antenna attitude detection capability.
- The **UPD ANTATTDETECTREF** command can be used to update the antenna attitude reference value. The **DSP RAESUBUNIT** and **DSP ALDSUBUNIT** commands can be used to query the reference value.

Impact on the Network

No impact

Impact on NEs

No impact

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following MML commands have changed.

Change Type	MML Command	NE	Description
Modified	DSP RAEFUNCTION	eGBTS NodeB eNodeB gNodeB	The command output now includes an antenna attitude detection function field.
Modified	DSP ALDFUNCTION	eGBTS NodeB eNodeB gNodeB	The command output now includes an antenna attitude detection function field when querying RAE devices.
Modified	DSP RAEDEVICEDATA	eGBTS NodeB eNodeB gNodeB	The command output now includes a roll angle field.
Modified	DSP ALDDEVICEDATA	eGBTS NodeB eNodeB gNodeB	The command output now includes a roll angle field when querying RAE devices.
New	UPD ANTATTDETECTREF	eGBTS NodeB eNodeB gNodeB	This command can now be used to update the detected antenna attitude reference value.
Modified	DSP RAESUBUNIT	eGBTS NodeB eNodeB gNodeB	This command can now be used to query the detected antenna attitude reference value.

Change Type	MML Command	NE	Description
Modified	DSP ALDSUBUNIT	eGBTS NodeB eNodeB gNodeB	This command can now be used to query the detected antenna attitude reference value for RAE devices.
Modified	SET EQUIPMENT	eGBTS NodeB eNodeB gNodeB	This command can now be used to set an antenna attitude detection switch.
Modified	LST EQUIPMENT	eGBTS NodeB eNodeB gNodeB	This command can now be used to query the status of the antenna attitude detection switch.

- Performance management
No impact
- Fault management

EVT-26764 Antenna Attitude Detection Event has been added. This event is triggered if the detected value is beyond or falls within the set internal range.

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

- Whether the change takes effect by default after an upgrade varies as follows:
 - Roll angle queries: Yes
 - Antenna attitude detection: No. You can run the **SET EQUIPMENT** command with the **ANT_ATT_DETECT_SW** option of the **EQUIPMENT.EQMOPTFEATSW** parameter selected, and then run the **UPD ANTATTDETECTREF** command to enable the antenna attitude detection function.

Requirement/Trouble Ticket Number

AR-HERT_26B-FuR- IR20250603002557-01-02

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.3.25 Digital Antennas Supporting the Quick Scanning Function

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The quick scanning function has been introduced to digital antennas to improve the O&M efficiency.

Implementation

The **SCN ALD** command now includes the **SCANMODE** parameter to specify the scanning mode.

Impact on the Network

No impact

Impact on NEs

No impact

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following MML commands have changed.

Change Type	MML Command	NE	Description
Modified	SCN ALD	eGBTS NodeB eNodeB gNodeB	This command now includes the SCANMODE parameter to specify the scanning mode.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *ALD Management*.

Requirement/Trouble Ticket Number

AR-HERT_26B-FuR- IR20250530000378-01-02

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.3.26 Digital Antennas Supporting Low-Temperature Reliability Design

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The low-temperature reliability mechanism is introduced for digital antennas. If an ALD detects that the ambient temperature is below -40°C , it forbids the adjustment of the azimuth of the antenna to protect the hardware.

Implementation

- An azimuth adjustable status field (**Azimuth Adjustable Status**) has been added to the **DSP RASSUBUNIT** and **DSP ALDSUBUNIT** command outputs.
- If the value of **Azimuth Adjustable Status** is **NONADJUSTABLE**, operators cannot adjust the azimuth using the **MOD RASSUBUNIT** or **MOD ALDSUBUNIT** command. Running these commands online will trigger an error message.

Impact on the Network

No impact

Impact on NEs

No impact

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following MML commands have changed.

Change Type	MML Command	NE	Description
Modified	DSP RASSUBUNIT	eGBTS NodeB eNodeB gNodeB	An azimuth adjustable status field has been added to the command output.
Modified	DSP ALDSUBUNIT	eGBTS NodeB eNodeB gNodeB	An azimuth adjustable status field has been added to the command output.
Modified	MOD RASSUBUNIT	eGBTS NodeB eNodeB gNodeB	If the value of Azimuth Adjustable Status is NONADJUSTABLE , operators cannot adjust the azimuth using this command. Running this command online will trigger an error message.

Change Type	MML Command	NE	Description
Modified	MOD ALDSUBUNIT	eGBTS NodeB eNodeB gNodeB	If the value of Azimuth Adjustable Status is NONADJUSTABLE , operators cannot adjust the azimuth using this command. Running this command online will trigger an error message.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
If an ALD detects that the ambient temperature is below -40°C , it forbids the adjustment of the azimuth. Running the **MOD RASSUBUNIT** or **MOD ALDSUBUNIT** command triggers an error message.

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

AR-HERT_26B-FuR-IR20250728000347-01-01

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.3.27 Digital Antennas Supporting Topology View on the Device Panel of the OSS

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The DSP commands for querying the new model are now compatible with the old model.

Implementation

The **DSP ALDSUBUNIT** and **DSP ALDDEVICEDATA** commands are now compatible with the old model.

Impact on the Network

No impact

Impact on NEs

No impact

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management

Change Type	MML Command	NE	Description
Modified	DSP ALDDEVICEDATA	eNodeB gNodeB	This command is used to query the dynamic information in the ALD device data.

Change Type	MML Command	NE	Description
Modified	DSP ALDSUBUNIT	eNodeB gNodeB	This command is used to query the dynamic information about an ALD subunit.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

AR-HERT_26B-FuR- IR20250603002448-01-01

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.3.28 Optimized Fronthaul Optical Path Fault Detection

NEs Involved

5900 series base stations

Feature/Function Description

This function optimizes the GUI display of fronthaul optical path fault detection.

Implementation

- **UP_LINK_RX_NOSIGNAL** has been added to the values of **Up Link Detect Fail Reason** for cases where the uplink is interrupted.

- **OPTICAL_PATH_END** has been added to the values of **Fault Type** for cases where the CPRI is abnormal but the reflection rate is normal.
- **Distances for the Latest Optical Path Quality Detection** has been added to the **STR OPTPATHQUALITYDET** command output to assist maintenance personnel in troubleshooting.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the Node.

Impact on Hardware Requirements

- Base station models: 3900 and 5900 series base stations (macro base stations) and DBS3900 and DBS5900 LampSite base stations.
- BBUs: BBU3900, BBU3910, BBU5900, and BBU5910
- Boards:
 - Main control boards: UMPTb, UMPTe, UMPTg, UMPTga, UMPTth, and UMPTtha series boards
 - Baseband processing units: UBBPd, UBBPe (except UBBPeb), UBBPg, and UBBPh series boards that support a CPRI port rate of 10 Gbit/s, 25 Gbit/s, or 50 Gbit/s
- RF modules: Support a CPRI port rate of 25 Gbit/s or 50 Gbit/s. You can run the **DSP RRU** or **DSP CXU** command to check whether the optical path quality detection function is supported.
- Optical modules: Support the OTDR function. (To check whether the OTDR function is supported, run the **DSP SFP** command and check the **OPTICAL_PATH_QUALITY_DETECTION** setting of the **Function Support** field in the command output.)

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following MML commands have changed.

Change Type	Command	NE	Description
Modified	STR OPTPATHQUALITYDET	Node	1. UP_LINK_RX_NOSIGNAL has been added to the values of the Up Link Detect Fail Reason parameter. This parameter indicates the reason why uplink optical path quality detection fails. The value of this parameter can be STARTUP_FAIL, DETECT_FAIL, OPTICAL_MODULE_ABNORMAL, OPTICAL_MODULE_NOT_SUPPORT, RF_MODULE_NOT_SUPPORT, RF_MODULE_NOT_CONFIGURED, RF_LINK_UNAVAILABLE, BOARD_UNAVAILABLE, or UP_LINK_RX_NOSIGNAL. If the value of this parameter is UP_LINK_RX_NOSIGNAL, the optical module at the uplink receive end does not receive signals, indicating that a fault occurs on the uplink. In this case, you are advised to check the uplink first. NULL is displayed if no failure cause is found. 2. OPTICAL_PATH_END has been added to the

Change Type	Command	NE	Description
			values of the Fault Type parameter. This parameter indicates the fault type corresponding to a node over an optical path in optical path quality detection. The value of this parameter can be DIRTY_CONNECT OR, BENT_FIBER, OPTICAL_PATH_INTERRUPTED, OPTICAL_PATH_END, or NORMAL. If the value of this parameter is OPTICAL_PATH_INTERRUPTED, a fault occurs on the optical path node, causing optical signals to be interrupted at this node. If the value of this parameter is OPTICAL_PATH_END, optical signals are interrupted at the optical path node; in this case, if other optical path nodes are normal while the CPRI link is still abnormal, you are advised to check the optical path node. 3. Distances for the Latest Optical Path Quality Detection has been added to the command output

Change Type	Command	NE	Description
			to display the uplink and downlink detection distances in the last OTDR detection result.

Change Type	Command	NE	Description
Modified	DSP OPTPATHQUALITYDET	Node	1. UP_LINK_RX_NOSIGNAL has been added to the values of the Up Link Detect Fail Reason parameter. This parameter indicates the reason why uplink optical path quality detection fails. The value of this parameter can be STARTUP_FAIL, DETECT_FAIL, OPTICAL_MODULE_ABNORMAL, OPTICAL_MODULE_NOT_SUPPORT, RF_MODULE_NOT_SUPPORT, RF_MODULE_NOT_CONFIGURED, RF_LINK_UNAVAILABLE, BOARD_UNAVAILABLE, or UP_LINK_RX_NOSIGNAL. If the value of this parameter is UP_LINK_RX_NOSIGNAL, the optical module at the uplink receive end does not receive signals, indicating that a fault occurs on the uplink. In this case, you are advised to check the uplink first. NULL is displayed if no failure cause is found. 2. OPTICAL_PATH_END has been added to the

Change Type	Command	NE	Description
			values of the Fault Type parameter. This parameter indicates the fault type corresponding to a node over an optical path in optical path quality detection. The value of this parameter can be DIRTY_CONNECT OR, BENT_FIBER, OPTICAL_PATH_INTERRUPTED, OPTICAL_PATH_END, or NORMAL. If the value of this parameter is OPTICAL_PATH_INTERRUPTED, a fault occurs on the optical path node, causing optical signals to be interrupted at this node. If the value of this parameter is OPTICAL_PATH_END, optical signals are interrupted at the optical path node; in this case, if other optical path nodes are normal while the CPRI link is still abnormal, you are advised to check the optical path node.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20251204000006-0180

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.4.3.29 Put iSDU06D/iSDU03D Used with a Third-Party Power System into Commercial Use

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The iSDU06D/iSDU03D used with a third-party power system function is changed from a trial function to a commercial function.

This change is irrelevant to LampSite base stations.

Implementation

For details, see *Green Site*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20251204001679-0020

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5 Operation and Maintenance

5.2.5.1 General Management

This section describes the changes in general management.

5.2.5.1.1 Added the High-Precision Measurement Function for Transmission Port Performance Monitoring

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Transmission port performance monitoring now supports high-precision measurement, allowing 1 ms or 10 ms traffic statistics on EthPort or EthTrunk ports carrying IP transmission.

Implementation

The RX/TX traffic of EthPort or EthTrunk ports carrying IP transmission is sampled at high-precision measurement intervals. At the end of the measurement, the following statistics are obtained: High-precision RX Maximum Rate of Link Layer(pps), High-precision RX Maximum Peak-to-Average Ratio of Pkts of Link Layer, High-precision RX Average Peak-to-Average Ratio of Pkts of Link Layer, High-precision TX Maximum Rate of Link Layer(bps), High-precision TX Maximum Rate of Link Layer(pps), High-precision TX Maximum Peak-to-Average Ratio of Rate of Link Layer, and High-precision TX Average Peak-to-Average Ratio of Rate of Link Layer.

Impact on the Network

No impact

Impact on NEs

No impact

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact

- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250526001584-0010

FeR-IR20251204000006-0120

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.1.2 Expanded the Value Range of Both TX Channel Number and Transmitter Number to 0 to 23.

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

- The value range of the TX channel number for the distance to fault (DTF) test is expanded to 0 to 23.
- The value range of the transmitter number for the Distance to Passive Inter Modulation (DTP) test is expanded to 0 to 23.

Implementation

- The value range of the TX channel number for the DTF test is expanded to 0 to 23. The delivered TX channel number must be within the channel number range of the corresponding RF unit. If it exceeds this range, the detection fails.
- The value range of the transmitter number for the DTP test is expanded to 0 to 23. The delivered transmitter number must be within the transmitter number range of the corresponding RF unit. If it exceeds this range, the detection fails.

Impact on the Network

No impact

Impact on NEs

No impact

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250711001672-0020

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.1.3 Enhanced Operation Log Recording for Asynchronous MML Commands

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Operation log recording has been enhanced for asynchronous MML commands. Specifically, operation logs are recorded for asynchronous MML commands when a client is disconnected from the base station. An asynchronous MML command does not take effect immediately upon execution. Instead, it requires some time to take effect. For example, when you run a command to upload a file, a message is displayed, indicating that the command is executed successfully. However, the file

upload is complete only after the file upload progress reaches 100%. Such a command is an asynchronous MML command.

Implementation

After this function is enabled, during the reporting of MML command progress messages, the base station can cache information about a maximum of three disconnections with the client, which is used to record operation logs for the unfinished MML commands. The historical connection information stored on the base station includes fields such as the source, username, domain, IP type, and client IP address.

If the number of disconnections exceeds three within a certain period, the cached information will be overwritten. As a result, in the MML command progress report, the connection information cannot be found and operation logs cannot be recorded.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the eGBTS, NodeB, eNodeB, and gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact

- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20250319000813

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.1.4 AdoptOpenJDK21 Recommended for the WebLMT

NEs Involved

BSC6910, 3900 series base stations, 5900 series base stations, USU3910

Feature/Function Description

The new version of third-party software in the Java open-source ecosystem does not support AdoptOpenJDK8. AdoptOpenJDK21 is recommended for the WebLMT in accordance with the JDK lifecycle development and open-source software lifecycle management requirements.

Implementation

From this version onwards, AdoptOpenJDK21 is recommended for the WebLMT. Before using AdoptOpenJDK21 and JRE 1.8.X of AdoptOpenJDK to open a JNLP file, install the OpenWebStart plug-in. Otherwise, the JNLP file cannot be opened.

Impact on the Network

No impact

Impact on NEs

No impact

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

26B_WEBLMT_001

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.1.5 Added Security Log Recording After a KMC-Managed Key File Is Damaged

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

A key management CBB (KMC) is a key management module that implements basic functions such as encrypted key storage, encryption, and decryption. When a KMC-managed key file is damaged, the base station attempts to restore it from the backup file. As an optimization, security log recording has been added to this restoration procedure.

Implementation

Security log recording is a common mechanism of the base station. A security log entry has been added to the processing procedure after a key file is damaged.

The security event "Restore a key file." is added to the **LST SECLOG** command. If a key file is damaged and an attempt is made to restore this key file, the corresponding security log is recorded. The execution result of the security event "Restore a key file." in the **LST SECLOG** command output indicates whether this key file has been successfully restored.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the eGBTS, NodeB, eNodeB, and gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

SR-IR20240902002286-0660

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.1.6 Risk Mitigation for the Default Password of the emscomm User

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

This function checks whether the password of the **emscomm** user is the factory password on the MAE and, if so, reminds the user to change the password. This mitigates security risks.

Implementation

This function involves the following optimizations:

- The check item "Factory password for the MAE emscomm user" is added to the MAE-Access to check whether the password of the **emscomm** user is the factory password for each NE.
- The check item "Access-Checking whether the global emscomm password equals the default password" is added to the MAE-OSMU to check whether the global password of the **emscomm** user equals the default password.
- A message is added to the MAE to remind users to change the global password of the **emscomm** user if it equals the default password.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the MAE.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20241227001926-0010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.1.7 Support for Displaying the Wavelength of the LampSite RHUB Optical Module on the MAE Device Panel

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

This function enables the MAE device panel to display the wavelength of the LampSite RHUB optical module.

Implementation

The LampSite reports the wavelength of the RHUB optical module to the MAE, which subsequently displays the wavelength on the device panel.

Impact on the Network

No impact

Impact on NEs

No impact

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration Management
No impact
- Performance Management
No impact
- Fault Management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250303001254-004

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS

5.2.5.2 Configuration Management

This section describes the changes in configuration management.

For details about changes in MML commands, MOs, and parameters, see *MML Command Changes* and *MO and Parameter Changes*, which are provided with the software version. You can obtain *MML Command Changes* and *MO and Parameter Changes* from the following directory: **BTS3900&BTS5900 VXXXRXXXCXXSPCXXX Release Documents-EN\01-Release Notes\Delta(MMI)**.

5.2.5.2.1 Optimized the DSP PDPATTR Command Output

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The **PDPATTR.PORTSTATUSCONDITION (5G gNodeB, LTE eNodeB)** parameter has been added to the **DSP PDPATTR** command to query the status of a specified power distribution port.

This change is irrelevant to LampSite base stations.

Implementation

The **PDPATTR.PORTSTATUSCONDITION (5G gNodeB, LTE eNodeB)** parameter has been added to the **DSP PDPATTR** command to indicate the status of a specified power distribution port. In the iSDU subrack, the port status indicates only the status of the internal contactors of circuit breaker output ports, excluding the status of the circuit breakers for the ports. Specifically:

- For non-intelligent power distribution ports, the value of this parameter is NULL.
- For intelligent power distribution ports:
 - **FAULT** indicates that the power distribution port is faulty.
 - **POWER_OFF** indicates that the contactor of the power distribution port is disconnected.
 - **POWER_ON** indicates that the contactor of the power distribution port is connected.
 - **NOT_IN_POSITION** indicates that no intelligent circuit breaker is installed for the power distribution port. (This value can be displayed only if a power system whose type is SMU is used.)

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following MML commands have changed.

Change Type	MML Command	NE	Description
Modified	DSP PDPATTR	eGBTS NodeB eNodeB gNodeB	The PDPATTR.PORTSTATUSCONDITION (5G gNodeB, LTE eNodeB) parameter has been added to the DSP PDPATTR command to query the status of a specified power distribution port.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250217003314-0010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.2.2 Optimized Display of the iSDU Power Supply Topology

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

For the RF modules in the iSDU topology, only their ESNs are displayed, and the information of the NEs where they are located and their cabinet, subrack, and slot information are not displayed. As a result, the cabinet, subrack, and slot numbers of the modules corresponding to the iSDU ports cannot be viewed, affecting O&M efficiency. To resolve this issue, the display of the power supply topology has been optimized, so that the NEs where the RF modules are located, and cabinet, subrack, and slot numbers of the RF modules are now displayed.

This change is irrelevant to LampSite base stations.

Implementation

The display of the power supply topology has been optimized, so that the NEs where the RF modules are located, and cabinet, subrack, and slot numbers of the RF modules are now displayed.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250918001890-0010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.2.3 New Fields in the DSP SWITCHINGPORT Command Output

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

This function optimizes the output of the **DSP SWITCHINGPORT** command to make the information more complete.

Implementation

This function adds the numbers of pre-FEC and post-FEC errored blocks to the output of the **DSP SWITCHINGPORT** command.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB and eNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following MML commands have changed.

Change Type	MML Command	NE	Description
Modified	DSP SWITCHING PORT	gNodeB eNodeB	The numbers of pre-FEC and post-FEC errored blocks have been added to the command output.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250806001609-0230

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD

5.2.5.2.4 Optimized Reliability Processing Upon an Upgrade After a Rollback

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

If an app with missing software packages is configured and then upgraded, the missing software packages are supplemented in the target version, and the corresponding configuration is added. After rolling back to the source version, the app will remain unavailable. However, the configuration file generated for the app in the upgraded version is retained internally. If the app is upgraded again, the NE attempts to restore the previous configuration data of the app.

This function optimizes the processing of configuration data during re-upgrade: the app no longer restores its previous configuration but starts with a fresh initial configuration.

Implementation

With this function, when an app is rolled back to the source version after an upgrade, any configuration file associated with the app—with missing software packages—is automatically cleared once the app reaches a stable state.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions

- No impact
- Impacted features/functions
- No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20250319000813

SR-IR20250319000813-0160

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.2.5 Enhanced Maintenance and Test Reliability of DSP CFGFILEACTSTAT

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

After the configuration file is activated, the NE is started, and the apps are started in sequence.

- Before the upgrade: **Success** is displayed when the **DSP CFGFILEACTSTAT** command is executed to query the configuration file activation result during the app startup.
- After the upgrade: The output of the **DSP CFGFILEACTSTAT** command is optimized during the app startup.

Implementation

With this function, **Activating** is displayed in the **DSP CFGFILEACTSTAT** command output during the app startup.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following MML commands have changed.

Change Type	MML Command	NE	Description
Modified	DSP CFGFILEACTSTAT	eGBTS NodeB eNodeB gNodeB	The display result during app startup has been modified.

- Performance management

No impact

- Fault management

No impact

Impact on Features/Functions

- Prerequisite features/functions

No impact

- Mutually exclusive features/functions

No impact

- Impacted features/functions

No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20250319000835

SR-IR20250319000835-0530

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.2.6 Optimized the Application Type Field in the DSP CFGFILEACTSTAT Command Output

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Before the upgrade, the **Application Type** field in the **DSP CFGFILEACTSTAT** command output is displayed as a character string only when the corresponding software packages are available; otherwise, it appears as a numeric value.

In this version, the display format of **Application Type** has been optimized, and a new field, **Extended Application Type**, has been introduced.

Implementation

In the **DSP CFGFILEACTSTAT** command output:

1. The **Application Type** field has been optimized:
 - For basic applications, the application type is displayed.
 - For extended applications, **EXTAPPTYPE** is always displayed.
2. The **Extended Application Type** field has been added:
 - For basic applications, **NULL** is displayed.
 - For extended applications, the actual extended application types configured by the customer are displayed.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following MML commands have changed.

Change Type	MML Command	NE	Description
Modified	DSP CFGFILEA CTSTAT	eGBTS NodeB eNodeB gNodeB	- The Application Type field has been optimized: - For basic applications, the application type is displayed. - For extended applications, EXTAPPTYPE is always displayed. - The Extended Application Type field has been added: - For basic applications, NULL is displayed. - For extended applications, the actual extended application types configured by the customer are displayed.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20250319000813

SR-IR20250319000813-0150

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.2.7 Added Support for Packet Loss Detection for IPv6 IP PM Sessions

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

As the base station is unaware of the transmission quality of the backhaul network, packet loss detection for IP PM sessions is important for service self-protection and is used to monitor network quality or guarantee the QoS. In IPv6 networking, packet loss detection for IPv6 IP PM sessions can be performed, effectively preventing the impact of backhaul network transmission exceptions on communications service transmission. This function is controlled by the **TRANSFUNCTIONSW.IPPMIPV6PKTLOSDETSW (5G gNodeB, LTE eNodeB)** parameter, whose default value is **DISABLE**.

This function takes effect only if both the local and peer ends of an IP PM session support IPv6 packet loss rate measurement.

Implementation

For details, see *IP Performance Monitor*. The specific sections are listed in the change history of the document.

Impact on the Network

After this function takes effect and packet loss is detected on the transport network, the base station proactively adjusts the transmit bandwidth to reduce packet loss, which may cause the transmit traffic to fluctuate.

For the base station configured with IPv6 IP PM sessions, the values of the following counters may change from zero to a non-zero one:

- **VS.IPPM.Forword.DropMeans (LTE eNodeB, 5G gNodeB)**
- **VS.IPPM.Forword.Peak.DropRates (LTE eNodeB, 5G gNodeB)**
- **VS.IPPM.Bits.MeansTx (LTE eNodeB, 5G gNodeB)**
- **VS.IPPM.Pkts.MeansTx (LTE eNodeB, 5G gNodeB)**
- **VS.IPPM.Peak.Bits.RateTx (LTE eNodeB, 5G gNodeB)**
- **VS.IPPM.Peak.Pkts.RateTx (LTE eNodeB, 5G gNodeB)**
- **VS.IPPM.Peer.Bits.MeansRx (LTE eNodeB, 5G gNodeB)**
- **VS.IPPM.Peer.Pkts.MeansRx (LTE eNodeB, 5G gNodeB)**
- **VS.IPPM.Peer.Peak.Bits.RateRx (LTE eNodeB, 5G gNodeB)**
- **VS.IPPM.Peer.Peak.Pkts.RateRx (LTE eNodeB, 5G gNodeB)**
- **VS.IPPM.Local.TxPkts (LTE eNodeB, 5G gNodeB)**
- **VS.IPPM.Peer.RxPkts (LTE eNodeB, 5G gNodeB)**

For the base station configured with IPv4 or IPv6 IP PM sessions, the values of the following counters may slightly increase:

- **VS.IPPM.HighLevel.Rtt.Means (LTE eNodeB, 5G gNodeB)**

- **VS.IPPM.HighLevel.MaxRttDelay** (LTE eNodeB, 5G gNodeB)
- **VS.IPPM.HighLevel.MinRttDelay** (LTE eNodeB, 5G gNodeB)
- **VS.IPPM.Rtt.Means** (LTE eNodeB, 5G gNodeB)
- **VS.IPPM.MaxRttDelay** (LTE eNodeB, 5G gNodeB)
- **VS.IPPM.MinRttDelay** (LTE eNodeB, 5G gNodeB)
- **VS.IPPM.Owd.Means** (LTE eNodeB, 5G gNodeB)
- **VS.IPPM.MaxOwdDelay** (LTE eNodeB, 5G gNodeB)
- **VS.IPPM.MinOwdDelay** (LTE eNodeB, 5G gNodeB)

Changing the setting of the **IP PM IPv6 Pkt Loss Detect Switch** parameter will cause service exceptions during IPv6 IP PM renegotiation, affecting all inter-base-station coordination and data split services established using IPv6. Services can be initiated again only after IPv6 IP PM negotiation is normal.

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

- Base station models
The BBU3910A does not support packet loss detection for IPv6 IP PM sessions.
- Boards
Main control boards: The UMPTb does not support packet loss detection for IPv6 IP PM sessions.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following MML commands have changed.

Change Type	MML Command	NE	Description
Modified	DSP IPPMSESSION	eGBTS NodeB eNodeB gNodeB	In the output of the DSP IPPMSESSION command, which is used to query the status of an IP PM session, the value of TX Loss Rate(per mill) changes from NULL to a specific value.

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	TRANSFUNC TIONSW. <i>IPP</i> <i>MIPV6PKTL</i> <i>OSDETSW</i>	SET TRANSFUNC TIONSW LST TRANSFUNC TIONSW	eGBTS NodeB eNodeB gNodeB	This parameter has been added to specify whether the base station supports packet loss detection for IPv6 IP PM sessions. The base station supports this function if this parameter is set to ENABLE and does not support this function if this parameter is set to DISABLE .

- Performance management
No impact
- Fault management
The following alarms have changed.

Change Type	Alarm Name	NE	Description
Modified	ALM-25959 User Plane Path Excessive Packet Loss Rate	eGBTS NodeB eNodeB gNodeB	If the ADD IPPMSESSION command is executed with IP Version set to IPv6 , the status of the IP PM session is now associated with ALM-25959 User Plane Path Excessive Packet Loss Rate. With IPPMSESSION.ASSOCIATEUPATHPLR (LTE eNodeB, 5G gNodeB) set to YES , if the packet loss rate exceeds the specified threshold, ALM-25959 User Plane Path Excessive Packet Loss Rate is reported.
Modified	ALM-25900 IP PM Activation Failure	eGBTS NodeB eNodeB gNodeB	Changing the setting of the IP PM IPv6 Pkt Loss Detect Switch parameter will cause ALM-25900 IP PM Activation Failure to be temporarily reported for the IP PM sessions added by running the ADD IPPMSESSION command. This alarm can be automatically cleared.

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *IP Performance Monitor*.

Requirement/Trouble Ticket Number

FeR-IR20250114003368-0010

FeR-IR20250213000671-0180

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.2.8 Support for Querying Whether a Site Is a POC Site

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The **Poc Flag** parameter has been added to specify whether the site is a proof-of-concept (POC) site. For a POC site, some functions are restricted.

Implementation

The **Poc Flag** parameter has been added to specify whether the site is a POC site. Its value can be obtained by running the **LST SYSOM** command and its default value is **NO**. For a POC site, the functions of exporting the summary data file for base station deployment, comparing parameters, and exporting the configuration template by right-clicking are restricted.

If the site purchased by a customer is a POC site based on the query result, contact Huawei engineers.

Impact on the Network

No impact

Impact on NEs

This change does not involve NE requirements and is implemented only on the MAE.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact

- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Run the **LST SYSOM** command to query the **Poc Flag** parameter.

Requirement/Trouble Ticket Number

FeR-IR20250114000986-0020

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.2.9 Support for Foolproof Detection for Parameters in the RST APP Command

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The optional parameters **APPLICATION.AT (LTE eNodeB, 5G gNodeB)** and **APPLICATION.EXTAPPTYPE (LTE eNodeB, 5G gNodeB)** have been added. The system uses these parameters to help check whether the application corresponding to the **APPLICATION.AID (LTE eNodeB, 5G gNodeB)** parameter is of the expected type. This provides foolproof detection for the parameters specified using the **RST APP** command.

Implementation

This function is implemented as follows:

- The optional parameter **APPLICATION.AT (LTE eNodeB, 5G gNodeB)** has been added to specify the application type.
- The optional parameter **APPLICATION.EXTAPPTYPE (LTE eNodeB, 5G gNodeB)** has been added to specify the extended application type.

The system checks whether the **APPLICATION.AID (LTE eNodeB, 5G gNodeB)** parameter value matches the specified application type or extended application

type when running the reset command. If yes, the reset will succeed; if not, the reset will fail.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	APPLICATION.AT (LTE eNodeB, 5G gNodeB)	ADD APP MOD APP RMV APP RST APP DSP APP LST APP	eGBTS NodeB eNodeB gNodeB	This parameter has been added to specify the application type to ensure the accuracy of the RAT to which the application to be reset belongs.
New	APPLICATION.EXTAPPTYPE (LTE eNodeB, 5G gNodeB)	ADD APP MOD APP RMV APP RST APP DSP APP LST APP	eGBTS NodeB eNodeB gNodeB	This parameter has been added to specify the extended application type to ensure the accuracy of the RAT to which the application to be reset belongs.

- Performance management

No impact

- Fault management

No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

Run the **RST APP** command with the **APPLICATION.AT (LTE eNodeB, 5G gNodeB)** or **APPLICATION.EXTAPPTYPE (LTE eNodeB, 5G gNodeB)** parameter specified.

Requirement/Trouble Ticket Number

FeR-IR20250115001908-0010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.2.10 Enhanced Health Check and Self-Healing for Self-Organizing Networks

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Fast check is now supported for self-organizing networks to check whether the configurations of key objects that are created automatically are consistent with user configurations. If they are inconsistent, the inconsistency is automatically rectified.

This function is controlled by the **SET HEALCHKSW** command, which can also be used to set the check period and check time.

Implementation

This function verifies the configurations for self-setup by checking whether, for an **EPGROUP** MO, the **SCTPHOST** and **SCTPPEER** MOs have corresponding **SCTPLNK** MOs, the **USERPLANEHOST** and **USERPLANEPEER** MOs have corresponding **EPPATHs**, and the **SECURITYHOST** and **SECURITYPEER** MOs have corresponding **IKEPEER** or **TUNNELITF** MOs.

If the configurations are incorrect, the system will automatically rectify them.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

Run the **SET HEALCHSW** command to set the health check switch. For detailed settings, contact Huawei engineers.

Requirement/Trouble Ticket Number

SR-IR20240902002296-0010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.2.11 Extended Software Version Included in Configuration Data

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

During the configuration data backup of an NE, if the extended software version of the NE (specified by the **NE.ExtendedSwVersion (LTE eNodeB, 5G gNodeB)** parameter) is configured, it is included in the configuration data.

When a deployment task is created using the configuration data, the extended software version is automatically filled in based on the configuration data.

This change is irrelevant to LampSite base stations.

Implementation

The extended software version of an NE (specified by the **NE.ExtendedSwVersion (LTE eNodeB, 5G gNodeB)** parameter) is now included in its configuration data.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	<i>NE.ExtendedSwVersion (LTE eNodeB, 5G gNodeB)</i>	LST NE	eGBTS NodeB eNodeB gNodeB	This parameter has been added to specify the extended software version of an NE.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250114001465-0010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.2.12 Optimized DHCP-based RAT Change Mechanism

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Before an upgrade, if DHCP packets carry incorrect RAT information in scenarios other than site deployment, the RAT information is modified. As a result, the base station may fail to start after being reset.

After the upgrade, the RAT information is not modified in scenarios other than site deployment.

Implementation

A configuration activation flag has been added to determine whether the current scenario is a site deployment scenario. Modifying RAT information is supported only in the site deployment scenario.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

SR-IR20240902002296-0480

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.2.13 Precise High-Risk Warnings Provided for Some Parameters in MML Commands

NEs Involved

3900 series base stations, 5900 series base stations, USU3910

Feature/Function Description

Precise high-risk warnings are now provided for some enumerated and bitmap parameters in MML commands.

Implementation

Precise high-risk warnings are provided for some enumerated and bitmap parameters in MML commands.

For example, if the **ADD FLOODDEFEND** command is executed with the **FLOODDEFEND.DFDSW (LTE eNodeB, 5G gNodeB)** parameters set to **DISABLE**, the following message is displayed: "If Defend Switch is set to DISABLE, flood attacks cannot be prevented, which poses security risks. Therefore, you are advised to set this parameter to ENABLE."

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact

- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

SR-IR20240902002286-0620

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.2.14 Permission Control for Adjusting RXU Maintenance Mode

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Permission control is now available for changing the maintenance mode of an RXU board (RRU, CXU, or RHUB) to an engineering state (that is, a non-normal state).

Implementation

This function is implemented as follows:

1. The base-station-level MML command **SET MNTMODEPLY** has been added to determine whether an RXU board (RRU, CXU, or RHUB) is allowed to enter an engineering state.
2. The group-based MML permission management function of the MAE is used. Specifically, only high-privilege users can execute the **SET MNTMODEPLY** command. They can run this command to set the **SYSTEMOM.RXUMNTENABLEPERMISSION (LTE eNodeB, 5G gNodeB)** parameter to **OFF**. As a result, low-privilege users are restricted from modifying the maintenance mode of an RXU board through commands such as **MOD RRU**, **MOD CXU**, and **MOD RHUB**. This prevents unauthorized alarm masking by low-privilege users.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

Only RRUs, CXUs, and RHUBs are involved.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following MML commands have changed.

Change Type	MML Command	NE	Description
New	SET MNTMODEPLY	eGBTS NodeB eNodeB gNodeB	This command has been added to set the maintenance mode adjustment policy configuration.
New	LST MNTMODEPLY	eGBTS NodeB eNodeB gNodeB	This command has been added to list the maintenance mode adjustment policy configuration.

NOTE

The parameters in new and deleted MML commands are no longer listed separately.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions

- No impact
- Mutually exclusive features/functions
- No impact
- Impacted features/functions
- No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

SR-IR20240902002286-0300

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.2.15 Optimized Value Policy for a Bit-Field-Type Parameter in an ADD Command

NEs Involved

3900 series base stations, 5900 series base stations, USU3910

Feature/Function Description

The value policy for bit-field-type parameters in ADD commands is now optimized, improving MML-based O&M efficiency.

Implementation

Before an upgrade, if a bit-field-type parameter in an ADD command has one bit field specified, either selected or deselected, the system sets all unspecified bit fields to 0 by default to indicate a disabled state.

After the upgrade, the **SYSTEMOM.BitFieldParaValueMode (LTE eNodeB, 5G gNodeB)** parameter has been added to specify the value policy for a bit-field-type parameter in an ADD command.

- If this parameter is set to **BIT_VALUE_0**, the pre-upgrade value policy remains unchanged.
- If this parameter is set to **BIT_VALUE_DEF** and a bit-field-type parameter in an ADD command has one bit field specified, either selected or deselected, the system sets all unspecified bit fields to their default values.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	SYSTEMOM.BitFieldParaValueMode (LTE eNodeB, 5G gNodeB)	SET SYSOM LST SYSOM	eGBTS NodeB eNodeB gNodeB	This parameter has been added to specify the value policy for bit-field-type parameters in ADD commands.

- Performance management

No impact

- Fault management

No impact

Impact on Features/Functions

- Prerequisite features/functions

No impact

- Mutually exclusive features/functions

No impact

- Impacted features/functions

No impact

Related Operations

By default, the change does not take effect after an upgrade.

Run the **SET SYSOM** command with the **SYSTEMOM.BitFieldParaValueMode (LTE eNodeB, 5G gNodeB)** parameter set to **BIT_VALUE_DEF**.

Requirement/Trouble Ticket Number

FeR-IR20250213000671-0150

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.2.16 SNR Query of 50G Backhaul Optical Modules

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

50G backhaul optical modules now supports querying signal-to-noise ratio (SNR) using **DSP SFP**, enabling rapid verification of optical link quality and clear identification of packet errors and losses. This enhances both engineering delivery and routine maintenance efficiency for backhaul optical paths.

Implementation

For 50G backhaul optical modules, the value of **FunSupport** is **SNR Function:SUPPORT**, and the **Signal To Noise Ratio(0.1dB)** field displays the SNR value after the **DSP SFP** command is executed.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB and eNodeB.

Impact on Hardware Requirements

This change is supported only when a 50G backhaul optical module is used and the main control board is a UMPTha/UMPTb series board.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact

- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250806001609-0370

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD

5.2.5.2.17 Enhanced Memory Resource Monitoring and Inspection

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

This function enables real-time online monitoring of board memory resource usage and logs events when the memory usage exceeds a certain percentage of the upper limit. Insufficient memory issues can be identified by inspecting these logs.

Implementation

This function works by setting a memory usage baseline for each board type. During operation, the memory usage of each board is periodically scanned and compared against its baseline. If the usage exceeds a certain percentage of the upper limit, the system records logs.

Impact on the Network

In no-load scenarios, the values of the **VS.BBUBoard.CPULoad.Max** and **VS.BBUBoard.CPULoad.Mean** counters slightly increase.

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20250319000813

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.2.18 Added the Function of Reporting Simplified Logs.

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Before an upgrade, the NE collects an excessive number of logs, making download and upload difficult and time-consuming. This severely compromises the comprehensiveness of early network evaluation in the new version.

After the upgrade, the NE can independently report simplified logs to help you preliminarily identify faulty base stations. Full logs or tracing data of identified base stations can then be collected to locate the root cause.

Implementation

This function is implemented as follows:

SLOG-NODE is added to the **SRCF** parameter in the **ULD FILE** command to facilitate collections of simplified logs, with a total size of up to 1 MB per base station.

Collecting node simplified logs is added to the **COMM_Logs and Alarms** category on the NIC collection page, facilitating network-level batch collection.

Impact on the Network

After an upgrade, the values of the **VS.BBUBoard.CPULoad.Max (LTE eNodeB, 5G gNodeB)** and **VS.BBUBoard.CPULoad.Mean (LTE eNodeB, 5G gNodeB)** counters slightly increase.

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	<i>SRCF</i>	ULD FILE	eGBTS NodeB eNodeB gNodeB	The enumerated value SLOG-NODE has been added to control the upload of simplified logs.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FuR-SAR-IR20241209002193-0030_0020-07

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.3 Performance Management

This section describes the changes in performance management.

For details about changes in performance counters and EBCs, see *Performance Counter Changes* and *EBC Changes*, which are provided with the software version. You can obtain *Performance Counter Changes* and *EBC Changes* from the following directory: **BTS3900&BTS5900 VXXXRXXCXXSPCXXX Release Documents-EN\01-Release Notes\Delta(MMI)**.

5.2.5.3.1 Optimized Frequency of Logging Internal Time Synchronization for Maintenance and Testing

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

After the base station succeeds in internal time synchronization, it records time change information. This function reduces the frequency of logging time synchronization records for maintenance and testing, simplifying maintenance and testing.

Implementation

A threshold is introduced for logging a time change caused by internal time synchronization, thereby reducing the frequency of logging time synchronization records for maintenance and testing. After internal time synchronization succeeds, the base station checks whether the time change exceeds the threshold before logging and logs the change only if the change exceeds the threshold.

Impact on the Network

This function reduces the frequency of logging time synchronization records for maintenance and testing. Under the same traffic model, device parameter configuration, and UE count, the CPU usage of the main control board and baseband processing unit may be lower than that in earlier versions, and the values of the following counters may decrease:

- VS.BBUBoard.CPULoad.Max
- VS.BBUBoard.CPULoad.Mean
- VS.BBUBoard.CPULoad.Over
- VS.BBUBoard.CPULoad.CumulativeHighloadCount

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
- No impact

- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20250319000835

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.3.2 Optimized the Method for Summarizing Performance Measurement Results of EOS Boards by the Main Control Board

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The method for summarizing performance counter measurement results of EOS boards by the applicable main control board has been optimized. This prevents fluctuations in the peak CPU usage of the main control board caused by inconsistencies in counter attributes between the main control board and EOS boards.

Implementation

When EOS boards report performance counter measurement results to the main control board, the main control board checks whether counter attributes are consistent between itself and the EOS boards. If consistent, the main control board directly copies the performance counter measurement results from EOS boards

during the next summarization, eliminating the need for extensive comparisons. In this way, performance is optimized.

Impact on the Network

When EOS boards report performance counter measurement results to the main control board and counter attributes are consistent between EOS boards and the main control board, the peak CPU usage of the main control board does not fluctuate.

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20240902002296

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.3.3 Support for Querying the Numbers of Started Trace and Monitoring Tasks

NEs Involved

3900 series base stations, 5900 series base stations, USU3910

Feature/Function Description

When creating trace or monitoring tasks of a base station on the MAE or WebLMT, users may find that the maximum allowed number of such tasks has been reached. Therefore, before conducting an important test, users need to check whether there are ongoing trace or monitoring tasks and whether the number of such tasks that can be created for the test is insufficient.

This function allows users to query the numbers of supported and started trace and monitoring tasks of the eNodeB, gNodeB, and Node function types on the base station by running MML commands.

Implementation

This function is implemented as follows:

- The **DSP TRCTASK** command has been added to query trace task information. Currently, **Trace Type ID**, **Trace Task ID**, **Max Task Num**, and **Started Task Num** can be queried. You can specify a function type to query the maximum task number and started task number of a trace type ID within the target range.
- The **DSP TSTTASK** command has been added to query monitoring task information. Currently, **Monitoring Type ID**, **Monitoring Task ID**, **Max Task Num**, and **Started Task Num** can be queried. You can specify a function type to query the maximum task number and started task number of a monitoring type ID within the target range.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the eNodeB and gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following MML command has changed.

Change Type	Command	NEs	Description
New	DSP TRCTASK	eNodeB gNodeB	This command has been added to query trace task information.
New	DSP TSTTASK	eNodeB gNodeB	This command has been added to query monitoring task information.

NOTE

The parameters in these new MML commands are no longer listed separately.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250114002394-0010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD

5.2.5.3.4 Optimized the Base Station Processes for Obtaining Counters

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

This function optimizes the O&M architecture of board software to improve software maintainability.

Implementation

This function optimizes the O&M architecture of board software by decoupling and optimizing the processes for obtaining counters.

Impact on the Network

The values of the following counters will decrease:

- VS.BBUBoard.CPULoad.Mean (LTE eNodeB, 5G gNodeB)
- VS.BBUBoard.CPULoad.CumulativeHighloadCount (LTE eNodeB, 5G gNodeB)
- VS.BBUBoard.CPULoad.Max (LTE eNodeB, 5G gNodeB)
- VS.BBUBoard.CPULoad.Over (LTE eNodeB, 5G gNodeB)

When the network is overloaded with traffic, the values of the following counters may decrease:

- VS.IPv6.TxPackets (LTE eNodeB, 5G gNodeB)
- VS.IPv6.TxBytes (LTE eNodeB, 5G gNodeB)
- VS.IP.TxPackets (LTE eNodeB, 5G gNodeB)
- VS.IP.TxBytes (LTE eNodeB, 5G gNodeB)

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

Only the main control board is involved.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management

- No impact
- Performance management
- No impact
- Fault management
- No impact

Impact on Features/Functions

- Prerequisite features/functions
- No impact
- Mutually exclusive features/functions
- No impact
- Impacted features/functions
- No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

SR-IR20240902002296-0130

SR-IR20240902002296-0120

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.4 Software Management

This section describes the changes in software management.

5.2.5.4.1 NE Upgrade Status Exception Processing Optimization

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

With this function, NEs requiring extended upgrade time will be moved to the timeout queue, preventing delays in subsequent operations of other NEs.

Implementation

With this function, an NE can be moved to the timeout queue if all of the following conditions are met: (1) The MAE runs in the risk check, download, or activation phase. (2) The corresponding task is in progress for more than 25 minutes. (3) The NE upgrade is not complete.

The NE in the timeout queue cannot be operated. The MAE keeps monitoring the NE until the task is complete and displays the NE status after the task is complete. To restart the NE's upgrade, creating an upgrade batch or moving the NE to another batch is required.

Impact on the Network

Moving an NE to the timeout queue has no impact on the network. However, the NE status will be retained after the movement. If an NE is moved to the timeout queue during the activation phase, the NE activation continues. In addition, the NE restarts in the last 2 to 6 minutes of the activation progress, during which the NE does not provide services.

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250523001024-0010

IR20250701003024

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.4.2 Automatic Selection of Associated Software Packages During an Upgrade

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Software packages vary between standard services and extended services of base stations. In addition, extended software increases with the surge of extended services. To simplify upgrades, this function allows associated software packages to be automatically selected. In other words, the operating personnel only need to select the basic software package.

Implementation

With this function, the basic software package integrates the version numbers of associated software packages; during the creation of an upgrade task, associated software packages are automatically selected on the MAE based on the associated version numbers. During the upgrade, the MAE dynamically selects software packages for NEs based on the actual app configurations of each NE. Only associated software packages will be downloaded to NEs.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the MAE.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact

- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250523001024-0020

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.4.3 Log Recording and Export After a Main Control Board Enters the BootROM Version

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

This function enables log recording while a main control board is operating in the BootROM version, and supports log export in **BRDLOG** format after the board exits the BootROM version.

Implementation

With this function, logs are recorded in a separate log file when the main control board enters the BootROM version due to an exception. After the main control board exits the BootROM version, logs can be exported by running the **ULD FILE** command (with **SRCF** set to **BRDLOG**).

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

BBU: Only the Book BBU series supports this function.

Main control board: Only UMPTg and later main control boards support this function.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20250319000835

SR-IR20250319000835-0010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.4.4 Automatic Generation of a Peer NE's Quality Report by the MAE During an Independent Upgrade of the Local NE in Separate-MPT Scenarios

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In separate-MPT RF-module-sharing scenarios, the shared RF modules will be upgraded during the independent upgrade of the local NE. In this case, although the peer NE is not upgraded, the software mapping between RF modules and BBUs changes. To facilitate the version quality check after mapping, this function enables the MAE to automatically generate a quality report of the peer NE during the independent upgrade of the local NE in separate-MPT scenarios.

Implementation

Collect Supplementary Upgrade Data is added to the shortcut menu for an upgrade batch.

1. Before the activation in this batch, click ... for the batch and choose **Collect Supplementary Upgrade Data** from the menu. (Before the upgrade, a **Collect before-upgrade peer NE status information** dialog box is displayed.) The MAE automatically finds the peer NE based on the topology of the separate-MPT multimode base station, collects alarms, cell status, and cell device resource status, as well as generates a collection report before the upgrade.
2. After the activation, click ... for the batch and choose **Collect Supplementary Upgrade Data** from the menu again. (After the upgrade, a **Compare peer NE status information after upgrade** dialog box is displayed.) The MAE repeats the preceding collection process and generates a comparison report based on the collection report before the upgrade.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the MAE.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-SAR-IR_MAE_26.1_RM_20250303003297-0001_0020-0010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.4.5 Support for Querying a Comparison Report of Cell Device Resources Before and After an Upgrade

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In a multi-RRU cell, if some devices are abnormal, the cell status may be displayed as normal. If the displayed cell status is normal before an NE upgrade, cell exceptions cannot be quickly identified by comparing pre- and post-upgrade cell states on the MAE software management expert (SME). To improve O&M efficiency, comparing cell device resources before and after an upgrade is introduced.

Implementation

Before an NE upgrade, the MAE uses the **DSP CELLPHYTOPO** or **DSP NRDUCELLANTENNA** command to query cell device status and saves the query result. After the upgrade, the MAE queries cell device status again, compares the query results, and generates a comparison report.

Impact on the Network

No impact

Impact on NEs

This change does not involve NE requirements and is implemented only on the MAE.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

On the MAE, choose **Software > Software Management Expert**. Find the relevant upgrade task on the displayed window, click the **Compare Cell Device Resources After Upgrade** button, and download the comparison report.

Requirement/Trouble Ticket Number

FeR-IR20240913004460-0030

IR20241101000992

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD

5.2.5.4.6 Support for Importing or Exporting the Maintenance Modes of Peer NEs (Excluded from Upgrade) of a Separate-MPT Base Station

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In a separate-MPT base station with shared RF modules, the RF modules shared by the local and peer NEs will be reset during the upgrade of the local NE. Therefore, alarms will also be generated on the peer NEs even if they are not upgraded. Currently, **Export Before-Upgrade Maintenance Mode Setting Table** and **Export After-Upgrade Maintenance Mode Setting Table** apply only to the local NE. The maintenance modes of the peer NEs are not included in the exported tables and need to be manually configured, which results in low O&M efficiency. To improve O&M efficiency, the maintenance modes of peer NEs (excluded from upgrade) of a separate-MPT base station can now be exported or imported in batches.

Implementation

On the MAE software management expert (SME), after users choose **Export Before-Upgrade Maintenance Mode Setting Table** or **Export After-Upgrade Maintenance Mode Setting Table**, a dialog box is displayed, in which the **Include peer end of the MBTS** and **Exclude peer end of the MBTS** options are provided. If the **Include peer end of the MBTS** option is selected, the exported data contains the maintenance modes of peer NEs of to-be-upgraded NEs.

Impact on the Network

No impact

Impact on NEs

This change does not involve NE requirements and is implemented only on the MAE.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

On the MAE, choose **Software > Software Management Expert**. Find the relevant upgrade task on the displayed window and choose **Export Before-Upgrade Maintenance Mode Setting Table** or **Export After-Upgrade Maintenance Mode Setting Table**. In the dialog box, select the **Include peer end of the MBTS** or **Exclude peer end of the MBTS** option as required.

Requirement/Trouble Ticket Number

FeR-IR20240913004460-0010

IR20241101000992

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.4.7 Unsupported Software Upgrades for EOS Baseband or RF Boards Using the Basic Software Package of V100R022C00

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The software of some baseband or RF hardware boards cannot be upgraded to V100R022C00 because these boards have reached the end of service and support (EOS) phase. In addition, these boards do not support feature or function changes in V100R022C00 and later versions.

Implementation

The EOS boards are removed from the basic software package of V100R022C00. The boards involved are those for which the value of "Last Base Station Version That Supports Board Software Mapping" is V100R021C10 in *3900 & 5900 Series Base Station Software and Hardware Compatibility Description*. The basic software package and BLS package of the same version must be loaded together during base station deployment or upgrade.

NOTE

If no EOS board is installed in the base station and the BLS package is installed, users can run the **LST VER** or **LST SOFTWARE** command to query the version of the BLS package after an upgrade. There is no impact on the current base station software.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

The software of EOS boards cannot be upgraded to V100R022C00. In addition, these boards do not support feature or function changes in V100R022C00 and later versions.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20240808004444-0030

FeR-IR20240806000931-0009

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.4.8 Support for Automatic Flash Memory Space Clearing by the UMPTe

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Automatic flash memory space clearing is now supported by the UMPTe to ensure sufficient flash memory resources for normal upgrades.

Since the ***SWCLRSW*** parameter has been added to the parameter disuse list in the current version, this function is no longer controlled by any parameter and is automatically configured based on the scenario.

Implementation

When an NE using the UMPTe is upgraded from a version earlier than V100R022C00 to V100R022C00 or later, the **SET BTSUPGPLY** command is automatically executed with the ***SWCLRSW*** parameter set to **ON**, enabling running software clearing during a base station upgrade. This process is triggered through the risk check package when the following conditions are met:

- The **DSP APP** command output indicates the presence of a co-MPT multimode base station with **Application Type** being **eNodeB** or **gNodeB**.
- The **LST BTSUPGPLY** command output shows that the value of the ***SWCLRSW*** parameter is **OFF**.

After an NE using the UMPTe is upgraded to V100R022C00 or later, executing the **SET BTSUPGPLY** command automatically sets the ***SWCLRSW*** parameter to **OFF**

for a single-mode base station and to **ON** for a multimode base station. From V100R022C00 onwards, modifying the ***SWCLRSW*** setting through the **SET BTSUPGPLY** command does not take effect.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

This function takes effect only on the UMPTe.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Disused	<i>SWCLRSW</i>	SET BTSUPGPLY LST BTSUPGPLY	eGBTS NodeB eNodeB gNodeB	This parameter has been added to the parameter disuse list in the current version. This parameter will be deleted in later versions. (In the current version, the setting of this parameter does not take effect. It is automatically set to OFF for single-mode base stations and ON for multimode base stations.)

- Performance management

No impact

- Fault management

No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20241219002172-0010

IR20250115001583

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.4.9 Support for AISU Software Upgrade Using the MAE SME

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Traditionally, antenna information sensor units (AISUs) are upgraded by running the **DLD ALDSW** command, which is inefficient. To simplify the upgrade operations and improve the upgrade efficiency, AISU software can now be upgraded using the MAE software management expert (SME).

Implementation

AISU software upgrade is carried out on the MAE SME by first uploading the AISU software and then initiating an upgrade task. GUI-based upgrade is supported, mirroring the process for basic-package-based upgrade. The upgrade is executed in wizard mode, comprising the following steps: selecting the software package, creating a task, creating a batch, downloading the AISU software package, activating the AISU software package, and generating a report.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250120002357-0001

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.4.10 Optimized the Alarm and Upgrade Record upon Base Station Upgrade Failures

NEs Involved

3900 series base stations, 5900 series base stations, USU3910

Feature/Function Description

The base station upgrade fails when the configuration data file of the main control board is lost. In this case, the alarm and upgrade record are optimized as follows:

- The value of **Detail Information** of ALM-26255 Automatic Version Rollback reported in this scenario is changed from **other** to **The Configuration Upgrade Failure**, and the value of **Specific Problem** of this alarm is changed to **Configuration Data**.
- The **LST UPGRADEREC** command can be used to query an upgrade record with the value of **Upgrade result** and **Upgrade Fail Cause** being **Fail** and **The Configuration Upgrade Failure**, respectively.

Implementation

After an upgrade, if the configuration data file of the main control board is detected as lost during the startup of the board, a version rollback is performed, ALM-26255 Automatic Version Rollback is reported, and the upgrade failure is recorded.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
The following alarms have changed.

Change Type	Alarm Name	NE	Description
Modified	ALM-26255 Automatic Version Rollback	eGBTS NodeB eNodeB gNodeB	The value of Detail Information of ALM-26255 Automatic Version Rollback is changed from other to The Configuration Upgrade Failure , and the value of Specific Problem of this alarm is changed to Configuration Data .

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20240902002286

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.5 Fault Management

This section describes the changes in fault management.

For details about changes in alarms and events, see *Alarm Changes* and *Event Changes*, which are provided with the software version. You can obtain *Alarm Changes* and *Event Changes* from the following directory: **BTS3900&BTS5900 VXXXRXXXCXXSPCXXX Release Documents-EN\01-Release Notes\Delta(MMI)**.

5.2.5.5.1 Support for Displaying Automatic Detection Results Collected by the NIC on the FMA's Information Collection Tab Page

NEs Involved

3900 and 5900 series base stations

Feature/Function Description

The automatic detection results collected by the network information collection (NIC) tool can now be displayed on the FMA's **Information Collection** tab page. Such automatic detection improves data collection efficiency.

Implementation

For details, see *FMA*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This change does not involve NE requirements and is implemented only on the MAE.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20250717005198
IR20250717005183
FeR-IR20250102000700-0022
FeR-IR20250717005198-0001

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD

5.2.5.5.2 Added RedCap-, Access-, and Data Transmission-related Counters for Network-wide Automatic Detection in All Scenarios

NEs Involved

3900 and 5900 series base stations

Feature/Function Description

RedCap-, access-, and data transmission-related counters listed in [Table 5-4](#) have been added for network-wide automatic detection in all scenarios.

Table 5-4 KPI scenarios defined by the MAE-Access

Service Type	Sub-Scenario	KPI	Formula
Basic (SA)	RedCap	RedCap QoS Flow Setup Success Rate	$\frac{N.QoSFlow.Est.Succ.RedCap}{N.QoSFlow.Est.Att.RedCap} \times 100\%$
		RedCap NGSIG Connection Setup Success Rate	$\frac{N.NGSig.ConnEst.Succ.RedCap}{N.NGSig.ConnEst.Att.RedCap} \times 100\%$
		RedCap RRC Setup Success Rate	$\frac{N.RRC.SetupReq.Succ.RedCap}{N.RRC.SetupReq.Att.RedCap} \times 100\%$
		RedCap Service Drop Rate	$\frac{N.QoSFlow.AbnormRel.RedCap}{(N.QoSFlow.AbnormRel.RedCap + N.QoSFlow.NormRel.RedCap)} \times 100\%$

Service Type	Sub-Scenario	KPI	Formula
		RedCap Handover Out Success Rate (NR to LTE)	$\frac{N.HO.InterRAT.N2E.ExecSuccOut.RedCap}{N.HO.InterRAT.N2E.PrepAttOut.RedCap} \times 100\%$
		RedCap Inter Freq HO Out Success Rate	$\frac{(N.HO.InterFreq.IntragNB.ExecSuccOut.RedCap + N.HO.InterFreq.Xn.IntergNB.ExecSuccOut.RedCap + N.HO.InterFreq.Ng.IntergNB.ExecSuccOut.RedCap)}{(N.HO.InterFreq.IntragNB.PrepAttOut.RedCap + N.HO.InterFreq.Xn.IntergNB.PrepAttOut.RedCap + N.HO.InterFreq.Ng.IntergNB.PrepAttOut.RedCap)} \times 100\%$
		RedCap Intra Freq HO Out Success Rate	$\frac{(N.HO.IntraFreq.Ng.IntergNB.ExecSuccOut.RedCap + N.HO.IntraFreq.Xn.IntergNB.ExecSuccOut.RedCap + N.HO.IntraFreq.IntragNB.ExecSuccOut.RedCap)}{(N.HO.IntraFreq.IntragNB.PrepAttOut.RedCap + N.HO.IntraFreq.Xn.IntergNB.PrepAttOut.RedCap + N.HO.IntraFreq.Ng.IntergNB.PrepAttOut.RedCap)} \times 100\%$
		Downlink Traffic Volume of PDCP Services of RedCap UEs	$N.PDCP.Vol.DL.TrfSDU.Rx.RedCap/1000/1000/8$
		Uplink Traffic Volume of PDCP Services of RedCap UEs	$N.PDCP.Vol.UL.TrfSDU.Tx.RedCap/1000/1000/8$
		RedCap User DL Experienced Rate	$\frac{(N.ThpVol.DL.RedCap - N.ThpVol.DL.LastSlot.RedCap)}{N.ThpTime.DL.RmvLastSlot.RedCap} \times 1000$
		RedCap User UL Experienced Rate	$\frac{(N.ThpVol.UL.RedCap - N.ThpVol.UE.UL.SmallPkt.RedCap)}{N.ThpTime.UE.UL.RmvSmallPkt.RedCap} \times 1000$

Service Type	Sub-Scenario	KPI	Formula
	Access	Ratio of Ngsig Setup	$\frac{N.NGSig.ConnEst.Att}{N.RRC.SetupReq.Att} \times 100\%$
		Number of Users	$N.User.RRCConn.Avg \times 100\%$
	Data Transmission	PDCP DL Packet Loss Rate	$\frac{(N.PDCP.DL.TrfSDU.TxPacket.Discard - N.PDCP.DL.TrfSDU.TxPacket.Discard.GTPU)}{N.PDCP.DL.TrfPDU.TxPackets} \times 100\%$
		PDCP UL Packet UU Loss Rate	$\frac{N.PDCP.UL.TrfSDU.RxPacket.Loss}{(N.PDCP.UL.TrfPDU.RxPackets + N.PDCP.UL.TrfSDU.RxPacket.Loss)} \times 100\%$

This function is supported only in SA networking.

Implementation

For details, see *FMA*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This change does not involve NE requirements and is implemented only on the MAE.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management

- No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20250529001344

FeR-IR20250529001344-0001

RAT

NR FDD, NR TDD

5.2.5.5.3 Optimized the KPI Statistics and Display on the All Scenario KPI Overview Tab Page

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The KPI measurement and display on the **All Scenario KPI Overview** tab page have been optimized as follows:

NR:

- Users can select a version group on the **All Scenario KPI Overview** tab page for analysis in the MOCN scenario.
- The **Analysis Period** parameter has been replaced by the **Analysis Date** parameter. Users can select the analysis date (the current day is selected by default). The analysis period is [00:00, 24:00), which is fixed to 24 hours. After the analysis is complete, the KPI statistics of the current day (corresponding to **today**), the previous day (corresponding to **yesterday**), and the same day of the previous week (corresponding to **last week**) are displayed by default for comparison.

- Users can specify a KPI period to view the statistical data in different KPI periods. The supported values on the GUI are **P5**, **P15**, **P30**, and **P60**, which correspond to 5 minutes, 15 minutes, 30 minutes, and 60 minutes, respectively. The default value is **P60**. Users need to ensure that the selected KPI period has been subscribed to.
- KPI statistics are displayed independently for each KPI or performance counter listed in [Table 5-5](#).

Table 5-5 KPIs or performance counters applicable to NR

Scenario	KPI/Performance Counter
NSA Signaling	SgNB Addition Success Rate(%)
	Inter-SgNB PSCell Change Success Rate(%)
	Intra-SgNB PSCell Change Success Rate(%)
	SgNB-triggered Abnormal SgNB Released Rate(Only gNB Triggered) (%)
	SgNB Abnormal Release Rate(Including eNB&gNB Triggered) (%)
SA Intra-RAT Signaling	RRC Setup Success Rate(SA) (%)
	NgSig Setup Success Rate(SA) (%)
	QosFlow Setup Success Rate(SA) (%)
	Intra-Frequency Handover Out Execution Success Rate(%)
	Intra-Frequency Handover Out Preparation Success Rate(%)
	Inter-Frequency Handover Out Execution Success Rate(%)
	Inter-Frequency Handover Out Preparation Success Rate(%)
	Handover In Execution Success Rate(%)
	Handover In Preparation Success Rate(%)
	Service Call Drop Rate(%)
	Service Call Drop Rate with AMF(%)

Scenario	KPI/Performance Counter
SA Inter-RAT Signaling	EPS Fallback Preparation Success Rate(%)
	Handover-based EPS Fallback Preparation Success Rate(%)
	Handover-based EPS Fallback Execution Success Rate(%)
	Coverage-based NR2L HO Preparation Success Rate(%)
	Coverage-based NR2L HO Execution Success Rate(%)
Voice	Voice Service Access Success Rate 5QI1(%)
	Voice Service Access Success Rate 5QI5(%)
	N.Paging.Dis.PchCong
	Voice Call Drop Rate 5QI1(%)
	Voice Call Drop Rate 5QI5(%)
	Voice Service Handover Success Rate 5QI1(%)
	Voice Service Handover Success Rate 5QI5(%)
Rate & Interference	Cell Downlink Average Throughput(Mbps)
	Cell Uplink Average Throughput(Mbps)
	User Downlink Average Throughput(Mbps)
	N.ThpVol.DL(Kbit)
	User Uplink Average Throughput(Mbps)
	N.ThpVol.UL(Kbit)
	N.UL.NI.Avg(dBm)
	N.UL.Last.Symbol13.Pwr(dBm)
	N.UL.Last.Symbol6.Pwr(dBm)
	N.GAP.LastSymbol.Pwr(dBm)

Scenario	KPI/Performance Counter
	VS.Gtpu.RxDropPkts
RedCap	RedCap RRC Setup Success Rate(%)
	RedCap NgSig Setup Success Rate(%)
	RedCap QosFlow Setup Success Rate(%)
	RedCap Service Drop Rate(%)
	RedCap Intra Freq HO Out Success Rate(%)
	RedCap Inter Freq HO Out Success Rate(%)
	RedCap Average User Number
	RedCap NR2L HO Out Success Rate(%)
	Downlink Traffic Volume of PDCP Services of RedCap Ues(GB)
	Uplink Traffic Volume of PDCP Services of RedCap Ues(GB)
	RedCap User DL Experienced Rate(Excluding Last Slot Packets) (Mbit/s)
	RedCap User UL Experienced Rate(Excluding Small Packets) (Mbit/s)
	Average Downlink PRB Rate of RedCap BWP(%)
	Average Uplink PRB Rate of RedCap BWP(%)
	Average Downlink PRB Rate of RedCap Ues(%)
	Average Uplink PRB Rate of RedCap Ues(%)
MOCN	SgNB Addition Success Rate PLMN(%)
	SgNB-triggered Abnormal SgNB Released Rate PLMN(%)

Scenario	KPI/Performance Counter
	Abnormal SgNB Released Rate(Include MeNB triggered) PLMN(%)
	SA NgSig Setup Success Rate PLMN(%)
	QoS Flow Setup Success Ratio PLMN(%)
	VONR QoS Flow Setup Success Ratio PLMN(%)
	SA QoSFlow Drop Rate PLMN(%)
	VONR SA QoSFlow Drop Rate PLMN(%)
	DL RLC Traffic PLMN(GB)
	UL RLC Traffic PLMN(GB)
	N.User.RRCCConn.Avg.Operator

LTE:

- Users can select a version group on the **All Scenario KPI Overview** tab page for analysis.
- The **Analysis Period** parameter has been replaced by the **Analysis Date** parameter. Users can select the analysis date (the current day is selected by default). The analysis period is [00:00, 24:00), which is fixed to 24 hours. After the analysis is complete, the KPI statistics of the current day (corresponding to **today**), the previous day (corresponding to **yesterday**), and the same day of the previous week (corresponding to **last week**) are displayed by default for comparison.
- Users can specify a KPI period to view the statistical data in different KPI periods. The supported values on the GUI are **P5**, **P15**, **P30**, and **P60**, which correspond to 5 minutes, 15 minutes, 30 minutes, and 60 minutes, respectively. The default value is **P60**. Users need to ensure that the selected KPI period has been subscribed to.

Implementation

For details, see *FMA*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This change does not involve NE requirements and is implemented only on the MAE.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20250717005209

IR20250717005183

FeR-IR20250717005209-0001

FeR-IR20250717005183-0001

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD

5.2.5.5.4 Support for DDR SDRAM Fault Isolation by Certain Boards

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Before an upgrade, certain UBBPg1, UBBPg2, UBBPg3, UBBPg2a, and UMPTg3 boards do not support DDR SDRAM fault isolation. After the upgrade, all UBBPg1, UBBPg2, UBBPg3, UBBPg2a, and UMPTg3 boards support DDR SDRAM fault isolation.

Implementation

When detecting a DDR SDRAM fault on the UBBPg1, UBBPg2, UBBPg3, UBBPg2a, or UMPTg3 board and determining that the fault may be rectified, the base station triggers a board reset for self-healing in the early morning to attempt to rectify the DDR SDRAM fault.

Impact on the Network

This optimization may trigger a board reset for self-healing in the early morning.

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

Boards: UBBPg1, UBBPg2, UBBPg3, UBBPg2a, and UMPTg3

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact

- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20250319000813
SR-IR20250319000813-0210
IR20250908001612
SR-IR20250908001612-0120
IR20250908001644
SR-IR20250908001644-0040

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.5.5 Support for Fast Alarm Clearance Due to the Awareness of Fault Rectification by the RHUB and pRRU

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The RHUB and pRRU now support the fast alarm clearance due to the awareness of fault rectification. The RHUB and pRRU that support this function before and after the upgrade are as follows:

- Before an upgrade
 - RHUBs: RHUB5x7x series
 - pRRUs: pRRU5x5x series, pRRU5x2x series, and pRRU5x3x series
- After an upgrade
 - RHUBs: all RHUBs except the RHUB3908, RHUB3909, RHUB3910, and RHUB3918
 - pRRUs: all pRRUs except the pRRU3901, pRRU3902, pRRU3907, pRRU3911, pRRU3912, and pRRU3916

Implementation

For details, see *Fault Management*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

RF modules:

- RHUBs: all RHUBs except the RHUB3908, RHUB3909, RHUB3910, and RHUB3918
- pRRUs: all pRRUs except the pRRU3901, pRRU3902, pRRU3907, pRRU3911, pRRU3912, and pRRU3916

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

Whether the change takes effect by default after an upgrade varies as follows:

- Yes if enhanced alarm toggling and filtering is enabled before an upgrade
- No in other scenarios To enable enhanced alarm toggling and filtering for a specific alarm, run the **SET ALMTOGGLERULE** command with **ALMTOGGLERULE.ALMTOGGLESWITCH** corresponding to the alarm ID set to **ON**.

Requirement/Trouble Ticket Number

FeR-IR20250303001254-010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.5.6 Retaining Alarms Uncleared Following a Base Station Reset

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The function of retaining alarms uncleared following a base station reset preserves alarms that are not cleared before a base station reset. By maintaining these alarms, this function reduces repeated alarm tickets. Before this function is enabled, the base station clears alarms that are not cleared before a base station reset after it is reset. Then, the base station detects and reports these alarms again. After this function is enabled, if any identical alarm is detected within 30 seconds after the EVT-26213 NE Startup event is reported following a base station reset, the alarm serial number remains unchanged from before the reset. If an alarm reported after a base station reset has the same alarm ID and location information as an uncleared alarm in the alarm log, the two alarms are regarded as identical.

This function is enabled by running the **SET ALMFUNCTIONSW** command with **ALMFUNCTIONSW.ALMRSTHOLDSWITCH** set to **ON**. This function takes effect once the base station is reset. If the base station is connected to the MAE, this function takes effect only after the MAE is upgraded to V100R026C10 or a later version.

Implementation

For details, see *Fault Management*. The specific sections are listed in the change history of the document.

Impact on the Network

This function keeps the serial numbers unchanged for certain alarms uncleared following a base station reset, instead of clearing them and reporting them again.

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following MML commands have changed.

Change Type	Command	NE	Description
New	SET ALMFUNCTI ONSW	eGBTS NodeB eNodeB gNodeB	This MML command has been added to set alarm function switch parameters.
New	LST ALMFUNCTI ONSW	eGBTS NodeB eNodeB gNodeB	This MML command has been added to list alarm function switch parameters.

NOTE

The parameters in these new MML commands are no longer listed separately.

- Performance management

No impact

- Fault management

This function keeps the serial numbers unchanged for certain alarms uncleared following a base station reset, instead of clearing them and reporting them again.

Impact on Features/Functions

- Prerequisite features/functions

No impact

- Mutually exclusive features/functions

No impact

- Impacted features/functions

No impact

Related Operations

By default, the change does not take effect after an upgrade.

To retain alarms uncleared following a base station reset, run the **SET ALMFUNCTIONSW** command with **ALMFUNCTIONSW.ALMRSTHOLDSWITCH (LTE eNodeB, 5G gNodeB)** set to **ON**.

Requirement/Trouble Ticket Number

FeR-IR20250518000326-0010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.5.7 Optimized MAE Alarm Aggregation

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

ALM-25954 User Plane Fault is reported for various types of services, such as those running over LTE X2, gNodeB X2, gNodeB Xn, and gNodeB eXn. As a result, a large number of alarms are reported. The alarm aggregation function on the MAE is optimized and provides built-in alarm aggregation rules for ALM-25954 User Plane Fault, eliminating the need to manually edit alarm aggregation rules.

Implementation

- Before an upgrade, alarm aggregation rules on the MAE need to be manually edited to reduce the number of alarms.
- After the upgrade, the alarm aggregation function on the MAE provides built-in alarm aggregation rules. To reduce the number of alarms, these rules can be enabled by default (can be set to be enabled by default), as described below:

This optimization aligns with the original alarm aggregation mechanism. That is, alarm reporting is delayed within the alarm aggregation period. For example, if the alarm aggregation period is 30 seconds, ALM-25954 User Plane Fault reported by the base station within 30 seconds after it is generated will be aggregated and displayed on the MAE 30 seconds later.

On the MAE, choose **Monitor > Alarm Settings > Aggregation Rules**. On the **Default** tab page, set **Rule Status** to **Enabled** (which is **Disabled** by default) for **User Plane Fault(25954)Aggregation_XXX**. Replace XXX with the actual NE type.

Impact on the Network

No impact

Impact on NEs

This change does not involve NE requirements and is implemented only on the MAE.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

On the MAE, choose **Monitor > Alarm Settings > Aggregation Rules**. On the **Default** tab page, set **Rule Status** to **Enabled** (which is **Disabled** by default) for **User Plane Fault(25954)Aggregation_XXX**. Replace **XXX** with the actual NE type.

Requirement/Trouble Ticket Number

FeR-IR20250518000324-0020

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.5.8 Optimized the TOD Clock Unavailable Alarm

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The cause value of **Clock Port Blocked** is added to ALM-26267 TOD Clock Unavailable, and alarm correlation suppression is performed based on the cause priority.

Implementation

For ALM-26267 TOD Clock Abnormal, if this alarm with a high-priority cause is reported before an alarm with a low-priority cause, the alarm with a low-priority cause is not reported; if an alarm with a low-priority cause is reported before an alarm with a high-priority cause, the previously reported alarm with a low-priority cause is cleared.

The priorities of the causes of this alarm are as follows:

- **Clock Port Blocked** takes precedence over **TOD Clock Output Abnormal** and **1PPS Signals for TOD Clock Unavailable**.
- **1PPS Signals for TOD Clock Unavailable** takes precedence over **TOD Clock Not Identified** and **Clock Quality Unavailable**.
- **TOD Clock Not Identified** takes precedence over **Clock Quality Unavailable**.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact

- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250806001609-0080

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.5.9 Optimized the ESN Information in the Query Result Sets of Alarms

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

After a board (such as the baseband processing unit or RF unit) is replaced, the ESN information queried by running the **LST ALMLOG** command is inaccurate. To address this issue, the ESN information in the additional alarm information in the **LST ALMLOG** command output is optimized for alarms with ESNs.

Implementation

For alarms with ESNs, the ESN field is recorded in alarm logs. When the **LST ALMLOG** command is executed to query alarm logs, the ESN field in the additional alarm information is preferentially obtained from the ESN field recorded in alarm logs. If the alarm logs do not contain this field, the latest ESN field of the corresponding equipment is used.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20250319000835

SR-IR20250319000835-0670

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.5.10 Optimized Board Memory Software Integrity Check

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The board memory software integrity check is enhanced to additionally support integrity checking for service processes

Implementation

If the board memory software integrity check for service processes fails, ALM-26203 Board Software Program Error is reported with the cause being "An integrity check on the board memory software fails."

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20250319000813

SR-IR20250319000813-0110

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.5.11 Optimized the Alarm Correlation Mechanism

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The current alarm correlation is subject to the following restrictions, which may cause the correlation failure of certain root and correlative alarms:

- When the parent and child faults are correlated but the root and correlative alarms are not generated simultaneously, then the root and correlative alarms cannot be correlated.
- For root and correlative alarms that are correlated, after the root alarm is cleared, the root alarm serial number of the correlative alarm is deleted and the alarm is changed.

Therefore, the alarm correlation mechanism is optimized.

Implementation

This function is implemented as follows:

- This function provides an alarm correlation mechanism based on the alarm lifecycle and extends the alarm correlation lifecycle. That is, after a correlative alarm is generated, the correlative alarm is associated with the root alarm by default. After the root alarm is cleared, the correlative alarm remains correlated.
- Alarm correlation is performed based on the following rules:
 - If the root and correlative alarms are not generated at the same time, but the faults have time intersection and are correlated, the root serial number of the root alarm can be displayed as the root serial number of the correlative alarm.
 - After the root alarm is cleared, the correlative alarm retains the root serial number of the root alarm for 3 minutes.

The principles for displaying the root serial number of the root alarm are as follows:

- A maximum of three serial numbers can be displayed.
- Serial numbers are sorted in ascending order of priority.
- Uncleared alarms take precedence over cleared alarms.

Impact on the Network

After the alarm correlation mechanism is optimized, the proportion of alarms that are not correlated may decrease.

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20250319000813

SR-IR20250319000813-0230

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.5.12 Optimized External Clock Reference Problem

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Base station clock alarms are reported as follows:

- The clock source faults are detected and related alarms (for example, ALM-26126 GNSS Locked Satellites Insufficient for GNSS clock source and ALM-26263 IP Clock Link Failure for IP clock source) are reported.
- The base station clock working status is detected and related alarms (for example, ALM-26262 External Clock Reference Problem) are reported.

Currently, ALM-26262 External Clock Reference Problem may be reported due to faults in the clock source rather than the working status of the base station clock. Therefore, the alarm reporting mechanism needs to be optimized to ensure appropriate alarm reporting.

Implementation

- The cause value of **TOD Clock Reference Unavailable due to Unacceptable Quality** is removed from ALM-26262 External Clock Reference Problem and the cause value of **Clock Quality Unavailable** is added to ALM-26267 TOD Clock Unavailable. That is, ALM-26262 External Clock Reference Problem is reported with the value of **Specific Problem** being **TOD Clock Reference Unavailable due to Unacceptable Quality** before an upgrade, and ALM-26267 TOD Clock Unavailable is reported with the value of **Specific Problem** being **Clock Quality Unavailable** after the upgrade.
- The cause value of **IPCLK Reference Unavailable due to Unacceptable Quality** is removed from ALM-26262 External Clock Reference Problem. Before an upgrade, ALM-26263 IP Clock Link Failure is reported with the value of **Specific Problem** being **Clock Quality Unavailable**. In this optimization, this cause value is removed.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
The following alarms have changed.

Change Type	Name	NE	Description
Modified	ALM-26262 External Clock Reference Problem	eGBTS NodeB eNodeB gNodeB	- The cause value of TOD Clock Reference Unavailable due to Unacceptable Quality is removed from ALM-26262 External Clock Reference Problem and the cause value of Clock Quality Unavailable is added to ALM-26267 TOD Clock Unavailable. That is, ALM-26262 External Clock Reference Problem is reported with the value of Specific Problem being TOD Clock Reference Unavailable due to Unacceptable Quality before an upgrade, and ALM-26267 TOD Clock Unavailable is reported with the value of Specific Problem being Clock Quality Unavailable after the upgrade. - The cause value of IPCLK Reference Unavailable due to Unacceptable Quality is removed from ALM-26262 External Clock Reference Problem. Before an upgrade, ALM-26263 IP Clock Link Failure is reported with
Modified	ALM-26267 TOD Clock Unavailable	eGBTS NodeB eNodeB gNodeB	

Change Type	Name	NE	Description
			the value of Specific Problem being Clock Quality Unavailable . In this optimization, this cause value is removed.

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250121001519-0030

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.5.13 Optimized OTDR-based Fronthaul Fault Root Cause Diagnosis

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

OTDR-based fronthaul fault root cause diagnosis has been optimized in the following scenarios:

- Scenarios where abnormal loss is caused by connector faults near the transmit end: Due to potential waveform distortion during OTDR detection, the loss of components may fail to be individually evaluated. Therefore, the concept of overall loss is introduced. When the overall loss of multiple components near the transmit end is abnormal, ALM-26409 Fronthaul Multiplexer/Demultiplexer or Connector Exception Alarm or ALM-26410

Fronthaul Optical Fiber Exception Alarm will be reported to indicate this abnormality.

- Scenarios where abnormal loss near the transmit end coexists with dirt or interruption at the trunk or remote node: Abnormal loss near the transmit end and dirt or interruption at the trunk or remote node can now be detected simultaneously. When dirt or interruption occurs at the trunk or remote node, ALM-26409 Fronthaul Multiplexer/Demultiplexer or Connector Exception Alarm or ALM-26410 Fronthaul Optical Fiber Exception Alarm will be reported.

Implementation

For details, see *Fault Management*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
The following alarms have changed.

Change Type	Name	NE	Description
Modified	ALM-26409 Fronthaul Multiplexer/ Demultiplexer or Connector Exception Alarm	eGBTS NodeB eNodeB gNodeB	- Scenarios where abnormal loss is caused by connector faults near the transmit end: Alarm reporting is now supported when the overall loss is abnormal. - Scenarios where abnormal loss near the transmit end coexists with dirt or interruption at the trunk or remote node: Alarm reporting is now supported when dirt or interruption occurs at the trunk or remote node.
Modified	ALM-26410 Fronthaul Optical Fiber Exception Alarm	eGBTS NodeB eNodeB gNodeB	

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250708001473-0120

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.5.14 Added Detailed Causes to OMCH Maintenance Info for the Transmission FMA

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

If no information is displayed in the **OMCH Maintenance Info** area in the diagnosis result of the transmission FMA, the detailed cause is now displayed to improve the maintainability of the online transmission FMA and fault locating efficiency. This cause is obtained by analyzing the packets returned by the NE and checking the DHCP port of the MAE.

The possible causes are as follows:

- The DHCP port of the OSS is not enabled.
- No DHCP packet is received from the NE.
- Other issues occur. Contact Huawei technical support.

Implementation

For details, see *FMA*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This change does not involve NE requirements and is implemented only on the MAE.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact

- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20241203000519-0010

IR20250115001586

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS

5.2.5.5.15 Optimized MAE Alarm Reporting Mechanism

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The generation and clearance mechanism is optimized for ALM-782 Trace Server Disconnected from the NMS and ALM-780 The Data Transmission Channel Between the Trace Server and the NE Is Disrupted when either of the following conditions is met: (1) The network transmission quality between the MAE and its upper-layer NMS or between the MAE and the lower-layer NE is poor. (2) The communication with the peer system is unstable. In addition, ALM-50 Task execution failure alarm is no longer reported when a northbound machine-machine interface task fails. This prevents alarms from being repeatedly generated and cleared within a short period of time, reducing the number of invalid alarms and improving O&M efficiency.

Implementation

This function is implemented as follows:

1. The clearance time is extended for ALM-782 Trace Server Disconnected from the NMS and ALM-780 The Data Transmission Channel Between the Trace Server and the NE Is Disrupted to reduce the number of invalid alarms when

either of the following conditions is met: (1) The network transmission quality between the MAE and its upper-layer NMS or between the MAE and the lower-layer NE is poor. (2) The communication with the peer system is unstable.

2. If a task invoked by the northbound machine-machine interface fails to be executed, a message indicating the failure is directly returned to the northbound invoker. ALM-50 Task execution failure alarm is not generated in this case, reducing the number of invalid alarms.

Impact on the Network

No impact

Impact on NEs

This change does not involve NE requirements and is implemented only on the MAE.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
The following alarms have changed.

Change Type	Name	NE	Description
Modified	ALM-782 Trace Server Disconnected from the NMS	eGBTS NodeB eNodeB gNodeB	The reporting mechanism is optimized for ALM-782 Trace Server Disconnected from the NMS, ALM-780 The Data Transmission Channel Between the Trace Server and the NE Is Disrupted, and ALM-50 Task execution failure alarm to reduce the number of invalid alarms.
Modified	ALM-780 The Data Transmission Channel Between the Trace Server and the NE Is Disrupted		
Modified	ALM-50 Task execution failure alarm		

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20240808004700-0040

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.5.16 Optimized BBU Power-Off Alarm Mechanism Through UBBP Mutual Aid Diagnosis

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Mutual aid diagnosis is now supported between the main control board and UBBPs to optimize alarm diagnosis for BBU power-off issues, thereby improving the efficiency of locating NE disconnection issues.

Implementation

This function is implemented as follows:

In scenarios where the main control board does not support power-off packet reporting while the UBBPs do, channels can be established between multiple UBBPs and the MAE for the transmission of power-off packets. When the base station disconnects from the MAE due to BBU power-off, the main control board can forward the power-off packets reported by the UBBPs to the MAE, which then diagnoses the NE disconnection as being caused by a power supply fault based on these packets.

In GSM, only co-MPT base stations involving GSM support this function.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

Main control boards: Only the UMPTb, UMPTe, UMPTga, and later boards support this function.

Baseband processing units: The UBBPd, UBBPe, UBBPg, UBBPh, and later boards (excluding the UBBPg2x, UBBPg3x, and UBBPg4) support this function. x indicates any letter.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20241202003015-0010
IR20250115001584

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.5.17 Optimized the Enhanced Alarm Toggling and Filtering Function

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The enhanced alarm toggling and filtering function is optimized with the alarms listed in [Table 5-6](#) supported. In addition, to reduce the impact of this function on alarm clearance, fast alarm clearance due to the awareness of fault rectification is introduced.

Table 5-6 Alarms for which fast alarm clearance due to the awareness of fault rectification can take effect

Alarm	Applicable Base Station Model
ALM-26235 RF Unit Maintenance Link Failure	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
ALM-26234 BBU CPRI Interface Error	
ALM-26504 RF Unit CPRI Interface Error	
ALM-26324 BBU IR Interface Error	3900 and 5900 series base stations (macro base stations)
ALM-26508 RF Unit IR Interface Error	

Fast alarm clearance due to the awareness of fault rectification accelerates alarm clearance. When this function is in effect, the base station automatically resets the value of **Window for Ceasing Toggle Alarm** upon detecting that the customer has performed any of the following typical alarm clearance operations.

- Configuration change, such as configuration object deletion and port unblocking
- Hardware change, such as the replacement, insertion/installation, or removal of an optical module, BBU board, RRU, RHUB, or pRRU
- Hardware reset, such as MML-based board reset, RRU power-off reset, and RHUB/pRRU power-off reset

Implementation

For details, see *Fault Management*. The specific sections are listed in the change history of the document.

Impact on the Network

Enabling the enhanced alarm toggling and filtering function may reduce how often the following alarms are reported:

- ALM-26235 RF Unit Maintenance Link Failure
- ALM-26234 BBU CPRI Interface Error
- ALM-26324 BBU IR Interface Error
- ALM-26504 RF Unit CPRI Interface Error
- ALM-26508 RF Unit IR Interface Error

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

Enhanced alarm toggle filtering: none

Fast alarm clearance due to the awareness of fault rectification:

- Boards:
 - Main control boards: no requirements
 - Baseband processing units: only UBBP series
- RF modules:
 - For macro base stations:
 - 5000 series RRUs and 5000 series AAUs
 - Certain 3000 series RRUs and 3000 series AAUs: RRU3236E, RRU3249, RRU3260, RRU3262, RRU3268, RRU3269, RRU3276, RRU3278, RRU3278u, RRU3279, RRU3281, RRU3668, RRU3832 (except the one with the SBOM code of 02310PTJ), RRU3931E, RRU3952, RRU3952m, RRU3953, RRU3953w, RRU3954, RRU3956, RRU3956w, RRU3958, RRU3959, RRU3959a, RRU3959w, RRU3962,

AAU3910 (SBOM code: 02312DAU), AAU3911 (except the one with the SBOM code of 02311DBG), AAU3920, AAU3940, and AAU3961

- For LampSite base stations:
 - RHUBs: RHUB5x7x series
 - pRRUs: pRRU5x5x series, pRRU5x2x series, and pRRU5x3x series

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management

The following alarms have changed.

Change Type	Name	NE	Description
Modified	ALM-26235 RF Unit Maintenance Link Failure	eGBTS NodeB eNodeB gNodeB	Enhanced alarm toggling and filtering is now supported.
Modified	ALM-26234 BBU CPRI Interface Error	eGBTS NodeB eNodeB gNodeB	
Modified	ALM-26324 BBU IR Interface Error	eGBTS NodeB eNodeB gNodeB	
Modified	ALM-26504 RF Unit CPRI Interface Error	eGBTS NodeB eNodeB gNodeB	

Change Type	Name	NE	Description
Modified	ALM-26508 RF Unit IR Interface Error	eGBTS NodeB eNodeB gNodeB	

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

Run the **SET ALMTOGGLERULE** command with the **ALMTOGGLERULE.ALMTOGGLESWITCH (LTE eNodeB, 5G gNodeB)** parameter set to **ON** to enable enhanced alarm toggling and filtering.

Requirement/Trouble Ticket Number

FeR-IR20240808004700-0010

FeR-IR20241101000989-001

IR20240910001328

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.5.18 Support for Alarm Reporting Based on the Bit Error Rate When the Receive Optical Power Is Low

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

One cause of ALM-26234 BBU CPRI Interface Error and ALM-26504 RF Unit CPRI Interface Error is that the receive power is too low, that is, the receive optical power of the optical module is too low. If one of them is reported due to too low receive power and no bit errors occur, services may not be affected. Under this

circumstance, the function of alarm reporting based on the bit error rate when the receive optical power is low is introduced.

After this function is enabled, these alarms are reported only when the receive optical power of the optical module on the CPRI port is too low and bit errors occur. That is, the alarms are reported only when the receive optical power is too low and services are affected. This function can be enabled if you do not want to handle alarms that do not affect services. It is enabled by running the **SET EQUIPMENT** command with the **LOW_RX_OP_PWR_ALM_OPT_SW** option of the **EQUIPMENT.EQMOPTFEATSW** parameter selected, and takes effect only when the **NEMNT.MNTMODE** parameter is set to **NORMAL** or **UPGRADE**.

This function takes effect only for ALM-26234 BBU CPRI Interface Error and ALM-26504 RF Unit CPRI Interface Error with the value of **Alarm Severity** being **Minor**.

Implementation

For details, see *Fault Management*. The specific sections are listed in the change history of the document.

Impact on the Network

After the function of alarm reporting based on the bit error rate when the receive optical power is low is enabled, ALM-26234 BBU CPRI Interface Error and ALM-26504 RF Unit CPRI Interface Error are reported less often. In extreme scenarios where the bit error rate is low and bit errors occur intermittently, these alarms are frequently reported and cleared.

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

- BBUs: non-integrated BBUs
- Boards:
 - Main control boards: no requirements
 - Baseband processing units: only UBBP series
- RF modules:
 - 5000 series RRUs and 5000 series AAUs
 - Certain 3000 series RRUs and 3000 series AAUs: RRU3236E, RRU3249, RRU3260, RRU3262, RRU3268, RRU3269, RRU3276, RRU3278, RRU3278u, RRU3279, RRU3281, RRU3668, RRU3832 (except the one with the SBOM code of 02310PTJ), RRU3931E, RRU3952, RRU3952m, RRU3953, RRU3953w, RRU3954, RRU3956, RRU3956w, RRU3958, RRU3959, RRU3959a, RRU3959w, RRU3962, AAU3910 (SBOM code: 02312DAU), AAU3911 (except the one with the SBOM code of 02311DBG), AAU3920, AAU3940, and AAU3961

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	EQUIPMENT .EQMOPTFE ATSW (LTE eNodeB, 5G gNodeB)	SET EQUIPMENT LST EQUIPMENT	eGBTS NodeB eNodeB gNodeB	The LOW_RX_OP_PWR_ ALM_OPT_SW option has been added to specify whether to report an alarm based on the bit error rate when the receive optical power is low.

- Performance management
No impact
- Fault management
The following alarms have changed.

Change Type	Name	NE	Description
Modified	ALM-26234 BBU CPRI Interface Error	eGBTS NodeB eNodeB gNodeB	After the function of reporting an alarm based on the bit error rate when the receive optical power is low is enabled, these alarms can be reported only when the receive optical power of the optical module on the CPRI port is too low and bit errors occur.
Modified	ALM-26504 RF Unit CPRI Interface Error	eGBTS NodeB eNodeB gNodeB	

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions

No impact

Related Operations

By default, the change does not take effect after an upgrade.

Run the **SET EQUIPMENT** command with the **LOW_RX_OP_PWR_ALM_OPT_SW** option of the **EQUIPMENT.EQMOPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter selected to enable the function of alarm reporting based on the bit error rate when the receive optical power is low.

Requirement/Trouble Ticket Number

FeR-IR20240808004700-0030

IR20241101000989

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.5.19 Support for OTDR-based Fronthaul Fault Root Cause Diagnosis

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

OTDR-based fronthaul fault root cause diagnosis is now supported. (OTDR is short for optical time-domain reflectometer.) When the CPRI port of the OTDR optical module is faulty and services are unavailable, an OTDR detection is performed and a fronthaul fault root cause alarm is reported based on the event location detected by the OTDR. Fronthaul fault root cause alarms include ALM-26409 Fronthaul Multiplexer/Demultiplexer or Connector Exception Alarm and ALM-26410 Fronthaul Optical Fiber Exception Alarm.

This function can be enabled only when both of the following conditions are met:

- On the MAE side: The MAE is upgraded to V100R026C00 or a later version. On the MAE, the enhanced fronthaul diagnosis capability sub-function of the intelligent alarm aggregation function is enabled.
On the MAE, click **iFaultCare**, choose **Intelligent Fault Incident Management > Incident Settings**, and turn on the **Incident Enable** switch to enable the intelligent alarm aggregation function. Then, choose **Troubleshooting Policies > Diagnosis Ability Settings** and turn on the **Enhance the fronthaul diagnosis capability** switch to enable the enhanced fronthaul diagnosis capability.
- On the base station side:
The fronthaul optical path topology reference information is built by running the **STR OPTPATHQUALITYDET** command.

LampSite base stations do not support this function.

Implementation

For details, see *Fault Management*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

- Base station models: 3900 and 5900 series base stations (macro base stations)
- BBUs: BBU3900, BBU3910, BBU5900, and BBU5910
- Boards:
 - Main control boards: UMPTe and later
 - Baseband processing units: UBBPd, UBBPe (except UBBPeb), UBBPg, UBBPh, and later boards that support a CPRI port rate of 10 Gbit/s, 25 Gbit/s, or 50 Gbit/s
- Optical modules: 25 Gbit/s or 50 Gbit/s OTDR optical modules

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following MML commands have changed.

Change Type	MML Command	NE	Description
New	DSP OPTPATHT OPO	eGBTS NodeB eNodeB gNodeB	This MML command has been added to display fronthaul optical path topology identification results.

 NOTE

The parameters in these new or deleted MML commands are no longer listed separately.

- Performance management

No impact

- Fault management

The following alarms and events have changed.

Change Type	Name	NE	Description
New	ALM-26409 Fronthaul Multiplexer/ Demultiplexer or Connector Exception Alarm	eGBTS NodeB eNodeB gNodeB	A fronthaul fault root cause alarm is reported based on the event location detected by the OTDR when OTDR-based fronthaul fault root cause diagnosis is enabled. This alarm is generated when the multiplexer/demultiplexer is detected as interrupted, dirty, or damaged through automatic OTDR detection triggered by fronthaul link disconnection.
New	ALM-26410 Fronthaul Optical Fiber Exception Alarm	eGBTS NodeB eNodeB gNodeB	A fronthaul fault root cause alarm is reported based on the event location detected by the OTDR when OTDR-based fronthaul fault root cause diagnosis is enabled. This alarm is generated when the fronthaul optical fiber is detected as interrupted, dirty, damaged, bent, or spliced through automatic OTDR detection triggered by a fronthaul-link-related fault.

Change Type	Name	NE	Description
New	EVT-26411 Fronthaul Optical Path Topology Reference Information Invalid Event	eGBTS NodeB eNodeB gNodeB	This event is generated when the current fronthaul optical path topology reference information is invalid.

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

- On the MAE, click **iFaultCare**, choose **Intelligent Fault Incident Management > Incident Settings**, and turn on the **Incident Enable** switch to enable the intelligent alarm aggregation function. Then, choose **Troubleshooting Policies > Diagnosis Ability Settings** and turn on the **Enhance the fronthaul diagnosis capability** switch to enable the enhanced fronthaul diagnosis capability.
- On the base station, run the **STR OPTPATHQUALITYDET** command to build fronthaul optical path topology reference information.

Requirement/Trouble Ticket Number

FeR-IR20240808004698-0010

FeR-IR20240808004700-0020

FeR-IR20250213000671-0190

FeR-IR20250528001008-0200

FeR-IR20250528001008-0270

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.5.20 Added the Mechanism of Prohibiting File Writing After Files Are Deleted Before Base Station Delivery

NEs Involved

3900 series base stations, 5900 series base stations, USU3910

Feature/Function Description

Before base station delivery, if the deletion of dynamic files is triggered, new alarm logs may be generated in subsequent test steps, causing residual alarm logs. This problem also occurs in other log files. Therefore, a mechanism has been added to prohibit file writing after a file is deleted before base station delivery. This prevents residual alarm logs and other logs.

Implementation

Before base station delivery, uninstalling the equipment application triggers the deletion of dynamic files and ends the equipment test process. Therefore, prohibiting log file writing after the equipment application is uninstalled prevents residual logs. After the base station is reset and restarted, log files can be recorded normally.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20240902002286

SR-IR20240902002286-0140

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.5.21 Optimized the Alarm Reporting Mechanism for Automatic Software Version Rollback

NEs Involved

3900 series base stations, 5900 series base stations, USU3910

Feature/Function Description

If the main control board fails to start due to a hardware fault after a base station software upgrade is complete, the base station automatically switches the software versions in the active and standby areas to roll back the software version and reports ALM-26255 Automatic Version Rollback. In this case, the hardware fault of the main control board may cause repeated switchovers between the software versions in the active and standby areas. The base station may finally switch back to the target software version. The function involved in this change optimizes the alarm reporting mechanism in the preceding scenario to prevent ALM-26255 Automatic Version Rollback from being reported unexpectedly.

Implementation

After a base station software upgrade is complete, the base station checks whether the target software version is the same as the software version in the active area to determine whether to report ALM-26255 Automatic Version Rollback. If the target software version is different from the software version in the active area, the upgrade fails, and ALM-26255 Automatic Version Rollback is reported. Otherwise, the alarm is not reported.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20240902002286

SR-IR20240902002286-0140

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.5.22 Optimized Post-processing Mechanism for Hardware Message Queue Faults

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In scenarios where UMPTe series or later main control boards are used, the function involved in this change will trigger process reset for self-healing if the hardware message queue is faulty (for example, the message queue is overloaded or full).

Implementation

- Overloaded hardware message queue: Maintenance and test logs are recorded first. If the hardware message queue is overloaded for more than 5 minutes, process reset is triggered for self-healing.
- Full hardware message queue:
 - For UMPTe boards: Process reset is directly triggered for self-healing.
 - For main control boards other than UMPTe boards: Maintenance and test logs are recorded first. If the hardware message queue is full for more than 5 minutes, process reset is triggered for self-healing.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB and eNodeB.

Impact on Hardware Requirements

- BBU: Only the BBU3900, BBU3910, BBU5900, and BBU5910 support this function.
- Main control board: UMPTe series and later main control boards support this function.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20240902002286

SR-IR20240902002286-0390

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD

5.2.5.5.23 Optimized Processing of the Power Supply DC Output Out of Range Alarm

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

This optimization consists of the following two functions:

- Function 1: Optimized the processing when file operations are prohibited in the case of DC undervoltage
If the lower threshold of the DC voltage (specified by the **PMU.DCVLTHD (5G gNodeB, LTE eNodeB)** parameter) is incorrectly configured for the PMU power system, ALM-25621 Power Supply DC Output Out of Range may be reported with the value of **Specific Problem** being **DC Undervoltage**. If this occurs, file operations are prohibited, and an error is reported when the lower threshold of the DC voltage is modified again.
The change involved in this change allows the **LOW_VOL_FORBID_FILE_OPR** option of the **PMU.FltPostProcSw (LTE eNodeB, 5G gNodeB)** parameter of the PMU to be temporarily deselected using the **CLR PMUALM** command. As a result, the lower threshold of the DC voltage can be modified as required.

- Function 2: Optimized the reporting of ALM-25621 Power Supply DC Output Out of Range
ALM-25621 Power Supply DC Output Out of Range can be reported when the PMU is unavailable.

Implementation

- Function 1:
If the **LOW_VOL_FORBID_FILE_OPR** option of the **PMU.FltPostProcSw (LTE eNodeB, 5G gNodeB)** parameter is selected, file operations are prohibited in the case of DC undervoltage. If this option is deselected, file operations are allowed in the case of DC undervoltage.
When ALM-25621 Power Supply DC Output Out of Range is reported with the value of **Specific Problem** being **DC Undervoltage** and the alarm is caused by incorrect configuration of the **PMU.DCVLTHD (5G gNodeB, LTE eNodeB)** parameter:
 - a. Run the **CLR PMUALM** command to temporarily deselect the **LOW_VOL_FORBID_FILE_OPR** option of the **PMU.FltPostProcSw (LTE eNodeB, 5G gNodeB)** parameter of the PMU, which allows file operations in the case of DC undervoltage.
 - b. Correct the **PMU.DCVLTHD (5G gNodeB, LTE eNodeB)** parameter setting.
 - c. Select the **LOW_VOL_FORBID_FILE_OPR** option of the **PMU.FltPostProcSw (LTE eNodeB, 5G gNodeB)** parameter to restore the prohibition of file operations in the case of DC undervoltage.
- Function 2:
If the DC output of the power system is abnormal and the PMU is unavailable, ALM-25621 Power Supply DC Output Out of Range is not reported before the upgrade but is reported after the upgrade.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	PMU.FltPostProcSw (LTE eNodeB, 5G gNodeB)	CLR PMUALM	eGBTS NodeB eNodeB gNodeB	This parameter has been added to specify the post-processing switch when a PMU-related fault occurs.

- Performance management

No impact

- Fault management

No impact

Impact on Features/Functions

- Prerequisite features/functions

No impact

- Mutually exclusive features/functions

No impact

- Impacted features/functions

No impact

Related Operations

By default, the change does not take effect after an upgrade.

Run the **CLR PMUALM** command to deselect the **LOW_VOL_FORBID_FILE_OPR** option of the **PMU.FltPostProcSw (LTE eNodeB, 5G gNodeB)** parameter of the PMU to optimize the processing when file operations are prohibited in the case of DC undervoltage. After the configuration of the **PMU.DCVLTHD (5G gNodeB, LTE eNodeB)** parameter is corrected, the **LOW_VOL_FORBID_FILE_OPR** option of the **PMU.FltPostProcSw (LTE eNodeB, 5G gNodeB)** parameter is selected to restore the prohibition of file operations in the case of DC undervoltage.

Requirement/Trouble Ticket Number

IR20240902002286

SR-IR20240902002286-0710

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.5.24 Support for Isolating Faulty Units on Boards.

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Isolating faulty units on boards is now supported to quickly restore services when a BBU board is faulty.

This function is enabled through the following parameter settings:

- Set the **MOCRSVDPARA.MOCNAME** parameter to **EQUIPMENT**.
- Set the **MOCRSVDPARA.MOIKEY** parameter to **0**.
- Set the **MOCRSVDPARA.RSVDPARAID** parameter to **BRDCHECKFUNC**.
- Set the **MOCRSVDPARA.RSVDPARAValue** parameter to the ID of the faulty unit. For details about how to configure the faulty unit ID, contact Huawei engineers.

Implementation

Isolating faulty units on boards can be used when a BBU board is faulty to isolate the faulty units that can be isolated, thereby restoring services quickly.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

To isolate a faulty unit, run the **ADD MOCRSVDPARA** command with **MOCRSVDPARA.MOCNAME** set to **EQUIPMENT**, **MOCRSVDPARA.MOIKEY** set to **0**, **MOCRSVDPARA.RSVDPARAID** set to **BRDCHECKFUNC**, and **MOCRSVDPARA.RSVDPARAValue** set to the ID of the faulty unit.

Requirement/Trouble Ticket Number

IR20240902002286

SR-IR20240902002286-0770

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.5.25 Optimized the CPU Fault Detection and Alarm Reporting Capabilities

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The maintenance and detection capabilities for component faults have been optimized. The system now supports alarm reporting for CPU component faults.

Implementation

ALM-26200 Board Hardware Fault is reported when the CPU detects that a component is faulty.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

- BBU: Book BBU
- Boards:
 - Baseband processing units: UBBPg series and UBBPh series
 - Main control boards: UMPTg series

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20250908001612

SR-IR20250908001612-0110

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.5.26 Support for Fault Detection and Alarm Reporting for Certain Components of a UPEU

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Fault detection and alarm reporting for certain components of a universal power and environment interface unit (UPEU) enables the eNodeB to report ALM-26200 Board Hardware Fault when detecting that the reported UPEU power consumption is abnormal. This facilitates board maintenance.

This function is controlled by the **UPEU_HW_FAULT_ALM_SW** option of the **EQUIPMENT.EQMOPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter.

Implementation

- If the fault detection and alarm reporting function is enabled for certain components of a UPEU, ALM-26200 Board Hardware Fault is reported when a hardware fault is detected on the UPEU due to abnormal power consumption reporting during operation.
- If the fault detection and alarm reporting function is disabled for certain components of a UPEU, the preceding faults are detected during operation. Once a fault is detected, ALM-26200 Board Hardware Fault is reported only when the function is enabled.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameter has changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	EQUIPMENT.EQMOPTFEATSW (LTE eNodeB, 5G gNodeB)	SET EQUIPMENT LST EQUIPMENT	eGBTS NodeB eNodeB gNodeB	The UPEU_HW_FAULT_ALM_SW option has been added to specify whether to detect faults on certain components of a UPEU and report related alarms in the case that such a fault is detected.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

You can run the **SET EQUIPMENT** command with the **UPEU_HW_FAULT_ALM_SW** option of the **EQUIPMENT.EQMOPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter selected to enable fault detection and alarm reporting for certain components of a UPEU.

Requirement/Trouble Ticket Number

IR20250908001644

SR-IR20250908001644-0050

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.5.27 Support for Alarm Reporting and Maintenance Mode Updating When a Base Station Continuously Encounters Exceptions

NEs Involved

3900 series base stations, 5900 series base stations, USU3910

Feature/Function Description

When a base station continuously encounters exceptions and cannot be remotely operated and maintained, it automatically uses minimum configuration. Now, after the function involved in this change is enabled, a base station not only uses minimum configuration, but also reports ALM-26255 Automatic Version Rollback with the alarm maintenance mode updated in the following scenarios:

- Scenario 1: The base station has been restarted abnormally for more than 10 consecutive times, and the MAE has logged in to the NE for less than 30 minutes each time during this period.
- Scenario 2: The base station fails to be started for more than 30 consecutive times.

Implementation

- Scenario 1: ALM-26255 Automatic Version Rollback is reported with the value of **Specific Problem** being **Consecutive Abnormal Resets** and the alarm maintenance mode being updated to the base station maintenance mode before the use of minimum configuration.
- Scenario 2: ALM-26255 Automatic Version Rollback is reported with the value of **Specific Problem** being **Startup Failure** and the alarm maintenance mode being updated to the base station maintenance mode before the use of minimum configuration.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact

- Configuration management
No impact
- Performance management
No impact
- Fault management
The following alarm has changed.

Change Type	Name	NE	Description
Modified	ALM-26255 Automatic Version Rollback	eGBTS NodeB eNodeB gNodeB	ALM-26255 Automatic Version Rollback is reported with the alarm maintenance mode being updated in either of the following scenarios: (1) The base station has been restarted abnormally for more than 10 consecutive times, and the MAE has logged in to the NE for less than 30 minutes each time during this period. (2) The base station fails to be started for more than 30 consecutive times. This alarm is an automatically detected and manually cleared (ADMC) one so that it needs to be manually cleared.

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20240902002286

SR-IR20240902002286-0750

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.5.28 Optimized the Reporting of Board Hardware Fault Alarm Based on Bad Flash Block Rate on the Book BBU

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The reporting of ALM-26200 Board Hardware Fault for bad flash block rate on the Book BBU has been optimized so that the bad flash block rate threshold (specified by the **MOCRSVDPARA.RSVDPARAVALUE** parameter) can be configured as required to specify the alarm reporting sensitivity.

NOTE

This change was first incorporated into V100R020C10SPC290 but is first described in the network impact report of V100R020C10SPC300.

Implementation

When the bad flash block rate is too high, the system clears the bad blocks in the early morning and then checks whether the bad flash block rate meets the alarm reporting conditions. If the conditions are met, ALM-26200 Board Hardware Fault is reported. This optimization takes effect through the following parameter settings:

- Set the **MOCRSVDPARA.MOCNAME** parameter to **EQUIPMENT**.
- Set the **MOCRSVDPARA.MOIKEY** parameter to **0**.
- Set the **MOCRSVDPARA.RSVDPARAID** parameter to **BRDFLASHBBDECTLV**.
- Set the **MOCRSVDPARA.RSVDPARAVALUE** parameter to **1**, **2**, or **3**. This parameter can be set as required to specify the alarm reporting sensitivity.
 - When this parameter is set to **0**, detection on the health status of the flash memory is disabled.
 - When this parameter is set to **1**, low-sensitivity detection on the health status of the flash memory is enabled.
 - When this parameter is set to **2**, medium-sensitivity detection on the health status of the flash memory is enabled.
 - When this parameter is set to **3**, high-sensitivity detection on the health status of the flash memory is enabled.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

BBU: Book BBU

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

Run the **ADD MOCRSVDPARA** command with **MOCRSVDPARA.MOCNAME** set to **EQUIPMENT**, **MOCRSVDPARA.MOIKEY** set to **0**, **MOCRSVDPARA.RSVDPARAID** set to **BRDFLASHBBDECTLV**, and **MOCRSVDPARA.RSVDPARAValue** set to **1**, **2**, or **3** to enable this optimization.

Requirement/Trouble Ticket Number

IR20230224002099

SR-IR20230224002099-1090

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.5.29 Alarm for a Clock Synchronization Mode Configuration Error in BBU-DCU Interconnection Scenarios

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In BBU-DCU interconnection scenarios, the base station clock synchronization mode must be set to time synchronization (that is, the **Tasm.ClkSyncMode (LTE eNodeB, 5G gNodeB)** parameter must be set to **TIME**) and loose time synchronization must be disabled (that is, the **Tasm.LooseTime (LTE eNodeB, 5G gNodeB)** parameter can only be set to **OFF**) on the BBU side.

ALM-26318 BBU-DCU Interconnection Configuration Failure has been added to notify users of a base station clock synchronization mode configuration error.

Implementation

With this function, ALM-26318 BBU-DCU Interconnection Configuration Failure is reported with the value of **Specific Problem** being **Base Station Clock Synchronization Mode Configuration Error** when **Clock Synchronization Mode** is not set to **TIME**, or this parameter is set to **TIME** but **Loose Time Synchronization** is set to **ON** on the BBU side.

If the preceding alarm is reported, you can run the **LST CLKSYNCMODE** command to check whether **Clock Synchronization Mode** is set to **TIME** and whether **Loose Time Synchronization** is set to **ON**.

If redundant MOs **RCSSERVER** and **EXTRRURES** exist, the alarm is falsely reported. In this case, you can run the **LST RCSSERVER** and **LST EXTRRURES** commands, and then run the **RMV RCSSERVER** and **RMV EXTRRURES** commands to delete all redundant objects to clear the alarm.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
The following alarms have changed.

Change Type	Name	NE	Description
New	ALM-26318 BBU-DCU Interconnection Configuration Failure	eGBTS NodeB eNodeB gNodeB	This alarm has been added to report that the BBU-DCU interconnection configuration is incorrect or the configuration is inconsistent with the actual situation.

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250305000198-001

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.6 Inventory Management

This section describes the changes in inventory management.

5.2.5.6.1 Support for Extracting BLS Version Information from Base Station Inventory Files

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The version numbers of BLS packages are now included in inventory management and can be obtained in batches, improving query efficiency.

Implementation

The **BLSVER** and **BLSBACKVER** fields have been added to the inventory management page, which indicate the version numbers of BLS packages in the active and standby areas, respectively. These fields can be displayed on the MAE and exported from device archives.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the eNodeB, gNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact

- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250726000136-0010

IR20250728002833

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.6.2 Optimized Display of Board Names on the Device Panel

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The display of board and module names on the interface of the device panel has been optimized to match physical labels.

Implementation

The display of board and module names on the interface of the device panel has been optimized by standardizing the letter case and shortening the names to ensure consistency with physical labels. Here are some examples of the optimizations:

1. Suffixes for some existing boards and modules have been simplified, and the part indicating the product subcategory has been reduced. For example, "UMPTg2BT" is now "UMPTg2".
2. The letter cases of suffixes in the silkscreens of some existing boards and modules have been standardized. For example, "LBBPD5" is now "LBBPd5".

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB, eNodeB, NodeB, and eGBTS.

Impact on Hardware Requirements

Only the UMPT, UBBP, WBBP, LBBP, GTMU, FAN, and UPEU series boards/modules are involved.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20250319000813

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.2.5.7 License Management

This section describes the changes in license management.

For details about changes in licenses, see *License Changes*, which is provided with the software version. You can obtain *License Changes* from the following directory: **BTS3900&BTS5900 VXXRXXXCXXSPCXXX Release Documents-EN\01-Release Notes\Delta(MMI).**

5.2.5.7.1 Optimized the Postpaid Statistics Rules of UMPT Multimode License Control Items

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The standalone license control item usage displayed in the postpaid statistics of license control items is the used value divided by the licensed value. A licensed value is the number of license units purchased by the customer, indicated in the LKF, and a used value is the number of license units actually used by the base station. There is a gifting mechanism for UMPT multimode license control items, which means that an operator can use a license control item without purchasing it, resulting in a licensed value of 0 and a used value of 1. In this context, the statistics rules of UMPT multimode license control items has been optimized to ensure more accurate statistics.

Implementation

If an operator uses a UMPT multimode license that is provided for free but does not purchase the corresponding UMPT multimode license control item, the licensed value and used value of the UMPT multimode license control item are both 0 in postpaid statistics.

Impact on the Network

No impact

Impact on NEs

No impact

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
If an operator uses a UMPT multimode license that is provided for free but does not purchase the corresponding UMPT multimode license control item, this license control item is not counted in this case. That is, both the licensed value and used value are counted as 0.
- Configuration management
No impact

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20240902002286

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD, UMTS, GSM

5.3 gNodeBFunction

5.3.1 New Features

5.3.1.1 FBFD-100100 Uplink Slot Configuration Optimization

5.3.1.1.1 Uplink Slot Configuration Optimization (Only in High-Frequency Bands)

NEs Involved

5900 series base stations

Feature/Function Description

Commonly used slot configurations (like the 4:1 slot configuration: 4_1_DDDSU) deliver limited uplink capacity due to a small proportion of UL-only slots. Uplink Slot Configuration Optimization uses the 2:3 slot configuration to increase the proportion of UL-only slots for a higher uplink capacity.

In high-frequency TDD where the SCS is 120 kHz, the 2:3 slot configuration (2_3_DSUUU) described in [Table 5-7](#) is supported. The 2:3 slot configuration

(2_3_DSUUU) comprises one DL-only slot, one self-contained slot, and three UL-only slots.

Table 5-7 Description of the 2:3 slot configuration (2_3_DSUUU) in high frequency bands

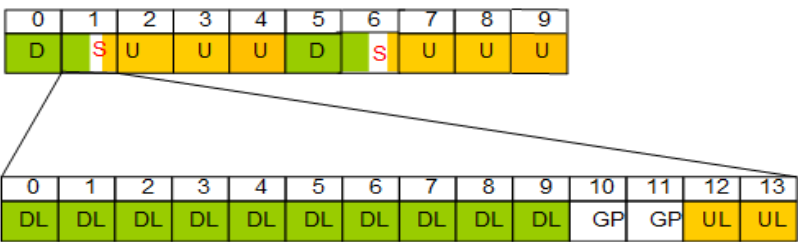
Slot Structure	GP	X (ms)	x1	x2	y2	y1
SS221	2	0.625	1	10	2	3

The explanations are as follows:

- X: indicates the configuration period.
- x1: indicates the number of DL-only slots.
- y1: indicates the number of UL-only slots.
- x2: indicates the number of downlink symbols following DL-only slots.
- y2: indicates the number of uplink symbols before UL-only slots.
- GP: indicates the number of guard period (GP) symbols between uplink and downlink data transmission.

A slot configuration is determined by its slot assignment and slot structure, which are specified by the **NRDUCell.SlotAssignment** and **NRDUCell.SlotStructure** parameters, respectively. **Figure 5-1** illustrates the slot configuration with **NRDUCell.SlotAssignment** set to **2_3_DSUUU** and **NRDUCell.SlotStructure** set to **SS221**. In this configuration, each configuration period is 0.625 ms and consists of five slots: 1 DL-only slot (**D**), 3 UL-only slots (**U**) and 1 self-contained slot (**S**). In a self-contained slot, there are 2 GP symbols (**GP**), 10 downlink symbols (**DL**), and 2 uplink symbols (**UL**).

Figure 5-1 2:3 slot configuration



Implementation

For details, see *Standards Compliance*. The specific sections are listed in the change history of the document.

Impact on the Network

Benefits:

Uplink Slot Configuration Optimization increases the number of uplink-only slots, lifting the uplink cell capacity.

This function is suitable only for scenarios where UEs are stationary. It does not ensure performance in mobile scenarios.

Impacts:

If intra-band or adjacent-band cells served by the gNodeBs differ in their slot configuration, interference arises between the cells, affecting the capacity and user experience of the gNodeBs. To address this issue, it is recommended that the slot configurations of neighboring cells be the same.

The 2:3 slot configuration has fewer downlink slots and therefore supports a smaller downlink cell capacity than the 4:1 slot configuration (4_1_DDDSU).

Impact on NEs

This feature is implemented on the gNodeB.

Impact on Hardware Requirements

- Base station models
5900 series base stations (macro base stations)
- Boards
TDD cells with different slot configurations must be bound to different BBPs. The slot configurations of all TDD cells on the same BBP must be the same.
In high-frequency TDD, a BBP can be configured with only one sector if 2:3 slot configuration is used. For details about sectors, see *Cell Management*.
The main control boards must be UMPTg or UMPTh series boards.
The BBPs must be UBBPg3, UBBPh3a, or UBBPh3b boards.
- RF modules
Macro base stations: HAAU5323, HAAU5233w, HAAU5235w and later versions
Pole sites: HAAU5222 only
If the NR carrier bandwidth is less than or equal to 400 MHz, one optical port on an RF module is used for installing optical fibers. If the NR carrier bandwidth is greater than 400 MHz, two optical ports on an RF module are required for installing optical fibers and the **RRUCHAIN.TT** parameter must be set to **LOADBALANCE** for the relevant RRU chain or ring.

NOTE

Among cells served by HAAU5233w and HAAU5235w, only those with the NR carrier bandwidth less than or equal to 800 MHz and **NRDUCellTrp.TxRxMode** set to **4T4R** support the 2:3 slot configuration (2_3_DSUUU).

NR cells served by the same RF module must use the same slot configuration and slot structure.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
This feature is not under license control.
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Downlink DMRS Maximum Length	NRDUCellPdsch.DlDmrsMaxLength	<i>Downlink Scheduling</i>	When NRDUCell.SlotAssignment is set to 2_3_DSUUU , NRDUCellPdsch.DlDmrsMaxLength must be set to 1SYMBOL .
High-frequency TDD	CSI-RS for BM in SSB slots	SSBSLOT_CARRY_BEAM_MEAS_CSI_SW option of the NRDUCellCsirs.CsiSwitch parameter	None	If the NRDUCell.SlotAssignment parameter is set to 2_3_DSUUU , the SSBSLOT_CARRY_BEAM_MEAS_CSI_SW option of the NRDUCellCsirs.CsiSwitch parameter must be selected.
High-frequency TDD	FR2 CA carrier management policy	NRDUCellCarrierMgmt.Fr2CaCarrierMgmtPolicy	<i>Carrier Aggregation</i>	When NRDUCell.SlotAssignment is set to 2_3_DSUUU , NRDUCellCarrierMgmt.Fr2CaCarrierMgmtPolicy must be set to USER_EXP_BASED .

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Short PUCCH Symbol Number	NRDUCellPucch.ShortPucchSymbolNum	None	When NRDUCell.SlotAssignment is set to 2_3_DSUUU , NRDUCellPucch.ShortPucchSymbolNum can only be set to SYMBOL1 or SYMBOL3 .
High-frequency TDD	Fixed Weight Type	NRDUCellPdsch.FixedWeightType	None	NRDUCellPdsch.FixedWeightType must be set to PMI_WEIGHT for a cell with NRDUCell.SlotAssignment set to 2_3_DSUUU .
High-frequency TDD	PDCCH symbol number configuration	NRDUCellPdcch.OccupiedSymbolNum	<i>Channel Management</i>	When NRDUCell.SlotAssignment is set to 2_3_DSUUU and NRCell.CellActiveState is set to CELL_ACTIVE , NRDUCellPdcch.OccupiedSymbolNum can only be set to PART_SLOT_3SYM .

- Mutually exclusive features/functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	BFR	CONTENTION_FREE_BFR_SW option of the NRDUCellAlgoSwitch.BeamOptSwitch parameter	None	If the NRDUCell.SlotAssignment parameter is set to 2_3_DSUUU , the CONTENTION_FREE_BFR_SW option of the NRDUCellAlgoSwitch.BeamOptSwitch parameter must be deselected.
High-frequency TDD	Wide TRS beam	NRDUCellCsirs.TrsBeamType	None	If the NRDUCell.SlotAssignment parameter is set to 2_3_DSUUU , the NRDUCellCsirs.TrsBeamType parameter must be set to NARROW_BEAM .
High-frequency TDD	LNR DRX synchronization	LNR_DRX_SYNC_HRONIZE_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch parameter	None	If the NRDUCell.SlotAssignment parameter is set to 2_3_DSUUU , the LNR_DRX_SYNC_HRONIZE_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch parameter must be deselected.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Uplink MU-MIMO for mmWave large-packet UEs	MMW_LRG_PKT_UL_MU_MIMO_SW option of the NRDUCellFeatureSw.MimoFeatureSwitch parameter	None	If the NRDUCell.SlotAssignment parameter is set to 2_3_DSUUU , the MMW_LRG_PKT_UL_MU_MIMO_SW option of the NRDUCellFeatureSw.MimoFeatureSwitch parameter must be deselected.
High-frequency TDD	PUSCH time domain enhancement	PUSCH_TIME_DOMAIN_ENH_SW option of the NRDUCellAlgoSwitch.FullChannelCovEnhSwitch parameter	None	If the NRDUCell.SlotAssignment parameter is set to 2_3_DSUUU , the PUSCH_TIME_DOMAIN_ENH_SW option of the NRDUCellAlgoSwitch.FullChannelCovEnhSwitch parameter must be deselected.
High-frequency TDD	Intelligent beam selection	INTELLIGENT_BEAM_SELECTION_SW option of the NRDUCellBeamAlgo.BeamOptAlgoSwitch parameter	None	If the NRDUCell.SlotAssignment parameter is set to 2_3_DSUUU , the INTELLIGENT_BEAM_SELECTION_SW option of the NRDUCellBeamAlgo.BeamOptAlgoSwitch parameter must be deselected.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Uplink fallback of data transmission to the MCG	UL_FALLBACK_TO_MCG_SW option of the NRCellNrDcConfig.NrDcAlgoSwitch parameter	None	If the NRDUCell.SlotAssignment parameter is set to 2_3_DSUUU , the UL_FALLBACK_TO_MCG_SW option of the NRCellNrDcConfig.NrDcAlgoSwitch parameter must be deselected.

- Impacted features/functions
None

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Standards Compliance*.

Requirement/Trouble Ticket Number

FeR-IR20250401000781-0001

RAT

NR TDD

5.3.1.1.2 Added Support for the Slot Structure SS05080402 by the Uplink Slot Configuration Optimization Feature (for High Frequency Bands Only)

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In high-frequency TDD with the SCS of 120 kHz, the 2:3 slot configuration (2_3_DSUUU) now supports the slot structure **SS05080402**, as listed in [Table 5-8](#).

Table 5-8 Description of the new slot structure SS05080402 for the 2:3 slot configuration (2_3_DSUUU) in high frequency bands

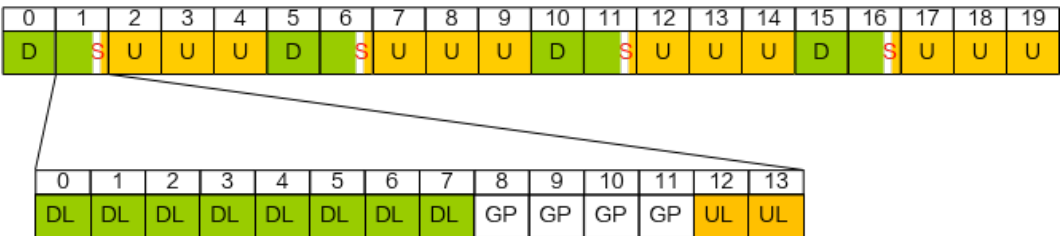
Slot Structure	GP	X (ms)	x1	x2	y2	y1
SS05080402	4	0.625	1	8	2	3

The explanations are as follows:

- X: indicates the configuration period.
- x1: indicates the number of DL-only slots.
- y1: indicates the number of UL-only slots.
- x2: indicates the number of downlink symbols following DL-only slots.
- y2: indicates the number of uplink symbols before UL-only slots.
- GP: indicates the number of guard period (GP) symbols between uplink and downlink data transmissions.

Figure 5-2 illustrates the slot configuration with **NRDUCell.SlotAssignment** set to **2_3_DSUUU** and **NRDUCell.SlotStructure** set to **SS05080402**. In this configuration, each configuration period is 0.625 ms and consists of five slots: 1 DL-only slot (**D**), 1 self-contained slot (**S**), and 3 UL-only slots (**U**). In a self-contained slot, there are 8 downlink symbols (**DL**), 4 GP symbols (**GP**), and 2 uplink symbols (**UL**).

Figure 5-2 Slot structure SS05080402 for the 2:3 slot configuration



Implementation

For details, see *Standards Compliance*. The specific sections are listed in the change history of the document.

Impact on the Network

After the 2:3 slot configuration (2_3_DSUUU) is configured with the slot structure **SS05080402**, the total number of downlink symbols decreases. This may increase the downlink PRB usage and downlink IBLER, decrease the average downlink MCS index, average downlink cell throughput, and average downlink UE throughput.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact.
- Configuration management
The following parameters have been added to accommodate the change in this feature.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCell.SlotStructure	ADD NRDUCELL MOD NRDUCELL LST NRDUCELL	gNodeB	The value SS05080402 has been added to this parameter, which specifies the slot structure. This value means that, when the ratio of downlink slots to uplink slots in an mmWave band is 2:3, the numbers of downlink symbols, GP symbols, and uplink symbols in the self-contained slot are 8, 4, and 2, respectively, and the corresponding parameters are X=0.625 ms, x1=1, x2=8, y2=2, and y1=3.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Standards Compliance*.

Requirement/Trouble Ticket Number

FeR-IR20250306002890-0003

RAT

NR TDD

5.3.1.1.3 Added the Relationship Between the PRACH Configuration Index and the Cell Radius for Uplink Slot Configuration Optimization (for High Frequency Bands Only)

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The following cell requirement has been added for the 2:3 slot configuration (2_3_DSUUU) in high-frequency TDD with the SCS of 120 kHz:

If the **NRDUCell.SubcarrierSpacing** parameter is set to **120KHZ** and the **NRDUCellPrach.PrachConfigurationIndex** parameter is set to **188** or **65535**, the **NRDUCell.CellRadius** parameter can be set to a value of **2410** or smaller.

The constraint relationships between **NRDUCellPrach.PrachConfigurationIndex** set to **65535** and the cell radius are as follows:

- The PRACH configuration index of **65535** is automatically mapped to **186** when the cell radius is set to a value of **1160** or smaller.

- The PRACH configuration index of **65535** is automatically mapped to **188** when the cell radius is set to a value between **1161** and **2410** (inclusive).

Implementation

For details, see *Standards Compliance*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature is not under license control.
- Configuration management
The change in this feature has no impact.
- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250306002890-0003

RAT

NR TDD

5.3.1.2 FOFD-071203 Adaptive Managed Latency

5.3.1.2.1 Added Adaptive Managed Latency

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The Adaptive Managed Latency feature has been introduced.

In NSA networking, this feature provides the following functions: uplink L4S congestion identification and downlink L4S congestion identification.

Uplink L4S congestion identification

- On the eNodeB side:
 - Select the **UL_CONGESTION_CTRL_SW** option of the **ENodeBAlgoExtSwitch.L4SAlgoSwitch** parameter to enable uplink congestion identification.
 - Set **QciPara.L4SFlag** to **TRUE** for a specific QCI.
- On the gNodeB side:
 - Select the **UL_CONGEST_CTRL_SW** option of the **gNodeBParam.L4sAlgoSwitch** parameter to enable uplink congestion identification.
 - Set **gNBQciBearer.L4sFlag** to **TRUE** for a specific QCI.
 - Set **gNBQciBearer.L4sCongestDetectAlgorithm** to **L4S**.

Downlink L4S congestion identification

- On the eNodeB side:
 - Select the **DL_CONGESTION_CTRL_SW** option of the **ENodeBAlgoExtSwitch.L4SAlgoSwitch** parameter to enable downlink congestion identification.
 - Set **QciPara.L4SFlag** to **TRUE** for a specific QCI.
- On the gNodeB side:
 - Select the **DL_CONGEST_CTRL_SW** option of the **gNodeBParam.L4sAlgoSwitch** parameter to enable downlink congestion identification.
 - Set **gNBQciBearer.L4sFlag** to **TRUE** for a specific QCI.
 - Set **gNBQciBearer.L4sCongestDetectAlgorithm** to **L4S**.

In SA networking, this feature provides the following functions: L4S congestion control based on local identification, enhanced L4S congestion control, and enhanced L4S congestion control based on rate adjustment.

L4S congestion control based on local identification is used to relieve congestion and improve user experience for services such as cloud gaming and extended reality (XR).

- Uplink L4S congestion control based on local identification is controlled by the **UL_CONGEST_CTRL_SW** option of the **gNodeBParam.L4sAlgoSwitch** parameter. This function also requires that the **gNBQciBearer.L4sCongestDetectAlgorithm** parameter be set to **L4S**. Before this function is enabled, the L4S function must be enabled by setting the **gNBQciBearer.L4sFlag** parameter to **TRUE**.
- Downlink L4S congestion control based on local identification is controlled by the **DL_CONGEST_CTRL_SW** option of the **gNodeBParam.L4sAlgoSwitch** parameter. This function also requires that the **gNBQciBearer.L4sCongestDetectAlgorithm** parameter be set to **L4S**. Before this function is enabled, the L4S function must be enabled by setting the **gNBQciBearer.L4sFlag** parameter to **TRUE**.

The overall process of **enhanced L4S congestion control** is the same as that of L4S congestion control based on local identification. The difference between these two functions lies in the delay evaluation method.

- Enhanced uplink L4S congestion control is controlled by the **UL_CONGEST_CTRL_SW** option of the **gNodeBParam.L4sAlgoSwitch** parameter. This function also requires that the **gNBQciBearer.L4sCongestDetectAlgorithm** parameter be set to **L4SPLUS**. Before this function is enabled, the L4S function must be enabled by setting the **gNBQciBearer.L4sFlag** parameter to **TRUE**.
- Enhanced downlink L4S congestion control is controlled by the **DL_CONGEST_CTRL_SW** option of the **gNodeBParam.L4sAlgoSwitch** parameter. This function also requires that the **gNBQciBearer.L4sCongestDetectAlgorithm** parameter be set to **L4SPLUS**. Before this function is enabled, the L4S function must be enabled by setting the **gNBQciBearer.L4sFlag** parameter to **TRUE**.

Enhanced L4S congestion control based on rate adjustment follows the same overall process as the enhanced L4S congestion control function. The difference is that enhanced L4S congestion control based on rate adjustment uses a virtual queue mechanism to adjust the transmission rate and ensures fairness in rate adjustment by considering the experience of large-packet UEs in the cell when calculating the maximum rate, fair rate, and target rate.

- If uplink congestion control based on rate adjustment is required, enable enhanced uplink L4S congestion control (by selecting the **UL_CONGEST_CTRL_SW** option of the **gNodeBParam.L4sAlgoSwitch** parameter and setting the **gNBQciBearer.L4sCongestDetectAlgorithm** parameter to **L4SPLUS**), and select the **UL_L4SPLUS_FOR_FAIR_SW** option of the **NRDUCellUISch.UISchAlgoSwitch** parameter.
- If downlink congestion control based on rate adjustment is required, enable enhanced downlink L4S congestion control (by selecting the **DL_CONGEST_CTRL_SW** option of the **gNodeBParam.L4sAlgoSwitch** parameter and setting the **gNBQciBearer.L4sCongestDetectAlgorithm**

parameter to **L4SPLUS**), and select the **DL_L4SPLUS_FOR_FAIR_SW** option of the **NRDUCellDISch.DISchAlgoEnhSwitch** parameter.

Implementation

For details about implementation in SA networking, see *Adaptive Managed Latency*. The specific sections are listed in the change history of the document.

For details about implementation in NSA networking, see *EPC-based NSA Basics*. The specific sections are listed in the change history of the document.

Impact on the Network

Benefits:

This feature reduces the uplink and downlink latency of UEs, improving user experience. In the case of congestion, enhanced L4S congestion control based on rate adjustment ensures that resources of large-packet UEs are not preempted by L4S UEs.

Impacts:

After this feature is enabled, the **Cell Uplink Average Throughput (DU)** and **Cell Downlink Average Throughput (DU)** decrease. In addition, the amount of data received by the base station from the UE and server may decrease.

Impact on NEs

This feature is implemented on the eNodeB and the gNodeB.

Impact on Hardware Requirements

All NR-capable main control boards and baseband processing units support this function. The LBBPd board on the LTE side does not support this function.

Impact on Inter-NE Interfaces

In NSA networking, L4S congestion identification is supported only in Option 3x networking.

Impact on Operation and Maintenance

- License management

This feature is controlled by the following new license control item.

Feature ID	Feature Name	Model	Sales Unit
FOFD-071203	Adaptive Managed Latency	NR0S00AML000	per gNodeB

- Configuration management

No impact

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
The following impact relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR DU cell resource management between network slices	RB_DYNAMIC_CONTROL_SW option of the NRDUCellAlgoSwitch.NetworkSliceAlgoSwitch parameter	<i>Network Slicing</i>	If both "NR DU cell resource management between network slices" and "enhanced L4S congestion control based on rate adjustment" are enabled, the gNodeB restricts the available RBs for L4S UEs based on the available resource range of the network slice group to which the UEs belong, which may cause the calculated maximum rate and fair rate to be inaccurate.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RAN sharing with common carrier	gNBSharingMode.gNBMultiOpSharingMode set to SHARED_FREQ	<i>Multi-Operator Sharing</i>	If enhanced L4S congestion control based on rate adjustment is enabled in RAN sharing with common carrier scenarios, the gNodeB restricts the available RBs for L4S UEs based on the available resource range of the UEs' operator, which may cause the calculated maximum rate and fair rate to be inaccurate.
Low-frequency TDD	User PRB control for FWA	PRB_CTRL_SW option of the NRDUCellAlgoSwitch.FwaAlgoSwitch parameter	<i>FWA</i>	If both "user PRB control for FWA" and "enhanced L4S congestion control based on rate adjustment" are enabled, the gNodeB restricts the available RBs for L4S UEs based on the available resource range of FWA UEs, which may cause the calculated maximum rate and fair rate to be inaccurate.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	QCI-based resource control	RB_DYNAMIC_CONTROL_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter	<i>QCI-based Resource Control</i>	If both "QCI-based resource control" and "enhanced L4S congestion control based on rate adjustment" are enabled, the gNodeB restricts the available RBs for L4S UEs based on the available resources of the QCI resource group associated with the UEs, which may cause the calculated maximum rate and fair rate to be inaccurate.
Low-frequency TDD	MU-MIMO	DL_MU_MIMO_SW and UL_MU_MIMO_SW options of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	If both MU-MIMO and "enhanced L4S congestion control based on rate adjustment" are enabled, the maximum rate and fair rate calculated by the gNodeB for L4S UEs may be inaccurate.

Related Operations

By default, the change does not take effect after an upgrade.

For details about related operations in SA networking, see the corresponding "Operation and Maintenance" section in *Adaptive Managed Latency*.

For details about related operations in NSA networking, see the corresponding "Operation and Maintenance" section in *EPC-based NSA Basics*.

Requirement/Trouble Ticket Number

FeR-IR20251211001076-0001

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD

5.3.1.3 FOFD-050201 Network Slicing Introduction

5.3.1.3.1 Support for Network Slicing in High Frequency Bands (for High Frequency Bands Only)

NEs Involved

5900 series base stations

Feature/Function Description

The following network slicing functions now apply to high frequency bands:

- Network slice deployment and awareness
- Network slice admission control
- NR cell resource management between network slices
- Basic resource management under NR DU cell resource management between network slices
- Network slice group configuration consistency query

LampSite base stations do not support this feature in high-frequency scenarios.

This feature is supported only in SA networking.

Implementation

For details, see *Network Slicing*. The specific sections are listed in the change history of the document.

Impact on the Network

Gains: Network slicing enables multiple logical networks to run on an operator's physical network. This function provides 5G network services tailored to different requirements, reduces costs for operators, and improves competitiveness of vertical industries.

Impacts:

- It is recommended that SR-based smart scheduling be enabled (by selecting the **SR_SMART_SCH_SW** option of the **NRDUCellPusch.UlPuschAlgoSwitch** parameter) when you enable NR DU cell resource management between network slices. If the former function is not enabled together with the latter one, the scheduling priorities of UEs outside slice groups may be higher than those of UEs inside slice groups, meaning that SLAs for UEs inside slice groups cannot be ensured.
- In high-frequency scenarios, the number of available beams in a slot is limited. In this context, if the minimum percentage of RBs (specified by the **NRDUCellRes.DlRbMinRatio** or **NRDUCellRes.UlRbMinRatio** parameter) is set to an excessively high value for a network slice group, the function has the following network impacts:
 - The resources available to UEs outside the network slice group are critically insufficient, causing repeated access attempts and frequent service drops for these UEs.

- If the carriers supported by a specific UE in the network slice group are fewer than the actual carriers in the sector, the RBs configured by the minimum RB percentage may fail to be reserved for this UE.
- The beam resources occupied by UEs fluctuate, resulting in an increased difference between the percentage of RBs used by UEs in each network slice group and the configured RB percentage.

Impact on NEs

This change is implemented on the gNodeB.

Impact on Hardware Requirements

- Base station models: 5900 series base stations (macro base stations)
- Boards: All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- RF modules: All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

This feature is controlled by the following new license control item.

RAT	Feature ID	Feature Name	Model	Sales Unit
High - frequency TDD	FOFD-050201	Network Slicing Introduction	NR0S00BNSL00	per Cell

- Configuration management

The following MML command has changed.

Change Type	MML Command	NE	Description
New	DSP NRDUCELL NSGRP	gNodeB	This command has been added to query the network slice group configurations of an NR DU cell.

 NOTE

The parameters in these new or deleted MML commands are no longer listed separately.

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellResId	ADD NRDUCELLRES LST NRDUCELLRES MOD NRDUCELLRES RMV NRDUCELLRES	gNodeB	This parameter now applies to high frequency bands.
Modified	NRDUCellResUsageObject	ADD NRDUCELLRES LST NRDUCELLRES MOD NRDUCELLRES	gNodeB	This parameter now applies to high frequency bands.
Modified	NRCelAlgoSwitch.NetworkSliceAlgoSwitch	MOD NRCELLALGO SWITCH LST NRCELLALGO SWITCH	gNodeB	The ADMISSION_BASED_ON_NS_ID_SW option of this parameter now applies to high frequency bands.
Modified	NRCelAlgoSwitch.NetworkSliceAlgoSwitch	MOD NRCELLALGO SWITCH LST NRCELLALGO SWITCH	gNodeB	The ENHANCED_NETWORKSLICE_SW option of this parameter now applies to high frequency bands.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellAIgoSwitch.NetworkSliceAIgoSwitch	MOD NRDUCELLALGOSWITCH LST NRDUCELLALGOSWITCH	gNodeB	The RB_DYNAMIC_CONTROL_SW option of this parameter now applies to high frequency bands.

- Performance management

The following counters have changed.

Change Type	Counter Name	Function Subset	NE	Description
Modified	N.PRBUUsed.DL.NsGrp0.Avg to N.PRBUUsed.DL.NsGrp31.Avg	PRB Measurement	gNodeB	These counters measure the downlink/uplink PRB usages in network slice groups of all NR DU cells within the same sector in high-frequency scenarios.
Modified	N.PRBUUsed.UL.NsGrp0.Avg to N.PRBUUsed.UL.NsGrp31.Avg	PRB Measurement	gNodeB	

- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions

The following dependency relationship has been added.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Network slicing enhancement	ENHANCED_NETWORKS_LICE_SW option of the NRCellAlgoSwitch . <i>NetworkSliceAlgoSwitch</i> parameter	<i>Network Slicing</i>	The dynamic RB control switch (that is, the RB_DYNAMIC_CONTROL_SW option of the NRDUCellAlgoSwitch . <i>NetworkSliceAlgoSwitch</i> parameter) can be turned on only after the network slicing enhancement switch is turned on.

- Mutually exclusive features/functions

The following mutually exclusive relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Operator-specific RB management	RB_DYNAMIC_SHARING_SW option of the NRDUCellAlgoSwitch . <i>RanSharingAlgoSwitch</i> parameter	<i>Multi-Operator Sharing</i>	"Operator-specific RB management" and "NR DU cell resource management between network slices" cannot be both enabled.
High-frequency TDD	User PRB control for FWA	PRB_CTRL_SW option of the NRDUCellAlgoSwitch . <i>FwaAlgoSwitch</i> parameter	<i>FWA</i>	"User PRB control for FWA" and "NR DU cell resource management between network slices" cannot be both enabled.
High-frequency TDD	Dynamic RB control for UEs requiring low latency and high reliability	RB_DYNAMIC_CONTROL_SW option of the NRDUCellAlgoSwitch . <i>HighReliabilitySwitch</i> parameter	<i>URLLC</i>	"Dynamic RB control for UEs requiring low latency and high reliability" and "NR DU cell resource management between network slices" cannot be both enabled.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	UL Grant-free	GRANT_FREE_SW option of the NRDUCellAlgoSwitch.LowLatencySwitch parameter	None	UL Grant-free and "NR DU cell resource management between network slices" cannot be both enabled.
High-frequency TDD	Low Latency High Reliability Level 2	LOW_LAT_HIGH_RELIABILITY_LEVEL2_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	None	Low Latency High Reliability Level 2 and NR DU cell resource management between network slices cannot be both enabled.
High-frequency TDD	mmWave level-1 low jitter and high reliability	L_JIT_H_RELIAB_MMW_LVL1_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	None	"mmWave level-1 low jitter and high reliability" and "NR DU cell resource management between network slices" cannot be both enabled.
High-frequency TDD	Low Latency High Reliab mmWave Adv	LOW_LAT_HIGH_RELIAB_MMW_ADV_SW option of the NRDUCellFeatureSw.HighReliabilityExtSwitch parameter	None	"Low Latency High Reliab mmWave Adv" and "NR DU cell resource management between network slices" cannot be both enabled.
High-frequency TDD	Submeter-level precise positioning	AOA_UTDOA_HYBRID_LOCATION_SW option of the NRCCellAlgoSwitch.LcsSwitch parameter	None	"Submeter-level precise positioning" and "NR DU cell resource management between network slices" cannot be both enabled.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	mmWave Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	None	mmWave Cell Combination and "NR DU cell resource management between network slices" cannot be both enabled.
High-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	None	"NR DU cell resource management between network slices" cannot be enabled in hyper cells.
High-frequency TDD	Cell coordination	INTRA_GNB_DL_JT_SW option of the NRDUCellaLGOSwitch.CompSwitch parameter	None	Cell coordination and "NR DU cell resource management between network slices" cannot be both enabled.

The following mutually exclusive relationship has been deleted.

RAT	Function Name	Function Switch	Reference
Low-frequency TDD	Inter-FR CA	INTER_FR_FR1TOFR2_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>

- Impacted features/functions

The following impact relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Frequency-based 5GC selection	gNBCUNgTrackingArea.gNBCuNgld and gNBCUNgTrackingArea.TrackingAreaId	<i>Multi-Operator Sharing</i>	When both Network Slicing Introduction and frequency-based 5GC selection are enabled, there must be an NG interface for which gNBCUNg.CuNgType is set to DEFAULT_CP or DEFAULT_CP_UP among an operator's NG interfaces corresponding to the same gNBCUNgTrackingArea.TrackingAreaId value. Otherwise, the gNodeB cannot select a default AMF for a network slice UE if the UE does not report either the S-NSSAIs supported by the gNodeB or the Temp ID, resulting in access failures. In live network planning, if a gNBCUNg MO of the default type (gNBCUNg.CuNgType set to DEFAULT_CP or DEFAULT_CP_UP) is configured, there must be an available NG link under the gNBCUNg MO of the default type. Otherwise, services will be affected.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	mmWave PUSCH MU-MIMO	UL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	When "NR DU cell resource management between network slices" and "mmWave PUSCH MU-MIMO" are both enabled, UEs in a network slice group cannot participate in pairing.
High-frequency TDD	Downlink intra-band CA	INTRA_BAND_CA_SW and INTRA_BAND_UL_CA_SW options of the NRDUCellAlgoSwitch.CarrierAggregation parameter	<i>Carrier Aggregation</i>	The network slices configured for an SCell must include UEs' network slices.
Low-frequency TDD High-frequency TDD	Inter-FR CA	INTER_FR_F1TOFR2_CA_SW option of the NRDUCellAlgoSwitch.CarrierAggregation parameter	<i>Carrier Aggregation</i>	The network slices configured for an SCell must include UEs' network slices.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Network Slicing*.

Requirement/Trouble Ticket Number

FeR-IR20250116001301-0001

RAT

NR TDD

5.3.1.4 FOFD-100200 iBeam 3.0

5.3.1.4.1 iBeam 3.0

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

iBeam 3.0 has been introduced to further increase the downlink user-perceived rate in medium- and heavy-load enhanced Mobile Broadband (eMBB) cells that offer contiguous coverage. This feature applies to TDD massive MIMO cells. It improves downlink performance and increases the downlink user-perceived rate through weight enhancement, scheduling enhancement, and interference suppression enhancement.

To enable iBeam 3.0 and its various functions for these enhancements, the **NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw** parameter must be set to **ON**.

Table 5-9 describes the functions of iBeam 3.0.

Table 5-9 iBeam 3.0 functions

Category	Function Name	Description	Parameter
Weight enhancement	PDSCH self-fusion weights	PDSCH self-fusion weights are generated by combining statistical SRS-based weights with instantaneous channel characteristics. These self-fusion weights are more accurate and thereby increase the downlink user-perceived rate.	DL_SELF_FUSION_WEIGHT_SW option of the NRDUCellPdschPre-code.PrecodingAlgoSwitch parameter
	PDCCH robust weights	By using the more accurate robust weights instead of DFT weights, the PDCCH can have its beams targeted at UEs more accurately, increasing the downlink user-perceived rate.	PDCCH_ROBUST_WEIGHT_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter

Category	Function Name	Description	Parameter
Scheduling enhancement	SINR optimization for MCS selection	SINR optimization for MCS selection adjusts the SINRs used to select MCSs for MU-MIMO UEs, thereby enhancing UEs' spectral efficiency and improving the downlink user-perceived rate.	DL_SMART_AMC_SW option of the NRDUCellDlMimo.DLMuMimoSchSupplementSw parameter
	Downlink rank robustness	Downlink rank robustness enables the base station to determine whether to use instantaneous UE-reported ranks based on filtered statistics. This function improves the accuracy of MCS selection and the downlink user-perceived rate.	ROBUST_RANK_SW option of the NRDUCellDlAmc.DlMcsSelAlgoSw parameter
	Rank robustness optimization in mobility scenarios	Rank robustness optimization in mobility scenarios enables the base station to refilter the rank reported by a UE after the CSI-RS beams of the UE change, improving the rank accuracy.	ROBUST_RANK_MOBILITY_OPT_SW option of the NRDUCellDlAmc.DlMcsSelAlgoSw parameter
	Downlink small packet shaping	Downlink small packet shaping adaptively selects UEs for scheduling based on the RB resources occupied by them, improving the user-perceived rate.	DL_SMALL_PACKET_SHAPE_SCH_SW option of the NRDUCellDlSch.DlSchAlgoSwitch parameter
	Power optimization for downlink retransmissions	Power optimization for downlink retransmissions reduces the power used for retransmissions and reallocates the freed-up power to initial transmissions, further improving spectral efficiency.	DL_RETRANS_PWR_OPT_SW option of the NRDUCellChnPwr.ChnPwrAlgoSwitch parameter

Category	Function Name	Description	Parameter
	Downlink retransmission scheduling priority adjustment	Downlink retransmission scheduling priority adjustment adjusts the scheduling priorities of UEs involved in retransmissions based on user experience, thereby improving the downlink user-perceived rate.	DL_RETRANS_PRI_ADJ_SW option of the NRDUCellDISch.DISchAlgoSwitch parameter
	Downlink slot awareness scheduling optimization	Downlink slot awareness scheduling optimization schedules UEs having large buffer sizes into slots with more symbols, thereby improving the downlink user-perceived rate.	DL_SLOT_AWARE_SCH_OPT_SW option of the NRDUCellDISch.DISchAlgoSwitch parameter
	Intelligent MCS optimization for downlink MU-MIMO	Intelligent MCS optimization for downlink MU-MIMO accurately calculates the MCSs for MU-MIMO UEs, thereby enhancing UEs' spectral efficiency and improving the downlink user-perceived rate.	DL_MU_MCS_INTEL_OPT_SW option of the NRDUCellAiAlgo.AiAmcAlgoSwitch parameter
Interference suppression enhancement	Blind SRS interference suppression enhancement	In contiguous networking, SRS interference changes rapidly and constantly. Blind SRS interference suppression enhancement uses the differences between interference signals and wanted signals to reduce or eliminate interference for UEs severely affected by SRS interference. This function improves the accuracy of SRS measurement and thereby increases the average downlink UE throughput.	SRS_ENH_BLIND_IS_SW option of the NRDUCellSrs.SrsDetectionAlgoSwitch parameter

This feature is supported only in low frequency bands.

Implementation

For details, see *Massive MIMO iBeam (Low-Frequency TDD)*. The specific sections are listed in the change history of the document.

Impact on the Network

iBeam 3.0 offers the following benefits:

iBeam 3.0 increases the **User Downlink Average Throughput (DU)**. The actual benefits of iBeam 3.0 depend on factors such as the traffic model, network load, and PCI planning. This feature is recommended for intra-frequency contiguous massive MIMO networks when the following conditions are met:

- The downlink PRB usage (equal to **N.PR.B.DL.Used.Avg / N.PR.B.DL.Avail.Avg**) ranges from 30% to 70%.
- The proportion of SRS PreSINRs less than 0 dB is greater than 50%. This proportion equals $(N.UL.SRS.PreSINR.Index0 + N.UL.SRS.PreSINR.Index1 + N.UL.SRS.PreSINR.Index2 + N.UL.SRS.PreSINR.Index3) / (N.UL.SRS.PreSINR.Index0 + N.UL.SRS.PreSINR.Index1 + N.UL.SRS.PreSINR.Index2 + N.UL.SRS.PreSINR.Index3 + N.UL.SRS.PreSINR.Index4 + N.UL.SRS.PreSINR.Index5)$.
- The downlink CCE allocation failure rate is greater than 5%. This rate equals $(N.CCE.DL.AllocReq.Num - N.CCE.DL.AggLvl1Num - N.CCE.DL.AggLvl2Num - N.CCE.DL.AggLvl4Num - N.CCE.DL.AggLvl8Num - N.CCE.DL.AggLvl16Num) / N.CCE.DL.AllocReq.Num$.

For other networks or networks that serve UEs moving at high speed (for example, on highways), some functions offer fewer or even no benefits. In addition, iBeam 3.0 offer fewer benefits if any of the functions for weight enhancement, scheduling enhancement, and interference suppression enhancement is disabled.

PDCCH robust weights offer fewer or no benefits if the average uplink CCE aggregation level is less than 3. The average uplink CCE aggregation level equals $(1 \times N.CCE.UL.AggLvl1Num + 2 \times N.CCE.UL.AggLvl2Num + 4 \times N.CCE.UL.AggLvl4Num + 8 \times N.CCE.UL.AggLvl8Num + 16 \times N.CCE.UL.AggLvl16Num) / (N.CCE.UL.AggLvl1Num + N.CCE.UL.AggLvl2Num + N.CCE.UL.AggLvl4Num + N.CCE.UL.AggLvl8Num + N.CCE.UL.AggLvl16Num)$.

When the PDSCH self-fusion weights function is enabled, it is recommended that downlink rank adaptation for cell edge UEs performing small-packet services also be enabled for greater benefits. (Downlink rank adaptation for cell edge UEs performing small-packet services is controlled by the **DL_SMALL_PKT_CEU_RANK_ADAPT_SW** option of the **NRDUCellDLRank.DlRankSelAlgoSw** parameter.)

When downlink rank robustness is enabled, it is recommended that the following functions also be enabled for greater benefits: CSI reporting period adaptation (controlled by the **CSI_REPORT_PERIOD_ADAPT_SWITCH** option of the

NRDUCellPucch.CsiResoureAlgoSwitch parameter) and PUCCH RB adaptation (controlled by the **PUCCH_RBRES_ADAPTIVE_SWITCH** option of the **NRDUCellPucch.PucchAlgoSwitch** parameter).

When downlink retransmission scheduling priority adjustment is enabled, it is recommended that resource-based adaptive scheduling for massive MIMO also be enabled for greater benefits. (Resource-based adaptive scheduling for massive MIMO is controlled by the **RES_BASED_MM_ADAPT_SCH_SW** option of the **NRDUCellServExp.ServiceExpAlgoSwitch** parameter.)

Blind SRS interference suppression enhancement may offer fewer benefits in contiguous networking where UEs exhibit concentrated distribution within the cells.

iBeam 3.0 may have the following impacts on the network:

- The downlink MCS index may increase.
- The MU-MIMO pairing rate may increase. For details about the impacts of the increase on related KPIs, see the "Impacts" section of MU-MIMO (low-frequency TDD) in *MIMO (TDD)*. (MU-MIMO pairing rate = $\text{N.ChMeas.MIMO.DL.Pair.PRB} / (\text{N.PR.BD.Used.Avg} \times \text{N.DL.PDSCH.Tti.Num})$)
- The average number of layers paired for downlink MIMO (indicated by the **N.ChMeas.MIMO.DL.Pair.Layer.Avg** counter) may change.
- The downlink PRB usage may decrease. (Downlink PRB usage = $\text{N.PR.BD.Used.Avg} / \text{N.PR.BD.Avail.Avg}$)
- The values of the following scheduling-related indicators may change:
 - Downlink average rank, which equals $(1 \times \text{N.PDSCH.InitTbDL.Rank1} + 2 \times \text{N.PDSCH.InitTbDL.Rank2} + 3 \times \text{N.PDSCH.InitTbDL.Rank3} + 4 \times \text{N.PDSCH.InitTbDL.Rank4}) / (\text{N.PDSCH.InitTbDL.Rank1} + \text{N.PDSCH.InitTbDL.Rank2} + \text{N.PDSCH.InitTbDL.Rank3} + \text{N.PDSCH.InitTbDL.Rank4})$
 - Downlink IBLER, which equals $(\text{N.DL.SCH.QPSK.ErrTB.Ibler} + \text{N.DL.SCH.16QAM.ErrTB.Ibler} + \text{N.DL.SCH.64QAM.ErrTB.Ibler} + \text{N.DL.SCH.256QAM.ErrTB.Ibler} + \text{N.DL.SCH.1024QAM.ErrTB.Ibler}) / (\text{N.DL.SCH.QPSK.TB} + \text{N.DL.SCH.16QAM.TB} + \text{N.DL.SCH.64QAM.TB} + \text{N.DL.SCH.256QAM.TB} + \text{N.DL.SCH.1024QAM.TB})$
 - Downlink RBLER, which equals $(\text{N.DL.SCH.QPSK.ErrTB.Rbler} + \text{N.DL.SCH.16QAM.ErrTB.Rbler} + \text{N.DL.SCH.64QAM.ErrTB.Rbler} + \text{N.DL.SCH.256QAM.ErrTB.Rbler} + \text{N.DL.SCH.1024QAM.ErrTB.Rbler}) / (\text{N.DL.SCH.QPSK.TB} + \text{N.DL.SCH.16QAM.TB} + \text{N.DL.SCH.64QAM.TB} + \text{N.DL.SCH.256QAM.TB} + \text{N.DL.SCH.1024QAM.TB})$
- The equivalent PDCCH CCE usage of a cell may change. (Equivalent PDCCH CCE usage of a cell = $(\text{N.CCE.ULDCI.Used.Equivalent} + \text{N.CCE.DLDCI.Used.Equivalent}) / \text{N.CCE.Avail.Equivalent}$)
- The average CCE aggregation level and CCE allocation failure rate may decrease for both the uplink and downlink.
 - Average uplink CCE aggregation level = $(1 \times \text{N.CCE.UL.AggLvl1Num} + 2 \times \text{N.CCE.UL.AggLvl2Num} + 4 \times \text{N.CCE.UL.AggLvl4Num} + 8 \times \text{N.CCE.UL.AggLvl8Num} + 16 \times \text{N.CCE.UL.AggLvl16Num}) / (\text{N.CCE.UL.AggLvl1Num} + \text{N.CCE.UL.AggLvl2Num} + \text{N.CCE.UL.AggLvl4Num} + \text{N.CCE.UL.AggLvl8Num} + \text{N.CCE.UL.AggLvl16Num})$

- Average downlink CCE aggregation level = $(1 \times \text{N.CCE.DL.AggLvl1Num} + 2 \times \text{N.CCE.DL.AggLvl2Num} + 4 \times \text{N.CCE.DL.AggLvl4Num} + 8 \times \text{N.CCE.DL.AggLvl8Num} + 16 \times \text{N.CCE.DL.AggLvl16Num}) / (\text{N.CCE.DL.AggLvl1Num} + \text{N.CCE.DL.AggLvl2Num} + \text{N.CCE.DL.AggLvl4Num} + \text{N.CCE.DL.AggLvl8Num} + \text{N.CCE.DL.AggLvl16Num})$
- Uplink CCE allocation failure rate = $(\text{N.CCE.UL.AllocReq.Num} - \text{N.CCE.UL.AggLvl1Num} - \text{N.CCE.UL.AggLvl2Num} - \text{N.CCE.UL.AggLvl4Num} - \text{N.CCE.UL.AggLvl8Num} - \text{N.CCE.UL.AggLvl16Num}) / \text{N.CCE.UL.AllocReq.Num}$
- Downlink CCE allocation failure rate = $(\text{N.CCE.DL.AllocReq.Num} - \text{N.CCE.DL.AggLvl1Num} - \text{N.CCE.DL.AggLvl2Num} - \text{N.CCE.DL.AggLvl4Num} - \text{N.CCE.DL.AggLvl8Num} - \text{N.CCE.DL.AggLvl16Num}) / \text{N.CCE.DL.AllocReq.Num}$
- The first packet delay (indicated by the **N.Traffic.DL.RlcFirstPktDelay.Time** counter) may change.
- The duration in which BWP2 is in effect (indicated by the **N.Bwp2.Dur.Total** counter) may increase.
- In light- and medium-load cells where tail packets account for an excessively large proportion of data volume, the **Cell Downlink Average Throughput (DU)** may increase but the **User Downlink Average Throughput (DU)** may decrease. The data volume proportion of tail packets equals $\text{N.ThpVol.DL.LastSlot} / \text{N.ThpVol.DL}$.

Impact on NEs

This feature is implemented on the gNodeB.

Impact on Hardware Requirements

- Boards
 - Main control boards: All NR TDD-capable low-frequency main control boards can be used.
 - Baseband processing units: All NR TDD-capable low-frequency UBBPg and later series baseband processing units can be used.
- RF modules

All NR TDD-capable low-frequency RF modules with at least 32T32R can be used.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

This feature is controlled by the following new license control items.

Feature ID	Feature Name	Model	Sales Unit
FOFD-100200	iBeam 3.0	NR0S00BEAM 30	Per Cell

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw	MOD NRDUCELLFEATURESW LST NRDUCELLFEATURESW	gNodeB	This parameter has been added to control iBeam 3.0 functions.
Modified	NRDUCellPdschPrecoding.PrecodingAlgoSwitch	MOD NRDUCELLPDSCHPRECODE LST NRDUCELLPDSCHPRECODE	gNodeB	The DL_SELF_FUSION_WEIGHT_SW option has been added to specify whether to enable PDSCH self-fusion weights.
New	NRDUCellPdschPrecoding.SelfFusionWtSrsPreSnrThld	MOD NRDUCELLPDSCHPRECODE LST NRDUCELLPDSCHPRECODE	gNodeB	This parameter specifies the SRS PreSINR threshold for PDSCH self-fusion weights.
New	NRDUCellPdschPrecoding.SelfFusionWtRhoThld	MOD NRDUCELLPDSCHPRECODE LST NRDUCELLPDSCHPRECODE	gNodeB	This parameter specifies the channel correlation threshold for PDSCH self-fusion weights.
Modified	NRDUCellPdcch.PdcchAlgoSwitch	MOD NRDUCELLPDCCH LST NRDUCELLPDCCH	gNodeB	The PDCCH_ROBUST_WEIGHT_SW option has been added to specify whether to enable PDCCH robust weights.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellPdcchAlgo.PdcchRobWtSrsPresinrThld	MOD NRDUCELLP DCCHALGO LST NRDUCELLP DCCHALGO	gNodeB	This parameter specifies the SRS PreSINR threshold for PDCCH robust weights.
New	NRDUCellPdcchAlgo.PdcchRobWtBFGainCoeff	MOD NRDUCELLP DCCHALGO LST NRDUCELLP DCCHALGO	gNodeB	This parameter specifies the beamforming gain coefficient for PDCCH robust weights.
Modified	NRDUCellDLMimo.DLMuMimoSchSupplementSw	MOD NRDUCELLD LMIMO LST NRDUCELLD LMIMO	gNodeB	The DL_SMART_AMC_SW option has been added to specify whether to enable SINR optimization for MCS selection.
Modified	NRDUCellDLAmc.DLMcsSelAlgoSw	MOD NRDUCELLD LAMC LST NRDUCELLD LAMC	gNodeB	The ROBUST_RANK_SW option has been added to specify whether to enable downlink rank robustness, and the ROBUST_RANK_MOBILITY_OPT_SW option to specify whether to enable "rank robustness optimization in mobility scenarios".
Modified	NRDUCellDISch.DISchAlgoSwitch	MOD NRDUCELLD LSCH LST NRDUCELLD LSCH	gNodeB	The DL_SMALL_PACKET_SHAPE_SCH_SW option has been added to specify whether to enable downlink small packet shaping.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellChnPwrAlgoSwitch	MOD NRDUCELLCHNPWR LST NRDUCELLCHNPWR	gNodeB	The DL_RETRANS_PWR_OPT_SW option has been added to specify whether to enable power optimization for downlink retransmissions.
Modified	NRDUCellsSrsDetectionAlgoSwitch	MOD NRDUCELLSRS LST NRDUCELLSRS	gNodeB	The SRS_ENH_BLIND_IS_SW option has been added to specify whether to enable blind SRS interference suppression enhancement.
Modified	NRDUCellDISch.DLSchAlgoSwitch	MOD NRDUCELLDLSCH LST NRDUCELLDLSCH	gNodeB	The DL_RETRANS_PRI_ADJ_SW option has been added to specify whether to enable downlink retransmission scheduling priority adjustment.
Modified	NRDUCellDISch.DLSchAlgoSwitch	MOD NRDUCELLDLSCH LST NRDUCELLDLSCH	gNodeB	The DL_SLOT_AWARE_SCH_OPT_SW option has been added to specify whether to enable downlink slot awareness scheduling optimization.
Modified	NRDUCellAiAlgo.AiAlgoSwitch	MOD NRDUCELLAIALGO LST NRDUCELLAIALGO	gNodeB	The DL_MU_MCS_INTEL_OPT_SW option has been added to specify whether to enable intelligent MCS optimization for downlink MU-MIMO.

- Performance management
No impact
- Fault management

No impact

Impact on Features/Functions

- Prerequisite features/functions
 - PDSCH self-fusion weights

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	iBeam 3.0	NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	PDSCH self-fusion weights can take effect only when the NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw parameter is set to ON .
Low-frequency TDD	Low-speed cell	NRDUCell.HighSpeedFlag set to LOW_SPEED	None	PDSCH self-fusion weights can be enabled only for low-speed cells.

- PDCCH robust weights

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	iBeam 3.0	NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	PDCCH robust weights can take effect only when the NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw parameter is set to ON .
Low-frequency TDD	PDSCH self-fusion weights	DL_SELF_FUSION_WEIGHT_SW option of the NRDUCellPdschPrecode.PrecodingAlgorithmSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	PDCCH robust weights can be enabled only when the PDSCH self-fusion weights function is enabled.

– SINR optimization for MCS selection

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	iBeam 3.0	NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	SINR optimization for MCS selection can take effect only when the NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw parameter is set to ON .
Low-frequency TDD	Downlink MU-MIMO	DL_MU_MIMO_SW option of the NRDUCellAlgorithmSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	SINR optimization for MCS selection can take effect only when downlink MU-MIMO is enabled.

– Downlink rank robustness

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	iBeam 3.0	NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Downlink rank robustness can take effect only when the NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw parameter is set to ON .

– Rank robustness optimization in mobility scenarios

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink rank robustness	ROBUST_RAN_K_SW option of the NRDUCellIDIA mc.DlMcsSelA lgoSw parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	"Rank robustness optimization in mobility scenarios" (controlled by the ROBUST_RAN_K_MOBILITY_OPT_SW option of the NRDUCellIDIA mc.DlMcsSelA lgoSw parameter) can be enabled only when downlink rank robustness is enabled.

– Downlink small packet shaping

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	iBeam 3.0	NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Downlink small packet shaping can take effect only when the NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw parameter is set to ON .

– Power optimization for downlink retransmissions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	iBeam 3.0	NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Power optimization for downlink retransmissions can take effect only when the NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw parameter is set to ON .

- Downlink retransmission scheduling priority adjustment

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	iBeam 3.0	NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Downlink retransmission scheduling priority adjustment can take effect only when the NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw parameter is set to ON .

- Downlink slot awareness scheduling optimization

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	iBeam 3.0	NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Downlink slot awareness scheduling optimization can take effect only when the NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw parameter is set to ON .

– Intelligent MCS optimization for downlink MU-MIMO

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	iBeam 3.0	NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Intelligent MCS optimization for downlink MU-MIMO can take effect only when the NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw parameter is set to ON .
Low-frequency TDD	Downlink MU-MIMO	DL_MU_MIMO_SW option of the NRDUCellAlgorithmSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	Intelligent MCS optimization for downlink MU-MIMO can take effect only when downlink MU-MIMO is enabled.

– Blind SRS interference suppression enhancement

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	iBeam 3.0	NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Blind SRS interference suppression enhancement takes effect only when the NRDUCellFeatureSw.HighPrecisionBeamPhase3Sw parameter is set to ON .
Low-frequency TDD	UE-characteristic-based SRS period adaptation	USER_CHARACTER_SRS_ADAPT_SW option of the NRDUCellSrs.SrsAlgorithm parameter	None	Blind SRS interference suppression enhancement can be enabled only when UE-characteristic-based SRS period adaptation is enabled.

- Mutually exclusive features/functions
 - PDCCH robust weights

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Tridimensional coverage mode	NRDUCell.TridCommMode set to TRID_COVERAGE_MODE	None	When the NRDUCell.TridCommMode parameter is set to TRID_COVERAGE_MODE , PDCCH robust weights cannot be enabled.

- Blind SRS interference suppression enhancement

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.Hig hSpeedFlag set to HIGH_SPEED	<i>High Speed Mobility</i>	When blind SRS interference suppression enhancement is enabled, the NRDUCell.Hig hSpeedFlag parameter cannot be set to HIGH_SPEED .

- Impacted features/functions
 - PDSCH self-fusion weights

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Adaptive filtering for SRS channel estimation	CHN_EST_AD APT_FLTR_SW option of the NRDUCellSrs.SrsDetection AlgoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	The PDSCH self-fusion weights function will yield fewer benefits when it is enabled together with adaptive filtering for SRS channel estimation.
Low-frequency TDD	Enhanced Co-MM	INTRA_GNB_S U_COMM_EN H_SW option of the NRDUCellColl abServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	The PDSCH self-fusion weights function will yield fewer benefits when it is enabled together with enhanced Co-MM.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Interference-based subband robust weights	INTRF_BASED_ROBUST_WEIGHT_SW option of the NRDUCellIntrIdent.ChnMeasCtrlSw parameter	<i>Interference Avoidance</i>	The PDSCH self-fusion weights function will yield fewer benefits when it is enabled together with interference-based subband robust weights.

– PDCCH robust weights

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast intra-gNodeB CA	CA_ULTRA_LOW_LATENCY_SPLIT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter	<i>CA Experience Improvement</i>	The PDCCH robust weights function will yield fewer benefits when it is enabled together with fast intra-gNodeB CA.
Low-frequency TDD	Fast inter-gNodeB CA	INTER_GNB_ULTRA_LOW_LATENCY_SPLIT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter	<i>CA Experience Improvement</i>	The PDCCH robust weights function will yield fewer benefits when it is enabled together with fast inter-gNodeB CA.

– Power optimization for downlink retransmissions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Precise RB control for downlink retransmissions	DL_RETRANS_PRECISE_RB_CTRL_SW option of the NRDUCellDisch.DISchAlgoSwitch parameter	<i>Massive MIMO AHR (Low-Frequency TDD)</i>	Precise RB control for downlink retransmissions preferentially takes effect when it is enabled together with power optimization for downlink retransmissions. Power optimization for downlink retransmissions can take effect only when precise RB control for downlink retransmissions is not in effect.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Massive MIMO iBeam (Low-Frequency TDD)*.

Requirement/Trouble Ticket Number

- FeR-IR20240629000373-0004
- FeR-IR20240629000373-0002
- FeR-IR20240629000373-0005
- FeR-IR20240629000373-0001
- FeR-IR20241211002485-0003
- FeR-IR20241211002485-0001
- FeR-IR20241211002485-0004
- FeR-IR20241211002485-0005
- FeR-IR20241211002485-0006

FeR-IR20250528001857-0045

RAT

NR TDD

5.3.1.5 FOFD-100201 Uplink Boosting 2.0

5.3.1.5.1 Added Uplink Boosting 2.0

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

As the number of 5G UEs increases rapidly, the uplink load in NR cells rises, leading to a decline in uplink user experience in medium- and heavy-load scenarios. Uplink Boosting 2.0 is designed to address this issue. This feature enhances the uplink performance of TDD massive MIMO cells through multi-cell power control for interference reduction, time-frequency scheduling for interference avoidance, and multi-domain demodulation for interference resistance, thereby increasing the average uplink UE throughput in medium- and heavy-load scenarios. This feature is controlled by the **UL_LOW_NOISE_PHASE2_SW** option of the **NRDUCellFeatureSw.MimoFeatureSwitch** parameter. This feature includes the following functions:

Functions of multi-cell power control for interference reduction:

- PUSCH coordinated power control: This function selects UEs that require transmit power adjustment in the serving cell and uses transmit power control (TPC) commands to adjust their PUSCH transmit power, thereby reducing interference between cells providing contiguous coverage.
 - This function is triggered only when both of the following conditions are met:
 - Interference over thermal (IoT, which indicates the increase in the uplink interference and noise) in the cell $\geq$ **NRDUCellUIPcConfig.PuschJointPcCellIotThld**
 - PUSCH PRB usage of the cell $\geq$ **NRDUCellUIPcConfig.PuschCoordPcPrbThld**
 - A target UE must meet all the following conditions:
 - Measured SSB RSRP of the UE in the serving cell $>$ **NRCeIlMeasConfig.UlCpcUeSsbRsrpThld**
 - Measured SRS RSRP of the UE in the serving cell $>$ **NRDUCellUIPcConfig.UlCpcUeSrsRsrpThld**
 - Uplink interference of the UE to neighboring cells $>$ **NRDUCellUIPcConfig.UlCpcUeIntrfRsrpThld**

This function is controlled by the **PUSCH_COORD_PWR_CTRL_SW** option of the **NRDUCellUIPcConfig.UlPwrCtrlAlgoExtSwitch** parameter and takes effect when the **UL_LOW_NOISE_PHASE2_SW** option of the **NRDUCellFeatureSw.MimoFeatureSwitch** parameter is selected.

LampSite base stations do not support PUSCH coordinated power control.

- Link-performance-based power control: This function enables the gNodeB to consider UEs' link performance and power headroom during power adjustment. Specifically, the gNodeB reduces the PUSCH transmit power of UEs based on their uplink RSRP to mitigate inter-UE interference.

This function is controlled by the **INTRF_SC_LINK_PERF_PC_SW** option of the **NRDUCellUIPcConfig.UlPwrCtrlAlgoExtSwitch** parameter and takes effect when the **UL_LOW_NOISE_PHASE2_SW** option of the **NRDUCellFeatureSw.MimoFeatureSwitch** parameter is selected.

LampSite base stations do not support link-performance-based power control.

Functions of time-frequency scheduling for interference avoidance:

- Precise RB control for uplink retransmissions: This function reduces the number of RBs for the first retransmission of UEs and reallocates the freed-up RBs to UEs performing initial transmissions on the PUSCH. This increases the average uplink UE throughput in medium- and heavy-load scenarios.

This function is controlled by the **UL_RETRANS_PREC_RB_CTRL_SW** option of the **NRDUCellPusch.PuschAlgoExtSwitch** parameter and takes effect when the **UL_LOW_NOISE_PHASE2_SW** option of the **NRDUCellFeatureSw.MimoFeatureSwitch** parameter is selected. The minimum ratio of RBs for first uplink retransmissions is specified by the **NRDUCellUISch.UlFirstRetransMinRbPct** parameter. This parameter must be set to a value other than **100**. Network performance is optimal when this parameter is set to **20**.

- Uplink precise MCS optimization: This function enables the gNodeB to use a predictive model to determine an intelligent-optimization-based uplink MCS for a UE, accounting for the UE's channel and scheduling features. If the MCS prediction error is less than or equal to the predefined upper limit for uplink precise MCS optimization, this predicted MCS is used for uplink scheduling to increase the average uplink UE throughput.

This function is controlled by the **UL_PRECISE_MCS_OPT_SW** option of the **NRDUCellAiAlgo.AiAmcAlgoSwitch** parameter and takes effect when the **UL_LOW_NOISE_PHASE2_SW** option of the **NRDUCellFeatureSw.MimoFeatureSwitch** parameter is selected. The **NRDUCellAiAlgo.UlMcsPredictErrUpLimit** parameter specifies the upper limit to the MCS prediction error for uplink precise MCS optimization. You can run the **DSP NRDUCELLAISCHMODEL** command to query model usage.

LampSite base stations do not support uplink precise MCS optimization.

- Adaptive awareness scheduling: This function enables the gNodeB to flexibly adjust the uplink scheduling priorities of UEs based on the RBs that they occupy and their scheduling delays. The purpose is to achieve consecutive scheduling for UEs whenever possible and minimize the data transmission delay, thereby increasing the average uplink UE throughput.

This function is controlled by the **ADAPT_AWARE_SCH_SW** option of the **NRDUCellUISch.UlSchAlgoSwitch** parameter and takes effect when the **UL_LOW_NOISE_PHASE2_SW** option of the

NRDUCellFeatureSw.MimoFeatureSwitch parameter is selected. The **NRDUCellUISch.AdaptAwareSchTimeThld** parameter specifies the duration threshold for adaptive awareness scheduling to take effect, and the **NRDUCellUISch.AdaptAwareSchPriFactor** parameter specifies the uplink data scheduling priority factor used in this function.

- Uplink-experience-based scheduling optimization: This function enables the gNodeB to flexibly adjust the uplink scheduling priorities of UEs based on UE types. Specifically:
 - If a small-packet UE's retransmission scheduling duration is less than the threshold for delayed scheduling of uplink retransmission, the gNodeB adaptively adjusts the UE's retransmission scheduling priority to delay its scheduling.
 - If a UE engaged in uplink preallocation has an average uplink MCS index less than the MCS threshold for preallocation period adjustment, the gNodeB adaptively adjusts the UE's preallocation period to speed up its scheduling.
 - If a UE with low channel quality experiences consecutive bit errors, the gNodeB updates the UE's data information in a timely manner to save uplink scheduling resources.

This function is controlled by the **UL_EXP_SCH_OPT_SW** option of the **NRDUCellUISch.UISchAlgoSwitch** parameter and takes effect when the **UL_LOW_NOISE_PHASE2_SW** option of the **NRDUCellFeatureSw.MimoFeatureSwitch** parameter is selected. The **NRDUCellUISch.UlRetransSchDelayThld** parameter specifies the threshold for delayed scheduling of uplink retransmission, and the **NRDUCellUISch.PreSchPeriodAdjMcsThld** parameter specifies the MCS index threshold for preallocation period adjustment.

- Probability-based uplink MU-MIMO MCS selection: During PUSCH MCS selection in MU-MIMO, this function estimates the extent of fluctuations in the channel transmission quality of UEs. It does so by analyzing the probability distribution of SINR deviations between the measurement moment and the scheduling moment. The estimation is then used to correct uplink MCS selection for UEs and thereby increase the average uplink UE throughput.

This function is controlled by the **UL_MU_PBMCSSEL_SW** option of the **NRDUCellUIAmc.UlAmcPerformanceSw** parameter and takes effect when the **UL_LOW_NOISE_PHASE2_SW** option of the **NRDUCellFeatureSw.MimoFeatureSwitch** parameter is selected.

LampSite base stations do not support probability-based uplink MU-MIMO MCS selection.

Functions of multi-domain demodulation for interference resistance:

- Multi-beam reception enhancement: In a massive MIMO cell with multiple receive channels, this function enables the gNodeB to adaptively adjust the time-domain filtering coefficient for the power delay profile (PDP) of PUSCH DMRS. This adjustment is based on the signal-to-interference ratio on receive beams, improving the accuracy of channel estimation and the uplink user-perceived rate.

This function is controlled by the **MULTI_BEAM_RX_ENH_SW** option of the **NRDUCellPusch.PuschAlgoExtSwitch** and takes effect when the **UL_LOW_NOISE_PHASE2_SW** option of the

NRDUCellFeatureSw.MimoFeatureSwitch parameter is selected. This function does not take effect for MU-MIMO UEs or UEs requiring connectivity with low latency and high reliability.

LampSite base stations do not support multi-beam reception enhancement.

- Interference-based precise frequency offset estimation: During uplink frequency offset estimation, this function enables the gNodeB to suppress interference in directions with strong interference and enhance wanted signals in directions with weak interference. This reduces the impact of interference on target beams, improving the accuracy of frequency offset estimation and uplink demodulation performance for UEs.

This function is controlled by the **INTRF_PREC_FREQ_OFS_EST_SW** option of the **NRDUCellPusch.PuschMeasSw** and takes effect when the **UL_LOW_NOISE_PHASE2_SW** option of the **NRDUCellFeatureSw.MimoFeatureSwitch** parameter is selected.

- Real-time-information-based precise channel estimation: This function enables the gNodeB to leverage current information on scheduling and interference to calculate real-time PreSINRs for uplink channel estimation, thereby improving estimation accuracy.

This function is controlled by the **PREC_CHANNEL_EST_SW** option of the **NRDUCellPusch.UlPuschAlgoSwitch** parameter and takes effect when the **UL_LOW_NOISE_PHASE2_SW** option of the **NRDUCellFeatureSw.MimoFeatureSwitch** parameter is selected.

This feature is supported only in low-frequency TDD.

Implementation

For details, see *Massive MIMO Uplink Boosting (Low-Frequency TDD)*. The specific sections are listed in the change history of the document.

Impact on the Network

Uplink Boosting 2.0 enhances uplink cell performance through multi-cell power control for interference reduction, time-frequency scheduling for interference avoidance, and multi-domain demodulation for interference resistance, thereby increasing the average uplink UE throughput.

The following impacts may be caused by all the functions of multi-cell power control for interference reduction, time-frequency scheduling for interference avoidance, and multi-domain demodulation for interference resistance:

- Due to improved uplink cell performance, the overall online duration of UEs is reduced, prolonging the DRX sleep time. This increases UEs' service delay and first packet delay, and consequently decreases the average downlink UE throughput. In addition, UEs performing periodic services will experience more re-synchronizations, potentially lowering their contention-based access success rate.
- In cells with only cell-center UEs or a large proportion of tail packet traffic, improved uplink cell performance may increase the uplink cell throughput and spectral efficiency while reducing the average uplink UE throughput.

Table 5-10 describes the impacts specific to each function on the network.

Table 5-10 Network impacts specific to each function

Function Type	Network Impact
Multi-cell power control for interference reduction	- Due to lower PUSCH transmit power in the serving cell, the following impacts will occur: - The RSRP and SINRs of SRSs and PUSCH in the cell may decrease. As a result, the average MCS index and IBLER in the uplink will change. In addition, more NR UEs will switch to LTE for uplink data transmission, increasing the value of the N.NsaDc.ULChangetoMeNB counter. - The uplink padding rate decreases. As a result, the uplink traffic volume and average uplink throughput at the MAC layer of the cell decrease, and the uplink PRB usage of the cell increases. - The uplink MU pairing rate changes. - The CPU usage may increase.
Time-frequency scheduling for interference avoidance	- As the retransmission priorities of UEs are adjusted in a cell, UEs may experience a few more retransmissions, and slightly higher uplink IBLER and RBLER. In addition, the uplink PRB usage and uplink MU pairing rate in the cell will change. - With UEs' scheduling priorities updated in a cell, the uplink PRB usage of the cell may change, and the average uplink throughput of cell-edge UEs may decrease. - The uplink padding rate in a cell decreases. As a result, the uplink traffic volume and average uplink throughput at the MAC layer of the cell decrease. - The average uplink MCS index may change. As a result, the uplink IBLER and RBLER may increase, and the uplink MU pairing rate may decrease, and consequently the uplink PRB usage decreases. - UE data is scheduled more quickly in a cell, reducing the number of BSRs. As a result, the average number of active UEs in the cell may decrease. - The CPU usage may increase.
Multi-domain demodulation for interference resistance	The average uplink MCS index, uplink IBLER, uplink RBLER, and PUSCH SINR in a cell may change.

Impact on NEs

This feature is implemented on the gNodeB.

Impact on Hardware Requirements

- Base station models
PUSCH coordinated power control, uplink precise MCS optimization, probability-based uplink MU-MIMO MCS selection, link-performance-based power control, and multi-beam reception enhancement:
3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
Other functions:
 - 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
 - DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.
- Boards
All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- RF modules
Real-time-information-based precise channel estimation and interference-based precise frequency offset estimation: All NR TDD-capable 32T32R and 64T64R RF modules that work in low frequency bands can be used. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.
Other functions: All NR TDD-capable low-frequency RF modules with at least 32T32R can be used. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
This feature is controlled by the following new license control items.

Feature ID	Feature Name	Model	Sales Unit
FOFD-100201	Uplink Boosting 2.0	NR0S00UPBT20	per Cell

- Configuration management
The following MML commands have changed.

Change Type	MML Command	NE	Description
Modified	DSP NRDUCELLAISCHMO- DEL	gNodeB	The value UL_PRECISE_MCS_OPT(UL Precise MCS Optimization) has been added to the AiCase parameter used in this command.

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellFeatureSw.MimoFeatureSwitch	MOD NRDUCELLF EATURESW LST NRDUCELLF EATURESW	gNodeB	The UL_LOW_NOISE_PHASE2_SW option has been added to specify whether to enable Uplink Boosting 2.0.
Modified	NRDUCellUIPcConfig.UlPwrCtrlAlgoExtSwitch	MOD NRDUCELLU LPCCONFIG LST NRDUCELLU LPCCONFIG	gNodeB	The PUSCH_COORD_PWR_CTRL_SW option has been added to specify whether to enable PUSCH coordinated power control.
Modified	NRDUCellUIPcConfig.UlPwrCtrlAlgoExtSwitch	MOD NRDUCELLU LPCCONFIG LST NRDUCELLU LPCCONFIG	gNodeB	The INTRF_SC_LINK_PERF_PC_SW option has been added to specify whether to enable link-performance-based power control.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellPusch.PuschAlgoExtSwitch	MOD NRDUCCELLPUSCH LST NRDUCCELLPUSCH	gNodeB	The UL_RETRANS_PREC_RB_CTRL_SW option has been added to specify whether to enable precise RB control for uplink retransmissions.
Modified	NRDUCellAiAlgo.AiAmcAlgoSwitch	MOD NRDUCCELLAIALGO LST NRDUCCELLAIALGO	gNodeB	The UL_PRECISE_MCS_OPT_SW option has been added to specify whether to enable uplink precise MCS optimization.
Modified	NRDUCellULSch.ULSchAlgoSwitch	MOD NRDUCCELLULSCH LST NRDUCCELLULSCH	gNodeB	The ADAPT_AWARE_SCH_SW option has been added to specify whether to enable adaptive awareness scheduling.
Modified	NRDUCellULSch.ULSchAlgoSwitch	MOD NRDUCCELLULSCH LST NRDUCCELLULSCH	gNodeB	The UL_EXP_SCH_OPT_SW option has been added to specify whether to enable uplink-experience-based scheduling optimization.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellUlamc.UlamcPerformanceSw	MOD NRDUCELLU LAMC LST NRDUCELLU LAMC	gNodeB	The UL_MU_PBMCSSEL_SW option has been added to specify whether to enable probability-based uplink MU-MIMO MCS selection.
Modified	NRDUCellPusch.PuschAlgoExtSwitch	MOD NRDUCELLP USCH LST NRDUCELLP USCH	gNodeB	The MULTI_BEAM_RX_ENH_SW option has been added to specify whether to enable multi-beam reception enhancement.
Modified	NRDUCellPusch.PuschMeasSw	MOD NRDUCELLP USCH LST NRDUCELLP USCH	gNodeB	The INTRF_PREC_FREQ_OFFSET_SW option has been added to specify whether to enable interference-based precise frequency offset estimation.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellPusch.U lPuschAlgoSwitc h	MOD NRDUCELLP USCH LST NRDUCELLP USCH	gNodeB	The PREC_CHANN EL_EST_SW option has been added to specify whether to enable real-time-information-based precise channel estimation.
New	NRDUCellUIPcCo nfig.UlCpcUeSrsR srpThld	MOD NRDUCELLU LPCCONFIG LST NRDUCELLU LPCCONFIG	gNodeB	This parameter has been added to specify the SRS RSRP threshold for PUSCH coordinated power control.
New	NRCellMeasConfi g.UlCpcUeSsbRsr pThld	MOD NRCELLMEA SCONFIG LST NRCELLMEA SCONFIG	gNodeB	This parameter has been added to specify the SSB RSRP threshold for PUSCH coordinated power control.
New	NRDUCellUIPcCo nfig.UlCpcUeIntrf RsrpThld	MOD NRDUCELLU LPCCONFIG LST NRDUCELLU LPCCONFIG	gNodeB	This parameter has been added to specify the RSRP threshold for interference from UEs involved in PUSCH coordinated power control to other UEs.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellULPcConfig.LinkPerfPcRsrpThldOffset	MOD NRDUCELLU LPCCONFIG LST NRDUCELLU LPCCONFIG	gNodeB	This parameter has been added to specify the offset to the RSRP threshold for link-performance-based power control.
New	NRDUCellAIAlgo.UIMcsPredictErrorUpLimit	MOD NRDUCELLAI ALGO LST NRDUCELLAI ALGO	gNodeB	This parameter has been added to specify the upper limit to the MCS prediction error for uplink precise MCS optimization.
New	NRDUCellULSch.AdaptAwareSchTimeThld	MOD NRDUCELLU LSCH LST NRDUCELLU LSCH	gNodeB	This parameter has been added to specify the scheduling duration threshold for adaptive awareness scheduling to take effect.
New	NRDUCellULSch.AdaptAwareSchPriorityFactor	MOD NRDUCELLU LSCH LST NRDUCELLU LSCH	gNodeB	This parameter has been added to specify the uplink scheduling priority factor for UEs in adaptive awareness scheduling.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellULSch.UlRetransSchDelayThld	MOD NRDUCELLU LSCH LST NRDUCELLU LSCH	gNodeB	This parameter has been added to specify the duration threshold for delayed scheduling of uplink retransmission UEs.
New	NRDUCellULSch.PreSchPeriodAdjMcsThld	MOD NRDUCELLU LSCH LST NRDUCELLU LSCH	gNodeB	This parameter has been added to specify the MCS index threshold for preallocation period adaptation to take effect for UEs.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added for functions of multi-cell power control for interference reduction:
PUSCH coordinated power control

Function Name	Function Switch	Reference	Description
Intra-frequency neighboring cell measurement	INTRA_FREQ_NCELL_STAT_SW option of the NRDUCellCollabServ.NeighborNrCellMgmtSw parameter	None	Enabling PUSCH coordinated power control requires that intra-frequency neighboring cell measurement be enabled.

Function Name	Function Switch	Reference	Description
SRS interference coordination based on self-contained slots	SRS_INTRF_COORD_S_SLOT_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter	<i>Channel Management</i>	Enabling PUSCH coordinated power control requires that SRS interference coordination based on self-contained slots be enabled.
Closed-loop power control for the PUSCH in dynamic scheduling mode	PUSCH_DYN_SCH_CLOSED_LOOP_SW option of the NRDUCellULPcConfig.ULPwrCtrlAlgoSwitch parameter	<i>Power Control</i>	"Closed-loop power control for the PUSCH in dynamic scheduling mode" must be enabled for PUSCH coordinated power control to take effect.

Link-performance-based power control

Function Name	Function Switch	Reference	Description
Closed-loop power control for the PUSCH in dynamic scheduling mode	PUSCH_DYN_SCH_CLOSED_LOOP_SW option of the NRDUCellULPcConfig.ULPwrCtrlAlgoSwitch parameter	<i>Power Control</i>	"Closed-loop power control for the PUSCH in dynamic scheduling mode" must be enabled for link-performance-based power control to take effect.

The following dependency relationships have been added for functions of time-frequency scheduling for interference avoidance:

Uplink precise MCS optimization

Function Name	Function Switch	Reference	Description
Intelligent uplink multi-layer SINR prediction	UL_SU_SINR_INTEL_PREDICT_SW option of the NRDUCellAiAlgo.AiAmcAlgoSwitch parameter	<i>MIMO (TDD)</i>	Intelligent uplink multi-layer SINR prediction must be enabled for uplink precise MCS optimization to take effect.

Function Name	Function Switch	Reference	Description
Additional DMRS	NRDUCellPusch.UAdditionalDmrsPos set to a value other than NOT_CONFIG	None	Uplink precise MCS optimization can be enabled only when NRDUCellPusch.UAdditionalDmrsPos is set to a value other than NOT_CONFIG .

Uplink-experience-based scheduling optimization

Function Name	Function Switch	Reference	Description
SR-based scheduling optimization	SR_SCH_OPT_SW option of the NRDUCellPusch.PuschPerformanceSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	SR-based scheduling optimization must be enabled for uplink-experience-based scheduling optimization to take effect.
Stopping uplink scheduling for uplink-abnormal UEs	ABN_UE_STOP_ULSCH_SW option of the NRDUCellPusch.UlPuschAlgoSwitch parameter	<i>Uplink Scheduling</i>	"Stopping uplink scheduling for uplink-abnormal UEs" must be enabled for uplink-experience-based scheduling optimization to take effect.
Uplink basic preallocation	UL_BASIC_PREALLOCATION_SWITCH option of the NRDUCellPusch.UlPreallocationSwitch parameter	<i>Uplink Scheduling</i>	Uplink-experience-based scheduling optimization can take effect only when uplink basic preallocation or uplink smart preallocation is enabled.
Uplink smart preallocation	UL_SMART_PREALLOCATION_SWITCH option of the NRDUCellPusch.UlPreallocationSwitch parameter	<i>Uplink Scheduling</i>	

Probability-based uplink MU-MIMO MCS selection

Function Name	Function Switch	Reference	Description
Probability-based uplink MCS selection	UL_PBMCSSEL_SW option of the NRDUCellULAmc.UlAmcPerformanceSw parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	Enabling probability-based uplink MU-MIMO MCS selection requires that probability-based uplink MCS selection be enabled.
PUSCH MU-MIMO	UL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	Probability-based uplink MU-MIMO MCS selection can take effect only when PUSCH MU-MIMO is enabled.

The following dependency relationships have been added for functions of multi-domain demodulation for interference resistance:

Multi-beam reception enhancement

Function Name	Function Switch	Reference	Description
Time-domain adaptive filtering	PUSCH_TIME_DOMAIN_ADAPT_FILTER_SW option of the NRDUCellChnCovalgo.UlCoverageAlgoSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	Time-domain adaptive filtering must be enabled for multi-beam reception enhancement to take effect.

Interference-based precise frequency offset estimation

Function Name	Function Switch	Reference	Description
Uplink DMRS Type	NRDUCellPusch.UlDmrsType	None	Interference-based precise frequency offset estimation requires that the NRDUCellPusch.UlDmrsType parameter be set to TYPE1 .

- Mutually exclusive features/functions

The following mutually exclusive relationships have been added for Uplink Boosting 2.0.

Function Name	Function Switch	Reference	Description
High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag set to HIGH_SPEED	<i>High Speed Mobility</i>	High-speed cells do not support Uplink Boosting 2.0.

The following mutually exclusive relationships have been added for functions of multi-cell power control for interference reduction:

PUSCH coordinated power control:

Function Name	Function Switch	Reference	Description
Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	Hyper cells do not support PUSCH coordinated power control.
Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTrp.DataTransmissionMode set to BASIC_MODE or DPS_MODE	<i>Cell Combination</i>	For an NR TDD combined cell in basic or dynamic point selection (DPS) transmission mode, PUSCH coordinated power control cannot be enabled.
Joint Transmission Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTrp.DataTransmissionMode set to JOINT_MODE	<i>Cell Combination</i>	For an NR TDD cell with Joint Transmission Cell Combination enabled, PUSCH coordinated power control cannot be enabled.
Fusion Cell	NRDUCell.NrDuCellNetworkingMode set to FUSION_CELL	<i>Fusion Cell (Low-Frequency TDD)</i>	PUSCH coordinated power control and Fusion Cell cannot be both enabled.

Function Name	Function Switch	Reference	Description
Positioning-dedicated SRS transmission in uplink slots	SRS_FOR_POSITION_ON_USLOT_SW option of the NRDUCellPosAlgo.LcsMeasureAlgoSw parameter	<i>Positioning</i>	PUSCH coordinated power control and positioning-dedicated SRS transmission in uplink slots cannot be both enabled.

The following mutually exclusive relationships have been added for functions of time-frequency scheduling for interference avoidance:

Uplink precise MCS optimization

Function Name	Function Switch	Reference	Description
Uplink scheduling on a per RB basis	UL_1RB_SCH_SWITCH option of the NRDUCellPusch.UlPuschAlgoSwitch parameter	<i>Uplink Scheduling</i>	Uplink scheduling on a per RB basis and uplink precise MCS optimization cannot be both enabled.

The following mutually exclusive relationships have been added for functions of multi-domain demodulation for interference resistance:

Real-time-information-based precise channel estimation

Function Name	Function Switch	Reference	Description
3D communication coverage mode	NRDUCell.TridCommMode set to TRID_COVERAGE_MODE and NRDUCell.AerialCellFlag set to GROUND_AERIAL_CELL	None	Real-time-information-based precise channel estimation cannot be enabled if both of the following conditions are met: • The NRDUCell.TridCommMode parameter is set to TRID_COVERAGE_MODE. • The NRDUCell.AerialCellFlag parameter is set to GROUND_AERIAL_CELL.

- Impacted features/functions

The following impact relationships have been added for functions of time-frequency scheduling for interference avoidance:

Uplink precise MCS optimization

Function Name	Function Switch	Reference	Description
Uplink PAMC	UL_PAMC_SWITCH option of the NRDUCellPusch.UlPuschAlgoSwitch parameter	<i>Uplink Scheduling</i>	Uplink precise MCS optimization preferentially takes effect when it is enabled together with uplink PAMC.
Uplink target IBLER adaptation	UL_IBLER_ADAPT_SW option of the NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch parameter	<i>Uplink Scheduling</i>	Uplink precise MCS optimization preferentially takes effect when it is enabled together with uplink target IBLER adaptation.

Function Name	Function Switch	Reference	Description
Closed-loop PUSCH antenna selection	PUSCH_CLOSED_LOOP_ANT_SEL_SW option of the NRDUCellChnCoVAlgo.UlCoverageAlgoSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	Closed-loop PUSCH antenna selection preferentially takes effect when it is enabled together with uplink precise MCS optimization.
Basic VoNR functions	VONR_SW option of the NRCelAlgoSwitch.VonrSwitch parameter	<i>VoNR</i>	Uplink precise MCS optimization does not take effect for UEs with VoNR in effect.
Identification of UEs requiring connectivity with low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCelAlgoSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	Uplink precise MCS optimization does not take effect for UEs with the low latency and high reliability function in effect.
Probability-based uplink MCS selection	UL_PBMCSSEL_SW option of the NRDUCelUlAmc.UlAmcPerformanceSw parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	Uplink precise MCS optimization preferentially takes effect when it is enabled together with probability-based uplink MCS selection.
Adaptive outer-loop adjustment for small-packet services	SMALL_PKT_OL_ADAPT_ADJ_SW option of the NRDUCelUlAmc.UlAmcPerformanceSw parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	Uplink precise MCS optimization preferentially takes effect when it is enabled together with "adaptive outer-loop adjustment for small-packet services".
Uplink 4-layer SU transmission	UL_SU_FOUR_LAYER_SW option of the NRDUCelUlRank.UlRankAlgoSw parameter	None	Uplink precise MCS optimization does not take effect for UEs with rank 3 or rank 4 in the uplink.

Adaptive awareness scheduling

Function Name	Function Switch	Reference	Description
Packet-length-based scheduling optimization	PKT_LEN_BASED_SCH_OPT_SW option of the NRDUCellPusch.PuschPerformanceSwitch parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	Packet-length-based scheduling optimization does not take effect when it is enabled together with adaptive awareness scheduling.

Probability-based uplink MU-MIMO MCS selection

Function Name	Function Switch	Reference	Description
Basic VoNR functions	VONR_SW option of the NRCeAlgoSwitch.VonrSwitch parameter	<i>VoNR</i>	Probability-based uplink MU-MIMO MCS selection does not take effect for UEs with VoNR in effect.
Identification of UEs requiring connectivity with low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCeAlgoSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	Probability-based uplink MU-MIMO MCS selection does not take effect for UEs with the low latency and high reliability function in effect.
Uplink PAMC	UL_PAMC_SWITCH option of the NRDUCePusch.UlPuschAlgoSwitch parameter	<i>Uplink Scheduling</i>	Probability-based uplink MU-MIMO MCS selection preferentially takes effect when it is enabled together with uplink PAMC.
Uplink target IBLER adaptation	UL_IBLER_ADAPT_SW option of the NRDUCeAlgoSwitch.AdaptiveEdgeExpEnhSwitch parameter	<i>Uplink Scheduling</i>	Probability-based uplink MU-MIMO MCS selection preferentially takes effect when it is enabled together with uplink target IBLER adaptation.

The following impact relationships have been added for functions of multi-domain demodulation for interference resistance:

Multi-beam reception enhancement

Function Name	Function Switch	Reference	Description
PUSCH MU-MIMO	UL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	Multi-beam reception enhancement does not take effect for UEs with PUSCH MU-MIMO in effect.
Identification of UEs requiring connectivity with low latency and high reliability	gNBQciBearer.ServiceType set to URLLC_HIGH_RELIABILITY_SERVICE	<i>URLLC</i>	Multi-beam reception enhancement does not take effect for UEs requiring connectivity with low latency and high reliability.

Interference-based precise frequency offset estimation

Function Name	Function Switch	Reference	Description
Frequency offset measurement optimization based on interference suppression	INTRF_FREQOFFSET_OPT_SW option of the NRDUCellPusch.PuschMeasSw parameter	<i>Interference Avoidance</i>	Frequency offset measurement optimization based on interference suppression does not take effect when it is enabled together with interference-based precise frequency offset estimation.

Real-time-information-based precise channel estimation

Function Name	Function Switch	Reference	Description
PUSCH beam domain enhancement	PUSCH_BEAM_DOMAIN_ENH_SW option of the NRDUCellAlgoSwitch.FullChannelCovEnhSwitch parameter	<i>MIMO (TDD)</i>	Real-time-information-based precise channel estimation does not take effect when it is enabled together with PUSCH beam domain enhancement.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Massive MIMO Uplink Boosting (Low-Frequency TDD)*.

Requirement/Trouble Ticket Number

FeR-IR20241211002489-0001

FeR-IR20241211002489-0002

FeR-IR20241211002489-0003

FeR-IR20240828003082-0001

FeR-IR20240828003082-0006

FeR-IR20240828003082-0003

FeR-IR20240828003082-0005

FeR-IR20241211002489-0004

FeR-IR20241211002489-0005

FeR-IR20241211002489-0007

FeR-IR20241211002489-0006

IR20241209002193

DTS2025050905172

RAT

NR TDD

5.3.1.6 FOFD-100205 Uplink 4 Layers for a Single FWA User (Trial)

5.3.1.6.1 Uplink 4-Layer Transmission for a Single FWA User

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Uplink 4-layer transmission for a single FWA user has been added. This function is designed for UEs capable of uplink four-layer transmission. When channel conditions permit, this function uses multi-antenna technologies to enable spatial multiplexing of time-frequency resources on the PUSCH for a single UE. A maximum of four layers of data can be transmitted in the uplink for the UE, increasing the single-UE peak rate. This function is controlled by the **UL_SU_FOUR_LAYER_SW** option of the **NRDUCellULRank**. *ULRankAlgoSw* parameter.

LampSite base stations do not support this feature.

The change in this feature is supported only in low frequency bands.

The change in this feature is supported only in SA networking.

Implementation

For details, see *FWA*. The specific sections are listed in the change history of the document.

Impact on the Network

- After an upgrade, more cell configuration information is exchanged over the eXn interface between base stations. As a result, a single message becomes longer and, in certain scenarios, may be broken into more fragments. This may increase the values of counters related to the received and transmitted traffic volumes in the **SCTP Link Measurement** function subset.
- Benefits
After this function is enabled:
 - UEs that use four TX antennas can obtain transmit diversity gains.
 - In locations with many multipath signals and good channel conditions, UEs supporting four-layer transmission use rank 2 or higher. This brings multi-layer gains.
 - UEs supporting uplink four-layer transmission near the cell center are scheduled with more layers than 2T UEs, leading to higher average uplink rank and **User Uplink Average Throughput (DU)** compared with a network with only 2T UEs. In addition, there is a high probability that the **Cell Uplink Average Throughput (DU)** increases because the number of scheduling layers for a single UE, spectral efficiency, and cell capacity increase. In peak rate scenarios, when channel conditions are good, the uplink peak rate of a single UE is higher than that of a 2T UE.
- Impacts
After this function is enabled:
 - A higher number of layers is mistakenly selected for UEs capable of uplink four-layer transmission at a very small number of locations where this number of layers is not permitted by the channel conditions. As a result, the uplink IBLER increases, and a fallback to the original number of layers is triggered for the UEs. In this case, the uplink UE throughput slightly decreases.
 - If UEs capable of uplink four-layer transmission are located in positions with good channel conditions, some UEs in the cell cannot participate in uplink MU pairing. In this case, the average uplink cell throughput may slightly decrease.

Impact on NEs

This feature is implemented on the gNodeB.

UEs must support uplink four-layer transmission. A UE supports uplink four-layer transmission if the **MIMO-LayersUL** field in **maxNumberMIMO-LayersCB-PUSCH** of **FeatureSetUplinkPerCC** is **fourLayers** for the corresponding frequency band.

Impact on Hardware Requirements

Base station models:

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards:

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF modules:

All NR TDD-capable 32T32R and 64T64R RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

This feature is controlled by the following new license control items.

Feature ID	Feature Name	Model	Sales Unit
FOFD-100205	Uplink 4 Layers for a Single FWA User (Trial)	NR0S00ULSF40	per Cell

- Configuration management

The following counters have changed.

Change Type	Counter Name	Function Subset	NE	Description
New	N.ChMeas.SU.MIMO.UL.RANK3.RB	MIMO Channel Measurement (MIMO.DuCell)	gNode B	This counter has been added to measure the number of RBs for uplink SU-MIMO in rank 3 mode.
New	N.ChMeas.SU.MIMO.UL.RANK4.RB	MIMO Channel Measurement (MIMO.DuCell)	gNode B	This counter has been added to measure the number of RBs for uplink SU-MIMO in rank 4 mode.

Change Type	Counter Name	Function Subset	NE	Description
New	N.PUSCH.TbUL.Rank3	MAC Measurement (MAC.DuCell)	gNode B	This counter has been added to measure the number of TBs transmitted with rank 3 in the uplink.
New	N.PUSCH.TbUL.Rank4	MAC Measurement (MAC.DuCell)	gNode B	This counter has been added to measure the number of TBs transmitted with rank 4 in the uplink.
New	N.PUSCH.InitTbUL.Rank3	MAC Measurement (MAC.DuCell)	gNode B	This counter has been added to measure the number of uplink TBs initially transmitted with rank 3.
New	N.PUSCH.InitTbUL.Rank4	MAC Measurement (MAC.DuCell)	gNode B	This counter has been added to measure the number of uplink TBs initially transmitted with rank 4.
New	N.User.RRCConn.UL.4Layers.Avg	Cell User Quantity Measurement (User.Cell)	gNode B	This counter has been added to measure the average number of UEs capable of uplink four-layer transmission in a cell.
New	N.User.RRCConn.UL.4Layers.Max	Cell User Quantity Measurement (User.Cell)	gNode B	This counter has been added to measure the maximum number of UEs capable of uplink four-layer transmission in a cell.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions

Function Name	Function Switch	Reference	Description
Uplink SU-MIMO	UL_SU_MULTI_LAYER_SW option of the NRDUCellAlgoSwitch.SuMimoMultipleLayerSw parameter	<i>MIMO (TDD)</i>	Uplink 4-layer transmission for a single FWA user can take effect only when uplink SU-MIMO is enabled.
Uplink Maximum MIMO Layer Count Quota	NRDUCellPusch.MaxMimoLayerCnt	<i>MIMO (TDD)</i>	Uplink 4-layer transmission for a single FWA user can be enabled only when Uplink Maximum MIMO Layer Count Quota is set to LAYER_4 or a higher value.
SRS resource adjustment for 4T UEs	UL_4T_SRS_RES_ADJ_SW option of the NRDUCellSrs.SrsAlgorithmSwitch parameter	<i>Channel Management</i>	Uplink 4-layer transmission for a single FWA user can take effect only when SRS resource adjustment for 4T UEs is enabled.

- Mutually exclusive features/functions

Function Name	Function Switch	Reference	Description
High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag set to HIGH_SPEED	<i>High Speed Mobility</i>	High-speed Railway Superior Experience cannot work with uplink 4-layer transmission for a single FWA user.
Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	Cell Combination cannot work with uplink 4-layer transmission for a single FWA user.
Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	Hyper Cell cannot work with uplink 4-layer transmission for a single FWA user.

Function Name	Function Switch	Reference	Description
Fusion Cell	NRDUCell.NrDuCellNetworkingMode set to FUSION_CELL	<i>Fusion Cell (Low-Frequency TDD)</i>	Fusion Cell cannot work with uplink 4-layer transmission for a single FWA user.

- Impacted features/functions

Function Name	Function Switch	Reference	Description
Co-MM	INTRA_GNB_SU_COMM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	If uplink 4-layer transmission for a single FWA user has taken effect, Co-MM does not take effect for the user.
Intra-base-station UL CoMP	INTRA_GNB_UL_COMP_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	If uplink 4-layer transmission for a single FWA user has taken effect, intra-base-station UL CoMP does not take effect for the user.
High-reliability uplink MCS selection	UL_HIGH_RELIAB_MCS_SEL_SW option of the NRDUCellULAmc.UlAmcPerformanceSw parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	If uplink 4-layer transmission for a single FWA user has taken effect, high-reliability uplink MCS selection does not take effect for the user.
Uplink PAMC	UL_PAMC_SWITCH option of the NRDUCellPusch.UlPuschAlgoSwitch parameter	<i>Uplink Scheduling</i>	If uplink 4-layer transmission for a single FWA user has taken effect, uplink PAMC does not take effect for the user.

Function Name	Function Switch	Reference	Description
Uplink scheduling on a per RB basis	UL_1RB_SCH_SW option of the NRDUCellRes.SchedulingPolicySwitch parameter	<i>Network Slicing</i>	If uplink 4-layer transmission for a single FWA user has taken effect, whether uplink scheduling on a per RB basis takes effect for the user is affected.
Full-power dual transmission	UL_RANK1_FULL_POWER_SW option of the NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	If uplink 4-layer transmission for a single FWA user has taken effect, full-power dual transmission does not take effect for the user.
PUSCH MU-MIMO	UL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	If uplink 4-layer transmission for a single FWA user has taken effect, PUSCH MU-MIMO does not take effect for the user.
Spatial channel estimation under weak uplink coverage	UL_WEAK_COV_SPACE_CHN_ESTI_SW option of the NRDUCellChnCoverageAlgo.UlCoverageAlgoSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	If uplink 4-layer transmission for a single FWA user has taken effect, spatial channel estimation under weak uplink coverage does not take effect for the user.

Function Name	Function Switch	Reference	Description
Closed-loop PUSCH antenna selection	PUSCH_CLOSED_LOOP_ANT_SEL_SW option of the NRDUCellChnCo vAlgo.UICoverageAlgoSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	If uplink 4-layer transmission for a single FWA user has taken effect, closed-loop PUSCH antenna selection does not take effect for the user.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *FWA*.

Requirement/Trouble Ticket Number

FeR-IR20240629001666-0001

IR20241209002193

FeR-IR20250528001857-0059

RAT

NR TDD

5.3.1.7 FOFD-100206 FWA Ultra-Large Coverage (Trial)

5.3.1.7.1 FWA Device-Pipe-based Uplink Dense Codebooks

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The FWA device-pipe-based uplink dense codebooks function has been added. This function is controlled by the **UL_DENSER_CODEBOOK_SW** option of the **NRDUCellUeCoop.SpecUeCooperationSwitch** parameter. This function enables the base station to instruct UEs to expand their codebook sets for the PUSCH beyond standard codebooks. This function applies to UEs that support uplink dense codebooks when channel conditions permit. It makes uplink data transmission more accurate and improves coverage for UEs at the cell edge.

LampSite base stations do not support this feature.

The change in this feature is supported only in low frequency bands.

The change in this feature is supported only in SA networking.

Implementation

For details, see *FWA*. The specific sections are listed in the change history of the document.

Impact on the Network

FWA Ultra-Large Coverage improves uplink coverage for 4T UEs supporting this feature and increases the **User Uplink Average Throughput (DU)** for such UEs at the cell edge compared to 2T UEs.

When FWA Ultra-Large Coverage is enabled in a cell, the following impacts occur:

- In poor channel conditions, incorrect beam directions may be selected for UEs at a few positions. As a result, the uplink IBLER increases and the uplink UE throughput slightly decreases for these UEs.
- The average PUSCH interference in the cell may increase. In addition, the average PUSCH interference in neighboring cells may also increase, causing the uplink experience of UEs in neighboring cells to slightly deteriorate.

Impact on NEs

This feature is implemented on the gNodeB.

This function requires that UEs have the capability of device-pipe-based uplink dense codebooks. This capability is indicated by the **support-DenserCodebook** field in the RRC_HPS_UE_CAP_INFO message. If the value of this field from a UE is **supported**, the UE has this capability.

Impact on Hardware Requirements

Base station models:

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards:

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF modules:

All NR TDD-capable 32T32R and 64T64R RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

This feature is controlled by the following new license control items.

Feature ID	Feature Name	Model	Sales Unit
FOFD-100206	FWA Ultra-Large Coverage (Trial)	NR0S00FULC30	per Cell

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellUeCoop.SpecUeCooperationSwitch	MOD NRDUCELLU ECOOP LST NRDUCELLU ECOOP	gNodeB	The UL_DENSER_CODEBOOK_SW option has been added to specify whether to enable FWA device-pipe-based uplink dense codebooks.
Modified	NRDUCellPusch.PuschMeasSw	MOD NRDUCELLP USCH LST NRDUCELLP USCH	gNodeB	The UL_4T_PREC_CHANNEL_ADAPT_SW option has been added to specify whether to enable uplink precise channel adaptation for 4T UEs. If this option is selected, multi-layer channel calculation is more accurate for 4T UEs, and the uplink user-perceived rate increases in multi-layer scenarios.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions

Function Name	Function Switch	Reference	Description
Device-pipe identification	SPEC_UE_IDENTIFY_SW option of the NRDUCellUeCoop.SpecUeCooperationSwitch parameter	<i>Device-Pipe Synergy</i>	Device-pipe identification must be enabled to enable FWA device-pipe-based uplink dense codebooks and FWA device-pipe-based uplink high power.
SRS resource adjustment for 4T UEs	UL_4T_SRS_RES_ADJ_SW option of the NRDUCellSrs.SrsAlgorithmSwitch parameter	<i>Channel Management</i>	SRS resource adjustment for 4T UEs must be enabled for FWA device-pipe-based uplink dense codebooks and FWA device-pipe-based uplink high power to take effect.

Function Name	Function Switch	Reference	Description
RRC-reconfiguration-based dynamic waveform switching	UL_WAVEFORM_ADAPT_SW option of the NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	Either RRC-reconfiguration-based dynamic waveform switching or DCI-based dynamic waveform switching must be enabled for FWA device-pipe-based uplink dense codebooks to take effect. DCI-based dynamic waveform switching has the following UE capability requirements: • If only one carrier is configured in a band, UEs must have the capability indicated by the dynamicWaveformSwitch-r18 field. • If multiple carriers are configured in a band, the value of the dynamicWaveformSwitchIntraCA-r18 field in the reported UE capability information must be greater than or equal to the number of carriers configured in the band.
DCI-based dynamic waveform switching	DYNAMIC_WAVEFORM_SWITCHING_SW option of the NRDUCellChnCovAlgo.UlCoverageAlgoSwitch parameter	<i>Uplink Scheduling</i>	

- Mutually exclusive features/functions

Function Name	Function Switch	Reference	Description
Fusion Cell	NRDUCell.NrDuCellINetworkingMode set to FUSION_CELL	<i>Fusion Cell (Low-Frequency TDD)</i>	Fusion Cell cannot work with FWA device-pipe-based uplink dense codebooks or FWA device-pipe-based uplink high power.
Hyper Cell	NRDUCell.NrDuCellINetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	Hyper Cell cannot work with FWA device-pipe-based uplink dense codebooks or FWA device-pipe-based uplink high power.
Cell Combination	NRDUCell.NrDuCellINetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	Cell Combination cannot work with FWA device-pipe-based uplink dense codebooks or FWA device-pipe-based uplink high power.

- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *FWA*.

Requirement/Trouble Ticket Number

FeR-IR20240629001666-0004

IR20241209002193

FeR-IR20251216000780-0001

RAT

NR TDD

5.3.1.7.2 FWA Device-Pipe-based Uplink High Power

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The FWA device-pipe-based uplink high power function has been added. This function is controlled by the **UL_HIGH_POWER_UE_SW** option of the **NRDUCellUeCoop.SpecUeCooperationSwitch** parameter. This function increases the transmit power of UEs that support uplink high power. It achieves this by enabling the gNodeB to instruct such UEs to use high power. This function improves coverage for UEs at the cell edge.

LampSite base stations do not support this feature.

The change in this feature is supported only in low frequency bands.

The change in this feature is supported only in SA networking.

Implementation

For details, see *FWA*. The specific sections are listed in the change history of the document.

Impact on the Network

FWA Ultra-Large Coverage improves uplink coverage for 4T UEs supporting this feature and increases the **User Uplink Average Throughput (DU)** for such UEs at the cell edge compared to 2T UEs.

When FWA Ultra-Large Coverage is enabled in a cell, the following impacts occur:

- In poor channel conditions, incorrect beam directions may be selected for UEs at a few positions. As a result, the uplink IBLER increases and the uplink UE throughput slightly decreases for these UEs.
- The average PUSCH interference in the cell may increase. In addition, the average PUSCH interference in neighboring cells may also increase, causing the uplink experience of UEs in neighboring cells to slightly deteriorate.

Impact on NEs

This feature is implemented on the gNodeB.

This function requires that UEs have the capability of device-pipe-based uplink high power. This capability is indicated by the **hps-ue-PowerClass** field in the **RRC_HPS_UE_CAP_INFO** message. If the value of this field from a UE is **pc1-32dbm-ext**, the UE has this capability.

Impact on Hardware Requirements

Base station models:

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards:

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF modules:

All NR TDD-capable 32T32R and 64T64R RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

This feature is controlled by the following new license control items.

Feature ID	Feature Name	Model	Sales Unit
FOFD-100206	FWA Ultra-Large Coverage (Trial)	NR0S00FULC30	per Cell

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCell UeCoop.SpecUeCooperationSwitch	MOD NRDUCELLU ECOOP LST NRDUCELLU ECOOP	gNodeB	The UL_HIGH_POWER_UE_SW option has been added to specify whether to enable FWA device-pipe-based uplink high power.
Modified	NRDUCell Pusch.PuschMeasSwitch	MOD NRDUCELLP USCH LST NRDUCELLP USCH	gNodeB	The UL_4T_PREC_CHANNEL_ADAPT_SW option has been added to specify whether to enable uplink precise channel adaptation for 4T UEs. If this option is selected, multi-layer channel calculation is more accurate for 4T UEs, and the uplink user-perceived rate increases in multi-layer scenarios.

- Performance management

No impact

- Fault management

No impact

Impact on Features/Functions

- Prerequisite features/functions

Function Name	Function Switch	Reference	Description
Device-pipe identification	SPEC_UE_IDENTIFY_SW option of the NRDUCellUeCoop.SpecUeCooperationSwitch parameter	<i>Device-Pipe Synergy</i>	Device-pipe identification must be enabled to enable FWA device-pipe-based uplink dense codebooks and FWA device-pipe-based uplink high power.
SRS resource adjustment for 4T UEs	UL_4T_SRS_RES_ADJ_SW option of the NRDUCellSrs.SrsAlgorithmSwitch parameter	<i>Channel Management</i>	SRS resource adjustment for 4T UEs must be enabled for FWA device-pipe-based uplink dense codebooks and FWA device-pipe-based uplink high power to take effect.

Function Name	Function Switch	Reference	Description
RRC-reconfiguration-based dynamic waveform switching	UL_WAVEFORM_ADAPT_SW option of the NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	Either RRC-reconfiguration-based dynamic waveform switching or DCI-based dynamic waveform switching must be enabled for FWA device-pipe-based uplink dense codebooks to take effect. DCI-based dynamic waveform switching has the following UE capability requirements: • If only one carrier is configured in a band, UEs must have the capability indicated by the dynamicWaveformSwitch-r18 field. • If multiple carriers are configured in a band, the value of the dynamicWaveformSwitchIntraCA-r18 field in the reported UE capability information must be greater than or equal to the number of carriers configured in the band.
DCI-based dynamic waveform switching	DYNAMIC_WAVEFORM_SWITCHING_SW option of the NRDUCellChnCovAlgo.UlCoverageAlgoSwitch parameter	<i>Uplink Scheduling</i>	

- Mutually exclusive features/functions

Function Name	Function Switch	Reference	Description
Fusion Cell	NRDUCell.NrDuCellINetworkingMode set to FUSION_CELL	<i>Fusion Cell (Low-Frequency TDD)</i>	Fusion Cell cannot work with FWA device-pipe-based uplink dense codebooks or FWA device-pipe-based uplink high power.
Hyper Cell	NRDUCell.NrDuCellINetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	Hyper Cell cannot work with FWA device-pipe-based uplink dense codebooks or FWA device-pipe-based uplink high power.
Cell Combination	NRDUCell.NrDuCellINetworkingMode set to HYPER_CELL_COMBINATION_MODE	<i>Cell Combination</i>	Cell Combination cannot work with FWA device-pipe-based uplink dense codebooks or FWA device-pipe-based uplink high power.

- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *FWA*.

Requirement/Trouble Ticket Number

FeR-IR20240629001666-0003

IR20241209002193

FeR-IR20251216000780-0001

RAT

NR TDD

5.3.1.8 FOFD-100210 Flexible Rate Grading

5.3.1.8.1 Support for the Flexible Rate Grading Feature

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The Flexible Rate Grading feature is introduced to support differentiated and customizable peak user-perceived rate guarantee.

This feature includes flexible experience differentiation. This function is controlled by the **FLEX_EXP_DIFF_SW** option of the **NRDUCellFeatureSw.QciBasedDiffServExpSw** parameter and includes the following sub-functions:

- Data rate limitation in flexible experience differentiation: This function allows differentiated data rate proportions to be configured for users with different privilege levels, thereby achieving differentiated and customizable peak user-perceived rates. This sub-function is controlled by the **UNLIMITED_RATE_SW** option of the **gNBFlexExpDiffCfg.EquityLevelAlgoSw** parameter. Data rate limitation in flexible experience differentiation is enabled only if this option is deselected. The parameters related to privilege levels are as follows:

Table 5-11 Flexible experience differentiation configurations

Parameter Name	Parameter ID	Description	Impact
Equity Level	NRDUCellQciBearer.EquityLevel	If this parameter is set to a value ranging from 0 to 5, a smaller value of this parameter indicates a higher privilege level.	More privilege levels lead to more user-perceived data rate levels.
		If this parameter is set to 255, flexible experience differentiation does not take effect for the QCI.	
Equity Level	gNBFlexExpDiffCfg.EquityLevel	Set this parameter to a value ranging from 0 to 5. A smaller value of this parameter indicates a higher privilege level.	

Parameter Name	Parameter ID	Description	Impact
UL Target Rate Ratio	gNBFlexExpDiffCfg.UlTargetRateRatio	Set this parameter to a value ranging from 1 to 100 in percentage. The value of this parameter is the result of the uplink available peak data rate of a QCI-specific user divided by the current network's maximum capability.	In conditions of the same network and UE capabilities, smaller data rate proportions of a privilege level result in: (1) more significant differences in the uplink or downlink peak data rate between this privilege level and higher privilege levels; (2) lower uplink or downlink peak data rate of users at this privilege level, potentially impacting user experience.
DL Target Rate Ratio	gNBFlexExpDiffCfg.DlTargetRateRatio	Set this parameter to a value ranging from 1 to 100 in percentage. The value of this parameter is the result of the downlink available peak data rate of a QCI-specific UE divided by the current network's maximum capability.	

Parameter Name	Parameter ID	Description	Impact
UL Min Guaranteed Bandwidth	gNBFlexExpDiffCfg.UlMinGteeBW	Set this parameter to a value ranging from 0 to 1000, which indicates the minimum uplink guaranteed bandwidth for the associated privilege level. If this parameter is set to 0, the UE and network capabilities are not verified.	Given the same network and UE capabilities, a larger minimum guaranteed bandwidth may result in less significant differences in the uplink or downlink peak data rate between privilege levels. • If the bandwidth configured for a user cannot reach the minimum guaranteed bandwidth of any level, the uplink or downlink peak data rate of each privilege level is the current maximum available data rate for the user, with no difference between privilege levels. • If this parameter is set to 0, UE and network capabilities are not verified. The uplink or downlink peak user-perceived rate proportions of different privilege levels are basically consistent with the configured target uplink or downlink data rate proportions. • If the minimum guaranteed bandwidths of different privilege levels are not configured based on the target uplink or downlink data rate proportions, the uplink or downlink peak user-perceived rate proportions of
DL Min Guaranteed Bandwidth	gNBFlexExpDiffCfg.DlMinGteeBW	Set this parameter to a value ranging from 0 to 1000, which indicates the minimum downlink guaranteed bandwidth for the associated privilege level. If this parameter is set to 0, the UE and network capabilities are not verified.	

Parameter Name	Parameter ID	Description	Impact
			different privilege levels may not be fully consistent with the configured target uplink or downlink data rate proportions.

- Automatic exit from flexible experience differentiation is introduced to ensure that the uplink or downlink experience of UEs with poor channel conditions does not deteriorate when data rate limitation in flexible experience differentiation is enabled. After "automatic exit from flexible experience differentiation" is enabled, the base station checks whether the uplink or downlink peak rate of UEs at their current locations falls below the corresponding threshold before rate control on these UEs in each period. The details are as follows:
 - If the **NRDUCellPusch.UIFlexExpDiffExitRateThld** parameter is set to a non-zero value, the base station compares the uplink peak rate of UEs at their current locations with the value of this parameter.
 - If the estimated uplink peak rate of UEs is greater than or equal to the rate threshold for exiting flexible experience differentiation in the uplink, the base station performs uplink rate control on the UEs based on the uplink target rate of the privilege level corresponding to the QCI.
 - If the estimated uplink peak rate of UEs is less than the rate threshold for exiting flexible experience differentiation in the uplink, the base station does not perform uplink rate control on the UEs within this period.
 - If the **NRDUCellRsvdExt04.RsvdParam56** parameter is set to a non-zero value, the base station compares the downlink peak rate of UEs at their current locations with the value of this parameter.
 - If the estimated downlink peak rate of UEs is greater than or equal to the rate threshold for exiting flexible experience differentiation in the downlink, the base station performs downlink rate control on the UEs based on the downlink target rate of the privilege level corresponding to the QCI.
 - If the estimated downlink peak rate of UEs is less than the rate threshold for exiting flexible experience differentiation in the downlink, the base station does not perform downlink rate control on the UEs within this period.
- Data rate reporting in flexible experience differentiation: This function allows the uplink or downlink data rate to be periodically reported to the core network. This sub-function is controlled by the **FLEX_EXP_DIFF_RPT_SW** option of the **NRDUCellFeatureSw.QciBasedDiffServExpSw** parameter.

The change in this feature is supported only in low frequency bands.

Implementation

For details, see *Service-differentiated Experience Guarantee*. The specific sections are listed in the change history of the document.

Impact on the Network

Benefits:

After this function is enabled, differentiated peak user-perceived rates can be customized for users at different privilege levels on the network.

- Given the same network conditions, the uplink or downlink peak user-perceived rates for different privilege levels of a single user match the associated flexible experience differentiation configurations. The details are as follows:
 - For each privilege level of a user, if the uplink or downlink minimum guaranteed bandwidth is set to 0, the uplink or downlink peak user-perceived rate meets the configured target uplink or downlink data rate proportion.
 - For a privilege level of a user, if the uplink or downlink minimum guaranteed bandwidth is not set to 0, the uplink or downlink peak user-perceived rate is determined based on (1) the target uplink or downlink data rate proportion and (2) the uplink or downlink minimum guaranteed bandwidth. In this case, the uplink or downlink peak user-perceived rate of each privilege level of the user may not match the configured target uplink or downlink data rate proportion.
- When users with different privilege levels perform full-buffer services simultaneously under the same network conditions, the users with different privilege levels have differentiated uplink or downlink peak user-perceived rates which cannot precisely match the associated proportions in flexible experience differentiation configurations. This is caused by multiple factors such as flexible experience differentiation configuration, user scheduling priority, and data rate estimation.
- In NSA networking, flexible experience differentiation does not take effect for LTE carriers. If a UE's uplink or downlink data is transmitted over LTE carriers, the UE's uplink or downlink peak user-perceived rate cannot precisely match the associated target uplink or downlink data rate proportion.

Impacts:

- In the following scenarios, the limited uplink or downlink peak data rate of some UEs prolongs the user scheduling duration in conditions of the same data volume. As a result, the uplink or downlink RB usage in a cell, average uplink or downlink UE throughput, average uplink or downlink cell throughput, uplink or downlink traffic volume in a cell, and other indicators may decrease. In addition, the average downlink packet delay at the MAC layer, average downlink packet delay at the RLC layer, and other indicators may increase. For the definitions of related indicators, see [Table 5-12](#). A higher penetration rate of UEs without rate limitation leads to a smaller impact of this function on KPIs.
 - Due to rapid changes in factors such as network loads and channel conditions, the uplink or downlink peak data rate of each privilege level,

which is calculated periodically during the execution of this function, becomes inaccurate.

- The configured target uplink and downlink data rate proportions are excessively low. In addition, the uplink and downlink minimum guaranteed bandwidths are lower than the bandwidths available for the UE before this function is enabled.

Table 5-12 Indicator definitions related to the impact analysis of data rate limitation in flexible experience differentiation

Indicator Name	Definition
Average downlink UE throughput	User Downlink Average Throughput (DU)
Average uplink UE throughput	User Uplink Average Throughput (DU)
Average downlink cell throughput	Cell Downlink Average Throughput (DU)
Average uplink cell throughput	Cell Uplink Average Throughput (DU)
Downlink RB usage	Downlink Resource Block Utilizing Rate (DU)
Uplink RB usage	Uplink Resource Block Utilizing Rate (DU)
Downlink traffic volume in a cell	Downlink Traffic Volume (DU)
Uplink traffic volume in a cell	Uplink Traffic Volume (DU)
Average downlink packet delay at the MAC layer	$(N.QoS.DL.PktDelayAirInterface.Time / N.QoS.DL.PktDelayAirInterface.Num) / 1000$
Average downlink packet delay at the RLC layer	$(N.QoS.DL.PktDelaygNBDU.Time / N.QoS.DL.PktDelaygNBDU.Num) / 1000$

- In NSA networking, the SCG may be removed for a UE if the NR traffic volume is lower than expected due to a low target uplink or downlink data rate proportion on the NR side. In this case, the values of the following counters may increase:
 - **N.NsaDc.SgNB.Add.Att** or **N.NsaDc.SgNB.Add.Succ**
 - **N.NsaDc.SgNB.Rel**
- Parameters related to flexible experience differentiation have the following impacts on the network.

Parameter Name	Parameter ID	Impact on the Network
Diff Target Rate Calculation Period	NRDUCellUuPrfmOpt.DiffTargetRateCalcPeriod	• A smaller value of this parameter improves the accuracy of differentiated target data rate calculation but increases the computational overhead. • A larger value of this parameter leads to slower updates of the differentiated target data rate and increases the calculation error of the target data rate.
UL Target Rate Ratio	gNBFlexExpDiffCfg.ULTargetRateRatio	• A larger value of this parameter leads to a higher uplink peak perceived rate of users at the associated privilege level. • A smaller value of this parameter leads to a lower uplink peak perceived rate of users at the associated privilege level.
DL Target Rate Ratio	gNBFlexExpDiffCfg.DLTargetRateRatio	• A larger value of this parameter leads to a higher downlink peak perceived rate of users at the associated privilege level. • A smaller value of this parameter leads to a lower downlink peak perceived rate of users at the associated privilege level.
UL Min Guarantee Bandwidth	gNBFlexExpDiffCfg.ULMinGteeBw	• A larger value of this parameter leads to a higher uplink peak perceived rate of users at the associated privilege level. • A smaller value of this parameter leads to a lower uplink peak perceived rate of users at the associated privilege level.
DL Min Guarantee Bandwidth	gNBFlexExpDiffCfg.DLMinGteeBw	• A larger value of this parameter leads to a higher downlink peak perceived rate of users at the associated privilege level. • A smaller value of this parameter leads to a lower downlink peak perceived rate of users at the associated privilege level.

Impact on NEs

This feature is implemented on the gNodeB.

Impact on Hardware Requirements

- Boards
All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- RF modules
All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
This feature is controlled by the following new license control item.

Feature ID	Feature Name	Model	Sales Unit
FOFD-100210	Flexible Rate Grading (trial)	NR0SFLEXGR10	per Cell

- Configuration management
The following MML commands have changed.

Change Type	Command	NE	Description
New	ADD GNBFLXEXPDIFFC FG	gNodeB	This MML command has been added to add flexible experience differentiation configurations to the gNodeB.
New	MOD GNBFLXEXPDIFFC FG	gNodeB	This MML command has been added to modify flexible experience differentiation configurations of the gNodeB.
New	LST GNBFLXEXPDIFFC FG	gNodeB	This MML command has been added to list flexible experience differentiation configurations of the gNodeB.

Change Type	Command	NE	Description
New	RMV GNBFLEXEXPDIFFCFG	gNodeB	This MML command has been added to remove flexible experience differentiation configurations of the gNodeB.

NOTE

The parameters in these new or deleted MML commands are no longer listed separately.

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellFeatureSw.QciBasedDiffServExpSw	MOD NRDUCELLFEATURESW LST NRDUCELLFEATURESW	gNodeB	The FLEX_EXP_DIFF_SW option has been added to specify whether to enable flexible experience differentiation.
New	NRDUCellQciBearer.EquityLevel	ADD NRDUCELLQCBEARER LST NRDUCELLQCBEARER MOD NRDUCELLQCBEARER RMV NRDUCELLQCBEARER	gNodeB	This parameter has been added to specify the privilege level of a bearer for experience differentiation.
Modified	NRDUCellFeatureSw.QciBasedDiffServExpSw	MOD NRDUCELLFEATURESW LST NRDUCELLFEATURESW	gNodeB	The FLEX_EXP_DIFF_RPT_SW option has been added to specify whether to enable data rate reporting in flexible experience differentiation.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellUuPrfmOpt.DiffTargetRateCalcPeriod	LST NRDUCELLU UPRFMOPT MOD NRDUCELLU UPRFMOPT	gNodeB	This parameter has been added to specify the interval at which the differentiated target data rate is calculated for flexible experience differentiation.
New	NRDUCellPusch.UlDiffTargetRateEstFactor	MOD NRDUCELLP USCH LST NRDUCELLP USCH	gNodeB	This parameter has been added to specify the factor for estimating the differentiated target uplink rate in flexible experience differentiation.
New	NRDUCellUuPrfmOpt.DiffTargetRateEstFactor	LST NRDUCELLU UPRFMOPT MOD NRDUCELLU UPRFMOPT	gNodeB	This parameter has been added to specify the factor for estimating the differentiated target downlink rate in flexible experience differentiation.
New	NRDUCellPusch.UlFlexExpDiffExitRateThld	MOD NRDUCELLP USCH LST NRDUCELLP USCH	gNodeB	This parameter has been added to specify the rate threshold for exiting flexible experience differentiation in the uplink.
Modified	NRDUCellRsvdExt04.RsvdParam56	MOD NRDUCELLRS VDEXT04 LST NRDUCELLRS VDEXT04	gNodeB	This reserved parameter has been activated to specify the rate threshold for exiting flexible rate grading.

 NOTE

NRDUCellRsvdExt04.RsvdParam56 is a reserved parameter.

For details about reserved parameters, see 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List.

- Performance management

No impact

- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions

RAT	Function Name	Function Switch	Reference	Description
Low - frequency TDD	Uplink UE-AMBR-based rate limitation for non-GBR services	UL_AMBR_RATE_CTRL_SWITCH . <i>ServiceDiffSwitch</i> parameter	<i>QoS Management</i>	After both "uplink UE-AMBR-based rate limitation for non-GBR services" and "flexible experience differentiation" are enabled, the UE's uplink peak data rate takes the value of the target uplink data rate calculated in "flexible experience differentiation" if the uplink UE-AMBR used for "uplink UE-AMBR-based rate limitation for non-GBR services" is greater than the aforementioned calculated target uplink data rate.

RAT	Function Name	Function Switch	Reference	Description
Low - frequency TDD	Downlink UE-AMBR-based rate limitation for non-GBR services	DL_AMBR_RATE_CTRL_SWITCH.ServiceDiffSwitch parameter	<i>QoS Management</i>	After both "downlink UE-AMBR-based rate limitation for non-GBR services" and "flexible experience differentiation" are enabled, the UE's downlink peak data rate takes the value of the target downlink data rate calculated in "flexible experience differentiation" if the downlink UE-AMBR used for "downlink UE-AMBR-based rate limitation for non-GBR services" is greater than the aforementioned calculated target downlink data rate.
Low - frequency TDD	Maximum rate limitation for GBR services	None	<i>QoS Management</i>	When a UE performs GBR services and "flexible experience differentiation" is enabled, the UE's uplink or downlink peak data rate takes the value of the target uplink or downlink data rate calculated in "flexible experience differentiation" if the UE's MBR or MFBR is greater than the aforementioned calculated target uplink or downlink data rate.

RAT	Function Name	Function Switch	Reference	Description
Low - frequency TDD	Minimum uplink rate guarantee for non-GBR services	gNBDUMacParamGroup.UlGuaranteedRate	<i>QoS Management</i>	After "flexible experience differentiation" is enabled, a UE's uplink peak data rate takes the value of the gNBDUMacParamGroup.UlGuaranteedRate parameter if this parameter is set to a non-zero value which exceeds the target uplink data rate calculated in "flexible experience differentiation". As a result, the UE's uplink data rate may not be consistent with the flexible experience differentiation configuration, which may lead to an actual data rate lower than the minimum guaranteed rate.

RAT	Function Name	Function Switch	Reference	Description
Low - frequency TDD	Minimum downlink rate guarantee for non-GBR services	gNBDUMacParamGroup.DIGuaranteedRate	<i>QoS Management</i>	After "flexible experience differentiation" is enabled, a UE's downlink peak data rate takes the value of the gNBDUMacParamGroup.DIGuaranteedRate parameter if this parameter is set to a non-zero value which exceeds the target downlink data rate calculated in "flexible experience differentiation". As a result, the UE's downlink data rate may not be consistent with the flexible experience differentiation configuration, which may lead to an actual data rate lower than the minimum guaranteed rate.

RAT	Function Name	Function Switch	Reference	Description
Low - frequency TDD	Minimum rate guarantee for GBR services	None	<i>QoS Management</i>	When a UE performs GBR services and "flexible experience differentiation" is enabled, the UE's uplink or downlink peak data rate takes the value of the GBR or GFBR configured on the core network if the UE's GBR or GFBR is greater than the target uplink or downlink data rate calculated in "flexible experience differentiation". As a result, the UE's uplink or downlink data rate may not be consistent with the flexible experience differentiation configuration, which may lead to an actual data rate lower than the minimum guaranteed rate.

RAT	Function Name	Function Switch	Reference	Description
Low - frequency TDD	Uplink experience-based scheduling	UL_EXP_BASED_SCH_SW option of the NRDUCellAlgoSwitch.FwaAlgoSwitch parameter	<i>FWA</i>	After both "uplink experience-based scheduling" and "flexible experience differentiation" take effect for a UE, the UE's uplink peak data rate takes the value of the target uplink data rate calculated in "flexible experience differentiation" if the value of the gNBDUMacParamGroup.UlExpBasedSchGuarUserRate or gNBDURfspConfig.UlExpBasedSchGuarUserRate parameter is greater than the aforementioned calculated target uplink data rate.
Low - frequency TDD	Downlink experience-based scheduling	DL_EXP_BASED_SCH_SW option of the NRDUCellAlgoSwitch.FwaAlgoSwitch parameter	<i>FWA</i>	After both "downlink experience-based scheduling" and "flexible experience differentiation" take effect for a UE, the UE's downlink peak data rate takes the value of the target downlink data rate calculated in "flexible experience differentiation" if the value of the gNBDUMacParamGroup.DlExpBasedSchGuarUserRate or gNBDURfspConfig.DlExpBasedSchGuarUserRate parameter is greater than the aforementioned calculated target downlink data rate.

RAT	Function Name	Function Switch	Reference	Description
Low - frequency TDD	Service Type	gNBQciBearer.ServiceType set to a value other than NORMAL	None	If the service type of a QCI-specific bearer is not common service and there are certain requirements on the uplink or downlink data rate, it is not recommended that data rate limitation in flexible experience differentiation be enabled for the QCI.
Low - frequency TDD	RedCap Introduction	REDCAP_SW option of the NRDUCellFeatureSw.UeCapBasedFunSw parameter	<i>RedCap</i>	Flexible experience differentiation does not take effect for RedCap UEs.
Low - frequency TDD	User PRB control for FWA	PRB_CTRL_SW option of the NRDUCellAlgorithmSwitch.FwaAlgorithmSwitch parameter	<i>FWA</i>	After both "user PRB control for FWA" and "flexible experience differentiation" are enabled, the base station estimates the uplink or downlink peak data rate based on the available resource range of FWA UEs and performs further data rate control. As a result, the difference between the estimated data rate and the actually measured data rate may increase.

RAT	Function Name	Function Switch	Reference	Description
Low - frequency TDD	NR DU cell resource management between network slices	RB_DYNAMIC_CONTROL_SW option of the NRDUCellAlgoSwitch . <i>NetworkSliceAlgoSwitch</i> parameter	<i>Network Slicing</i>	After both "NR DU cell resource management between network slices" and "flexible experience differentiation" are enabled, the base station estimates the uplink or downlink peak data rate based on the available resource range of the slice to which the UE belongs and performs further data rate control. As a result, the difference between the estimated data rate and the actually measured data rate may increase.
Low - frequency TDD	RFSP-group-based RB resource control	RB_DYNAMIC_CONTROL_SW option of the NRDUCellCommonSw . <i>RfspAlgoSwitch</i> parameter	<i>Resource Control in NSA Networking</i>	After both "RFSP-group-based RB resource control" and "flexible experience differentiation" are enabled, the base station estimates the uplink or downlink peak data rate based on the available resource range of the RFSP resource group to which the UE belongs and performs further data rate control. As a result, the difference between the estimated data rate and the actually measured data rate may increase.

RAT	Function Name	Function Switch	Reference	Description
Low - frequency TDD	RAN sharing with common carrier	gNBSharingMode.gNBMultiOpSharingMode set to SHARED_FREQ	<i>Multi-Operator Sharing</i>	After "flexible experience differentiation" is enabled in RAN sharing with common carrier scenarios, the base station estimates the uplink or downlink peak data rate based on the available resource range of the UE's operator and performs further data rate control. As a result, the difference between the estimated data rate and the actually measured data rate may increase.

RAT	Function Name	Function Switch	Reference	Description
Low - frequency TDD	Intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter INTRA_BAND_UL_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	After both "CA" and "flexible experience differentiation" are enabled, the base station estimates the uplink or downlink peak data rate of each carrier of a UE. After both "CA" and "flexible experience differentiation" are enabled, the following rules also apply: (1) If the SCell activation threshold is set to a small value, the estimated uplink or downlink peak UE data rate is more accurate, but the UE power consumption increases. (2) If an SCell cannot be activated because the SCell activation threshold is set to a large value, the estimated uplink or downlink peak UE data rate may be inaccurate. It is recommended that the NRDUCellCarrMgmt.CaDisCellActThld-SlotNum and NRDUCellCarrMgmt.CaUISCellActThld-SlotNum parameters be set to 0 in this scenario.
Low - frequency TDD	Intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter INTRA_FR_INTER_BAND_UL_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	
Low - frequency TDD	Inter-gNodeB CA	INTER_GNODEB_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>CA Experience Improvement</i>	

RAT	Function Name	Function Switch	Reference	Description
Low - frequency TDD	Inter-FR CA (trial)	INTER_FR_FR1TOFR2_CA_SW option of the NRDUCellAlgoSwitch . <i>CaAlgoSwitch</i> parameter	<i>Carrier Aggregation</i>	Flexible experience differentiation does not take effect for inter-FR CA UEs.
Low - frequency TDD	Super uplink	SUPER_UL_TDM_SCH_SW , FLEX_SUPER_UPLINK_SW , and INTER_GNB_SUPER_UPLINK_SW options of the NRDUCellAlgoSwitch . <i>SuperUplinkSwitch</i> parameter	<i>Super Uplink</i>	After both "super uplink" and "flexible experience differentiation" are enabled, the base station estimates the uplink peak data rate of each carrier of a UE. If a UE is configured with an SUL carrier which is not activated, the UE's estimated uplink peak data rate may be inaccurate.

RAT	Function Name	Function Switch	Reference	Description
Low - frequency TDD	Uplink consecutive scheduling and non-consecutive scheduling	UL_NON_CON_SCH_SW option of the NRDUCellAlgoSwitch . UllInconsecutiveSchSwitch parameter	<i>Uplink Scheduling</i>	After the UL_NON_CON_SCH_SW option of the NRDUCellAlgoSwitch . UllInconsecutiveSchSwitch parameter is deselected, only scheduling of contiguous RBs is supported in the uplink. In this case, if uplink RB resources are truncated or fragmented, UEs' maximum target uplink data rates may not be reached, resulting in insignificant or no differentiation in uplink data rates among users with different privilege levels. Therefore, it is recommended that the UL_NON_CON_SCH_SW option of the NRDUCellAlgoSwitch . UllInconsecutiveSchSwitch parameter be selected if significant differentiation in uplink user experience among users is required when flexible experience differentiation is enabled.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Service-differentiated Experience Guarantee*.

Requirement/Trouble Ticket Number

FeR-IR20250626000485-0001

FeR-IR20241209002193-0117

FeR-IR20250724000484-0004

FeR-IR20250528001857-0117

RAT

NR TDD

5.3.1.9 FOFD-100252 AMC-based Precise Power Control Energy Saving

5.3.1.9.1 Added the AMC-based Precise Power Control Energy Saving Feature

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

AMC-based precise power control energy saving is introduced in light- and moderate-load scenarios. It enables refined power allocation for initial transmission scheduling and retransmission scheduling by using the initial transmission power reduction function and retransmission power reduction function. The two functions reduce the redundant power for PDSCH data transmission during downlink initial MCS index selection and downlink retransmission, thereby achieving energy saving for base stations.

The initial transmission power reduction function and retransmission power reduction function are controlled by the **INIT_TRANS_POWER_REDUCE_SW** and **RETRANS_POWER_REDUCE_SW** options of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter, respectively. The parameters in the **NRDUCellPowerSaving** MO can be configured to specify the effective period, load threshold, and other information under an energy saving policy of initial transmission power reduction or retransmission power reduction.

In addition, initial transmission power reduction and retransmission power reduction can also be configured to not take effect for specific service types, by using the **SERV_TYPE_DYN_EXIT_PWR_SAVE_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter and the **INIT_TRANS_PWR_RD_EXIT_SW** and **RETRANS_PWR_RD_EXIT_SW** options of the **NRDUCellPsPolicyGrp.ServTypeDynExitPwrSavingSw** parameters.

This feature is supported only in low frequency bands.

Implementation

For details, see *Energy Conservation and Emission Reduction*. The specific sections are listed in the change history of the document.

Impact on the Network

When the network is under low or moderate load, initial transmission power reduction and retransmission power reduction dynamically lower the transmit power of cells. This reduces the overall energy consumption of the base station. In other load scenarios, the energy saving gains are limited.

Energy saving gains may also decrease when some of the UEs on the network are not target UEs of initial transmission power reduction and retransmission power reduction. If none of the UEs are target UEs of the two functions, there will be no energy saving gains.

Service-type-based dynamic exit from initial transmission power reduction can guarantee the performance of UEs of specific service types but reduces energy saving gains.

Service-type-based dynamic exit from retransmission power reduction can guarantee the performance of UEs of specific service types but reduces energy saving gains.

In addition, initial transmission power reduction and retransmission power reduction also have the following impacts:

- The initial block error rate (IBLER) and residual block error rate (RBLER) increase.
- The MAC-layer processing delay in the gNodeB may increase. This delay is calculated as follows: **Average Downlink Packet Delay in MAC** = $(N.QoS.DL.PktDelayAirInterface.Time / N.QoS.DL.PktDelayAirInterface.Num) / 1000$.
- The average MCS index for downlink scheduling decreases.
- The downlink user-perceived rate decreases.
- The number of retransmissions and the retransmission rate increase.
- In cells exposed to strong interference from neighboring cells, when initial transmission power reduction and retransmission power reduction take effect in neighboring cells, interference from neighboring cells decreases.
 - The average CQI may increase. $Average\ CQI = ((0 \times N.ChMeas.CQI.SingleCW.0 + 1 \times N.ChMeas.CQI.SingleCW.1 + 2 \times N.ChMeas.CQI.SingleCW.2 + 3 \times N.ChMeas.CQI.SingleCW.3 + \dots + 13 \times N.ChMeas.CQI.SingleCW.13 + 14 \times N.ChMeas.CQI.SingleCW.14 + 15 \times N.ChMeas.CQI.SingleCW.15) / (N.ChMeas.CQI.SingleCW.0 + N.ChMeas.CQI.SingleCW.1 + N.ChMeas.CQI.SingleCW.2 + N.ChMeas.CQI.SingleCW.3 + \dots + N.ChMeas.CQI.SingleCW.13 + N.ChMeas.CQI.SingleCW.14 + N.ChMeas.CQI.SingleCW.15))$
 - The values of the **N.PRB.DL.DrbUsed.Avg** and **N.PRB.DL.Used.Avg** counters may decrease.
- The values of the following counters related to downlink PRB usage may increase in common cells:
 - **N.PRB.DL.DrbUsed.Avg**
 - **N.PRB.DL.ExtendUsed.Avg**
 - **N.PRB.DL.Used.Avg**
- The values of the following counters related to the downlink PRB usage of TRPs may increase in hyper cells or combined cells:

- **N.HyperCell.TRP.PRB.DL.Used.Avg**
- **N.HyperCell.TRP.PRB.DL.ExtendUsed.Avg**

Impact on NEs

This feature is implemented on the gNodeB.

Impact on Hardware Requirements

- Boards
All NR-capable main control boards and baseband processing units support this feature. For details, see the related BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- RF modules
TDD: All RF modules can be used.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
A license control item has been added to accommodate this feature.

Feature ID	Feature Name	Model	Sales Unit
FOFD-100252	AMC-based Precise Power Control Energy Saving	NR0SPCPWMG0 0	per Cell

- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellPowerSaving.PowerProtectionMargin	MOD NRDUCCELLPW RSVEALGO LST NRDUCCELLPW RSVEALGO	gNodeB	The parameter has been added to specify the power protection margin for initial transmission power reduction and retransmission power reduction.
Modified	NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch	MOD NRDUCCELLPW RSVEALGO LST NRDUCCELLPW RSVEALGO	gNodeB	The INIT_TRANS_POWER_REDUCE_SW and RETRANS_POWER_REDUCE_SW options have been added to specify whether to enable initial transmission power reduction and retransmission power reduction.
Modified	NRDUCellPowerSaving.PowerSavingType	MOD NRDUCCELLPW RSVEALGO LST NRDUCCELLPW RSVEALGO	gNodeB	The INIT_TRANS_POWER_REDUCE and RETRANS_POWER_REDUCE options have been added to specify the power reduction type for initial transmission and retransmission.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellPsPolicy Grp.ServTypeDynExitPwrSavingSw	ADD NRDUCCELLPSP OLICYGRP MOD NRDUCCELLPSP OLICYGRP LST NRDUCCELLPSP OLICYGRP	gNodeB	The INIT_TRANS_PWR_RD_EXIT_SW and RETRANS_PWR_RD_EXIT_SW options have been added to specify whether to enable service-type-based dynamic exit from power reduction for initial transmission and retransmission.

- Performance management
The following counters have changed.

Change Type	Counter Name	Function Subset	NE	Description
New	N.PowerSaving.PrecisePwrMgmt.Duration	DU Cell Algorithm Measurement	gNodeB	This counter has been added to measure the duration in which AMC-based Precise Power Control Energy Saving is in effect.

- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Initial transmission power reduction	INIT_TRAN S_POWER_REDUCE_SW option of the NRDUCellPwrSaveAlgo.PowerSavingsAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Service-type-based dynamic exit from initial transmission power reduction can take effect only when initial transmission power reduction is enabled.
Low-frequency TDD	Retransmission power reduction	RETRANS_POWER_REDUCE_SW option of the NRDUCellPwrSaveAlgo.PowerSavingsAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Service-type-based dynamic exit from retransmission power reduction can take effect only when retransmission power reduction is enabled.

- Mutually exclusive features/functions
- No impact
- Impacted features/functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch . <i>PowerSavingSwitch</i> parameter	<i>Energy Conservation and Emission Reduction</i>	When both initial transmission power reduction and timing carrier shutdown take effect, the former function does not deliver any gains. When both retransmission power reduction and timing carrier shutdown take effect, the former function does not deliver any gains.
Low-frequency TDD	Intelligent carrier shutdown	INTRA_GNB_MULTICARRIER_SHUTDOWN_SW or INTER_GNB_MULTICARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch . <i>PowerSavingSwitch</i> parameter	<i>Energy Conservation and Emission Reduction</i>	When both initial transmission power reduction and intelligent carrier shutdown take effect, the former function does not deliver any gains. When both retransmission power reduction and intelligent carrier shutdown take effect, the former function does not deliver any gains.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATS parameter	<i>Multi-RAT Coordinated Energy Saving</i>	When both initial transmission power reduction and RF module fast deep dormancy take effect, the former function does not deliver any gains. When both retransmission power reduction and RF module fast deep dormancy take effect, the former function does not deliver any gains.
Low-frequency TDD	RF module deep dormancy	RRU.DORMANCYSW set to ON or ON_DEEP_SUPERR	<i>Multi-RAT Coordinated Energy Saving</i>	Initial transmission power reduction does not take effect when RF module deep dormancy is in effect. Retransmission power reduction does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF module super dormancy	RRU.DORMANCY set to ON_SUPER or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	Initial transmission power reduction does not take effect when RF module super dormancy is in effect. Retransmission power reduction does not take effect when RF module super dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast carrier shutdown	- Capacity-layer cells: INTRA_GNB_MULTI_CARR_SD_SW option (intra-base-station multi-carrier scenarios) or INTER_GNB_MULTI_CARR_SD_SW option (inter-base-station multi-carrier scenarios) of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter selected FAST_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter selected - Basic-layer cells: FAST_CARRIER_SHUT_SERV_ASSUR_SW	<i>Energy Conservation and Emission Reduction</i>	When both initial transmission power reduction and fast carrier shutdown take effect, the former function does not deliver any gains. When both retransmission power reduction and fast carrier shutdown take effect, the former function does not deliver any gains.

RAT	Function Name	Function Switch	Reference	Description
		option of the NRDUCellAlgoSwitch. <i>PowerSavingExtSwitch</i> parameter selected		

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Speed turbo UE identification	gNodeBParam.StUeIdentSw	None	Initial transmission power reduction and retransmission power reduction fail to take effect in cells served by a base station, resulting in a decrease in the energy saving gains, when any of the following configuration conditions is met and the configured function takes effect:
Low-frequency TDD	gNodeB acceleration service application	• gNBAsAppParam.AppDomainName • gNBAsAppParam.Applpv4 • gNBAsAppParam.Applpv6	None	• Any option of the gNodeBParam.StUeIdentSw parameter is selected. • The gNBAsAppParam.AppDomainName parameter is set. • The gNBAsAppParam.Applpv4 parameter is set. • The gNBAsAppParam.Applpv6 parameter is set. The following methods can be used to address the preceding issue:

RAT	Function Name	Function Switch	Reference	Description
				• Select the SPEED_TURBO_PWR_SAVING_SW option of the gNBPwrSavingAlgo.PwrSavingAlgoSwitch parameter. Initial transmission power reduction and retransmission power reduction can take effect for cells served by the base station in some of the preceding scenarios within the period specified by the gNBPwrSavingAlgo.SpeedTurboPwrSavingStartTime and gNBPwrSavingAlgo.SpeedTurboPwrSavingEndTime parameters, increasing the energy saving gains. • Select the STATIC_STATIC_PWR_SAVING_SW option of the gNBPwrSavingAlgo.PwrSavingAlgoSwitch parameter. Initial

RAT	Function Name	Function Switch	Reference	Description
				transmission power reduction and retransmission power reduction can take effect for cells served by the base station in some of the preceding scenarios within the period specified by the gNBWPwrSavin gAlgo.StaticS TPwrSavStartTime and gNBWPwrSavin gAlgo.StaticS TPwrSavEndTime parameters, increasing the energy saving gains.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Energy Conservation and Emission Reduction*.

Requirement/Trouble Ticket Number

- FeR-IR20240717000223-0003
- FeR-IR20240730001143-0004
- FeR-IR20250213000671-0020
- FeR-IR20240729003544-0060
- FeR-IR20250102000700-0093

RAT

NR FDD, NR TDD

5.3.1.10 FOFD-101202 Smooth Video Interaction

5.3.1.10.1 Support for the Smooth Video Interaction Feature

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Video interaction services (such as OTT voice calls, OTT video calls, and OTT video conferences) have high requirements on uplink and downlink rates and latency. The video interaction experience guarantee function is introduced to further guarantee the experience of these services. This function is controlled by the **VIDEO_INTERACT_EXP_GUAR_SW** option of the **NRDUCellFeatureSw.QciBasedDiffServExpSw** parameter, and includes the following sub-functions.

- "Outer-loop robustness improvement for PDCCH aggregation level adaptation": This function takes effect at the cell level and cannot differentiate between different types of UEs. To address this issue, "QCI-specific outer-loop robustness improvement for PDCCH aggregation level adaptation" is introduced. This function is controlled by the **PUSCH_HARQ_DTX_PDCCH_ADJ_SW** option of the **gNBBDUSchParamGroup.ScheduleOptSwitch** parameter. After this function is enabled, when UEs with QCI-specific bearers perform downlink retransmissions, DTXs over the PUSCH can be used for outer-loop adjustment of the PDCCH aggregation level, accelerating aggregation level convergence and decreasing the downlink IBLER and residual block error rate (RBLER) for these UEs.
- QCI-specific uplink MCS adjustment adaptation: To improve the link reliability of UEs with QCI-specific bearers, "QCI-specific scheduling based on the uplink target IBLER" and "QCI-specific uplink MCS selection optimization" can be enabled. In this way, selected uplink MCS indexes are more conservative. However, if UEs are under weak coverage or the traffic volume exceeds the air interface capability, a conservative MCS index will cause UE-side data not to be scheduled in a timely manner, thereby affecting user experience. To address this issue, uplink MCS adjustment adaptation is introduced. This function is controlled by the **UL_MCS_ADJ_ALGO_ADAPT_SW** option of the **gNBBDUSchParamGroup.ScheduleOptSwitch** parameter. After this function is enabled, the following rules apply to UEs with QCI-specific bearers:
 - If the estimated uplink latency of UEs is greater than the value of **gNBBDUSchParamGroup.UlMcsAdjAdaptThld**, "QCI-specific scheduling based on the uplink target IBLER" and "QCI-specific uplink MCS selection optimization" do not take effect for the UEs, thereby improving service experience.
 - If the estimated uplink latency of UEs is less than the value of **gNBBDUSchParamGroup.UlMcsAdjAdaptThld** minus a fixed offset, "QCI-

specific scheduling based on the uplink target IBLER" and "QCI-specific uplink MCS selection optimization" are allowed to take effect for the UEs, thereby improving link reliability.

- Differentiated FDM between PUSCH data and DMRS: "FDM between PUSCH data and DMRS" affects uplink demodulation performance and increases the BER, thereby degrading the experience of delay-sensitive services. To address this issue, "differentiated FDM between PUSCH data and DMRS" is introduced. This function is controlled by the **DIFF_PUSCH_DMRS_FDM_SW** option of the **gNBDUSchParamGroup.ScheduleOptSwitch** parameter. After this function is enabled, "FDM between PUSCH data and DMRS" does not take effect for UEs with QCI-specific bearers when any of the following conditions is met, thereby improving uplink demodulation performance, reducing the uplink BER, and shortening the delay for these UEs:
 - Considering the user data volume, data transmission cannot be completed at one time by enabling "FDM between PUSCH data and DMRS".
 - Considering the user data volume, data transmission can be completed at one time without enabling "FDM between PUSCH data and DMRS".

Implementation

For details, see *Service-differentiated Experience Guarantee*. The specific sections are listed in the change history of the document.

Impact on the Network

Benefits:

- After "QCI-specific outer-loop robustness improvement for PDCCH aggregation level adaptation" is enabled, (1) the PDCCH aggregation level increases, the PDCCH BLER decreases, and the downlink IBLER and RBLER decrease for UEs with QCI-specific bearers, and (2) the associated 5QI-level average downlink delay at the MAC layer decreases for the QCI-specific bearers.
- After QCI-specific uplink MCS adjustment adaptation is enabled, the uplink spectral efficiency of UEs with QCI-specific bearers increases.
- After differentiated FDM between PUSCH data and DMRS is enabled, the uplink BER of UEs with QCI-specific bearers decreases, thereby reducing the associated 5QI-level average uplink delay at the MAC layer for QCI-specific bearers.
- If the **gNBQciBearer.ServiceType** parameter is set to **VIDEO_INTERACTIVE** for a QCI of an SA UE, the QCI can be configured as a flexible performance counter object. After the functions of the Smooth Video Interaction feature are enabled, the uplink and downlink RAN delay distributions for specific services of a flexible object change for the SA UE due to the shortened uplink and downlink RAN delays. To configure a QCI as a flexible performance counter object, configure a **gNBFlexCounterObjCfg** MO with (1) **gNBFlexCounterObjCfg.FlexCounterObjType** set to **NRCELL_FLEX_AND_NRDUCCELL_FLEX** or **NRDUCCELL_FLEX**, and (2) **gNBFlexCounterObjCfg.Qci** set to this QCI. For details, see the section "Custom Object Type" in "Object Type" of *Performance Management*.

Impacts:

- After "QCI-specific outer-loop robustness improvement for PDCCH aggregation level adaptation" is enabled, the following impacts apply:
 - Assume that the average uplink or downlink PDCCH CCE aggregation level of UEs with QCI-specific bearers is low before this function takes effect. After this function takes effect, (1) the average uplink or downlink PDCCH CCE aggregation level of UEs with QCI-specific bearers increases, (2) the uplink or downlink BLER decreases for these UEs, and (3) the uplink or downlink rate may decrease for specific QCIs/5QIs in that the aggregation level increase may cause a decrease in the uplink or downlink CCE allocation success rate.
 - Assume that the average uplink or downlink PDCCH CCE aggregation level of UEs with QCI-specific bearers is high before this function takes effect. After this function takes effect, (1) the average uplink or downlink PDCCH CCE aggregation level of UEs with QCI-specific bearers decreases, and (2) the uplink or downlink rate may increase for specific QCIs/5QIs in that the aggregation level decrease may cause an increase in the uplink or downlink CCE allocation success rate.
- After QCI-specific uplink MCS adjustment adaptation is enabled, the uplink MCS index and uplink IBLER increase.
- After differentiated FDM between PUSCH data and DMRS is enabled, the value of **Cell Uplink Average Throughput (DU)** decreases.

Impact on NEs

This feature is implemented on the gNodeB.

Impact on Hardware Requirements

- Boards
All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- RF modules
All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
This feature is controlled by the following new license control item.

Feature ID	Feature Name	Model	Sales Unit
FOFD-101 202	Smooth Video Interaction	NR0SSMTHVN10	per Cell

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellFeatureSw.QciBasedDiffServExpSw	MOD NRDUCELLFEATURESW LST NRDUCELLFEATURESW	gNodeB	The VIDEO_INTERACT_EXP_GUAR_SW option has been added to specify whether to enable video interaction experience guarantee.
Modified	gNBDUSchParamGroup.ScheduleOptSwitch	ADD GNBUSCHPARAMGROUP MOD GNBUSCHPARAMGROUP LST GNBUSCHPARAMGROUP	gNodeB	• The PUSCH_HARQ_DTX_PDCCH_ADJ_SW option has been added to specify whether to enable "QCI-specific outer-loop robustness improvement for PDCCH aggregation level adaptation". • The UL_MCS_ADJ_ALGO_ADAPT_SW option has been added to specify whether to enable QCI-specific uplink MCS adjustment adaptation. • The DIFF_PUSCH_DMRS_FDM_SW option has been added to specify whether to enable differentiated FDM between PUSCH data and DMRS.

Change Type	Parameter ID	MML Command	NE	Description
New	gNBDUSchParamGroup.UIMcsAdaptThld	ADD GNBDUSCHP ARAMGROUP MOD GNBDUSCHP ARAMGROUP LST GNBDUSCHP ARAMGROUP	gNodeB	This parameter has been added to specify the threshold used in uplink MCS adjustment adaptation.

- Performance management

The following function subsets have changed.

Change Type	Function Subset	NE	Description
New	Cell Flexible Object Downlink RAN Delay Measurement	gNodeB	This function subset has been added to measure the downlink RAN delay distribution of a flexible object in a cell.
New	Cell Flexible Object Uplink RAN Delay Measurement	gNodeB	This function subset has been added to measure the uplink RAN delay distribution of a flexible object in a cell.

- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PUSCH DTX detection	PUSCH_ACK_DTX_DETECT_SW option of the NRDUCellPusch.UlPuschAlgoSwitch parameter	<i>Downlink Scheduling</i>	"QCI-specific outer-loop robustness improvement for PDCCH aggregation level adaptation" requires "PUSCH DTX detection" to be enabled.
Low-frequency TDD	QCI-specific scheduling based on the uplink target IBLER	UL_IBLER_QCI_SW option of the gNBDUSchParamGroup.ScheduleOptSwitch parameter	<i>Service-differentiated Experience Guarantee</i>	QCI-specific uplink MCS adjustment adaptation requires "QCI-specific scheduling based on the uplink target IBLER" or "QCI-specific uplink MCS selection optimization" to be enabled.
Low-frequency TDD	QCI-specific uplink MCS selection optimization	SCH_MCS_SELECT_OPT_SW option of the gNBDUSchParamGroup.ScheduleOptSwitch parameter	<i>Service-differentiated Experience Guarantee</i>	
Low-frequency TDD	FDM between PUSCH data and DMRS	NRDUCellPusch.PuschDmrsFdmSwitch	<i>Uplink Scheduling</i>	"Differentiated FDM between PUSCH data and DMRS" requires "FDM between PUSCH data and DMRS" to be enabled.

- Mutually exclusive features/functions
No impact
- Impacted features/functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Outer-loop robustness improvement for PDCCH aggregation level adaptation	PUSCH_HARQ_DTX_AGG_LVL_OPT_SW option of the NRDUCellPdcch.h.PdcchAlgoEnhSwitch parameter	<i>Channel Management</i>	When "QCI-specific outer-loop robustness improvement for PDCCH aggregation level adaptation" is enabled together with "outer-loop robustness improvement for PDCCH aggregation level adaptation", "outer-loop robustness improvement for PDCCH aggregation level adaptation" takes effect at the cell level.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Service-differentiated Experience Guarantee*.

Requirement/Trouble Ticket Number

FeR-IR20250724000484-0001

FeR-IR20250528001857-0075

FeR-IR20250528001857-0082

RAT

NR FDD, NR TDD

5.3.1.11 FOFD-101225 L/NR Inter-band Deep Coordination

5.3.1.11.1 Added L/NR Inter-band Deep Coordination

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

L/NR Inter-band Deep Coordination enables power sharing between several LTE cells and two NR cells sharing the same RF channel.

This function requires the following options on both the LTE and NR sides to be selected:

- LTE: **LNR_INTER_BAND_DEEP_COORD_SW** option of the **CellDynPowerSharing.DynamicPowerSharingSwitch** parameter
- NR: **LNR_INTER_BAND_DEEP_COORD_SW** option of the **NRDUCellResShrCfg.DynPwrShrAlgoSwitch** parameter

If an AAU5685md is used and the **NR_PWR_SHARE_CAP_ENH_SW** option of the **NRDUCellResShrCfg.DynPwrShrAlgoSwitch** parameter is selected, the value of **NRDUCellTrpDynPwrShr.MaxPowerSharingRate** can reach **200** and takes effect, increasing the power sharing percentage.

LampSite base stations do not support this feature.

Implementation

For details, see *LTE and NR Power Allocation*. The specific sections are listed in the change history of the document.

Impact on the Network

L/NR Inter-band Deep Coordination gains are listed in [Table 5-13](#).

Table 5-13 L/NR Inter-band Deep Coordination gains

Scenario Description	Gain on the NR Side	Gain on the LTE Side
Power sharing between a single LTE cell and a single NR cell	• The average downlink UE throughput in the NR cell increases by 2% to 8%. • The average downlink throughput of UEs in the overlapping areas in the NR cell increases by 5% to 10%.	• The average downlink UE throughput in the LTE cell increases by 0% to 4%. • The average downlink throughput of CEUs in the LTE cell increases by 0% to 5%.
Power sharing between multiple LTE cells and a single NR cell	• The average downlink UE throughput in the NR cell increases by 2% to 10%. • The average downlink throughput of UEs in the overlapping areas in the NR cell increases by 5% to 15%.	• The average downlink UE throughput in LTE cells increases by 2% to 17%. • The average downlink throughput of CEUs in LTE cells increases by 5% to 25%.

Scenario Description	Gain on the NR Side	Gain on the LTE Side
Power sharing between multiple LTE cells and two NR cells	• The average downlink UE throughput in NR cells increases by 2% to 10%. • The average downlink throughput of UEs in the overlapping areas in NR cells increases by 5% to 15%.	• The average downlink UE throughput in LTE cells increases by 0% to 17%. • The average downlink throughput of CEUs in LTE cells increases by 0% to 25%.

This function has the following impacts on LTE and NR network performance:

- This function increases the power consumed by the PDSCH, possibly increases the average MCS index, and reduces the downlink PRB usage.
- The impact of this function on the downlink packet delay (LTE: **L.Traffic.DL.PktDelay.Time**; NR: **N.QoS.DL.PktDelaygNBDU.Time**) is related to services. When the tail packet percentage (the proportion of the last TTI before the downlink buffer is emptied) is low, the downlink packet delay decreases. When the tail packet percentage is high, the downlink packet delay increases.
- When this function is enabled, the average uplink and downlink UE throughput may decrease for cells that constantly share out power.
 - LTE: The average downlink UE throughput in the cell is indicated by **User Downlink Average Throughput**, and the average uplink UE throughput in the cell is indicated by **User Uplink Average Throughput**.
 - NR: The average downlink UE throughput in the cell is indicated by **User Downlink Average Throughput (DU)**, and the average uplink UE throughput in the cell is indicated by **User Uplink Average Throughput (DU)**.
- Function activation or deactivation involves the adjustment of system parameters, which causes service rate fluctuations for approximately 2 minutes and affects the cell throughput. Therefore, it is recommended that you enable this function when the traffic is light.
- When this function is enabled, the RF module power utilization increases. Base stations consume more power, but the total power consumption does not exceed the maximum power consumed when all cells work with a full load.
- This function requires information exchanges between cells. This may increase the CPU load of baseband processing units (**VS.BBUBoard.CPULoad.Mean**) by less than 5%.
- This function increases the instantaneous power used for PDSCH, which may increase interference to neighboring cells. It is recommended that this function be enabled in contiguous coverage areas. If the average cell throughput decreases, or if the interference to neighboring cells significantly increases after this function is enabled, you can decrease the maximum power sharing rate to counteract the negative impacts. (Setting the maximum power sharing rate to half of its recommended value is recommended.) If the gains

are not noticeable, you can increase the value of the maximum power sharing rate to achieve higher function gains.

- LTE: The maximum power sharing rate is specified by the **CellDynPowerSharing.MaxPowerSharingRate** parameter.
- NR: The maximum power sharing rate is specified by the **NRDUCellTrpDynPwrShr.MaxPowerSharingRate** parameter.
- It is recommended that this function be disabled before downlink peak rate tests on an eNodeB or gNodeB.
- When the **LNR_POWER_SHARING_RATIO_OPT_SW** option of the **CellDynPowerSharing.DynamicPowerSharingSwitch** parameter is selected, as scheduling is performed in advance on the LTE side, the random access response success rate slightly decreases, and the UE access delay slightly increases. The average uplink and downlink UE throughputs slightly decrease as the LTE cell has its power shared.

Impact on NEs

This feature is implemented on the eNodeB and the gNodeB.

Impact on Hardware Requirements

- Boards
 - Main control boards: all main control boards that support 8T8R and eCPRI-capable main control boards that support 32T32R or higher
 - Baseband processing units: UBBPe and later baseband processing units
- RF modules

RF modules that meet both of the following conditions support this function:

 - The RF modules are 8T8R RRUs or eCPRI-capable 32T32R or higher AAUs.
 - The RF modules must support LTE and NR concurrency.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

This feature is controlled by the following new license control items.

RAT	Feature ID	Feature Name	Model	Sales Unit
LTE TDD	TDLCOFD-221611	L/NR Inter-band Deep Coordination (LTE TDD)	LT4SNTBADC00	per Cell
NR TDD	FOFD-101225	L/NR Inter-band Deep Coordination	NR0SNTBACD00	per Cell

- Configuration management

The following MML commands have changed.

Change Type	MML Command	NE	Description
Modified	DSP NRDUCELLT RPDYNPWR SHR	gNodeB	The LNR Inter-band Deep Coord Valid Status field has been added to the output of this command.
Modified	DSP CELLDYNP WRSHASTATUS	eNodeB	The LNR Inter-Band Deep Coord Status field has been added to the output of this command.

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	CellDynPowerSharing.DynPowerSharingSwitch	MOD CELLDYNPOWERSHARING LST CELLDYNPOWERSHARING	eNodeB	The LNR_INTER_BAND_DEEP_COORD_SW option has been added to specify whether to enable L/NR Inter-band Deep Coordination.
Modified	NRDUCellResShrCfg.DynPwrShrAlgoSwitch	MOD NRDUCELLRESHRCFG LST NRDUCELLRESHRCFG	gNodeB	The LNR_INTER_BAND_DEEP_COORD_SW option has been added to specify whether to enable L/NR Inter-band Deep Coordination.
Modified	NRDUCellResShrCfg.DynPwrShrAlgoSwitch	MOD NRDUCELLRESHRCFG LST NRDUCELLRESHRCFG	gNodeB	The NR_PWR_SHARE_CAP_ENH_SW option has been added to specify whether to increase the maximum proportion of shared power received by NR carriers.

- Performance management
No impact

- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.
The prerequisite functions on the LTE side are listed below.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	Dynamic Power Sharing Between LTE Carriers	LTE_DYN_POWER_SHARING_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter	<i>Dynamic Power Sharing Between LTE Carriers</i>	L/NR Inter-band Deep Coordination requires LTE dynamic power sharing to be enabled. LTE cells enabled with Dynamic Power Sharing Between LTE Carriers automatically form an LTE power sharing group when the conditions for triggering Dynamic Power Sharing Between LTE Carriers are met. When the LTE cells included in the L/NR Inter-band Deep Coordination group are the same as those included in the LTE power sharing group, both of these two functions take effect. Otherwise, only Dynamic Power Sharing Between LTE Carriers takes effect, and L/NR Inter-band Deep Coordination does not take effect. For details about Dynamic

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentat ion)	Description
				Power Sharing Between LTE Carriers, see <i>Dynamic Power Sharing Between LTE Carriers</i> .

The prerequisite functions on the NR side are listed below.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCellAI goSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	L/NR Inter-band Deep Coordination requires NR inter-carrier dynamic power sharing to be enabled. Cells enabled with NR inter-carrier dynamic power sharing automatically form an NR power sharing group when the conditions for triggering NR inter-carrier dynamic power sharing are met. When the LTE cells included in the L/NR Inter-band Deep Coordination group are the same as those included in the NR power sharing group, both of these two functions take effect. Otherwise, only NR inter-carrier dynamic power sharing takes effect, and L/NR Inter-band Deep Coordination does not take effect.

- Mutually exclusive features/functions
The following mutually exclusive relationships have been added.
The mutually exclusive functions on the LTE side are listed below.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentat ion)	Description
LTE TDD	LTE TDD and NR Flash Dynamic Power Sharing	LTE_NR_DYN_POWER_SHARING_SWITCH option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	L/NR Inter-band Deep Coordination cannot be enabled together with LTE TDD and NR Flash Dynamic Power Sharing.
LTE TDD	PHICH duration	PHICHCfg.PhighDuration set to EXTENDED	<i>Physical Channel Resource Management</i>	L/NR Inter-band Deep Coordination cannot be enabled together with PHICH duration.
LTE TDD	PDCCH spare power allocation optimization	CellPdcchAlgorithm.PdcchSparePowerAllocationStrage	<i>Physical Channel Resource Management</i>	L/NR Inter-band Deep Coordination cannot be enabled together with PDCCH spare power allocation optimization.
LTE TDD	PDCCH power check	CellPdcchAlgorithm.PdcchPowerCheckSwitch	None	L/NR Inter-band Deep Coordination cannot be enabled together with PDCCH power check.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	Soft split resource duplex	None	<i>Soft Split Resource Duplex (TDD)</i>	If Cell.MultiCellShareMode is set to SPLIT for a cell and the SsrdCellGroupMO with SsrdCellGroup.PrimaryLocalCellID or SsrdCellGroup.SsrdLocalCellID set to the value of Cell.LocalCellId of the cell, L/NR Inter-band Deep Coordination must be disabled.
LTE TDD	CSPC	CellCspcPara.CellCspcSwitch	<i>CSPC</i>	L/NR Inter-band Deep Coordination cannot be enabled together with CSPC.
LTE TDD	Enhanced coordinated scheduling based power control	CellCspcPara.CelleCspcSwitch	<i>Massive MIMO Enhancements (TDD)</i>	L/NR Inter-band Deep Coordination cannot be enabled together with enhanced coordinated scheduling based power control.
LTE TDD	PHICH inner-loop power control	PhichInnerLoopPcSwitch option of the CellAlgoSwitch.DIPcAlgoSwitch parameter	<i>Power Control</i>	L/NR Inter-band Deep Coordination cannot be enabled together with PHICH inner-loop power control.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	PDCCH SDMA enhancement	CellPdcchAlgorithm.PdcchSdmaWeightType set to POWER_SHARING_AMONG_3_BEAMS , POWER_SHARING_AMONG_4_BEAMS , or POWER_SHARING_AMONG_5_BEAMS	<i>Massive MIMO Basics (TDD)</i>	If PDCCH SDMA (controlled by the MASSIVE_MIMO_PDCCH_SDMA_SW option of CellPdcchAlgorithm.SplitBeamPdcchSdmaSw) and L/NR Inter-band Deep Coordination are both enabled, PDCCH SDMA enhancement cannot be enabled.
LTE TDD	B-Trunc mode	eNodeBFunction.eNodeBMode set to BTRUNC_MODE	None	L/NR Inter-band Deep Coordination cannot be enabled together with B-Trunc mode.
LTE TDD	Intelligent Multi-Beam of 8T8R	None	<i>Intelligent Multi-Beam of 8T8R (TDD)</i>	If Cell.TxRxMode is set to 8T8R and a SectorSplitCell or SectorSplitCellExt MO has been configured, L/NR Inter-band Deep Coordination cannot be enabled.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	Flag of Multi-RRU Cell	Cell.MultiRruCellFlag set to BOOLEAN_TRUE	<i>SFN</i>	If Cell.TxRxMode is set to 8T8R and Cell.MultiRruCellFlag is set to BOOLEAN_TRUE , L/NR Inter-band Deep Coordination cannot be enabled.

The mutually exclusive functions on the NR side are listed below.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	If L/NR Inter-band Deep Coordination is enabled together with High-speed Railway Superior Experience, power estimation may be inaccurate. Therefore, they cannot be both enabled.
Low-frequency NR TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	L/NR Inter-band Deep Coordination cannot be enabled for hyper cells.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentat ion)	Description
Low-frequenc y NR TDD	Fusion Cell	NRDUCell.N rDuCellNet workingMode set to FUSION_CEL L	<i>Fusion Cell (Low-Frequency TDD)</i>	L/NR Inter-band Deep Coordination cannot be enabled together with Fusion Cell.
Low-frequenc y NR TDD	Bi-Carrier	BI_CARRIER_SW option of the NRDUCellFe atureSw.Mul tiCarrierSwit ch parameter	None	L/NR Inter-band Deep Coordination cannot be enabled together with Bi-Carrier.
Low-frequenc y NR TDD	LTE TDD and NR Flash Dynamic Power Sharing	LTE_NR_DYN _POWER_SH ARING_SW option of the NRDUCellAl goSwitch.DynPowerShari ngSwitch parameter	<i>LTE and NR Power Allocation</i>	L/NR Inter-band Deep Coordination cannot be enabled together with LTE TDD and NR Flash Dynamic Power Sharing.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentat ion)	Description
Low-frequenc y NR TDD	NR smart power lock	AVERAGE_P OWER_CON TROL_SW option of the NRDUCellS martPwrLoc k . <i>SmartPow erLockSw</i> parameter	None	If enhanced NR power sharing is enabled by selecting the NR_PWR_SHARE _CAP_ENH_SW option of NRDUCellResSh rCfg . <i>DynPwrShr AlgoSwitch</i> , the power sharing capability exceeds 100%. In this case, the power is more likely to be limited when smart power lock is also enabled, affecting cell performance. Therefore, these two functions cannot be both enabled.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentat ion)	Description
Low-frequenc y NR TDD	NR smart power scale	AVERAGE_3 DEIRP_CON TROL_SW option of the NRDUCellS martPwrLoc k.SmartPow erLockSw parameter	None	If enhanced NR power sharing is enabled by selecting the NR_PWR_SHARE _CAP_ENH_SW option of NRDUCellResSh rCfg.DynPwrShr AlgoSwitch , the power sharing capability exceeds 100%. In this case, the power is more likely to be limited when smart power lock is also enabled, affecting cell performance. Therefore, these two functions cannot be both enabled.

- Impacted features/functions
The following impact relationships have been added.
The function impacts on the NR side are listed below.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	Scalable bandwidth	None	<i>Cell Management</i>	10 MHz LTE cells that use default channel configurations seldom share power with NR cells as these LTE cells have only few power resources left after power reservation for paging channels, control channels, and traffic bursts.
LTE TDD	DL 256QAM	CellAlgoSwitch.DL256QAMAlgoSwitch	<i>Modulation Schemes</i>	When UEs for which DL 256QAM has taken effect are scheduled in a TTI on the LTE side, LTE does not use the power shared by NR.
LTE TDD	RF channel dynamic muting	RF_CHN_DYNAMIC_MUTING_SW option of the CellRfChnDynamicMuting.RfChannelDynamicMutingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	L/NR Inter-band Deep Coordination does not take effect when RF channel dynamic muting is in effect.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SW option of the CellPwrSavingAlgo.PwrSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Energy saving based on flexible frequency-domain scheduling is not performed in subframes where received shared power is used.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	Intelligent power-off of carriers in the same coverage	CellShutdownSwitch set to ON	<i>Energy Conservation and Emission Reduction</i>	• A cell with the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter deselected does not participate in L/NR Inter-band Deep Coordination if it has entered the energy saving state. The cell will attempt to participate in L/NR Inter-band Deep Coordination every 10 minutes. • When the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter is selected, L/NR Inter-band Deep Coordination can take effect, but the energy saving gains may decrease.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	Cell Switch-off Based on Automatic Co-coverage Identification	CellShutdownSwitch set to ON_CO_COV_AUTO_IDENT	<i>Energy Conservation and Emission Reduction</i>	• A cell with the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter deselected does not participate in L/NR Inter-band Deep Coordination if it has entered the energy saving state. The cell will attempt to participate in L/NR Inter-band Deep Coordination every 10 minutes. • When the LTE_CARR_SD_PWR_SHR_OPT_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter is selected, L/NR Inter-band Deep Coordination can take effect, but the energy saving gains may decrease.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentat ion)	Description
LTE TDD	Multi-carrier coordinated energy saving	CellShutdown . CellShutdo wnSwitch set to ON_MULTI_CARRIER_HI ER_SHUTDO WN , ON_MUL_CA RR_HIER_M ANUAL_SHU TDN , or ON_CARRIE R_PRECISE_S HUTDOWN	<i>Energy Conservation and Emission Reduction</i>	• A cell with the LTE_CARR_SD_PWR_SHR_O PT_SW option of the CellDynPowe rSharing.Dyn amicPowerSh aringSwitch parameter deselected does not participate in L/NR Inter-band Deep Coordination if it has entered the energy saving state. The cell will attempt to participate in L/NR Inter-band Deep Coordination every 10 minutes. • When the LTE_CARR_SD_PWR_SHR_O PT_SW option of the CellDynPowe rSharing.Dyn amicPowerSh aringSwitch parameter is selected, L/NR Inter-band Deep Coordination can take effect, but the energy saving gains may decrease.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	Automatic cell shutdown	CellAutoShutdown.CellAutoShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	• A cell with the LTE_CARR_SD_PWR_SHR_OPTION_SW option of the CellDynamicPowerSharing.DynamicPowerSharingSwitch parameter deselected does not participate in L/NR Inter-band Deep Coordination if it has entered the energy saving state. The cell will attempt to participate in L/NR Inter-band Deep Coordination every 10 minutes. • When the LTE_CARR_SD_PWR_SHR_OPTION_SW option of the CellDynamicPowerSharing.DynamicPowerSharingSwitch parameter is selected, L/NR Inter-band Deep Coordination can take effect, but the energy saving gains may decrease.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	Dynamic carrier shutdown phase 2	CellPwrSavingsAlgo.CarrierDynShutdownPh2Switch set to CAPACITY_CELL	<i>Energy Conservation and Emission Reduction</i>	• A cell with the LTE_CARR_SD_PWR_SHR_OPTION_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter deselected does not participate in L/NR Inter-band Deep Coordination if it has entered the energy saving state. The cell will attempt to participate in L/NR Inter-band Deep Coordination every 10 minutes. • When the LTE_CARR_SD_PWR_SHR_OPTION_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter is selected, L/NR Inter-band Deep Coordination can take effect, but the energy saving gains may decrease.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE TDD	RF channel intelligent shutdown	CellRfShutdown.RfShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	Cells that have entered the energy saving state cannot participate in L/NR Inter-band Deep Coordination. These cells will attempt to participate in L/NR Inter-band Deep Coordination every 10 minutes.
LTE TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATSW parameter	<i>Multi-RAT Coordinated Energy Saving</i>	RF module fast deep dormancy takes precedence over L/NR Inter-band Deep Coordination when both functions are enabled. L/NR Inter-band Deep Coordination does not take effect when RF module fast deep dormancy is in effect.

The function impacts on the NR side are listed below.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	DL 256QAM	NRDUCellAIgoSwitch.DL256QamSwitch	<i>Modulation Schemes</i>	When UEs for which DL 256QAM has taken effect are scheduled in a slot on the NR side, NR does not use the power shared by LTE.
Low-frequency NR TDD	Power saving scheduling	POWER_SAVING_SCH_OPTION option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When L/NR Inter-band Deep Coordination and power saving scheduling are both enabled, the effects of power saving scheduling are affected. It is recommended that L/NR Inter-band Deep Coordination not be enabled with this function when the traffic of LTE and NR cells is light.
Low-frequency NR TDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SWITCH option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When L/NR Inter-band Deep Coordination takes effect together with energy saving based on flexible frequency-domain scheduling, the benefits of both functions decrease.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentat ion)	Description
Low-frequency NR TDD	Multi-position PDCCH scheduling	ANY_OCCASION_PDCCH_SW option of the NRDUCellAI goSwitch.LowLatencySwitch parameter	None	When L/NR Inter-band Deep Coordination and multi-position PDCCH scheduling are both enabled, NR cells cannot estimate the slot and power resources to be scheduled for UEs for which multi-position PDCCH scheduling takes effect. Consequently, NR cells can receive shared power, and do not share power to other cells.
Low-frequency NR TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAI goSwitch.CompSwitch parameter	<i>CoMP</i>	When L/NR Inter-band Deep Coordination and intra-base-station DL CoMP are both enabled, the power sharing gains on the NR side may decrease.
Low-frequency NR TDD	Co-MM	INTRA_GNB_SU_COMM_SW option of the NRDUCellCo llabServ.Mul tiCellMIMOS witch parameter	<i>Co-MM (Low-Frequency TDD)</i>	When L/NR Inter-band Deep Coordination and Co-MM are both enabled, the power sharing gains on the NR side may decrease.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentat ion)	Description
Low-frequenc y NR TDD	Coordinated interference management	INTRA_GNB_DL_CS_CBF_SW or INTER_GNB_DL_CS_CBF_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>Coordinated Interference Management (Low-Frequency TDD)</i>	When L/NR Inter-band Deep Coordination and coordinated interference management are both enabled, the power sharing gains on the NR side may decrease.
Low-frequenc y NR TDD	Power reduction for CCUs	None	<i>Power Control</i>	When L/NR Inter-band Deep Coordination and power reduction for CCUs are both enabled, the power sharing gains on the NR side may decrease.
Low-frequenc y NR TDD	Interference coordination between public and dedicated networks	HIGH_SPEED_INTRF_AVOID_SW or INTRF_COORD_ALWAYS_WORKING_SW option of the NRCCellAlgoSwitch.HighSpeedDedPolicySw parameter	<i>High Speed Mobility</i>	When L/NR Inter-band Deep Coordination and "interference coordination between public and dedicated networks" are both enabled, the power sharing gains on the NR side may decrease.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentat ion)	Description
Low-frequenc y NR TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAI goSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	• If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellRes ShrCfg.DynP wrShrAlgoSwitch parameter is deselected for a carrier, the carrier exits a power sharing group when the effective period of timing carrier shutdown arrives, and power sharing does not take effect for the carrier during this period. • If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellRes ShrCfg.DynP wrShrAlgoSwitch parameter is selected for a carrier, the carrier exits a power sharing group once it is shut down. When a carrier enters or exits the energy saving state, the coverage of SSB,

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentat ion)	Description
				CSI-RS, PDCCH, and control messages changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	Intelligent carrier shutdown	INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	- If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResShrCfg.DynPwrShrAlgoSwitch parameter is deselected for a carrier, the carrier exits a power sharing group when the effective period of intelligent carrier shutdown arrives, and power sharing does not take effect for the carrier during this period. - If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResShrCfg.DynPwrShrAlgoSwitch parameter is selected for a carrier, the carrier exits a power sharing group once it is shut down. When a carrier enters or exits the energy saving state, the

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentat ion)	Description
				coverage of SSB, CSI-RS, PDCCH, and control messages changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentat ion)	Description
Low-frequenc y NR TDD	RF channel intelligent shutdown	RF_SHUTDO WN_SW option of the NRDUCellAI goSwitch.PowerSavingS witch parameter	<i>Energy Conservation and Emission Reduction</i>	• If the PWR_SHR_IN _PWR_SAVE_ SW option of the NRDUCellRes ShrCfg.DynP wrShrAlgoSw itch parameter is deselected for a cell, power sharing is not performed on the cell within the period specified for RF channel intelligent shutdown. • If the PWR_SHR_IN _PWR_SAVE_ SW option of the NRDUCellRes ShrCfg.DynP wrShrAlgoSw itch parameter is selected for a cell, power sharing is not performed on the cell when the cell is in the energy saving state. When a cell enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control messages

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentat ion)	Description
				changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	Fast carrier shutdown	- Capacity-layer cells: - In intra-base-station multi-carrier scenarios: INTRA_GNB_MULTI_CARR_SD_SW option of the NRDUCellCfg.PwrSavingsSwitch parameter and FAST_CARRIER_SHUTDOWN_SW option of the NRDUCellCfg.PwrSavingsExtSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	- If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResShrCfg.DynPwrShrAlgoSwitch parameter is deselected for a carrier, the carrier exits a power sharing group when the effective period of fast carrier shutdown arrives, and power sharing does not take effect for the carrier during this period. - If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResShrCfg.DynPwrShrAlgoSwitch parameter is selected for a carrier, the carrier exits a power sharing group once it is shut down. When a carrier enters or exits the energy saving state, the coverage of SSB,

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentat ion)	Description
		- In inter-base-station multi-carrier scenarios: INTER_GNB_MULTI_CARRIER_SD_SWITCH option of the NRDU CellAI goSwitch.PowerSaverSwitch parameter and FAST_CARRIER_SHUTDOWN_SWITCH option of the NRDU CellAI goSwitch.PowerSaverSwitchExtSwitch parameter • Basic-layer cells: FAST_CARRIER_SHUT_		CSI-RS, PDCCH, and control messages changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
		SERV_ASSUR_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter		
Low-frequency NR TDD	TTI-level channel shutdown	NRDUCellPowerSaving.ComChnDynShutdownSwitch parameter and TTI_RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When L/NR Inter-band Deep Coordination takes effect together with TTI-level channel shutdown, the energy saving benefits of TTI-level channel shutdown decrease and downlink UE throughput may decrease.
Low-frequency NR TDD	PDCCH aggregation level reduction	PDCCH_AGG_LVL_COMP_R_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	When L/NR Inter-band Deep Coordination and PDCCH aggregation level reduction are both enabled, only L/NR Inter-band Deep Coordination takes effect.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	DL 1024QAM	DL_1024QAM_SW option of the NRDUCellDLAmc.DLModulationSchemeSw parameter	<i>Modulation Schemes</i>	If there are UEs for which DL 1024QAM is being used, the proportion of occasions where L/NR Inter-band Deep Coordination and LTE and NR power sharing optimization take effect decreases.
Low-frequency NR TDD	NR low power consumption mode	LOW_PWR_MODE_SW option of the NRDUCellAIalgoSwitch.PowerSavingSwitch parameter	<i>Green Site</i>	Only low power consumption mode takes effect when low power consumption mode and L/NR Inter-band Deep Coordination are both enabled and a customized backup power saving policy is used in low power consumption mode.
Low-frequency NR TDD	Maximum SSB power offset	NRDUCellTrpBeam.MaxSsbPwrOffset	<i>Power Control</i>	When the SSB power offset is set to a value greater than 3 dB and L/NR Inter-band Deep Coordination is in effect, the SSB power aggregation capability decreases.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	CCE partial symbol adaptation	CCE_PARTIAL_SYM_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgorithmSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	When L/NR Inter-band Deep Coordination takes effect, the benefits of CCE partial symbol adaptation will decrease.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATSW parameter	<i>Multi-RAT Coordinated Energy Saving</i>	- If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResShrCfg.DynPwrShrAlgoSwitch parameter is deselected for a carrier, the carrier exits a power sharing group when the effective period of RF module fast deep dormancy arrives, and power sharing does not take effect for the carrier during this period. - If the PWR_SHR_IN_PWR_SAVE_SW option of the NRDUCellResShrCfg.DynPwrShrAlgoSwitch parameter is selected for a carrier, the carrier exits a power sharing group once it is shut down. When a carrier enters or exits the energy saving state, the

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentat ion)	Description
				coverage of SSB, CSI-RS, PDCCH, and control information changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *LTE and NR Power Allocation*.

Requirement/Trouble Ticket Number

FeR-IR20250321002349-0001

FeR-IR20250321002349-0002

RAT

NR TDD

5.3.1.12 FOFD-101240 Ultra Uplink

5.3.1.12.1 Uplink Waveform Adaptation (for High Frequency Bands Only)

NEs Involved

5900 series base stations

Feature/Function Description

The NR system supports Cyclic Prefix-Orthogonal Frequency Division Multiplexing (CP-OFDM) and DFT-spread OFDM (DFT-s-OFDM).

- CP-OFDM allows for the use of non-contiguous frequency-domain resources and flexible resource allocation. It brings a high frequency-diversity gain but produces a high peak to average power ratio (PAPR).
- DFT-s-OFDM is based on discrete fourier transform (DFT) and requires contiguous frequency-domain resources. It ensures low PAPR (which is similar to that of a single carrier) and high transmit power.

Uplink waveform adaptation supports adaptation between the CP-OFDM and DFT-s-OFDM waveforms based on the radio environment (such as coverage) of UEs to improve user experience. Uplink waveform adaptation includes RRC-reconfiguration-based dynamic waveform switching and DCI-based dynamic waveform switching.

- RRC-reconfiguration-based dynamic waveform switching: This function is controlled by the **UL_WAVEFORM_ADAPT_SW** option of the **NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch** parameter.
- DCI-based dynamic waveform switching: This function is controlled by the **DYNAMIC_WAVEFORM_SWITCHING_SW** option of the **NRDUCellChnCovAlgo.UlCoverageAlgoSwitch** parameter.

When both RRC-reconfiguration-based dynamic waveform switching and DCI-based dynamic waveform switching are enabled, DCI-based dynamic waveform switching preferentially takes effect for UEs capable of dynamic waveform switching, using DCI to indicate the waveform.

NOTE

In versions earlier than 3GPP R18, the switching between the two waveforms can be implemented only through the *transformPrecoder* IE in the RRCReconfiguration message. For details, see section 6.1.3 "UE procedure for applying transform precoding on PUSCH" in 3GPP TS 38.214 V15.5.0.

According to 3GPP Release 18, the switching between the two waveforms is dynamically indicated by the **Transform precoder indicator** field in the DCI, for details, see section 6.1.3 "UE procedure for applying transform precoding on PUSCH" in 3GPP TS 38.214 V18.1.0 and section 7.3.1 "DCI formats" in 3GPP TS 38.212 V18.1.0.

Before this function takes effect, the network side instructs UEs to always select CP-OFDM.

After this function takes effect, the network side adaptively instructs UEs to select CP-OFDM or DFT-s-OFDM waveform based on the channel quality and the threshold specified by the **NRDUCellPusch.SinrThldforWaveformSel** parameter.

After this function takes effect, if the estimated channel capacity of DFT-s-OFDM is higher than that of CP-OFDM, the uplink SINR is less than the threshold, and single-carrier scheduling is performed on the PUSCH, the network side instructs UEs to select DFT-s-OFDM. In other cases, the network side instructs UEs to select CP-OFDM.

According to section 6.2.2 "UE maximum output power reduction" in 3GPP TS 38.101-2 g50, the maximum power reduction (MPR) of UEs varies with positions of allocated RBs: When the PUSCH waveform remains unchanged, the MPR of a

UE to which the inner RB frequency-domain resources are allocated is less than the MPR of a UE to which the outer or edge RB frequency-domain resources are allocated. Therefore, when all the following conditions are met, the spectrum resources at the inner RB position can be preferentially allocated to UEs to obtain the UE power benefits.

- The uplink waveform adaptation function takes effect and the DFT-s-OFDM waveform is used.
- Single-carrier scheduling is performed on the PUSCH, and the carrier bandwidth is 100 MHz, 200 MHz, or 400 MHz.
- The number of UEs to be scheduled in the current TTI is 1.

If the preceding conditions are not met, frequency-domain resources are allocated in the same way as PUSCH time-frequency resource allocation. For details, see *Uplink Scheduling*.

NOTE

The network side adds a hysteresis range of ± 2 dB to the threshold to prevent ping-pong waveform switching.

The frequency-domain positions of inner RBs, outer RBs, and edge RBs are related to the configured carrier bandwidth, number of RBs scheduled, and UE power class. For details, see sections 6.2.2 "UE maximum output power reduction" and 6.2A.2 "UE maximum output power reduction for CA" in 3GPP TS 38.101-2 g50.

LampSite base stations do not support this function.

Implementation

For details, see *Ultra Uplink (High-Frequency TDD)*.

Impact on the Network

- Benefits

This function increases the UE transmit power and can increase the average uplink UE throughput when the DFT-s-OFDM waveform is selected and inner RB frequency-domain resources are scheduled.

This function does not produce any gains in single-UE multi-stream transmission scenarios, where DFT-s-OFDM is not applicable and the UE still works with CP-OFDM.

DCI-based dynamic waveform switching uses DCI to indicate waveform switching, reducing RRC signaling overhead and mitigating the risk of service drops introduced by RRCReconfiguration.

NOTE

The 3GPP-defined MPR is the upper limit of the maximum power reduction value for UEs. The MPRs of UEs provided by different vendors on the live network meet the protocol requirements as long as their values are less than or equal to this upper limit. As such, the transmit power of different UEs on the live network may differ, so do the benefits delivered after this function takes effect.

For high-frequency cells, it is recommended that the switches for uplink waveform adaptation be configured consistently across all carriers within the same sector. Otherwise, uplink waveform adaptation may not take effect, resulting in no or reduced benefits.

- Impacts

CP-OFDM can use non-contiguous frequency-domain resources while DFT-s-OFDM can use only contiguous frequency-domain resources. If UEs select DFT-s-OFDM rather than CP-OFDM in non-contiguous frequency-domain scenarios, the scheduling bandwidth is smaller, and the single-UE throughput may not increase or may decrease. The specific negative results depend on the changes in (1) scheduled bandwidth and power spectral density and (2) increase in UE transmit power after the waveform mode is switched from CP-OFDM to DFT-s-OFDM.

After uplink waveform adaptation takes effect, the maximum transmit power of a UE changes abruptly when the UE switches between CP-OFDM and DFT-s-OFDM. Specifically, the UE throughput increases significantly when the UE switches from CP-OFDM to DFT-s-OFDM, and conversely, decreases significantly when the UE switches from DFT-s-OFDM to CP-OFDM.

After the uplink waveform adaptation function is enabled, compatibility issues may occur on UEs not supporting the DFT-s-OFDM waveform, causing service drops.

Impact on NEs

This feature is implemented on the gNodeB.

DCI-based dynamic waveform switching requires that UEs report their dynamic waveform switching capabilities through the **dynamicWaveformSwitch-r18** and **dynamicWaveformSwitchIntraCA-r18** fields.

Impact on Hardware Requirements

Base station models:

5900 series base stations (macro base stations)

Boards:

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF modules:

All NR TDD-capable AAUs that work in high frequency bands support this function. For details, see the technical specifications of AAUs in *3900 & 5900 Series Base Station Product Documentation*.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

This feature is controlled by the following new license control item.

Feature ID	Feature Name	Model	Sales Unit
FOFD-101240	Ultra Uplink	NR0S00ULUL00	Per Cell

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellAIdgoSwitch.AdaptiveEdgeExpEnhSwitch	MOD NRDUCELLA LGOSWITCH LST NRDUCELLA LGOSWITCH	gNodeB	The UL_WAVEFORM_ADAPT_SW option has been added to specify whether to enable RRC-reconfiguration-based dynamic waveform switching.
Modified	NRDUCellChnCovAlgo.UlCoverageAlgorithmSwitch	MOD NRDUCELLC HNCVALGO LST NRDUCELLC HNCVALGO	gNodeB	The DYNAMIC_WAVEFORM_SWITCHING_SW option has been added to specify whether to enable DCI-based dynamic waveform switching.
Modified	NRDUCellPusch.SinrThldforWaveformSel	MOD NRDUCELLP USCH LST NRDUCELLP USCH	gNodeB	The NRDUCellPusch.SinrThldforWaveformSel parameter now additionally applies to high frequency bands.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Ultra Uplink	ULTRA_ULINK_SW option of the NRDUCellChnCovAlgo.UlCoverageAlgoSwitch parameter	<i>Ultra Uplink (High-Frequency TDD)</i>	Uplink waveform adaptation can take effect only if Ultra Uplink is enabled.

- Mutually exclusive features/functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Uplink DMRS Type	NRDUCellPusch.UlDmrsType	<i>Uplink Scheduling</i>	When NRDUCellPusch.UlDmrsType is set to TYPE2 in a cell, the DYNAMIC_WAVEFORM_SWITCHING_SW option of the NRDUCellChnCovAlgo.UlCoverageAlgoSwitch parameter must be deselected.

- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Ultra Uplink (High-Frequency TDD)*.

Requirement/Trouble Ticket Number

IR20250815001172

RAT

NR TDD

5.3.1.12.2 Uplink Data Compression (for High Frequency Bands Only)

NEs Involved

5900 series base stations

Feature/Function Description

Uplink data compression (also referred to as UDC) uses the 3GPP Deflate compression algorithm to compress uplink data packets. This reduces resource consumption over the air interface and improves uplink data transmission capability. This function applies to the PDCP layers of the UE and gNodeB. After this function is enabled, the UE compresses uplink data packets that are not compressed or encrypted and then the gNodeB decompresses the compressed packets to reduce the amount of data transmitted over the air interface.

Uplink data compression can be configured at the cell level or QCI level.

- Cell-level parameter configuration
 - **UDC_SW** option of the **NRDUCellChnCovAlgo.UlCoverageAlgoSwitch** parameter: If this option is selected, uplink data compression can take effect for all uplink data packets used for data transmission at the NAS layer in a cell.
 - **UPLINK_SIG_COMPRESS_SW** option of the **NRDUCellChnCovAlgo.UlCoverageAlgoSwitch** parameter: If this option is selected, uplink data compression can take effect for all uplink data packets used for signaling transmission at the NAS layer in a cell.
 - **NRCeCellServExp.UdcBufferSize**: This parameter specifies the buffer size on the gNodeB side. The actual buffer size is equal to the value of this parameter or the buffer size reported by the UE, whichever is smaller. A smaller buffer size results in a lower compression efficiency but a shorter decompression delay. A larger buffer size produces the opposite effects.
- QCI-level parameter configuration
 - **QCI_UDC_SW** option of the **gNBQciBearer.PdcpAlgoSwitch** parameter: This option specifies whether to enable QCI-specific uplink data compression. If this option is selected, uplink data compression can take effect for bearers with the specified QCI.
 - For such bearers used for data transmission at the NAS layer, if the **UDC_SW** option of the cell-level parameter **NRDUCellChnCovAlgo.UlCoverageAlgoSwitch** is selected, this cell-level parameter setting takes effect; otherwise, the QCI-level parameter setting takes effect.
 - For such bearers used for signaling transmission at the NAS layer, if the **UPLINK_SIG_COMPRESS_SW** option of the cell-level parameter

NRDUCellChnCovAlgo.UlCoverageAlgoSwitch is selected, this cell-level parameter setting takes effect; otherwise, the QCI-level parameter setting takes effect.

LampSite base stations do not support this function.

The change in this feature is supported only in SA networking.

Implementation

For details, see *Ultra Uplink (High-Frequency TDD)*.

Impact on the Network

- Benefits

This function improves the compression efficiency of uplink data packets, decreases the amount of data transmitted over the air interface, and reduces the transmission delay for CEUs. The higher the compression efficiency of uplink data packets, the greater the decrease in the amount of data transmitted over the air interface.

 NOTE

Compression efficiency = $(1 - \frac{\text{N.PDCP.Vol.Ul.UDC.BefDecomp}}{\text{N.PDCP.Vol.Ul.UDC.AftDecomp}}) \times 100\%$

- Impacts

After this function takes effect:

- Compressed data packets carry compression indication information, causing extra overhead. Therefore, the size of data packets after compression may be greater than that before compression and there may be negative impacts if the repetition rate of uplink data is not high, for example, if uplink data packets have been compressed or encrypted.
- The average gNodeB board user-plane CPU usage (indicated by **VS.NRBoard.UPlane.CPULoad.Avg**) increases, the uplink PRB usage (indicated by **Uplink Resource Block Utilizing Rate (DU)**) may decrease, and the uplink MCS index fluctuates.

Impact on NEs

This feature is implemented on the gNodeB.

This function requires UDC capabilities of UEs, which are reported through the **udc-r17** field.

Impact on Hardware Requirements

Base station models:

5900 series base stations (macro base stations)

Boards:

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF modules:

All NR TDD-capable AAUs that work in high frequency bands support this function. For details, see the technical specifications of AAUs in *3900 & 5900 Series Base Station Product Documentation*.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

This feature is controlled by the following new license control item.

Feature ID	Feature Name	Model	Sales Unit
FOFD-101240	Ultra Uplink	NR0S00ULUL00	Per Cell

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellChnCovAlgo.UlCoverageAlgoSwitch	MOD NRDUCELLC HNCVALG O LST NRDUCELLC HNCVALG O	gNodeB	The UDC_SW and UPLINK_SIG_COMPRESS_SW options now additionally applies to high-frequency TDD.
Modified	gNBQciBearer.PdcpAlgoSwitch	ADD GNBQCIBEA RER MOD GNBQCIBEA RER LST GNBQCIBEA RER	gNodeB	The QCI_UDC_SW option has been added to specify whether to enable QCI-specific uplink data compression.

- Performance management

The following counters have changed.

Change Type	Counter Name	Function Subset	NE	Description
New	N.PDCP.Vol. UL.UDC.Bef Decomp	PDCP Measuremen t	gNodeB	This counter has been added to measure the data volume of SDUs received at the PDCP layer before UDC is successfully performed in a cell.
New	N.PDCP.Vol. UL.UDC.AftD ecom	PDCP Measuremen t	gNodeB	This counter has been added to measure the data volume of SDUs received at the PDCP layer after UDC is successfully performed in a cell.

- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Ultra Uplink	ULTRA_UP LINK_SW option of the NRDUCellC hnCovAlgo .ULCoverag eAlgoSwitc h parameter	<i>Ultra Uplink (High- Frequency TDD)</i>	Uplink data compression can take effect only if Ultra Uplink is enabled.

- Mutually exclusive features/functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	QoS Class Identifier	gNBQciBearer.Qci	<i>QoS Management</i>	When the gNBQciBearer.Qci parameter is set to 1 , 65 , or 66 , the QCI_UDC_SW option of the gNBQciBearer.PdcpAlgoSwitch parameter cannot be selected.

- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Ultra Uplink (High-Frequency TDD)*.

Requirement/Trouble Ticket Number

IR20250815001172

RAT

NR TDD

5.3.1.12.3 Full-Power Dual Transmission (for High Frequency Bands Only)

NEs Involved

5900 series base stations

Feature/Function Description

Full-power dual transmission is introduced to improve the performance of CEUs. This function allows UEs to transmit at full power and achieve better performance when the UEs work in rank 1 and have two or more transmit antennas and SRS ports. This function is controlled by the **UL_RANK1_FULL_POWER_SW** option of the **NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch** parameter. After full-power dual transmission is enabled, it works as follows:

1. The base station checks the *ul-FullPwrMode-r16*, *ul-FullPwrMode1-r16*, and *ul-FullPwrMode2-TPMIGroup-r16* fields in the *FeatureSetUplink* IE to determine whether a UE supports full-power dual transmission.
2. If the UE supports this function, the base station sets the *ul-FullPowerTransmission-r16* IE in *PUSCH-Config*.

3. After the UE completes configuration, full-power dual transmission takes effect.

LampSite base stations do not support this function.

Implementation

For details, see *Ultra Uplink (High-Frequency TDD)*.

Impact on the Network

This function increases the uplink transmit power and uplink user-perceived rate at the coverage edge if UEs support full-power transmission in the uplink and work in rank 1.

After this function takes effect, the transmit power of CEUs increases, slightly increasing interference.

Impact on NEs

This feature is implemented on the gNodeB.

This function requires that UEs support the transmission mode of FullPowerMode1, FullPowerMode2, or FullPower. The specific transmission mode that a UE supports is indicated by the *ul-FullPwrMode-r16*, *ul-FullPwrMode1-r16*, or *ul-FullPwrMode2-TPMIGroup-r16* field in the *FeatureSetUplink* IE. If the UE reports *ul-FullPwrMode-r16*, it supports FullPower. If the UE reports *ul-FullPwrMode2-TPMIGroup-r16* while not reporting *ul-FullPwrMode-r16*, it supports FullPowerMode2. Otherwise, it supports FullPowerMode1.

Impact on Hardware Requirements

Base station models:

5900 series base stations (macro base stations)

Boards:

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF modules:

All NR TDD-capable AAUs that work in high frequency bands support this function. For details, see the technical specifications of AAUs in *3900 & 5900 Series Base Station Product Documentation*.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

This feature is controlled by the following new license control item.

Feature ID	Feature Name	Model	Sales Unit
FOFD-101240	Ultra Uplink	NR0S00ULUL00	Per Cell

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellA goSwitch.Ad aptiveEdgeE xpEnhSwitc h	MOD NRDUCELLA LGOSWITCH LST NRDUCELLA LGOSWITCH	gNodeB	The UL_RANK1_FULL_POWER_SW option has been added to support full-power dual transmission.

- Performance management

No impact

- Fault management

No impact

Impact on Features/Functions

- Prerequisite features/functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Ultra Uplink	ULTRA_UP LINK_SW option of the NRDUCellC hnCovAlgo .ULCoverageAlgoSwitch parameter	<i>Ultra Uplink (High-Frequency TDD)</i>	Full-power dual transmission can take effect only if Ultra Uplink is enabled.

- Mutually exclusive features/functions

No impact

- Impacted features/functions

No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Ultra Uplink (High-Frequency TDD)*.

Requirement/Trouble Ticket Number

IR20250815001172

RAT

NR TDD

5.3.1.12.4 PUSCH Time Domain Enhancement (for High Frequency Bands Only)

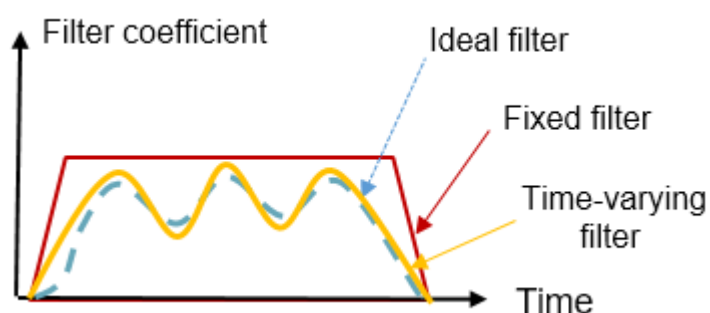
NEs Involved

5900 series base stations

Feature/Function Description

The complexity of the radio propagation environment means that signals arrive at the receiver after a series of time-variant fading and delay. A simple fixed-coefficient channel filter design cannot track channel changes in a time-varying manner, impairing performance as a result. As such, the PUSCH time domain enhancement function is introduced. Based on the power delay profile (PDP) of PUSCH DMRS, the base station selects the time-domain filtering coefficient that matches the channel changes based on the time-domain changes of the wanted signal strength, as shown in [Figure 5-3](#). This effectively improves the channel estimation accuracy and uplink coverage. This function is controlled by the **PUSCH_TIME_DOMAIN_ENH_SW** option of the **NRDUCellAlgoSwitch.FullChannelCovEnhSwitch** parameter.

Figure 5-3 Principles of PUSCH time domain enhancement



LampSite base stations do not support this function.

Implementation

For details, see *Ultra Uplink (High-Frequency TDD)*.

Impact on the Network

PUSCH time domain enhancement increases the **User Uplink Average Throughput (DU)** when QPSK is used for uplink scheduling.

Impact on NEs

This feature is implemented on the gNodeB.

Impact on Hardware Requirements

Base station models:

5900 series base stations (macro base stations)

Boards:

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF modules:

All NR TDD-capable AAUs that work in high frequency bands support this function. For details, see the technical specifications of AAUs in *3900 & 5900 Series Base Station Product Documentation*.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

This feature is controlled by the following new license control item.

Feature ID	Feature Name	Model	Sales Unit
FOFD-101240	Ultra Uplink	NR0S00ULUL00	Per Cell

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellAlgoSwitch.FullChannelCoverageEnhSwitch	MOD NRDUCELLA LGOSWITCH LST NRDUCELLA LGOSWITCH	gNodeB	The PUSCH_TIME_DOMAIN_ENH_SW option has been added to support PUSCH time domain enhancement.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Ultra Uplink	ULTRA_ULINK_SW option of the NRDUCellCnCoverageAlgoSwitch parameter	<i>Ultra Uplink (High-Frequency TDD)</i>	PUSCH time domain enhancement can be enabled only if the Ultra Uplink switch is turned on.

- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Ultra Uplink (High-Frequency TDD)*.

Requirement/Trouble Ticket Number

IR20250815001172

RAT

NR TDD

5.3.1.13 FOFD-101241 Uplink IAS

5.3.1.13.1 Added the Uplink IAS Feature (High Frequency Bands Only)

NEs Involved

5900 series base stations

Feature/Function Description

The Uplink IAS feature is controlled by the **FR2_EXP_IMP_SW** option of the **NRDUCellPusch.PuschPerformanceSwitch** parameter. The Uplink IAS feature includes the following functions:

- Uplink packet bundled scheduling:
When a cell has medium or heavy load in the uplink (for example, when uplink PRB usage is higher than 60%), beam resources usually become insufficient earlier than PRB resources. Therefore, when UEs running small-packet services occupy beam resources exclusively, PUSCH PRBs may not be fully utilized due to beam resource insufficiency. To address this issue, uplink packet bundled scheduling is introduced. This function is controlled by the **UL_PACKET_AGGREGATED_SCH_SW** option of the **NRDUCellULSch.UISchAlgoSwitch** parameter. After this function takes effect, data of UEs running large-packet services and that of UEs running small-packet services are aggregated for scheduling, enabling beam resources to be fully utilized and thereby improving uplink PRB usage. This function applies only to 4T4R or 8T8R cells.
- Packet-length-based scheduling optimization:
In cells with medium or heavy load in the uplink (for example, with uplink PRB usage being higher than 60%), UEs' varied signal quality and service models make it possible to reduce the total transmission delay of UEs. To achieve this, packet-length-based scheduling optimization is introduced. This function is controlled by the **PKT_LEN_BASED_SCH_OPT_SW** option of the **NRDUCellPusch.PuschPerformanceSwitch** parameter. Aimed at minimizing the total transmission delay of UEs to be scheduled, this function adaptively selects appropriate UEs for scheduling based on their traffic models and signal quality. This increases the average uplink UE throughput. After this function takes effect, small-packet-service UEs with a smaller amount of buffered data volume and better signal quality are preferentially scheduled.
- Dual PRACH periods:
The "dual PRACH periods" function extends the physical random access channel (PRACH) period for contention-based random access (CBRA) and this allows the number of slots occupied by the PRACH within a given time segment to be reduced, thereby increasing the number of slots available for

the physical uplink shared channel (PUSCH). In addition, this function enables the base station to receive preambles for contention-free random access (CFRA) in specific slots, further reducing the amount of time-frequency resources occupied by the PRACH. The "dual PRACH periods" function allows the PRACH period for CBRA and that for CFRA to be separately configured using the **NRDUCellPrach.PrachConfigurationIndex** and **NRDUCellPrach.CfraPrachConfigIndex** parameters, respectively, only under identical settings of the preamble format, PUSCH subcarrier spacing (SCS), and slot configuration.

- PUCCH receive channel adaptation:

The PUCCH receive channel adaptation function adaptively adjusts the allocation of PUCCH receive channels based on the coverage and channel quality of UEs. The **NRDUCellPucch.PucchRxAllocMode** parameter specifies whether this function is enabled.

- When this parameter is set to **ADAPTIVE**, this function is enabled, and the number of PUCCH receive channels is adaptively adjusted based on the coverage and channel quality of UEs. For example, fewer PUCCH receive channels are used for cell-center UEs with good signal quality than for other UEs, reducing uplink resource consumption. Conversely, more PUCCH receive channels are used for cell-edge UEs with poor signal quality, improving user experience.
- When this parameter is set to **OFF**, this function is disabled and the receive channel allocation mode is the same for all UEs.

- Uplink IBLER adaptation optimization:

A fixed target IBLER cannot always match the need for SINR correction. As a result, the channel quality indicated by the SINR used for MCS selection is significantly different from the actual channel quality and this restricts throughput. To select a more appropriate target IBLER and thus enable UEs to have higher uplink throughput, uplink IBLER adaptation optimization is introduced. This function is controlled by the **UL_IBLER_ADAPT_OPT_SW** option of the **NRDUCellUIMc.UIMcAlgoSw** parameter. After uplink IBLER adaptation optimization takes effect, the gNodeB determines the channel quality change for individual UEs based on the standard deviation of SINR fluctuations. This way, the gNodeB can obtain more accurate information about channel quality changes for UEs and select a more appropriate target IBLER accordingly.

After uplink IBLER adaptation optimization is enabled, the following parameters can be used to set the thresholds for the adaptation:

- **NRDUCellUIMc.UIMcsIdxForLowIblerTarget**. This parameter specifies the MCS index threshold for the low target IBLER to take effect in the uplink. When the MCS index is greater than or equal to the value of this parameter, the low target IBLER takes effect.
- **NRDUCellUIMc.UISinrAvgLowThld**. This parameter specifies the low threshold expressed as the average uplink SRS SINR. If the average uplink SINR of a UE is greater than this threshold and less than or equal to the SINR value corresponding to **NRDUCellUIMc.UIMcsIdxForLowIblerTarget**, the UE is considered to be of slightly poor channel quality in terms of the average SINR. If the average uplink SINR of a UE is less than or equal to this threshold, the UE is considered to be of poor channel quality in terms of the average SINR.

- **NRDUCellUplAmc.UlSnrAvgHighThld.** This parameter specifies the high threshold expressed as the average uplink SRS SINR. If the average uplink SINR of a UE is greater than this threshold, the UE is considered to be of good channel quality in terms of the average SINR. If the average uplink SINR of a UE is less than or equal to this threshold and greater than the SINR value corresponding to **NRDUCellUplAmc.UlMcsIdxForLowBlrTarget**, the UE is considered to be of slightly good channel quality in terms of the average SINR.
- **NRDUCellUplAmc.UlSnrDiffLowThld.** This parameter specifies the lower threshold of the uplink SRS SINR standard deviation. If the uplink SINR standard deviation of a UE is less than or equal to this threshold, the UE is considered to be of high channel quality in terms of SINR standard deviation. If the uplink SINR standard deviation of a UE is greater than this threshold and less than or equal to **NRDUCellUplAmc.UlSnrDiffHighThld**, the UE is considered to be of medium channel quality in terms of SINR standard deviation.
- **NRDUCellUplAmc.UlSnrDiffHighThld.** This parameter specifies the upper threshold of the uplink SRS SINR standard deviation. If the uplink SINR standard deviation of a UE is greater than this threshold, the UE is considered to be of poor channel quality in terms of SINR standard deviation. If the uplink SINR standard deviation of a UE is less than or equal to this threshold and greater than **NRDUCellUplAmc.UlSnrDiffLowThld**, the UE is considered to be of medium channel quality in terms of SINR standard deviation.

This feature is supported only in high-frequency TDD.

LampSite base stations do not support this feature.

Implementation

For details, see *Intelligent Aggregate Scheduling (High-Frequency TDD)*.

Impact on the Network

- Benefits

After Uplink IAS takes effect, **Cell Uplink Average Throughput (DU)** and **User Uplink Average Throughput (DU)** increase. The specific benefits depend on factors such as the traffic model and network load.

Table 5-14 describes the benefits of each sub-function.

Table 5-14 Benefits of each sub-function

Sub-function Name	Benefits
Packet-length-based scheduling optimization	In medium- or heavy-load scenarios (for example, uplink PRB usage is higher than 60%), packet-length-based scheduling optimization can shorten the service waiting time of UEs, and increase average uplink UE throughput while keeping Cell Uplink Average Throughput (DU) almost unchanged. When there are multiple high-frequency cells, it is recommended that the average uplink UE throughput of all cells in a sector be checked. For example, a sector has two cells: cell 1 and cell 2. Assume that the values of N.ThpVol.UL, N.ThpVol.UE.UL.SmallPkt, and N.ThpTime.UE.UL.RmvSmallPkt of cell 1 are a1, b1, and c1, respectively, and the values of N.ThpVol.UL, N.ThpVol.UE.UL.SmallPkt, and N.ThpTime.UE.UL.RmvSmallPkt of cell 2 are a2, b2, and c2, respectively. Check the average value of the two cells, that is, result of $[(a1 - b1) + (a2 - b2)]/(c1+c2)$.
Uplink packet bundled scheduling	In medium- or heavy-load scenarios (for example, uplink PRB usage is higher than 60%), uplink packet bundled scheduling increases uplink PRB usage and Cell Uplink Average Throughput (DU).
Dual PRACH periods	In heavy-load scenarios (for example, uplink PRB usage is higher than 75%), the "dual PRACH periods" function reduces PRACH resource consumption, increases the amount of available PUSCH resources, and therefore increases User Uplink Average Throughput (DU).
PUCCH receive channel adaptation	PUCCH receive channel adaptation increases Cell Uplink Average Throughput (DU) and User Uplink Average Throughput (DU). More cell-center UEs and higher uplink PRB usage can lead to larger increases.
Uplink IBLER adaptation optimization	Uplink IBLER adaptation optimization increases User Uplink Average Throughput (DU) when there are some cell-center UEs and cell-edge UEs (the average uplink MCS index for cell-center UEs ranges from 20 to 27 and that for cell-edge UEs ranges from 0 to 6).

- Impacts

After Uplink IAS takes effect, there are fluctuations in **Cell Downlink Average Throughput (DU)** and **User Downlink Average Throughput (DU)**.

Table 5-15 describes the impacts of each sub-function.

Table 5-15 Network impacts of the sub-functions

Sub-function Name	Impact on the Network
Packet-length-based scheduling optimization	When there is a large number of UEs running small-packet services in a cell, packet-length-based scheduling optimization increases the probability that these UEs are scheduled in advance. As a result, uplink PRB usage and Cell Uplink Average Throughput (DU) decrease. In addition, after this function takes effect, if the number of uplink scheduling occasions increases, the number of downlink response packets increases. This may cause Cell Downlink Average Throughput (DU) and User Downlink Average Throughput (DU) to decrease.
Uplink packet bundled scheduling	Uplink packet bundled scheduling causes fluctuations in User Uplink Average Throughput (DU) . These fluctuations vary depending on the carrier configuration and inbound traffic model of UEs.

Sub-function Name	Impact on the Network
Dual PRACH periods	This function requires that the related parameters be set to identical values for NR cells that are served by the same RF module. Otherwise, UE access failures or service drops may occur. After this function takes effect in heavy-load scenarios (for example, uplink PRB usage is higher than 75%), there will be fluctuations in the values of the following indicators: • Cell Uplink Average Throughput (DU) and Cell Downlink Average Throughput (DU) • Uplink PRB usage (N.PR.B.UL.Used.Avg divided by N.PR.B.UL.Avail.Avg) and downlink PRB usage (Downlink Resource Block Utilizing Rate (DU)) • Number of online UEs in a cell • Average downlink MCS index and average uplink MCS index - Average DownLink MCS=$\frac{(0 \times \text{N.ChMeas.PDSCH.MCS.0} + 1 \times \text{N.ChMeas.PDSCH.MCS.1} + 2 \times \text{N.ChMeas.PDSCH.MCS.2} + \dots + 28 \times \text{N.ChMeas.PDSCH.MCS.28})}{(\text{N.ChMeas.PDSCH.MCS.0} + \text{N.ChMeas.PDSCH.MCS.1} + \text{N.ChMeas.PDSCH.MCS.2} + \dots + \text{N.ChMeas.PDSCH.MCS.28})}$ - Average UpLink MCS=$\frac{(0 \times \text{N.ChMeas.PUSCH.MCS.0} + 1 \times \text{N.ChMeas.PUSCH.MCS.1} + 2 \times \text{N.ChMeas.PUSCH.MCS.2} + \dots + 28 \times \text{N.ChMeas.PUSCH.MCS.28})}{(\text{N.ChMeas.PUSCH.MCS.0} + \text{N.ChMeas.PUSCH.MCS.1} + \text{N.ChMeas.PUSCH.MCS.2} + \dots + \text{N.ChMeas.PUSCH.MCS.28})}$ • Average downlink rank and average uplink rank - Average DownLink RANK=$\frac{(1 \times \text{N.PDSCH.InitTbDL.Rank1} + 2 \times \text{N.PDSCH.InitTbDL.Rank2} + 3 \times \text{N.PDSCH.InitTbDL.Rank3} + 4 \times \text{N.PDSCH.InitTbDL.Rank4})}{(\text{N.PDSCH.InitTbDL.Rank1} + \text{N.PDSCH.InitTbDL.Rank2} + \text{N.PDSCH.InitTbDL.Rank3} + \text{N.PDSCH.InitTbDL.Rank4})}$ - Average UpLink RANK=$\frac{(1 \times \text{N.PUSCH.InitTbUL.Rank1} + 2 \times \text{N.PUSCH.InitTbUL.Rank2})}{(\text{N.PUSCH.InitTbUL.Rank1} + \text{N.PUSCH.InitTbUL.Rank2})}$ • Proportions of uplink and downlink scheduling occasions using different modulation schemes

Sub-function Name	Impact on the Network
	There will be the following changes in the values of access delay indicators: • The delay for the IDLE2ACTIVE state transition increases. • The attach delay increases.
PUCCH receive channel adaptation	After PUCCH receive channel adaptation takes effect, Cell Uplink Average Throughput (DU) and User Uplink Average Throughput (DU) increase; however, the number of downlink radio link control (RLC) status reports or Transmission Control Protocol (TCP) response packets increases, which results in fluctuations in Cell Downlink Average Throughput (DU) and User Downlink Average Throughput (DU).
Uplink IBLER adaptation optimization	In cells with heavy downlink load (for example, downlink PRB usage is higher than 70%) and a high proportion of cell-edge UEs, uplink IBLER adaptation optimization enables UEs to provide more timely feedback on downlink data transmission. This may increase downlink traffic at the Packet Data Convergence Protocol (PDCP) layer. However, User Downlink Average Throughput (DU) may decrease due to limited downlink air interface capabilities.

Impact on NEs

This feature is implemented on the gNodeB.

Impact on Hardware Requirements

- Base station models
 - 5900 series base stations (macro base stations)
- Boards
 - All NR TDD-capable main control boards and baseband processing units that work in mmWave frequency bands support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- RF modules
 - Uplink packet bundled scheduling
 - The RF module must be a 4T4R or 8T8R AAU that supports NR TDD high frequency bands.
 - PUCCH receive channel adaptation
 - The RF module must be the HAAU5233w.
 - Other sub-functions

All NR TDD-capable AAUs that work in high frequency bands support this function. For details, see the technical specifications of AAUs in *3900 & 5900 Series Base Station Product Documentation*.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
This feature is controlled by the following new license control items.

Feature ID	Feature Name	Model	Sales Unit
FOFD-101241	Uplink IAS	NR0S00PCDW00	per Cell

- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	FR2 uplink experience improvement	FR2_EXP_IM P_SW option of the NRDUCellPusch.PuschPerformanceSwitch parameter	<i>Intelligent Aggregate Scheduling (High-Frequency TDD)</i>	- FR2 uplink experience improvement must be enabled before the "dual PRACH periods" and PUCCH receive channel adaptation functions can be enabled. - Uplink packet bundled scheduling, packet-length-based scheduling optimization, and uplink IBLER adaptation optimization can take effect only after FR2 uplink experience improvement is enabled.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	PRACH Configuration Index	NRDUCellPrach.PrachConfigurationIndex	<i>Random Access Control</i>	- The NRDUCellPrach.CfraPrachConfigIndex parameter can be set to 188 only when the NRDUCellPrach.PrachConfigurationIndex parameter is set to 192. - The NRDUCellPrach.CfraPrachConfigIndex parameter can be set to 74 only when the NRDUCellPrach.PrachConfigurationIndex parameter is set to 81. - In other cases, the NRDUCellPrach.CfraPrachConfigIndex parameter must be set to 65535.
High-frequency TDD	PUCCH symbol number adaptation	PUCCH_RES_ADAPTIVE_SWITCH option of the NRDUCellPucch.PucchAlgoSwitch parameter	<i>Channel Management</i>	PUCCH receive channel adaptation can take effect only after PUCCH symbol number adaptation is enabled.

- Mutually exclusive features/functions
The following mutually exclusive relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	mmWave 3D coverage pattern	NRDUCellTrpMmwavBeam.CoverageScenario	<i>mmWave Beam Management (High-Frequency TDD)</i>	PUCCH receive channel adaptation cannot be enabled when the coverage scenario is single-beam coverage (that is, when the NRDUCellTrpMmwavBeam.CoverageScenario parameter is set to SCENARIO_103).
High-frequency TDD	Increased downlink symbol proportion in self-contained slots	S_SLOT_DL_SYM_RATIO_INC_SW option of the NRDUCellDLISchRes.DischResAlgorithm parameter	None	PUCCH receive channel adaptation cannot be enabled together with the "increased downlink symbol proportion in self-contained slots" function.

- Impacted features/functions
The following impact relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Maximum Preamble Transmission Count	NRDUCellPrach.MaxPreambleTransCnt	<i>Random Access Control</i>	If the "dual PRACH periods" function is enabled and the Maximum Preamble Transmission Count parameter is set to a value less than the scenario-specific recommended values, there is a low probability that UEs experience service drops. The recommended values are as follows: • N50 if the FLEXIBLE_DENSE_BEAM_SW option of the NRDUCellTrpMmwavBeam.BeamPerformanceSw parameter is selected • N20 if the FLEXIBLE_DENSE_BEAM_SW option of the NRDUCellTrpMmwavBeam.BeamPerformanceSw parameter is deselected

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	PUCCH grouping adaptation enhancement for inter-FR CA	PUCCH_GROUP_ADAPT_ENH_SW option of the NRDUCellChnCovAlgo.DlCoverageAlgoSwitch parameter	<i>Carrier Aggregation</i>	PUCCH receive channel adaptation does not take effect for UEs with PUCCH grouping adaptation enhancement for inter-FR CA in effect.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Intelligent Aggregate Scheduling (High-Frequency TDD)*.

Requirement/Trouble Ticket Number

IR20250815001172

RAT

NR TDD

5.3.1.14 FOFD-101242 mmWave Super Coverage

5.3.1.14.1 Support for mmWave Super Coverage (for High Frequency Bands Only)

NEs Involved

5900 series base stations

Feature/Function Description

The mmWave super coverage function is designed to cope with the weaknesses of mmWave bands in an NR system—such as weaker coverage than low frequency bands due to high propagation loss and susceptibility to obstacles. This feature improves downlink cell coverage and user experience using beam directivity accuracy enhancement, adaptive optimization, channel noise reduction, channel prediction, and spectral efficiency optimization. This feature applies to high-frequency TDD cells with coverage requirements or scenarios with user experience improvement requirements.

mmWave super coverage is controlled by the **SUPER_COVERAGE_SW** option of the **NRDUCellChnCovAlgo.DlCoverageAlgoSwitch** parameter and includes the following functions:

- Intelligent beam selection
Intelligent beam selection is controlled by the **INTELLIGENT_BEAM_SELECTION_SW** option of the **NRDUCellBeamAlgo.BeamOptAlgoSwitch** parameter. As an enhancement to Flexible Dense Beam, intelligent beam selection uses the neural network technology and more CSI-RS for BM beams to help the gNodeB accurately direct CSI-RS for BM beams at UEs, thereby increasing average uplink and downlink UE throughputs.
Intelligent beam selection takes effect only if the switches for mmWave super coverage and intelligent beam selection are both turned on for all activated cells in the same sector.
- Beam tracking
This function is controlled by the **BEAM_TRACKING_SW** option of the **NRDUCellBeamAlgo.BeamOptAlgoSwitch** parameter. Using a reinforcement learning model and CSI-RS for BM beam measurements, the gNodeB predicts the trajectory of a moving UE. Based on the result, the gNodeB predicts the best beam to serve the UE for the next moment. The gNodeB finally selects the best CSI-RS for BM beam as the serving beam for the UE, based on the prediction and UE-reported measurements.
Beam tracking takes effect only if the beam tracking switch is turned on for all activated cells in the same sector.
- SRS-based digital weight
This function includes fixed SRS-based digital weight, SRS- and PMI-based weight adaptation, downlink single-UE mobility precoding, and SRS filtering-based downlink beamforming weight optimization. Fixed SRS-based digital weight and SRS- and PMI-based weight adaptation do not take effect when the sector-level downlink bandwidth (sum of **NRDUCell.DLBandwidth** for all cells in a sector) exceeds 400 MHz in FWA and eMBB scenarios.
 - Fixed SRS-based digital weight
This function has taken effect if the **DL_PMI_SRS_ADAPT_SW** option of the **NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch** parameter is deselected, and the **NRDUCellPdsch.FixedWeightType** parameter is set to **SRS_WEIGHT**. The gNodeB uses a fixed SRS-based weight.
 - SRS- and PMI-based weight adaptation
This function can be enabled by selecting the **DL_PMI_SRS_ADAPT_SW** option of the **NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch** parameter. The gNodeB adaptively selects the SRS-based digital weight or PMI-based weight based on the spectral efficiency.
 - Downlink single-UE mobility precoding
This function has been enabled when the **NRDUCellPdschPrecode.DLSuMimoMobPrecodeMode** parameter is set to **ROBUST_PHASEIII**. With this function, a prediction model is trained through a neural network and then used to predict channels, thereby alleviating the loss caused by channel changes and increasing the average downlink UE throughput. This function takes effect only for UEs performing downlink large-packet services.
 - SRS filtering-based downlink beamforming weight optimization
This function can be enabled by selecting the **DL_BF_WEIGHTFILTER_OPT_SW** option of the

NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch parameter. In FWA scenarios, this function implements smooth noise reduction on channels at several moments, thereby increasing the SNR and average downlink UE throughput.

- SRS bandwidth adaptation

This function is controlled by the **SRS_BANDWIDTH_ADAPT_SW** option of the **NRDUCellSrs.SrsAlgoExtSwitch** parameter. With this function, the SRS transmission mode is adaptively switched based on channel quality measurements for the SRS. SRS frequency-hopping transmission can be used to increase the SRS SNR and proportion of cases that the SRS-based digital weight takes effect. In this way, the SRS quality is improved and the accuracy of the SRS-based weight is enhanced, thereby increasing the average downlink throughput of non-cell-center UEs. In this process, SRS frequency hopping enables a UE to transmit an SRS on a small bandwidth in one symbol and to transmit an SRS on another bandwidth in the next symbol. The same rule applies to the rest SRSs. In this way, the gNodeB achieves sounding on the entire channel bandwidth. Frequency hopping offers channel quality measurement across the entire bandwidth using a small sounding bandwidth. SRS bandwidth adaptation takes effect only for one UE performing downlink large-packet services in a sector at a time.

SRS bandwidth adaptation can take effect only if all activated cells in the same sector meet the following conditions:

- The SRS bandwidth adaptation switch is turned on for all activated cells.
- Fixed SRS-based digital weight is configured for all activated cells, or SRS- and PMI-based weight adaptation is configured for all activated cells.

- Precise power control

Precise power control includes two sub-functions: downlink PDSCH coverage enhancement and PDSCH DMRS power adaptation. It is recommended that both sub-functions be enabled. These two sub-functions take effect only for inter-FR CA CEUs using extremely low MCS indexes.

- Downlink PDSCH coverage enhancement

This function is controlled by the **PDSCH_COV_ENHANCE_SW** option of the **NRDUCellChnPwr.ChnPwrAlgoSwitch** parameter. With this function, the PDSCH PSD is adaptively adjusted based on the MCS indexes and the number of RBs at the cell reference power to support higher MCS indexes. This improves the spectral efficiency of UEs, thereby increasing the downlink cell coverage and average downlink UE throughput. For details about the cell reference power, see *Power Control*.

- PDSCH DMRS power adaptation

This function is controlled by the **DMRS_PWR_ADAPT_SW** option of the **NRDUCellChnPwr.ChnPwrAlgoSwitch** parameter. With this function, the transmission power of PDSCH data which is on the same symbol with the PDSCH DMRS is adaptively allocated to the PDSCH DMRS to improve the demodulation performance of CEUs and therefore support higher MCS indexes. This improves the spectral efficiency of UEs, thereby improving the downlink cell coverage and average downlink UE throughput.

- Downlink IBLER adaptation for CEUs

This function is controlled by the **DL_EDGE_IBLER_ADAPT_SW** option of the **NRDUCellDLAmc.DIMcsSelAlgoSw** parameter. With this function, the

gNodeB adaptively selects the downlink target IBLER and a higher average MCS index for downlink scheduling based on (1) channel condition fluctuations and (2) whether CEUs use extremely low MCS indexes. This improves their spectral efficiency, thereby increasing their average downlink throughput. This function takes effect only for CEUs using extremely low MCS indexes and experiencing significant channel condition fluctuations.

LampSite base stations do not support this feature.

Implementation

For details, see *mmWave Super Coverage (High-Frequency TDD)*.

Impact on the Network

[Table 5-16](#) lists the benefits of each function after the mmWave Super Coverage feature is activated.

Table 5-16 Benefits

Function Name	Benefit
Intelligent beam selection	- After intelligent beam selection is enabled, the values of User Downlink Average Throughput (DU) and User Uplink Average Throughput (DU) increase for the UEs with this function in effect in the cell. The benefits are related to the locations of UEs with this function in effect: - There are greater benefits when UEs with this function in effect are closer to the dents of Flexible Dense Beams. - There are less or even no benefits when UEs with this function in effect are closer to the wave peak of Flexible Dense Beams. - After intelligent beam selection is enabled, the gNodeB collects data and trains the neural network model. During this period, no benefits are provided. When the status of the mmWave Super Coverage or intelligent beam selection switch changes from on to off, the neural network model's related data is cleared. In this case, if the switch is turned on again, the gNodeB collects data and trains the neural network model again. During this period, no benefits are provided. Benefit monitoring is recommended when the neural network model is in the In Use state. Run the DSP NRDUCELLBMAIMODEL command with the NRDUCellBmAiModel.AiCase parameter set to INTELLIGENT_BEAM_SELECTION. Observe the value of AI Model Status. The value In Use indicates that the neural network model is in use. NOTE After intelligent beam selection is enabled, run the DSP NRDUCELLBMAIMODEL command with the NRDUCellBmAiModel.AiCase parameter set to INTELLIGENT_BEAM_SELECTION. Observe the value of AI Model Status. The possible values include Building, In Use, and Updating. Intelligent beam selection provides benefits only when the value is In Use. When the neural network model is in use, the gNodeB monitors the model performance in real time. If the gNodeB detects that the model parameters are aged, it automatically collects data and trains the neural network model again. During this process, the value of AI Model Status is Updating, and no benefits are provided.

Function Name	Benefit
Beam tracking	- After beam tracking is enabled, the average uplink and downlink throughputs increase for the UEs with this function in effect in the cell. This function is recommended in eMBB scenarios. The benefits are related to the following factors: - A higher UE moving speed results in a longer beam selection lag and greater benefits. A lower UE moving speed results in a shorter beam selection lag and less benefits. - A higher proportion of LOS UEs results in a higher accuracy rate of UE trajectory prediction and greater benefits. A higher proportion of NLOS UEs results in a lower accuracy rate of UE trajectory prediction and less benefits. - After beam tracking is enabled, the gNodeB collects data and trains the reinforcement learning model. During this period, less or even no benefits are provided. When the status of the mmWave Super Coverage or beam tracking switch changes from on to off, the reinforcement learning model's related data is cleared. In this case, if the switch is turned on again, the gNodeB collects data and trains the reinforcement learning model again. During this period, less or even no benefits are provided. Benefit monitoring is recommended when the reinforcement learning model is in the In Use state. Run the DSP NRDUCELLBMAIMODEL command with the NRDUCellBmAiModel.AiCase parameter set to BEAM_TRACKING. Observe the value of AI Model Status. The value In Use indicates that the reinforcement learning model is in use.

Function Name	Benefit
SRS-based digital weight	- Fixed SRS-based digital weight: The average downlink UE throughput increases. A higher proportion of NLOS scenarios results in greater benefits. A higher proportion of LOS scenarios results in fewer or even no benefits. - SRS- and PMI-based weight adaptation: The average downlink UE throughput increases. The benefits are related to the following factors: - A higher UE moving speed results in faster channel changes, lower weight correlation, and less benefits. - A larger number of UEs results in a longer interval of aperiodic SRS reporting, and a longer weight update interval. As a result, the weight correlation decreases, causing reduced benefits. - Downlink single-UE mobility precoding: The gNodeB predicts channel changes to alleviate the loss caused by channel changes, thereby increasing the average downlink UE throughput. This function is recommended in eMBB scenarios. The benefits are related to the following factors: - A longer SRS period results in lower SRS channel correlation and higher benefits from downlink single-UE mobility precoding. - A larger SRS bandwidth results in more frequent channel sampling, higher accuracy of "downlink single-UE mobility precoding", and greater benefits. - A higher UE moving speed results in faster channel changes, lower SRS channel correlation, and higher benefit potential from downlink single-UE mobility precoding. However, an extremely high moving speed leads to overly fast channel changes and thereby lower SRS channel correlation. As a result, accurate channel prediction cannot be realized even through "downlink single-UE mobility precoding", thereby reducing the benefits or even causing no benefits. A lower UE moving speed results in slower channel changes, higher SRS channel correlation, and lower benefit potential or even no benefits from downlink single-UE mobility precoding. - A larger number of antennas in a cell results in denser channel sampling in the space domain, higher accuracy of "downlink single-UE mobility precoding", and greater benefits. - In NLOS scenarios, the aging of SRS-based digital weights occurs, resulting in a high benefit potential of "downlink single-UE mobility precoding". In LOS scenarios, SRS-based digital weights do not age, resulting in a low benefit potential or even no benefits of "downlink single-UE mobility precoding".

Function Name	Benefit
	- SRS filtering-based downlink beamforming weight optimization: The gNodeB performs noise reduction on channels to increase the downlink UE throughput. This function is recommended in FWA scenarios. The benefits are related to the following factors: - When the SNR of a UE is less than or equal to -20 dB, "SRS filtering-based downlink beamforming weight optimization" does not produce benefits. - When the SNR of a UE is greater than -20 dB and less than 10 dB, "SRS filtering-based downlink beamforming weight optimization" has obvious noise reduction effect and produces high benefits. - When the SNR of a UE is greater than or equal to 10 dB, the impact of noise is small and "SRS filtering-based downlink beamforming weight optimization" produces low benefits.
SRS bandwidth adaptation	The average downlink throughput of non-cell-center UEs increases. The benefits are related to the following factors: - A lower UE moving speed results in slower channel changes, less loss caused by the prolonged SRS frequency hopping period, and greater benefits of SRS bandwidth adaptation. A higher UE moving speed results in faster channel changes, greater loss caused by the prolonged SRS frequency hopping period, and less benefits of SRS bandwidth adaptation. - Greater benefit potential of SRS-based weights in NLOS scenarios results in greater benefits of SRS bandwidth adaptation. Less benefit potential of SRS-based weights in LOS scenarios results in less or even no benefits of SRS bandwidth adaptation. - SRS bandwidth adaptation provides greater benefits if there is a larger number of UEs meeting both of the following requirements: (1) "SRS-based digital weight" does not take effect before SRS frequency hopping is used. (2) "SRS-based digital weight" takes effect after SRS frequency hopping is used.

Function Name	Benefit
Precise power control	- After downlink PDSCH coverage enhancement is enabled, the average downlink throughput increases for UEs with this function in effect in the cell and the proportion^b of low rates^a decreases for the cell. This function is recommended in eMBB scenarios. It is recommended that PUCCH grouping adaptation enhancement for inter-FR CA be enabled (by selecting the PUCCH_GROUP_ADAPT_ENH_SW option of the NRDUCellCarrMgmt.InterFrCaEnhSw parameter) to achieve optimal PDSCH coverage, thereby further improving the average downlink throughput of these UEs. A higher proportion of inter-FR CA UEs using extremely low MCS indexes results in greater benefits. - After PDSCH DMRS power adaptation is enabled, the average downlink throughput increases for UEs with this function in effect in the cell and the proportion^b of low rates^a decreases for the cell. This function is recommended in eMBB scenarios. It is recommended that PUCCH grouping adaptation enhancement for inter-FR CA be enabled (by selecting the PUCCH_GROUP_ADAPT_ENH_SW option of the NRDUCellCarrMgmt.InterFrCaEnhSw parameter) to achieve optimal PDSCH coverage, thereby further improving the average downlink throughput of these UEs. A higher proportion of inter-FR CA UEs using extremely low MCS indexes results in greater benefits.
Downlink IBLER adaptation for CEUs	After downlink IBLER adaptation for CEUs is enabled, if the proportion of CEUs on the PDSCH is extremely large and their downlink signal quality fluctuates greatly, the average downlink throughput of such UEs may increase and the proportion ^b of low rates ^a in the cell decreases. This function is recommended in eMBB scenarios.
a: A rate is considered low if it is less than the result of "(Lower boundary of the throughput range corresponding to the non-zero counter with the largest index ID from N.Thp.DL.Samp.Index0 to N.Thp.DL.Samp.Index14) x 5%". For instance, if the value of N.Thp.DL.Samp.Index14 is 0, the counter with the highest index ID is N.Thp.DL.Samp.Index13 which corresponds to a throughput range of [512000 kbit/s, 1024000 kbit/s). In this case, a rate is considered low if it falls below 25600 kbit/s (25600 kbit/s = 512000 kbit/s x 5%). b: Proportion of low rates = Number of samples with low rates/Number of samples with all rates. If low rates refer to the rates below 25600 kbit/s and the throughput range associated with N.Thp.DL.Samp.Index6 is [25600 kbit/s, 30720 kbit/s), the number of samples with rates below 25600 kbit/s is the sum of counters ranging from N.Thp.DL.Samp.Index0 to N.Thp.DL.Samp.Index5. The proportion of low rates is calculated as follows: Proportion of low rates = (N.Thp.DL.Samp.Index0 + N.Thp.DL.Samp.Index1 + ... + N.Thp.DL.Samp.Index5)/(N.Thp.DL.Samp.Index0 + N.Thp.DL.Samp.Index1 + ... + N.Thp.DL.Samp.Index14).	

The following lists the network impacts of mmWave Super Coverage functions:

- After intelligent beam selection is enabled, the average uplink and downlink throughputs of UEs with this function not in effect may fluctuate due to possible changes in scheduling resources.
- After fixed SRS-based digital weight is enabled, the average downlink throughput of CEUs may decrease.
- After downlink PDSCH coverage enhancement is enabled, the proportion may increase, the downlink PRB usage may decrease, and the average downlink throughput may fluctuate for CEUs on the PDSCH.
- After downlink IBLER adaptation for CEUs is enabled, the downlink IBLER (**DL IBLER**) may increase. When the proportion of CEUs on the PDSCH is high, downlink data retransmissions increase due to UE distribution on different CSI-RS for BM beams or SSB beams. As a result, the average uplink or downlink UE throughput may fluctuate. When the proportion of CEUs is low, the average uplink and downlink UE throughputs may remain unchanged.

Impact on NEs

This feature is implemented on the gNodeB.

Intelligent beam selection has the following requirements on UEs:

- The value of **maxNumberCSI-RS-Resource** is greater than or equal to n32.
- The value of **maxNumberNonGroupBeamReporting** is n4.

For details about definitions of **maxNumberCSI-RS-Resource** and **maxNumberNonGroupBeamReporting**, see section 4.2.7.2 "BandNR parameters" in 3GPP TS 38.306 V17.4.0.

Impact on Hardware Requirements

Base Station Models

5900 series base stations (macro base stations). These base stations must be equipped with BBUs other than Book BBUs when intelligent beam selection, beam tracking, SRS-based digital weight, or SRS bandwidth adaptation is used.

Boards

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

HAAU5235w, HAAU5323, HAAU5233w

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

This feature is controlled by the following new license control item.

Feature ID	Feature Name	Model	Sales Unit
FOFD-101242	mmWave Super Coverage	NR0S00MWBM00	per Cell

- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
 - Intelligent beam selection

RAT	Function Name	Function Switch	Reference	Description
High - frequency TDD	Flexible Dense Beam	FLEXIBLE_DENSE_BEAM_SW option of the NRDUCellTrpMmwavBeam.BeamPerformanceSw parameter	<i>mmWave Beam Management (High-Frequency TDD)</i>	Intelligent beam selection can be enabled only if Flexible Dense Beam is enabled.
High - frequency TDD	mmWave super coverage	SUPER_COVERAGE_SW option of the NRDUCellCnncovAlgo.DlCoverageAlgoSwitch parameter	<i>mmWave Super Coverage (High-Frequency TDD)</i>	Intelligent beam selection can take effect only if mmWave super coverage is enabled.

- Beam tracking

RAT	Function Name	Function Switch	Reference	Description
High - frequency TDD	mmWave super coverage	SUPER_COVERAGE_SW option of the NRDUCellC hnCovAlgo.DlCoverageAlgoSwitch parameter	<i>mmWave Super Coverage (High-Frequency TDD)</i>	Beam tracking can be enabled only if mmWave super coverage is enabled.

– SRS-based digital weight

RAT	Function Name	Function Switch	Reference	Description
High - frequency TDD	mmWave super coverage	SUPER_COVERAGE_SW option of the NRDUCellC hnCovAlgo.DlCoverageAlgoSwitch parameter	<i>mmWave Super Coverage (High-Frequency TDD)</i>	- SRS- and PMI-based weight adaptation and SRS filtering-based downlink beamforming weight optimization can take effect only if mmWave super coverage is enabled. - Fixed SRS-based digital weight can take effect only if mmWave super coverage is enabled and SRS- and PMI-based weight adaptation is disabled.

RAT	Function Name	Function Switch	Reference	Description
High - frequency TDD	SRS- and PMI-based weight adaptation	DL_PMI_SRS_ADAPT_SW option of the NRDUCellAdvancedEdgeExpEnhSwitch parameter	<i>mmWave Super Coverage (High-Frequency TDD)</i>	- Fixed SRS-based digital weight can take effect only if mmWave super coverage is enabled and SRS- and PMI-based weight adaptation is disabled. - Downlink single-UE mobility precoding can be enabled only if fixed SRS-based digital weight or SRS- and PMI-based weight adaptation is enabled.
High - frequency TDD	Fixed SRS-based digital weight	NRDUCellPdsch.FixedWeightType set to SRS_WEIGHT	<i>mmWave Super Coverage (High-Frequency TDD)</i>	Downlink single-UE mobility precoding can be enabled only if fixed SRS-based digital weight or SRS- and PMI-based weight adaptation is enabled.
High - frequency TDD	SRS resource allocation for UEs performing downlink large-packet services	DL_BIG_PKT_SRS_RESOURCE_ALLOCATION_SW option of the NRDUCellSrsSrsAlgoExtSwitch parameter	None	Downlink single-UE mobility precoding can be enabled only if SRS resource allocation for UEs performing downlink large-packet services is enabled.

- SRS bandwidth adaptation

RAT	Function Name	Function Switch	Reference	Description
High - frequency TDD	mmWave super coverage	SUPER_COVERAGE_SW option of the NRDUCellChnCovAlgo.DlCoverageAlgoSwitch parameter	<i>mmWave Super Coverage (High-Frequency TDD)</i>	SRS bandwidth adaptation can be enabled only if all the following conditions are met: • mmWave super coverage is enabled. • Fixed SRS-based digital weight or SRS- and PMI-based weight adaptation is enabled. • SRS resource allocation for UEs performing downlink large-packet services is enabled.
High - frequency TDD	SRS- and PMI-based weight adaptation	DL_PMI_SRS_ADAPT_SW option of the NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch parameter	<i>mmWave Super Coverage (High-Frequency TDD)</i>	
High - frequency TDD	Fixed SRS-based digital weight	NRDUCellPdsch.FixedWeightType set to SRS_WEIGHT	<i>mmWave Super Coverage (High-Frequency TDD)</i>	
High - frequency TDD	SRS resource allocation for UEs performing downlink large-packet services	DL_BIG_PKT_SRS_RES_ALLOC_SW option of the NRDUCellSrsSrsAlgoExtSwitch parameter	None	

- Precise power control

RAT	Function Name	Function Switch	Reference	Description
High - frequency TDD	mmWave super coverage	SUPER_COVERAGE_SW option of the NRDUCellChnCovAlgo.DlCoverageAlgoSwitch parameter	<i>mmWave Super Coverage (High-Frequency TDD)</i>	- Downlink PDSCH coverage enhancement can take effect only if all of the following conditions are met: - mmWave super coverage is enabled. - The NRDUCellChnPwr.MaxPdschConvPwrOffset parameter is set to a value other than 0. - PUCCH grouping adaptation for inter-FR CA takes effect and the PCell and SCells use a single set of PUCCH-based feedback mechanisms. - PDSCH DMRS power adaptation can take effect only if both of the following conditions are met: - mmWave super coverage is enabled. - PUCCH grouping adaptation for inter-FR CA takes effect and the PCell and SCells use a single set of PUCCH-based feedback mechanisms.
High - frequency TDD	PUCCH grouping adaptation for inter-FR CA	PUCCH_GROUP_ADAPT_SW option of the NRDUCellCarrierMgmt.InterFrCaEnhSw parameter	<i>Carrier Aggregation</i>	

RAT	Function Name	Function Switch	Reference	Description
High - frequency TDD	Maximum Converged PDSCH Power	NRDUCellChnPwr.MaxPdschConvPwrOffset	<i>Power Control</i>	Downlink PDSCH coverage enhancement can take effect only if all of the following conditions are met: • mmWave super coverage is enabled. • The NRDUCellChnPwr.MaxPdschConvPwrOffset parameter is set to a value other than 0. • PUCCH grouping adaptation for inter-FR CA takes effect and the PCell and SCells use a single set of PUCCH-based feedback mechanisms.

– Downlink IBLER adaptation for CEUs

RAT	Function Name	Function Switch	Reference	Description
High - frequency TDD	mmWave super coverage	SUPER_COVERAGE_SW option of the NRDUCellChnCovAlgo.DlCoverageAlgoSwitch parameter	<i>mmWave Super Coverage (High-Frequency TDD)</i>	Downlink IBLER adaptation for CEUs can take effect only if mmWave super coverage is enabled.

- Mutually exclusive features/functions
 - Intelligent beam selection

RAT	Function Name	Function Switch	Reference	Description
High - frequency TDD	mmWave 3D coverage pattern	NRDUCellTrpMmwavBeam.CoverageScenario	<i>mmWave Beam Management (High-Frequency TDD)</i>	In the single-beam scenario of mmWave 3D coverage pattern (with the NRDUCellTrpMmwavBeam.CoverageScenario parameter set to SCENARIO_103(Single Beam Scenario)), intelligent beam selection cannot be enabled.
High - frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	None	Intelligent beam selection cannot be enabled when the NRDUCell.NrDuCellNetworkingMode parameter is set to HYPER_CELL .
High - frequency TDD	Combined cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	None	Intelligent beam selection cannot be enabled when the NRDUCell.NrDuCellNetworkingMode parameter is set to HYPER_CELL_COMBINE_MODE .

– Beam tracking

RAT	Function Name	Function Switch	Reference	Description
High - frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	None	Beam tracking cannot be enabled when the NRDUCell.NrDuCellNetworkingMode parameter is set to HYPER_CELL .
High - frequency TDD	Combined cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	None	Beam tracking cannot be enabled when the NRDUCell.NrDuCellNetworkingMode parameter is set to HYPER_CELL_COMBINE_MODE .

– SRS-based digital weight

RAT	Function Name	Function Switch	Reference	Description
High - frequency TDD	Hyper Cell	NRDUCell.NrDuCellNet workingMode set to HYPER_CELL	None	Fixed SRS-based digital weight and SRS- and PMI-based weight adaptation cannot be enabled when the NRDUCell.NrDuCellNet workingMode parameter is set to HYPER_CELL .

– SRS bandwidth adaptation

RAT	Function Name	Function Switch	Reference	Description
High - frequency TDD	PUCCH resource reduction for heavy-traffic UEs	NRDUCellPucch.PucchResrcRednMthd set to MASS_FLOW	<i>Channel Management</i>	SRS bandwidth adaptation cannot be enabled when the NRDUCellPucch.PucchResrcRednMthd parameter is set to MASS_FLOW .
High - frequency TDD	Increased downlink symbol proportion in self-contained slots	S_SLOT_DL_SYM_RATIO_INC_SW option of the NRDUCellDLSchRes.DLSchResAlgoSw parameter	None	SRS bandwidth adaptation cannot be enabled together with increased downlink symbol proportion in self-contained slots.
High - frequency TDD	AoA- and UTDOA-based hybrid positioning	AOA_UTDOA_HYBRID_LOCATION_SW option of the NRCCellAlgoSwitch.LcsSwitch parameter	None	SRS bandwidth adaptation cannot be enabled together with AoA- and UTDOA-based hybrid positioning.

– Precise power control

None

– Downlink IBLER adaptation for CEUs

None

- Impacted features/functions
 - Intelligent beam selection

RAT	Function Name	Function Switch	Reference	Description
High - frequency TDD	PUCCH grouping adaptation for inter-FR CA	PUCCH_GROUP_ADAPT_SW option of the NRDUCellCarrierMgmt.InterFrCaEnhSw parameter	<i>Carrier Aggregation</i>	Intelligent beam selection does not take effect when PUCCH grouping adaptation for inter-FR CA is in effect, in which case a single set of PUCCH-based feedback mechanisms is used in the PCell and SCells of a UE in the inter-FR CA state.
High - frequency TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Intelligent beam selection does not take effect when RF channel intelligent shutdown has shut down RF channels to have 2T left.

- Beam tracking
None
- SRS-based digital weight
None
- SRS bandwidth adaptation
None
- Precise power control

RAT	Function Name	Function Switch	Reference	Description
High - frequency TDD	PDCCH rate matching	• PDCCH_RATEMATCH_SWITCH option of the NRDUCellPdsch.RateMatchSwitch parameter • PDCCH_NO_PATTERN_RATEMATCH_SWITCH option of the NRDUCellPdcch.PdcchAlgorithmSwitch parameter	<i>Downlink Scheduling</i>	Rate-matching-pattern-configuration-based PDCCH rate matching and rate-matching-pattern-configuration-free PDCCH rate matching do not take effect in a TTI when downlink PDSCH coverage enhancement is in effect and the transmit PSD of the base station is adjusted.

- Downlink IBLER adaptation for CEUs
None

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *mmWave Super Coverage (High-Frequency TDD)*.

Requirement/Trouble Ticket Number

- FeR-IR20230914002788-0001
- FeR-IR20230615001550-0002
- FeR-IR20230615001550-0003
- FeR-IR20230615001550-0004
- FeR-IR20230615001555-0004
- IR20250815001172

RAT

NR TDD

5.3.1.15 MRFD-221261 RF Module Fast Deep Dormancy (NR)

5.3.1.15.1 Added Support for NR RF Module Fast Deep Dormancy

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

It takes minutes for an RF module in traditional deep dormancy to wake up, making it more suitable for activation at night. To enhance energy savings during the daytime, the RF module fast deep dormancy function, which enables rapid wakeup from deep dormancy, has been proposed.

This function enables RF modules to enter dormancy to reduce power consumption when the network is under light or no load, while maintaining RF module reliability. When the network load is high, cells can be quickly woken without reestablishment, minimizing impact on user experience while saving energy.

The involved sub-functions include NR load-based RF module fast deep dormancy and LNR load-based RF module fast deep dormancy.

NR load-based RF module fast deep dormancy:

In addition to providing greater network capacity, NR multi-carrier co-coverage networking delivers higher data throughput on a heavily-loaded network. However, it wastes base station energy when the network is under light or no load. To address this, NR load-based RF module fast deep dormancy is introduced. The RF module shutdown measures are optimized to allow cells to exit dormancy without reestablishment, enabling faster wakeup.

NR load-based RF module fast deep dormancy enables the gNodeB to transfer UEs from capacity-layer cells to basic-layer cells and instruct the RF modules serving the capacity-layer cells to enter fast deep dormancy within a specified period when the network is under light or no load. When the load of the basic-layer cells increases to a specified threshold, the dormant RF modules are automatically woken and the capacity-layer cells are restored.

This function requires the following switches to be configured:

- RF module-level switch
FAST_DEEP_DORM_SW of the **RRUEXTINFO.RRUDOROPTFEATSW** parameter is selected.
- NR cell-level switch
NR_FAST_DORMANCY_BY_LOAD_SW of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter is selected.

The effective period and threshold parameters for NR load-based RF module fast deep dormancy can be uniquely determined by using the setting combinations of

the **NRDUCellPowerSaving.PowerSavingPolicyIndex** and **NRDUCellPowerSaving.PowerSavingType** parameters in the **NRDUCellPowerSaving** MOs on the NR side. Where,

- The **NRDUCellPowerSaving.PowerSavingPolicyIndex** parameter specifies an energy saving policy index. It uniquely identifies an energy saving policy within the local cell. With the index specified, the corresponding policy can be added, modified, queried, or removed at the local cell level.
- The **NRDUCellPowerSaving.PowerSavingType** parameter specifies the energy saving type indicated by an energy saving policy index. NR load-based RF module fast deep dormancy requires this parameter to be set to **NR_FAST_DORMANCY_BY_LOAD**. Then under the corresponding energy saving policy index, the local cell performs load-based fast deep dormancy during the specified period when certain load conditions are met.

NR load-based RF module fast deep dormancy assurance:

To reduce the impact of NR load-based RF module fast deep dormancy on user-perceived rates, RF module fast deep dormancy assurance is introduced. This function includes two sub-functions: NR load-based RF module fast deep dormancy service assurance and NR load-based RF module fast deep dormancy load assurance. After NR load-based RF module fast deep dormancy is enabled, the following two sub-functions can be enabled.

- NR load-based RF module fast deep dormancy service assurance (by selecting the **NR_FAST_DORM_SERVICE_ASSUR_SW** option of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter for capacity-layer cells and the **PWR_SAVE_SERVICE_ASSUR_SW** option of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter for basic-layer cells). Once enabled, the cell determines whether to exit dormancy for user experience assurance at an interval of 5s as follows:
 - If the downlink PRB usage of each basic-layer cell is less than the PRB usage threshold for fast evaluation on carrier shutdown (specified by the **NRDUCellMobility.CarrShutdFastEvalPrbThld** parameter) and any basic-layer cell serves UEs with heavy downlink traffic, the capacity-layer cell exits fast deep dormancy.
 - If the downlink PRB usage of any basic-layer cell is greater than or equal to the PRB usage threshold for fast evaluation on carrier shutdown (specified by the **NRDUCellMobility.CarrShutdFastEvalPrbThld** parameter), the capacity-layer cell exits fast deep dormancy based on the setting of the **NRDUCellPowerSaving.PwrSaveServiceAssurLevel** parameter. A larger value of this parameter results in a smaller impact on user experience and fewer energy saving gains.
- NR load-based RF module fast deep dormancy load assurance (by selecting the **NR_FAST_DORM_LOAD_ASSUR_SW** option of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter for capacity-layer cells). Once enabled, the downlink PRB usage of each basic-layer cell is evaluated per second. If the downlink PRB usage of any basic-layer cell is greater than the threshold specified by the **NRDUCellMobility.CarrShdnEventEvalPrbThld** parameter in the current period but does not exceed this threshold in the previous period, the capacity-layer cell exits fast deep dormancy. NR load-based RF module fast deep dormancy load assurance takes effect only when the **NRDUCellMobility.CarrShdnEventEvalPrbThld** parameter is set to a value

other than 255. A smaller value of this parameter results in a smaller impact on user experience and fewer energy saving gains.

After the capacity-layer cell exits fast deep dormancy, a procedure is immediately triggered for each UE with heavy downlink traffic in the basic-layer cells to evaluate whether a frequency-priority-based handover can be performed, as long as this type of handover is enabled in the basic-layer cells. This enables such UEs to be transferred back to the capacity-layer cell promptly. The time-to-trigger in this procedure is specified by the

NRCellMobilityConfig.CarrShutUserMeasTimeToTrig parameter. Frequency-priority-based handover will not be triggered in a basic-layer cell with this type of handover disabled.

For details about frequency-priority-based handover, see the section on frequency-priority-based inter-frequency PSCell change in *NSA Mobility Management* and the section on frequency-priority-based inter-frequency handover in *SA Mobility Management in Connected Mode* in *5G RAN Feature Documentation*.

LNR load-based RF module fast deep dormancy:

NR cells in operation continue to consume energy even under light or no traffic in areas covered by both LTE and NR, resulting in unnecessary base station energy consumption. To address this, LNR load-based RF module fast deep dormancy is introduced. By optimizing RF module shutdown measures, cells can exit dormancy without reestablishment, enabling faster wakeup.

LNR load-based RF module fast deep dormancy enables the gNodeB to transfer UEs from NR cells to LTE cells and instruct the RF modules serving the NR cells to enter fast deep dormancy within a specified period when the network is under light or no load. When the load of the LTE cells increases to a specified threshold, the dormant RF modules are automatically woken and the NR cells are restored.

This function requires the following switches to be configured:

- RF module-level switch
FAST_DEEP_DORM_SW of the **RRUEXTINFO.RRUDOROPTFEATSW** parameter is selected.
- LTE cell-level switch
LNR_LOAD_BASED_DORMANCY_SW of the **CellPwrSavingAlgo.PwrSavingAlgoSwitch** parameter is selected.
- NR cell-level switch
LNR_FAST_DORMANCY_BY_LOAD_SW of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter is selected.

The effective period for LNR load-based RF module fast deep dormancy can be uniquely determined by using the setting combinations of the **NRDUCellPowerSaving.PowerSavingPolicyIndex** and **NRDUCellPowerSaving.PowerSavingType** parameters in the **NRDUCellPowerSaving** MOs on the NR side. Where,

- The **NRDUCellPowerSaving.PowerSavingPolicyIndex** parameter specifies an energy saving policy index. It uniquely identifies an energy saving policy within the local cell. With the index specified, the corresponding policy can be added, modified, queried, or removed at the local cell level.
- The **NRDUCellPowerSaving.PowerSavingType** parameter specifies the energy saving type indicated by an energy saving policy index. LNR load-

based RF module fast deep dormancy requires that this parameter be set to **LNR_FAST_DORMANCY_BY_LOAD**. Then under the corresponding energy saving policy index, the local cell performs LNR load-based RF module fast deep dormancy during the specified period when certain load conditions are met.

LampSite base stations do not support this function.

Implementation

For details, see *Multi-RAT Coordinated Energy Saving*. The specific sections are listed in the change history of the document.

Impact on the Network

Positive impacts:

It is recommended that RF module fast deep dormancy be enabled during fixed daily periods with light or no load to reduce base station power consumption. To ensure that the actual operating duration of an RF module meets lifecycle management requirements in typical scenarios, the module can enter dormancy when its reliability requirements (such as temperature and humidity) are met. If the RF module is reset during the dormancy period for purposes such as upgrade, the module and its cells will return to the normal state, which leads to reduced energy saving gains.

When the network is under heavy load, the RF module and its served cells can be quickly woken. This mitigates the impacts on user experience while achieving energy saving effects.

Negative impacts:

- RF module fast deep dormancy triggers handovers or redirections for active UEs, increasing the numbers of handover requests and redirections from the cell.
- In areas served by an RF module in fast deep dormancy, signal quality deteriorates because the module does not transmit signals. As a result, UEs may experience service drops or access failures.
- If a batch processing task on the OSS contains an eNodeBFunction or gNodeBFunction MML command that affects cell setup, or a Node MML command, and an NE of the base station is in fast deep dormancy, the OSS can execute the related configurations only after the base station exits fast deep dormancy. This results in increased waiting time before configuration execution. On average, the base station exits fast deep dormancy in less than 10 seconds.
- If a power-related command is executed on a base station with an NE in fast deep dormancy, the base station first exits fast deep dormancy and then executes the command. On average, the base station exits fast deep dormancy in less than 10 seconds.
- Before an RF module enters fast deep dormancy, new UEs are prohibited from accessing the NR cells of the module during NR carrier shutdown. If UEs attempt to be handed over to the NR cells, the values of the counters related to handover preparation failures (such as **N.HO.IntraFreq.FailOut.PrepFailure**) slightly increase. Such counters are not

used for calculating Huawei-defined KPIs outgoing intra- and inter-frequency handover success rates, but may be used for calculating user-defined KPIs. When dormancy is set to take effect on the hour, no UEs can be handed over to the deactivated NR cell. Therefore, even slight increases in the counter values may significantly affect the user-defined outgoing handover success rates. If fast deep dormancy is scheduled to take effect on the hour, you are advised to set the dormancy start time 5 minutes earlier, and set the lengths of penalty timers for handover failures to 30s. After penalty timers for handover failures are set, UEs may experience service drops due to delayed handovers. The penalty timers for handover failures are specified by the **gNBMobilityCommParam.NecHoPrepFailPunishTimer** and **gNBMobilityCommParam.OptHoPrepFailPunishTimer** parameters.

- Assuming that the neighboring base stations are in the dormancy state, CEUs in weak coverage areas may access the cells served by the local base station. This affects KPIs related to the aspects such as access and service drop of the local base station. For example, the value of **N.NsaDc.DRB.AbnormRel** may increase.

For NR load-based RF module fast deep dormancy:

- When NR load-based RF module fast deep dormancy takes effect, counters related to inter-frequency handovers and redirections are updated.
- If a UE attempts to hand over to a capacity-layer cell that is about to enter fast deep dormancy during UE transfer (lasting up to 30s), the source cell does not initiate handover execution. In this case, the values of the counters related to outgoing and incoming handover attempts increase in the source and capacity-layer cells, respectively, while the values of the counters related to handover executions and successful handovers do not increase. As Huawei-defined KPIs **Inter-Frequency Handover Out Success Rate (CU)** and **Intra-RAT Handover In Success Rate (CU)** are calculated using the numbers of outgoing and incoming handover executions as their denominators, these KPIs are not affected in this scenario. However, if the user-defined KPIs are calculated using the numbers of outgoing and incoming handover attempts as their denominators, these KPIs may be slightly affected in this scenario.

For LNR load-based RF module fast deep dormancy:

- When LNR load-based RF module fast deep dormancy is in effect, performance counters on the LTE and NR sides are affected. The affected performance counters are the same as those described in the "Impacts" section under "LTE and NR Intelligent Carrier Shutdown".
- In inter-base-station SA networking, X2 link setup may be triggered for energy saving purposes, leading to reduced EN-DC X2 link capacity and degraded experience of NSA UEs. The probability of ALM-29225 Maximum Number of X2 Interfaces Reached being reported also increases. Therefore, this function is not recommended when the X2 link capacity is limited.
- In inter-base-station SA networking, if the **LNR_INTER_BS_PWR_SAVE_SW** option of the **EnodebPwrSavingAlgo.PwrSavingAlgoSwitch** parameter or the **LNR_SA_GNB_X2_SETUP_SW** option of the **eNodeBOamParam.InterfaceSonConfigSwitch** parameter is selected but the **X2_SETUP_FOR_SA_SW** option of the **gNBInterfaceParam.X2ConfigSw** parameter is deselected on the gNodeB side, ALM-29204 X2 Interface Fault and ALM-25888 SCTP Link Fault may be reported.

- In inter-base-station SA networking, the configurations of the neighboring NR cell with overlapping coverage areas on the LTE side must be the same as those on the NR side, such as the cell ID, base station ID, and operator information. Otherwise, the SA X2 link may fail to be set up.

Impact on NEs

This feature is implemented on the gNodeB.

Impact on Hardware Requirements

- Main control boards: Only the UMPTe, UMPTg, UMPTga, UMPTHa, and UMPTHb series boards support this function.
- Baseband processing units: Only the UBBPg and UBBPh series boards support this function. The UBBPg1 and UBBPg1a do not support this function.
- RF modules:
The AAU5685md is compatible with RF module fast deep dormancy.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
This feature is controlled by the following new license control item.

RAT	Feature ID	Feature Name	Model	NE	Sales Unit
NR	MRFD-221 261	RF Module Fast Deep Dormancy (NR)	NR0SRFMF DD10	gNodeB	per RU

- Configuration management
The following MML commands have changed.

Change Type	MML Command	NE	Description
Modified	DSP BRDDORMANCY	NodeB eNodeB gNodeB	- The value Fast Deep Dormancy has been added to the Board Dormancy State parameter in the MML command output. - The Reason for Not Entering Fast Deep Dormancy parameter has been added to the MML command output.
Modified	DSP NRDUCELL	gNodeB	NR Load-based Fast Dormancy and its status have been added to the Extension Power Saving Status field in the MML command output. LNR Load-based Fast Dormancy and its status have been added to the Extension Power Saving Status field in the MML command output.

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellPowerSaving.PwrSaveServiceAssurLevel	ADD NRDUCELLPOWERSAVING MOD NRDUCELLPOWERSAVING LST NRDUCELLPOWERSAVING	gNodeB	This parameter has been added to specify the level of user service experience assurance after NR load-based fast dormancy is enabled.
Modified	RRUEXTINFO.RRUDOROPTFEATSW (5G gNodeB, LTE eNodeB)	ADD RRUEXTINFO MOD RRUEXTINFO LST RRUEXTINFO	No deB eNo deB gNodeB	The FAST_DEEP_DORM_SW option has been added to specify whether to enable RF module fast deep dormancy.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch	MOD NRDUCELLPWRS AVEALGO LST NRDUCELLPWRS AVEALGO	gNodeB	- The NR_FAST_DORMANCY_BY_LOAD_SW option has been added to specify whether to enable NR load-based fast dormancy. - The LNR_FAST_DORMANCY_BY_LOAD_SW option has been added to specify whether to enable LNR load-based fast dormancy. - The NR_FAST_DORM_SERVICE_ASSUR_SW option has been added to specify whether service assurance upon NR fast dormancy can take effect. - The PWR_SAVE_SERVICE_ASSUR_SW option has been added to specify whether to allow energy saving service assurance to take effect in capacity-layer cells when the local cell serves as a basic-layer cell. - The NR_FAST_DORM_LOAD_ASSUR_SW option has been added to specify whether load assurance upon NR fast dormancy can take effect.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellPowerSaving.PowerSavingType	ADD NRDUCELLPOWERSAVING MOD NRDUCELLPOWERSAVING DSP NRDUCELLPOWERSAVING LST NRDUCELLPOWERSAVING	gNodeB	- The NR_FAST_DORMANCY_BY_LOAD option has been added to enable the local cell to perform fast dormancy in the preset duration based on the load of the local cell and that of inter-frequency neighboring NR cells. - The LNR_FAST_DORMANCY_BY_LOAD option has been added to enable the local cell to perform fast dormancy in the preset duration based on the local cell load and the neighboring LTE cell load.
Modified	NRDUCellPowerSaving.UlPrbThld	ADD NRDUCELLPOWERSAVING MOD NRDUCELLPOWERSAVING LST NRDUCELLPOWERSAVING	gNodeB	This parameter now supports NR/LNR load-based fast dormancy.
Modified	NRDUCellPowerSaving.DlPrbThld	ADD NRDUCELLPOWERSAVING MOD NRDUCELLPOWERSAVING LST NRDUCELLPOWERSAVING	gNodeB	This parameter now supports NR/LNR load-based fast dormancy.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellPowerSaving.UserNumThld	ADD NRDUCCELLPOW ERSAVING MOD NRDUCCELLPOW ERSAVING LST NRDUCCELLPOW ERSAVING	gNodeB	This parameter now supports NR/LNR load-based fast dormancy.
Modified	NRDUCellPowerSaving.PowerSavingPenaltyTime	ADD NRDUCCELLPOW ERSAVING MOD NRDUCCELLPOW ERSAVING LST NRDUCCELLPOW ERSAVING	gNodeB	This parameter now supports NR/LNR load-based fast dormancy.
Modified	NRDUCellPowerSaving.UserNumOffset	ADD NRDUCCELLPOW ERSAVING MOD NRDUCCELLPOW ERSAVING LST NRDUCCELLPOW ERSAVING	gNodeB	This parameter now supports NR/LNR load-based fast dormancy.
Modified	NRDUCellPowerSaving.DLPrbOffset	ADD NRDUCCELLPOW ERSAVING MOD NRDUCCELLPOW ERSAVING LST NRDUCCELLPOW ERSAVING	gNodeB	This parameter now supports NR/LNR load-based fast dormancy.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellPowerSaving.NCellULPrbOffset	ADD NRDUCELLPOWERSAVING MOD NRDUCELLPOWERSAVING LST NRDUCELLPOWERSAVING	gNodeB	This parameter now supports NR/LNR load-based fast dormancy.
Modified	NRDUCellPowerSaving.EutranCarrShutDownExitOffset	ADD NRDUCELLPOWERSAVING MOD NRDUCELLPOWERSAVING LST NRDUCELLPOWERSAVING	gNodeB	This parameter now supports LNR load-based fast dormancy.
Modified	NRDUCellPowerSaving.EutranNrUserDLPrbThldOffset	ADD NRDUCELLPOWERSAVING MOD NRDUCELLPOWERSAVING LST NRDUCELLPOWERSAVING	gNodeB	This parameter now supports LNR load-based fast dormancy.
Modified	NRDUCellPowerSaving.EutranNrUserULPrbThldOffset	ADD NRDUCELLPOWERSAVING MOD NRDUCELLPOWERSAVING LST NRDUCELLPOWERSAVING	gNodeB	This parameter now supports LNR load-based fast dormancy.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellPowerSaving.EutranDlPrbOffset	ADD NRDUCELLPOWERSAVING MOD NRDUCELLPOWERSAVING LST NRDUCELLPOWERSAVING	gNodeB	This parameter now supports LNR load-based fast dormancy.
Modified	NRDUCellPowerSaving.EutranNrUserULPrbThld	ADD NRDUCELLPOWERSAVING MOD NRDUCELLPOWERSAVING LST NRDUCELLPOWERSAVING	gNodeB	This parameter now supports LNR load-based fast dormancy.
Modified	NRDUCellPowerSaving.EutranCarrShutDownRatio	ADD NRDUCELLPOWERSAVING MOD NRDUCELLPOWERSAVING LST NRDUCELLPOWERSAVING	gNodeB	This parameter now supports LNR load-based fast dormancy.
Modified	NRDUCellPowerSaving.EutranNrUserDLPrbThld	ADD NRDUCELLPOWERSAVING MOD NRDUCELLPOWERSAVING LST NRDUCELLPOWERSAVING	gNodeB	This parameter now supports LNR load-based fast dormancy.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellPowerSaving.EutranDlPrbThld	ADD NRDUCELLPOWERSAVING MOD NRDUCELLPOWERSAVING LST NRDUCELLPOWERSAVING	gNodeB	This parameter now supports LNR load-based fast dormancy.
Modified	NRDUCellPowerSaving.EutranNrUserNumThld	ADD NRDUCELLPOWERSAVING MOD NRDUCELLPOWERSAVING LST NRDUCELLPOWERSAVING	gNodeB	This parameter now supports LNR load-based fast dormancy.
Modified	NRDUCellPowerSaving.EutranNrUserNumThldOfs	ADD NRDUCELLPOWERSAVING MOD NRDUCELLPOWERSAVING LST NRDUCELLPOWERSAVING	gNodeB	This parameter now supports LNR load-based fast dormancy.
Modified	NRDUCellPowerSaving.NCellUlPrbThld	ADD NRDUCELLPOWERSAVING MOD NRDUCELLPOWERSAVING LST NRDUCELLPOWERSAVING	gNodeB	This parameter now supports LNR load-based fast dormancy.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRCellEutraNRelation.PowerSavingOverlapInd	ADD NRCELLEUTRAN RELATION MOD NRCELLEUTRAN RELATION LST NRCELLEUTRAN RELATION	gNodeB	This parameter now supports LNR load-based fast dormancy.
Modified	gNBpwrSavingsAlgo.PwrSwitch	MOD GNBPWRSAVING GALGO LST GNBPWRSAVING GALGO	gNodeB	The LNR_LOAD_DORM_PRIM_OP_RECOG_SW option of this parameter now supports LNR load-based fast dormancy.
Modified	NRCellRelation.PowerSavingCellFlag	ADD NRCELLRELATION MOD NRCELLRELATION LST NRCELLRELATION	gNodeB	This parameter now supports LNR load-based fast dormancy.
Modified	NRDUCellMobility.CarrShutdFastEvalPrbThld	MOD NRDUCELLMOBILITY LST NRDUCELLMOBILITY	gNodeB	This parameter now supports NR load-based fast dormancy.
Modified	NRDUCellMobility.CarrShdnEventEvalPrbThld	MOD NRDUCELLMOBILITY LST NRDUCELLMOBILITY	gNodeB	This parameter now supports NR load-based fast dormancy.

- Performance management
The following counters have changed.

Change Type	Counter Name	Function Subset	NE	Description
New	VS.RADIOEQM.FastDeepDormancy.Dur (LTE eNodeB, 5G gNodeB)	RF Unit Power Saving Measurement	NodeB eNodeB gNodeB	This counter has been added to measure the duration for which an RF unit remains in fast deep dormancy.
New	N.Cell.Unavail.Dur.RFModuleFastDormancy	Cell Algorithm Measurement	gNodeB	This counter has been added to measure the duration of cell unavailability due to RF module fast dormancy.

- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	The time period configured for timing carrier shutdown cannot overlap with the period configured for NR or LNR load-based RF module fast deep dormancy, which is a sub-function of RF module deep dormancy of the local cell. You are advised to preferentially plan the periods during which timing carrier shutdown takes effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Deployment Position	NRDUCell.DeployPosition set to DEPLOY_IAB	None	If this parameter is set to DEPLOY_IAB , NR or LNR load-based RF module fast deep dormancy cannot be enabled.
NR FDD	MBS broadcast	MBS_BROADCAST_SWITCH option of the NRCellFeatureSwitch.MbsSwitch parameter	None	When the MBS broadcast switch is on, NR or LNR load-based RF module fast deep dormancy cannot be enabled.
LTE FDD LTE TDD	Flag of Multi-RRU Cell	Cell.MultiRruCellFlag set to BOOLEAN_TRUE	None	LTE load-based RF module fast deep dormancy is incompatible with multi-RRU cells.
NR FDD Low-frequency NR TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	<i>URLLC in 5G RAN Feature Documentation</i>	NR or LNR load-based RF module fast deep dormancy is not supported in low latency and high reliability scenarios.
Low-frequency NR TDD	Time-domain reservation mode	NRDUCellTimeDomainReservationMode	None	RF module fast deep dormancy cannot be enabled when the NRDUCellTimeDomainReservationMode parameter is set to a value other than DEFAULT .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Dual-BBU backup in dual-network redundancy networking	DUAL_BBU_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	Dual-BBU backup in dual-network redundancy networking is incompatible with NR or LNR load-based RF module fast deep dormancy.
Low-frequency NR TDD	Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios	DUAL_PLANE_TOB_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	<i>Base Station Equipment Reliability</i>	"Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "NR or LNR load-based RF module fast deep dormancy" cannot be both enabled.
NR FDD Low-frequency NR TDD	Primary/Backup function in intra-frequency dual-network redundancy networking	PRIMARY_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	Primary/Backup function in intra-frequency dual-network redundancy networking is incompatible with NR or LNR load-based RF module fast deep dormancy.
NR FDD	Dual-plane function in intra-frequency dual-network redundancy networking	DUAL_PLANE_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	"Dual-plane function in intra-frequency dual-network redundancy networking" and "NR or LNR load-based RF module fast deep dormancy" cannot be both enabled.

- Impacted features/functions

When RF module fast deep dormancy and another energy saving function related to RF modules (such as carrier shutdown, symbol power saving, and channel shutdown) are both enabled, RF module fast deep dormancy preferentially takes effect.

When RF module fast deep dormancy and RF module deep dormancy (or RF module super dormancy) are both enabled, RF module deep dormancy (or RF module super dormancy) preferentially takes effect.

Cells do not provide services after entering dormancy, and related features do not take effect in these cells.

The following table lists other function impacts.

Table 5-17 Function impacts of NR load-based RF module fast deep dormancy

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Downlink intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAIgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation in 5G RAN Feature Documentation</i>	After RF module fast deep dormancy takes effect, the capacity-layer cells that have been shut down cannot be configured as SCells for UEs.
NR FDD Low-frequency NR TDD	Downlink intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAIgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation in 5G RAN Feature Documentation</i>	After RF module fast deep dormancy takes effect, the capacity-layer cells that have been shut down cannot be configured as SCells for UEs.
NR FDD Low-frequency NR TDD	Inter-gNodeB CA	INTER_GNODEB_CA_SW option of the NRDUCellAIgoSwitch.CaAlgoSwitch parameter	<i>CA Experience Improvement in 5G RAN Feature Documentation</i>	After RF module fast deep dormancy takes effect, the capacity-layer cells that have been shut down cannot be configured as SCells for UEs.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Uplink symbol power saving	UL_SYMBOL_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Uplink symbol power saving does not take effect when RF module fast deep dormancy is in effect.
NR FDD Low-frequency NR TDD	Symbol power saving	NRDUCellAIgoSwitch.SymbolShutdownSwitch	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over symbol power saving when both functions are enabled. Symbol power saving does not take effect when RF module fast deep dormancy is in effect.
NR FDD Low-frequency NR TDD	Deep symbol power saving	DEEP_SYMBOL_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over deep symbol power saving when both functions are enabled. Deep symbol power saving does not take effect when RF module fast deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Power saving scheduling	POWER_SAVING_SCHEDULE_SWITCH option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over power saving scheduling when both functions are enabled. Power saving scheduling does not take effect when RF module fast deep dormancy is in effect.
Low-frequency NR TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SWITCH option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over RF channel intelligent shutdown when both functions are enabled. RF channel intelligent shutdown does not take effect when RF module fast deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over energy saving based on flexible frequency-domain scheduling when both functions are enabled. Energy saving based on flexible frequency-domain scheduling does not take effect when RF module fast deep dormancy is in effect.
NR FDD Low-frequency NR TDD	TTI-level channel shutdown	TTI_RF_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over TTI-level channel shutdown when both functions are enabled. TTI-level channel shutdown does not take effect when RF module fast deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Low-power RF channel dynamic shutdown	LOW_POWER_RF_SHUTDOWN_SWITCH option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over low-power RF channel dynamic shutdown when both functions are enabled. Low-power RF channel dynamic shutdown does not take effect when RF module fast deep dormancy is in effect.
NR FDD Low-frequency NR TDD	Intelligent power control energy saving	SMART_POWER_CTRL_ENERGY_SAVE_SWITCH option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over intelligent power control energy saving when both functions are enabled. Intelligent power control energy saving does not take effect when RF module fast deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Dynamic RF channel shutdown	RF_CHN_DYN_SHUTDOWN_SW option of the NRDUCellParameters.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over dynamic RF channel shutdown when both functions are enabled. Dynamic RF channel shutdown does not take effect when RF module fast deep dormancy is in effect.
NR FDD	Uplink TTI-level RF channel dynamic muting	UL_TTI_CHANNEL_DYNAMICSW option of the NRDUCellParameters.PowerSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over uplink TTI-level RF channel dynamic muting when both functions are enabled. Uplink TTI-level RF channel dynamic muting does not take effect when RF module fast deep dormancy is in effect.
NR FDD Low-frequency NR TDD	Initial transmission power reduction	INIT_TRANSMISSION_POWER_REDUCTION_SW option of the NRDUCellParameters.PowerSavingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over initial transmission power reduction when both functions are enabled. Initial transmission power reduction does not take effect when RF module fast deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Retransmission power reduction	RETRANS_POWER_REDUCTION_SW option of the NRDUCellPowerSavingAlgorithm parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over retransmission power reduction when both functions are enabled. Retransmission power reduction does not take effect when RF module fast deep dormancy is in effect.
NR FDD Low-frequency NR TDD	Intelligent carrier shutdown in precise mode	INTRANB_MULTICARRIER_SHUTDOWN_SW or INTERNANB_MULTICARRIER_SHUTDOWN_SW option of the NRDUCellPowerSavingSwitch parameter MULTICARRIER_PRECISE_SHUTDOWN_SW option of the NRDUCellPowerSavingSwitch parameter selected	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over intelligent carrier shutdown when both functions are enabled. Intelligent carrier shutdown does not take effect when RF module fast deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Intelligent carrier shutdown in non-precise mode	• INTRA_GNB_MULTI_CARRIER_SHUTDOWN_SW or INTER_GNB_MULTI_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter • MULTI_CARRIER_PRECISE_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter deselected	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over intelligent carrier shutdown when both functions are enabled. Intelligent carrier shutdown does not take effect when RF module fast deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Cell load simulation	STR NRDUCELLS IMULATELO AD	None	• If the SIMU_LOAD_DORMANCY_IMP_SW option of the gNBWPwrSavin gAlgo.PwrSav ingAlgoSwit ch parameter is selected, and cell load simulation takes effect, the RF module serving the local cell is not allowed to enter the dormancy state, but the RF modules serving the cells not enabled with cell load simulation are allowed to enter the dormancy state. • If the SIMU_LOAD_DORMANCY_IMP_SW option of the gNBWPwrSavin gAlgo.PwrSav ingAlgoSwit ch parameter is deselected, and cell load simulation takes effect, all RF modules of the base station are not allowed

RAT	Function Name	Function Switch	Reference	Description
				to enter the dormancy state.
NR FDD Low-frequency NR TDD	Automatic removal of neighboring cells	NR_NR_ANR_DEL_SW option of the NRCellAlgoSwitch.AnrSwitch parameter	<i>ANR in 5G RAN Feature Documentation</i>	When automatic removal of neighboring cells in ANR takes effect and the neighbor relationships with the basic-layer cells are automatically removed for capacity-layer cells, RF module fast deep dormancy does not take effect.
NR FDD Low-frequency NR TDD	CMAS	gNBSibConfig.SibType set to SIB_TYPE_8	<i>CMAS in 5G RAN Feature Documentation</i>	When RF module fast deep dormancy is in effect, the corresponding NR cell is blocked and CMAS warning notifications cannot be properly sent in this cell. However, UEs can still receive CMAS warning notifications from other carriers.
Low-frequency NR TDD	Super uplink with SUL and NR FDD co-deployment	FLEX_SUPER_UPLINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter	<i>Super Uplink in 5G RAN Feature Documentation</i>	After RF module fast deep dormancy takes effect, SUL scheduling cannot be performed for super uplink UEs.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Basic VoNR functions	VONR_SW option of the NRCellAlgoSwitch.VonrSwitch parameter	<i>VoNR in 5G RAN Feature Documentation</i>	RF module fast deep dormancy does not take effect if any VoNR UE fails to be handed over from a capacity-layer cell during UE transfer.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Virtual Grid-based Multi-Frequency Coordination	• RSRP prediction model – VG_MODEL_ALLOW_BUILDFLAG option of the NRCellFreqRelationAggregationAttribute parameter – NR_IF_HO_MEAS_QTY_PRED_FUN_SW option of the NRCellSmartMultiCarr.NrMultiCarrierAlgoSwitch parameter • Downlink intra-frequency spectral efficiency prediction model	<i>Virtual Grid-based Multi-Frequency Coordination in 5G RAN Feature Documentation</i>	After RF module fast deep dormancy takes effect, the building of virtual grid models will be suspended and existing virtual grid models will go offline and cannot be used.

RAT	Function Name	Function Switch	Reference	Description
		– DL_IN TRAF_SE_MODEL_BUILT_ALLOW option of the NRCellSmartMultiCarr. <i>ModelBuildingSwitch</i> parameter – DL_IN TRAF_REQ_SPECT_EFFECTED_SWITCH option of the NRCellSmartMultiCarr. <i>NrMultiCarrierAlgorithmSwitch</i> parameter • Intra-frequency SRS beam prediction model INTRAF_SRS_MODEL_BUILT		

RAT	Function Name	Function Switch	Reference	Description
		LD_ALLO W option of the NRCells martMul tiCarr.M odelBuil dingSwit ch paramete r		
NR FDD Low- frequency NR TDD	Channel calibration	Macro base station: None LampSite base station: INTER_RRU _CHN_CALI B_SW option of the NRDUCellAl goSwitch.C hnCalibSwi tch parameter	None	If channel calibration is being performed when RF module fast deep dormancy takes effect, channel calibration fails and will not be started again. You can run the DSP NRDUCELLCHN CALIB command to check whether channel calibration is successful.
NR FDD Low- frequency NR TDD	RB allocation for downlink experience optimization	NRDUCellD lSchRes.DIE xpOptRblnc reasePct	<i>Downlink Scheduling in 5G RAN Feature Documentat ion</i>	When both RB allocation for downlink experience optimization and RF module fast deep dormancy take effect, an exit from RF module fast deep dormancy may be performed earlier than expected.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Cell-level MRO	MRO_SCENARIO_IDENTIFIER_SWITCH option of the gNB-MRO-ScenarioIdentifierSwitch parameter and INTRA_FREQ_MRO_SWITCH option of the NRCell-MRO-MroSwitch parameter	<i>MRO in 5G RAN Feature Documentation</i>	When both RF module fast deep dormancy and intra-frequency MRO are enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, MRO may take effect later than expected.
NR FDD Low-frequency NR TDD	UE-level MRO	UE_MRO_SWITCH option of the NRCell-MRO-MroSwitch parameter	<i>MRO in 5G RAN Feature Documentation</i>	When both RF module fast deep dormancy and UE-level MRO are enabled, the number of handovers within a period decreases if energy saving lasts for a long time. As a result, the penalty mechanism accuracy of UE-level MRO may be affected.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCellAIgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation in 5G RAN Feature Documentation</i>	A carrier exits a power sharing group once RF module fast deep dormancy takes effect for this carrier. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control messages changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	NR multi-band power adaptive scheduling	NR_MULTI_BAND_PWR_ADAPT_SWITCH_SW option of the NRDUCellAllgoSwitch.DynamicPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation in 5G RAN Feature Documentation</i>	A carrier exits a power sharing group once RF module fast deep dormancy takes effect for this carrier. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control messages changes due to parameter adjustment. This may cause changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.
NR FDD Low-frequency NR TDD	Inter-RAT redirection back to the HPLMN based on roaming UE identification	VOICE_CAPB_MOB_TO_HPLMN_SW option of the gNBOperator.HplmnAllgoSwitch parameter	<i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i>	When RF module fast deep dormancy is enabled, only blind redirection can be performed to transfer VoNR-incapable roaming UEs from the visited 5G network to the home 5G network.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Clock out-of-synchronization detection	CLK_DETECTION_SW option of the gNodeBParameter.ClkOutofsyncDetSwitch parameter	<i>Synchronization</i>	If clock out-of-synchronization occurs during the dormancy period, this issue may fail to be detected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Ultra-long holdover for sites with abnormal clock sources	CLK_OTASY NC_PROC_SW option of the TASM.MSCLKDIAGPROCESSSW parameter	<i>Synchronization</i>	Assume that the ultra-long hold switch is turned on but the ultra-long hold enhancement switch is turned off. If sequence detection is performed on a to-be-detected base station during the period when RF module fast deep dormancy takes effect for all cells of the base station, the sequence detection fails. In this case, the base station cannot enter the ultra-long holdover state even if its clock is faulty, or it exits the ultra-long holdover state as sequence detection cannot be performed for a long time due to energy saving. If the ultra-long hold enhancement switch is turned on, RF module fast deep dormancy does not take effect for the base station in the ultra-long holdover state and its reference

RAT	Function Name	Function Switch	Reference	Description
				neighboring base stations.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Voice-capability-based inter-frequency handover back to the HPLMN	VOICE_CAP B_MOB_TO_ HPLMN_SW option of the gNBOperat or.HplmnAl goSwitch parameter	<i>SA Mobility Managemen t in Connected Mode in 5G RAN Feature Documentat ion</i>	Assume that voice-capability-based inter-frequency handover back to the HPLMN is enabled on the gNodeB for the home operator of a VoNR-incapable roaming UE, the INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter is selected for the cell serving the UE, handover evaluation for RF module fast deep dormancy is ongoing, and the related timer has not expired. When the gNBFreqPriorityGroup.RoamingUeHplmnInterFHoFlag parameter is set to TRUE for the target cell's frequency in the frequency list group associated with the roaming UE, the following mechanism is applied:

- If the roaming UE supports handovers, the gNodeB instructs the roaming UE to immediately perform a handover to the target cell.
- If the roaming UE does not support handovers, inter-frequency handover for RF module fast deep dormancy is not performed for the roaming UE.
- When the **gNBFreqPriorityGroup.RoamingUeHplmnInterFHoFlag** parameter is set to **FALSE** for the target cell's frequency in the frequency list group associated with the

RAT	Function Name	Function Switch	Reference	Description
				roaming UE, inter-frequency handover for RF module fast deep dormancy is not performed for the roaming UE.

RAT	Function Name	Function Switch	Reference	Description
NR FDD	PIM interference cancellation (FDD) (trial)	PIM_INTERF_CANCELLATION_SWITCH option of the NRDUCellPimIntrfAvd.PimAlgoSwitch parameter	<i>Interference Avoidance in 5G RAN Feature Documentation</i>	After RF module fast deep dormancy takes effect in an interfering or victim cell, the gNodeB evaluates whether to continue PIM interference cancellation in each cell. • If PIM interference persists after RF module fast deep dormancy takes effect, PIM interference cancellation continues. • If PIM interference disappears after RF module fast deep dormancy takes effect, PIM interference cancellation is stopped to reduce resource consumption caused by periodic PIM interference characteristic training and interference characteristic extraction.

RAT	Function Name	Function Switch	Reference	Description
NR FDD	RF PIM interference cancellation	RF_PIM_CANN_SW option of the RRUEXTINFO.RruOptInterfCancellationSw parameter	<i>Interference Avoidance in 5G RAN Feature Documentation</i>	For RF PIM interference cancellation, after RF module fast deep dormancy takes effect in an interfering or victim cell, PIM interference characteristics change. In this case, the PIM interference cancellation process is restarted, causing fluctuations in PIM interference cancellation performance.
NR FDD	Intelligent PIM interference avoidance	PIM_INTERF_SMART_AVOID_SW option of the NRDUCellPimIntrfAvoid.PimAlgoSwitch parameter	<i>Interference Avoidance in 5G RAN Feature Documentation</i>	RF module fast deep dormancy is terminated during PIM interference detection.

RAT	Function Name	Function Switch	Reference	Description
NR FDD	Time-frequency two-dimensional unified scheduling optimization	TIME_FREQ_2D_UNIF_OPT_SCH_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter	<i>Multi-Frequency Smart Aggregation in 5G RAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over time-frequency two-dimensional unified scheduling optimization when both functions are enabled. Time-frequency two-dimensional unified scheduling optimization does not take effect when RF module fast deep dormancy is in effect.
Low-frequency NR TDD	CA Performance Boost Based on Multi-dimensional Optimization	CA_PRFM_BOOST_BSD_ON_MD_OPT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter	<i>Multi-Frequency Smart Aggregation in 5G RAN Feature Documentation</i>	RF module fast deep dormancy takes precedence over CA Performance Boost Based on Multi-dimensional Optimization when both functions are enabled. CA Performance Boost Based on Multi-dimensional Optimization does not take effect when RF module fast deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Co-MM	INTRA_GNB_SU_COMM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD) in 5G RAN Feature Documentation</i>	When RF module fast deep dormancy takes effect, the probability that Co-MM takes effect decreases.
Low-frequency NR TDD	LTE and NR multi-band power adaptive scheduling	- LTE: LNR_MULTIBAND_POWER_ADPT_SWITCH option of the CellDynamicPowerSharing.DynamicPowerSharingSwitch parameter - NR: LNR_MULTIBAND_POWER_ADPT_SWITCH option of the NRDUCellAlgoSwitch.DynamicPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	A carrier exits a power adaptive scheduling group once RF module fast deep dormancy takes effect for this carrier. When a carrier enters or exits the energy saving state, the coverage of SSB, CSI-RS, PDCCH, and control messages changes due to system parameter adjustment. This causes changes in the access success rate, handover success rate, service drop rate, and paging success rate, leading to traffic fluctuations for about 2 minutes.

RAT	Function Name	Function Switch	Reference	Description
NR FDD	LTE FDD and NR Flash Dynamic Power Sharing	LTE_NR_DYNAMIC_POWER_SHARING_SWITCH option of the NRDUCellAlgorithmSwitch.DynamicPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	RF module fast deep dormancy takes precedence over LTE FDD and NR Flash Dynamic Power Sharing when both functions are enabled. LTE FDD and NR Flash Dynamic Power Sharing does not take effect when RF module fast deep dormancy is in effect.
Low-frequency NR TDD	LTE TDD and NR Flash Dynamic Power Sharing	LTE_NR_DYNAMIC_POWER_SHARING_SWITCH option of the NRDUCellAlgorithmSwitch.DynamicPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	RF module fast deep dormancy takes precedence over LTE TDD and NR Flash Dynamic Power Sharing when both functions are enabled. LTE TDD and NR Flash Dynamic Power Sharing does not take effect when RF module fast deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	L/NR Inter-band Deep Coordination	LNR_INTER_BAND_DEEP_COORD_SW option of the NRDUCellResShrCfg.DynPwrShrAlgorithmSwitch parameter	<i>LTE and NR Power Allocation</i>	RF module fast deep dormancy takes precedence over L/NR Inter-band Deep Coordination when both functions are enabled. L/NR Inter-band Deep Coordination does not take effect when RF module fast deep dormancy is in effect.
NR FDD	LTE-NR multi-dimensional scheduling optimization	LNR_MULTID_OPT_SWITCH option of the NRDUCellGigaBand.GigaBandCellAlgorithmSw parameter	<i>LTE FDD and NR Inter-RAT Unified Scheduling</i>	LTE-NR multi-dimensional scheduling optimization does not take effect for cells in the RF module fast deep dormancy state.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Speed turbo UE identification	gNodeBParam.StUelidentSw	None	When any of the following configuration operations is performed and the configured function takes effect on the base station, the served cells already in the dormancy state remain in this state but other served cells will not enter this state, resulting in a decrease in the energy saving gains:
NR FDD Low-frequency NR TDD	gNodeB acceleration service application	• gNBAsAppParam.AppDomainName • gNBAsAppParam.AppIpv4 • gNBAsAppParam.AppIpv6	None	• Any option of the gNodeBParam.StUelidentSw parameter is selected. • The gNBAsAppParam.AppDomainName parameter is set. • The gNBAsAppParam.AppIpv4 parameter is set. • The gNBAsAppParam.AppIpv6 parameter is set. The following methods can be used to address the preceding issue:

RAT	Function Name	Function Switch	Reference	Description
				• Select the SPEED_TURBO_PWR_SAVING_SW option of the gNBWPwrSavingsAlgo.PwrSavingAlgoSwitch parameter. Dormancy can take effect for cells served by the base station in some of the preceding scenarios within the period specified by the gNBWPwrSavingsAlgo.SpeedTurboPwrSavingsStartTime and gNBWPwrSavingsAlgo.SpeedTurboPwrSavingsEndTime parameters, increasing the energy saving gains. • Select the STATIC_STATIC_PWR_SAVING_SW option of the gNBWPwrSavingsAlgo.PwrSavingAlgoSwitch parameter. Dormancy can take effect for cells served by the base station in some of the preceding

RAT	Function Name	Function Switch	Reference	Description
				scenarios within the period specified by the gNBPwrSavin gAlgo.StaticS TPwrSavStartTime and gNBPwrSavin gAlgo.StaticS TPwrSavEndTime parameters, increasing the energy saving gains.
NR FDD Low-frequency NR TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag set to HIGH_SPEED	<i>High Speed Mobility in 5G RAN Feature Documentation</i>	When the high-speed cell function in high speed specified policy management is enabled, you are advised not to enable NR load-based RF module fast deep dormancy or LNR load-based RF module fast deep dormancy.

In addition to the function impacts of NR load-based RF module fast deep dormancy (except the impact relationship with intra-RAT ANR), LNR load-based RF module fast deep dormancy has the following function impacts.

Table 5-18 Function impacts of LNR load-based RF module fast deep dormancy

RA T	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Work mode	Cell.<i>WorkMode</i>	None	- An LTE FDD cell with the Cell.<i>WorkMode</i> parameter set to DL_ONLY or LAA cannot be configured as an overlapping neighboring cell of an NR cell. - An LTE TDD cell with the Cell.<i>WorkMode</i> parameter set to LAA cannot be configured as an overlapping neighboring cell of an NR cell.
LTE FDD LTE TDD	Cell System Information Switch	CellSiMap.<i>SiSwitch</i>	None	A cell whose ForbidCellSiSwitch option of the CellSiMap.<i>SiSwitch</i> parameter is selected cannot be configured as a basic-layer cell.
LTE FDD LTE TDD	eMTC introduction	EMTC_SWITCH option of the CellEmtcAlgo.<i>EmtcAlgoSwitch</i> parameter	<i>eMTC</i> in <i>eRAN Feature Documentation</i>	An eMTC-only cell (specified by the EMTC_SWITCH and EMTC_ONLY_CELL_SWITCH options of the CellEmtcAlgo.<i>EmtcAlgoSwitch</i> parameter) cannot be configured as a basic-layer cell.

RA T	Function Name	Function Switch	Reference	Description
LTE FDD	Static Multiple Beam	SectorSplitGroup.BeamTemplateId set to a value greater than 1	<i>Massive MIMO Basics (FDD) in eRAN Feature Documentation</i>	For LTE FDD, if a Static Multiple Beam cell is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgorithm.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in NR cells, the user-perceived rate may decrease in the split cell that serves as a basic-layer cell.
LTE TDD	Intelligent Multi-Beam of 8T8R	None	<i>Intelligent Multi-Beam of 8T8R (TDD) in eRAN Feature Documentation</i>	For LTE TDD, if a cell enabled with Intelligent Multi-Beam of 8T8R is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgorithm.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in NR cells, the user-perceived rate may decrease in the split cell that serves as a basic-layer cell.
LTE FDD	Static Multiple Beam	None	<i>Smart 8T8R (FDD) in eRAN Feature Documentation</i>	For LTE FDD, if a Static Multiple Beam cell is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgorithm.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in NR cells, the user-perceived rate may decrease in the split cell that serves as a basic-layer cell.

RA T	Function Name	Function Switch	Reference	Description
LTE TDD	Intra-frequency split	None	<i>Soft Split Resource Duplex (TDD) in eRAN Feature Documentat ion</i>	For LTE TDD, if an intra-frequency split cell is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in NR cells, the user-perceived rate may decrease in the split cell that serves as a basic-layer cell.
LTE FDD	RuralLink antenna solution (LTE)	None	<i>RuralLink Solution</i>	For LTE FDD, if a split cell established through the RuralLink antenna solution (LTE) is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in NR cells, the user-perceived rate may decrease in the split cell that serves as a basic-layer cell.
LTE FDD	LumiLink solution (LTE)	• SectorSplitGroup.BeamTemplateId set to 0 • SectorSplitGroup.BeamDeploymentMode set to MODE_5 • SectorSplitCell.CellBeamIndex set to 0, 1, or 2	<i>LumiLink Solution</i>	For LTE FDD, if a split cell established through the LumiLink solution (LTE) is configured as a basic-layer cell, the SPLIT_CELL_RF_DORMANCY_SW option of the EnodebPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected. If load-based dormancy takes effect in NR cells, the user-perceived rate may decrease in the split cell that serves as a basic-layer cell.

RA T	Func ti on Name	Function Switch	Reference	Description
LTE FD D LTE TD D	RAN sharin g	RAN sharing with common carrier: ENodeBSharing Mode.ENodeBS haringMode set to SHARED_FREQ RAN sharing with dedicated carrier: ENodeBSharing Mode.ENodeBS haringMode set to SEPARATED_FR EQ Hybrid RAN sharing: ENodeBSharing Mode.ENodeBS haringMode set to HYBRID_SHARE D	<i>RAN Sharing in eRAN Feature Documentat ion</i>	In RAN sharing scenarios, the PLMN IDs of a capacity-layer cell must be a subset of the PLMN IDs of its co-coverage basic- layer cells. Otherwise, the capacity-layer cell cannot enter the dormancy state.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	NSA anchor carrier shutdown	NSA_ANCHOR_CARR_SHUTDOWN_SW option of the CellShutdown.CarrierShutdownEnhancedSw parameter	<i>Energy Conservation and Emission Reduction in eRAN Feature Documentation</i>	In NSA or NSA and SA hybrid networking, it is recommended that the INTER_RAT_GRAD_PWR_SAVING_SW option of the gNBpwrSavingAlgo.PwrSavingAlgoSwitch parameter be selected when NSA anchor carrier shutdown is enabled for the LTE basic-layer cell and NSA anchor cells and LNR load-based RF module fast deep dormancy is enabled for NR capacity-layer cells and NSA anchor cells. • If NSA anchor carrier shutdown takes effect first, carriers are shut down in NSA anchor cells. If NR capacity-layer cells do not have other neighboring LTE cells, LNR load-based RF module fast deep dormancy does not take effect. • If LNR load-based RF module fast deep dormancy takes effect first, this function can take effect in NR capacity-layer cells.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Multi-RAT Coordinated Energy Saving*.

Requirement/Trouble Ticket Number

IR20250210002499

RAT

NR FDD, NR TDD

5.3.2 Enhanced Features

5.3.2.1 FBFD-010014 Mobility Management

5.3.2.1.1 RSRP- and SINR-based Joint Handover

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

RSRP- and SINR-based joint handover is designed to ensure that both the RSRP and SINR of the selected target cell meet specific requirements during a handover, thereby improving user experience afterward.

In specific scenarios, if (1) the cell-specific switch for RSRP- and SINR-based joint handover is turned on, and (2) the bearer-specific switch for RSRP- and SINR-based joint handover is turned on for any bearer of a UE, RSRP- and SINR-based joint handover will take effect for the UE. With this function, the gNodeB uses UE-reported SINR measurements to evaluate ground neighboring cells and, as a result, filters out those whose SINRs fall below a specific threshold. The specific scenarios and related parameters are explained as follows:

- Specific scenarios refer to handover or redirection scenarios other than the following:
 - Blind handovers
 - Blind redirections
 - Handovers in which the gNodeB performs evaluations based on measurement reports—for example, in a handover scenario where experience-based smart carrier selection is applied, the gNodeB evaluates user experience after receiving the reports.
 - Grid-based handovers
- Related parameters:
 - Cell-specific RSRP- and SINR-based joint handover switch: **RSRP_SINR_HO_SW** option of the **NRCellMobilityConfig.InterFreqHoExtSwitch** parameter
 - Bearer-specific RSRP- and SINR-based joint handover switch: **RSRP_SINR_HO_SW** option of the **gNBQciBearer.ServiceDiffAlgoSwitch** parameter
 - Ground neighboring cell flag: **NRExternalNCell.AerialCellFlag** set to **GROUND_CELL**
 - Threshold: Indicates the threshold corresponding to the bearer with the highest handover priority on a UE. For bearer-specific handover priorities in SA networking, refer to the descriptions of QCI priorities for handovers in the "QCI-specific Measurement Event Configuration" section of *SA Mobility Management in Connected Mode*. For bearer-specific handover priorities in NSA networking, refer to the "QCI-specific Measurement Parameter Configuration" section in *NSA Mobility Management*.

- In the case of necessary handovers, the threshold is specified by the **NRCellInterFHoMeaGrp.RsrpSinrBasedNecSinrThld** parameter.
- In the case of unnecessary handovers, the threshold is specified by the **NRCellInterFHoMeaGrp.RsrpSinrBasedUnecSinrThld** parameter.

The change in this feature is supported only in low-frequency bands.

Implementation

For details, see *SA Mobility Management in Connected Mode* and *NSA Mobility Management*. The specific sections are listed in the change history of the documents.

Impact on the Network

When (1) the **RSRP_SINR_HO_SW** option of the **NRCellMobilityConfig.InterFreqHoExtSwitch** parameter is selected, and (2) the **RSRP_SINR_HO_SW** option of the **gNBQciBearer.ServiceDiffAlgoSwitch** parameter is selected for any bearer of a UE, the ground neighboring cells with SINRs below the threshold will be filtered out. In this way, the UE can select a ground neighboring cell with a better SINR as the target cell, thereby improving user experience after it is handed over to the target cell. Note that, if the SINR threshold is too high, there will be no available ground neighboring cell for handovers, increasing the service drop rate.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRCeIIInterFHoM eaGrp.RsrpSinrBas edNecSinrThld	ADD NRCELLINTER FHOMEAGRP MOD NRCELLINTER FHOMEAGRP LST NRCELLINTER FHOMEAGRP	gNodeB	This parameter has been added to specify the SINR threshold for necessary handovers in the RSRP- and SINR-based joint handover function.
New	NRCeIIInterFHoM eaGrp.RsrpSinrBas edUnecSinrThld	ADD NRCELLINTER FHOMEAGRP MOD NRCELLINTER FHOMEAGRP LST NRCELLINTER FHOMEAGRP	gNodeB	This parameter has been added to specify the SINR threshold for unnecessary handovers in the RSRP- and SINR-based joint handover function.
Modified	NRCeIIIMobilityCo nfig.InterFreqHoE xtSwitch	MOD NRCELLMOBIL ITYCONFIG LST NRCELLMOBIL ITYCONFIG	gNodeB	The RSRP_SINR_H O_SW option has been added to specify whether to enable RSRP- and SINR-based joint handover.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNBQciBearer.ServiceDiffAlgoSwitch	ADD GNBQCIBEARER MOD GNBQCIBEARER LST GNBQCIBEARER	gNodeB	The RSRP_SINR_HO_SW option has been added to specify whether to enable RSRP- and SINR-based joint handover for UEs using specific service bearers.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationship has been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RSRP- and SINR-based joint handover	RSRP_SINR_HO_SW option of the NRCellMobilityConfig.InterFreqHoExtSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	Bearer-specific RSRP- and SINR-based joint handover (controlled by the RSRP_SINR_HO_SW option of the gNBQciBearer.ServiceDiffAlgoSwitch parameter) can take effect only when cell-specific RSRP- and SINR-based joint handover has been enabled (by selecting the RSRP_SINR_HO_SW option of the NRCellMobilityConfig.InterFreqHoExtSwitch parameter).

- Mutually exclusive features/functions
The change in this feature has no impact.

- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *SA Mobility Management in Connected Mode*.

Requirement/Trouble Ticket Number

FeR-IR20250430001361-0001

RAT

NR FDD, NR TDD

5.3.2.1.2 Optimized Non-Overlapping Between Gap and NR Resources

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The uplink and downlink throughput of UEs decrease in the measurement gaps for inter-frequency measurements because UEs cannot transmit uplink signals or receive downlink signals on the serving frequency throughout the measurement gaps. If the SRS period is shorter than the gap period, specific time-domain positions of SRSs and gap-assisted measurements will conflict. This affects the accuracy of MCS index selection and the timeliness of measurement reporting, resulting in deteriorated service experience for moving UEs. To address this issue, the "optimized non-overlapping between gap and NR resources" function is introduced. This function staggers the channel resources for specific signals (such as SRS, PUCCH SR, CSI-RS for CM, and CSI-RS for CM measurement reporting) from measurement gaps in the time domain, when the SRS period is shorter than the gap period.

This function is controlled by the **GAP_AVOID_NR_RES_OPT_SW** option of the **NRDUCellMeas.MeasAlgoSwitch** parameter. It is recommended that this function be enabled when the channel quality fluctuates significantly.

- If this option is deselected, the channel resources for the aforesaid signals are not staggered from measurement gaps in the time domain. In this case, measurement gaps preferentially take effect on UEs, which do not transmit or receive signals throughout measurement gaps.
- If this option is selected, the channel resources for the aforesaid signals are staggered from measurement gaps in the time domain. This lowers the uplink IBLER and uplink RBLER, reduces the number of delayed measurement reports, but may increase the number of reconfiguration signaling messages. (IBLER: initial block error rate; RBLER: residual block error rate)

The change in this feature is supported only in low-frequency bands.

Implementation

For details, see *SA Mobility Management in Connected Mode* and *NSA Mobility Management*. The specific sections are listed in the change history of the documents.

Impact on the Network

When "optimized non-overlapping between gap and NR resources" is enabled after an upgrade, the channel resources for specific signals are staggered from measurement gaps in the time domain. This lowers the uplink IBLER and uplink RBLER, reduces the loss in uplink and downlink UE throughput caused by gap-assisted measurement, reduces the number of delayed measurement reports, but may increase the number of reconfiguration signaling messages. The values of the following indicators may change:

- The values of **N.ThpTime.UL.Cell** and **N.ThpTime.DL.Cell** increase.
- **Cell Uplink Average Throughput (DU)** and **Cell Downlink Average Throughput (DU)** decrease.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact.
- Configuration management
The following parameter has changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellMeas.MeasAlgoSwitch	MOD NRDUCELLMEAS LST NRDUCELLMEAS	gNodeB	The GAP_AVOID_NR_RES_OPT_SW option has been added to this parameter to specify whether to enable optimized non-overlapping between gap and NR resources.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *SA Mobility Management in Connected Mode*.

Requirement/Trouble Ticket Number

FeR-IR20250528001856-0015

RAT

NR FDD, NR TDD

5.3.2.1.3 Service-based Inter-Frequency Handover Enhancement

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

With service-based inter-frequency handover in effect, the gNodeB does not consider the serving cell's signal quality when deciding whether to trigger a service-based inter-frequency handover. However, this approach may impact UEs who already enjoy good service experience with their RSRPs higher than a specific threshold in the serving cell, so handovers should be limited for them. To improve this, service-based inter-frequency handover enhancement is introduced.

This function is jointly controlled by the **NRCeIIInterFHoMeaGrp.InterFreqServHoA2ThldRsrp** parameter and the **SERV_BASED_ENH_HO** option of the **NRCeIIIMobilityConfig.InterFreqHoExtSwitch** parameter. This function consists of two parts:

- Event A2 can be used in the handover function start decision for service-based inter-frequency handover.
 - If the aforesaid option is selected, event A2 is used in the handover function start decision for service-based inter-frequency handover, and the RSRP threshold for event A2 related to service-based inter-frequency handover takes the value of the **NRCeIIInterFHoMeaGrp.InterFreqServHoA2ThldRsrp** parameter.
 - If the aforesaid option is deselected, event A2 is not used in the handover function start decision for service-based inter-frequency handover.
- The **NRCeIIInterFHoMeaGrp.InterFreqServHoA2ThldRsrp** parameter can be used to configure RSRP threshold 1 for event A5 related to service-based inter-frequency handover.

Implementation

For details, see *SA Mobility Management in Connected Mode*. The specific sections are listed in the change history of the document.

Impact on the Network

When service-based inter-frequency handover enhancement takes effect after an upgrade, the number of outgoing service-based inter-frequency handovers from the serving cell decreases and the unnecessary measurement overheads decline, resulting in fluctuations in the number of UEs served by a cell and the values of mobility-related performance counters. This is because a service-based inter-frequency handover can be triggered only when the serving cell's signal quality measured by a UE meets the event A2 threshold. A smaller value of the event A2 threshold results in a lower probability of triggering event A2 reports and inter-frequency measurements. A larger value of the event A2 threshold produces the opposite effects.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRCellInterFHomeGrp.InterFreqServHoA2ThldRsrp	ADD NRCELLINTER FHOMEAGRP MOD NRCELLINTER FHOMEAGRP LST NRCELLINTER FHOMEAGRP	gNodeB	This parameter has been added to indicate the RSRP threshold for event A2 related to service-based inter-frequency handover and RSRP threshold 1 for event A5.
Modified	NRCellMobilityConfig.InterFreqHoeExtSwitch	MOD NRCELLMOBILITYCONFIG LST NRCELLMOBILITYCONFIG	gNodeB	The SERV_BASED_ENH_HO option has been added to this parameter to specify whether to enable service-based inter-frequency handover enhancement.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationship has been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Service-based inter-frequency handover	SERVICE_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	Service-based inter-frequency handover enhancement (controlled by the SERV_BASED_ENH_HO option of the NRCellMobilityConfig.InterFreqHoExtSwitch parameter) requires service-based inter-frequency handover to be enabled.
Low-frequency TDD	Service-based inter-frequency handover enhancement	SERV_BASED_ENH_HO option of the NRCellMobilityConfig.InterFreqHoExtSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	The RSRP threshold for event A2 related to service-based inter-frequency handovers takes the value of the NRCellInterFHoMeaGrp.InterFreqServHoA2ThldRsrp parameter only after service-based inter-frequency handover enhancement is enabled.

- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *SA Mobility Management in Connected Mode*.

Requirement/Trouble Ticket Number

FeR-IR20250408001259-0001

RAT

NR FDD, NR TDD

5.3.2.1.4 DRB Session Forwarding

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The source gNodeB does not forward the packets that have been sent but not acknowledged over the air interface to the target gNodeB in a specific scenario where both of the following conditions apply: (1) There is an inter-base-station handover that involves abnormal UEs or interconnection across different vendors. (2) The "full configurations during inter-base-station handovers" function is enabled (by selecting the **INTER_GNB_HO_FULL_CONFIG_SW** option of the **NRCellAlgoSwitch.HoCompatibilitySwitch** parameter). As a result, those packets are lost, negatively impacting user experience. To address this, the DRB session forwarding function is introduced.

DRB session forwarding is controlled by the **DRB_SESSION_FWD_SW** option of the **gNBMobilityCommParam.MobilityAlgoSwitch** parameter. When this function is enabled, the source gNodeB can forward the packets that have been sent but not acknowledged over the air interface to the target gNodeB. This reduces packet loss and enhances user experience. It is recommended that this function be enabled when "full configurations during inter-base-station handovers" is enabled.

The change in this feature is supported only in SA networking.

Implementation

For details, see *SA Mobility Management in Connected Mode*. The specific sections are listed in the change history of the document.

Impact on the Network

With this function enabled after an upgrade, the loss of packets that have been sent but not acknowledged over the air interface is mitigated, enhancing user experience.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact.
- Configuration management
The following parameter has changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNB MobilityCom mParam. <i>MobilityA lgoSwitch</i>	MOD GNBMOBILITY COMMPARAM LST GNBMOBILITY COMMPARAM	gNodeB	The DRB_SESSION_FWD_SW option has been added to this parameter to specify whether to enable DRB session forwarding.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *SA Mobility Management in Connected Mode*.

Requirement/Trouble Ticket Number

FeR-IR20250103001850-0001

FeR-IR20250528001857-0003

RAT

NR FDD, NR TDD

5.3.2.1.5 Full Configuration Optimization

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Full configuration optimization is designed to disable the "full configurations during inter-base-station handovers" function when all of the following conditions are met in the same scenario: (i) an Xn-based handover occurs in SA networking, (ii) the target and source gNodeBs are provided by the same vendor, and (iii) "full configurations during inter-base-station handovers" is enabled (by selecting the **INTER_GNB_HO_FULL_CONFIG_SW** option of the **NRCellAlgoSwitch.HoCompatibilitySwitch** parameter). Disabling the function under these conditions prevents the loss of specific packets for the UE involved in the handover, thereby ensuring user experience. Full configuration optimization is controlled by the **FULL_CONFIG_OPT_SW** option of the **NRCellFeatureSwitch.NrHoCompatibilitySwitch** parameter. When this optimization is enabled along with the "full configurations during inter-base-station handovers" function, the latter will not apply to Xn-based handovers between base stations of the same vendor.

The change in this feature is supported only in SA networking.

Implementation

For details, see *SA Mobility Management in Connected Mode*. The specific sections are listed in the change history of the document.

Impact on the Network

With the "full configurations during inter-base-station handovers" function and full configuration optimization both enabled after an upgrade, the former does not apply to Xn-based handovers between base stations of the same vendor. This reduces packet loss associated with the "full configurations during inter-base-station handovers" function and ensures user experience.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact.
- Configuration management
The following parameter has changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRCellFeatureSwitch.NrHoCompatibilitySwitch	MOD NRCELLFEATURSWITCH LST NRCELLFEATURSWITCH	gNodeB	The FULL_CONFIG_OPT_SW option of this parameter specifies whether to enable full configuration optimization.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *SA Mobility Management in Connected Mode*.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0003

RAT

NR FDD, NR TDD

5.3.2.1.6 Gap-assisted Scheduling Compatibility

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

If the measurement gap timing advance (MGTA) value delivered by the gNodeB is not 0, UEs with compatibility issues fail to decode information in the first slot following the measurement gap. This triggers RLC retransmissions, resulting in delay performance degradation for these UEs. To mitigate this, the gap-assisted scheduling compatibility function is introduced. This function is controlled by the **GAP_SCH_COMP_SW** option of the **gNodeBParam.CompatibilityAlgoSwitch** parameter. When this option is selected, the function takes effect, and no service or signaling data is transmitted in the first slot following the measurement gap, improving the performance and experience of UEs with compatibility issues. The change in this feature is supported only in low-frequency bands.

NOTE

For details about MGTA, see section 6.3.5 "Measurement information elements" in 3GPP TS 38.331 V16.7.0.

Implementation

For details, see *SA Mobility Management in Connected Mode* and *NSA Mobility Management*. The specific sections are listed in the change history of the documents.

Impact on the Network

This function reduces RLC retransmissions triggered by decoding failures experienced by UEs with compatibility issues, thereby minimizing delay performance degradation caused by retransmissions. This function reduces the **User Uplink Average Throughput (DU)** and **User Downlink Average Throughput (DU)** for UEs without compatibility issues.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact.
- Configuration management
The following parameter has changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNodeBParam.CompatibilityAlgorithmSwitch	MOD GNODEBPARAM LST GNODEBPARAM	gNodeB	The GAP_SCH_COMP_SW option has been added to specify whether to enable gap-assisted scheduling compatibility.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *SA Mobility Management in Connected Mode*.

Requirement/Trouble Ticket Number

IR20241209002193
DTS2025082805385

RAT

NR TDD

5.3.2.1.7 High-Load-specific Handover in SA Networking

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In scenarios where UEs are largely stationary, network load is high, and overlapping cell coverage occurs (for example, during concerts), fluctuations in channel quality may lead to frequent variations in the measured signal quality of neighboring cells. This may trigger ping-pong handovers, which in turn degrade the user experience for services. To mitigate this, the high-load-specific handover function is introduced. This function binds different intra- or inter-frequency measurement parameter groups to UEs with different QCI, meeting diverse user requirements and ensuring differentiated user experience.

This function is controlled by the **HIGH_LOAD_DIFF_HO_POL_SW** option of the **NRCellMobilityConfig.HoMeasAlgoSwitch** parameter. When this option is selected and all required conditions are met, high-load-specific handover takes effect. The **gNBScenarioDiffGrp** MO can be used to bind different intra- or inter-frequency measurement parameter groups to UEs with different QCI and from different operators, ensuring differentiated user experience.

- Cell requirements:
 - Number of RRC_CONNECTED UEs in the cell $\geq$ **NRCellMobilityConfig.HighLoadDiffHoUeNumThld**
 - Downlink PRB usage of the cell $\geq$ **NRCellMobilityConfig.HighLoadDiffHoDlPrbThld**
 - Uplink PRB usage of the cell $\geq$ **NRCellMobilityConfig.HighLoadDiffHoUlPrbThld**
- UE requirements: The **gNBScenarioDiffGrp** MO associated with the QCI specific to the UE's highest handover priority is present, and either the **NRCellIntraFHoMeaGrp** or **NRCellInterFHoMeaGrp** MO associated with the **gNBScenarioDiffGrp** MO is also present.

A UE with high-load-specific handover in effect also supports the anti-ping-pong function based on the QCI-specific cell-load-dependent inter-frequency handover waiting time. If the UE is handed over to a cell (the current serving cell) through any handover function, a timer (specified by **gNBScenarioDiffGrp.InterFreqHoWaitTime**) is started when the incoming handover is complete. The UE cannot be handed over back to the original serving cell through an unnecessary handover function to which the anti-ping-pong function is applied before the timer expires. In this case, an unnecessary handover function to which the anti-ping-pong function is applied can be:

- Operator-specific-priority-based inter-frequency handover
- Uplink-interference-based inter-frequency handover
- SSB SINR-based inter-frequency handover (for LampSite)

- RFSP-based inter-frequency handover back to the HPLMN
- Experience-based smart carrier selection
- Uplink multi-frequency coordination
- MLB by transferring UEs in connected mode
- Uplink-service-based carrier selection in high-speed scenarios
- Downlink-service-based carrier selection in high-speed scenarios

The change in this feature is supported only in SA networking.

The change in this feature is supported only in low frequency bands.

Implementation

For details, see *SA Mobility Management in Connected Mode*. The specific sections are listed in the change history of the document.

Impact on the Network

With this function enabled after an upgrade, the number of handovers changes in a high-load cell, as the intra- or inter-frequency measurement parameter groups bound to the gNodeB scenario differentiation group apply to UEs in the cell. Any larger value of the following thresholds results in a lower probability that high-load-specific handover takes effect and a smaller change in the number of handovers in the high-load cell. Any smaller value of the following thresholds produces the opposite effects.

- **NRCellMobilityConfig.HighLoadDiffHoUeNumThld** (indicating the UE number threshold for high-load-specific handover)
- **NRCellMobilityConfig.HighLoadDiffHoDlPrbThld** (indicating the downlink PRB usage threshold for high-load-specific handover)
- **NRCellMobilityConfig.HighLoadDiffHoUlPrbThld** (indicating the uplink PRB usage threshold for high-load-specific handover)

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact.
- Configuration management
The following MML commands have changed.

Change Type	MML Command	NE	Description
New	ADD GNBSCENARIODIFFGRP	gNodeB	This command has been added to add a gNodeB scenario differentiation group.
New	MOD GNBSCENARIODIFFGRP	gNodeB	This command has been added to modify a gNodeB scenario differentiation group.
New	LST GNBSCENARIODIFFGRP	gNodeB	This command has been added to list gNodeB scenario differentiation groups.
New	RMV GNBSCENARIODIFFGRP	gNodeB	This command has been added to remove a gNodeB scenario differentiation group.

NOTE

The parameters in these new or deleted MML commands are no longer listed separately.

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRCeIIQciBearer.S cenarioDiffGrpId	ADD NRCELLQCIBER ARER MOD NRCELLQCIBER ARER LST NRCELLQCIBER ARER	gNodeB	This parameter has been added to specify the ID of a scenario differentiation parameter group.

Change Type	Parameter ID	MML Command	NE	Description
New	NRCellMobilityConfig.HoMeasAlgoSwitch	MOD NRCELLMOBILITYCONFIG LST NRCELLMOBILITYCONFIG	gNodeB	The HIGH_LOAD_DIFF_HO_POL_SW option of this parameter specifies whether to enable high-load-specific handover.
New	NRCellMobilityConfig.HighLoadDiffHoUeNumThld	MOD NRCELLMOBILITYCONFIG LST NRCELLMOBILITYCONFIG	gNodeB	This parameter has been added to specify the UE number threshold for high-load-specific handover.
New	NRCellMobilityConfig.HighLoadDiffHoDlPrbThld	MOD NRCELLMOBILITYCONFIG LST NRCELLMOBILITYCONFIG	gNodeB	This parameter has been added to specify the downlink PRB usage threshold for high-load-specific handover.
New	NRCellMobilityConfig.HighLoadDiffHoUlPrbThld	MOD NRCELLMOBILITYCONFIG LST NRCELLMOBILITYCONFIG	gNodeB	This parameter has been added to specify the uplink PRB usage threshold for high-load-specific handover.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *SA Mobility Management in Connected Mode*.

Requirement/Trouble Ticket Number

FeR-IR20241209001765-0003

RAT

NR FDD, NR TDD

5.3.2.1.8 Added the High-Load-specific Handover Function in NSA Networking

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In certain scenarios where UEs are largely stationary, network load is high, and overlapping cell coverage occurs (such as during concerts), channel variations may lead to frequent fluctuations in the measured signal quality of neighboring cells. This may trigger ping-pong handovers, subsequently resulting in degraded user experience for services. To mitigate this, the high-load-specific handover function is introduced.

This function is controlled by the **HIGH_LOAD_DIFF_HO_POL_SW** option of the **NRCellMobilityConfig.HoMeasAlgoSwitch** parameter. When this option is selected and all the conditions for this function to take effect are met, high-load-specific handover takes effect. The **gNBScenarioDiffGrp** MO can be used for binding of different intra- or inter-frequency measurement parameter groups to UEs with different QCI and operators, ensuring differentiated user experience. The conditions for the function to take effect are as follows:

- Cell requirements:
 - Number of RRC_CONNECTED UEs in the cell $\geq$ **NRCellMobilityConfig.HighLoadDiffHoUeNumThld**
 - Downlink PRB usage of the cell $\geq$ **NRCellMobilityConfig.HighLoadDiffHoDlPrbThld**

- Uplink PRB usage of the cell $\geq$
NRCellMobilityConfig.HighLoadDiffHoUlPrbThld
- UE requirements: The **gNBScenarioDiffGrp** MO associated with the QCI of the UE's highest handover priority is present, and either the **NRCellIntraFHoMeaGrp** or **NRCellInterFHoMeaGrp** MO associated with the **gNBScenarioDiffGrp** MO is also present. [Table 5-19](#) describes the parameters in the **gNBScenarioDiffGrp** MO.

Table 5-19 Parameters in the **gNBScenarioDiffGrp** MO

Parameter	Description
gNBScenarioDiffGrp <i>.OperatorId</i>	Operator of the UE.
gNBScenarioDiffGrp <i>.ScenarioType</i>	Scenario type. Set this parameter to HIGH_LOAD .
gNBScenarioDiffGrp <i>.ScenarioDiffGrpId</i>	Set this parameter to the same value as the NRCellQciBearer.ScenarioDiffGrpId parameter. Then, the QCI is associated with the gNBScenarioDiffGrp MO.
gNBScenarioDiffGrp <i>.IntraFreqHoMeasGroupId</i>	Set this parameter to the same value as the NRCellIntraFHoMeaGrp.IntraFreqHoMeasGroupId parameter. Then, the NRCellIntraFHoMeaGrp MO is associated with the gNBScenarioDiffGrp MO.
gNBScenarioDiffGrp <i>.NsaInterFreqHoMeasGroupId</i>	Set this parameter to the same value as the NRCellInterFHoMeaGrp.InterFreqHoMeasGroupId parameter. Then, the NRCellInterFHoMeaGrp MO is associated with the gNBScenarioDiffGrp MO.

The change in this feature is supported only in low-frequency bands.

Implementation

For details, see *NSA Mobility Management*. The specific sections are listed in the change history of the document.

Impact on the Network

After this function takes effect, the values of the following counters related to PSCell changes will change:

- **N.NsaDc.InterSgNB.PSCell.Change.Att**
- **N.NsaDc.InterSgNB.PSCell.Change.Succ**
- **N.NsaDc.IntraSgNB.PSCell.Change.Att**
- **N.NsaDc.IntraSgNB.PSCell.Change.Succ**
- **N.NsaDc.InterSgNB.SgNB.Add.Att**
- **N.NsaDc.InterSgNB.SgNB.Add.Ack**

- N.NsaDc.InterSgNB.SgNB.Add.Succ
- N.NsaDc.InterSgNB.PSCell.Change.UeContextRel
- N.NsaDc.InterSgNB.IntraFreq.PSCell.Change.Att
- N.NsaDc.InterSgNB.IntraFreq.PSCell.Change.Succ
- N.NsaDc.IntraSgNB.IntraFreq.PSCell.Change.Att
- N.NsaDc.IntraSgNB.IntraFreq.PSCell.Change.Succ

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature is not under license control.
- Configuration management
The following MML commands have changed.

Change Type	Command	NE	Description
New	ADD GNBSCENARIODIFFGRP	gNodeB	This MML command has been added to add a gNodeB scenario differentiation group.
New	MOD GNBSCENARIODIFFGRP	gNodeB	This MML command has been added to modify a gNodeB scenario differentiation group.
New	LST GNBSCENARIODIFFGRP	gNodeB	This MML command has been added to list gNodeB scenario differentiation groups.
New	RMV GNBSCENARIODIFFGRP	gNodeB	This MML command has been added to remove a gNodeB scenario differentiation group.

 NOTE

The parameters in these new MML commands are no longer listed separately.
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRCellQciBearer.ScenarioDiffGrpld	ADD NRCELLQCIBERARER MOD NRCELLQCIBERARER LST NRCELLQCIBERARER	gNodeB	This parameter has been added to specify the ID of a scenario differentiation parameter group.
New	NRCellMobilityConfig.HoMeasAlgoSwitch	MOD NRCELLMOBILITYCONFIG LST NRCELLMOBILITYCONFIG	gNodeB	This parameter has been added. The HIGH_LOAD_DIFF_HO_POL_SW option of this parameter specifies whether to enable high-load-specific handover.
New	NRCellMobilityConfig.HighLoadDiffHoUeNumThld	MOD NRCELLMOBILITYCONFIG LST NRCELLMOBILITYCONFIG	gNodeB	This parameter has been added to specify the UE number threshold for high-load-specific handover.
New	NRCellMobilityConfig.HighLoadDiffHoDlPrbThld	MOD NRCELLMOBILITYCONFIG LST NRCELLMOBILITYCONFIG	gNodeB	This parameter has been added to specify the downlink PRB usage threshold for high-load-specific handover.

Change Type	Parameter ID	MML Command	NE	Description
New	NRCellMobilityConfig.HighLoadDiffHoUlPrbThld	MOD NRCELLMOBILITYCONFIG LST NRCELLMOBILITYCONFIG	gNodeB	This parameter has been added to specify the uplink PRB usage threshold for high-load-specific handover.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The following impact relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	QCI-specific measurement parameter configuration	NRCellQciBearer.IntraFreqHoMeasGroupId or NRCellQciBearer.NsaInterFreqHoMeasGroupId	<i>NSA Mobility Management</i>	When both QCI-specific measurement parameter configuration and high-load-specific handover are configured, the latter function preferentially takes effect.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *NSA Mobility Management*.

Requirement/Trouble Ticket Number

FeR-IR20241209001765-0003

RAT

NR FDD, NR TDD

5.3.2.1.9 Switch Control for UE-Layering-based Differentiated Cell Reselection Priority Configuration

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

UE-layering-based differentiated cell reselection priority configuration can now be disabled for all QCI by selecting the **DIFF_UE_RESEL_PRI_SUPPR_SW** option of the **NRCellMobilityConfig.IdleReselPolicySw** parameter.

The change in this feature is supported only in SA networking.

Implementation

For details, see *SA Mobility Management in Idle Mode*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact.
- Configuration management
The following parameter has changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRCellMobilityConfig.IdleReselPrioritySw	MOD NRCELLMOBILITYCONFIG LST NRCELLMOBILITYCONFIG	gNodeB	The DIFF_UE_RESEL_PRI_SUPPR_SW option specifies whether to enable UE-layering-based differentiated cell reselection priority configuration.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *SA Mobility Management in Idle Mode*.

Requirement/Trouble Ticket Number

FeR-IR20250116002382-0010

RAT

NR FDD, NR TDD

5.3.2.1.10 Optimized SMTTC Offset Configuration

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The SMTC offset configuration mechanism has been optimized to enable a flexible SMTC offset for handover and reselection.

- The **NRCellFreqRelation.SaSmtcOffset** parameter has been introduced to provide a configurable inter-frequency SMTC offset. When the parameter is set to a value other than **255**, the SMTC offset takes the parameter's value. When the parameter is set to **255**, the gNodeB calculates the SMTC offset.
- If an operator configures different frame offsets (**gNBFreqBandConfig.FrameOffset**) for gNodeBs operating on the same frequency band, each gNodeB applies its own frame offset to calculate the SMTC offset. In this scenario, when both (1) UEs are located in the overlapping area of gNodeBs with different frame offsets, and (2) the serving gNodeB's frame offset exceeds that of a neighboring gNodeB, the intra- and inter-frequency measurements by UEs on the cells of neighboring gNodeBs will be incomplete, leading to an increased service drop rate. To address this issue, an additional frame offset (**gNBFreqBandConfig.AdditionalFrameOffset**) is introduced to a frequency band. When calculating the SMTC offset, the gNodeB checks for its own valid additional frame offset for the frequency band. If there is a valid additional frame offset, the gNodeB uses the smaller value between the frame offset and additional frame offset as the input for the SMTC offset calculation. This ensures the complete intra- and inter-frequency measurements by UEs on the cells of neighboring gNodeBs operating on the frequency band.

The change in this feature is supported only in SA networking.

Implementation

For details, see *SA Mobility Management in Connected Mode* and *SA Mobility Management in Idle Mode*. The specific sections are listed in the change history of the documents.

Impact on the Network

The SMTC offset can be parameter-configurable, improving configuration flexibility. However, inappropriate parameter settings may lead to incomplete inter-frequency measurements, rendering specific non-serving frequencies unavailable for handovers and thereby increasing the service drop rate.

If both (i) different frame offsets are configured for base stations operating on the same frequency band, and (ii) UEs are located in the overlapping area of the base stations with different frame offsets, then configuring an additional frame offset for these base stations in the overlapping area can reduce the service drop rate.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management

The change in this feature has no impact.

- Configuration management

The following parameter has changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRCellFreqRelation.SaSmtcOffset	ADD NRCELLFREQR ELATION MOD NRCELLFREQR ELATION LST NRCELLFREQR ELATION	gNodeB	This parameter has been added to specify the SMTc offset for inter-frequency handover and inter-frequency reselection in SA networking.

- Performance management

The change in this feature has no impact.

- Fault management

The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

The change in this feature has no impact.

- Mutually exclusive features/functions

The change in this feature has no impact.

- Impacted features/functions

The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *SA Mobility Management in Connected Mode*.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0011

RAT

NR FDD, NR TDD

5.3.2.1.11 Optimized Mechanism for Selecting the Target AMF in Xn-based Handovers

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

During handover preparation, if a handover request is initiated over the Xn interface, the target AMF selection process proceeds as follows:

- The target gNodeB preferentially selects the UE's serving AMF as the target AMF. However, an AMF with the *Relative AMF Capacity* IE set to 0 cannot be selected. To allow a UE to select the serving AMF with the *Relative AMF Capacity* IE set to 0 as the target AMF for a handover, you can enable the path switch AMF selection function (by selecting the **PATH_SWITCH_AMF_SEL_SW** option of the **gNodeBParam.NgInterfaceAlgoExtSw** parameter). Path switch AMF selection is enabled by default. For details about the *Relative AMF Capacity* IE, see section 9.3.1.32 "Relative AMF Capacity" in 3GPP TS 38.413 V17.6.0.
- If (1) the target gNodeB cannot select the UE's serving AMF and instead selects a backup AMF, and (2) the target gNodeB and the selected AMF are supplied by different vendors, the UE may encounter a handover failure, resulting in a service drop. To mitigate this risk, the path switch backup AMF selection function can be enabled to prevent selection of a backup AMF. This function is controlled by the **PATH_SWITCH_BACKUP_AMF_SEL_SW** option of the **gNodeBParam.NgInterfaceAlgoExtSw** parameter. When this option is selected, the target gNodeB prevents selection of a backup AMF. Consequently, if handover preparation over the Xn interface fails, the UE can perform handover preparation over the NG interface instead. When this option is deselected, the target gNodeB can select a backup AMF. Path switch backup AMF selection is enabled by default.

Before an upgrade, path switch AMF selection is controlled by the **RSVDSWPARAM0_BIT4** option of the **gNBRsvdExt01.RsvdSwParam0** parameter, and path switch backup AMF selection is controlled by the **RSVDSWPARAM0_BIT5** option of the **gNBRsvdExt01.RsvdSwParam0** parameter. After this change takes effect, these reserved parameters are added to the disuse list and replaced by formal parameters.

The change in this feature is supported only in SA networking.

Implementation

For details, see *SA Mobility Management in Connected Mode*. The specific sections are listed in the change history of the document.

Impact on the Network

Path switch AMF selection: This function may increase the Xn-based handover success rate.

Path switch backup AMF selection: This function may increase the Xn-based handover success rate and may reduce the service drop rate.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact.
- Configuration management
The following parameter has changed.

Change Type	Parameter ID	MML Command	NE	Description
New	gNodeBParam.NgInterfaceAlgoExtSw	MOD GNODEBPARAM LST GNODEBPARAM	gNodeB	This parameter and its options have been added: • The PATH_SWITCH_AMF_SEL_SW option has been added to specify whether to enable the path switch AMF selection function. • The PATH_SWITCH_BACKUP_AMF_SEL_SW option has been added to specify whether to enable the path switch backup AMF selection function.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

Whether the change takes effect by default after an upgrade varies as follows:

- In site deployment scenarios, path switch AMF selection and path switch backup AMF selection take effect by default.
- In site upgrade scenarios, if the **RSVDSWPARAM0_BIT4** option of the **gNBRsvdExt01.RsvdSwParam0** parameter is selected before the upgrade, path switch AMF selection takes effect by default after the upgrade; otherwise, this function does not take effect by default after the upgrade. Similarly, if the **RSVDSWPARAM0_BIT5** option of the **gNBRsvdExt01.RsvdSwParam0** parameter is selected before the upgrade, path switch backup AMF selection takes effect by default after the upgrade; otherwise, this function does not take effect by default after the upgrade.

For details, see the corresponding "Operation and Maintenance" section in *SA Mobility Management in Connected Mode*.

Requirement/Trouble Ticket Number

DTS2025121532270

RAT

NR FDD, NR TDD

5.3.2.2 FOFD-010010 MU-MIMO Basic Pairing

5.3.2.2.1 mmWave PUSCH MU-MIMO (for High Frequency Bands Only)

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

mmWave PUSCH MU-MIMO is now supported. With this function, the gNodeB selects appropriate UEs for pairing according to pairing rules and schedules the UEs. The pairing rules can be set using parameters and are as follows:

- When a UE initially accesses a cell, measurement of the spatial isolation between beams is started if the SRS SINR before equalization of the UE is greater than the threshold specified by **NRDUCellUIMimo.UlMuSrsPreSnrThld** and the transmitted traffic volume for the UE is greater than the threshold specified by **NRDUCellUIMimo.UlMulIsolationMeasThld**; otherwise, the measurement is not performed and the UE is not selected for pairing.
- This UE can be successfully paired with another UE in a TTI only if the following conditions are met: (1) the spatial isolation between their optimal beams is greater than the threshold specified by **NRDUCellUIMimo.UlMuSpacelsoThld**; (2) the gain in the post-pairing throughput relative to the pre-pairing throughput is greater than the threshold specified by **NRDUCellUIMimo.UlMuBackToSuSeGainThld**.

This function is controlled by the **UL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch.MuMimoSwitch** parameter. Currently, mmWave PUSCH MU-MIMO supports spatial multiplexing for multiple UEs only in 4T4R or 8T8R FWA scenarios. It is recommended that this function be enabled when the total bandwidth of active cells served by a single RF module is less than or equal to 200 MHz. In other scenarios, enabling this function cannot ensure performance.

After an upgrade from a version earlier than V100R022C00SPC010 to V100R022C00SPC010 or later, the **UL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch.MuMimoSwitch** parameter will be deselected by default.

The maximum number of spatial multiplexing layers for multiple UEs is four, which can be specified by the **NRDUCellPusch.MaxMimoLayerCnt** parameter.

LampSite base stations do not support this function.

Implementation

For details, see *MIMO (TDD)*. The specific sections are listed in the change history of the document.

Impact on the Network

This function has the following impacts on the network:

- In light- or medium-load cells where resources are sufficient, mmWave PUSCH MU-MIMO may cause fluctuations in the average uplink UE and cell throughputs.
- In heavy-load cells where resources are insufficient, mmWave PUSCH MU-MIMO increases the average uplink UE and cell throughputs.
- It is recommended that mmWave PUSCH MU-MIMO be enabled when the uplink load of a cell is heavy. The uplink load of a cell is heavy when the uplink PRB usage is greater than or equal to 50%.
- Under the same traffic volume, the increases in the average uplink UE and cell throughputs are related to the number of paired RBs and the number of paired layers. The greater the numbers, the larger the increases.
- After an increase in the uplink MU-MIMO pairing rate, there may be a decrease in the average uplink UE throughput in either of the following situations: (1) the data volume of tail packets accounts for a large proportion in light- or medium-load scenarios; (2) the channel quality of UEs is good and UEs are concentrated.
- After an increase in the uplink MU-MIMO pairing rate, there will be increases in the number of scheduled UEs in the uplink and the consumption of CCE resources. If CCE resources for the uplink are limited, there will be an increase in the CCE allocation failure rate for uplink transmission.
- After an increase in the uplink MU-MIMO pairing rate, there will be more downlink response packets. If PDSCH PRB resources are insufficient, there will be changes in the average downlink cell and UE throughputs, average downlink MCS index, average rank, and values of scheduling-related indicators such as the downlink BLER.
- After an increase in the uplink MU-MIMO pairing rate, there will be an increase in the interference between paired UEs and accordingly an increase in the uplink BLER in the cell.

- After an increase in the uplink MU-MIMO pairing rate, the number of times each UE is scheduled will change, causing changes in the average uplink MCS index, average rank, and values of scheduling-related indicators such as the uplink IBLER.
- After an increase in the uplink MU-MIMO pairing rate, the downlink transmit power may decrease.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

- Boards
 - Main control boards:
All NR-capable main control boards support this function.
 - Baseband processing units:
The UBBPg and later baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- RF modules:
RF modules must be the HAAU5323 or later HAAU series modules. For details about HAAU series modules, see the AAU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCell ULMimo.Ul MuSrsPreS inrThld	MOD NRDUCELLU LMIMO LST NRDUCELLU LMIMO	gNodeB	This parameter has been added to specify the threshold of the SRS SINR before equalization (SRS PreSINR) for UEs to enter uplink MU-MIMO scheduling.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellULMimo.UIMuIsolationMeasurement	MOD NRDUCELLU LMIMO LST NRDUCELLU LMIMO	gNodeB	This parameter has been added to specify the UE traffic volume threshold for triggering uplink MU-MIMO isolation measurement.
New	NRDUCellULMimo.UIMuSpatialIsolationThreshold	MOD NRDUCELLU LMIMO LST NRDUCELLU LMIMO	gNodeB	This parameter has been added to specify the spatial isolation threshold for uplink MU-MIMO to take effect for UEs.
New	NRDUCellULMimo.UIMuBackToSuSeGainThreshold	MOD NRDUCELLU LMIMO LST NRDUCELLU LMIMO	gNodeB	This parameter has been added to specify the spectral efficiency gain threshold for switching uplink MU-MIMO back to SU-MIMO.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The following mutually exclusive relationships have been added.

Function Name	Function Switch	Reference	Description
mmWave 3D coverage pattern	NRDUCellTrpMmwavBeam.CoverageScenario	<i>mmWave Beam Management (High-Frequency TDD)</i>	mmWave PUSCH MU-MIMO cannot be enabled if the NRDUCellTrpMmwavBeam.CoverageScenario parameter is set to SCENARIO_103 or CUSTOMIZED_BEAM_SCENARIO_106 .

Function Name	Function Switch	Reference	Description
Adaptive HARQ-ACK feedback on downlink transmission	ACK_NACK_REPORTING_PUSCH_SW option of the NRDUCellAlgoSwitch.FullChannelCovEnhSwitch parameter	<i>Uplink Scheduling</i>	If adaptive HARQ-ACK feedback on downlink transmission is enabled, mmWave PUSCH MU-MIMO cannot be enabled.
DMRS spatial multiplexing for uplink MU-MIMO	MU_DMRS_SPATIAL_MULTIPLEX_SW option of the NRDUCellDmrs.UIDmrsSwitch parameter	None	If DMRS spatial multiplexing for uplink MU-MIMO is enabled, mmWave PUSCH MU-MIMO cannot be enabled.
Uplink MU-MIMO for large-packet UEs in mmWave cells	MMW_LRG_PKT_UL_MU_MIMO_SW option of the NRDUCellFeatureSw.MimoFeatureSwitch parameter	None	If uplink MU-MIMO for large-packet UEs in mmWave cells is enabled, mmWave PUSCH MU-MIMO cannot be enabled.

- Impacted features/functions

The following impact relationships have been added.

Function Name	Function Switch	Reference	Description
Flexible Dense Beam	FLEXIBLE_DENSE_BEAM_SW option of the NRDUCellTrpMmwav Beam.BeamPerformanceSw parameter	<i>mmWave Beam Management (High-Frequency TDD)</i>	After Flexible Dense Beam is enabled together with mmWave PDSCH MU-MIMO or mmWave PUSCH MU-MIMO, the cell throughput changes because the serving beams of UEs and the measured analog beam interference between UEs change.
Operator-specific RB management	RB_DYNAMIC_SHARING_SW option of the NRDUCellAlgoSwitch.RanSharingAlgoSwitch parameter	<i>Multi-Operator Sharing</i>	In RAN sharing with common carrier, mmWave PDSCH MU-MIMO and mmWave PUSCH MU-MIMO do not take effect if operator-specific RB management is enabled.

Function Name	Function Switch	Reference	Description
mmWave 3D coverage pattern	NRDUCellTrpMmwavBeam.CoverageScenario set to SCENARIO_104	<i>mmWave Beam Management (High-Frequency TDD)</i>	In the wide-beam scenario of mmWave 3D coverage pattern (with the NRDUCellTrpMmwavBeam.CoverageScenario parameter set to SCENARIO_104), the gains of mmWave PDSCH MU-MIMO and mmWave PUSCH MU-MIMO may decrease because coverage areas of SSB and CSI-RS beams may change.
Inter-FR CA	INTER_FR_FR1TOFR2_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	mmWave PUSCH MU-MIMO does not take effect for inter-FR CA UEs.

Related Operations

By default, the change does not take effect after an upgrade.
For details, see the corresponding "Operation and Maintenance" section in *MIMO (TDD)*.

Requirement/Trouble Ticket Number

FeR-IR20240628000590-0001

RAT

NR TDD

5.3.2.3 FOFD-010205 UL and DL Decoupling

5.3.2.3.1 UL and DL Decoupling with SUL and NR FDD Co-Deployment

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

With UL and DL Decoupling with SUL and NR FDD co-deployment enabled, the SUL function is provided by NR FDD cells. UL and DL Decoupling is implemented

using NR TDD cells and NR FDD cells. UL and DL Decoupling supports one-to-one mapping between intra-gNodeB and inter-gNodeB NR TDD cells and SUL cells (NR FDD cells).

UL and DL Decoupling with SUL and NR FDD co-deployment is enabled by setting the **NRDUCellAlgoSwitch.UlDlDecouplingSwitch** parameter to **ON** in both NR TDD and NR FDD cells and setting the following parameters and options to specific values.

For TDD cells:

- **NRDUCellSulRelation.SulType** is set to **DYN_SUL**.
- **NRDUCellCollabServ.NulToSulMappingType** is set to **1TO1**.
- The NCGI-related parameters (including **NRDUCellCollabServ.SulMcc**, **NRDUCellCollabServ.SulMnc**, **NRDUCellCollabServ.SulGnodebId**, and **NRDUCellCollabServ.SulCellId**) are configured for the NR FDD cell in SUL and NR FDD co-deployment mode. NCGI stands for NR cell global identifier.

For FDD cells:

- **NRDUCellCollabCoop.SulToNulMappingType** is set to **1TO1**.
- The NCGI-related parameters (including **NRDUCellCollabCoop.NulMcc**, **NRDUCellCollabCoop.NulMnc**, **NRDUCellCollabCoop.NulGnodebId**, and **NRDUCellCollabCoop.NulCellId**) are configured for the NUL cell bound in 1:1 mode.

If the gNodeB is shared by multiple operators, that is, the **gNBSharingMode.gNBMultiOpSharingMode** parameter is set to a value other than **INDEPENDENT**, the **OP_SUL_SWITCH** option of the **NRDUCellOp.PolicySwitch** parameter must be selected in the NR TDD cell for the operator that requires UL and DL Decoupling.

Implementation

For details, see *Super Uplink* and *UL and DL Decoupling*. The specific sections are listed in the change history of each document.

Impact on the Network

Basic functions:

- When UL and DL Decoupling is enabled, only one RRC connection is established between a UE and the network. The "RRC Connected User License" is consumed only in the NR TDD cell. However, each UE occupies a set of hardware resources in both the NR TDD cell and NR FDD cell. Therefore, the actual number of RRC_CONNECTED UEs supported on the network is halved in extreme scenarios (where all UEs on the network support SUL).
- When UL and DL Decoupling is enabled, uplink resources of the NR FDD cell can be fully utilized. Therefore, the uplink PRB usage of the NR FDD cell increases. When the NR FDD cell load is high, the uplink throughput of NR FDD UEs in the cell decreases.
- When UL and DL Decoupling is enabled, event A5 measurement configurations are delivered in the NR TDD cell. Since event A5 measurement

configurations require measurement gaps, uplink and downlink throughput of the UE in the NR TDD cell decreases.

- The number of the RBs occupied by SIB1 may slightly increase and the downlink peak throughput slightly decreases because the gNodeB broadcasts the SUL information through SIB1.
- Uplink scheduling consumes CCE resources of the NR TDD cell after UL and DL Decoupling is enabled. When CCE resources are insufficient in the NR TDD cell, uplink or downlink CCE resource application is more likely to fail. As a result, the uplink or downlink throughput of the NR TDD cell may decrease, and the downlink user-plane delay may increase.
- It is recommended that the **UL_DL_CCE_SHARING_SW** or **UL_DL_CCE_RATIO_ADAPT_SW** option of the **NRDUCellPdcch.PdcchAlgoSwitch** parameter be selected to reduce the probability of CCE resource application failures.
- In an NR TDD cell enabled with uplink MU-MIMO, if scheduling time alignment for SUL is enabled (that is, the **UL_SCH_ALIGN_SW** option of the **NRDUCellPusch.PuschPerformanceSwitch** parameter is selected), the average uplink cell throughput and the **N.CCE.UL.Avail.Equivalent** and **N.CCE.ULDCI.Avail.Equivalent** counter values increase, as does the average delay of non-SUL UEs by up to 1%.
- The uplink throughput of UEs that switch between carriers in 0 μs decreases when UE compatibility handling is enabled with the **NRDUCellCollabServ.Nul1TxToSul1TxSwitchTime** parameter set to a value other than 0 μs.
- When flow control against multi-carrier operations at the gNodeB is triggered and the CPU usage reaches the load threshold corresponding to the flow control level, SUL will no longer take effect for newly admitted UEs. As a result, the average uplink UE throughput (**User Uplink Average Throughput (DU)**) and average downlink UE throughput (**User Downlink Average Throughput (DU)**) decrease. When the **gNodeBAlgo.MultiCarrierFlowCtrlLevel** parameter is set to **HIGH**, the cell service drop rate (**Service Call Drop Rate (CU, Inactive)**) increases as well.
- When inter-gNodeB UL and DL Decoupling is enabled and the NR FDD cell load is high, SUL UEs may not benefit from SUL.
- When inter-gNodeB UL and DL Decoupling is enabled, more cell configuration information is exchanged over the eXn interface between gNodeBs. Therefore, the values of counters related to the received and transmitted traffic volumes in the **SCTP Link Measurement** function subset increase.
- When UL and DL Decoupling is in effect, UEs randomly access an SUL cell. Random access-related counters of the NR TDD cell are measured, which may cause the values to fluctuate. The values of random access-related counters of the SUL cell co-deployed with the NR FDD cell are not measured (except for **N.RA.Contention.Att.Max**, **N.RA.Dedicated.PreambleAssign.Num**, and **N.RA.Dedicated.PreambleAssign.Num**).
- The UE random access control based on cell radius function enables the gNodeB to detect UE access beyond the cell radius, which increases the consumption of physical resources. Since contention-based random access is prohibited for UEs whose distance to the gNodeB exceeds the cell radius, the random access success rate and RRC connection setup success rate may decrease, and the cell traffic, cell rate, and PRB usage may decrease. In

addition, enabling "optimized measurement of RRC connection setup success rate based on cell radius" can prevent the RRC connection setup success rate from decreasing.

Enhanced functions:

- Enabling the symbol-level scheduling function affects the uplink throughput of the NR FDD cell.
- When the SUL CCE adjustment policy function is enabled and uplink-to-downlink CCE ratio adaptation (controlled by the **UL_DL_CCE_RATIO_ADAPT_SW** option of the **NRDUCellPdcch.PdcchAlgoSwitch** parameter) is enabled:
 - If the proportion of SUL UEs is low, the uplink user-perceived rates of these UEs increase while those of other UEs in the NR TDD cell are not affected.
 - If the proportion of SUL UEs is high, the probability of uplink CCE resource application failures increases and scheduling opportunities for other UEs in the NR TDD cell decrease.
- When the SUL CCE adjustment policy function is enabled and uplink-to-downlink CCE ratio adaptation is disabled, there is no network impact.

The simultaneous use with super uplink has the following impacts:

If either of the following conditions is met, UL and DL Decoupling is disabled for UEs in a cell, and super uplink is disabled for UEs in the bound neighboring cell.

- UL and DL Decoupling is disabled in the NR TDD cell, or any of the **NRDUCellCollabServ.SulMcc**, **NRDUCellCollabServ.SulMnc**, **NRDUCellCollabServ.SulGnodebId**, and **NRDUCellCollabServ.SulCellId** parameters is modified for the NR FDD cell bound to the NR TDD cell.
- UL and DL Decoupling is disabled in the NR FDD cell, or any of the **NRDUCellCollabCoop.NulMcc**, **NRDUCellCollabCoop.NulMnc**, **NRDUCellCollabCoop.NulGnodebId**, and **NRDUCellCollabCoop.NulCellId** parameters is modified for the NR TDD cell bound to the NR FDD cell.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

All NR-capable main control boards support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

Baseband processing units:

- Macro base stations:
 - For NR TDD cells, all NR TDD-capable baseband processing units can be used.
 - For NR FDD cells:
 - Baseband processing units that support 32T32R/64T64R and those that support 2T2R/4T4R/8T8R cannot be both used.

- NR TDD-capable baseband processing units and NR FDD-capable baseband processing units cannot be both used.
 - Baseband processing units that can serve a maximum of 24 cells can be used.
 - LampSite base stations:
 - For NR TDD cells, baseband processing units must support distributed massive MIMO. For details, see *Distributed Antenna Solutions (Low-Frequency TDD)*.
 - For NR FDD cells, all NR FDD-capable baseband processing units can be used.
 - RF modules
- Table 5-20** lists the RF modules supported by macro base stations.

Table 5-20 RF modules supported by macro base stations

Frequency Band	Supported Module	Bandwidth Requirement
n78 (3.5/3.7 GHz)	Modules supporting 32T32R and 64T64R	Same as the bandwidths supported by the RF modules. For details, see related technical specifications in <i>3900 & 5900 Series Base Station Product Documentation</i> .
n40 (2.3 GHz)	Modules supporting 32T32R and 64T64R	
n41 (2.6 GHz)	Modules supporting 32T32R and 64T64R	
n3/n80 (1.8 GHz)	Modules supporting NR FDD	10 MHz, 15 MHz, and 20 MHz
n20/n82 (800 MHz)	Modules supporting NR FDD	5 MHz, 10 MHz, 15 MHz, and 20 MHz
n28/n83 (700 MHz)	Modules supporting NR FDD	10 MHz, 15 MHz, 20 MHz, and 30 MHz
n1/n84 (2.1 GHz)	Modules supporting NR FDD	10 MHz, 15 MHz, 20 MHz, 30 MHz, 40 MHz, and 50 MHz
n8/n81 (900 MHz)	Modules supporting NR FDD	10 MHz, 15 MHz, and 20 MHz

Table 5-21 lists the RF modules supported by LampSite base stations.

Table 5-21 RF modules supported by LampSite base stations

Frequency Band	Supported Module	TX/RX Mode	Bandwidth Requirement
n78 (3.5/3.7 GHz)	pRRU56xx series, pRRU596x series ^a , pRRU5973, and pRRU5973H	4T4R	20 MHz, 40 MHz, 60 MHz, 80 MHz, and 100 MHz
n41 (2.6 GHz)	pRRU596x series	4T4R	60 MHz, 80 MHz, and 100 MHz
n3/n80 (1.8 GHz)	pRRU56xx series and pRRU596x series	2R	10 MHz, 15 MHz, and 20 MHz
n1/n84 (2.1 GHz)	pRRU56xx series and pRRU596x series	2R	10 MHz, 15 MHz, 20 MHz, 30 MHz, 40 MHz, and 50 MHz
a: x in pRRU56xx series and pRRU596x series can be a digit or a combination of a digit and a letter (for example, pRRU5961).			

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management

The following feature license is required by UL and DL Decoupling with SUL and NR FDD co-deployment. The number of required license units is calculated based on the NR TDD cells but not NR FDD cells.

Feature ID	Feature Name	Model	License Control Item Name	Sales Unit
FOFD-010205	UL and DL Decoupling	NR0SULADL D00	UL and DL Decoupling (NR)	Per Cell

Feature ID	Feature Name	Model	License Control Item Name	Sales Unit
For the license control item NR0SULADLD00: • If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. • If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation: – When the license sales number pre-check function is enabled, the NRDUCellAlgoSwitch.UlDlDecouplingSwitch parameter cannot be set to ON due to license insufficiency, preventing cell activation failures. – When the license sales number pre-check function is disabled, the cell cannot be activated again due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified. The license sales number pre-check function is controlled by the LIC_SALE_NUM_PRE_CHECK_SW option of the gNBLicenseAlmThld.LicenseChkSw parameter.				

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellCollabCoop.SulToNulMappingType	MOD NRDUCELLCOLLABC OOP LST NRDUCELLCOLLABC OOP	gNo dB	This parameter has been added to specify the SUL and NUL mapping type for an FDD cell with UL and DL Decoupling enabled in SUL and NR FDD co-deployment mode.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellCollabCoop.NulCellId	MOD NRDUCELLCOLLABC OOP LST NRDUCELLCOLLABC OOP	gNodeB	This parameter has been added to specify the ID of the NUL cell bound to the FDD cell in UL and DL Decoupling with SUL and NR FDD co-deployment.
New	NRDUCellCollabCoop.NulMcc	MOD NRDUCELLCOLLABC OOP LST NRDUCELLCOLLABC OOP	gNodeB	This parameter has been added to specify the mobile country code (MCC) of the NUL cell bound to the FDD cell in UL and DL Decoupling with SUL and NR FDD co-deployment.
New	NRDUCellCollabCoop.NulMnc	MOD NRDUCELLCOLLABC OOP LST NRDUCELLCOLLABC OOP	gNodeB	This parameter has been added to specify the mobile network code (MNC) of the NUL cell bound to the FDD cell in UL and DL Decoupling with SUL and NR FDD co-deployment.
New	NRDUCellCollabCoop.NulGnodebId	MOD NRDUCELLCOLLABC OOP LST NRDUCELLCOLLABC OOP	gNodeB	This parameter has been added to specify the ID of the base station serving the NUL cell bound to the FDD cell in UL and DL Decoupling with SUL and NR FDD co-deployment.
New	NRDUCellCollabServ.NulTtoSulMappingType	MOD NRDUCELLCOLLABSERV LST NRDUCELLCOLLABSERV	gNodeB	This parameter has been added to specify the NUL and SUL mapping type for a TDD cell in UL and DL Decoupling with SUL and NR FDD co-deployment.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellCol labServ.SulCellID	MOD NRDUCELLCOLLABS ERV LST NRDUCELLCOLLABS ERV	gNo deB	This parameter has been added to specify the ID of the SUL cell bound to the TDD cell in UL and DL Decoupling with SUL and NR FDD co-deployment.
New	NRDUCellCol labServ.SulMcc	MOD NRDUCELLCOLLABS ERV LST NRDUCELLCOLLABS ERV	gNo deB	This parameter has been added to specify the MCC of the SUL cell bound to the TDD cell in UL and DL Decoupling with SUL and NR FDD co-deployment.
New	NRDUCellCol labServ.SulMnc	MOD NRDUCELLCOLLABS ERV LST NRDUCELLCOLLABS ERV	gNo deB	This parameter has been added to specify the MNC of the SUL cell bound to the TDD cell in UL and DL Decoupling with SUL and NR FDD co-deployment.
New	NRDUCellCol labServ.SulGnodebId	MOD NRDUCELLCOLLABS ERV LST NRDUCELLCOLLABS ERV	gNo deB	This parameter has been added to specify the ID of the base station serving the SUL cell bound to the TDD cell in UL and DL Decoupling with SUL and NR FDD co-deployment.
Modified	NRDUCellCol labServ.NulSulMaxInterGnbDelay	MOD NRDUCELLCOLLABS ERV LST NRDUCELLCOLLABS ERV	gNo deB	The description of inter-gNodeB UL and DL Decoupling has been added to the meaning of this parameter.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellCollabCoop.NulSulMaxInterGnbDelay	MOD NRDUCELLCOLLABCOOP LST NRDUCELLCOLLABCOOP	gNodeB	The description of inter-gNodeB UL and DL Decoupling has been added to the meaning of this parameter.
Modified	NRDUCellCollabCoop.InterGnbSulMaxRbPct	MOD NRDUCELLCOLLABCOOP LST NRDUCELLCOLLABCOOP	gNodeB	The meaning of this parameter has been changed to the maximum percentage of RBs that can be used by SUL UEs in SUL and NR FDD co-deployment mode.
Modified	NRDUCellCollabCoop.InterGnbSulMinRbPct	MOD NRDUCELLCOLLABCOOP LST NRDUCELLCOLLABCOOP	gNodeB	The meaning of this parameter has been changed to the minimum percentage of RBs that can be used by SUL UEs in SUL and NR FDD co-deployment mode.
Modified	NRDUCellSulRelation.SulType	ADD NRDUCELLSULRELATION MOD NRDUCELLSULRELATION LST NRDUCELLSULRELATION	gNodeB	The description of UL and DL Decoupling has been added to the meaning of this parameter.
Modified	NRDUCellAlgoSwitch.UlDlDecouplingSwitch	MOD NRDUCELLALGOSWITCH LST NRDUCELLALGOSWITCH	gNodeB	The applicable modes of this parameter have been revised. This parameter applies only to low-frequency TDD and FDD cells.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRCellRelation.InterGnodebSulFlag	ADD NRCELLRELATION MOD NRCELLRELATION LST NRCELLRELATION	gNodeB	The applicable modes of this parameter have been revised. This parameter applies only to low-frequency TDD and FDD cells.
Modified	NRCellSulFrequency.SulFreqIndex	ADD NRCELLSULFREQUENCY LST NRCELLSULFREQUENCY MOD NRCELLSULFREQUENCY RMV NRCELLSULFREQUENCY	gNodeB	FOFD-010205 UL and DL Decoupling has been added to the Feature ID/Feature Name field in the reference of this parameter.
Modified	NRCellSulFrequency.SsbFreqPos	ADD NRCELLSULFREQUENCY MOD NRCELLSULFREQUENCY LST NRCELLSULFREQUENCY	gNodeB	FOFD-010205 UL and DL Decoupling has been added to the Feature ID/Feature Name field in the reference of this parameter.
Modified	NRCellSulFrequency.SulInterFreqA5RsrpThld2	ADD NRCELLSULFREQUENCY MOD NRCELLSULFREQUENCY LST NRCELLSULFREQUENCY	gNodeB	FOFD-010205 UL and DL Decoupling has been added to the Feature ID/Feature Name field in the reference of this parameter.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellPdcch.SulCceAdjPolicy	MOD NRDUCCELLPDCCH LST NRDUCCELLPDCCH	gNodeB	- The following description has been added to the meaning of this parameter: This parameter takes effect only when the NRDUCellSulRelation.SulType parameter is set to DYN_SUL or ELAS_SUL, or the NRDUCell.SubcarrierSpacing parameter is set to 30KHZ for the corresponding SUL cell. - FOFD-010205 UL and DL Decoupling has been added to the Feature ID/Feature Name field in the reference of this parameter.
Modified	NRDUCellSulRelation.FddUlnrArfcn	ADD NRDUCCELLSULRELATION MOD NRDUCCELLSULRELATION LST NRDUCCELLSULRELATION	gNodeB	- The meaning of this parameter has been changed to the uplink NR-ARFCN of the FDD cell on the SUL carrier in SUL and NR FDD co-deployment mode. - FOFD-010205 UL and DL Decoupling has been added to the Feature ID/Feature Name field in the reference of this parameter.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellSulRelation.FddUlBandwidth	ADD NRDUCELLSULRELATION MOD NRDUCELLSULRELATION LST NRDUCELLSULRELATION	gNodeB	FOFD-010205 UL and DL Decoupling has been added to the Feature ID/Feature Name field in the reference of this parameter.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Intra-base-station UL CoMP	INTRA_GNB_UL_COMP_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	SUL CoMP requires that intra-base-station UL CoMP be enabled in NR FDD cells when UL and DL Decoupling with SUL and NR FDD co-deployment is enabled.
Low-frequency NR TDD	PDCCH uplink and downlink CCE sharing	UL_DL_CCE_SHARING_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter	<i>Channel Management</i>	Only the SUL CCE adjustment policy function requires that the switch for uplink and downlink CCE sharing be turned on.

- Mutually exclusive features/functions
The following mutually exclusive relationships have been added.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag set to HIGH_SPEED	<i>High Speed Mobility</i>	UL and DL Decoupling cannot be enabled in high-speed cells.
Low-frequency NR TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	UL and DL Decoupling cannot be enabled in hyper cells.
Low-frequency NR TDD	Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	UL and DL Decoupling cannot be enabled in combined cells.
Low-frequency NR TDD	Channel decoupling	SUL_CHANNEL_DECODUPLE_SW option of the NRDUCellCollabServ.SulAlgoSwitch parameter	<i>Super Uplink</i>	Channel decoupling cannot be enabled when UL and DL Decoupling is enabled.
Low-frequency NR TDD	SRS interference coordination based on self-contained and uplink slots	SRS_INTRFCOORD_SU_SLOT_SW option of the NRDUCellSrsSrsAlgoExtSwitch parameter	<i>Channel Management</i>	When SRS interference coordination based on self-contained and uplink slots is enabled, the NRDUCellCollabServ.Nul1TxToSul1TxSwitchTime parameter cannot be set to MICROSECOND 140 .

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	Fusion Cell	NRDUCell.NrDuCellNetworkingMode set to FUSION_CELL	<i>Fusion Cell (Low-Frequency TDD)</i>	UL and DL Decoupling cannot be enabled in Fusion Cell networking.
Low-frequency NR TDD	Pattern-free PDCCH rate matching activation rate increase	PDCCH_NO_PAT_RM_RATIO_INC_S option of the NRDUCellservExp.SpdTurboAlgoSwitch parameter	<i>Downlink Scheduling</i>	Pattern-free PDCCH rate matching activation rate increase cannot be enabled when UL and DL Decoupling is enabled.
Low-frequency NR TDD	5QI-specific UE-number-based connected mode MLB	INTER_FREQ_CONNECTED_MLB option of the NRCCellAlgoSwitch.MlbAlgoSwitch parameter selected, and NRCCellMlb.MlbTriggerMode set to SPECIFIED_5QI_USER_NUM	<i>Mobility Load Balancing in Connected Mode</i>	5QI-specific UE-number-based connected mode MLB cannot be enabled when UL and DL Decoupling is enabled.
Low-frequency NR TDD	PUSCH scheduling based on SRS interference avoidance	SRS_INTRF_AVOID_PUSCH_S option of the NRDUCellSrs.SrsAlgoExtSwitch parameter	None	PUSCH scheduling based on SRS interference avoidance cannot be enabled when UL and DL Decoupling is enabled.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	Cell deployment in IAB mode	NRDUCell.D eployPosition set to DEPLOY_IAB	None	UL and DL Decoupling cannot be enabled in cells deployed in IAB mode.
Low-frequency NR TDD	Cooperation between joint transmission and pattern-free PDCCH rate matching	JT_PDCCH_NO_PAT TERM_COORD_SW option of the NRDUCellPdcch PdcchAlgoSwitch parameter	None	"UL and DL Decoupling" and "cooperation between joint transmission and pattern-free PDCCH rate matching" cannot be both enabled.
Low-frequency NR TDD	Uplink assistance	UL_ASSISTANCE_SW option of the NRDUCellCollabServ MultiCellMimoSwitch parameter	None	UL and DL Decoupling and uplink assistance cannot be both enabled.
NR FDD	Hybrid DSS Based on Asymmetric Bandwidth	LTE_NR_FDD_SPCT_SHR_SW option of the NRDUCellAlgoSwitch SpectrumCloudSwitch parameter	<i>LTE FDD and NR Hybrid Dynamic Spectrum Sharing</i>	When UL and DL Decoupling with SUL and NR FDD co-deployment is enabled, Hybrid DSS Based on Asymmetric Bandwidth cannot be enabled.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documenta tion)	Description
NR FDD	LTE FDD and NR Flash Dynamic Spectrum Sharing	LTE_NR_FDD_SPCT_SHR_SW option of the NRDUCellAI goSwitch.SpectrumCloudSwitch parameter	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>	When UL and DL Decoupling with SUL and NR FDD co-deployment is enabled, LTE FDD and NR Flash Dynamic Spectrum Sharing cannot be enabled.

- Impacted features/functions
The following impact relationships have been added.

RA T	Function Name	Function Switch	Reference	Description
LTE FD D LTE TD D Lo w- fre qu enc y NR TD D	Uplink data transmissio n path selection	• LTE NSA_DC _UL_PAT H_SELEC TION_SW option of the NsaDcM gmtConf ig.NsaDc AlgoSwi tch paramet er • NR NSA_DC _UL_PAT H_SELEC TION_SW option of the NRCellN saDcConf ig.NsaDc AlgoSwi tch paramet er	<i>EPC-based NSA Performanc e Enhanceme nt</i>	Uplink data transmission path selection does not take effect in an NR cell enabled with UL and DL Decoupling.

RA T	Function Name	Function Switch	Reference	Description
LTE FD D LTE TD D Lo w- fre qu enc y NR TD D	Network- coordinated dynamic UE power sharing	• LTE NSA_DC_COOPERATION_DPS_SW option of the NsaDcMgmtConfig.NsaDcAlgoSwitch parameter • NR NSA_DC_COOPERATION_DPS_SW option of the NRCellNsaDcConfig.NsaDcAlgoSwitch parameter	<i>EPC-based NSA Performance Enhancement</i>	If UL and DL Decoupling has taken effect and the base station performs uplink scheduling in TDM mode, network-coordinated dynamic UE power sharing does not take effect.
Lo w- fre qu enc y NR TD D	Power saving BWP	BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSwitch parameter	<i>UE Power Saving</i>	UEs working in BWP2 for power saving do not support SUL CoMP.

RA T	Function Name	Function Switch	Reference	Description
Lo w- fre qu enc y NR TD D	Low latency and high reliability	HIGH_RELI ABILITY_BA SIC_SW option of the NRDUCella lgoSwitch. HighReliab ilitySwitch parameter	<i>URLLC</i>	The low latency and high reliability function does not take effect for UEs using UL and DL Decoupling.
Lo w- fre qu enc y NR TD D	Spectral- efficiency- based connected mode MLB	INTER_FRE Q_CONNEC TED_MLB_S W option of the NRCelAlgo Switch.Mlb AlgoSwitch parameter, and NRCelMlb. MlbTrigger Mode set to SPEC_EFF_B ASED_USER _NUM	<i>Mobility Load Balancing in Connected Mode</i>	The gNodeB does not select UEs for which UL and DL Decoupling has taken effect as the UEs to be handed over.
Lo w- fre qu enc y NR TD D	Intra-band CA (TDD)	INTRA_BA ND_CA_SW option of the NRDUCella lgoSwitch. CaAlgoSwit ch parameter	<i>Carrier Aggregatio n</i>	For UEs that support both UL and DL Decoupling and intra-band CA, either UL and DL Decoupling or intra-band CA can take effect at a time.
Lo w- fre qu enc y NR TD D	Intra-FR inter-band CA	INTRA_FR_I NTER_BAN D_CA_SW option of the NRDUCella lgoSwitch. CaAlgoSwit ch parameter	<i>Carrier Aggregatio n</i>	For UEs that support both UL and DL Decoupling and intra-FR inter-band CA, either UL and DL Decoupling or intra-FR inter-band CA can take effect at a time.

RA T	Function Name	Function Switch	Reference	Description
Lo w- fre qu enc y NR TD D	NR DU cell resource manageme nt between network slices	RB_DYNAM IC_CONTR OL_SW option of the NRDUCellA lgoSwitch . NetworkSli ceAlgoSwit ch parameter	<i>Network Slicing</i>	When both the NR DU cell resource management between network slices function and UL and DL Decoupling are enabled, and the RB_DYNAMIC_CONTROL_ SW option of the NRDUCellAlgoSwitch . Net workSliceAlgoSwitch parameter is selected for the SUL cell, this option must also be selected for the corresponding TDD cell, and the proportion of RB resources configured for the TDD cell takes effect in the SUL cell.
Lo w- fre qu enc y NR TD D	RFSP-based RB resource control	RB_DYNAM IC_CONTR OL_SW option of the NRDUCellC ommonSw . RfspAlgoS witch parameter	<i>Resource Control in NSA Networking</i>	Assume that both RFSP- based RB resource control and UL and DL Decoupling are enabled. If RFSP-based RB resource control is enabled in the SUL cell, it must also be enabled in the associated TDD cell, and the proportion of RB resources configured for the TDD cell takes effect in the SUL cell.
Lo w- fre qu enc y NR TD D	Uplink multi- frequency coordinatio n	MULTI_FRE Q_UL_COO RD_SW option of the NRDUCellF eatureSw . MultiCarrie rSwitch parameter	<i>Multi- Frequency Smart Selection</i>	The gNodeB does not select UEs for which UL and DL Decoupling has taken effect.

RA T	Function Name	Function Switch	Reference	Description
Lo w- fre qu enc y NR TD D	Multi- frequency beam coordinatio n	MULTI_FRE Q_BEAM_C OORD_SW option of the NRDUCellF eatureSw. MultiCarrie rSwitch parameter	<i>Multi- Frequency Smart Selection</i>	The gNodeB does not select UEs for which UL and DL Decoupling has taken effect.
Lo w- fre qu enc y NR TD D	PDCCH skipping	PDCCH_SKI P_SW option of the NRDUCellU ePwrSavin g.SchPwrSa vingSw parameter	<i>UE Power Saving</i>	When both SUL CoMP and PDCCH skipping are enabled, PDCCH skipping does not take effect for SUL UEs.
Lo w- fre qu enc y NR TD D	Downlink massive CA	NRDUCellC arrMgmt.C aDLMaxCc Num set to DL4CC or DL5CC .	<i>CA Experience Improveme nt</i>	UL and DL Decoupling and downlink massive CA cannot take effect at the same time.
Lo w- fre qu enc y NR TD D	Inter-FR CA	INTER_FR_F R1TOFR2_C A_SW option of the NRDUCellA lgoSwitch. CaAlgoSwit ch parameter	<i>Carrier Aggregatio n</i>	Inter-FR CA and UL and DL Decoupling cannot take effect at the same time.
Lo w- fre qu enc y NR TD D	3GPP R15- compliant uplink slot aggregation	VONR_UL_ SLOT_AGG REGATION_ SW option of the NRDUCellU lSch.VoiceU lSchSwitch parameter	<i>VoNR</i>	3GPP R15-compliant uplink slot aggregation does not take effect on UEs with UL and DL Decoupling in effect.

RA T	Function Name	Function Switch	Reference	Description
Lo w- fre qu enc y NR TD D	3GPP R16- compliant uplink slot aggregation	VOICE_R16_UL_SLOT_AGG_SW option of the NRDUCellULSch.VoiceULSchSwitch parameter	<i>VoNR</i>	3GPP R16-compliant uplink slot aggregation does not take effect on UEs with UL and DL Decoupling in effect.
Lo w- fre qu enc y NR TD D	QCI-based resource control	RB_DYNAMIC_CONTROL_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter	<i>QCI-based Resource Control</i>	When QCI-based resource control (controlled by the RB_DYNAMIC_CONTROL_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter) is enabled together with UL and DL Decoupling and is enabled in the SUL cell, the proportion of RB resources configured for the TDD cell takes effect in the SUL cell.
Lo w- fre qu enc y NR TD D	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	After the NR FDD carrier is shut down, SUL scheduling cannot be performed for SUL UEs.
Lo w- fre qu enc y NR TD D	Intelligent carrier shutdown	INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	After the NR FDD carrier is shut down, SUL scheduling cannot be performed for SUL UEs.

RA T	Function Name	Function Switch	Reference	Description
Lo w- fre qu enc y NR TD D	Fast carrier shutdown	For capacity- layer cells: • INTRA_G NB_MUL TI_CARR _SD_SW option of the NRDUce lAlgoS witch.PowerSavi ngSwitc h paramet er selected (intra- base- station multi- carrier scenarios) or INTER_G NB_MUL TI_CARR _SD_SW option of the NRDUce lAlgoS witch.PowerSavi ngSwitc h paramet er selected (inter- base- station multi- carrier scenarios)	<i>Energy Conservatio n and Emission Reduction</i>	After the NR FDD carrier is shut down, SUL scheduling cannot be performed for SUL UEs.

RA T	Function Name	Function Switch	Reference	Description
		FAST_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter For basic-layer cells: FAST_CARRIER_SHUT_SERVICE_ASSUR_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter selected		
Low-frequency NR TD	Rate-matching-pattern-configuration-based PDCCH rate matching	PDCCH_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	UL and DL Decoupling cannot take effect when rate-matching-pattern-configuration-based PDCCH rate matching is enabled and the NRDUCellPdcch.OccupiedRbNum parameter is set to 1 or 2.
Low-frequency NR TD	WUS	DRX_ADAPTATION_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch parameter	<i>UE Power Saving</i>	WUS does not take effect on a UE with inter-gNodeB SUL in effect. If WUS has taken effect on a UE before inter-gNodeB SUL, the UE exits WUS before inter-gNodeB SUL can take effect.

RA T	Function Name	Function Switch	Reference	Description
Lo w- fre qu enc y NR TDD	Time- domain power saving BWP (low- frequency TDD)	TDM_BWP 2_SWITCH option of the NRDUCellU ePwrSavin g.BwpPwrS avingSw parameter	<i>UE Power Saving</i>	UEs on the NUL carrier can be switched to BWP2 while UEs on the SUL carrier cannot be switched to BWP2. The scheduling delay of BWP2 UEs on the NUL carrier increases when the SUL carrier is activated.
Lo w- fre qu enc y NR TDD	Secondary harmonic interference avoidance	SEC_HARM ONIC_INTR F_AVOID_S W option of the NRDUCellC ollabServ.S ulAlgoSwit ch parameter	None	Secondary harmonic interference avoidance does not take effect for UEs enabled with UL and DL Decoupling with SUL and NR FDD co- deployment.
Lo w- fre qu enc y NR TDD	RedCap Introduction	REDCAP_S W option of the NRDUCellF eatureSw.U eCapbBase dFunSw parameter	<i>RedCap</i>	UL and DL Decoupling does not take effect on RedCap UEs when proactive avoidance of remote interference is enabled.
Lo w- fre qu enc y NR TDD	Proactive avoidance of remote interference	SRS_RIM_I NTRF_AVOI D_SW and SRS_INTRF_ COORD_S_ SLOT_SW options of the NRDUCellS rs.SrsAlgoE xtSwitch parameter	<i>Remote Interference Manageme nt (Low- Frequency TDD)</i>	

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	User-rate-based dynamic network slice resource management	DYNAMIC_NETWORK_SLICE_SW option of the NRDUCellCommonSw.ServiceDiffAlgoExtSwitch parameter	<i>Network Slicing</i>	When user-rate-based dynamic network slice resource management is enabled for an SUL cell with UL and DL Decoupling in effect, you are advised to also enable it for the associated TDD cell. Otherwise, the gNodeB may perform inaccurate dynamic resource adjustments.
Low-frequency NR TDD	User-rate-based dynamic QCI resource management	DYNAMIC_QCI_GROUP_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter	<i>QCI-based Resource Control</i>	When user-rate-based dynamic QCI resource management is enabled for an SUL cell with UL and DL Decoupling in effect, you are advised to also enable it for the associated TDD cell. Otherwise, the gNodeB may perform inaccurate dynamic resource adjustments.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *UL and DL Decoupling*.

Requirement/Trouble Ticket Number

FeR-IR20250711002400-0010

IR20250910001876

RAT

NR TDD

5.3.2.3.2 Impact Relationships of UL and DL Decoupling with RedCap and Proactive Avoidance of Remote Interference

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Impact relationships of UL and DL Decoupling with RedCap and proactive avoidance of remote interference have been added. UL and DL Decoupling does not take effect on RedCap UEs when proactive avoidance of remote interference is enabled.

Implementation

For details, see *UL and DL Decoupling, RedCap, and Remote Interference Management (Low-Frequency TDD)*. The specific sections are listed in the change history of the documents.

Impact on the Network

When UL and DL Decoupling, proactive avoidance of remote interference, and RedCap are all enabled, the **N.RRC.ReEst.Att.RedCap** counter value may decrease.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this function has no impact. It is controlled by the existing license.
- Configuration management
The change in this feature has no impact.
- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The following impact relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	RedCap Introduction	REDCAP_SW option of the NRDUCellFeatureSw.UeCapBase dFunSw parameter	<i>RedCap</i>	UL and DL Decoupling does not take effect on RedCap UEs when proactive avoidance of remote interference is enabled.
Low-frequency NR TDD	Proactive avoidance of remote interference	SRS_RIM_INTERF_AVOID_SW and SRS_INTERF_COORD_SLOT_SW options of the NRDUCellsrs.SrsAlgoExtSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

DTS2025122721356
DTS2025122719590
IR20251204000006

RAT

NR TDD

5.3.2.3.3 Compatibility of UE Random Access Control Based on Cell Radius with SUL UEs

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

SUL UEs support the "UE random access control based on cell radius" function. This function is controlled by the **ACCESS_BYND_CELL_RAD_PROHIB_SW** option of the **NRDUCellPrach.PrachAlgoExtSwitch** parameter.

- In independent SUL deployment mode, when the distance between an SUL UE and an SUL cell is greater than the SUL cell radius, UEs for which UL and DL Decoupling has taken effect are not allowed to access the cell.
- In SUL and NR FDD co-deployment mode, when the distance between an SUL UE and an NR FDD cell is greater than the NR FDD cell radius, UEs for which UL and DL Decoupling has taken effect are not allowed to access the cell.

For details about the "UE random access control based on cell radius" function, see *Random Access Control*.

This change is irrelevant to LampSite base stations.

Implementation

For details, see *UL and DL Decoupling*. The specific sections are listed in the change history of the document.

Impact on the Network

This function enables the gNodeB to detect UE access beyond the cell radius, which increases the consumption of physical resources. Since contention-based random access is prohibited for UEs whose distance to the gNodeB exceeds the cell radius, the random access success rate and RRC connection setup success rate may decrease, and the cell traffic, cell rate, and PRB usage may decrease.

In addition, enabling "optimized measurement of RRC connection setup success rate based on cell radius" can prevent the RRC connection setup success rate from decreasing.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this function has no impact. It is controlled by the existing license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellPrach.PrachAlgoExtSwitch	MOD NRDUCELLPRACH LST NRDUCELLPRACH	gNodeB	The ACCESS_BYND_CELL_RAD_PROHIB_SW option of this parameter is now applicable to SUL cells.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *UL and DL Decoupling*.

Requirement/Trouble Ticket Number

FeR-IR20250711002400-0010

RAT

NR TDD

5.3.2.3.4 Removed Mutually Exclusive Relationship Between UL and DL Decoupling and Proactive Avoidance of Remote Interference

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The mutually exclusive relationships of proactive avoidance of remote interference with UL and DL Decoupling and NUL 1Tx to SUL 1Tx Switching Time have been removed.

This change is irrelevant to LampSite base stations.

Implementation

For details, see *UL and DL Decoupling*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this function has no impact. It is controlled by the existing license.
- Configuration management
The change in this feature has no impact.
- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The following mutually exclusive relationship has been removed.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	Proactive avoidance of remote interference	SRS_RIM_INTRF_AVOID_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	Proactive avoidance of remote interference cannot be enabled when UL and DL Decoupling is enabled.

The following description has been deleted from section "Data Preparation":
When SRS remote interference avoidance (controlled by the **SRS_RIM_INTRF_AVOID_SW** option of the **NRDUCellSrs.SrsAlgoExtSwitch** parameter) or proactive avoidance of remote interference (controlled by the **SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs.SrsAlgoExtSwitch** parameter) is enabled on the NR side, the **NRDUCellCollabServ.Nul1TxToSul1TxSwitchTime** parameter cannot be set to **MICROSECOND140**.

- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250711002400-0010

RAT

NR TDD

5.3.2.4 FOFD-021203 gNodeB Power Saving

5.3.2.4.1 Added Support for the Precise Backup Power Saving Policy by Third-Party Power Systems

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

When low power consumption mode is used with a third-party power system, Huawei lithium batteries (used to provide backup power), and a precise backup power saving policy, the following requirements must be met:

- The third-party power system supports connection through a dry contact.
- A PMU 15A(Ver.B) is configured.

This change is irrelevant to LampSite base stations.

Implementation

For details, see *Green Site*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact.
- Configuration management
The change in this feature has no impact.
- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20240812001108-0050

RAT

NR FDD, NR TDD

5.3.2.4.2 Added Support for Load-based Evaluation for Entering Timing Carrier Shutdown

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

After timing carrier shutdown (controlled by the **TIMING_CARRIER_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch.PowerSavingSwitch** parameter) is selected, the **NRDUCellPowerSaving.CarrierShutdownJudgeMode** parameter can be used to specify the evaluation mode for a cell to enter the timing carrier shutdown state.

- If this parameter is set to **CARR_SHDN_BY_TIME**, time-based evaluation applies. The cell enters the carrier shutdown state within the configured power saving period.
- If this parameter is set to **CARR_SHDN_BY_LOAD**, load-based evaluation applies. The cell enters the carrier shutdown state when the load conditions are met within the configured power saving period. In addition, the following parameters are used to configure the relevant load conditions and evaluation period.
 - **NRDUCellPowerSaving.DlPrbThld**
 - **NRDUCellPowerSaving.UlPrbThld**
 - **NRDUCellPowerSaving.UserNumThld**
 - **NRDUCellPowerSaving.EvalPeriodGranularity**
 - **NRDUCellPowerSaving.ShutdownTrigJudgePeriod**

The change in this feature is supported only in low frequency bands.

Implementation

For details, see *Energy Conservation and Emission Reduction*. The specific sections are listed in the change history of the document.

Impact on the Network

When the **NRDUCellPowerSaving.CarrierShutdownJudgeMode** parameter is set to **CARR_SHDN_BY_LOAD**, timing carrier shutdown takes effect only when the

specified time and cell load requirements are both met. This reduces the probability of triggering carrier shutdown and may deliver fewer energy saving gains compared to using **CARR_SHDN_BY_TIME**. In addition, the impact of energy saving on the base station may decrease. For details about the impacts of timing carrier shutdown on the network, see *Energy Conservation and Emission Reduction*.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellPowerSaving.CarrierShutdownJudgeMode	ADD NRDUCELLPO WERSAVING MOD NRDUCELLPO WERSAVING LST NRDUCELLPO WERSAVING	gNodeB	This parameter has been added to specify the evaluation mode for a cell to enter the timing carrier shutdown state.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.

- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Energy Conservation and Emission Reduction*.

Requirement/Trouble Ticket Number

FeR-IR20250526000353-0010

RAT

NR FDD, NR TDD

5.3.2.5 FOFD-030207 High-speed Railway Superior Experience

5.3.2.5.1 Added Inter-Frequency Directional Handover for High-Speed UEs (TDD)

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

With the "Inter-frequency directional handover for high-speed UEs" function enabled, high-speed UEs in a high-speed cell on a high-speed-railway dedicated network are not allowed to be handed over to neighboring cells working on frequencies for low speed mobility.

Before enabling this function, ensure that the local cell is configured with inter-frequency neighboring cells and at least one of operating frequencies of these neighboring cells is configured as non-low-speed neighboring frequency through the **NRCellFreqRelation.HighSpeedFlag** parameter. If this parameter is set to **HIGH_SPEED** or **NOT_CONFIG**, the frequency is not a low-speed frequency. If this parameter is set to **LOW_SPEED**, the frequency is not a low-speed frequency.

After "inter-frequency directional handover for high-speed UEs" is enabled, the base station delivers measurement configurations only to operating frequencies of neighboring cells for which the **NRCellFreqRelation.HighSpeedFlag** parameter is set to **NOT_CONFIG** or **HIGH_SPEED** not **LOW_SPEED** when high-speed UEs on a high-speed-railway dedicated network meet the triggering conditions of inter-frequency handovers. In this way, handovers to inter-frequency neighboring cells working on only non-low-speed frequencies are performed.

This function is controlled by the **HS_INTER_FREQ_DIR_HO_SW** option of the **NRCellMobilityConfig.InterFreqHoExtSwitch** parameter.

Implementation

For details, see *High Speed Mobility*. The specific sections are listed in the change history of the document.

Impact on the Network

"Inter-frequency directional handover for high-speed UEs" provides the following benefits:

- Only inter-frequency handovers to frequencies for high speed mobility are allowed, which prevents handovers to frequencies for low speed mobility. This improves the user experience of high-speed UEs and may increase the **User Downlink Average Throughput (DU)** and **User Uplink Average Throughput (DU)**.
- For high-speed UEs in the downlink CA state, if all non-low-speed neighboring frequencies have been added as SCCs, gap-assisted measurements are not started on added SCCs. This reduces the gap-assisted measurement overhead and improves the user experience of high-speed UEs.

"Inter-frequency directional handover for high-speed UEs" has the following impacts:

- In scenarios where high-speed railways and highways run parallel, if UEs on highway are incorrectly identified as high-speed UEs and high-speed UEs change to low-speed UEs, the UEs may fail to be handed over to low-speed cells because the base station does not deliver neighboring frequencies for which the **NRCellFreqRelation.HighSpeedFlag** parameter set to **LOW_SPEED** for measurement. As a result, the handover failure rate and service drop rate increase. Therefore, you are advised not to enable "inter-frequency directional handover for high-speed UEs" in the preceding scenarios.
- In scenarios where high-speed trains decelerate in urban areas, if the UE attribute changes from high speed to low speed due to train deceleration and weak coverage occurs on the high-speed-railway dedicated network, high-speed UEs may fail to be handed over to low-speed cells because the base station does not deliver neighboring frequencies for which the **NRCellFreqRelation.HighSpeedFlag** parameter is set to **LOW_SPEED** for measurement. As a result, the handover failure rate and service drop rate increase. Therefore, you are advised not to enable "inter-frequency directional handover for high-speed UEs" in the weak coverage scenarios.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRCellMobilityConfig.InterFreqHoExtSwitch	MOD NRCELLMOBILITYCONFIG LST NRCELLMOBILITYCONFIG	gNodeB	The HS_INTER_FREQ_DIR_HO_SW option has been added to specify whether to enable "inter-frequency directional handover for high-speed UEs".

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	The "inter-frequency directional handover for high-speed UEs" function requires that NRDUCell.HighSpeedFlag be set to HIGH_SPEED .

- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *High Speed Mobility*.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0015

RAT

NR TDD

5.3.2.5.2 Added "SRS Resource Extension in a High-Speed Cell"

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In high-speed railway scenarios, channel time variation is severe due to the high-speed movement of UEs. As a result, SRS resource demand increases significantly, which may lead to SRS resource insufficiency. SRS resource extension in a high-speed cell is introduced to increase UE capacity. After this function is enabled, the base station can allocate SRS resources to more UEs through optimized SRS resource allocation.

This function is controlled by the **SRS_RES_EXT_SW** option of the **NRDUCellHighSpeedMob.HighSpeedAlgoOptSw** parameter.

Implementation

For details, see *High Speed Mobility*. The specific sections are listed in the change history of the document.

Impact on the Network

Benefits:

In heavy-load scenarios, enabling SRS resource extension in a high-speed cell increases the available SRS resources for UEs and allows the high-speed cell to support more UEs.

Impacts:

In heavy-load scenarios, enabling SRS resource extension in a high-speed cell may reduce the average uplink and downlink throughputs of UEs in neighboring cells.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management

The change in this feature has no impact. It is controlled by the existing feature license.

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellHighSpeedMob.HighSpeedAlgoOptSw	MOD NRDUCELLHIGHSPEDMOB LST NRDUCELLHIGHSPEDMOB	gNodeB	The SRS_RES_EXT_SW option has been added to control whether to enable "SRS resource extension in a high-speed cell".

- Performance management

The change in this feature has no impact.

- Fault management

The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	SRS resource extension in a high-speed cell takes effect only when the NRDUCell.HighSpeedFlag parameter is set to HIGH_SPEED .
Low-frequency TDD	SRS period and CDM adaptation	SRS_PERIOD_AND_CDM_ADAPT_SW option of the NRDUCellHighSpeedMob.HighSpeedAlgoOptSw parameter	<i>High Speed Mobility</i>	SRS resource extension in a high-speed cell requires that "SRS period and CDM adaptation" be enabled.

- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *High Speed Mobility*.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0073

RAT

NR TDD

5.3.2.5.3 Added Inter-TRP PDCCH Interference Avoidance for VoNR (TDD)

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In high-speed scenarios, after cell combination networking is used or the Hyper Cell function is enabled, each TRP performs independent scheduling, yielding spatial multiplexing gains. However, this also introduces PDCCH interference and can degrade the MOS of VoNR UEs. To address this issue, the inter-TRP PDCCH interference avoidance for VoNR function is introduced. After this function is enabled, interference between the PDCCHs of VoNR UEs served by different TRPs is reduced by staggering time-domain and frequency-domain resources, thereby reducing the packet loss rate and ensuring VoNR voice quality.

This function is controlled by the **PDCCH_VONR_INTF_AVOID_SW** option of the **NRDUCellPdcch.PdcchAlgoExtSwitch** parameter. It is also recommended that the **TOP_VONR_UE_CCE_ALLOC_OPT_SW** option of the same parameter be selected so that CCE allocation for the top-priority VoNR UEs is not constrained by the CCE allocation proportion, further reducing interference between the PDCCHs of VoNR UEs.

NOTE

In earlier versions, inter-TRP PDCCH interference avoidance for VoNR is controlled by **RSVDSWPARAM1_BIT14** of the reserved parameter **NRDUCellRsvdExt01.RsvdSwParam1**, and this function does not take effect by default. In the current version, **RSVDSWPARAM1_BIT14** of the reserved parameter **NRDUCellRsvdExt01.RsvdSwParam1** is replaced by the **PDCCH_VONR_INTF_AVOID_SW** and **TOP_VONR_UE_CCE_ALLOC_OPT_SW** options of the **NRDUCellPdcch.PdcchAlgoExtSwitch** parameter.

Implementation

For details, see *High Speed Mobility*. The specific sections are listed in the change history of the document.

Impact on the Network

In high-speed scenarios, after cell combination network is used or the Hyper Cell function is enabled, inter-TRP PDCCH interference avoidance for VoNR can be enabled to reduce interference between the PDCCHs of VoNR UEs, thereby reducing the VoNR packet loss rate and improving VoNR user experience.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellPdcch.PdcchAlgoExtSwitch	MOD NRDUCELLP DCCH LST NRDUCELLP DCCH	gNodeB	The PDCCH_VONR_INTF_AVOID_SW option has been added to specify whether to enable inter-TRP PDCCH interference avoidance for VoNR.
Modified	NRDUCellPdcch.PdcchAlgoExtSwitch	MOD NRDUCELLP DCCH LST NRDUCELLP DCCH	gNodeB	The TOP_VONR_UE_CCE_ALLOC_OPT_SW option has been added to specify whether to enable CCE allocation optimization for top-priority VoNR UEs.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellRsvdExt01.RsvdSwParam1	MOD NRDUCELLR SVDEXT01 LST NRDUCELLR SVDEXT01	gNodeB	RSVDSWPARAM1_BIT14 of this reserved parameter has been replaced by the PDCCH_VONR_INTF_AVOID_SW and TOP_VONR_UE_CCE_ALLOC_OPT_SW options of the NRDUCellPdcch.PdcchAlgoExtSwitch parameter and therefore has been added to the disuse list. Although the configuration interface in this version supports synchronization and delivery of the setting of this bit, the system does not use this bit.

NOTE

NRDUCellRsvdExt01.RsvdSwParam1 is a reserved parameter.

For details about reserved parameters, see 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *High Speed Mobility*.

Requirement/Trouble Ticket Number

FeR-IR20250102000700-0024

RAT

NR TDD

5.3.2.5.4 Added Downlink Priority Enhancement for Voice Service UEs in a High-Speed Cell (TDD)

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In high-speed railway scenarios, a large number of UEs are handed over to and from cells at a time, triggering massive signaling configurations. In the traditional scheduling mechanism, the priority of signaling is higher than that of voice packets. As a result, voice packets may not be scheduled in a timely manner. To reduce the scheduling delay of VoNR services, downlink priority enhancement for voice service UEs in a high-speed cell is introduced. After this function is enabled, the downlink scheduling priority of VoNR UEs is increased to prevent voice packets from failing to be scheduled due to signaling impact.

This function is controlled by the **VONR_DL_PRI_SCH_ENH_SW** option of the **NRDUCellDISch.VoiceDISchSwitch** parameter. Before enabling this function, "downlink scheduling with the same priority as signaling scheduling" needs to be enabled (by the selecting the **VONR_DL_PRI_SCH_WITH_SIG_SW** option of the **NRDUCellDISch.VoiceDISchSwitch** parameter). For details about "downlink scheduling with the same priority as signaling scheduling", see "Scheduling Guarantee" in *VoNR*.

NOTE

In earlier versions, "downlink priority enhancement for voice service UEs in a high-speed cell" is controlled by **RSVDSWPARAM1_BIT16** of the reserved parameter **NRDUCellRsvdExt01.RsrdSwParam1**, and this function does not take effect by default. In the current version, **RSVDSWPARAM1_BIT16** of the reserved parameter **NRDUCellRsvdExt01.RsrdSwParam1** has been replaced by the **VONR_DL_PRI_SCH_ENH_SW** option of the **NRDUCellDISch.VoiceDISchSwitch** parameter.

Implementation

For details, see *High Speed Mobility*. The specific sections are listed in the change history of the document.

Impact on the Network

Downlink priority enhancement for voice service UEs in a high-speed cell is recommended when the maximum number of UEs in a high-speed cell is greater

than 100. After this function is enabled, the downlink packet loss rate of voice services and the downlink BLER of voice services decrease in the cell.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCelIDLSch.VoiceDischSwitch	MOD NRDUCELLD LSCH LST NRDUCELLD LSCH	gNodeB	The VONR_DL_PRI_SCH_ENH_SW option has been added to control whether to enable downlink priority enhancement for voice service UEs in a high-speed cell.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellRs vdExt01.Rsv dSwParam1	MOD NRDUCELLR SVDEXT01 LST NRDUCELLR SVDEXT01	gNodeB	RSVDSWPARAM1_B IT16 of this reserved parameter has been replaced by the VONR_DL_PRI_SCH_ENH_SW option of the NRDUCellDLSch.VoiceDLSchSwitch parameter and therefore has been added to the disuse list. Although the configuration interface in this version supports synchronization and delivery of the setting of this bit, the system does not use this bit.

NOTE

NRDUCellRsvdExt01.RsvdSwParam1 is a reserved parameter.

For details about reserved parameters, see 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *High Speed Mobility*.

Requirement/Trouble Ticket Number

FeR-IR20250102000700-0024

RAT

NR TDD

5.3.2.5.5 Added Downlink-Service-based Carrier Selection in High-Speed Scenarios (TDD)

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

On a high-speed railway multi-carrier network, a base station triggers handovers or carrier changes for UEs mainly based on UE-reported RSRP, and selects carriers for UEs to camp on. As carrier difference is not considered, the carriers selected by the base station may not be able to provide the optimal experience for certain UEs running different services. Downlink-service-based carrier selection in high-speed scenarios is introduced to enable differentiated selection of optimal carriers for UEs with different preferred services. This function is controlled by the **HIGH_SPD_DL_SERV_CARR_SEL_SW** option of the **NRCellMobilityConfig.InterFreqHoExtSwitch** parameter.

Assume that downlink-service-based carrier selection in high-speed scenarios is enabled. Once identifying downlink-preferred UEs, the base station instructs these UEs to perform inter-frequency measurements. In this case, the base station performs carrier selection based on the frequency priorities for downlink services. The frequency priority for downlink services is specified by the **NRCellFreqRelation.DlTrafficFreqPriority** parameter.

Implementation

For details, see *High Speed Mobility*. The specific sections are listed in the change history of the document.

Impact on the Network

Benefits: After downlink-service-based carrier selection in high-speed scenarios is enabled and takes effect in high-speed railway multi-carrier scenarios, the base station identifies UEs with downlink-preferred services and selects frequencies for handovers based on the frequency priorities for downlink services, improving downlink user experience in high-speed railway scenarios. In addition, UE handovers may change the downlink traffic distribution across cells, and the average downlink UE throughput (indicated by the **User Downlink Average Throughput (DU)**) may increase.

Impacts: After downlink-service-based carrier selection in high-speed scenarios takes effect, unnecessary inter-frequency handovers are triggered. As a result, UE distribution among frequency bands changes, air interface signaling overhead increases, and the number of handovers increases. In addition, the values of accessibility and retainability indicators in high-speed cells may slightly fluctuate. In addition, after this function is enabled, the uplink user-perceived data rate in a cell (indicated by the **User Uplink Average Throughput (DU)**) may fluctuate.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRCellMobilityConfig.InterFreqHoExtSwitch	MOD NRCELL MOBILITYCONFIG LST NRCELL MOBILITYCONFIG	gNodeB	The HIGH_SPD_DL_SERV_CARR_SEL_SW option has been added to specify whether to enable downlink-service-based carrier selection in high-speed scenarios.

Change Type	Parameter ID	MML Command	NE	Description
New	NRCellFreqRelation.DlTrafficFreqPriority	ADD NRCELL FREQRELATION MOD NRCELL FREQRELATION LST NRCELL FREQRELATION	gNodeB	This parameter has been added to specify the frequency priority for downlink services. Set this parameter based on the network plan.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	Downlink-service-based carrier selection in high-speed scenarios requires that the NRDUCell.HighSpeedFlag parameter be set to HIGH_SPEED .

- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *High Speed Mobility*.

Requirement/Trouble Ticket Number

FeR-IR20250321000501-0001

RAT

NR TDD

5.3.2.5.6 Replaced the Reserved-Parameter Bit That Controls "Uplink Frequency Offset Range Estimation Optimization for High-speed Cells" with a Formal-Parameter Option (TDD)

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

When uplink frequency offset range estimation is enabled in a high-speed cell, uplink frequency offset range estimation optimization is introduced to further improve user experience with voice services. This function is controlled by the **UL_FREQ_OFS_INTVL_EST_OPT_SW** option of the **NRDUCellHighSpeedMob.HighSpeedAlgoOptSw** parameter.

RSVDSWPARAM1_BIT22 of the reserved parameter **NRDUCellRsvdExt01.RsvdSwParam1** that controls this function has been replaced by the **UL_FREQ_OFS_INTVL_EST_OPT_SW** option of the **NRDUCellHighSpeedMob.HighSpeedAlgoOptSw** parameter.

Implementation

For details, see *High Speed Mobility*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellHighSpeedMob.HighSpeedAlgoOptSw	MOD NRDUCELLHIGH SPEEDMOB LST NRDUCELLHIGH SPEEDMOB	gNodeB	The UL_FREQ_OFS_INTVL_EST_OPT_SW option has been added to specify whether to enable uplink frequency offset range estimation optimization.
Modified	NRDUCellRsvdExt01.RsvdSwParam1	MOD NRDUCELLRSVDEXT01 LST NRDUCELLRSVDEXT01	gNodeB	RSVDSWPARAM1_BIT22 of this reserved parameter has been added to the disuse list and replaced with the UL_FREQ_OFS_INTVL_EST_OPT_SW option of the NRDUCellHighSpeedMob.HighSpeedAlgoOptSw parameter. Although the configuration interface in this version supports synchronization and delivery of the setting of this bit, the system does not use this bit.

NOTE

NRDUCellRsvdExt01.RsvdSwParam1 is a reserved parameter.

For details about reserved parameters, see 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List.

- Performance management
The change in this feature has no impact.

- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *High Speed Mobility*.

Requirement/Trouble Ticket Number

FeR-IR20250102000700-0024

RAT

NR TDD

5.3.2.5.7 Replaced the Reserved-Parameter Bit That Controls "Service-based Carrier Selection Enhancement in High-Speed Scenarios" with a Formal-Parameter Option (TDD)

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

After "uplink-service-based carrier selection in high-speed scenarios" is enabled, the "service-based carrier selection enhancement in high-speed scenarios" function is introduced to better identify UEs running uplink large-packet services. This function is controlled by the **HIGH_SPD_SERV_CARR_SEL_ENH_SW** option of the **NRCellFeatureSwitch.MultiFreqAlgoSwitch** parameter.

RSVDSWPARAM1_BIT11 of the reserved parameter **NRCellRsvd.RsvdSwParam1** that controls this function has been replaced by the **HIGH_SPD_SERV_CARR_SEL_ENH_SW** option of the **NRCellFeatureSwitch.MultiFreqAlgoSwitch** parameter.

Implementation

For details, see *High Speed Mobility*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRCellFeatureSwitch.MultiFreqAlgorithm	MOD NRCELLFEAT URESWITCH LST NRCELLFEAT URESWITCH	gNodeB	The HIGH_SPD_SERV_CARR_SEL_ENH_SW option has been added to specify whether to enable service-based carrier selection enhancement in high-speed scenarios.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRCeIIrSvd. RsvdSwParam1	MOD NRCELLRSVD LST NRCELLRSVD	gNodeB	RSVDSWPARAM1_BIT11 of this reserved parameter has been added to the disuse list and replaced with the HIGH_SPD_SERV_CARR_SEL_ENH_SW option of the NRCeIIFeatureSwitch.MultiFreqAlgoSwitch parameter. Although the configuration interface in this version supports synchronization and delivery of the setting of this bit, the system does not use this bit.

NOTE

NRCeIIrSvd.RsvdSwParam1 is a reserved parameter.

For details about reserved parameters, see 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *High Speed Mobility*.

Requirement/Trouble Ticket Number

IR20251204000006

RAT

NR TDD

5.3.2.6 FOFD-030211 Super Uplink

5.3.2.6.1 2CC SRS Carrier Switching for SUL and CA

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

SRS transmission can use the SUL carrier, PCC, and SCC in turn when both super uplink using band combinations in compliance with R16 and later and downlink 3CC aggregation take effect. This function is controlled by the **SUL_CA_2CC_SRS_SWITCHING_SW** option of **NRDUCellCollabServ.SulAlgoSwitch**, improving downlink user experience.

Implementation

For details, see *Super Uplink*. The specific sections are listed in the change history of the document.

Impact on the Network

When the **SUL_CA_2CC_SRS_SWITCHING_SW** option of the **NRDUCellCollabServ.SulAlgoSwitch** parameter is selected, UEs that support downlink 3CC aggregation, SUL in compliance with R16, and SRS carrier switching can obtain the following gains:

- Compared with UEs for which only downlink CA and SRS carrier switching take effect, the average uplink throughput of UEs that also support super uplink is improved.
- Compared with UEs for which only downlink CA and super uplink take effect, the average downlink throughput of UEs that also support SRS carrier switching is improved. The average downlink throughput is indicated by **User Downlink Average Throughput (DU)**.

The average uplink throughput of UEs that also support SRS carrier switching decreases by about 7%, as time needs to be reserved for switching SRS transmission between SUL and PCC carriers. The average uplink throughput is indicated by **User Uplink Average Throughput (DU)**.

Impact on NEs

UEs must support downlink 3CC aggregation, SRS carrier switching, and SUL in compliance with R16.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this function has no impact. It is controlled by the existing license.
- Configuration management

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellCollabServ.SulAlgoSwitch	MOD NRDUCellCollabServ LST NRDUCellCollabServ	gNodeB	The SUL_CA_2CC_SRS_SWITCHING_SW option has been added to control 2CC SRS carrier switching for SUL and CA.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

The **MOD NRDUCellCollabServ** command can be executed to select the **SUL_CA_2CC_SRS_SWITCHING_SW** option of the **NRDUCellCollabServ.SulAlgoSwitch** parameter.

Requirement/Trouble Ticket Number

FeR-IR20250711002400-0010

RAT

NR TDD

5.3.2.6.2 Enhanced SRS Carrier Switching for SUL and Downlink CA

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Enhanced SRS carrier switching is now supported for SUL and downlink CA. The enhancements are as follows:

Enhancement 1: SRS carrier switching is now supported for SUL in compliance with R15 and downlink CA.

SRS transmission can use the SUL carrier, PCC, and SCC in turn when both super uplink using band combinations in compliance with R15 and downlink 2CC or 3CC aggregation take effect. This function is controlled by the **SUL_CA_SRS_SWITCHING_EXT_SW** option of **NRDUCellCollabServ.SulAlgoSwitch**, improving downlink user experience.

Enhancement 2: SRS carrier switching for SUL in compliance with R16 or later and downlink CA is modified.

- The meaning of the **SUL_CA_SRS_SWITCHING_SW** option of **NRDUCellCollabServ.SulAlgoSwitch** has been changed to the following:
Indicates whether to enable SRS transmission using the SUL carrier, PCC, and SCC in turn when both super uplink using band combinations in compliance with R16 or later and downlink CA take effect. The base station supports this function only if this option is selected.
- When downlink 2CC aggregation and super uplink are both enabled, SRS transmission can use the SUL carrier, PCC, and SCC in turn, which is consistent with a band combination in compliance with R16 or later. Verification on band combinations used by SUL is stricter on the network side if the following conditions are met:
 - The **SUL_CA_SRS_SWITCHING_SW** option of the **NRDUCellCollabServ.SulAlgoSwitch** parameter is selected.
 - The **SUL_CA_SRS_SWITCHING_EXT_SW** option of the **NRDUCellCollabServ.SulAlgoSwitch** parameter is deselected.
 - UEs supports only downlink CA, SUL using band combinations in compliance with R15, and SRS carrier switching.

Before the modification of SRS carrier switching for SUL in compliance with R16 or later and downlink CA takes effect, UEs' band combination capabilities

that support downlink 2CC aggregation and SUL in compliance with R15 but not SRS carrier switching are likely to be selected and enabled based on the traffic model and configuration policy on the network side. As a result, uplink user experience is improved.

After the modification of SRS carrier switching for SUL in compliance with R16 or later and downlink CA takes effect, UEs' band combination capabilities that support downlink 2CC aggregation and SRS carrier switching but not CA and SUL in compliance with R15 are likely to be selected and enabled based on the traffic model and configuration policy on the network side. As a result, downlink user experience is improved.

Implementation

For details, see *Super Uplink*. The specific sections are listed in the change history of the document.

Impact on the Network

When the **SUL_CA_SRS_SWITCHING_EXT_SW** option of the **NRDUCellCollabServ.SulAlgoSwitch** parameter is selected, UEs that support downlink 2CC or 3CC aggregation, SUL in compliance with R15, and SRS carrier switching can obtain the following gains:

- Compared with UEs for which only downlink CA and SRS carrier switching take effect, the average uplink throughput of UEs that also support super uplink is improved.
- Compared with UEs for which only downlink CA and super uplink take effect, the average downlink throughput of UEs that also support SRS carrier switching is improved. The average downlink throughput is indicated by **User Downlink Average Throughput (DU)**.

The average uplink throughput of UEs that also support SRS carrier switching decreases by about 7%, as time needs to be reserved for switching SRS transmission between SUL and PCC carriers. The average uplink throughput is indicated by **User Uplink Average Throughput (DU)**.

The values of the following indicators change in the case that SRS carrier switching for SUL in compliance with R16 or later and downlink CA is modified:

- The values of the following indicators may increase:
 - **Cell Downlink Average Throughput (DU)**
 - **User Downlink Average Throughput (DU)**
- The values of the following indicators may decrease:
 - **Cell Uplink Average Throughput (DU)**
 - **User Uplink Average Throughput (DU)**
 - **N.SUL.IntergNB.SulCarrier.Rmv.Succ**
 - **N.SUL.IntergNB.SulCarrier.Rmv.Att**
 - **N.SUL.IntergNB.SulCarrier.Add.Succ**
 - **N.SUL.IntergNB.SulCarrier.Add.Att**
 - **N.SUL.SulCarrier.Rmv.Succ**

- **N.SUL.SulCarrier.Rmv.Att**
- **N.SUL.SulCarrier.Add.Succ**
- **N.SUL.SulCarrier.Add.Att**

Impact on NEs

UEs must support downlink 2CC or 3CC aggregation, SRS carrier switching, and SUL in compliance with R15.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this function has no impact. It is controlled by the existing license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellCollabServ <i>.SulAlgoSwitch</i>	MOD NRDUCellCollabServ LST NRDUCellCollabServ	gNodeB	The SUL_CA_SRS_SWITCHING_EXT_SW option has been added to specify whether to enable SRS carrier switching for downlink 2CC or 3CC aggregation and SUL in compliance with R15. The meaning of the SUL_CA_SRS_SWITCHING_SW option has been changed to the following: Indicates whether to enable SRS transmission using the SUL carrier, PCC, and SCC in turn when both super uplink using band combinations in compliance with R16 or later and downlink CA take effect. The base station supports this function only if this option is selected.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions

The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Super Uplink*.

Requirement/Trouble Ticket Number

FeR-IR20250708001473-0050

RAT

NR TDD

5.3.2.6.3 Added Support for the Frequency Band Combination n78+n82 in SUL and NR FDD Co-deployment Mode

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The frequency band combination n78+n82, as listed in [Table 5-22](#), is now supported in SUL and NR FDD co-deployment mode.

Table 5-22 New combination of NR TDD, NR FDD, and SUL frequency bands supported by super uplink with SUL and NR FDD co-deployment

NR TDD Frequency Band	Base Station Model Applicable to NR TDD Frequency Bands	SUL Frequency Band	NR FDD Frequency Band	Base Station Model Applicable to SUL Frequency Bands
n78	Macro and LampSite base stations	n82	n20	Macro base station

This change is irrelevant to LampSite base stations.

Implementation

For details, see *Super Uplink*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRCeSulFrequency.SsbFreqPos	ADD NRCELLSULF REQUENCY MOD NRCELLSULF REQUENCY LST NRCELLSULF REQUENCY	gNodeB	The frequency band combination n78+n82 is now supported, and the range of SSB frequency-domain positions corresponding to the SUL frequency has been changed from 1901–2002 to 1901–2047.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions

The change in this feature has no impact.

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20240904002794-0010

IR20241101001016

RAT

NR TDD

5.3.2.6.4 Added Support for Fast SUL in SUL and NR FDD Co-deployment Mode

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Fast SUL is introduced to improve the uplink throughput in scenarios where inter-base-station super uplink is enabled. This function is controlled by the **FAST_SUL_SW** option of the **NRDUCellQciBearer.ServiceDiffAlgoSwitch** parameter. With fast SUL, the SUL carrier can be enabled more quickly in inter-gNodeB SUL scenarios.

Implementation

For details, see *Super Uplink*. The specific sections are listed in the change history of the document.

Impact on the Network

Fast SUL enables earlier scheduling for UEs on the SUL carrier, increasing the uplink throughput of inter-gNodeB SUL UEs.

Impact on NEs

The change in this feature is implemented on the gNodeB.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellQciBearerServiceDiffAlgoSwitch	ADD NRDUCELLQ CIBEARER MOD NRDUCELLQ CIBEARER LST NRDUCELLQ CIBEARER	gNode B	The FAST_SUL_SW option has been added to specify whether to enable fast SUL.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Super Uplink*.

Requirement/Trouble Ticket Number

FeR-IR20240904002794-0010

IR20241101001016

RAT

NR TDD

5.3.2.6.5 Replaced the Reserved Parameter for Configuration Optimization for SUL UEs with a Formal Parameter in SUL and NR FDD Co-deployment Mode

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The reserved parameter for configuration optimization for SUL UEs has been replaced with a formal parameter in SUL and NR FDD co-deployment mode. "Configuration optimization for SUL UEs" used to be controlled by **RSVDSWPARAM1_BIT1** of the reserved parameter **NRDUCellRsvdExt01.RsvdSwParam1**. This bit has now been replaced with the **SUL_CONFIG_OPT_SW** option of the **NRDUCellCollabServ.SulAlgoSwitch** parameter.

Implementation

For details, see *Super Uplink*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this function has no impact. It is controlled by the existing license.
- Configuration management

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellRsvdExt01.<i>RsvdSwParam1</i>	MOD NRDUCELLRS VDEXT01 LST NRDUCELLRS VDEXT01	gNodeB	RSVDSWPARAM1_BIT1 of this parameter has been replaced by the SUL_CONFIG_OPT_SW option of the NRDUCellCollabServ.SulAlgoSwitch parameter and therefore has been added to the parameter disuse list. This bit will be deleted in later versions. Although the configuration interface in this version supports synchronization and delivery of the setting of this bit, the system does not use this bit.
Modified	NRDUCellCollabServ.SulAlgoSwitch	MOD NRDUCELLC OLLABSERV LST NRDUCELLC OLLABSERV	gNodeB	The SUL_CONFIG_OPT_SW option has been added to this parameter to control configuration optimization for SUL UEs.

 NOTE

NRDUCellRsvdExt01.*RsvdSwParam1* is a reserved parameter.

For details about reserved parameters, see 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250711002400-0010

RAT

NR TDD

5.3.2.6.6 Replaced the Option of a Reserved Parameter Controlling Uplink Optimal Carrier Selection for SUL and CA with an Option of a Formal Parameter

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The option of a reserved parameter controlling uplink optimal carrier selection for SUL and CA has been replaced with an option of a formal parameter. The **RSVDSWPARAM1_BIT3** option of the reserved parameter **NRCellRsvd.RsvdSwParam1**, which controls uplink optimal carrier selection for SUL and CA, has been added to the parameter disuse list and replaced by the **SUL_CA_BASED_UL_CARR_SEL_SW** option of the **NRCellCaMgmtConfig.CaStrategySwitch** parameter.

Implementation

For details, see *Super Uplink*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

UEs must support both the combination of downlink 3CC aggregation and uplink intra-band contiguous 2CC aggregation and that of downlink 2CC aggregation, uplink intra-band contiguous 2CC aggregation, and SUL.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management

The change in this feature has no impact. It is controlled by the existing feature license.

- Configuration management

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRCellCaMgmtConfig.CaStrategySwitch	MOD NRCELLCAM GMTCONFIG LST NRCELLCAM GMTCONFIG	gNodeB	The SUL_CA_BASED_UL_CARR_SEL_SW option has been added to specify whether to enable uplink optimal carrier selection for SUL and CA.
Disused	NRCellRsvd.RsvdSwParam1	MOD NRCELLRSVD LST NRCELLRSVD	gNodeB	RSVDSWPARAM1_BIT3 of this parameter has been replaced by the SUL_CA_BASED_UL_CARR_SEL_SW option of the NRCellCaMgmtConfig.CaStrategySwitch parameter and therefore has been added to the parameter disuse list. This bit will be deleted in later versions. Although the configuration interface in this version supports synchronization and delivery of the setting of this bit, the system does not use this bit.

 NOTE

NRCeIRsvd.RsvdSwParam1 is a reserved parameter.

For details about reserved parameters, see 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20250213000671

RAT

NR TDD

5.3.2.7 FOFD-051204 Experience-based FWA

5.3.2.7.1 Added the Downlink Experience Satisfaction Guarantee Function

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

FWA operators are launching data plans that guarantee a certain level of downlink experience satisfaction. However, downlink experience satisfaction is at risk of declining as both the committed downlink target rates in the plans and cell loads are rising. The downlink experience satisfaction guarantee function can improve downlink experience satisfaction in cells.

Downlink experience satisfaction guarantee is controlled by the **DL_EXP_SATIS_GUAR_SW** option of the **NRDUCellFwaQos.SchAlgoSwitch** parameter. It is applicable to UEs having non-GBR bearers with target guaranteed rates configured through the **gNBDUMacParamGroup.DIExpBasedSchGuarUserRate** parameter. If the rate of such a UE is close to the target, the base station preferentially increases the UE's priority to maximize downlink experience satisfaction in the serving cell. Downlink experience satisfaction is calculated using this formula: Number of samples with downlink rates greater than or equal to the target downlink rate/Total number of samples. The numbers of these samples can be calculated based on the values of the following counters.

Application Scenario	Counter Set	Counter Name
NSA network	QCI-specific RLC-Layer Downlink UE Throughput Distribution	N.Thp.DL.Samp.QCI.Index0 to N.Thp.DL.Samp.QCI.Index14
SA network	5QI-specific Downlink Throughput Distribution at the RLC Layer	N.Thp.DL.Samp.DuCell5QI.Index0 to N.Thp.DL.Samp.DuCell5QI.Index14

To ensure more accurate calculation of downlink experience satisfaction, you are advised to select the **DL_THPT_L_SLOT_STATS_POL_SW** option of the **NRDUCellOamParam.StatisticsStrategySwitch** parameter. When this option is selected, the duration and traffic of downlink tail data are not excluded from the measurement of the preceding counters. Tail data is the data transmitted in the last slot during which the downlink buffer becomes empty.

Implementation

For details, see *FWA*. The specific sections are listed in the change history of the document.

Impact on the Network

Benefits: After this function is enabled, downlink experience satisfaction increases for UEs configured with target guaranteed rates.

Impacts:

- When downlink experience satisfaction guarantee is enabled, the first packet delay increases, and the average downlink cell throughput, downlink spectral efficiency, and average downlink user-perceived rate may slightly change.
- Downlink experience satisfaction guarantee preferentially assures the priorities of UEs whose rates are close to the target. As a result, service experience will deteriorate for UEs that have already reached the downlink

target rates (for example, cell-center UEs) and cell-edge UEs that struggle to reach the downlink target rates.

- When downlink experience satisfaction guarantee is enabled, the service experience will deteriorate for UEs without the **gNBDUMacParamGroup.DIExpBasedSchGuarUserRate** parameter configured. The deterioration degree depends on factors such as the cell load and traffic model.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellFwaQos.SchAlgoSwitch	MOD NRDUCELLF WAQOS LST NRDUCELLF WAQOS	gNodeB	This parameter has been added to specify whether to enable optimization functions related to the downlink scheduling algorithm. In addition, the DL_EXP_SATIS_GUAR_SW option has been added to specify whether to enable downlink experience satisfaction guarantee.
Modified	NRDUCellOamParameter.StatisticsStrategySwitch	MOD NRDUCELLO AMPARAM LST NRDUCELLO AMPARAM	gNodeB	The DL_THPT_L_SLOT_STATS_POL_SW option has been added to specify the last-slot statistical policy used to measure the downlink throughput of target-rate-guaranteed UEs.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The following mutually exclusive relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNe tworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	Downlink experience satisfaction guarantee cannot work with Hyper Cell.
Low-frequency TDD	Cell Combination	NRDUCell.NrDuCellNe tworkingMode set to HYPER_CELL_COMBIN E_MODE	<i>Cell Combination</i>	Downlink experience satisfaction guarantee cannot work with Cell Combination.

- Impacted features/functions
The following impact relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink experience-based scheduling	DL_EXP_BASED_SC H_SW option of the NRDUCellAlgoSwitc h.FwaAlgoSwitch parameter	<i>FWA</i>	When downlink experience satisfaction guarantee and downlink experience-based scheduling are both enabled, both functions can take effect, but downlink experience satisfaction guarantee may produce fewer benefits.

RA T	Function Name	Function Switch	Referen ce	Description
Lo w- fre qu enc y TD D	FWA private line	PRIVATE_LINE_SW option of the NRDUCellAlgoSwitch h.FwaAlgoSwitch parameter	<i>FWA</i>	When downlink experience satisfaction guarantee and basic service assurance for FWA private line are both enabled, and the gNBDUMacParamGroup.DlExpBasedSchGuarUserRate and gNBDUMacParamGroup.DlGuaranteedRate parameters are both configured for FWA private line users, the following mechanism applies: • When the rate of a private line user is lower than the value of gNBDUMacParamGroup.DlGuaranteedRate, absolute priority scheduling is used for the private line user. Downlink experience satisfaction guarantee does not take effect for the user. • When the rate of a private line user is greater than or equal to the value of gNBDUMacParamGroup.DlGuaranteedRate, downlink experience satisfaction guarantee can take effect for the user.

RA T	Function Name	Function Switch	Referen ce	Description
Lo w- fre qu enc y TD D	Downlink throughput transmissio n duration	DL_THPT_TRANS_TI ME_SW option of the NRDUCellOamPara m.StatisticsStrate- gySwitch parameter	None	Downlink experience satisfaction guarantee cannot take effect when the DL_THPT_TRANS_TI ME_SW option is selected.
Lo w- fre qu enc y TD D	Load-based adaptive downlink scheduling	NRDUCellDISch.DIS chPolicy set to ADAPTIVE	<i>Massive MIMO AHR (Low- Frequenc y TDD)</i>	When downlink experience satisfaction guarantee is in effect, the number of available downlink RBs in a cell decreases. This makes it more likely that the downlink PRB usage falls below the threshold (that determines whether a cell is heavily loaded) for load-based adaptive downlink scheduling to take effect. Consequently, there are fewer opportunities for load-based adaptive downlink scheduling to produce its intended benefits. If the downlink PRB usage reaches the threshold for load- based adaptive downlink scheduling to take effect, load- based adaptive downlink scheduling can take effect, but the benefits of downlink experience satisfaction guarantee may decrease.

RA T	Function Name	Function Switch	Referen ce	Description
Lo w- fre qu enc y TD D	Downlink resource isolation	DL_EMBB_PREFERE NTIAL_SCH_SW option of the NRDUCellAlgoSwitc h.FwaAlgoSwitch parameter	<i>FWA</i>	If downlink experience satisfaction guarantee has taken effect for FWA users, enabling downlink resource isolation may degrade FWA user experience.
Lo w- fre qu enc y TD D	Rate adaptation for device- pipe synergy	RATE_ADAPTION_S W option of the NRDUCellQciBear- er.ServiceDiffAlgoS- witch parameter	<i>FWA</i>	Downlink experience satisfaction guarantee preferentially takes effect when it is enabled together with rate adaptation for device-pipe synergy.
Lo w- fre qu enc y TD D	Flexible experience differentiat ion	FLEX_EXP_DIFF_SW option of the NRDUCellFeaturesS w.QciBasedDiffSer- vExpSw parameter	<i>Service- differenti ated Experien ce Guarant ee</i>	Assume that both downlink experience satisfaction guarantee and flexible experience differentiation are in effect for a UE. If the value of the gNBDUMacParamGr oup.DIExpBasedSch GuarUserRate or gNBDURfspConfig.D IExpBasedSchGuar- UserRate parameter exceeds the downlink target rate determined by the flexible experience differentiation function, the UE's downlink peak rate will be capped at this downlink target rate determined by the flexible experience differentiation function.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *FWA*.

Requirement/Trouble Ticket Number

FeR-IR20241211002563-0001

RAT

NR FDD, NR TDD

5.3.2.8 FOFD-051212 Converged Load Balancing

5.3.2.8.1 Idle-Mode MLB Frequency Optimization

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In the procedure of low-frequency dedicated-priority-based idle mode MLB, if (1) the **NRCellRelation.MlbHoFlag** parameter is set to **FALSE** for all cells on a neighboring frequency, and (2) any cell on the neighboring frequency is lightly loaded, then the neighboring frequency is considered as a low-load frequency and UEs may be reselected to that frequency. In this context, idle-mode MLB frequency optimization can be used to avoid cell reselections to the neighboring frequency through idle mode MLB. This function is controlled by the **IDLE_MLB_FREQ_OPT_SW** option of the **NRCellMlb.MlbAlgoEnhSwitch** parameter. When idle-mode MLB frequency optimization is enabled, the aforesaid neighboring frequency will be considered as a high-load frequency, and UEs will not be reselected to that frequency through idle mode MLB.

This change is supported only in SA networking.

Implementation

For details, see *Mobility Load Balancing in Idle Mode*. The specific sections are listed in the change history of the document.

Impact on the Network

After idle-mode MLB frequency optimization is enabled, the number of target frequencies available for idle mode MLB may decrease, which in turn may cause low-frequency dedicated-priority-based idle mode MLB to yield fewer gains.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRCeLLMLb .MlbAlgoEnhSwitch	MOD NRCELLMLB LST NRCELLMLB	gNodeB	The IDLE_MLB_FREQ_OPT_SW option has been added to specify whether to enable idle-mode MLB frequency optimization.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Mobility Load Balancing in Idle Mode*.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0048

RAT

NR FDD, NR TDD

5.3.2.8.2 Support for Transfer of Downlink CA UEs by Connected Mode MLB

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

UE-number-based connected mode MLB, spectral-efficiency-based connected mode MLB, and inter-vendor UE-number-based connected mode MLB now support the transfer of downlink CA UEs.

- A downlink CA UE (configured with downlink SCC(s) only, but not configured with any uplink SCC) having no activated SCC can be selected for load redistribution when the serving cell meets all of the following conditions:
 - The **NRCellMlb.MlbTransCA** parameter is set to **DL_CA** for the serving cell.
 - The **NRCellMlb.MlbDirection** parameter is set to either **DUAL** and the MLB procedure is triggered only by uplink MLB meeting its entering condition, or **UPLINK**.
- During serving cell determination, if the **NRCellMlb.MlbTransCA** parameter is set to **DL_CA** and the selected UE is a CA UE, the gNodeB also checks whether a candidate cell's CA configuration remains applicable to the CA UE following a handover to the candidate cell. If the configuration is no longer applicable, the candidate cell will not be selected as the target cell of the CA UE's handover.

Implementation

For details, see *Mobility Load Balancing in Connected Mode*. The specific sections are listed in the change history of the document.

Impact on the Network

After downlink CA UEs are transferred to other cells through the uplink MLB procedure of UE-number-based or spectral-efficiency-based connected mode MLB, the number of UEs in the uplink and the uplink air-interface load of the serving cell decrease. Although the gNodeB checks in advance whether a candidate cell's CA configuration remains applicable to the CA UEs, the applicability of the CA configuration also depends on the CA UEs' capabilities and service model. If the CA configuration is no longer applicable to the CA UEs following handovers to the candidate cell, the uplink and downlink user-perceived rates of these UEs will decrease.

Successful MLB-based incoming handovers of CA UEs to a target cell may increase the PRB usage of the target cell. Excessively high PRB usage may indicate that the target cell is heavily loaded.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameter has changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRCeLLMLb .MLbTrans CA	MOD NRCELLMLB LST NRCELLMLB	gNodeB	This parameter has been added to specify whether connected mode MLB supports transfer of downlink CA UEs.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationship has been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	MLB Trigger Mode	NRCellMlb.MlbTriggerMode	<i>Mobility Load Balancing in Connected Mode</i>	The NRCellMlb.MlbTriggerMode parameter can be set to DL_CA only when the NRCellMlb.MlbTriggerMode parameter is set to CELL_USER_NUM or SPEC_EFF_BASED_USER_NUM .

- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The following impact relationships have been modified.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-band CA	INTRABAND_CASW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	During the procedure of UE-number-based or spectral-efficiency-based connected mode MLB, the gNodeB can select CA UEs to transfer only when NRCellMlb.MlbTriggerMode is set to DL_CA , and can only select downlink CA UEs to transfer for the purpose of uplink MLB. After a CA UE is transferred to the target cell, the CA configuration may no longer be applicable, resulting in reduced uplink and downlink user-perceived rates of the UE.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink intra-band CA	INTRA_BAND_UL_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	When selecting UEs to transfer during the procedure of UE-number-based or spectral-efficiency-based connected mode MLB, the gNodeB excludes those with uplink intra-band CA in effect.
Low-frequency TDD	Downlink intra-FR inter-band CA	INTRA_FR_INTE_R_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	During the procedure of UE-number-based or spectral-efficiency-based connected mode MLB, the gNodeB can select CA UEs to transfer only when NRCellMLb.MlbTransCA is set to DL_CA , and can only select downlink CA UEs to transfer for the purpose of uplink MLB. After a CA UE is transferred to the target cell, the CA configuration may no longer be applicable, resulting in reduced uplink and downlink user-perceived rates of the UE.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink intra-FR inter-band CA	INTRA_FR_INTE R_BAND _UL_CA _SW option of the NRDUC ellAlgoS witch.C aAlgoS witch parameter	<i>Carrier Aggregation</i>	When selecting UEs to transfer during the procedure of UE-number-based or spectral-efficiency-based connected mode MLB, the gNodeB excludes those with uplink intra-FR inter-band CA in effect.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Mobility Load Balancing in Connected Mode*.

Requirement/Trouble Ticket Number

FeR-IR20250221001097-0001

RAT

NR FDD, NR TDD

5.3.2.9 FOFD-061224 Fast CA

5.3.2.9.1 Added Uplink Fast Intra-gNodeB CA

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

After an uplink SCell is activated, the first scheduling can be performed only after the gNodeB determines that the channel quality of the SCell meets the transmission requirements. Without uplink fast intra-gNodeB CA, determining the SCell's channel quality is relatively time-consuming, delaying the first scheduling in the SCell. In uplink intra-band CA and uplink intra-FR inter-band CA scenarios, uplink fast intra-gNodeB CA is introduced to optimize the procedure for determining the SCell channel quality and shorten the decision-making time. This

enables faster first scheduling in the SCell, improving CA user experience. This function is controlled by the **UL_CA_FAST_SPLIT_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoSwitch** parameter.

The change in this feature is supported only in low-frequency bands.

Implementation

For details, see *CA Experience Improvement*. The specific sections are listed in the change history of the document.

Impact on the Network

Uplink fast intra-gNodeB CA enables faster first scheduling in SCells so that data can be scheduled earlier. This may increase the average throughput of SCell-activated uplink CA UEs. This average throughput equals $(N.CA.SccActiveUser.ThpVol.UL - N.CA.ThpVol.UL.SmallPkt) / N.CA.ThpTime.UL.RmvSmallPkt$. The gains in throughput improvement decrease as the continuous scheduling duration lasts longer.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.

- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellCarrMgmt.CaEnhancedAlgoSwitch	MOD NRDUCELLCAR RMGMT LST NRDUCELLCAR RMGMT	gNodeB	The UL_CA_FAST_SPLIT_SW option has been added to control uplink fast intra-gNodeB CA.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationship has been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink intra-band CA	INTRA_BAND_UL_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	Uplink intra-band fast intra-gNodeB CA (controlled by the UL_CA_FAST_SPLIT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter) requires that uplink intra-band CA be enabled.
Low-frequency TDD	Uplink intra-FR inter-band CA	INTRA_FR_INTER_BAND_UL_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	Uplink intra-FR inter-band fast intra-gNodeB CA (controlled by the UL_CA_FAST_SPLIT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter) requires that uplink intra-FR inter-band CA be enabled.

- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *CA Experience Improvement*.

Requirement/Trouble Ticket Number

FeR-IR20250211000074-0002

RAT

NR FDD, NR TDD

5.3.2.10 FOFD-070206 Inter-FR CA (Trial)

5.3.2.10.1 Added the "FR2 Primary-SCell-Inclusive Activation and Deactivation" Function

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In the case of FR2 SCell deactivation for inter-FR CA, the primary FR2 SCell stays activated to prevent FR2 UEs from being out of synchronization. This leads to rapid drain of UE power. To address this, the "FR2 primary-SCell-inclusive activation and deactivation" function is introduced to allow the primary FR2 SCell to be deactivated and activated again. This function includes the following sub-functions:

- FR2 SCell activation and deactivation: controlled by the **FR2_SCELL_ACTIV_DEACTV_SW** option of the **NRDUCellCarrMgmt.InterFrCaEnhSw** parameter.
- FR2 SCell deactivation and fast activation: controlled by the **FR2_SCELL_FAST_ACTIV_DEACTV_SW** option of the **NRDUCellCarrMgmt.InterFrCaEnhSw** parameter.

The two sub-functions differ in support for fast activation. With "FR2 SCell deactivation and fast activation", fast activation is supported, whereby the gNodeB directly sends a MAC CE for activating the primary FR2 SCell to a UE and then the UE sends a related measurement report to fast trigger such activation. With "FR2 SCell activation and deactivation", fast activation is not supported. Instead, the gNodeB first sends an RRC reconfiguration message to instruct a UE to measure the primary FR2 SCell; after the UE sends a related measurement report, the gNodeB sends the MAC CE for activating the primary FR2 SCell.

The switches of the preceding two sub-functions are configured on the PCell.

Table 5-23 describes how the "FR2 primary-SCell-inclusive activation and deactivation" function works in different scenarios.

Table 5-23 "FR2 primary-SCell-inclusive activation and deactivation" in different scenarios

FR2_SCELL_ACTIV_DEACTV_SW Option	FR2_SCELL_FAST_ACTIV_DEACTV_SW Option	Implementation of "FR2 Primary-SCell-Inclusive Activation and Deactivation"
Deselected	Deselected	The "FR2 primary-SCell-inclusive activation and deactivation" function does not take effect.
Selected	Deselected	The primary FR2 SCell can be deactivated or activated again based on traffic volume ^a .

FR2_SCELL_ACTIVATION_DEACTIVATION_Switch Option	FR2_SCELL_FAST_ACTIVATION_DEACTIVATION_Switch Option	Implementation of "FR2 Primary-SCell-Inclusive Activation and Deactivation"
Selected	Selected	- For UEs^b that support measurement reporting for unknown SCell activation as defined in 3GPP Release 18, their primary FR2 SCells can be deactivated or fast activated again based on traffic volume. - For other UEs, their primary FR2 SCells can be deactivated or activated based on traffic volume (fast activation is not supported).
Deselected	Selected	- For UEs^b that support measurement reporting for unknown SCell activation as defined in 3GPP Release 18, their primary FR2 SCells can be deactivated or fast activated again based on traffic volume. - For other UEs, the "FR2 primary-SCell-inclusive activation and deactivation" function does not take effect.
a: For details about SCell deactivation and activation based on traffic volume, see sections "Traffic Volume-triggered SCell Deactivation" and "SCell Activation Through a MAC CE" in <i>Carrier Aggregation</i> b: The l3-MeasUnknownSCellActivation-r18 field in the UE-reported capability information indicates whether the UE supports measurement reporting for unknown SCell activation, as detailed in section 4.2.9 in 3GPP TS 38.306 V18.3.0.		

The "FR2 primary-SCell-inclusive activation and deactivation" function is recommended only when power saving is required for UEs. It is not recommended when power saving is not strongly required, such as in FWA or toB scenarios.

A certain time interval must be maintained between the activation and deactivation of a primary FR2 SCell to prevent ping-pong issues.

- After being activated, the primary FR2 SCell can be deactivated based on traffic volume only when the timer specified by **NRDUCellCarrMgmt.Fr2ScellDeactvProhibitTmr** expires.
- After being deactivated, the primary FR2 SCell cell can be activated or fast activated again based on traffic volume only after the interval specified by **NRDUCellCarrMgmt.Fr2ScellActvInterval** elapses.

Both **NRDUCellCarrMgmt.Fr2ScellDeactvProhibitTmr** and **NRDUCellCarrMgmt.Fr2ScellActvInterval** are configured on the PCell.

Implementation

For details, see *Carrier Aggregation*. The specific sections are listed in the change history of the document.

Impact on the Network

The "FR2 primary-SCell-inclusive activation and deactivation" function shortens the time for the primary FR2 SCell to stay activated, reducing UE power consumption.

This function has the following network impacts:

- The uplink and downlink traffic volumes carried by FR1 and FR2 cells change, causing the average uplink and downlink throughputs of UEs and PCells to change.
 - Average uplink UE throughput: **User Uplink Average Throughput (DU)**
 - Average downlink UE throughput: **User Downlink Average Throughput (DU)**
 - Average uplink cell throughput: **Cell Uplink Average Throughput (DU)**
 - Average downlink cell throughput: **Cell Downlink Average Throughput (DU)**
- The uplink and downlink traffic volumes carried by FR1 and FR2 cells change, causing the air interface scheduling indicators of these cells to fluctuate, including the PRB usage, CCE usage, IBLER, RBLER, MCS, and rank.
- The number of RRC reconfiguration messages increases, which may increase the service drop rate and the probability of RRC connection reestablishment.
- The number of MAC CEs delivered in FR1 cells increases. When there are a large number of inter-FR CA UEs, the average uplink and downlink throughputs of UEs in the FR1 cells decrease.
- Uplink random access increases in FR2 cells. When there are a large number of inter-FR CA UEs, uplink access to FR2 cells is delayed or even fails. This causes removal of FR2 SCells, reducing the number of inter-FR CA UEs. Counters related to the number of inter-FR CA UEs:
 - **N.User.InterFRCA.PCell.Avg**
 - **N.User.InterFRCA.PCell.Max**
 - **N.User.InterFRCA.PUCCHSCell.Avg**
 - **N.User.InterFRCA.PUCCHSCell.Max**
- As there are more SCell activation and deactivation scenarios, the values of the following counters increase:
 - **N.CA.SCell.Deact.Att**
 - **N.CA.SCell.Deact.Succ**
 - **N.CA.SCell.Act.Att**
 - **N.CA.SCell.Act.Succ**
- In inter-gNodeB scenarios, gNodeBs need to exchange configuration information related to this function through the eXn interface. As such, a single message becomes longer and, in certain scenarios, may be broken into more fragments. This may increase the values of counters related to the

received and transmitted traffic volumes in the **SCTP Link Measurement** function subset.

Impact on NEs

As for fast activation of the primary FR2 SCell in the change in this feature, UEs must support measurement reporting for unknown SCell activation in accordance with 3GPP Release 18. This capability is indicated by the **I3-MeasUnknownSCellActivation-r18** field in the UE-reported capability information, as detailed in section 4.2.9 in 3GPP TS 38.306 V18.3.0.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellCarrMgmt.InterFrCaEnhSw	MOD NRDUCELL CARRMGMT LST NRDUCELL CARRMGMT	gNodeB	The FR2_SCELL_ACTV_DEACTV_SW and FR2_SCELL_FAST_ACTV_DEACTV_SW options have been added to specify whether to enable the "FR2 primary-SCell-inclusive activation and deactivation" function.
New	NRDUCellCarrMgmt.Fr2ScellDeactvProhibitTmr	MOD NRDUCELL CARRMGMT LST NRDUCELL CARRMGMT	gNodeB	This parameter has been added to specify the length of the timer for prohibiting deactivation of FR2 SCells.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellCarrMgmt.Fr2ScellActivationInterval	MOD NRDUCELL CARRMGMT LST NRDUCELL CARRMGMT	gNodeB	This parameter has been added to specify the interval for activating FR2 SCells.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	FR2 primary-SCell-inclusive activation and deactivation	Either the FR2_SCELL_ACTIV_DEACTIV_SW or FR2_SCELL_FAST_ACTIV_DEACTIV_SW option of the NRDUCellCarrMgmt.InterFrCaEnhSw parameter selected	<i>Carrier Aggregation</i>	The NRDUCellCarrMgmt.Fr2ScellDeactivationProhibitTmr and NRDUCellCarrMgmt.Fr2ScellActivationInterval parameters take effect only when the "FR2 primary-SCell-inclusive activation and deactivation" function is enabled.

- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Carrier Aggregation*.

Requirement/Trouble Ticket Number

FeR-IR20240815001162-0001

RAT

NR TDD

5.3.2.10.2 Support for a Maximum of Six High-Frequency CCs in the Downlink for Inter-FR CA in the Case of 200 MHz per High-Frequency Cell

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Before the upgrade: A maximum of four high-frequency CCs can be aggregated in the downlink for inter-FR CA in the case of 200 MHz per high-frequency cell. The supported bandwidth combinations are as follows: $m \times 200$ MHz in the downlink and $n \times 200$ MHz in the uplink ($1 \leq n \leq m \leq 4$)

After the upgrade: A maximum of six high-frequency CCs can be aggregated in the downlink for inter-FR CA in the case of 200 MHz per high-frequency cell. The supported bandwidth combinations are as follows:

- If the **NRDUCellCarrMgmt.InterFrCa200MFr2CcMaxNum** parameter is set to **DL6CC_UL4CC**: $m \times 200$ MHz in the downlink and $n \times 200$ MHz in the uplink ($1 \leq m \leq 6$; $1 \leq n \leq 4$; $n \leq m$)
- If the **NRDUCellCarrMgmt.InterFrCa200MFr2CcMaxNum** parameter is set to **DL4CC_UL4CC**: $m \times 200$ MHz in the downlink and $n \times 200$ MHz in the uplink ($1 \leq n \leq m \leq 4$).

The **NRDUCellCarrMgmt.InterFrCa200MFr2CcMaxNum** parameter is configured on the PCell.

The change in this feature is supported only in SA networking.

Implementation

For details, see *Carrier Aggregation*. The specific sections are listed in the change history of the document.

Impact on the Network

Inter-FR CA now allows for up to six high-frequency CCs to be aggregated in the downlink in the case of 200 MHz per high-frequency cell. This fully utilizes high-frequency spectrum resources and increases downlink UE throughput.

Impact on NEs

The change in this feature requires that the number of CCs supported by the UE's CA capability, as reported in the **BandCombination** IE, be at least six.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellCarrMgmt.InterFrCa200MFr2CcMaxNum	MOD NRDUCELLCAR RMGMT LST NRDUCELLCAR RMGMT	gNodeB	This parameter has been added to specify the maximum number of high-frequency CCs that can be aggregated in the downlink for inter-FR CA in the case of 200 MHz per high-frequency cell.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.

- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The following impact relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	PUCCH grouping adaptation for inter-FR CA	PUCCH_GROUP_ADAPT_SW option of the NRDUCellCarrierMgmt.InterFrCa200MaEnhSw parameter	<i>Carrier Aggregation</i>	When PUCCH grouping adaptation for inter-FR CA is enabled, no more than four high-frequency CCs can be aggregated even if the NRDUCellCarrierMgmt.InterFrCa200MFr2CcMaxNum parameter is set to DL6CC_UL4CC . When two sets of PUCCH-based feedback mechanisms are changed to one set for an inter-FR CA UE with six high-frequency CCs, the maximum high-frequency bandwidth that can be reached in the downlink decreases from (6CC x 200 MHz) to (4CC x 200 MHz). This decreases the downlink data rate of the UE.

Related Operations

By default, the change does not take effect after an upgrade.
For details, see the corresponding "Operation and Maintenance" section in *Carrier Aggregation*.

Requirement/Trouble Ticket Number

FeR-IR20240626001884-0001

RAT

NR TDD

5.3.2.11 FOFD-070316 RedCap Introduction

5.3.2.11.1 Added RedCap Power Aggregation

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

A RedCap BWP1, which has a small bandwidth (20 MHz), is more likely to be affected by CCE allocation failures caused by insufficient PDCCH CCE resources

than the cell BWP1, which has a large bandwidth (up to 100 MHz for non-RedCap).

This feature now supports RedCap power aggregation in NR TDD cells. After this function takes effect, the gNodeB checks the uplink CCE usage in each TTI. If the uplink CCE usage is less than the CCE threshold for RedCap power aggregation, the gNodeB allows the remaining power on the BWP1 of the cell to be used by the RedCap BWP1 and reduces the PDCCH CCE aggregation level for RedCap UEs, thereby increasing the uplink CCE allocation success rate of RedCap UEs.

This function is controlled by the **REDCAP_POWER_AGGREGATION_SW** option of the **NRDUCellPdcch.PdcchAlgoExtSwitch** parameter. The CCE threshold for RedCap power aggregation is specified by the **NRDUCellPdcchAlgo.RedCapPwrBoostingCceThld** parameter.

The change in this feature is supported only in low-frequency SA networking.

Implementation

For details, see *RedCap*. The specific sections are listed in the change history of the document.

Impact on the Network

The uplink CCE allocation success rate of RedCap UEs increases, uplink CCE aggregation level and PDCCH CCE usage may decrease, and uplink RB usage fluctuates as more PDCCH CCE resources are allocated to RedCap UEs.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellPdcchAlgo.RedCapPowerBoostingCceThld	MOD NRDUCCELLP DCCHALGO LST NRDUCCELLP DCCHALGO	gNodeB	This parameter has been added to specify the CCE threshold for power aggregation to take effect for RedCap UEs.
Modified	NRDUCellPdcch.PdcchAlgoExtSwitch	MOD NRDUCCELLP DCCH LST NRDUCCELLP DCCH	gNodeB	The REDCAP_POWER_AGGREGATION_SW option has been added to specify whether to enable RedCap power aggregation.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The following impact relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDCCH power control for CEUs	NRDUCellPdcchAlgo.PdcchEdgePwrCtrlPolicy set to BASED_ON_AGG_LVL or BASED_ON_CSI_RPT	<i>Power Control</i>	When PDCCH power control for CEUs and RedCap power aggregation are both enabled, the power consumption of individual UEs increases, leading to power insufficiency in the cell. This, in turn, decreases the CCE allocation success rate and may reduce overall benefits.
Low-frequency TDD	PDCCH aggregation level reduction	PDCCH_AGG_LVL_COMPRESS option of the NRDUCellPdcchAlgoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	When PDCCH aggregation level reduction and RedCap power aggregation are both enabled, the overall benefits provided by these two functions decrease.
Low-frequency TDD	PDCCH resource allocation enhancement	PDCCH_RESOURCE_ALLOCATION_ENHANCE option of the NRDUCellPdcchAlgoSwitch parameter	<i>Channel Management</i>	When PDCCH resource allocation enhancement and RedCap power aggregation are both enabled, RedCap power aggregation does not take effect.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *RedCap*.

Requirement/Trouble Ticket Number

FeR-IR20240626002501-0004

RAT

NR TDD

5.3.2.11.2 Added RedCap BWP Frequency-domain Staggering

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

If the RedCap BWP1s with the NCD-SSBs of the local cell and its neighboring cells are completely aligned in the frequency domain, the PUSCHs of these cells interfere with each other, affecting the average uplink cell throughput.

To address this issue, RedCap BWP frequency-domain staggering is introduced to stagger the frequency-domain positions of RedCap BWP1s with the NCD-SSBs among cells, reducing inter-cell PUSCH interference.

This function is controlled by the **REDCAP_BWP_NON_OVERLAP_SW** option of the **NRDUCellRedCap.RedCapAlgoExtSwitch** parameter.

The change in this feature is supported only in low-frequency TDD in SA networking.

Implementation

For details, see *RedCap*. The specific sections are listed in the change history of the document.

Impact on the Network

With this function, the frequency-domain positions of RedCap BWP1s with the NCD-SSBs among different cells are staggered, reducing inter-cell PUSCH interference and increasing the average uplink throughput of RedCap UEs. In addition, the values of some indicators may decrease or fluctuate.

- The values of the following indicators may decrease:
 - Average throughput of eMBB UEs
 - Average uplink MCS index
 - Average number of available PRBs on the PUSCH and average number of actually available PRBs on the PUSCH
- The values of the following indicators may fluctuate:
 - Uplink and downlink IBLERs
 - Uplink and downlink RB usages
 - Uplink CCE allocation success rate

- Average number of layers paired for uplink MIMO, average number of RBs paired for uplink MIMO, and total number of RBs paired for uplink MU-MIMO
- Average uplink interference (average interference and noise received on each uplink PRB)
- The number of available uplink PRBs within the BWP may increase.

Definitions of specific indicators are as follows.

Table 5-24 Indicators involved after RedCap Introduction is enabled

Indicator	Formula
Average uplink throughput of RedCap UEs	$\frac{(N.ThpVol.UL.RedCap - N.ThpVol.UE.UL.SmallPkt.RedCap)}{N.ThpTime.UE.UL.RmvSmallPkt.RedCap}$
Average uplink MCS index	$\frac{(0 \times N.ChMeas.PUSCH.MCS.0 + 1 \times N.ChMeas.PUSCH.MCS.1 + 2 \times N.ChMeas.PUSCH.MCS.2 + 3 \times N.ChMeas.PUSCH.MCS.3 + \dots + 27 \times N.ChMeas.PUSCH.MCS.27 + 28 \times N.ChMeas.PUSCH.MCS.28)}{(N.ChMeas.PUSCH.MCS.0 + N.ChMeas.PUSCH.MCS.1 + N.ChMeas.PUSCH.MCS.2 + N.ChMeas.PUSCH.MCS.3 + \dots + N.ChMeas.PUSCH.MCS.27 + N.ChMeas.PUSCH.MCS.28)}$
Average number of available uplink PUSCH PRBs	$N.PR.B.UL.PUSCH.Avail.Avg$
Average number of actually available PRBs for the PUSCH	$N.PR.B.UL.PUSCH.Actual.Avail.Avg$
Uplink IBLER	$\frac{(N.UL.SCH.HalfPiBPSK.ErrTB.Ibler + N.UL.SCH.QPSK.ErrTB.Ibler + N.UL.SCH.16QAM.ErrTB.Ibler + N.UL.SCH.64QAM.ErrTB.Ibler + N.UL.SCH.256QAM.ErrTB.Ibler)}{(N.UL.SCH.HalfPiBPSK.TB + N.UL.SCH.QPSK.TB + N.UL.SCH.16QAM.TB + N.UL.SCH.64QAM.TB + N.UL.SCH.256QAM.TB)} \times 100\%$

Indicator	Formula
Downlink IBLER	$\frac{(N.DL.SCH.QPSK.ErrTB.Ibler + N.DL.SCH.16QAM.ErrTB.Ibler + N.DL.SCH.64QAM.ErrTB.Ibler + N.DL.SCH.256QAM.ErrTB.Ibler)}{(N.DL.SCH.QPSK.TB + N.DL.SCH.16QAM.TB + N.DL.SCH.64QAM.TB + N.DL.SCH.256QAM.TB)} \times 100\%$
Uplink RB usage	Uplink Resource Block Utilizing Rate (DU)
Downlink RB usage	Downlink Resource Block Utilizing Rate (DU)
Number of available uplink PRBs within the BWP	N.PR.B.UL.RedCap.BWP.Avail.Avg
Uplink CCE allocation success rate	$\frac{(N.CCE.UL.AggLvl1Num + N.CCE.UL.AggLvl2Num + N.CCE.UL.AggLvl4Num + N.CCE.UL.AggLvl8Num + N.CCE.UL.AggLvl16Num)}{N.CCE.UL.AllocReq.Num} \times 100\%$
Average number of layers paired for uplink MIMO	$N.ChMeas.MIMO.UL.Pair.Layer / N.ChMeas.MIMO.UL.Pair.PRB$

Indicator	Formula
Average number of RBs paired for uplink MIMO	$\frac{(1 \times \text{N.ChMeas.MIMO.UL.MuPairing.1Layer.RB} + 2 \times \text{N.ChMeas.MIMO.UL.MuPairing.2Layers.RB} + 3 \times \text{N.ChMeas.MIMO.UL.MuPairing.3Layers.RB} + 4 \times \text{N.ChMeas.MIMO.UL.MuPairing.4Layers.RB} + 5 \times \text{N.ChMeas.MIMO.UL.MuPairing.5Layers.RB} + 6 \times \text{N.ChMeas.MIMO.UL.MuPairing.6Layers.RB} + 7 \times \text{N.ChMeas.MIMO.UL.MuPairing.7Layers.RB} + 8 \times \text{N.ChMeas.MIMO.UL.MuPairing.8Layers.RB})}{(\text{N.ChMeas.MIMO.UL.MuPairing.1Layer.RB} + \text{N.ChMeas.MIMO.UL.MuPairing.2Layers.RB} + \text{N.ChMeas.MIMO.UL.MuPairing.3Layers.RB} + \text{N.ChMeas.MIMO.UL.MuPairing.4Layers.RB} + \text{N.ChMeas.MIMO.UL.MuPairing.5Layers.RB} + \text{N.ChMeas.MIMO.UL.MuPairing.6Layers.RB} + \text{N.ChMeas.MIMO.UL.MuPairing.7Layers.RB} + \text{N.ChMeas.MIMO.UL.MuPairing.8Layers.RB})}$
Total number of RBs paired for uplink MU-MIMO	$\text{N.ChMeas.MIMO.UL.MuPairing.1Layer.RB} + \text{N.ChMeas.MIMO.UL.MuPairing.2Layers.RB} + \text{N.ChMeas.MIMO.UL.MuPairing.3Layers.RB} + \text{N.ChMeas.MIMO.UL.MuPairing.4Layers.RB} + \text{N.ChMeas.MIMO.UL.MuPairing.5Layers.RB} + \text{N.ChMeas.MIMO.UL.MuPairing.6Layers.RB} + \text{N.ChMeas.MIMO.UL.MuPairing.7Layers.RB} + \text{N.ChMeas.MIMO.UL.MuPairing.8Layers.RB}$

Indicator	Formula
Average interference and noise received on each uplink PRB	N.UL.NI.Avg

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellRedCap.RedCapAlgoExtSwitch	MOD NRDUCELLR EDCAP LST NRDUCELLR EDCAP	gNodeB	The REDCAP_BWP_NON_OVERLAP_SW option has been added to specify whether to enable RedCap BWP frequency-domain staggering.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RedCap Introduction	REDCAP_SW option of the NRDUCellFeatureSw.UeCapbBasedFunSw parameter	<i>RedCap</i>	RedCap BWP frequency-domain staggering can be enabled only when RedCap Introduction is enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Multiple BWP1s for RedCap	NRDUCellRedCap.RedCapUeMaxAvailBandwidth set to a value greater than 20M	<i>RedCap</i>	RedCap BWP frequency-domain staggering can be enabled when any of the following conditions is met: • The NRDUCell.DlBandwidth parameter is set to CELL_BW_100M, and the NRDUCellRedCap.RedCapUeMaxAvailBandwidth parameter is set to 40M, 60M, 80M, or 100M. • The NRDUCell.DlBandwidth parameter is set to CELL_BW_90M, and the NRDUCellRedCap.RedCapUeMaxAvailBandwidth parameter is set to 40M, 60M, or 80M. • The NRDUCell.DlBandwidth parameter is set to CELL_BW_80M, and the NRDUCellRedCap.RedCapUeMaxAvailBa

RAT	Function Name	Function Switch	Reference	Description
				<i>ndwidth</i> parameter is set to 40M, 60M, or 80M. • The NRDUCell.Dl Bandwidth parameter is set to CELL_BW_70 M, and the NRDUCellRed Cap.RedCapUeMaxAvailBandwidth parameter is set to 40M or 60M. • The NRDUCell.Dl Bandwidth parameter is set to CELL_BW_60 M, and the NRDUCellRed Cap.RedCapUeMaxAvailBandwidth parameter is set to 40M or 60M.

- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The following impact relationship has been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	BWP frequency-domain adaptation for RedCap UEs	BWP_FREQ_A DAPT_SW option of the NRDUCellRedCap.RedCapAlgoSwitch parameter	<i>RedCap</i>	When RedCap BWP frequency-domain staggering and "BWP frequency-domain adaptation for RedCap UEs" are both enabled, the activation sequence of RedCap BWP1s is changed because of RedCap BWP frequency-domain staggering. As a result, RedCap BWP frequency-domain adaptation takes effect based on the new activation sequence of RedCap BWP1s.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE-number-based BWP1 adaptation	USER_NUM_B ASED_BWP_A CTV_SW option of the NRDUCellRed Cap.RedCapAl goSwitch parameter	<i>RedCap</i>	When both RedCap BWP frequency-domain staggering and UE-number-based BWP1 adaptation are enabled, the activation sequence of RedCap BWP1s is changed because of RedCap BWP frequency-domain staggering. As a result, UE-number-based BWP1 adaptation takes effect based on the new activation sequence of RedCap BWP1s.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PRB-and-CCE-resource-based BWP1 adaptation	RESOURCE_B ASED_BWP_A CTV_SW option of the NRDUCellRedCap.RedCapAlgoSwitch parameter	<i>RedCap</i>	When RedCap BWP frequency-domain staggering and PRB-and-CCE-resource-based BWP1 adaptation are both enabled, the activation sequence of RedCap BWP1s is changed because of RedCap BWP frequency-domain staggering. As a result, PRB-and-CCE-resource-based BWP1 adaptation takes effect based on the new activation sequence of RedCap BWP1s.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RedCap NCD-SSB measurement	REDCAP_NCD_SSB_MEAS_SW option of the NRCellRedCap.RedCapAlgoSwitch parameter	<i>RedCap</i>	When both RedCap BWP frequency-domain staggering and RedCap NCD-SSB measurement are enabled, the probability that intra-frequency NCD-SSBs are measured decreases. If intra-frequency NCD-SSBs cannot be measured, gap-assisted inter-frequency CD-SSB measurement is initiated, which decreases the average uplink and downlink throughputs of RedCap UEs. You can reduce the value of NRCellInterFHoMeaGrp.RedCapNcdSsbA2RsptHld to reduce the impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *RedCap*.

Requirement/Trouble Ticket Number

FeR-IR20240626002501-0002

RAT

NR TDD

5.3.2.11.3 Added Format-1 RB Adaptation

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Format-1 RB adaptation is now supported in this feature. With this function, the gNodeB adaptively adjusts the format-1 RB resources based on the format-1 resource usage on the RedCap BWP1.

This function is controlled by the **PUCCH_F1_RB_ADAPT_SW** option of the **NRDUCellRedCap.RedCapAlgoExtSwitch** parameter. It is recommended that this function be used when the **NRDUCellRedCap.RedCapUeMaxAvailBandwidth** parameter is set to a value greater than **20M**.

The change in this feature is supported only in low-frequency SA networking.

Implementation

For details, see *RedCap*. The specific sections are listed in the change history of the document.

Impact on the Network

When a large number of UEs access a cell, the PUCCH resources increase, the number of admitted RedCap UEs may increase, the available PUSCH resources decrease, and the average uplink UE throughput may decrease.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellRedCap.RedCapAlgoExtSwitch	MOD NRDUCELLREDCAP LST NRDUCELLREDCAP	gNodeB	The PUCCH_F1_RB_ADAPT_SW option has been added to specify whether to enable format-1 RB adaptation.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RedCap Introduction	REDCAP_SW option of the NRDUCellFeatureSw.UeCapbBasedFunSw parameter	<i>RedCap</i>	Format-1 RB adaptation can be enabled only when both RedCap Introduction and PUCCH RB adaptation are enabled.
Low-frequency TDD	PUCCH RB adaptation	PUCCH_RBRES_ADAPTIVE_SWITCH option of the NRDUCellPucch.PucchAlgoSwitch parameter	<i>Channel Management</i>	

- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *RedCap*.

Requirement/Trouble Ticket Number

FeR-IR20240626002501-0005

RAT

NR TDD

5.3.2.11.4 Added CSI RB Adaptation.

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

CSI RB adaptation is now supported in this feature. With this function, the gNodeB adaptively adjusts the CSI RB resources based on the CSI resource usage on the RedCap BWP1.

This function is controlled by the **PUCCH_CSI_RB_ADAPT_SW** option of the **NRDUCellRedCap.RedCapAlgoExtSwitch** parameter. It is recommended that this function be used when the **NRDUCellRedCap.RedCapUeMaxAvailBandwidth** parameter is set to a value greater than **20M**.

The change in this feature is supported only in low-frequency SA networking.

Implementation

For details, see *RedCap*. The specific sections are listed in the change history of the document.

Impact on the Network

When a large number of UEs access a cell, more PUCCH resources can be used for CSI reporting, the available PUSCH resources become fewer, the uplink UE throughput decreases, and the downlink UE throughput increases.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management

The change in this feature has no impact. It is controlled by the existing feature license.

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellRedCap.RedCapAlgoExtSwitch	MOD NRDUCELLREDCAP LST NRDUCELLREDCAP	gNodeB	The PUCCH_CSI_RB_ADAPT_SW option has been added to specify whether to enable CSI RB adaptation.

- Performance management

The change in this feature has no impact.

- Fault management

The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RedCap Introduction	REDCAP_SW option of the NRDUCellFeatureSw.UeCapbBasedFunSw parameter	<i>RedCap</i>	CSI RB adaptation can be enabled only when both RedCap Introduction and PUCCH RB adaptation are enabled.
Low-frequency TDD	PUCCH RB adaptation	PUCCH_RBRES_ADAPTIVE_SWITCH option of the NRDUCellPucch.PucchAlgoSwitch parameter	<i>Channel Management</i>	

- Mutually exclusive features/functions

The change in this feature has no impact.

- Impacted features/functions

The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *RedCap*.

Requirement/Trouble Ticket Number

FeR-IR20240626002501-0005

RAT

NR TDD

5.3.2.11.5 Added Format-1 SR Resource Optimization

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Format-1 SR resource optimization is introduced in NR TDD to ensure that more UEs can access a cell. With this function, 18 SR code channels are configured for each format-1 RB, and more format-1 SR resources can be used in the cell

This function is controlled by the **FORMAT1_SR_RES_OPT_SW** option of the **NRDUCellPucch.PucchAlgoSwitch** parameter.

The change in this feature is supported only in low-frequency SA networking.

Implementation

For details, see *RedCap*. The specific sections are listed in the change history of the document.

Impact on the Network

The number of UEs that a cell can serve increases. When the number of UEs in a cell increases, the probability of HARQ resource conflicts increases. As a result, the number of downlink scheduling times decreases, and the average downlink UE throughput and average downlink cell throughput may decrease.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management

The change in this feature has no impact. It is controlled by the existing feature license.

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellPucch.PucchAlgoSwitch	MOD NRDUCCELLPUCCH LST NRDUCCELLPUCCH	gNodeB	The FORMAT1_SR_RES_OPT_SW option has been added to specify whether to enable format-1 SR resource optimization.

- Performance management

The change in this feature has no impact.

- Fault management

The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RedCap Introduction	REDCAP_SW option of the NRDUCellFeatureSw.UeCapbBasedFunSw parameter	<i>RedCap</i>	Format-1 SR resource optimization can be enabled only when RedCap Introduction is enabled.

- Mutually exclusive features/functions

The change in this feature has no impact.

- Impacted features/functions

The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *RedCap*.

Requirement/Trouble Ticket Number

IR20241209002193

DTS2025032905800

RAT

NR TDD

5.3.2.11.6 Added PDCCH Symbol Adaptation for RedCap BWP with the CD-SSB

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The "PDCCH symbol adaptation for RedCap BWP with the CD-SSB" function allows RedCap BWP1 with the CD-SSB to be considered in PDCCH symbol number adaptation. If RedCap BWP1 with the CD-SSB meets the adaptation conditions, the number of PDCCH symbols can be scaled up or down. For details about the conditions for triggering PDCCH symbol number adaptation, see "adaptive configuration" in "PDCCH Symbol Number Configuration" described in *Channel Management*.

This function is controlled by the **CD_BWP_PDCCH_SYM_ADAPTIVE_SW** option of the **NRDUCellRedCap.RedCapAlgoExtSwitch** parameter.

The change in this feature is supported only in low-frequency TDD in SA networking.

Implementation

For details, see *RedCap*. The specific sections are listed in the change history of the document.

Impact on the Network

The number of PDCCH symbols in the RedCap BWP with the CD-SSB increases, and more UEs can be scheduled. The number of PDSCH symbols decreases due to an increase in the number of PDCCH symbols. As a result, the average cell throughput may fluctuate.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellRedCap.RedCapAlgoExtSwitch	MOD NRDUCELLREDCAP LST NRDUCELLREDCAP	gNodeB	The CD_BWP_PDCCH_SYM_ADAPTIVE_SW option has been added to specify whether to enable PDCCH symbol adaptation for RedCap BWP with the CD-SSB.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RedCap Introduction	REDCAP_SW option of the NRDUCellFeatureSw.UseCapBasedFunctionSw parameter	<i>RedCap</i>	PDCCH symbol adaptation for RedCap BWP with the CD-SSB requires that RedCap Introduction be enabled.

- Mutually exclusive features/functions
The change in this feature has no impact.

- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *RedCap*.

Requirement/Trouble Ticket Number

FeR-IR20241212002296-0002

RAT

NR TDD

5.3.2.11.7 Added "CSI RB and Format-1 RB Reposition"

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The "CSI RB and format-1 RB reposition" function adjusts the positions of the extended format-1 RBs to within the first activated RedCap BWP1 to increase the number of RedCap UEs supported on the first activated RedCap BWP1.

This function is controlled by the **PUCCH_RB_REPOSITION_SW** option of the **NRDUCellRedCap.RedCapAlgoExtSwitch** parameter.

The change in this feature is supported only in low-frequency TDD in SA networking.

Implementation

For details, see *RedCap*. The specific sections are listed in the change history of the document.

Impact on the Network

More RedCap UEs can access the first activated RedCap BWP1. When a large number of UEs access a cell, more PUCCH resources are used for CSI reporting, the available PUSCH resources become fewer, the uplink UE throughput may decrease, and the downlink UE throughput may increase.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellRedCap.RedCapAlgoExtSwitch	MOD NRDUCELLREDCAP LST NRDUCELLREDCAP	gNodeB	The PUCCH_RB_REPOSITION_SW option has been added to specify whether to enable the "CSI RB and format-1 RB reposition" function.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RedCap Introduction	REDCAP_SW option of the NRDUCellFeatureSw.UseCapBasedFunSw parameter	<i>RedCap</i>	CSI RB and format-1 RB reposition requires RedCap Introduction to be enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Multiple BWP1s for RedCap	NRDUCellRedCap.RedCapUeMaxAvailBandwidth	<i>RedCap</i>	CSI RB and format-1 RB reposition requires the NRDUCellRedCap.RedCapUeMaxAvailBandwidth parameter to be set to 20M .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CSI measurement optimization	CSI_MEAS_OPT_SW option of the NRDUCellCsiRs.CsiRsExtSwitch parameter	<i>Channel Management</i>	If "CSI RB and format-1 RB reposition " is enabled in any of the following scenarios, aperiodic CSI-RS resources may be insufficient for UEs in non-overlapping areas, and therefore CSI-RS measurement may fail. Therefore, in any of the following scenarios, "CSI RB and format-1 RB reposition" depends on CSI-RS measurement optimization. Adaptive adjustment of periodic CSI-RS resources ensures that UEs in non-overlapping areas can perform CSI-RS measurement. • The DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter is selected. • The NRDUCellAlgoSwitch.VmimoSwitch parameter is set to ON. • The RF_COMBINING_NOISE_SUPPR_SW option of the NRDUCellULSch.RfCombSwitch parameter is selected.

RAT	Function Name	Function Switch	Reference	Description
				The DISTRIBUTED_ANT_PMI_WT_OPTION_SW option of the NRDUCellAlgoSwitch.BeamOptSwitch parameter is selected.

- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *RedCap*.

Requirement/Trouble Ticket Number

FeR-IR20250717005585-0001

RAT

NR TDD

5.3.2.11.8 Added Support for "Scheduling in Common PDCCH Symbols for RedCap UEs" for BWP2 UEs

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The change in this feature enables "scheduling in common PDCCH symbols for RedCap UEs" for BWP2 UEs. In addition, **RSVDSWPARAM1_BIT23** of the reserved parameter **NRDUCellRsvdExt03.RsvdSwParam1** that controls this function has been added to the disuse list and replaced by the **BWP2_COM_PDCCH_SCH_SW** option of the **NRDUCellUePwrSaving.BwpPwrSavingSw** parameter.

The change in this feature is supported only in low-frequency TDD in SA networking.

Implementation

For details, see *RedCap*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellUePwrSaving.BwpPwrSavingSw	MOD NRDUCELLU EPWRSAVING LST NRDUCELLU EPWRSAVING	gNodeB	The BWP2_COM_PDCCH_SCH_SW option has been added to specify whether "scheduling in common PDCCH symbols for RedCap UEs" takes effect on BWP2 UEs.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellRsvdExt03.RsvdSwParam1	MOD NRDUCELLR SVDEXT03 LST NRDUCELLR SVDEXT03	gNodeB	RSVDSWPARAM1_BIT23 of this reserved parameter has been replaced by the BWP2_COM_PDCCH_SCH_SW option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter and therefore has been added to the disuse list. Although the configuration interface in this version supports synchronization and delivery of the setting of this bit, the system does not use this bit.

NOTE

NRDUCellRsvdExt03.RsvdSwParam1 is a reserved parameter.

For details about reserved parameters, see 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250528001856-0001

RAT

NR TDD

5.3.2.11.9 Added RedCap UE Weight Optimization

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In TDD, commonly, UEs can send SRS signals using antenna switching so that the base station can obtain the full downlink channel state information based on uplink and downlink channel reciprocity. However, RedCap UEs can send SRS signals through only fixed antennas. Therefore, the base station cannot obtain full downlink channel state information, affecting downlink experience of RedCap UEs. To address this issue, RedCap UE weight optimization is introduced.

After this function is enabled, the base station allocates 1T2R SRS resources to RedCap UEs that are identified as downlink large-packet UEs. In this case, the base station can receive SRS signals sent by these UEs using antenna switching and obtain full downlink channel state information. This improves the accuracy of channel estimation and downlink user experience of RedCap UEs. Specifically,

- A RedCap UE supports antenna switching if the value of the reported *supportedSRS-TxPortSwitch* IE is **t1r2**.
- The UE is identified as a downlink large-packet UE if it meets the following condition at least five times within 10s: The downlink traffic volume of the UE in 1s is greater than or equal to the threshold specified by the **gNBRedCapServIdenCfg.DlBigPktDataVolThld** parameter in the **gNBRedCapServIdenCfg** MO associated with the **NRDUCellRedCap.DlExpSrsWeightParamGrpld** parameter.

RedCap UE weight optimization is controlled by the **REDCAP_SRS_WEIGHT_SW** option of the **NRDUCellQciBearer.ServiceDiffAlgoSwitch** parameter. This function is not applicable to FWA users. For details about FWA user identification, see *FWA*.

In addition, to increase the number of RedCap UEs for which RedCap UE weight optimization takes effect, the function of increasing the number of RedCap UEs using SRS-based weights is introduced. This function is controlled by the **SRS_WEIGHT_UE_NUM_INCR_SW** option of the **NRDUCellRedCap.RedCapAlgoExtSwitch** parameter.

Implementation

For details, see *RedCap*. The specific sections are listed in the change history of the document.

Impact on the Network

The RedCap UE weight optimization function has the following impacts on the network:

- The average uplink and downlink throughputs of eMBB UEs may decrease.
- The average downlink throughput of RedCap UEs may increase, and the average uplink throughput of RedCap UEs may fluctuate.
- The downlink RB usage and value of service delay-related indicators may decrease. The average uplink MCS index, average downlink MCS index, average downlink rank, uplink IBLER, uplink RBLER, downlink IBLER, downlink RBLER, and values of downlink MU-related indicators may fluctuate.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	gNBRedCapServIdenCfg.DlBigPktDataVolThld	ADD GNBREDCAP SERVIDENCEF G LST GNBREDCAP SERVIDENCEF G MOD GNBREDCAP SERVIDENCEF G RMV GNBREDCAP SERVIDENCEF G	gNodeB	This parameter has been added to specify the data volume threshold for downlink large-packet services.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellRedCap.DlExpSrsWeightParamGrpId	MOD NRDUCELLREDCAP LST NRDUCELLREDCAP	gNodeB	This parameter has been added to specify the ID of a parameter group for RedCap downlink-experience-based SRS-based weights.
Modified	NRDUCellQciBearer.ServiceDiffAlgoSwitch	ADD NRDUCELLQCIBEARER MOD NRDUCELLQCIBEARER LST NRDUCELLQCIBEARER	gNodeB	The REDCAP_SRS_WEIGHT_SW option has been added to specify whether to enable RedCap UE weight optimization.
Modified	NRDUCellRedCap.RedCapAlgoExtSwitch	MOD NRDUCELLREDCAP LST NRDUCELLREDCAP	gNodeB	The SRS_WEIGHT_UE_NUM_INCR_SW option has been added to specify whether to enable the function of increasing the number of RedCap UEs using SRS-based weights.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RedCap Introduction	REDCAP_SW option of the NRDUCellFeatureSw.UeCapBasedFunctionSw parameter	<i>RedCap</i>	RedCap UE weight optimization requires that RedCap Introduction be enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS interference coordination based on self-contained slots	SRS_INTRF_COORD_S_S LOT_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter	<i>Channel Management</i>	RedCap UE weight optimization requires that SRS interference coordination based on self-contained slots be enabled.
Low-frequency TDD	Scenario-specific SRS resource allocation policy	NRDUCellSrs.ScenarioSrsResAllocPol	None	RedCap UE weight optimization requires that the NRDUCellSrs.ScenarioSrsResAllocPol parameter be set to CAPC_PRI .

- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *RedCap*.

Requirement/Trouble Ticket Number

FeR-IR20241212002296-0001

RAT

NR TDD

5.3.2.11.10 Added "BWP Scheduling Optimization for RedCap UEs" (TDD)

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

"BWP scheduling optimization for RedCap UEs" is now supported. After this function takes effect, RedCap UEs in the RRC_CONNECTED state are preferentially scheduled on light-load RedCap BWP1s to achieve load balancing.

This function is controlled by the **REDCAP_UE_BWP_SCH_OPT_SW** option of the **NRDUCellRedCap.RedCapSchAlgoSwitch** parameter.

The change in this feature is supported only in low-frequency SA networking.

Implementation

For details, see *RedCap*. The specific sections are listed in the change history of the document.

Impact on the Network

The average uplink and downlink throughputs of RedCap UEs may increase, the number of RRC connection reconfigurations increases, and the service drop rate in the cell may increase.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellRedCap.RedCapSchAlgoSwitch	MOD NRDUCELLR EDCAP LST NRDUCELLR EDCAP	gNodeB	The REDCAP_UE_BWP_SCH_OPT_SW option has been added to specify whether to enable BWP scheduling optimization for RedCap UEs.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RedCap Introduction	REDCAP_SW option of the NRDUCellFeatureSw.UeCapBasedFunSw parameter	<i>RedCap</i>	BWP scheduling optimization for RedCap UEs requires that RedCap Introduction be enabled.

- Mutually exclusive features/functions

The change in this feature has no impact.

- Impacted features/functions

The following impact relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	QCI-specific UE-number-based periodic BWP load balancing	QCI_UE_NUM_BASED_BLB_SW option of the NRDUCellRedCap.RedCapAlgoExtSwitch parameter and QCI_UE_NUM_BASED_P RDC_BLB_SW option of the gNBDRedCapBlbGrp.RedCapBlbAlgorithmSwitch parameter	<i>RedCap</i>	When both BWP scheduling optimization for RedCap UEs and QCI-specific UE-number-based periodic BWP load balancing are enabled, BWP scheduling optimization for RedCap UEs does not take effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE-number-based periodic BWP load balancing	UE_NUM_BASED_PRDC_BLB_SW option of the NRDUCellRedCap.RedCapAlgoExtSwitch parameter	<i>RedCap</i>	When both BWP scheduling optimization for RedCap UEs and UE-number-based periodic BWP load balancing are enabled, BWP scheduling optimization for RedCap UEs does not take effect.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *RedCap*.

Requirement/Trouble Ticket Number

IR20250528001857

DTS2025120509381

RAT

NR TDD

5.3.2.11.11 Added Support for PUCCH CSI RB Adjustment for the RedCap-dedicated BWP in NR TDD

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The change in this feature supports PUCCH CSI RB adjustment for the RedCap-dedicated BWP in NR TDD. With this function, the gNodeB optimizes the PUCCH resource allocation method. When there are a large number of RedCap UEs, the PUCCH CSI RB resources are adaptively adjusted on the RedCap-dedicated BWP (BWP1 that is not deployed at the band edges and does not contain the CD-SSB) based on the CSI resource usage.

This function is controlled by the **REDCAP_PUCCH_CSI_RB_ADJ_SW** option of the **NRDUCellRedCap.RedCapAlgoExtSwitch** parameter. This function is recommended when the **NRDUCellRedCap.RedCapUeMaxAvailBandwidth**

parameter is set to a value greater than **20M** and the number of UEs on the RedCap-dedicated BWP exceeds 100.

The change in this feature is supported only in low-frequency SA networking.

Implementation

For details, see *RedCap*. The specific sections are listed in the change history of the document.

Impact on the Network

When a large number of UEs access a RedCap-dedicated BWP, more PUCCH resources are used, the available PUSCH resources become fewer, the uplink UE throughput may decrease, and the downlink UE throughput may increase.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCell RedCap.RedCapAlgoExtSwitch	MOD NRDUCELLR EDCAP LST NRDUCELLR EDCAP	gNodeB	The REDCAP_PUCCH_CSI_RB_ADJ_SW option now supports NR TDD.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RedCap Introduction	REDCAP_SW option of the NRDUCellFeatureSw.UeCapbBasedFunSw parameter	<i>RedCap</i>	PUCCH CSI RB adjustment for the RedCap-dedicated BWP can take effect only when RedCap Introduction is enabled.

- Mutually exclusive features/functions

The following mutually exclusive relationships have been removed.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CSI RB resource extension	CSI_RES_EXT_SW option of the gNodeBAlgo.PucchSwitch parameter	None	For an NR TDD cell, if the REDCAP_SW option of the NRDUCellFeatureSw.UeCapbBasedFunSw parameter is selected and the NRDUCellRedCap.RedCapUeMaxAvailBandwidth parameter is set to a value other than 20M and 40M , CSI RB resource extension must be disabled.

- Impacted features/functions

The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *RedCap*.

Requirement/Trouble Ticket Number

FeR-IR20250528001856-0009

RAT

NR TDD

5.3.2.11.12 Replaced the Reserved-Parameter Bit That Controls RedCap UE Device-Pipe Identification Optimization with a Formal-Parameter Option (TDD)

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The reserved-parameter bit that controls RedCap UE device-pipe identification optimization has been replaced by a formal-parameter option.

RSVDSWPARAM2_BIT23 of the reserved parameter **gNBRsvd.RsvdSwParam2** has been added to the disuse list and replaced by the **REDCAP_SPEC_UE_IDENTIFY_OPT_SW** option of the **gNBMobilityCommParam.MobilityAlgoSwitch** parameter.

The change in this feature is supported only in low-frequency SA networking.

Implementation

For details, see *RedCap*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management

The change in this feature has no impact. It is controlled by the existing feature license.

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNBMobilityCommParam.MobilityAlgoSwitch	MOD GNBMOBILITYCOMMPARAM LST GNBMOBILITYCOMMPARAM	gNodeB	The REDCAP_SPEC_UE_IDENTIFY_OPT_SW option has been added to specify whether to enable RedCap UE device-pipe identification optimization.
Modified	gNBRSvd.RsvdSwParam2	MOD GNBRSVD LST GNBRSVD	gNodeB	RSVDSWPARAM2_BIT23 of this reserved parameter has been replaced by the REDCAP_SPEC_UE_IDENTIFY_OPT_SW option of the gNBMobilityCommParam.MobilityAlgoSwitch parameter and therefore has been added to the disuse list. Although the configuration interface in this version supports synchronization and delivery of the setting of this parameter, the system does not use this parameter.

NOTE

gNBRSvd.RsvdSwParam2 is a reserved parameter.

For details about reserved parameters, see 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.

- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250528001856-0001

RAT

NR TDD

5.3.2.12 FOFD-081260 Uplink Multi-band Flexible Scheduling

5.3.2.12.1 Support for 0- μ s Interval Between Two Consecutive FSA Band Combination Switches

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

UEs capable of FSA introduced in 3GPP Release 18 support a 0- μ s interval between two consecutive switches. The interval is indicated by the **uplinkTxSwitchingMinimumSeparationTime-r18** field, and explanations of this field can be found in section 4.2.7.1 "BandCombinationList parameters" in 3GPP TS 38.306 (V18.0.0).

LampSite base stations do not support the change in this feature.

Implementation

For details, see *Uplink Flexible Spectrum Scheduling*. The specific sections are listed in the change history of the document.

Impact on the Network

Base stations exchange the information of **uplinkTxSwitchingMinimumSeparationTime-r18=0 μ s** over the eXn interface. Consequently, in the **SCTP Link Measurement** function subset, the values of counters related to received and transmitted traffic volumes increase.

Impact on NEs

The change in this feature is implemented on the gNodeB.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The change in this feature has no impact.
- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20240904002804-0010

IR20241101001017

RAT

NR TDD

5.3.2.12.2 Optimized Carrier Combination Switching When No SUL PHR Is Received in 1 TDD+2 FDD FSA Scenarios

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The time for carrier combination change triggered by SUL PHR assurance is always 1s. When the NR TDD+NR FDD carrier combination has strong capabilities, frequent carrier combination changes will affect UEs' peak rates. Therefore, the carrier combination switching is optimized when no SUL power headroom report (PHR) is received in 1 TDD+2 FDD FSA scenarios.

If an FSA UE is using the NR TDD+NR FDD carrier combination and this combination has strong capabilities, and no SUL PHR is received for a period of time longer than the value of the **NRDUCellCarrMgmt.FlexSpctAcsSulPhrCarrIntvl** parameter, a carrier combination switch is triggered and the UE switches to the NR TDD+SUL carrier combination to obtain its capabilities and assess the SUL carrier performance. If the NR TDD+SUL carrier combination has stronger capability than the NR TDD+NR FDD carrier combination, the UE uses the NR TDD+SUL carrier combination. Otherwise, the UE switches back to the NR TDD+NR FDD carrier combination.

For an upgrade from a version earlier than V100R022C00 to V100R022C00 or later, the value of the **NRDUCellCarrMgmt.FlexSpctAcsSulPhrCarrIntvl** parameter is 1 after the upgrade.

LampSite base stations do not support the change in this feature.

Implementation

For details, see *Uplink Flexible Spectrum Scheduling*. The specific sections are listed in the change history of the document.

Impact on the Network

A smaller value of the **NRDUCellCarrMgmt.FlexSpctAcsSulPhrCarrIntvl** parameter results in more frequent carrier combination changes triggered by SUL PHR guarantee for FSA UEs. In addition, the peak rate of FSA UEs slightly fluctuates. A larger value of this parameter results in less frequent carrier combination changes triggered by SUL PHR guarantee for FSA UEs.

Impact on NEs

The change in this feature is implemented on the gNodeB.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Remarks
New	NRDUCellCarrMgmt.FlexSpectAccSulPhrCarrIntvl	MOD NRDUCELLCAR RMGMT LST NRDUCELLCAR RMGMT	gNodeB	This parameter specifies the interval at which the carrier combination of an FSA UE is changed due to SUL PHR assurance.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Uplink Flexible Spectrum Scheduling*.

Requirement/Trouble Ticket Number

FeR-IR20250528001008-0020

DTS2025041730406

RAT

NR TDD

5.3.2.13 FOFD-091253 Uplink Service Guarantee

5.3.2.13.1 Added the Uplink Differentiated Service Guarantee Function (TDD)

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In medium- and heavy-load scenarios, the guaranteed bit rate (GBR) of UEs with scheduling requests (SRs) may fail to be met. To ensure the GBR of specific UEs, the uplink differentiated service guarantee function is introduced. After this function is enabled, UEs are classified into common UEs and QCI-specific UEs for which differentiated service guarantee is performed (referred to as guaranteed UEs).

For guaranteed UEs, their scheduling priorities are not lowered. For common UEs, the base station dynamically adjusts the SR-based scheduling priorities, identifies UEs with zero buffer status reports (BSRs) and lowers their scheduling priorities to reserve some scheduling resources for guaranteed UEs when the PRB usage and uplink CCE allocation failure rate meet the conditions.

The PRB usage and uplink CCE allocation failure rate meet the conditions when:

- Value of the **NRDUCellPusch.SrSchOptPrbLowThld** parameter < Uplink PUSCH PRB usage < Value of the **NRDUCellPusch.SrSchOptPrbHighThld** parameter
- Uplink CCE allocation failure rate > Value of the **NRDUCellPusch.SrSchOptCceFailRateThld** parameter

This function is controlled by the **SERVICE_DIFF_GUARANTEE_UE_FLAG** option of the **NRDUCellQciBearer.ServiceDiffAlgoSwitch** parameter. Before enabling this function, you need to select the **UL_SERV_DIFF_EXP_GUAR_SW** option of the **NRDUCellFeatureSw.QciBasedDiffServExpSw** parameter.

Implementation

For details, see *Ultimate Video Experience*. The specific sections are listed in the change history of the document.

Impact on the Network

Benefits:

After uplink differentiated service guarantee is enabled, the average uplink throughput of QCI-specific services increases if (1) the PRB usage and uplink CCE allocation failure rate meet the related conditions and (2) the minimum rate guarantee for non-GBR services or GBR services takes effect. (Average uplink throughput of specific services = $(N.ThpVolForRate.UL.DuCell5QI - N.ThpVolForRate.UL.SmallPkt.DuCell5QI) / N.ThpTimeForRate.UL.RmvSmallPkt.DuCell5QI$)

Impacts:

- After uplink differentiated service guarantee takes effect, uplink scheduling resources are preempted by guaranteed UEs, which decreases the uplink traffic volumes of common UEs.
- In uplink heavy-load scenarios ($(\text{N.PRB.PUSCH.Used.Avg} / \text{N.PRB.UL.Avail.Avg}) \times 100\% > 60\%$), the average uplink UE throughput in a cell indicated by **User Uplink Average Throughput (DU)** may fluctuate, and the uplink user-perceived rates of guaranteed UEs fluctuate. The specific fluctuation depends on the UE distribution.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellFeatureSw.QciBasedDiffServExpSw	MOD NRDUCELLF EATURESW LST NRDUCELLF EATURESW	gNodeB	The UL_SERV_DIFF_EXP_GUAR_SW option has been added to specify whether to enable uplink differentiated service guarantee.
Modified	NRDUCellQciBearer.ServiceDiffAlgoSwitch	MOD NRDUCELLQ CIBEARER LST NRDUCELLQ CIBEARER	gNodeB	The SERVICE_DIFF_GUARANTEE_UE_FLAG option has been added to specify whether to enable uplink differentiated service guarantee.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellPusch.SrSc hOptCceFailureRateThld	MOD NRDUCCELLP USCH LST NRDUCCELLP USCH	gNodeB	This parameter has been added to specify the CCE failure rate threshold for SR-based scheduling optimization.
New	NRDUCellPusch.SrSc hOptPrbLowThld	MOD NRDUCCELLP USCH LST NRDUCCELLP USCH	gNodeB	This parameter has been added to specify the lower PRB threshold for SR-based scheduling optimization.
New	NRDUCellPusch.SrSc hOptPrbHighThld	MOD NRDUCCELLP USCH LST NRDUCCELLP USCH	gNodeB	This parameter has been added to specify the upper PRB threshold for SR-based scheduling optimization.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Turbo Uplink Switch	TURBO_UPLINK_SW option of the NRDUCellC hnCovAlgo. UICoverage AlgoSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	The uplink differentiated service guarantee function takes effect only when the turbo uplink function is enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SR-based scheduling optimization	SR_SCH_OP T_SW option of the NRDUCellPusch.PuschPerformanceSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	The uplink differentiated service guarantee function takes effect only when SR-based scheduling optimization is enabled.

- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Ultimate Video Experience*.

Requirement/Trouble Ticket Number

FeR-IR20240626000421-0002

IR20241209002193

RAT

NR TDD

5.3.2.13.2 Added Differentiated Uplink MU-MIMO Scheduling (TDD)

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

As the number of UEs requiring differentiated service guarantee (UEs with specific QCI) increases, differentiated uplink MU-MIMO scheduling is introduced to ensure user experience in medium- and heavy-load scenarios. This function includes the "RB policy for large-packet scheduling in uplink MU-MIMO pairing" and "RB control for SR-based scheduling in uplink MU-MIMO grouping and pairing" sub-functions.

Before enabling those sub-functions, the **UL_SERV_DIFF_EXP_GUAR_SW** option of the **NRDUCellFeatureSw.QciBasedDiffServExpSw** parameter must be selected.

RB policy for large-packet scheduling in uplink MU-MIMO pairing:

In medium- and heavy-load scenarios with uplink scheduling enhancement enabled, the RB resource size available for pairing may not match the size required by UEs enabled with differentiated service guarantee during uplink MU pairing, resulting in pairing failure. To improve the experience of those UEs, "RB policy for large-packet scheduling in uplink MU-MIMO pairing" is introduced. After this function is enabled, the base station evaluates the RB resource size available for pairing, the RB resource size required by UEs enabled with differentiated service guarantee, and the remaining RB resource size of the cell, and determines whether to perform MU-MIMO pairing.

This function is controlled by the **UL_MU_BIG_PKT_PAIR_RB_POLI_SW** option of the **NRDUCellQciBearer.ServiceDiffAlgoSwitch** parameter.

RB control for SR-based scheduling in uplink MU-MIMO grouping and pairing:

In medium- or heavy-load scenarios with uplink MU grouping and pairing enabled, common UEs may occupy a large number of RB resources during SR-based scheduling. As a result, the resources available for UEs enabled with differentiated service guarantee decrease. To improve the experience of those UEs, "RB control for SR-based scheduling in uplink MU-MIMO grouping and pairing" is introduced. After this function is enabled, the RB resources occupied by common UEs during SR-based scheduling are reduced.

This function is controlled by the **UL_MU_GRP_PAIR_RB_TILT_SW** option of the **NRDUCellQciBearer.ServiceDiffAlgoSwitch** parameter and the **UL_MU_GRP_PAIR_SR_RB_CTRL_SW** option of the **NRDUCellULMimo.UlMuMimoAlgoSwitch** parameter.

This function takes effect only when the uplink CCE allocation failure rate of the cell is less than or equal to the **NRDUCellULMimo.GrpPairSrRbCtrlCceFailThld** parameter value.

Implementation

For details, see *Ultimate Video Experience*. The specific sections are listed in the change history of the document.

Impact on the Network

Benefits:

After differentiated uplink MU-MIMO scheduling is enabled in medium- and heavy-load scenarios, the 5QI- and QCI-specific average uplink rates of UEs with specific QCIs increase, the number of RBs paired for uplink MU-MIMO in the cell increases, and both the total and average numbers of layers paired for uplink MU-MIMO increase.

- Average uplink rate of 5QI-specific services =
$$\frac{(N.ThpVolForRate.UL.DuCell5QI - N.ThpVolForRate.UL.SmallPkt.DuCell5QI)}{N.ThpTimeForRate.UL.RmvSmallPkt.DuCell5QI}$$
- Average uplink rate of QCI-specific services =
$$\frac{(N.ThpVolForRate.UL.QCI - N.ThpVolForRate.UL.SmallPkt.QCI)}{N.ThpTimeForRate.UL.RmvSmallPkt.QCI}$$

- Number of RBs paired for uplink MU-MIMO: **N.ChMeas.MIMO.UL.Pair.PR**
- Total number of layers paired for uplink MU-MIMO:
N.ChMeas.MIMO.UL.Pair.Layer
- Average number of layers paired for uplink MU-MIMO:
N.ChMeas.MIMO.UL.Pair.Layer.Avg

Impacts:

- After differentiated uplink MU-MIMO scheduling takes effect, uplink scheduling resources are preempted by guaranteed UEs, which decreases the uplink traffic volumes of common UEs.
- The average uplink UE throughput in a cell indicated by **User Uplink Average Throughput (DU)** may fluctuate, and the uplink user-perceived rates of guaranteed UEs fluctuate. The specific fluctuation depends on the UE distribution.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellQciBearer.ServiceDiffAlgoSwitch	MOD NRDUCELLQ CIBEARER LST NRDUCELLQ CIBEARER	gNodeB	The UL_MU_BIG_PKT_PAIR_RB_POLI_SW option has been added to specify whether to enable "RB policy for large-packet scheduling in uplink MU-MIMO pairing".

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellQciBearer.ServiceDiffAlgoSwitch	MOD NRDUCELLQ CIBEARER LST NRDUCELLQ CIBEARER	gNodeB	The UL_MU_GRP_PAIR_RB_TILT_SW option has been added to specify whether to enable "RB control for SR-based scheduling in uplink MU-MIMO grouping and pairing".
Modified	NRDUCellULMimo.UlMuMimoAlgoSwitch	MOD NRDUCELLU LMIMO LST NRDUCELLU LMIMO	gNodeB	The UL_MU_GRP_PAIR_SR_RB_CTRL_SW option has been added to specify whether to enable "RB control for SR-based scheduling in uplink MU-MIMO grouping and pairing".
New	NRDUCellULMimo.GrpPairSrRbCtrlCceFailThld	MOD NRDUCELLU LMIMO LST NRDUCELLU LMIMO	gNodeB	This parameter has been added to specify the CCE allocation failure rate threshold for "RB control for SR-based scheduling in uplink MU grouping and pairing" to take effect.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink scheduling enhancement	UL_SCHEDULE_ENH_SW option of the NRDUCellPusch.UlPuschAlgoSwitch parameter	<i>Uplink Scheduling</i>	"RB policy for large-packet scheduling in uplink MU-MIMO pairing" takes effect only when uplink scheduling enhancement is enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink MU grouping and pairing	UL_MU_GRP_PAIR_SW option of the NRDUCellUIMimo.UIMuMimoAlgoSwitch parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	"RB control for SR-based scheduling in uplink MU-MIMO grouping and pairing" takes effect only when uplink MU grouping and pairing is enabled.

- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Ultimate Video Experience*.

Requirement/Trouble Ticket Number

FeR-IR20241209001765-0001

RAT

NR TDD

5.3.2.14 MRFD-131161 LTE and NR X2 Interface Self-Configuration (NR)

5.3.2.14.1 Added Support for SA X2 Link Setup

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

This function now supports X2 link setup between eNodeBs and SA gNodeBs. The eNodeB automatically initiates X2 self-setup when the following conditions apply:
(1) The **LNR_SA_GNB_X2_SETUP_SW** option of the **eNodeBOamParam.InterfaceSonConfigSwitch** parameter on the LTE side and the **X2_SETUP_FOR_SA_SW** option of the **gNBInterfaceParam.X2ConfigSw**

parameter on the NR side are selected; (2) inter-base-station coordination services are performed for UEs.

Implementation

For details, see *X2 Self-Management in SA Networking*. The specific sections are listed in the change history of the document.

Impact on the Network

This function provides the following benefits:

SA X2 self-management simplifies configuration operations, reduces operating expense (OPEX) for operators, improves the eNodeB/gNodeB resource usage rate, and enables inter-base-station coordination services more efficiently.

This function has the following impacts:

- When the mis-deletion prevention mechanism for X2 self-removal triggered by link faults is in effect, the number of X2 interface setup attempts at the local eNodeB (indicated by the **L.Sig.X2.SendSetup.Att** counter) and the number of successful X2 interface setups at the local eNodeB (indicated by the **L.Sig.X2.SendSetup.Succ** counter) may decrease.
- SA X2 link setup attempts may cause the following alarms to be reported:
 - ALM-29801 gNodeB X2 Interface Full Specification
 - ALM-29225 Maximum Number of X2 Interfaces Reached
 - ALM-29800 gNodeB X2 Interface Fault
 - ALM-29204 X2 Interface Fault
- SA X2 link setup attempts may cause the values of the counters related to the following function subsets to fluctuate: SCTP Link Measurement, eNodeB Transmission Interface Measurement, and X2 Link Measurement. For descriptions of the preceding counters, see *3900 & 5900 Series Base Station Performance Counter Reference* in *3900 & 5900 Series Base Station Product Documentation*.
- SA X2 link setups occupy the X2 specifications of the main control board. If the X2 specifications are insufficient, the counters related to NSA handovers and number of UEs may be affected. For details about the impacts, see "Impacts" in *X2 and S1 Self-Management in NSA Networking*.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

- Main control boards: UMPTb and later main control boards
- Baseband processing units: UBBPe and later baseband processing units

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature is not under license control.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNBInterfaceParam.X2ConfigSw	MOD GNBINTERF ACEPARAM LST GNBINTERF ACEPARAM	gNodeB	The X2_SETUP_FOR_SA_SW option has been added to specify whether the base station allows X2 link setup in SA networking.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The following mutually exclusive relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD High-frequency NR TDD	Operator-specific NCI configuration	gNBSharingMode.MultipleNCISwitch	<i>Multi-Operator Sharing</i>	The SA X2 setup switch must be turned off when operator-specific NCI configuration is enabled.

- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *X2 Self-Management in SA Networking*.

Requirement/Trouble Ticket Number

FeR-IR20250103001347-0010

RAT

NR FDD, NR TDD

5.3.2.14.2 Added Support for Prohibition of Multi-Operator Sharing During X2-U Self-Setup

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In RAN sharing with common carrier scenarios, the "prohibition of multi-operator sharing during X2-U self-setup" function has been added to enable separated X2-U links for different operators for service isolation. This function is controlled by the **X2_UP_SELF_SETUP_NO_MOCN_SW** option of the **gNBInterfaceParam.InterfaceSelfCfgOptSw** parameter. After "prohibition of multi-operator sharing during X2-U self-setup" is enabled, separated X2-U links are used by multiple operators, and operators do not share user-plane transmission resources.

Implementation

For details, see *X2 and S1 Self-Management in NSA Networking* and *X2 Self-Management in SA Networking*. The specific sections are listed in the change history of the documents.

Impact on the Network

If an operator experiences X2 transmission faults, the operator cannot use the user-plane transmission resources of other operators. In this case, services may be affected.

Impact on NEs

The change in this feature is implemented on the gNodeB.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
 - The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNBInterfaceParam. InterfaceSelfCfgOptSw	MOD GNBINTERF ACEPARAM LST GNBINTERF ACEPARAM	gNodeB	The X2_UP_SELF_SETUP_NO_MOCN_SW option has been added to support separated X2-U links for multiple operators in RAN sharing with common carrier scenarios.

- Performance management

The change in this feature has no impact.
- Fault management

The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

The change in this feature has no impact.
- Mutually exclusive features/functions

The change in this feature has no impact.
- Impacted features/functions

The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *X2 and S1 Self-Management in NSA Networking* and *X2 Self-Management in SA Networking*.

Requirement/Trouble Ticket Number

FeR-IR20251205000509-0010

RAT

NR FDD, NR TDD

5.3.2.15 MRFD-160263 LTE and NR Intelligent Carrier Shutdown (NR)

5.3.2.15.1 Support for LTE and NR Intelligent Carrier Shutdown and LNR Load-based Dormancy in Inter-Base-Station SA Networking

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

LTE and NR intelligent carrier shutdown and LNR load-based dormancy are supported in inter-base-station SA networking. This change involves the LTE and NR intelligent carrier shutdown, LNR load-based RF module deep dormancy, and LNR load-based RF module super dormancy functions.

In SA networking, when you configure a neighbor relationship between an LTE cell and an NR cell, the **LNR_CARR_SHUTDOWN_OVERLAP_IND** option of the **NRCellEutraNRelation.PowerSavingOverlapInd** parameter on the NR side can be selected to configure the LTE cell as an overlapping neighboring cell of the NR cell. In inter-base-station SA networking, when you configure a neighbor relationship between an LTE cell and an NR cell, the **CARRIER_SHUTDOWN_OVERLAP_FLAG** option of the **NrNRelationship.AggregationAttribute** parameter on the LTE side can be selected to configure the LTE cell as an overlapping neighboring cell of the target NR cell. When manually configuring the neighbor relationship on the LTE and NR sides, set these parameters as planned for all cells to be enabled with LTE and NR intelligent carrier shutdown.

For LTE and NR intelligent carrier shutdown or LNR load-based dormancy to take effect in inter-base-station SA networking, X2 links need to be set up.

X2 links can be set up automatically or manually. Automatic setup is recommended. For details about X2 configurations, see *X2 Self-Management in SA Networking*. X2 self-setup is expected to be triggered on the LTE side within half an hour if the following conditions are met:

- The **CARRIER_SHUTDOWN_OVERLAP_FLAG** option of the **NrNRelationship.AggregationAttribute** parameter is selected for LTE cells.
- LTE and NR intelligent carrier shutdown or LNR load-based dormancy is enabled.
- The switches for LTE-NR inter-base-station power saving and X2 setup in SA networking are turned on. The specific switches are as follows:
 - LTE-NR SA gNodeB X2 setup switch on the LTE side, controlled by the **LNR_SA_GNB_X2_SETUP_SW** option of the **eNodeBOamParam.InterfaceSonConfigSwitch** parameter
 - LTE-NR SA inter-base-station power saving switch on the LTE side, controlled by the **LNR_INTER_BS_PWR_SAVE_SW** option of the **EnodebPwrSavingAlgo.PwrSavingAlgoSwitch** parameter

- SA X2 setup switch on the NR side, controlled by the **X2_SETUP_FOR_SA_SW** option of the **gNBInterfaceParam.X2ConfigSw** parameter
- LTE-NR SA inter-base-station power saving switch on the NR side, controlled by the **SA_LNR_INTER_BS_PWR_SAVE_SW** option of the **gNBPwrSavingAlgo.PwrSavingAlgoSwitch** parameter
- No X2 link is available between the eNodeB and the target gNodeB.

LampSite base stations do not support this change.

This change is supported only in low frequency bands.

This change is supported only in SA networking.

Implementation

For details, see *Multi-RAT Coordinated Energy Saving*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

This function is implemented on the gNodeB and eNodeB.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	eNodeBOamParam.InterfaceSonConfigSwitch	MOD ENODEBOAMPARAM LST ENODEBOAMPARAM	eNodeB	The LNR_SA_GNB_X2_SETUP_SW option has been added to specify whether to enable X2 setup between the eNodeB and gNodeBs that do not serve any NSA cells.
Modified	EnodebPwrSavingAlgo.PwrSavingAlgoSwitch	MOD ENODEBPWRSAVINGALGO LST ENODEBPWRSAVINGALGO	eNodeB	The LNR_INTER_BS_PWR_SAVE_SW option has been added to specify whether the eNodeB with LTE and NR power saving enabled can initiate an X2 setup with a gNodeB that meets the following conditions: (1) No X2 interface has been set up between the eNodeB and gNodeB. (2) The gNodeB serves cells with the overlapping coverage indicator for carrier shutdown configured.
Modified	gNBInterfaceParam.X2ConfigSw	MOD GNBINTERFACEPARAM LST GNBINTERFACEPARAM	gNodeB	The X2_SETUP_FOR_SA_SW option has been added to specify whether the base station allows X2 link setup in SA networking.
Modified	gNBPwrSavingAlgo.PwrSavingAlgoSwitch	MOD GNBPWRSAVINGALGO LST GNBPWRSAVINGALGO	gNodeB	The SA_LNR_INTER_BS_PWR_SAVE_SW option has been added to specify whether SA NR cells can be engaged in inter-base-station LTE and NR energy saving.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Multi-RAT Coordinated Energy Saving*.

Requirement/Trouble Ticket Number

FeR-IR20250103001347-0010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD

5.3.2.16 MRFD-161263 RF Module Deep Dormancy (NR)

5.3.2.16.1 Support for LTE and NR Intelligent Carrier Shutdown and LNR Load-based Dormancy in Inter-Base-Station SA Networking

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

LTE and NR intelligent carrier shutdown and LNR load-based dormancy are supported in inter-base-station SA networking. This change involves the LTE and NR intelligent carrier shutdown, LNR load-based RF module deep dormancy, and LNR load-based RF module super dormancy functions.

In SA networking, when you configure a neighbor relationship between an LTE cell and an NR cell, the **LNR_CARR_SHUTDOWN_OVERLAP_IND** option of the **NRCeUtraNRelation.PowerSavingOverlapInd** parameter on the NR side can be selected to configure the LTE cell as an overlapping neighboring cell of the NR cell. In inter-base-station SA networking, when you configure a neighbor relationship between an LTE cell and an NR cell, the **CARRIER_SHUTDOWN_OVERLAP_FLAG** option of the **NrNRelationship.AggregationAttribute** parameter on the LTE side can be selected to configure the LTE cell as an overlapping neighboring cell of the target NR cell. When manually configuring the neighbor relationship on the LTE and NR sides, set these parameters as planned for all cells to be enabled with LTE and NR intelligent carrier shutdown.

For LTE and NR intelligent carrier shutdown or LNR load-based dormancy to take effect in inter-base-station SA networking, X2 links need to be set up.

X2 links can be set up automatically or manually. Automatic setup is recommended. For details about X2 configurations, see *X2 Self-Management in SA Networking*. X2 self-setup is expected to be triggered on the LTE side within half an hour if the following conditions are met:

- The **CARRIER_SHUTDOWN_OVERLAP_FLAG** option of the **NrNRRelationship.AggregationAttribute** parameter is selected for LTE cells.
- LTE and NR intelligent carrier shutdown or LNR load-based dormancy is enabled.
- The switches for LTE-NR inter-base-station power saving and X2 setup in SA networking are turned on. The specific switches are as follows:
 - LTE-NR SA gNodeB X2 setup switch on the LTE side, controlled by the **LNR_SA_GNB_X2_SETUP_SW** option of the **eNodeBOamParam.InterfaceSonConfigSwitch** parameter
 - LTE-NR SA inter-base-station power saving switch on the LTE side, controlled by the **LNR_INTER_BS_PWR_SAVE_SW** option of the **EnodebPwrSavingAlgo.PwrSavingAlgoSwitch** parameter
 - SA X2 setup switch on the NR side, controlled by the **X2_SETUP_FOR_SA_SW** option of the **gNBInterfaceParam.X2ConfigSw** parameter
 - LTE-NR SA inter-base-station power saving switch on the NR side, controlled by the **SA_LNR_INTER_BS_PWR_SAVE_SW** option of the **gNBPwrSavingAlgo.PwrSavingAlgoSwitch** parameter
- No X2 link is available between the eNodeB and the target gNodeB.

LampSite base stations do not support this change.

This change is supported only in low frequency bands.

This change is supported only in SA networking.

Implementation

For details, see *Multi-RAT Coordinated Energy Saving*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

This function is implemented on the gNodeB and eNodeB.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	eNodeBOamParam.InterfaceSonConfigSwitch	MOD ENODEBOAMPARAM LST ENODEBOAMPARAM	eNodeB	The LNR_SA_GNB_X2_SETUP_SW option has been added to specify whether to enable X2 setup between the eNodeB and gNodeBs that do not serve any NSA cells.
Modified	EnodebPwrSavingAlgo.PwrSavingAlgoSwitch	MOD ENODEBPWRSAVINGALGO LST ENODEBPWRSAVINGALGO	eNodeB	The LNR_INTER_BS_PWR_SAVE_SW option has been added to specify whether the eNodeB with LTE and NR power saving enabled can initiate an X2 setup with a gNodeB that meets the following conditions: (1) No X2 interface has been set up between the eNodeB and gNodeB. (2) The gNodeB serves cells with the overlapping coverage indicator for carrier shutdown configured.
Modified	gNBInterfaceParam.X2ConfigSw	MOD GNBINTERFACEPARAM LST GNBINTERFACEPARAM	gNodeB	The X2_SETUP_FOR_SA_SW option has been added to specify whether the base station allows X2 link setup in SA networking.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNBPwrSavingAlgo.PwrSavingAlgoSwitch	MOD GNBPWRS AVINGALGO LST GNBPWRS AVINGALGO	gNodeB	The SA_LNR_INTER_BS_PWR_SAVE_SW option has been added to specify whether SA NR cells can be engaged in inter-base-station LTE and NR energy saving.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Multi-RAT Coordinated Energy Saving*.

Requirement/Trouble Ticket Number

FeR-IR20250103001347-0010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD

5.3.2.17 MRFD-171263 NSA Carrier Combination Selection Based on User Experience (NR)

5.3.2.17.1 NSA Carrier Combination Selection Triggered by Coverage-based NR SCC Release

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

NSA carrier combination selection triggered by coverage-based NR SCC release is now supported. With this function in effect, NSA carrier combination selection based on user experience now supports band combination change during coverage-based NR SCC release.

NSA carrier combination selection triggered by coverage-based NR SCC release takes effect when all of the following conditions are met:

- The **NR_SCC_REL_NSA_CARR_SEL_SW** option of the **NRCellCaMgmtConfig.CaStrategySwitch** parameter is selected on the NR side.
- The **NR_SCC_REL_NSA_CARR_SEL_SW** option of the **NsaDcMgmtConfig.NsaDcAlgoEnhSwitch** parameter is selected on the LTE side.
- The frequency numbers specified by **NrScgFreqConfig.ScgDLArfcn** for which the **NR_SCC_REL_NSA_CARR_SEL_FLAG** option of the **NrScgFreqConfig.AggregationAttribute** parameter on the LTE side is selected contain the frequency number of an NR SCC that is released based on coverage.

After NSA carrier combination selection triggered by coverage-based NR SCC release takes effect, the carrier selection penalty timer upon NR SCC release (specified by the **NsaDcMgmtConfig.NrSccRelCarrSelPunishTimer** parameter) is started on the LTE side. Before this timer expires, the band combination change procedure during coverage-based NR SCC release will not be triggered again for the NSA UE.

The change in this feature is supported only in NSA networking.

Implementation

For details, see *EPC-based NSA Experience Improvement*. The specific sections are listed in the change history of the document.

Impact on the Network

When NSA carrier combination selection triggered by coverage-based NR SCC release is enabled, the user-perceived rate of NSA UEs increases, and the values of indicators such as the NSA handover success rate, NSA service drop rate, random access success rate, and uplink and downlink cell throughput on the LTE and NR sides will fluctuate. When this function is disabled, network performance is not affected.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRCeMgmtConfig.CaStrategySwitch	MOD NRCELLCAM GMTCONFIG LST NRCELLCAM GMTCONFIG	gNodeB	The NR_SCC_REL_NSA_CARR_SEL_SW option has been added to specify whether to enable NSA carrier combination selection triggered by coverage-based NR SCC release.
Modified	NsaDcMgmtConfig.NsaDcAlgoEnhSwitch	MOD NSADCMGMTCONFIG LST NSADCMGMTCONFIG	eNodeB	The NR_SCC_REL_NSA_CARR_SEL_SW option has been added to specify whether to enable NSA carrier combination selection triggered by coverage-based NR SCC release.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NrScgFreqConfig.AggregationAttribute	ADD NRSCGFREQ CONFIG MOD NRSCGFREQ CONFIG LST NRSCGFREQ CONFIG	eNodeB	The NR_SCC_REL_NSA_CARR_SEL_FLAG option has been added to specify whether to enable NSA carrier combination selection triggered by coverage-based NR SCC release. If this option is selected, NSA carrier combination selection can be triggered during coverage-based release of an NR SCC whose frequency number is specified by the NrScgFreqConfig.ScgDLArfcn parameter.
New	NsaDcMgmtConfig.NrSccRelCarrSelPenaltyTimer	MOD NSADCMGMT CONFIG LST NSADCMGMT CONFIG	eNodeB	This parameter has been added to specify the duration of penalty on NSA carrier combination selection triggered by coverage-based NR SCC release in NSA networking.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	NSA carrier combination selection based on user experience	NSA_LTE_FDD_SEL_OPT_SW or NSA_LTE_TDD_SEL_OPT_SW option of the EnodebAlgoExtSwitch . <i>MultiNet workingOption OptSw</i> parameter	<i>EPC-based NSA Experience Improvement</i>	If the NR_SCC_REL_NS_A_CARR_SEL_SW option of the NsaDcMgmtConfig . <i>NsaDcAlgoEnhSwitch</i> parameter is selected, the NSA_LTE_FDD_SEL_OPT_SW or NSA_LTE_TDD_SEL_OPT_SW option of the EnodebAlgoExtSwitch . <i>MultiNet workingOption OptSw</i> parameter must be selected.
NR FDD Low-frequency NR TDD	NSA carrier combination selection based on user experience	NSA_LTE_SEL_OPT_SW option of the gNodeBParam . <i>NetworkingOption OptSw</i> parameter	<i>EPC-based NSA Experience Improvement</i>	If the NR_SCC_REL_NS_A_CARR_SEL_SW option of the NRCellCaMgmtConfig . <i>CaStrategySwitch</i> parameter is selected, the NSA_LTE_SEL_OPT_SW option of the gNodeBParam . <i>NetworkingOption OptSw</i> parameter must be selected.

- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *EPC-based NSA Experience Improvement*.

Requirement/Trouble Ticket Number

FeR-IR20241216002543-0010

IR20250115001578

RAT

NR FDD, NR TDD

5.3.2.17.2 Optimized NSA Carrier Combination Selection Based on User Experience

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

NSA carrier combination selection based on user experience has been optimized as follows:

- The "B1 measurement optimization for the serving NR frequency in NSA networking" function has been introduced. Once this function is enabled and the eNodeB receives an A3 or A5 measurement report containing measurement results of the NR PSCell or SCell along with the strongest intra-frequency neighboring cell of the NR PSCell or SCell prior to delivering B1 measurement configurations, then the eNodeB will not deliver B1 measurement configurations for the NR PSCell or SCell frequency. This reduces the number of NR B1 measurements. The "B1 measurement optimization for the serving NR frequency in NSA networking" function is controlled by the **NSA_NR_SER_FREQ_B1_MEAS_OPT_SW** option of the **EnodebAlgoExtSwitch.MultiNetworkingOptionOptSw** parameter.
- The downlink load factor is considered during user experience evaluation on the LTE side only when the **NsaDcMgmtConfig.NsaMinDLAvgToBeSchedUeNum** parameter is set to a non-zero value.
- The "blind SCG addition for NSA UEs without B1 measurement results on NR frequencies" function has been introduced. Once this function is enabled, if no measurement report on any frequency is received when the measurement waiting timer expires, and the frequency with the highest SCG frequency priority can only be added based on measurement while a lower-priority frequency can be added blindly, the eNodeB can blindly add the NR frequency that does not have the highest priority. The "blind SCG addition for NSA UEs without B1 measurement results on NR frequencies" function is controlled by the **NSA_NO_NR_B1_BLIND_ADD_SCG_SW** option of the **EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch** parameter.
- The "experience evaluation for the neighboring candidate NR PCell with the highest RSRP in NSA networking" function has been introduced. Once this function is enabled, the gNodeB returns only the evaluation result of the NR PSCell with the highest RSRP. This function is controlled by the

NSA_NR_STRONGEST_PCELL_EVAL_SW option of the
ENodeBAlgoExtSwitch.NsaDcAlgoExtSwitch.

- If the signal levels of the target cell for a handover and its intra-frequency neighboring cell are similar, the handover may fail. If NSA PCC anchoring for NSA carrier selection optimization based on user experience is triggered for NSA UEs and the **NsaDcAlgoParam.NsaGFactorThldForHoTrgCell** parameter is set to a value other than **32767**, an LTE inter-frequency neighboring cell can be selected as the target cell for the handover only when it meets the conditions specified by the parameter, thereby reducing the handover failure probability.
- Counters have been added to measure the changes in the frequency band combinations of UEs.
- The "optimized NSA NR SCC grid-based evaluation" function has been introduced. Once this function is enabled, if the RSRP of an NR SCell is predicted based on the virtual grid model, the measurement result of the NR SCell will be included in the X2 proprietary message even if the NR SCell RSRP does not meet the NR PSCell good-point threshold for grid-based prediction. The "optimized NSA NR SCC grid-based evaluation" function is controlled by the **NSA_NR_SCC_GRID_EVAL_OPT_SW** option of the **ENodeBAlgoExtSwitch.NsaDcAlgoExtSwitch** parameter.
- The mutually exclusive relationship between NSA smart carrier selection based on service characteristic awareness and NSA carrier combination selection based on user experience has been removed.
- The function "penalty on virtual-grid-based addition for SCG addition failures when NSA/LTE selection is enabled" has been introduced. If a UE reports an SCGFailureInformationNR message with the cause value "randomAccessProblem" or "synchReconfigFailureSCG" and the corresponding NR cell is predicted as a good candidate by virtual grid models, there remains a high probability that the cell will again fail to be added to the SCG, resulting in an increased SCG service drop rate. To mitigate this, you are advised to enable the function "penalty on virtual-grid-based addition for SCG addition failures when NSA/LTE selection is enabled". Once the function takes effect, the NR cell associated with the reported SCGFailure message will be excluded from virtual-grid-based SCG addition within the penalty period. The function "penalty on virtual-grid-based addition for SCG addition failures when NSA/LTE selection is enabled" is controlled by the **NSA_SEL_SCG_FAIL_VG_PUNISH_SW** option of the **ENodeBAlgoExtSwitch.NsaDcAlgoExtSwitch** parameter.
- The function allowing NR SCCs configured for NSA UEs to participate in experience evaluation has been introduced. Once this function is enabled, an NR SCC configured for a UE will be included in experience evaluation under either of the following conditions: (1) The NSA UE is configured with the NR SCC, and the X2 proprietary message does not contain the measurement result of the NR SCC. (2) The **NSA_BLIND_CFG_SCC_NOT_EVAL_SW** option of the **NRCeMgmtConfig.CaStrategySwitch** parameter is selected, and the NR SCC is blindly configured for the UE. This function is controlled by the **NSA_ACTUAL_CFG_SCC_EVAL_SW** option of the **gNodeBParam.NetworkingOptionOptSw** parameter.

The change in this feature is supported only in NSA networking.

Implementation

For details, see *EPC-based NSA Experience Improvement*. The specific sections are listed in the change history of the document.

Impact on the Network

- After the "B1 measurement optimization for the serving NR frequency in NSA networking" function is enabled, the number of times B1 measurement configurations are delivered and the number of times B1 measurements time out decrease, and the uplink and downlink throughputs of NSA UEs increase.
- After the "blind SCG addition for NSA UEs without B1 measurement results on NR frequencies" function is enabled, the number of NSA SCG addition attempts increases, the NSA SCG addition success rate decreases, and the proportion of NR CA UEs in NSA networking increases.
- After the "experience evaluation for the neighboring candidate NR PCell with the highest RSRP in NSA networking" function is enabled, the EN-DC band combination with the NR cell having the highest RSRP can be selected for NSA DC UEs. This improves downlink user experience and increases the NR handover and SCG addition success rates.
- After the **NsaDcAlgoParam.NsaGFactorThldForHoTrgCell** parameter is set to a value other than **32767**, the number of NSA PCC anchoring times, the proportion of LTE RRC connection reestablishments, the number of times NSA carrier selection based on user experience takes effect, and the throughput gains brought by NSA carrier selection based on user experience decrease.
- After the "optimized NSA NR SCC grid-based evaluation" function is enabled, the proportion of NR CA UEs increases.
- After the function "penalty on virtual-grid-based addition for SCG addition failures when NSA/LTE selection is enabled" is enabled, the number of virtual-grid-based SCG addition failures and the number of SCG additions decrease, while the success rate of virtual-grid-based SCG additions increases.
- After the function allowing NR SCCs configured for NSA UEs to participate in experience evaluation is enabled, the average downlink throughput of NSA UEs increases.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	EnodebAlgoExtSwitch.MultiNetworkingOptionOptSw	MOD ENODEBALG OEXTSWITCH LST ENODEBALG OEXTSWITCH	eNodeB	The NSA_NR_SER_FREQ_B1_MEAS_OPT_SW option has been added to specify whether to enable the "B1 measurement optimization for the serving NR frequency in NSA networking" function.
New	NsaDcMgmtConfig.NsaMinDLAvgToBeSchedUeNum	MOD NSADCMGMTCONFIG LST NSADCMGMTCONFIG	eNodeB	This parameter has been added to specify the minimum allowed number of UEs to be scheduled in the downlink for the formula used in "NSA carrier combination selection based on user experience" for evaluating the cell capability.
Modified	EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch	MOD ENODEBALG OEXTSWITCH LST ENODEBALG OEXTSWITCH	eNodeB	The NSA_NO_NR_B1_BLIND_ADD_SCG_SW option has been added to specify whether to enable the "blind SCG addition for NSA UEs without B1 measurement results on NR frequencies" function.
Modified	EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch	MOD ENODEBALG OEXTSWITCH LST ENODEBALG OEXTSWITCH	eNodeB	The NSA_NR_STRONGEST_PCELL_EVAL_SW option has been added to specify whether to enable the "experience evaluation for the neighboring candidate NR PCell with the highest RSRP in NSA networking" function.

Change Type	Parameter ID	MML Command	NE	Description
New	NsaDcAlgoParam.NsaGFactorThldForHoTrgCell	MOD NSADCALGO PARAM LST NSADCALGO PARAM	eNodeB	This parameter has been added to specify the G factor threshold for target cells during handovers in NSA carrier combination selection based on user experience.
Modified	EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch	MOD ENODEBALG OEXTSWITCH LST ENODEBALG OEXTSWITCH	eNodeB	The NSA_NR_SCC_GRID_EVAL_OPT_SW option has been added to specify whether to enable the "optimized NSA NR SCC grid-based evaluation" function.
Modified	EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch	MOD ENODEBALG OEXTSWITCH LST ENODEBALG OEXTSWITCH	eNodeB	The NSA_SEL_SCG_FAIL_VG_PUNISH_SW option has been added to specify whether to enable the function "penalty on virtual-grid-based addition for SCG addition failures when NSA/LTE selection is enabled".
Modified	gNodeBParam.NetworkSharingOptionOptSw	MOD GNODEBPARAM LST GNODEBPARAM	gNodeB	The NSA_ACTUAL_CFG_SCC_EVAL_SW option has been added to specify whether to enable the function allowing NR SCCs configured for NSA UEs to participate in experience evaluation.

- Performance management
The following function subsets have changed.

Change Type	Counter Name	Function Subset	NE	Description
New	L.NsaDc.BcSeI Opt.BcChange.LteCaToLteCa.DL	Number of Band Combination Change Attempts for Registered NSA-DC-capable UEs	eNode B	This counter has been added to measure the number of attempts to change an LTE CA band combination to another such combination for NSA DC UEs in downlink band combination selection optimization.
New	L.NsaDc.BcSeI Opt.BcChange.LteCaToENDC.DL	Number of Band Combination Change Attempts for Registered NSA-DC-capable UEs	eNode B	This counter has been added to measure the number of attempts to change an LTE CA band combination to an EN-DC band combination for NSA DC UEs in downlink band combination selection optimization.
New	L.NsaDc.BcSeI Opt.BcChange.ENDCToLteCa.DL	Number of Band Combination Change Attempts for Registered NSA-DC-capable UEs	eNode B	This counter has been added to measure the number of attempts to change an EN-DC band combination to an LTE CA band combination for NSA DC UEs in downlink band combination selection optimization.
New	L.NsaDc.BcSeI Opt.BcChange.ENDCToENDC.DL	Number of Band Combination Change Attempts for Registered NSA-DC-capable UEs	eNode B	This counter has been added to measure the number of attempts to change an EN-DC band combination to another such combination for NSA DC UEs in downlink band combination selection optimization.

- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
LTE FD D LTE TD D NR FD D Low - freq uen cy NR TD D	Optimized experience evaluation for NSA carrier combination selection based on user experience	NSA_NR_EVAL_REQ_NR_MEAS_I ND_SW option of the EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch parameter on the LTE side and the EXP_BASED_NSA_PCC_SEL_OPT_SW option of the gNodeBAlgo.NrSmartMultiCarrSw parameter on the NR side	<i>EPC-based NSA Experience Improvement</i>	To enable experience evaluation for the neighboring candidate NR PCell with the highest RSRP in NSA networking, optimized experience evaluation for NSA carrier combination selection based on user experience must be enabled.
LTE FD D LTE TD D NR FD D Low - freq uen cy NR TD D	Optimized experience evaluation for NSA carrier combination selection based on user experience	NSA_NR_EVAL_REQ_NR_MEAS_I ND_SW option of the EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch parameter on the LTE side and the EXP_BASED_NSA_PCC_SEL_OPT_SW option of the gNodeBAlgo.NrSmartMultiCarrSw parameter on the NR side	<i>EPC-based NSA Experience Improvement</i>	To enable the "optimized NSA NR SCC grid-based evaluation" function, optimized experience evaluation for NSA carrier combination selection based on user experience must be enabled.

- Mutually exclusive features/functions

The following mutually exclusive relationship has been removed.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	NSA smart carrier selection based on service characteristic awareness	NSA_TRAFFIC_AWARENESS_SCS_SW option of the MultiCarrierUnifiedSchEnS parameter	None	NSA smart carrier selection based on service characteristic awareness does not work with NSA carrier combination selection based on user experience.

- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *EPC-based NSA Experience Improvement*.

Requirement/Trouble Ticket Number

FeR-IR20250708001473-0260

RAT

NR FDD, NR TDD

5.3.2.18 MRFD-200261 RF Module Super Deep Dormancy (NR)

5.3.2.18.1 Support for LTE and NR Intelligent Carrier Shutdown and LNR Load-based Dormancy in Inter-Base-Station SA Networking

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

LTE and NR intelligent carrier shutdown and LNR load-based dormancy are supported in inter-base-station SA networking. This change involves the LTE and NR intelligent carrier shutdown, LNR load-based RF module deep dormancy, and LNR load-based RF module super dormancy functions.

In SA networking, when you configure a neighbor relationship between an LTE cell and an NR cell, the **LNR_CARR_SHUTDOWN_OVERLAP_IND** option of the

NRCellEutraNRelation.PowerSavingOverlapInd parameter on the NR side can be selected to configure the LTE cell as an overlapping neighboring cell of the NR cell. In inter-base-station SA networking, when you configure a neighbor relationship between an LTE cell and an NR cell, the **CARRIER_SHUTDOWN_OVERLAP_FLAG** option of the **NrNRelationship.AggregationAttribute** parameter on the LTE side can be selected to configure the LTE cell as an overlapping neighboring cell of the target NR cell. When manually configuring the neighbor relationship on the LTE and NR sides, set these parameters as planned for all cells to be enabled with LTE and NR intelligent carrier shutdown.

For LTE and NR intelligent carrier shutdown or LNR load-based dormancy to take effect in inter-base-station SA networking, X2 links need to be set up.

X2 links can be set up automatically or manually. Automatic setup is recommended. For details about X2 configurations, see *X2 Self-Management in SA Networking*. X2 self-setup is expected to be triggered on the LTE side within half an hour if the following conditions are met:

- The **CARRIER_SHUTDOWN_OVERLAP_FLAG** option of the **NrNRelationship.AggregationAttribute** parameter is selected for LTE cells.
- LTE and NR intelligent carrier shutdown or LNR load-based dormancy is enabled.
- The switches for LTE-NR inter-base-station power saving and X2 setup in SA networking are turned on. The specific switches are as follows:
 - LTE-NR SA gNodeB X2 setup switch on the LTE side, controlled by the **LNR_SA_GNB_X2_SETUP_SW** option of the **eNodeBOamParam.InterfaceSonConfigSwitch** parameter
 - LTE-NR SA inter-base-station power saving switch on the LTE side, controlled by the **LNR_INTER_BS_PWR_SAVE_SW** option of the **EnodebPwrSavingAlgo.PwrSavingAlgoSwitch** parameter
 - SA X2 setup switch on the NR side, controlled by the **X2_SETUP_FOR_SA_SW** option of the **gNBInterfaceParam.X2ConfigSw** parameter
 - LTE-NR SA inter-base-station power saving switch on the NR side, controlled by the **SA_LNR_INTER_BS_PWR_SAVE_SW** option of the **gNBPwrSavingAlgo.PwrSavingAlgoSwitch** parameter
- No X2 link is available between the eNodeB and the target gNodeB.

LampSite base stations do not support this change.

This change is supported only in low frequency bands.

This change is supported only in SA networking.

Implementation

For details, see *Multi-RAT Coordinated Energy Saving*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

This function is implemented on the gNodeB and eNodeB.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	eNodeBOamParam.InterfaceSonConfigSwitch	MOD ENODEBO AMPARAM LST ENODEBO AMPARAM	eNodeB	The LNR_SA_GNB_X2_SETUP_SW option has been added to specify whether to enable X2 setup between the eNodeB and gNodeBs that do not serve any NSA cells.
Modified	EnodebPwrSavingAlgo.PwrSavingAlgoSwitch	MOD ENODEBP WRSAVING ALGO LST ENODEBP WRSAVING ALGO	eNodeB	The LNR_INTER_BS_PWR_SAVE_SW option has been added to specify whether the eNodeB with LTE and NR power saving enabled can initiate an X2 setup with a gNodeB that meets the following conditions: (1) No X2 interface has been set up between the eNodeB and gNodeB. (2) The gNodeB serves cells with the overlapping coverage indicator for carrier shutdown configured.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNBInterfaceParam.X2ConfigSw	MOD GNBINTERFACEPARAM LST GNBINTERFACEPARAM	gNodeB	The X2_SETUP_FOR_SA_SW option has been added to specify whether the base station allows X2 link setup in SA networking.
Modified	gNBPwrSavingAlgo.PwrSavingAlgoSwitch	MOD GNBPWRSAVINGALGO LST GNBPWRSAVINGALGO	gNodeB	The SA_LNR_INTER_BS_PWR_SAVE_SW option has been added to specify whether SA NR cells can be engaged in inter-base-station LTE and NR energy saving.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Multi-RAT Coordinated Energy Saving*.

Requirement/Trouble Ticket Number

FeR-IR20250103001347-0010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD

5.3.2.18.2 Added Wake-up Algorithm Optimization for RF Module Super Dormancy

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Wake-up algorithm optimization for RF module super dormancy is controlled by the **RF_SDWAO_SW** option of the **EQUIPMENT.EQMDOROPTFEATSW** parameter. If this option is selected, the base station optimizes the algorithm for RF module wake-up. When an RF module exits the dormancy state, services can be restored without restarting the RF module, thereby shortening the service restoration duration of the RF module. This function must be configured under the guidance of Huawei engineers.

LampSite base stations do not support this feature.

Implementation

For details, see *Multi-RAT Coordinated Energy Saving*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

[Table 5-25](#) lists the RF modules that are compatible with wake-up algorithm optimization for RF module super dormancy.

Table 5-25 RF modules compatible with wake-up algorithm optimization for RF module super dormancy

RF Module	BOM Number
AAU5685md	02315FWJ

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following MML commands have changed.

Change Type	MML Command	NE	Description
Modified	DSP HWCAP	gNodeB eNodeB NodeB eGBTS	The Board Quick Wake-up Capb field has been added to display RF module information.

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	EQUIPMENT .EQMDORO PTFEATSW (LTE eNodeB, 5G gNodeB)	SET EQUIPMENT LST EQUIPMENT	gNodeB eNodeB NodeB eGBTS	The RF_SDWAO_SW option has been added to specify whether to enable wake-up algorithm optimization for RF module super dormancy. You are advised to perform configuration under the guidance of Huawei engineers.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.

- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Multi-RAT Coordinated Energy Saving*.

Requirement/Trouble Ticket Number

FeR-IR20250117000492-0010

RAT

NR FDD, NR TDD

5.3.3 Revised Features

5.3.3.1 FBFD-010007 Scalable Bandwidth

5.3.3.1.1 Support for the Setup of 30 MHz NR TDD Cells in the n38 Frequency Band

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

30 MHz NR TDD cells now can be set up in the n38 frequency band.

Only macro base stations support this change.

Implementation

For details, see *Scalable Bandwidth*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

For details about whether RF modules support the setup of 30 MHz NR TDD cells in the n38 frequency band, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature is not under license control.
- Configuration management
The change in this feature has no impact.
- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Scalable Bandwidth*.

Requirement/Trouble Ticket Number

FeR-IR20251211000241-0001

RAT

NR TDD

5.3.3.2 FBFD-010009 Channel Management

5.3.3.2.1 Added mmWave PDCCH Coverage Enhancement (for High Frequency Bands Only)

NEs Involved

5900 series base stations

Feature/Function Description

If the **NRDUCell.SlotAssignment** parameter is set to **2_3_DSUUU** for a high-frequency TDD cell under weak coverage and light load, UEs will experience PDCCH miss-detection, resulting in limited PDCCH coverage. To address this issue, mmWave PDCCH coverage enhancement is introduced and is controlled by the **MMW_PDCCH_COVERAGE_ENH_SW** option of the **NRDUCellPdcch.PdcchAlgoExtSwitch** parameter. With this function enabled, the number of configured blind detection times for the maximum PDCCH CCE aggregation level of 8 is increased, enhancing PDCCH coverage performance.

This function applies only to macro base stations, not to LampSite base stations. This function is supported only in high frequency bands.

Implementation

For details, see *Channel Management*. The specific sections are listed in the change history of the document.

Impact on the Network

After mmWave PDCCH coverage enhancement takes effect in weak-coverage and light-load scenarios, the performance of UEs under weak coverage improves, but the uplink and downlink CCE allocation success rates and the average uplink and downlink cell throughputs may decrease.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature is not under license control.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellPdcch.PdcchAlgoExtSwitch	MOD NRDUCELLPDCCH LST NRDUCELLPDCCH	gNodeB	The MMW_PDCCH_COVERAGE_ENH_SW option has been added to specify whether to enable mmWave PDCCH coverage enhancement.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Number of RBs for common control resources	NRDUCellCoreset.CommonCtrlResRbNum	None	mmWave PDCCH coverage enhancement (controlled by the MMW_PDCCH_COVERAGE_ENH_SW option of the NRDUCellPdcch.PdcchAlgoExtSwitch parameter) takes effect only when the NRDUCellCoreset.CommonCtrlResRbNum parameter is set to RB24 or RB48 .

- Mutually exclusive features/functions
The change in this feature has no impact.

- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Channel Management*.

Requirement/Trouble Ticket Number

DTS2025062711473

IR20241209002208

RAT

NR TDD

5.3.3.2.2 Added UE-specific CSI-RS Reconfiguration Reduction

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In low frequency bands, changes in the CSI-RS measurement results carried on the PUCCH during UE handovers can trigger repeated reconfigurations of UE-specific CSI-RS resources, increasing the uplink BLER and affecting user experience. To address this issue, UE-specific CSI-RS reconfiguration reduction is introduced, and is controlled by the **USER_CSI_RECFG_TIMES_RDC_SW** option of the **NRDUCellCsirs.CsRsExtSwitch** parameter. When enabled, this function reduces the number of UE-specific CSI-RS reconfigurations caused by changes in the $n1-n2$ value for the 8-port Type I codebook. For details about the $n1-n2$ value for the 8-port Type I codebook, see section 5.2.2.2.1 "Type I Single-Panel Codebook" in 3GPP TS 38.214 V15.16.0.

UE-specific CSI-RS reconfiguration reduction requires that the **NRDUCell.NrDuCellNetworkingMode** parameter not be set to **HYPER_CELL_COMBINE_MODE** or **HYPER_CELL**.

Implementation

For details, see *Channel Management*. The specific sections are listed in the change history of the document.

Impact on the Network

After this function takes effect, uplink retransmissions due to reconfigurations decrease, along with the uplink IBLER and RBLER.

- Uplink RBLER = $(N.UL.SCH.HalfPiBPSK.ErrTB.Rbler + N.UL.SCH.QPSK.ErrTB.Rbler + N.UL.SCH.16QAM.ErrTB.Rbler + N.UL.SCH.64QAM.ErrTB.Rbler + N.UL.SCH.256QAM.ErrTB.Rbler) / (N.UL.SCH.HalfPiBPSK.TB + N.UL.SCH.QPSK.TB + N.UL.SCH.16QAM.TB + N.UL.SCH.64QAM.TB + N.UL.SCH.256QAM.TB) \times 100\%$
- Uplink IBLER = $(N.UL.SCH.HalfPiBPSK.ErrTB.Ibler + N.UL.SCH.QPSK.ErrTB.Ibler + N.UL.SCH.16QAM.ErrTB.Ibler + N.UL.SCH.64QAM.ErrTB.Ibler + N.UL.SCH.256QAM.ErrTB.Ibler) / (N.UL.SCH.HalfPiBPSK.TB + N.UL.SCH.QPSK.TB + N.UL.SCH.16QAM.TB + N.UL.SCH.64QAM.TB + N.UL.SCH.256QAM.TB) \times 100\%$

Impact on NEs

UEs must support 8-port CSI-RS measurement on SCCs by reporting the *maxNumberTxPortsPerResource* IE with values that include **p8**.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature is not under license control.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellCsirs.CsRsExtSwitch	MOD NRDUCELLCSIRS LST NRDUCELLCSIRS	gNodeB	The USER_CSI_RECFG_TIMES_REDC_SW option has been added to specify whether to enable UE-specific CSI-RS reconfiguration reduction.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Channel Management*.

Requirement/Trouble Ticket Number

FeR-IR20241209002193-0095

RAT

NR FDD, NR TDD

5.3.3.2.3 Added Multi-BWP1 Ratio Statistics Optimization

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In low frequency bands, when the multi-BWP1 PDCCH transmission algorithm (controlled by the **PDCCH_MULTI_BWP1_ALGO_SW** option of the **NRDUCellPdcch.PdcchAlgoEnhSwitch** parameter) is enabled, the number of PDCCH symbols is adaptively adjusted based on the CCE usages corresponding to individual BWP1s. To prevent UE data scheduling failures in a cell serving a small number of UEs with limited service types (for example, only uplink or downlink services), multi-BWP1 ratio statistics optimization can be enabled by selecting the **MULTI_BWP1_RATIO_STAT_OPT_SW** option of the **NRDUCellPdcch.PdcchAlgoSwitch** parameter. This function optimizes the method of collecting statistics for PDCCH CCE ratio adaptation in multi-BWP1 scenarios.

Implementation

For details, see *Channel Management*. The specific sections are listed in the change history of the document.

Impact on the Network

After this function takes effect, the uplink and downlink CCE allocation success rates and the average uplink and downlink UE throughputs increase in cells with limited service types but may fluctuate in other scenarios.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellPdcch.PdcchAlgoSwitch	MOD NRDUCCELLP DCCH LST NRDUCCELLP DCCH	gNodeB	The MULTI_BWP1_RATIO_STAT_OPT_SW option has been added to specify whether to enable multi-BWP1 ratio statistics optimization.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Multi-BWP1 PDCCH transmission algorithm	PDCCH_MULTI_BWP1_ALGO_SW option of the NRDUCellPdcch.AlgoEnhSwitch parameter	<i>Channel Management</i>	Multi-BWP1 ratio statistics optimization (controlled by the MULTI_BWP1_RATIO_STAT_OPT_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter) takes effect only when the multi-BWP1 PDCCH transmission algorithm is enabled.

- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Channel Management*.

Requirement/Trouble Ticket Number

DTS2025082227892

IR20250528001857

RAT

NR FDD, NR TDD

5.3.3.2.4 Added Refined Control of Format-1 Resources

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In low frequency bands, to optimize format-1 RB allocation on a BBP by freeing up some RBs previously occupied by format 1, refined control of format-1 resources can be enabled by selecting the **F1_RES_REFINED_CTRL_SW** option of the **NRDUCellPucch.PucchAlgoExtSwitch** parameter. This function enables refined management of format-1 resources based on the number of activated cells and the number of configured format-1 RBs on the BBP.

Table 5-26 describes the upgrade rules for refined control of format-1 resources.

Table 5-26 Function upgrade rules

NRDUCellRsvdExt01.RsvdSwParam2	Upgrade from a Version Between V100R021C10SPC210 (Inclusive) and V100R022C00SPC010 (Exclusive) to V100R022C10SPC010 or Later	Upgrade from V100R022C00SPC100 or Later to V100R022C10SPC010 or Later	Other Upgrade Scenarios
RSVDSWPAR AM2_BIT7 set to 1	The function is enabled after the upgrade.	The function is enabled after the upgrade.	The function is disabled after the upgrade.
RSVDSWPAR AM2_BIT7 set to 0	The function is disabled after the upgrade.	The function is disabled after the upgrade.	The function is disabled after the upgrade.

Implementation

For details, see *Channel Management*. The specific sections are listed in the change history of the document.

Impact on the Network

After this function takes effect, some RBs previously occupied by format 1 are freed up, reducing the RBs used to transmit HARQ-ACKs with two or fewer bits. This allows the cell to serve more UEs transmitting HARQ-ACKs with more than

two bits, increasing the number of successful PUCCH HARQ-ACK resource requests (**N.PUCCH.ACK.AllocReq.Succ**) and the average downlink cell and UE throughputs.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellRsvdExt01.RsvdSwParam2	LST NRDUCELLR SVDEXT01 MOD NRDUCELLR SVDEXT01	gNodeB	RSVDSWPARAM2_BIT7 of the reserved parameter NRDUCellRsvdExt01.RsvdSwParam2 has been added to the parameter disuse list and replaced with the F1_RES_REFINED_CTRL_SW option of the NRDUCellPucch.PucchAlgoExtSwitch parameter.
Modified	NRDUCellPucch.PucchAlgoExtSwitch	MOD NRDUCELLP UCCH LST NRDUCELLP UCCH	gNodeB	The F1_RES_REFINED_CTRL_SW option has been added to specify whether to enable refined control of format-1 resources.

NOTE

NRDUCellRsvdExt01.RsvdSwParam2 is a reserved parameter.
For details about reserved parameters, see 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Channel Management*.

Requirement/Trouble Ticket Number

FeR-IR20250528001856-0001

RAT

NR FDD, NR TDD

5.3.3.2.5 Added HARQ Resource Increase

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In low frequency bands, to increase the number of format-3 and format-4 RBs in a cell serving UEs that are transmitting a high volume of data, HARQ resource increase can be enabled by selecting the **HARQ_SPEC_INCREASE_SW** option of the **NRDUCellPucch.PucchAlgoExtSwitch** parameter. This function allows the cell to use more format-3 and format-4 resources for HARQ feedback.

[Table 5-27](#) describes the upgrade rules for HARQ resource increase.

Table 5-27 Function upgrade rules

NRDUCellRsvdExt01.RsvdSwParam0	Upgrade from a Version Between V100R021C10SPC210 (Inclusive) and V100R022C00SPC010 (Exclusive) to V100R022C10SPC010 or Later	Upgrade from V100R022C00SPC100 or Later to V100R022C10SPC010 or Later	Other Upgrade Scenarios
RSVDSWPAR AMO_BIT21 set to 1	The function is enabled after the upgrade.	The function is enabled after the upgrade.	The function is disabled after the upgrade.
RSVDSWPAR AMO_BIT21 set to 0	The function is disabled after the upgrade.	The function is disabled after the upgrade.	The function is disabled after the upgrade.

Implementation

For details, see *Channel Management*. The specific sections are listed in the change history of the document.

Impact on the Network

After this function takes effect, additional format-3 and format-4 resources are available for HARQ feedback, increasing the RBs used to transmit HARQ-ACKs with more than two bits. This allows the cell to serve more UEs transmitting such HARQ-ACKs, increasing the number of successful PUCCH HARQ-ACK resource requests (**N.PUCCH.ACK.AllocReq.Succ**) and the average downlink cell and UE throughputs.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

- Base station models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

- Boards
All NR-capable main control boards and baseband processing units (except UBBPfw series) support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- RF modules
None

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellRsvdExt01.RsvdSwParam0	MOD NRDUCELLR SVDEXT01 LST NRDUCELLR SVDEXT01	gNodeB	RSVDSWPARAM0_BIT21 of the reserved parameter NRDUCellRsvdExt01.RsvdSwParam0 has been added to the parameter disuse list and replaced with the HARQ_SPEC_INCREASE_SW option of the NRDUCellPucch.PucchAlgoExtSwitch parameter.
Modified	NRDUCellPucch.PucchAlgoExtSwitch	MOD NRDUCELLP UCCH LST NRDUCELLP UCCH	gNodeB	The HARQ_SPEC_INCREASE_SW option has been added to specify whether to enable HARQ resource increase.

NOTE

NRDUCellRsvdExt01.RsvdSwParam0 is a reserved parameter.

For details about reserved parameters, see 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Channel Management*.

Requirement/Trouble Ticket Number

FeR-IR20250528001856-0001

RAT

NR FDD, NR TDD

5.3.3.2.6 Added Maximized Search Within HARQ Resource Set 1

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In low frequency bands, to increase the probability of successful format-3 and format-4 RB allocation for ACKs in a cell with limited RB resources, maximized search within HARQ resource set 1 can be enabled by selecting the **HARQ_RES_SET1_MAX_SRCH_SW** option of the **NRDUCellPucch.PucchAlgoExtSwitch** parameter. With this function, the base station searches for resources in HARQ resource set 1 for multiple times for UEs requesting format-3 and format-4 resources for HARQ feedback, increasing the request success rate.

[Table 5-28](#) describes the upgrade rules for maximized search within HARQ resource set 1.

Table 5-28 Function upgrade rules

NRDUCellRsvdExt01.RsvdSwParam1	Upgrade from a Version Between V100R021C10SPC210 (Inclusive) and V100R022C00SPC010 (Exclusive) to V100R022C10SPC010 or Later	Upgrade from V100R022C00SPC100 or Later to V100R022C10SPC010 or Later	Other Upgrade Scenarios
RSVDSWPAR AM1_BIT32 set to 1	The function is enabled after the upgrade.	The function is enabled after the upgrade.	The function is disabled after the upgrade.
RSVDSWPAR AM1_BIT32 set to 0	The function is disabled after the upgrade.	The function is disabled after the upgrade.	The function is disabled after the upgrade.

Implementation

For details, see *Channel Management*. The specific sections are listed in the change history of the document.

Impact on the Network

After this function takes effect, the probability of successful format-3 and format-4 RB allocation for ACKs increases, resulting in more successful PUCCH HARQ-ACK resource requests (**N.PUCCH.ACK.AllocReq.Succ**) and more RBs used to transmit HARQ-ACKs with more than two bits. This allows the cell to serve more UEs transmitting such HARQ-ACKs, increasing the average downlink cell and UE throughputs.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management

The change in this feature has no impact. It is controlled by the existing feature license.

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellRsvdExt01.RsvdSwParam1	MOD NRDUCELLR SVDEXT01LST NRDUCELLR SVDEXT01	gNodeB	RSVDSWPARAM1_BIT32 of the reserved parameter NRDUCellRsvdExt01.RsvdSwParam1 has been added to the parameter disuse list and replaced with the HARQ_RES_SET1_MAX_SRCH_SW option of the NRDUCellPucch.PucchAlgoExtSwitch parameter.
Modified	NRDUCellPucch.PucchAlgoExtSwitch	MOD NRDUCELLP UCCH LST NRDUCELLP UCCH	gNodeB	The HARQ_RES_SET1_MAX_SRCH_SW option has been added to specify whether to enable maximized search within HARQ resource set 1.

NOTE

NRDUCellRsvdExt01.RsvdSwParam1 is a reserved parameter.

For details about reserved parameters, see 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List.

- Performance management

The change in this feature has no impact.

- Fault management

The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

The change in this feature has no impact.

- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Channel Management*.

Requirement/Trouble Ticket Number

FeR-IR20250528001856-0001

RAT

NR FDD, NR TDD

5.3.3.2.7 Added CORESET 0 Offset Policy Configuration

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In low-frequency TDD, to flexibly configure the frequency-domain position of CORESET 0 relative to the SSB when uplink RBs are insufficient, the CORESET 0 offset policy can be configured by setting the **NRDUCellPdcchAlgo.Coreset0OffsetPol** parameter to **EDGE_FIRST**. With this parameter setting, the offset of the frequency-domain position of CORESET 0 relative to the center frequency of the SSB is adjusted to a protocol-defined value that makes CORESET 0 closest to the frequency-domain edge of the cell.

Implementation

For details, see *Channel Management*. The specific sections are listed in the change history of the document.

Impact on the Network

After this function takes effect, the offset of the frequency-domain position of CORESET 0 relative to the center frequency of the SSB is adjusted to a protocol-defined value that makes CORESET 0 closest to the frequency-domain edge of the cell. This enables PUCCH position multiplexing, eliminating the need for extra PUCCH resources and preserving available PUSCH resources. As a result, the average uplink and downlink cell throughputs may fluctuate.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellPdcchAlgo .Coreset0OffsetPol	MOD NRDUCCELLP DCCHALGO LST NRDUCCELLP DCCHALGO	gNodeB	This parameter has been added to configure the CORESET 0 offset policy.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The following mutually exclusive relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell-level PUSCH interference avoidance	DL_UE_LEVEL_INTRF_AVOID and UL_CELL_LEVEL_INTRF_AVOID options of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter	<i>Interference Avoidance</i>	When the CORESET 0 offset policy is configured (with the NRDUCellPdcchAlgo.Coreset0OffsetPol parameter set to EDGE_FIRST), neither of the following functions can be enabled: • Uplink cell-level interference avoidance (controlled by the UL_CELL_LEVEL_INTRF_AVOID option of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter) • Downlink UE-level interference avoidance (controlled by the DL_UE_LEVEL_INTRF_AVOID option of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter)

- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Channel Management*.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0071

RAT

NR TDD

5.3.3.2.8 Added CSI Measurement Optimization

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In low frequency bands, when certain functions (such as distributed massive MIMO and VMIMO in DAS-based indoor coverage) take effect and aperiodic CSI-RS resources are insufficient for UEs in non-overlapping areas, CSI measurement optimization can be enabled to adaptively adjust periodic CSI-RS resources, preventing CSI measurement failures for UEs in these areas. This function is controlled by the **CSI_MEAS_OPT_SW** option of the **NRDUCellCsirs.CsiRsExtSwitch** parameter.

It is recommended that CSI measurement optimization be enabled under any of the following conditions:

- The **DM_MIMO_SERVICE_SWITCH** option of the **NRDUCellAlgoSwitch.DmMimoSwitch** parameter is selected.
- The **NRDUCellAlgoSwitch.VmimoSwitch** parameter is set to **ON**.
- The **RF_COMBINING_NOISE_SUPPR_SW** option of the **NRDUCellUISch.RfCombSwitch** parameter is selected.
- The **DISTRIBUTED_ANT_PMI_WT_OPT_SW** option of the **NRDUCellAlgoSwitch.BeamOptSwitch** parameter is selected.

Implementation

For details, see *Channel Management*. The specific sections are listed in the change history of the document.

Impact on the Network

After this function takes effect, the base station adaptively allocates aperiodic or periodic CSI-RS resources to UEs in non-overlapping areas, leading to fluctuations in the CSI measurement result and average downlink UE throughput.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCell Csirs.Csirs ExtSwitch	MOD NRDUCELLC SIRS LST NRDUCELLC SIRS	gNodeB	The CSI_MEAS_OPT_SW option has been added to specify whether to enable CSI measurement optimization.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Channel Management*.

Requirement/Trouble Ticket Number

IR20250528001857

DTS2025091929206

RAT

NR FDD, NR TDD

5.3.3.2.9 Added Equivalent-CCE-Usage-based PDCCH Symbol Adaptation

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In low frequency bands, when PDCCH symbol number adaptation (controlled by the **UE_PDCCH_SYM_NUM_ADAPT_SW** option of the **NRDUCellPDCCH.PdcchAlgoExtSwitch** parameter) is enabled and CCE resources are insufficient, equivalent-CCE-usage-based PDCCH symbol adaptation can be enabled by selecting the **EQUIV_CCE_RATIO_SYM_ADPT_SW** option of the **NRDUCellPDCCH.PdcchAlgoExtSwitch** parameter. This function adaptively adjusts the number of PDCCH symbols based on the equivalent CCE usage. If the equivalent CCE usage in the slots where DCI for uplink scheduling is transmitted is greater than or equal to the equivalent PDCCH CCE usage threshold, PDCCH symbol adaptation is triggered to increase the number of PDCCH symbols. The equivalent PDCCH CCE usage threshold is specified by the **NRDUCellPdcchAlgo.PdcchEquivCceRatioThld** parameter, and the equivalent CCE usage in the slots where DCI for uplink scheduling is transmitted is equal to **N.CCE.UL.Used.Equivalent** divided by **N.CCE.UL.Avail.Equivalent**.

Implementation

For details, see *Channel Management*. The specific sections are listed in the change history of the document.

Impact on the Network

After this function takes effect, an increase in PDCCH symbols allows more UEs to be scheduled, increasing the uplink and downlink CCE allocation success rates. However, this comes at the cost of fewer PDSCH symbols, which may lead to fluctuations in the average uplink and downlink cell throughputs.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management

The change in this feature has no impact. It is controlled by the existing feature license.

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellPdcchAlgo <i>.PdcchEquivalentCceRatioThld</i>	MOD NRDUCCELLP DCCHALGO LST NRDUCCELLP DCCHALGO	gNodeB	This parameter has been added to specify the threshold expressed as an equivalent PDCCH CCE usage used to activate equivalent-CCE-usage-based PDCCH symbol adaptation.
Modified	NRDUCellPdcch.PdcchAlgoSwitch	MOD NRDUCCELLP DCCH LST NRDUCCELLP DCCH	gNodeB	The EQUIV_CCE_RATIO_SYM_ADPT_SW option has been added to specify whether to enable equivalent-CCE-usage-based PDCCH symbol adaptation.

- Performance management

The change in this feature has no impact.

- Fault management

The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

The change in this feature has no impact.

- Mutually exclusive features/functions

The change in this feature has no impact.

- Impacted features/functions

The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Channel Management*.

Requirement/Trouble Ticket Number

FeR-IR20241212002296-0002

RAT

NR FDD, NR TDD

5.3.3.2.10 Added Fast Aggregation Level Reduction

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In low frequency bands, if CCE resources are insufficient, fast aggregation level reduction can be enabled by selecting the **AGG_LEVEL_FAST_DECREASE_SW** option of the **NRDUCellPdcch.PdcchAlgoSwitch** parameter. This function quickly reduces the aggregation levels when the increase in the PDCCH power is less than or equal to the power threshold for fast aggregation level reduction. This allows lower aggregation levels to be selected for UEs in a cell, improving the PDCCH capacity of the cell.

The power threshold for fast aggregation level reduction is specified by the **NRDUCellPdcchAlgo.AggLvlFastDcrPwrThld** parameter, and the increase in the PDCCH power is determined by the PDCCH CCE aggregation level and PDCCH detection result.

Implementation

For details, see *Channel Management*. The specific sections are listed in the change history of the document.

Impact on the Network

After this function takes effect, UEs are more likely to be assigned lower aggregation levels. This increases the uplink and downlink CCE allocation success rates, but may also increase the uplink and downlink RBLERs and decrease the average uplink and downlink cell throughputs.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellPdcchAlgo <i>.AggLvlFastDcrPwrThld</i>	MOD NRDUCCELLP DCCHALGO LST NRDUCCELLP DCCHALGO	gNodeB	This parameter has been added to specify the threshold expressed as an increase in the PDCCH power for fast aggregation level reduction.
Modified	NRDUCellPdcch <i>.PdchAlgoSwitch</i>	MOD NRDUCCELLP DCCH LST NRDUCCELLP DCCH	gNodeB	The AGG_LEVEL_FAST_DECREASE_SW option has been added to specify whether to enable fast aggregation level reduction.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Channel Management*.

Requirement/Trouble Ticket Number

FeR-IR20241212002296-0002

RAT

NR FDD, NR TDD

5.3.3.2.11 Added DTX Filtering

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In low frequency bands, if CCE resources are insufficient, DTX filtering can be enabled by selecting the **DTX_FILTER_SW** option of the **NRDUCellPdcch.PdcchAlgoSwitch** parameter. This function excludes certain DTX occasions from aggregation level adjustment, allowing lower aggregation levels to be selected for UEs in a cell and improving the PDCCH capacity of the cell.

Implementation

For details, see *Channel Management*. The specific sections are listed in the change history of the document.

Impact on the Network

After this function takes effect, UEs are more likely to be assigned lower aggregation levels. This increases the uplink and downlink CCE allocation success rates, but may also increase the uplink and downlink RBLERs and decrease the average uplink and downlink cell throughputs.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellPdcch.PdcchAlgoSwitch	MOD NRDUCELLP DCCH LST NRDUCELLP DCCH	gNodeB	The DTX_FILTER_SW option has been added to specify whether to enable DTX filtering.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Channel Management*.

Requirement/Trouble Ticket Number

FeR-IR20241212002296-0002

RAT

NR FDD, NR TDD

5.3.3.2.12 Added PUCCH Symbol Length Number Increase

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

To minimize signal interference, PUCCH signals can be punctured through resource mapping. For details about puncturing, see section 6.3.2.4.1 "Sequence

modulation" in 3GPP TS 38.211 V18.6.0. As more features and functions evolve, the base station is expected to support additional puncturing schemes. To enable this, the PUCCH symbol length number increase function is introduced. This function is controlled by the **PUCCH_SYM_TYPE_NUM_INCR_SW** option of the **NRDUCellPucch.PucchAlgoSwitch** parameter. It increases the number of configurable PUCCH puncturing schemes in each TTI, allowing support for more PUCCH symbol lengths per TTI.

Implementation

For details, see *Channel Management*. The specific sections are listed in the change history of the document.

Impact on the Network

After this function takes effect, more PUCCH puncturing schemes are allowed in each TTI, preventing some UEs from ACK feedback or CSI reporting failures and causing the following impacts on indicators.

- The **N.RLC.DL.Cell5QI.Delay**, **N.MAC.DL.Cell5QI.Delay**, and **N.DL.PDSCH.Tti.Num** counter values and the uplink and downlink IBLERs and RBLERs remain unchanged or slightly decrease.
- The number of times the measured PUCCH SINR values fall into different ranges fluctuates.
- The average uplink and downlink cell throughputs and average uplink and downlink UE throughputs remain unchanged or slightly increase.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellPucch.PdcchAlgoSwitch	MOD NRDUCCELLPUCCH LST NRDUCCELLPUCCH	gNodeB	The PUCCH_SYM_TYPE_NUM_INCR_SW option has been added to specify whether to enable PUCCH symbol length number increase.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Channel Management*.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0110

RAT

NR FDD, NR TDD

5.3.3.2.13 Added Optimized PDCCH Symbol Adaptation

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In low frequency bands, when PDCCH symbol number adaptation (controlled by the **UE_PDCCH_SYM_NUM_ADAPT_SW** option of the **NRDUCellPDCCH.PdcchAlgoExtSwitch** parameter) is enabled in extreme CCE

allocation scenarios, it may be difficult to adaptively adjust the number of PDCCH symbols. In such cases, optimized PDCCH symbol adaptation can be enabled by selecting the **PDCCH_SYM_ADAPT_OPT_SW** option of the **NRDUCellPDCCH.PdcchAlgoExtSwitch** parameter. This function optimizes the triggering conditions for PDCCH symbol number adaptation to increase the probability that PDCCH symbol number adaptation takes effect in preceding extreme scenarios under light load.

Implementation

For details, see *Channel Management*. The specific sections are listed in the change history of the document.

Impact on the Network

After this function takes effect in extreme CCE allocation scenarios under light load, the probability of reducing the number of PDCCH symbols increases. This reduction frees up more PDSCH symbols, which may lead to decreased CCE allocation success rate, increased CCE usage, and higher average downlink UE throughput.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellPDCCH.PdcchAlgoExtSwitch	MOD NRDUCCELLPDCCH LST NRDUCCELLPDCCH	gNodeB	The PDCCH_SYM_ADAPT_OPT_SW option has been added to specify whether to enable optimized PDCCH symbol adaptation.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Channel Management*.

Requirement/Trouble Ticket Number

IR20250102000700

DTS2025121508767

RAT

NR FDD, NR TDD

5.3.3.2.14 Removed the Cell Bandwidth Restriction on Format-1 SR Code Channel Allocation Optimization

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In low-frequency TDD, the restriction requiring the cell bandwidth to be at least 60 MHz has been removed from format-1 SR code channel allocation optimization.

Implementation

For details, see *Channel Management*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature is not under license control.
- Configuration management
The change in this feature has no impact.
- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250717005585-0001

RAT

NR TDD

5.3.3.3 FBFD-010011 Scheduling

5.3.3.3.1 DCCH Type1 Scheduling

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In low-frequency scenarios, DCCH type 1 scheduling (controlled by the **DCCH_TYPE1_SCH_SW** option of the **NRDUCellDISchRes.DISchResAlgoSw** parameter) is introduced. When the number of UEs accessing a cell is greater than the UE number threshold for DCCH type 1 scheduling to take effect (specified by the **NRDUCellDISchRes.DcchType1SchUserNumThld** parameter), downlink resources can be allocated to small-packet MAC CEs using type 1 resource allocation. This function applies to low-frequency TDD cells. It is recommended that this function be enabled in heavy-load scenarios.

Implementation

For details, see *Downlink Scheduling*. The specific sections are listed in the change history of the document.

Impact on the Network

- Benefits
In low-frequency heavy-load scenarios, this function may increase the **Cell Downlink Average Throughput (DU)** and **User Downlink Average Throughput (DU)**.
- Impacts
After this function is enabled, the downlink IBLER may increase, and the downlink PRB usage may decrease.
If this function is enabled in light-load scenarios, the values of the following indicators may decrease:
 - Average downlink cell throughput
 - **User Downlink Average Throughput (DU)**

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management

The change in this feature has no impact.

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellIDLSchRes.DLSchResAlgoSw	MOD NRDUCELLDLSCHRES LST NRDUCELLDLSCHRES	gNodeB	The DCCH_TYPE1_SCH_SW option has been added to specify whether to enable DCCH type 1 scheduling.
New	NRDUCellIDLSchRes.DcchType1SchUserNumThld	MOD NRDUCELLDLSCHRES LST NRDUCELLDLSCHRES	gNodeB	This parameter has been added to specify the UE number threshold for DCCH type 1 scheduling to take effect. When the number of UEs admitted to a cell is greater than this threshold, DCCH type 1 scheduling takes effect in the cell.

- Performance management

The change in this feature has no impact.

- Fault management

The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

The following dependency relationship has been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DCCH type 1 scheduling	DCCH_TYPE1_SCH_SW option of the NRDUCellIDLSchRes.DLSchResAlgoSw parameter	<i>Downlink Scheduling</i>	The NRDUCellIDLSchRes.DcchType1SchUserNumThld parameter specifies the UE number threshold for DCCH type 1 scheduling to take effect. This parameter takes effect only when DCCH type 1 scheduling is enabled.

- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Downlink Scheduling*.

Requirement/Trouble Ticket Number

FeR-IR20250528001856-0012

RAT

NR FDD, NR TDD

5.3.3.3.2 Optimized Scheduling of Tail RBGs

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In low-frequency scenarios, optimized scheduling of tail RBGs (controlled by the **TAIL_RBG_SCH_OPT_SW** option of the **NRDUCellDLSchRes.DLSchResAlgoSw** parameter) is introduced to allocate the last RBG of the PDSCH to an appropriate UE, thereby reducing data packet truncation caused by insufficient resources. This function applies to low-frequency TDD cells.

Implementation

For details, see *Downlink Scheduling*. The specific sections are listed in the change history of the document.

Impact on the Network

- Benefits
After this function is enabled in low-frequency scenarios, the proportion of samples with low downlink average UE throughput in the cell may decrease. (This proportion = $(N.Thp.DL.Samp.Index0 + N.Thp.DL.Samp.Index1) / \sum(N.Thp.DL.Samp.Indexk)$, where k is in the range of 0 to 14)
- Impacts
After this function is enabled, the delay of the first packet may increase, and the downlink PRB usage may decrease.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact.
- Configuration management
The following parameter has changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellIDLSchRes.DLSchResAlgoSw	MOD NRDUCELLD LSCHRES LST NRDUCELLD LSCHRES	gNodeB	The TAIL_RBG_SCH_OPT_SW option has been added to specify whether to enable optimized scheduling of tail RBGs.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Downlink Scheduling*.

Requirement/Trouble Ticket Number

FeR-IR20250528001856-0001

RAT

NR FDD, NR TDD

5.3.3.3 Uplink Scheduling Gain Optimization

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Uplink scheduling gain optimization is introduced to allow uplink UEs to be scheduled on the optimal RBs, further increasing the average uplink UE throughput. After this function is enabled, the gNodeB adjusts the resource allocation mode of UEs and can allocate better RBs to UEs, thereby increasing the average uplink UE throughput. Uplink scheduling gain optimization is controlled by the **UL_SCH_GAIN_OPT_SW** option of the **NRDUCellPusch.PuschAlgoExtSwitch** parameter.

Implementation

For details, see *Uplink Scheduling*. The specific sections are listed in the change history of the document.

Impact on the Network

After this function takes effect, better RBs can be allocated to uplink UEs, increasing the average uplink UE throughput.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact.

- Configuration management

The following parameter has changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellPusch.PuschAlgoExtSwitch	MOD NRDUCCELLPUSCH LST NRDUCCELLPUSCH	gNodeB	The UL_SCH_GAIN_OPT_SW option has been added to specify whether to enable uplink scheduling gain optimization.

- Performance management

The change in this feature has no impact.

- Fault management

The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

The change in this feature has no impact.

- Mutually exclusive features/functions

The change in this feature has no impact.

- Impacted features/functions

The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Uplink Scheduling*.

Requirement/Trouble Ticket Number

IR20250102000700

RAT

NR FDD, NR TDD

5.3.3.4 FBFD-010020 Synchronization

5.3.3.4.1 Added Support for Clock Out-of-Synchronization Detection in Cells Covering Subways or High-Speed Railways

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Clock out-of-synchronization detection is now supported in cells covering subways or high-speed railways. This function can be enabled by selecting the **CLK_DETECT_IN_COMPLEX_SW** option of the **gNodeBParam.ClkOutofsyncDetSwitch** parameter.

Implementation

For details, see *Synchronization*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature is not under license control.
- Configuration management
The following parameter has changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNodeBParam.ClkOutofsyncDetSwitch	MOD GNODEBPARAM LST GNODEBPARAM	gNodeB	The CLK_DETECT_IN_COMPLEX_SW option has been added to specify whether to enable clock out-of-synchronization detection for cells covering subways or high-speed railways.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The following mutually exclusive relationship has been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Clock out-of-synchronization detection in complex network scenarios	CLK_DETECT_IN_COMPLEX_SW option of the gNodeBParam.ClkOutofsyncDetSwitch parameter	<i>Synchronization</i>	The switch controlling clock out-of-synchronization detection and the switch controlling clock out-of-synchronization detection in complex network scenarios cannot be both turned on.

- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Synchronization*.

Requirement/Trouble Ticket Number

FeR-IR20250410000439-0008

RAT

NR TDD

5.3.3.5 FBFD-010023 Security Mechanism

5.3.3.5.1 Support for Defense Against DDoS Attacks Triggered by SRB Scheduling Feedback Exceptions

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The function of defense against DDoS attacks triggered by SRB scheduling feedback exceptions is controlled by the **SRB_SCH_EXP_SW** option of the **gNBAirIntfSecParam.AirIntfAntiDdosSw** parameter. After this function is enabled, the gNodeB identifies UEs having SRB scheduling feedback exceptions, with the methods different between the detection and defense procedures of air interface DDoS attack that involves a single UE in connected mode and the detection procedures of air interface DDoS attack that involves multi-UE random access. The gNodeB then stops scheduling or releases the UEs having SRB scheduling feedback exceptions to implement the defense against DDoS attacks.

- In the detection and defense process of air interface DDoS attack that involves a single UE in connected mode, with the function of defense against DDoS attacks triggered by SRB scheduling feedback exceptions enabled, the gNodeB determines whether the UE has consecutive SRB scheduling feedback exceptions before setting up a DRB for the UE. If yes, the gNodeB considers the UE having SRB scheduling feedback exceptions and stops scheduling the UE.
- In the detection process of air interface DDoS attack that involves multi-UE random access, with the function of defense against DDoS attacks triggered by SRB scheduling feedback exceptions enabled, the gNodeB periodically checks the number of UEs having SRB scheduling feedback exceptions in a cell (the method for identifying UEs having SRB scheduling feedback exceptions is the same as that for the detection and defense procedures of air interface DDoS attack that involves a single UE in connected mode). When the number exceeds a certain threshold, the gNodeB considers that the cell is under a DDoS attack. When the number is below a certain threshold, the cell exits the DDoS attack state. When a cell is under a DDoS attack, the gNodeB performs the following operations on UEs in the cell: The gNodeB releases all UEs having SRB scheduling feedback exceptions. For a UE that subsequently initiates a random access procedure with Msg5 already sent to the gNodeB, the gNodeB checks whether the UE has SRB scheduling feedback exceptions based on the UE's "DedicatedNAS-Message" IE exchange with the CN. If yes, the gNodeB releases the UE.

Implementation

For details, see *Radio Security*. The specific sections are listed in the change history of the document.

Impact on the Network

Defense against DDoS attacks triggered by SRB scheduling feedback exceptions:
After this function is enabled, the UEs considered having SRB scheduling feedback exceptions may experience longer access delay and even service drops.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature is not under license control.
- Configuration Management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNBAirIntfSecParam.AirIntfAntiDdosSw	MOD GNBAIRINTFSE CPARAM LST GNBAIRINTFSE CPARAM	gNodeB	The SRB_SCH_EXP_SW option has been added to control the function of defense against DDoS attacks triggered by SRB scheduling feedback exceptions.

- Performance management
The change in this feature has no impact.

- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Radio Security*.

Requirement/Trouble Ticket Number

FeR-IR20250208000127-0032

RAT

NR FDD, NR TDD

5.3.3.5.2 Modified the Method of Calculating the Ratio of Emergency UEs in a Cell for the Function of "Preemption Ratio Restriction for Multi-UE DDoS Attacks over the Air Interface"

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

After the function of "preemption ratio restriction for multi-UE DDoS attacks over the air interface" is enabled (by selecting the **EMC_UE_PREEMPT_RATIO_LIMIT_SW** option of the **gNB AirIntfSecParam.AirIntfAntiDdosSw** parameter), if an emergency UE attempts to access a cell of a gNodeB and the number of admitted UEs in the cell has reached the upper limit, the gNodeB calculates the ratio of emergency UEs in the cell. If the ratio is greater than a specified threshold, the gNodeB rejects the UE's access requests.

The method of calculating the ratio of emergency UEs in a cell is now changed.

Before: Ratio of emergency UEs in a cell = Total number of emergency UEs in the cell / Total number of existing UEs in the cell

After: Ratio of emergency UEs in a cell = Total number of emergency UEs in the cell / Cell-level admission user number threshold for emergency UEs in the cell

Implementation

For details, see *Radio Security*. The specific sections are listed in the change history of the document.

Impact on the Network

After the change in this feature takes effect, the denominator for calculating the ratio of emergency UEs in a cell is smaller, and the preemption ratio restriction for multi-UE DDoS attacks over the air interface is more likely to take effect.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature is not under license control.
- Configuration Management
The change in this feature has no impact.
- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250528001856-0041

RAT

NR FDD, NR TDD

5.3.3.6 FBFD-020101 Reliability

5.3.3.6.1 Added the Function of Cell Outage Detection Based on Comprehensive Features

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Cell outage detection based on comprehensive features helps improve the accuracy of identifying an outage cell when the cell meets both of the following conditions: (1) there is no Msg3 message reported during a sustained period; (2) the criteria for cell outage detection based on zero access do not apply. This function is controlled by the **COMPR_FEATURE_DETECT_SW** option of the **gNBOamParam.CodrAlgoSwitch** parameter and applies to SA networking.

When cell outage detection based on comprehensive features is enabled, the gNodeB determines cell outage as follows:

1. The gNodeB filters ranges of hours during which a cell remains fully available across five consecutive days. (A cell is considered fully available if it is available for at least 3,595s per hour.) If the number of Msg3 messages (**N.RRC.SetupReq.Att**, representing the number of RRC connection setup requests) is 0 in the cell during the ranges of filtered hours, the cell is considered a suspected outage cell.
2. The gNodeB further determines whether a suspected outage cell is an outage cell based on the features listed in [Table 5-29](#).

Table 5-29 Feature items used for determination of an outage cell

Feature Item	Related Counter
System information	N.SI.Uu.Num
Cell uplink interference	N.UL.NI.Avg
Average downlink transmit power in a cell	N.DL.Pwr.Avg
Uplink sector RSSI	N.UL.RSSI.Avg

Feature Item	Related Counter
Downlink transmit power of RF units	VS.RADIOEQM.DL.Pwr.Avg
Number of seconds when CPRI port detection results are invalid	max(VS.RADIOEQM.CPRI0.UIVS, VS.RADIOEQM.CPRI0.DIVS, VS.RADIOEQM.CPRI1.UIVS, VS.RADIOEQM.CPRI1.DIVS)
Duration of successful channel calibration	N.TRP.Succ.Calib.Dur

The outage status of an outage cell is reported to the CODR module of the MAE-Access, with an outage reason of "Internal fault cell outage."

The gNodeB checks the number of Msg5 messages (**N.RRC.SetupReq.Succ**) every 5 minutes in the outage cell. If the number is greater than 0, the gNodeB determines that the outage cell has recovered. In this case, the MAE-Access removes the cell from the outage cell list.

Implementation

For details, see *Cell Outage Detection and Recovery*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature is implemented on the gNodeB and MAE-Access.

Impact on Hardware Requirements

- Boards
 - Baseband processing units: All NR-capable baseband processing units can be used.
 - Main control board: All NR-capable UMPTE, UMPTg, UMPTga, and UMPTha series boards can be used. In addition, UMPTE series boards must work in NR-only mode.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature is not under license control.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNBOampParam.CodrAlgoSwitch	MOD GNBOAMPA RAM LST GNBOAMPA RAM	gNodeB	The COMPR_FEATURE_DETECT_SW option has been added to specify whether to enable cell outage detection based on comprehensive features.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Cell Outage Detection and Recovery*.

Requirement/Trouble Ticket Number

FeR-IR20240815002936-0009

RAT

NR FDD, NR TDD

5.3.3.6.2 Added the Cell Outage Detection Based on Zero Access Function in NSA Networking

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The gNodeB identifies outage cells by detecting SgNB addition exceptions. This function is controlled by the **NSA_ZERO_ACCESS_DETECT_SW** option of the **gNB0amParam.CodrAlgoSwitch** parameter and applies to NSA networking.

When this option is selected, the function of cell outage detection based on zero access takes effect for the gNodeB. If the criteria described in [Table 5-30](#) are met, the gNodeB determines that the cell is in outage. The MAE-Access adds the outage cell to the outage cell list. Users can view this list on the MAE-Access. The cell is considered to be in outage before the gNodeB considers the cell returns to normal.

Table 5-30 Criteria for determining zero access

Scenario	Criteria
There are SgNB addition attempts but the number of SgNB addition acknowledgements is 0.	The gNodeB determines that the cell is in outage if all the following conditions are met for one consecutive hour during busy hours ^(a): • The number of SgNB addition attempts (indicated by the result of N.NsaDc.DRB.Add.Att + N.NsaDc.InterSgNB.SgNB.Add.Att) is greater than the threshold calculated by the gNodeB based on the number of SgNB addition attempts during busy hours in the cell. • Number of SgNB addition acknowledgements (N.NsaDc.SgNB.Add.Ack + N.NsaDc.InterSgNB.SgNB.Add.Ack) = 0 • Duration of cell availability ≥ 3595s
a: Busy hours are automatically calculated by the gNodeB based on N.PDCP.Vol.UL.TrfSDU.Tx and N.PDCP.Vol.DL.TrfSDU.Rx .	

When a cell is in outage, the gNodeB checks the number of SgNB addition acknowledgements (**N.NsaDc.SgNB.Add.Ack** + **N.NsaDc.InterSgNB.SgNB.Add.Ack**) in real time. If the number is greater than 0, the gNodeB determines that the outage cell has recovered. In this case, the MAE-Access removes the cell from the outage cell list.

Implementation

For details, see *Cell Outage Detection and Recovery*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature is not under license control.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNBOampParam.CodrAlgoSwitch	MOD GNBOAMPA RAM LST GNBOAMPA RAM	gNodeB	The NSA_ZERO_ACCESS_DETECT_SW option has been added to specify whether to enable cell outage detection based on zero access in NSA networking.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Cell Outage Detection and Recovery*.

Requirement/Trouble Ticket Number

FeR-IR20240815002936-0010

RAT

NR FDD, NR TDD

5.3.3.6.3 Optimized the Function of Cell Outage Detection Based on Neighboring-Cell UE Cooperation

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The function of "cell outage detection based on neighboring-cell UE cooperation" (controlled by the **NCELL_UE_COOP_DETECT_SW** option of the **gNBOamParam.CodrAlgoSwitch** parameter) accurately determines whether a cell without Msg1 for a long time (also called a suspected outage cell) is indeed in outage. For the function of "cell outage detection based on neighboring-cell UE cooperation" to determine whether a suspected outage cell is in outage, it is necessary to select the six intra-frequency neighboring cells with the greatest number of historical handovers to the suspected outage cell.

The algorithm for selecting those intra-frequency neighboring cells now changes to increase the selection success rate, thereby allowing for a higher success rate of determining outage cells.

For details about "cell outage detection based on neighboring-cell UE cooperation", see *Cell Outage Detection and Recovery*.

Implementation

The algorithm for selecting those intra-frequency neighboring cells now changes to increase the selection success rate, thereby allowing for a higher success rate of determining outage cells.

Impact on the Network

After the change in this feature takes effect, the function of "cell outage detection based on neighboring-cell UE cooperation" has a higher success rate of selecting the six intra-frequency neighboring cells with the greatest number of historical handovers to a suspected outage cell, allowing for a higher success rate of determining outage cells, while the value of the **Intra-Frequency Handover Out**

Success Rate (CU) counter may decrease slightly for the coordinating neighboring cells.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact.
- Configuration management
The change in this feature has no impact.
- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

Whether the change takes effect by default after an upgrade varies as follows:

- Yes: when the function of "cell outage detection based on neighboring-cell UE cooperation" (controlled by the **NCELL_UE_COOP_DETECT_SW** option of the **gNBOamParam.CodrAlgoSwitch** parameter) has taken effect before the upgrade
- No: when the function of "cell outage detection based on neighboring-cell UE cooperation" (controlled by the **NCELL_UE_COOP_DETECT_SW** option of the **gNBOamParam.CodrAlgoSwitch** parameter) has not taken effect before the upgrade

Requirement/Trouble Ticket Number

IR20250528001859

IR20250528001857

DTS2026010817550

DTS2026010817013

RAT

NR FDD, NR TDD

5.3.3.7 FBFD-020102 Radio QoS Management

5.3.3.7.1 PDU Session Resource Modify QCI Mapping Optimization

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

When the PDU Session Resource Modify compatibility optimization function (controlled by the **PDU_SESSION_MOD_COMPAT_OPT_SW** option of the **gNodeBParam.CompatibilityAlgoSwitch** parameter) is enabled, the gNodeB re-maps 5QIs to QCIs even if the 5QI values do not change after the core network delivers the *QoS Flow Add or Modify Request Item* IE to modify the QoS flow attributes. The remapping may interrupt some assurance features. Against this backdrop, the PDU Session Resource Modify QCI mapping optimization function (controlled by the **PDU_SESSION_MOD_QCI_MAP_OPT_SW** option of the **gNodeBParam.CompatibilityAlgoSwitch** parameter) is introduced. After this function is enabled and the *QoS Flow Add or Modify Request Item* IE is modified for a QoS flow, QCI mapping is re-performed only if the 5QI value changes, thereby avoiding service interruption caused by remapping.

The change in this feature is supported only in SA networking.

Implementation

For details, see *QoS Management*. The specific sections are listed in the change history of the document.

Impact on the Network

PDU Session Resource Modify QCI mapping optimization avoids service interruptions caused by remapping.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature is not under license control.
- Configuration management
The following parameter has changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNodeBParameter.CompatibilityAlgoSwitch	MOD GNODEBPARAM LST GNODEBPARAM	gNodeB	The PDU_SESSION_MOD_QCI_MAP_OPT_SW option has been added to specify whether to enable PDU Session Resource Modify QCI mapping optimization.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationship has been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	PDU Session Resource Modify compatibility optimization	PDU_SESSION_MOD_COMPAT_OPTION_SW option of the gNodeBParameter.CompatibilityAlgoSwitch parameter	<i>QoS Management</i>	PDU Session Resource Modify QCI mapping optimization can be enabled only when PDU Session Resource Modify compatibility optimization is enabled.

- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *QoS Management*.

Requirement/Trouble Ticket Number

IR20251204000006

DTS2025092641851

RAT

NR FDD, NR TDD

5.3.3.7.2 Support for Downlink PDB Guarantee Status Check in the Notification Control Function

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The notification control function (controlled by the **NOTIF_CTRL_SW** option of the **gNB5qiConfig.Nr5qiAlgoSwitch** parameter) enables the estimated guarantee status of GBR services to be reported to the core network, thereby supporting the core network in meeting differentiated requirements of GBR UEs.

Before the upgrade: In the notification control function, only the uplink and downlink GFBR guarantee status is checked, rather than the downlink PDB guarantee status, during the guarantee status check of GBR services.

After the upgrade: In the notification control function, the guarantee status of uplink and downlink GFBRs and the downlink PDB is checked during the guarantee status check of GBR services. If any of the uplink GFBR, downlink GFBR, or downlink PDB is estimated not to be guaranteed, the gNodeB reports a message indicating an estimated guarantee failure to the core network, thereby supporting the core network in meeting differentiated requirements of GBR UEs. The downlink PDB guarantee status check requires that **gNBQciBearer.QncArrangeType** be set to **DELAY_SENSITIVE**.

The change in this feature is supported only in low frequency bands in SA networking.

Implementation

For details, see *QoS Management*. The specific sections are listed in the change history of the document.

Impact on the Network

When the **gNBQciBearer.QncArrangeType** parameter is set to **DELAY_SENSITIVE**, the downlink PDB guarantee status can be checked. If any of the uplink GFBR, downlink GFBR, or downlink PDB cannot be guaranteed, the gNodeB reports a message indicating an estimated guarantee failure to the core network.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature is not under license control.
- Configuration management
The following parameter has changed.

Change Type	Parameter ID	MML Command	NE	Description
New	gNBQciBearer.QncArrangeType	ADD GNBQCIBEARER MOD GNBQCIBEARER LST GNBQCIBEARER	gNodeB	This parameter has been added. If this parameter is set to DELAY_SENSITIVE , the downlink PDB guarantee status is checked during the guarantee status check of GBR services in the notification control function.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *QoS Management*.

Requirement/Trouble Ticket Number

FeR-IR20251215002079-0010

RAT

NR FDD, NR TDD

5.3.3.8 FBFD-021104 SA Option 2 Architecture

5.3.3.8.1 Optimized Control-Plane Dual-Stack Alarm Reporting for the Xn Interface

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In transmission reconstruction scenarios (for example, a reconstruction from IPv4/IPv6 single-stack transmission to IPv4/IPv6 dual-stack transmission), the gNodeB may attempt to set up a second Xn control-plane link in addition to the existing Xn control-plane link. As a result, ALM-29810 gNodeB Xn Interface Fault is generated. To improve O&M efficiency and reduce unnecessary alarm reporting, control-plane dual-stack check for the Xn interface is introduced to optimize the Xn control-plane link setup mechanism in IPv4/IPv6 dual-stack scenarios.

When receiving an XN SETUP REQUEST message, the gNodeB automatically checks whether Xn control-plane links are available between the gNodeB and the source gNodeB. If Xn control-plane links are available, the gNodeB rejects the Xn control-plane link setup request of this time and notifies the source gNodeB that the Xn link setup fails. The source gNodeB then removes the corresponding Xn control-plane links, thereby avoiding the reporting of ALM-29810 gNodeB Xn Interface Fault.

Control-plane dual-stack check for the Xn interface is controlled by the **XN_CP_DUALSTACK_CHK_SW** option of the **gNBInterfaceParam.InterfaceOmSw** parameter.

Implementation

When receiving an XN SETUP REQUEST message, the gNodeB automatically checks whether Xn control-plane links are available between the gNodeB and the source gNodeB. If Xn control-plane links are available, the gNodeB rejects the Xn control-plane link setup request of this time and notifies the source gNodeB that the Xn link setup fails. The source gNodeB then removes the corresponding Xn control-plane links, thereby avoiding the reporting of ALM-29810 gNodeB Xn Interface Fault.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature is not under license control.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNBInterfaceParam.InterfaceOmSw	MOD GNBINTERFACEPARAM LST GNBINTERFACEPARAM	gNodeB	The XN_CP_DUALSTACK_CHK_SW option has been added to specify whether to enable control-plane dual-stack check for the Xn interface.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Xn self-removal	XN_SON_DEL_SW option of the gNBXnSonConfigSwitch parameter	<i>NG and Xn Self-Management</i>	Control-plane dual-stack check for the Xn interface can be enabled (by selecting the XN_CP_DUALSTACK_CHK_SW option of the gNBInterfaceParam.InterfaceOmSw parameter) only when Xn self-removal is enabled.

- Mutually exclusive features/functions

The change in this feature has no impact.

- Impacted features/functions

The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the "Operation and Maintenance" section in *NG and Xn Self-Management*.

Requirement/Trouble Ticket Number

FeR-IR20250117001898-0010

RAT

NR FDD, NR TDD

5.3.3.9 FBFD-050102 Basic Positioning

5.3.3.9.1 Addition of a Reserved Parameter Controlling the Support for NR T_{ADV} Measurement and Reporting in Uplink E-CID-based Positioning to the Disuse List

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Before an upgrade:

When the **RSVDSWPARAM2_BIT19** option of the **NRCeIIrsvd.RsvdSwParam2** parameter is selected, uplink E-CID-based positioning does not support NR T_{ADV} measurement and reporting.

When the **RSVDSWPARAM2_BIT19** option of the **NRCeIIrsvd.RsvdSwParam2** parameter is deselected, uplink E-CID-based positioning supports NR T_{ADV} measurement and reporting.

After the upgrade:

The **RSVDSWPARAM2_BIT19** option of the reserved parameter **NRCeIIrsvd.RsvdSwParam2** has been added to the disuse list. This is because the default support for NR T_{ADV} measurement and reporting in uplink E-CID-based positioning, which is controlled by the preceding option, has been disused.

Implementation

For details, see *Positioning*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact.
- Configuration management
The following parameter has changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRCeIlRsVd.RsVdSwParam2	LST NRCELLRSVD MOD NRCELLRSVD	gNodeB	The RSVDSWPARAM2_BIT19 option of the reserved parameter NRCeIlRsVd.RsVdSwParam2 has been added to the disuse list. This is because the default support for NR T _{ADV} measurement and reporting in uplink E-CID-based positioning, which is controlled by the preceding option, has been disused. Although the configuration interface in this version supports synchronization and delivery of the setting of this parameter, the system does not use this parameter.

 NOTE

NRCeIRsvd.RsvdSwParam2 is a reserved parameter.

For details about reserved parameters, see 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250528001856-0001

RAT

NR FDD, NR TDD

5.3.3.10 FBFD-050103 Flexible User Steering

5.3.3.10.1 RFSP-based Flexible Configuration of Intra-RAT Handover Measurement Parameters for Voice and Data Service UEs

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Voice and data service UEs require different handover measurement parameters. The change in this feature now supports RFSP-based flexible configuration of intra-RAT handover measurement parameters for voice and data service UEs. In this context, voice service UEs refer to UEs performing VoNR services, while data service UEs refer to those not performing VoNR services. "Flexible configuration of handover measurement parameters for voice and data service UEs" applies only to SA networking.

"Flexible configuration of handover measurement parameters for voice and data service UEs" is implemented as follows:

For a specific RFSP index, the RFSP-specific handover measurement parameter group ID can be configured using the parameters in the **gNBRfspConfig** MO. The handover measurement parameters associated with the parameter group ID are the handover measurement parameters associated with the general handover measurement parameter group ID that has the same value as the RFSP-specific handover measurement parameter group ID.

If the RFSP-specific handover measurement parameter group ID is set to **255** or there is no general handover measurement parameter group ID with the same value in the serving cells of the UEs, the RFSP-specific handover measurement parameter group ID does not take effect.

Table 5-31 lists the parameters of intra-RAT handover measurement parameter group IDs for different types of UEs in different scenarios.

Table 5-31 Parameters of intra-RAT handover measurement parameter group IDs for different types of UEs in different scenarios

Sce nar io	UE Type	RFSP-specific Handover Measurement Parameter Group ID Parameter	General Handover Measurement Parameter Group ID Parameter
Inte r- freq uen cy	Voic e servi ce UE	gNBRfspConfig.VolInterFreqHoGroupId	NRCelInterFHoMeaGrp.InterFreqHoMeasGroupId
Inte r- freq uen cy	Data servi ce UE	gNBRfspConfig.DataInterFreqHoGroupId	NRCelInterFHoMeaGrp.InterFreqHoMeasGroupId
Intr a- freq uen cy	Voic e servi ce UE	gNBRfspConfig.VolIntraFreqHoGroupId	NRCelIntraFHoMeaGrp.IntraFreqHoMeasGroupId
Intr a- freq uen cy	Data servi ce UE	gNBRfspConfig.DataIntraFreqHoGroupId	NRCelIntraFHoMeaGrp.IntraFreqHoMeasGroupId
Note: For details about the general handover measurement parameter group ID and its involved handover measurement parameters in intra-RAT handover scenarios, see <i>SA Mobility Management in Connected Mode</i> .			

Implementation

For details, see *Flexible User Steering*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature is not under license control.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	gNB RfSPConfig.VoInterFreqHoGroupId	ADD GNBRFSPCO NFIG MOD GNBRFSPCO NFIG LST GNBRFSPCO NFIG	gNodeB	This parameter has been added to specify the ID of an inter-frequency handover parameter group for voice services.
New	gNB RfSPConfig.DataInterFreqHoGroupId	ADD GNBRFSPCO NFIG MOD GNBRFSPCO NFIG LST GNBRFSPCO NFIG	gNodeB	This parameter has been added to specify the ID of an inter-frequency handover parameter group for data services.

Change Type	Parameter ID	MML Command	NE	Description
New	gNBfscConfig.VolIntraFreqHoGroupId	ADD GNBfscPCO NFIG MOD GNBfscPCO NFIG LST GNBfscPCO NFIG	gNodeB	This parameter has been added to specify the ID of an intra-frequency handover parameter group for voice services.
New	gNBfscConfig.DataIntraFreqHoGroupId	ADD GNBfscPCO NFIG MOD GNBfscPCO NFIG LST GNBfscPCO NFIG	gNodeB	This parameter has been added to specify the ID of an intra-frequency handover parameter group for data services.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

You can run the **MOD GNBfscPCONFIG** command to configure the **gNBfscConfig.VolInterFreqHoGroupId**, **gNBfscConfig.DataInterFreqHoGroupId**, **gNBfscConfig.VolIntraFreqHoGroupId**, and **gNBfscConfig.DataIntraFreqHoGroupId** parameters to implement the flexible configuration of intra-RAT handover measurement parameters for voice and data service UEs.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0120

RAT

NR FDD, NR TDD

5.3.3.10.2 Flexible Configuration of Freq Priority HO Trigger Mode Based on RFSP

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In intra-RAT frequency-priority-based inter-frequency handover scenarios, the handover event triggering mode can be customized for UEs based on RFSPs. The general working principles are as follows: On the gNodeB, differentiated values of **Freq Priority HO Trigger Mode Based on RFSP** (**gNBRfspConfig.RfspFreqPriHoTrigMode**) are configured for UEs with different RFSP indexes. In this way, differentiated triggering modes of frequency-priority-based inter-frequency handovers can take effect for different types of UEs.

Specifically:

1. The gNodeB obtains the RFSP indexes of UEs from the core network.
2. On the gNodeB, differentiated values of **Freq Priority HO Trigger Mode Based on RFSP** are configured for UEs with different RFSP indexes. Then, the gNodeB determines the handover event triggering modes for UEs with specific RFSP indexes according to the values of this parameter.
 - a. Differentiated values of **Freq Priority HO Trigger Mode Based on RFSP** are configured for UEs with different RFSP indexes.

Parameter Name	Parameter ID	Description
Operator ID	gNBRfspConfig.OperatorId	This parameter specifies the operator of a UE.
Index to RAT/ Frequency Selection Priority	gNBRfspConfig.RfspIndex	This parameter specifies the RFSP index of a UE.
Freq Priority HO Trigger Mode Based on RFSP	gNBRfspConfig.RfspFreqPriHoTrigMode	This parameter specifies the triggering mode of frequency-priority-based handover for UEs with a specific RFSP index.

- b. The gNodeB determines the triggering mode of frequency-priority-based handover for UEs with a specific RFSP index.
 - If the corresponding **Freq Priority HO Trigger Mode Based on RFSP** is set to **A1_BASED** for an RFSP index, event A1 related to frequency-

priority-based inter-frequency handover is triggered for UEs with this RFSP index.

- If the corresponding **Freq Priority HO Trigger Mode Based on RFSP** is set to **A2_BASED** for an RFSP index, event A2 related to frequency-priority-based inter-frequency handover is triggered for UEs with this RFSP index.
- If the corresponding **Freq Priority HO Trigger Mode Based on RFSP** is set to **NULL** for an RFSP index, the **NRCellMobilityConfig.FreqPriHoTriggerMode** parameter is used to specify the frequency-priority-based inter-frequency handover event type for UEs with this RFSP index.

Implementation

For details, see *Flexible User Steering*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature is not under license control.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	gNB RfSpConfig.RfspFreqPriHoTrigMode	MOD GNB RFSPCO NFIG LST GNB RFSPCO NFIG	gNodeB	This parameter has been added to specify the triggering mode of frequency-priority-based handovers for UEs with a valid RFSP.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

You can run the **MOD GNB RFSPCONFIG** command to configure the **gNB RfSpConfig.RfspFreqPriHoTrigMode** parameter.

Requirement/Trouble Ticket Number

FeR-IR20241209002193-0052

RAT

NR FDD, NR TDD

5.3.3.10.3 Flexible Configuration of Thresholds for Events A4 and A5 of Voice Service UEs in the Policy of RFSP-based Inter-Frequency Handover Back to the HPLMN

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In an RFSP-based inter-frequency handover back to the HPLMN, the flexible configuration of thresholds for events A5 and A4 is now supported for voice service UEs (that is, UEs performing VoNR services) during measurement configuration delivery. The following lists the thresholds for events A5 and A4 of voice service UEs:

- Threshold 1 for event A5: **gNB RfspConfig.BackToHplmnHoA5RsrpThld1 + gNB RfspConfig.VoBackHplmnA5RsrpThld1Ofs**
- Threshold 2 for event A5 and threshold for event A4: **gNB RfspConfig.BackToHplmnHoA5RsrpThld2 + gNB RfspConfig.VoBackHplmnA5RsrpThld2Ofs**

The change in this feature is supported only in SA networking.

Implementation

For details, see *Flexible User Steering*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature is not under license control.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	gNB RfspConfig.VoBackHplmnA5RsrpThld1Ofs	ADD GNB RSPCO NFIG MOD GNB RSPCO NFIG LST GNB RSPCO NFIG	gNodeB	This parameter has been added to specify the offset used to calculate RSRP threshold 1 for event A5 related to handovers back to the HPLMN for voice services.
New	gNB RfspConfig.VoBackHplmnA5RsrpThld2Ofs	ADD GNB RSPCO NFIG MOD GNB RSPCO NFIG LST GNB RSPCO NFIG	gNodeB	This parameter has been added to specify the offset used to calculate RSRP threshold 2 for event A5 related to handovers back to the HPLMN for voice services.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

You can run the **MOD GNB RSPCONFIG** command to configure the **gNB RfspConfig.VoBackHplmnA5RsrpThld1Ofs** and

gNBRfspConfig.VoBackHplmnA5RsrpThld2Ofs parameters to enable the flexible configuration of the thresholds for events A4 and A5 of voice service UEs.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0120

RAT

NR FDD, NR TDD

5.3.3.10.4 Flexible Configuration of Threshold for Event A4 and Threshold 2 for Event A5 in Service-based Inter-Frequency Handover Scenarios

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In RFSP-specific service-based inter-frequency handover scenarios, the threshold for event A4 and threshold 2 for event A5 related to service-based inter-frequency handover can be customized for UEs based on RFSPs.

- Threshold for event A4 = **NRCellInterFHoMeaGrp.SrvInterFreqA4RsrpThld** + **gNBRfspConfig.ServBasedInterFHoA4RsrpOfs**
- Threshold 2 for event A5 = **NRCellInterFHoMeaGrp.SrvInterFreqA4RsrpThld** + **gNBRfspConfig.ServBasedInterFHoA4RsrpOfs**

NOTE

Threshold 2 for event A5 related to service-based inter-frequency handover, which is customized for UEs based on RFSPs, reuses the parameters of the threshold for event A4.

Implementation

For details, see *Flexible User Steering*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature is not under license control.
- Configuration management
The following parameter has changed.

Change Type	Parameter ID	MML Command	NE	Description
New	gNBRfspConfig.ServBasedInterFHoA4RsrpOfs	MOD GNBRFSPCO NFIG LST GNBRFSPCO NFIG	gNodeB	This parameter is used for the flexible configuration in service-based inter-frequency handover scenarios.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

You can run the **MOD GNBRFSPCONFIG** command to configure the **gNBRfspConfig.ServBasedInterFHoA4RsrpOfs** parameter.

Requirement/Trouble Ticket Number

FeR-IR20241209002193-0081

RAT

NR FDD, NR TDD

5.3.3.11 FBFD-051102 Access Control

5.3.3.11.1 Admission Guarantee for Voice Service UEs

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

- Checking whether UE admission is limited

Insufficient UE number resources due to license or base station configuration restrictions can degrade user experience for both calling and called parties in MCPTT and VoNR services. To address this issue, admission guarantee for voice services is introduced. It ensures that voice service UEs are successfully admitted and can preempt resources from low-priority UEs. This function is enabled by setting the **NRCellPri.VoiceUeNumOfs** parameter to a non-zero value. It applies only to low-frequency TDD and FDD.

The three levels of UE-number-based admission thresholds are changed to the four levels of UE-number-based admission thresholds, where the level-3 threshold UE refers to the voice service UE and the original level-3 threshold UE is upgraded to the level-4 threshold UE. For details about the admission thresholds and other related information, see [Table 5-32](#). The gNodeB calculates the total number of UEs (including admitted UEs of all types and UEs of this type that are accessing the network). If the total number of UEs is greater than the level-3 admission threshold, the gNodeB determines that UE admission fails.

Table 5-32 Four levels of UE-number-based admission thresholds

UE Type	UE Type Definition	Admission Threshold
Level-4 threshold UE	Any of the following conditions is met: • UE performing an intra-base-station incoming handover, with the cause value in the RRCSetupRequest message being Emergency or highPriorityAccess^a • UE transiting from the inactive state to the connected state, with the cause value in the RRCResumeRequest1 message being Emergency or highPriorityAccess • UE performing an RRC initial access, with the cause value in the RRCSetupRequest message being Emergency^b or highPriorityAccess • Emergency call UE^c in the following scenarios: – Inter- or intra-base-station incoming handover – Transition from the inactive state to connected state – RRC connection reestablishment	The smaller of the following two values is used as the level-4 admission threshold: • $\text{NRCellPri. UserNumAdmThld} + \text{NRCellPri. HoReestUeNumOfs} + \text{NRCellPri. VoiceUeNumOfs} + \text{NRCellPri. HighPriUeNumOfs}$ • Maximum number of UEs that can access a cell under the limitation of hardware capabilities^d

UE Type	UE Type Definition	Admission Threshold
Level-3 threshold UE	Any of the following conditions is met: • UE involved in an inter-base-station incoming handover or intra-base-station incoming handover, with a 5QI 1/65/66 bearer established • UE transiting from the inactive state to the connected state, with the cause value in the RRCResumeRequest1 message being MO-VoiceCall • UE performing an RRC initial access, with the cause value in the RRCSetupRequest message being MO-VoiceCall • UE performing an RRC connection reestablishment, with a 5QI 1/65/66 bearer established • UE having a bearer with 5QI 1/65/66^e set up before the timer specified by NRCCellPri.TempAdmTimer expires, with a cause value other than Emergency, highPriorityAccess, and MO-VoiceCall	The smaller of the following two values is used as the level-3 admission threshold: • NRCCellPri.UserNumAdmThld + NRCCellPri.HoReestUeNumOfs + NRCCellPri.VoiceUeNumOfs • Maximum number of UEs that can access a cell under the limitation of hardware capabilities^d
Level-2 threshold UE	Any of the following conditions is met: • Non-level-4 or non-level-3 threshold UE involved in an inter-base-station incoming handover • Non-level-4 or non-level-3 threshold UE involved in an intra-base-station incoming handover • Non-level-4 or non-level-3 threshold UE that transits from the inactive state to the connected state • Non-level-4 or non-level-3 threshold UE involved in an RRC connection reestablishment	The smaller of the following two values is used as the level-2 admission threshold: • NRCCellPri.UserNumAdmThld + NRCCellPri.HoReestUeNumOfs • Maximum number of UEs that can access a cell under the limitation of hardware capabilities^d

UE Type	UE Type Definition	Admission Threshold
Level-1 threshold UE	Common UE (not level-4, level-3, or level-2 threshold UE)	The smaller of the following two values is used as the level-1 admission threshold: • NRCellPri.UserNumAdmThld • Maximum number of UEs that can access a cell under the limitation of hardware capabilities^d
a: During an inter-base-station handover, the target cell cannot inherit the access cause value from the source cell, resulting in the failure to inherit the UE's threshold attributes. This may cause a level-4 threshold UE to become a level-2 threshold UE after the handover. b: If a UE performs an RRC initial access with the cause value Emergency in the RRCSetupRequest message, its admission threshold is the smaller of the following two values: (1) the result of "NRCellPri.TempAdmUeNumThld + NRCellPri.UserNumAdmThld + NRCellPri.HoReestUeNumOfs + NRCellPri.VoiceUeNumOfs + NRCellPri.HighPriUeNumOfs" and (2) the maximum number of UEs that can access a cell under the limitation of hardware capabilities^d. c: Emergency call UEs are those for which the gNBOperatorParam.EmcBearerArp parameter is set to the emergency call ARP configured on the core network when the EMERGENCY_CALL_UE_ARP_IDENT_SW option of the gNB5qiConfig.Nr5qiAlgoSwitch parameter is selected. d: For details about the maximum number of UEs that can access a cell under the limitation of hardware capabilities, see <i>BBU Technical Specifications in 3900 & 5900 Series Base Station Product Documentation</i>. e: If admission guarantee for voice service UEs is enabled (by setting the NRCellPri.VoiceUeNumOfs parameter to a non-zero value) for a cell, and the number of UEs in the cell is greater than or equal to the value of NRCellPri.UserNumAdmThld, then the UE with an access cause value other than Emergency, highPriorityAccess, and MO-VoiceCall is processed as follows: • If the number of temporarily admitted UEs is less than the value of NRCellPri.TempAdmUeNumThld, the UE is temporarily admitted and a temporary RRC connection is set up. Before the timer specified by NRCellPri.TempAdmTimer expires, the following logic applies: – If a bearer with 5QI 1/65/66 is set up for the UE, the UE is identified as a voice service UE and the level-3 admission threshold is applied. – If no such bearer is set up, the level-1 admission threshold is applied. • If the number of temporarily admitted UEs is greater than or equal to the value of NRCellPri.TempAdmUeNumThld, the temporary RRC connection fails to be set up and the UE admission fails.		

- Processing Preemption

When the **NRCellPri.VoiceUeNumOfs** parameter is set to a non-zero value, the following logic applies:

- If the access cause value is **Emergency**, **highPriorityAccess**, or **MO-VoiceCall** for a UE, the gNodeB attempts to set up a temporary RRC connection with the UE to obtain the UE's ARP attributes for user preemption decision.
 - If the temporary RRC connection is successfully set up, the gNodeB obtains the ARP attributes of the first bearer set up for the UE within 5s.
 - If the UE's ARP attributes are successfully obtained within 5s, the gNodeB proceeds to the procedure for checking whether the UE meets the preemption conditions.
 - The gNodeB determines that UE admission fails if the UE's ARP attributes fail to be obtained within 5s.
 - If the temporary RRC connection fails to be set up, UE admission fails.
- If the access cause value is not **Emergency**, **highPriorityAccess**, or **MO-VoiceCall** and a temporary RRC connection is set up for a UE, the gNodeB obtains the ARP attributes of the highest-priority bearer of the UE after the timer specified by **NRCellPri.TempAdmTimer** expires.
 - If the ARP attributes are successfully obtained, the gNodeB promptly proceeds to the procedure for checking whether the UE meets the preemption conditions.
 - If no bearer is set up for the UE after the timer specified by **NRCellPri.TempAdmTimer** expires, the gNodeB determines that the UE admission fails.

If UE admission fails, the gNodeB immediately releases the temporary RRC connection. If UE admission succeeds, the temporary RRC connection is changed to a formal one.

The change in this feature is supported only in low frequency bands.

NOTE

ARP attributes are defined in section 9.3.1.19 "QoS Flow Level QoS Parameters" in 3GPP TS 38.413 V15.4.0 and are separately represented by the following IEs:

- *Priority Level*: The IE value ranges from 1 to 15 with **1** as the highest level of priority. A high-ARP-priority UE can preempt the resources of a low-ARP-priority UE.
- *Pre-emption Capability*
 - If the IE value is **may trigger pre-emption**, the UE can preempt resources of low-priority UEs.
 - If the IE value is **shall not trigger pre-emption**, the UE cannot preempt resources of low-priority UEs.
- *Pre-emption Vulnerability*
 - If the IE value is **pre-emptable**, the resources allocated to the UE can be preempted by other UEs.
 - If the IE value is **not pre-emptable**, the resources allocated to the UE cannot be preempted by other UEs.

Implementation

For details, see *Admission Control*. The specific sections are listed in the change history of the document.

Impact on the Network

- Benefits

When the **NRCellPri.VoiceUeNumOfs** parameter is set to a non-zero value, the access success rate of voice service UEs increases.

- Impacts

When the **NRCellPri.VoiceUeNumOfs** parameter is set to a non-zero value, the access success rate of non-voice service UEs decreases and the service drop rate increases. Although the access success rate of voice service UEs increases, the values of the following indicators change when the cell or network slice is heavily loaded:

- The number of RRC connection setup attempts (**N.RRC.SetupReq.Att** + **N.RRC.ResumeReq.Att**) decreases.
- The number of bearer setup attempts (**N.QosFlow.Est.Att** + **N.QosFlow.Resume.Att**) decreases.
- The values of the following counters related to the number of online UEs decrease:
 - **N.User.RRCConn.Max**, which indicates the maximum number of UEs in RRC_CONNECTED mode in a cell
 - **N.User.RRCConn.Avg**, which indicates the average number of UEs in RRC_CONNECTED mode in a cell
 - **N.User.NsaDc.PSCell.Avg**, which indicates the average number of LTE-NR NSA DC users using the current cell as PSCell
- Service drop rates (such as **Service Call Drop Rate (CU)** and **Service Call Drop Rate (CU, Inactive)**) increase.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRCellPri.VoiceUseNumOfs	MOD NRCCELLPRI LST NRCCELLPRI	gNodeB	This parameter has been added to specify the admission offset for VoNR and MCPTT UEs.
New	NRCellPri.TempAdmTimer	MOD NRCCELLPRI LST NRCCELLPRI	gNodeB	This parameter has been added to specify the timer threshold for temporary admission of VoNR and MCPTT UEs. UEs whose admission is limited will occupy the specifications of UEs that can be temporarily admitted for a period no longer than the value of this parameter.

Change Type	Parameter ID	MML Command	NE	Description
New	NRCellPri. <i>TempAdmUeNumThld</i>	MOD NRCELLPRI LST NRCELLPRI	gNodeB	This parameter has been added to specify the UE number threshold for temporary admission of VoNR and MCPTT UEs. If the number of temporarily admitted UEs does not exceed the value of this parameter, these UEs will occupy the specifications of UEs that can be temporarily admitted.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationship has been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Voice UE Number Offset	NRCellPri. <i>VoiceUeNumOfs</i>	<i>Admission Control</i>	The NRCellPri. <i>TempAdmUeNumThld</i> parameter can be set to a value other than 20 only when the NRCellPri. <i>VoiceUeNumOfs</i> parameter is set to a value other than 0.

- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Admission Control*.

Requirement/Trouble Ticket Number

FeR-IR20241211002479-0001

RAT

NR FDD, NR TDD

5.3.3.11.2 Configuration of the Admission Threshold for Emergency UEs Performing an RRC Initial Access

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

If a UE performs an RRC initial access with the cause value **Emergency** in the RRCSetupRequest message, its admission threshold is the smaller of the following two values: (1) the result of "**NRCellPri.TempAdmUeNumThld** + **NRCellPri.UserNumAdmThld** + **NRCellPri.HoReestUeNumOfs** + **NRCellPri.VoiceUeNumOfs** + **NRCellPri.HighPriUeNumOfs**" and (2) the maximum number of UEs that can access a cell under the limitation of hardware capabilities.

The change in this feature is supported only in low frequency bands.

NOTE

For details about the maximum number of UEs that can access a cell under the limitation of hardware capabilities, see *BBU Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*.

Implementation

For details, see *Admission Control*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact.
- Configuration management
The change in this feature has no impact.
- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Admission Control*.

Requirement/Trouble Ticket Number

FeR-IR20241211002479-0001

DTS2025092248816

RAT

NR FDD, NR TDD

5.3.3.11.3 Parameter Settings for Measurement Mode of the Number of RRC_CONNECTED UEs in Dynamic Access Class Control

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Before an upgrade, within dynamic access class control, the gNodeB determines whether to trigger UAC based on the number of RRC_CONNECTED UEs, which is fixed to be calculated on a per operator basis. When there are a large number of RRC_CONNECTED UEs in a cell, calculation only on a per operator basis may lead to a situation in which the cell is congested but some operators do not enable UAC based on the number of RRC_CONNECTED UEs, thereby affecting the effect of access class control.

After the upgrade, the **NRDUCellUacBar.RrcUserNumCountMode** parameter specifies the measurement mode of the number of RRC_CONNECTED UEs when the gNodeB determines whether to trigger UAC based on the number of RRC_CONNECTED UEs in dynamic access class control.

- If this parameter is set to **PLMN_MODE**, the number of RRC_CONNECTED UEs is calculated on a per operator basis in a cell.
- If this parameter is set to **CELL_MODE**, the number of RRC_CONNECTED UEs is calculated on a per cell basis, including the number of RRC_CONNECTED UEs of all operators in the cell.

The **NRDUCellUacBar.RrcUserNumCountMode** parameter takes effect only when UAC based on the number of RRC_CONNECTED UEs is in effect.

For an upgrade from a version earlier than V100R022C10SPC010 to V100R022C10SPC010 or later:

- Before the upgrade, the **NRDUCellUacBar.AccessControlType** parameter is set to **DYNAMIC_UAC**, and the **NRDUCellUacBar.OperatorId**, **NRDUCellUacBar.RrcUserNumUacEnterThld**, and **NRDUCellUacBar.RrcUserNumUacExitThld** parameters are not set to 255. After the upgrade, the **NRDUCellUacBar.RrcUserNumCountMode** parameter is set to **PLMN_MODE**.
- In other scenarios, the **NRDUCellUacBar.RrcUserNumCountMode** parameter is set to **CELL_MODE** after the upgrade.

After the change in this feature takes effect and the access barring factor adaptation function is enabled (by selecting the **BARRING_FACTOR_ADAPT_SW** option of the **NRDUCellAlgoSwitch.RacAlgoSwitch** parameter), if the **NRDUCellUacBar.AccessControlParamGrpId** parameter (specifying an access control parameter group ID) is set to the same value in different **NRDUCellUacBar** MO instances for a cell, the following parameters must be set to the same values: **NRDUCellUacBar.RrcUserNumCountMode**, **NRDUCellUacBar.FlowCtrlUacEnterThld**, **NRDUCellUacBar.RrcUserNumUacEnterThld**, **NRDUCellUacBar.FlowCtrlUacExitThld**, and **NRDUCellUacBar.RrcUserNumUacExitThld**.

Implementation

For details, see *Access Class Control*. The specific sections are listed in the change history of the document.

Impact on the Network

In multi-operator scenarios, if the **NRDUCellUacBar.RrcUserNumCountMode** parameter is set to **CELL_MODE**, UAC based on the number of RRC_CONNECTED

UEs is more likely to be triggered, leading to a more significant effect of relieving system congestion but a greater impact on user experience; if the **NRDUCellUacBar.RrcUserNumCountMode** parameter is set to **PLMN_MODE**, UAC based on the number of RRC_CONNECTED UEs is less likely to be triggered, leading to a less significant effect of relieving system congestion but a smaller impact on user experience. In single-operator scenarios, the values **CELL_MODE** and **PLMN_MODE** provide the same effects.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature is not under license control.
- Configuration management
The following parameter has changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellUacBar.RrcUserNumCountMode	ADD NRDUCELLU ACBAR MOD NRDUCELLU ACBAR LST NRDUCELLU ACBAR	gNodeB	This parameter has been added to specify the measurement mode of the number of RRC_CONNECTED UEs in dynamic access class control.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.

- Mutually exclusive features/functions

The following mutually exclusive relationship has been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Operator ID	NRDUCellUacBar.OperatorId	<i>Access Class Control</i>	When the NRDUCellUacBar.OperatorId parameter is set to 255 , the NRDUCellUacBar.RrcUserNumCountMode parameter must be set to CELL_MODE .

- Impacted features/functions

The change in this feature has no impact.

Related Operations

By default, the change takes effect after an upgrade.

To change the measurement mode of the number of RRC_CONNECTED UEs in dynamic access class control, run the **MOD NRDUCELLUACBAR** command with the **NRDUCellUacBar.RrcUserNumCountMode** parameter set to a desired value.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0111

RAT

NR FDD, NR TDD

5.3.3.12 FBFD-091100 ANBR

5.3.3.12.1 Support for ANBR Resource Factor Configuration

NEs Involved

3900 and 5900 series base stations

Feature/Function Description

In some scenarios, such as the scenario with a small cell bandwidth, the remaining RB resources may not be fully scheduled due to limited CCE resources. As a result, the estimated uplink and downlink recommended rates in "MAC CE-based rate adjustment for data services" are higher than expected. To address this issue, the resource factors used to estimate the capabilities of uplink and downlink rate guarantee in "MAC CE-based rate adjustment for data services" now can be flexibly configured to adjust the uplink and downlink resource ranges for estimation. In this way, the recommended uplink and downlink rates better match the actual available uplink and downlink rates.

- The uplink resource range for estimation by the gNodeB can be set by using the **NRDUCellUISch.AnbrResFactor** parameter.
- The downlink resource range for estimation by the gNodeB can be set by using the **NRDUCellDISch.AnbrResFactor** parameter.

The preceding parameter values are expressed as percentages. If this parameter is set to 100, the estimation is performed based on all resources allocated to the bearer in the cell. If this parameter is set to a value below 100 (such as 80), the estimation is performed based on 80% of the resources allocated to the bearer in the cell.

The preceding parameters take effect only when the **ANBR_SW** option of the **NRDUCellQciBearer.ServiceDiffAlgoSwitch** parameter is selected; they do not take effect for bearers with QCI 1 or 2.

LampSite base stations support this function.

The change in this feature is supported only in low frequency bands.

The change in this feature is supported only in SA networking.

Implementation

For details, see *Service-differentiated Experience Guarantee*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellDLSch.A <i>nbrResFactor</i>	LST NRDUCELLDLS CH MOD NRDUCELLDLS CH	gNodeB	This parameter has been added to specify the resource factor for downlink ANBR.
New	NRDUCellULSch.A <i>nbrResFactor</i>	LST NRDUCELLULS CH MOD NRDUCELLULS CH	gNodeB	This parameter has been added to specify the resource factor for uplink ANBR.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Service-differentiated Experience Guarantee*.

Requirement/Trouble Ticket Number

IR20241209002193/DTS2025042223987

RAT

NR FDD, NR TDD

5.3.3.13 FOFD-020205 Intra-Band CA

5.3.3.13.1 Added the CSI-RS Beam Sweeping Optimization Function

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Without this function, CSI-RS beam sweeping is performed in both periodic and aperiodic modes. After the sweeping succeeds, the UE reports the optimal beam in an SCell. Resources must be reserved for such information reporting in both periodic and aperiodic sweeping scenarios (or only the supported sweeping scenario if the UE does not support both). The resources available for UEs to report information are limited. If resources are insufficient, other features that require UE information reporting for CA carriers may fail to take effect. Considering that periodic CSI-RS beam sweeping has a high success rate, CSI-RS beam sweeping optimization is introduced to reduce the resource consumption by UE information reporting. CSI-RS beam sweeping optimization is controlled by the **CSI_BEAM_SWEEPING_OPT_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoSwitch** parameter.

- If this option is selected:
 - If a UE supports periodic sweeping, such sweeping is performed and resources are reserved only for UE information reporting for periodic sweeping.
 - If a UE does not support periodic sweeping, aperiodic sweeping is performed and resources are reserved only for UE information reporting for aperiodic sweeping.
- If this option is deselected, CSI-RS beam sweeping optimization does not take effect.

The change in this feature is supported only in low frequency bands.

Implementation

For details, see *Carrier Aggregation*. The specific sections are listed in the change history of the document.

Impact on the Network

CSI-RS beam sweeping optimization optimizes the allocation of CA UE reporting resources among multiple carriers, improving CA UE performance.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management

The change in this feature has no impact. It is controlled by the existing feature license.

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellCarrMgmt.CaEnhancedAlgoSwitch	MOD NRDUCELLCAR RMGMT LST NRDUCELLCAR RMGMT	gNodeB	The CSI_BEAM_SWEEPING_OPTION_SW option has been added to specify whether to enable CSI-RS beam sweeping optimization.

- Performance management

The change in this feature has no impact.

- Fault management

The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

The change in this feature has no impact.

- Mutually exclusive features/functions

The change in this feature has no impact.

- Impacted features/functions

The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Carrier Aggregation*.

Requirement/Trouble Ticket Number

IR20241209002193

DTS2025021835017

RAT

NR FDD, NR TDD

5.3.3.13.2 Added a Method for Observing the Activation and Running State of Intra-Band CA

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The change in this feature involves two parts:

- The **N.User.CA.PCell.DL.Max** counter has been added for monitoring the activation and running state of downlink intra-band CA. This counter measures the maximum number of downlink CA UEs that treat the local cell as their PCells.
- The **N.User.CA.PCell.UL.Max** counter has been added for monitoring the activation and running state of uplink intra-band CA. This counter measures the maximum number of uplink CA UEs that treat the local cell as their PCells.

Implementation

For details, see *Carrier Aggregation*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The change in this feature has no impact.
- Performance management
The following counters have changed.

Change Type	Counter Name	Function Subset	NE	Description
New	N.User.CA.P Cell.DL.Max	Cell User Quantity Measurement	gNode B	This counter has been added to measure the maximum number of downlink CA UEs that treat the local cell as their PCells.
New	N.User.CA.P Cell.UL.Max	Cell User Quantity Measurement	gNode B	This counter has been added to measure the maximum number of uplink CA UEs that treat the local cell as their PCells.

- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0023

RAT

NR FDD, NR TDD

5.3.3.13.3 Added the SRS and PUSCH/PUCCH Concurrency Compatibility Function

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

According to 3GPP specifications, SRS and PUSCH/PUCCH concurrency is supported in intra-band non-contiguous CA, but not in intra-band contiguous CA. In a CA scenario, SRS and PUSCH/PUCCH concurrency means that a PCC and an SCC work in the same symbols where one CC sends SRSs and the other CC sends PUSCH/PUCCH signals.

For intra-band non-contiguous CA, SRS and PUSCH/PUCCH concurrency requires that UEs support this capability. Huawei gNodeBs comply with the protocol specifications to support SRS and PUSCH/PUCCH concurrency, while a UE supports this capability only if it reports the **parallelTxSRS-PUCCH-PUSCH-intraBand-r17** field.

Before the SRS and PUSCH/PUCCH concurrency compatibility function is introduced, the gNodeB by default considers that all UEs support the concurrency. If a UE does not support this capability, the gNodeB demodulates SRSs or PUSCH/PUCCH signals at incorrect time-domain positions, which compromises uplink performance of CA UEs. To address this issue, the SRS and PUSCH/PUCCH concurrency compatibility function is introduced. This function is controlled by the **SRS_PUSCH_CONCURRENCY_COMPT_SW** option of the **NRDUCellCarrMgmt.CaCompatibilitySwitch** parameter. After the SRS and PUSCH/PUCCH concurrency compatibility function is enabled, the gNodeB checks whether a UE supports this concurrency so that the processing can vary with the UE capability:

- If the UE supports SRS and PUSCH/PUCCH concurrency, the UE can concurrently send SRSs and PUSCH/PUCCH signals.
- If the UE does not support SRS and PUSCH/PUCCH concurrency, the UE does not send SRSs and PUSCH/PUCCH signals concurrently on the PCC and SCCs. Instead, in the corresponding symbols, the UE sends SRSs only but not PUSCH/PUCCH signals.

The change in this feature is supported only in low frequency bands.

Implementation

For details, see *Carrier Aggregation*. The specific sections are listed in the change history of the document.

Impact on the Network

After SRS and PUSCH/PUCCH concurrency compatibility takes effect, the average throughput of SCell-activated uplink CA UEs ($(N.CA.SccActiveUser.ThpVol.UL - N.CA.ThpVol.UL.SmallPkt)/N.CA.ThpTime.UL.RmvSmallPkt$) increases.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellCa rrMgmt.CaC ompatibility Switch	MOD NRDUCELLC ARRMGMT LST NRDUCELLC ARRMGMT	gNodeB	The SRS_PUSCH_CONCURREN CY_COMPT_SW option has been added to specify whether to enable the SRS and PUSCH/PUCCH concurrency compatibility function.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS interference coordination based on self-contained slots	SRS_INTRF_CO ORD_S_SLOT_S W option of the NRDUCellSrs.S rsAlgoExtSwitc h parameter	<i>Channel Management</i>	The SRS_INTRF_COORD_S_SLOT_SW option must be selected if the SRS and PUSCH/PUCCH concurrency compatibility function is enabled and the SRS remote interference avoidance switch is turned on.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS remote interference avoidance	SRS_RIM_INTRF_AVOID_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter	None	
Low-frequency TDD	PUCCH symbol length number increase	PUCCH_SYM_TYPE_NUM_INCR_SW option of the NRDUCellPucch.PucchAlgoSwitch parameter	<i>Channel Management</i>	When the SRS and PUSCH/PUCCH concurrency compatibility function is enabled, the PUCCH_SYM_TYPE_NUM_INCR_SW option must be selected.

- Mutually exclusive features/functions

The following mutually exclusive relationship has been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Light-load time sequence optimization	NRDUCellAlgoSwitch.LIGHT_LOAD_TIME_SEQ_OPT_SW option of the LowLatencySwitch parameter	None	When the SRS and PUSCH/PUCCH concurrency compatibility function is enabled, the light-load time sequence optimization function must be disabled.
Low-frequency TDD	Positioning-dedicated SRS transmission in uplink slots	SRS_FOR_POSITION_ON_USLOT_SW option of the NRDUCellPosAlgo.LcsMeasureAlgoSwitch parameter	<i>Positioning</i>	When the SRS and PUSCH/PUCCH concurrency compatibility function is enabled, positioning-dedicated SRS transmission in uplink slots cannot be enabled together with proactive avoidance of remote interference.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Proactive avoidance of remote interference	SRS_RIM_INTERF_AVOID_SW and SRS_INTERF_COORD_S_SLOT_SW options of the NRDUCellSrsResourcesAlgoExtSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	

- Impacted features/functions
The following impact relationships have been modified.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Proactive avoidance of remote interference	SRS_RIM_INTRF_AVOID_SW and SRS_INTRF_COORD_S_SLOT_SW options of the NRDUCellSrs.SrsAlgoExtSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	- When the uplink CA switch is on, the SRS_INTRF_AVOID_PUSCH_SCH_SW setting of the NRDUCellSrs.SrsAlgoExtSwitch parameter must be the same for the PCell and the SCell. Otherwise, the cells differ in their scheduling in self-contained slots in the uplink, affecting the reporting of the power headroom in the PCell and SCell. As a result, adjustments for TPC are performed frequently and the average uplink throughput of uplink CA UEs changes greatly. - It is good practice to keep the "proactive avoidance of remote interference" switch status consistent between the PCell and the SCell. Otherwise, the bit errors of some UEs may increase due to inconsistent PUSCH

RAT	Function Name	Function Switch	Reference	Description
				scheduling status of the two cells in the same slot. In the event of inconsistent switch status between the two cells, it is good practice to enable the "SRS and PUSCH/PUCCH concurrency compatibility" function to reduce bit errors. For details about the "SRS and PUSCH/PUCCH concurrency compatibility" function, see <i>Carrier Aggregation</i> .

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Carrier Aggregation*.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0110

RAT

NR FDD, NR TDD

5.3.3.13.4 Support for a Maximum of Six High-Frequency CCs in the Downlink for Intra-Band CA in the Case of 200 MHz per High-Frequency Cell (for High Frequency Bands Only)

NEs Involved

5900 series base stations

Feature/Function Description

Before the upgrade: A maximum of four high-frequency CCs can be aggregated in the downlink for intra-band CA in the case of 200 MHz per high-frequency cell. The supported bandwidth combinations are as follows: $m \times 200$ MHz in the downlink and $n \times 200$ MHz in the uplink ($1 \leq n \leq m \leq 4$)

After the upgrade: A maximum of six high-frequency CCs can be aggregated in the downlink for intra-band CA in the case of 200 MHz per high-frequency cell. The supported bandwidth combinations are as follows:

- If the **gNodeBCarrMgmt.Fr2IntraBandCa200MCcMaxNum** parameter is set to **DL6CC_UL4CC**: $m \times 200$ MHz in the downlink and $n \times 200$ MHz in the uplink ($1 \leq m \leq 6$; $1 \leq n \leq 4$; $n \leq m$)
- If the **gNodeBCarrMgmt.Fr2IntraBandCa200MCcMaxNum** parameter is set to **DL4CC_UL4CC**: $m \times 200$ MHz in the downlink and $n \times 200$ MHz in the uplink ($1 \leq n \leq m \leq 4$).

The change in this feature is supported only by macro base stations in SA networking in high frequency bands.

Implementation

For details, see *Carrier Aggregation*. The specific sections are listed in the change history of the document.

Impact on the Network

Intra-band CA now allows for up to six high-frequency CCs to be aggregated in the downlink in the case of 200 MHz per high-frequency cell. This fully utilizes high-frequency spectrum resources and increases downlink UE throughput.

Impact on NEs

The change in this feature requires that the number of CCs supported by the UE's CA capability, as reported in the **BandCombination** IE, be at least six.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	gNodeBCarrMgmt. Fr2IntraBandCa20 OMCcMaxNum	MOD GNODEBCARR MGMT LST GNODEBCARR MGMT	gNodeB	This parameter has been added to specify the maximum number of high-frequency CCs that can be aggregated in the downlink for intra-band CA in the case of 200 MHz per high-frequency cell.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Carrier Aggregation*.

Requirement/Trouble Ticket Number

FeR-IR20240626001884-0001

RAT

NR TDD

5.3.3.13.5 Deleted the Mutually Exclusive Relationship Between Uplink Intra-band CA and PUSCH Scheduling in Self-Contained Slots

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The change in this feature has deleted the mutually exclusive relationship between uplink intra-band CA (controlled by the **INTRA_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch.CaAlgoSwitch** parameter) and PUSCH scheduling in self-contained slots (controlled by the **PUSCH_SCH_IN_S_SLOT_SW** option of the **NRDUCellPusch.UlPuschAlgoSwitch** parameter).

The change in this feature is supported only in low frequency bands.

Implementation

For details, see *Carrier Aggregation*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The change in this feature has no impact.
- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The following mutually exclusive relationships have been removed.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PUSCH scheduling in self-contained slots	PUSCH_SCH_IN_S_SLOT_SW option of the NRDUCellPusch.UlPuschAlgoSwitch parameter	None	When uplink intra-band CA is enabled, the PUSCH_SCH_IN_S_SLOT_SW option must be deselected.

- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250529001962-0017

RAT

NR TDD

5.3.3.14 FOFD-021204 Automatic Neighbour Relation (ANR)

5.3.3.14.1 Added Optimized Measurement of the Success Rate of Handovers to Neighboring Cells in SA Networking

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In intra-RAT neighbor cell class label management, automatic optimization of neighbor relationship attributes based on the handover success rate, and automatic optimization of neighbor relationship attributes based on identification of over-distant neighboring cells, optimized measurement of the success rate of handovers to neighboring NR cells in SA networking is controlled by the **SA_NR_NCELL_SUCC_RT_STAT_SW** option of the **NRCellAnr.AnrAlgoSwitch** parameter. If this option is selected, the number of handover attempts is measured

in SA networking. If this option is deselected, the number of handover executions is measured in SA networking. The handover success rate in SA networking is equal to the result of the following: Number of successful handovers/Number of handovers x 100%.

In inter-RAT automatic optimization of neighbor relationship attributes based on the handover success rate and automatic optimization of neighbor relationship attributes based on identification of over-distant neighboring cells, optimized measurement of the success rate of handovers to neighboring LTE cells in SA networking is controlled by the **SA_LTE_NCELL_SUCC_RT_STAT_SW** option of the **NRCellAnr.AnrAlgoSwitch** parameter. If this option is selected, the number of handover attempts is measured. If this option is deselected, the number of handover executions is measured. The handover success rate is equal to the result of the following: Number of successful handovers/Number of handovers x 100%.

The change in this feature is supported only in low-frequency SA networking.

Implementation

For details, see *ANR*. The specific sections are listed in the change history of the document.

Impact on the Network

If the **SA_NR_NCELL_SUCC_RT_STAT_SW** option of the **NRCellAnr.AnrAlgoSwitch** parameter is selected, a neighboring NR cell is identified as an abnormal neighboring cell in SA networking if preparations of a handover to this NR cell fail, increasing the handover success rate. Specifically:

- Intra-frequency handover success rate =
$$\frac{(N.HO.IntraFreq.IntragNB.ExecSuccOut + N.HO.IntraFreq.Ng.IntergNB.ExecSuccOut + N.HO.IntraFreq.Xn.IntergNB.ExecSuccOut)}{(N.HO.IntraFreq.IntragNB.PrepAttOut + N.HO.IntraFreq.Ng.IntergNB.PrepAttOut + N.HO.IntraFreq.Xn.IntergNB.PrepAttOut)} \times 100\%$$
- Inter-frequency handover success rate =
$$\frac{(N.HO.InterFreq.IntragNB.ExecSuccOut + N.HO.InterFreq.Ng.IntergNB.ExecSuccOut + N.HO.InterFreq.Xn.IntergNB.ExecSuccOut)}{(N.HO.InterFreq.IntragNB.PrepAttOut + N.HO.InterFreq.Ng.IntergNB.PrepAttOut + N.HO.InterFreq.Xn.IntergNB.PrepAttOut)} \times 100\%$$

If the **SA_LTE_NCELL_SUCC_RT_STAT_SW** option of the **NRCellAnr.AnrAlgoSwitch** parameter is selected, a neighboring LTE cell is identified as an abnormal neighboring cell if preparations of a handover to this LTE cell fail, increasing the handover success rate. Specifically, the handover success rate is equal to the result of the following:
$$N.HO.InterRAT.N2E.ExecSuccOut / N.HO.InterRAT.N2E.PrepAttOut \times 100\%.$$

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRCellAnr.AnrAlgoSwitch	MOD NRCELLANR LST NRCELLANR	gNodeB	- The SA_NR_NCELL_SUCC_RT_STAT_SW option has been added to specify whether to enable optimized measurement of the success rate of handovers to neighboring NR cells in SA networking. - The SA_LTE_NCELL_SUCC_RT_STAT_SW option has been added to specify whether to enable optimized measurement of the success rate of handovers to neighboring LTE cells in SA networking.

- Performance management
The change in this feature has no impact.
- Fault management

The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details about related operations, see the corresponding "Operation and Maintenance" section in *ANR*.

Requirement/Trouble Ticket Number

FeR-IR20240920002087-0003

RAT

NR FDD, NR TDD

5.3.3.14.2 Added Optimized Measurement for Automatic Removal of Neighboring Cells

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

After intra-RAT automatic removal of neighboring cells is enabled, the **NR_NCELL_DEL_STAT_OPT_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter can be used to control optimized measurement for automatic removal of neighboring NR cells.

- If this option is selected, the measured number of times a neighbor relationship with an NR cell is used is the sum of the numbers of the following attempts performed by the gNodeB using this neighbor relationship: handover attempts, SCC addition/change attempts, and SUL carrier addition/change attempts.
- If this option is deselected, the measured number of times such a neighbor relationship is used is the number of times the gNodeB detects the corresponding neighboring NR cell based on the PCI contained in measurement reports sent by UEs.

After inter-RAT automatic removal of neighboring cells is enabled, the **LTE_NCELL_DEL_STAT_OPT_SW** option of the **gNodeBAlgo.SelfConfigSwitch**

parameter can be used to control optimized measurement for automatic removal of neighboring LTE cells.

- If this option is selected, the measured number of times a neighbor relationship with an LTE cell is used is the actual number of handover attempts the gNodeB performs using this neighbor relationship.
- If this option is deselected, the measured number of times such a neighbor relationship is used is the number of times the gNodeB detects the corresponding neighboring LTE cell based on the PCI contained in measurement reports sent by UEs.

The change in this feature is supported only in low frequency bands.

Implementation

For details, see *ANR*. The specific sections are listed in the change history of the document.

Impact on the Network

With automatic removal of neighboring cells enabled, if the **NR_NCELL_DEL_STAT_OPT_SW** or **LTE_NCELL_DEL_STAT_OPT_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter is selected, unused neighbor relationships are more likely to be removed.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNodeBAlgo.Sel fConfigSwitch	MOD GNODEBA LGO LST GNODEBA LGO	gNodeB	- The NR_NCELL_DEL_STAT_OPT_SW option has been added to specify whether to enable optimized measurement for automatic removal of neighboring NR cells. - The LTE_NCELL_DEL_STAT_OPT_SW option has been added to specify whether to enable optimized measurement for automatic removal of neighboring LTE cells.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details about related operations, see the corresponding "Operation and Maintenance" section in *ANR*.

Requirement/Trouble Ticket Number

FeR-IR20240920002087-0002

RAT

NR FDD, NR TDD

5.3.3.14.3 Added UE Blacklist Management in Intra-RAT ANR

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The gNodeB now supports UE blacklist management in intra-RAT ANR. With this function, the gNodeB can take differentiated measures for UEs. This function is controlled by the **NR_NR_ANR_SW_OFF** option of the **gNBUECompatSw.BlacklistCtrlExtSwitch** parameter.

- If this option is selected, the gNodeB disables intra-RAT ANR for specified UEs and does not deliver reportCGI measurement configurations to these UEs.
- If this option is deselected, the gNodeB does not take differentiated measures for UEs and delivers reportCGI measurement configurations to all UEs.

The change in this feature is supported only in low frequency bands.

NOTE

This change was first incorporated into V100R022C00SPC010 but is first described in the network impact report of V100R022C00SPC050.

Implementation

For details, see *ANR*. The specific sections are listed in the change history of the document.

Impact on the Network

After UE blacklist management in intra-RAT ANR is enabled, intra-RAT ANR cannot take effect for blacklisted UEs, which may increase the service drop rate of these UEs.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNBUECompatSw.BlacklistCtrlExtSwitch	ADD GNBUECO MPATSW MOD GNBUECO MPATSW LST GNBUECO MPATSW	gNodeB	The NR_NR_ANR_SW_OFF option has been added to specify whether to enable UE blacklist management in intra-RAT ANR.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details about related operations, see the corresponding "Operation and Maintenance" section in *ANR*.

Requirement/Trouble Ticket Number

IR20241209002193

RAT

NR FDD, NR TDD

5.3.3.14.4 Intra-RAT PLMN Selection Optimization

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In the case of automatic intra-RAT external-cell PLMN list addition, the gNodeB adds the PLMNs in measurement reports from UEs or the query result from the MAE-Access to the external-cell PLMN list and now supports PLMN selection optimization. This new function is controlled by the **ANR_PLMN_SEL_OPT_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter.

- When this option is selected, PLMNs (configured by using the **gNBOperator MO**) supported by the local gNodeB in measurement reports from UEs or in the query result from the MAE-Access are preferentially selected with the corresponding information added to the external-cell PLMN list.
- When this option is deselected, the gNodeB selects PLMNs according to their sequence in measurement reports from UEs or the query result from the MAE-Access and adds the corresponding information to the external-cell PLMN list.

The change in this feature is supported only in low frequency bands.

Implementation

For details, see *ANR*. The specific sections are listed in the change history of the document.

Impact on the Network

In the case of automatic intra-RAT external-cell PLMN list addition, with PLMN selection optimization enabled, PLMNs supported by the local gNodeB are preferentially selected. This reduces the impact of PLMN addition failure due to insufficient capacity of the external-cell PLMN list.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNodeBAlgo.Sel fConfigSwitch	MOD GNODEBA LGO LST GNODEBA LGO	gNodeB	The ANR_PLMN_SEL_OPT_SW option has been added to specify whether to enable PLMN selection optimization.

- Performance management

The change in this feature has no impact.

- Fault management

The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

The change in this feature has no impact.

- Mutually exclusive features/functions

The change in this feature has no impact.

- Impacted features/functions

The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details about related operations, see the corresponding "Operation and Maintenance" section in *ANR*.

Requirement/Trouble Ticket Number

IR20250528001857

RAT

NR FDD, NR TDD

5.3.3.14.5 Support for Forbidding Automatic Neighboring Frequency Configuration for a Specific Neighboring Frequency

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

After intra-RAT automatic neighboring frequency configuration is enabled, forbidding automatic neighboring frequency configuration for a specific neighboring NR frequency is supported. This function is controlled by the **AUTO_NR_FREQ_REL_FORBID_SW** option of the **NRCellFreqRelation.FrequencyCtrlSwitch** parameter.

- If this option is selected, the gNodeB cannot automatically adjust the **NRCellFreqRelation.MeasActvFlag** parameter value for the corresponding neighboring frequency based on measurement reports.
- If this option is deselected, forbidding automatic neighboring frequency configuration does not take effect.

After inter-RAT automatic neighboring frequency configuration is enabled, forbidding automatic neighboring frequency configuration for a specific neighboring LTE frequency is supported. This function is controlled by the **AUTO_EUTRAN_FREQ_REL_FORBID_SW** option of the **NRCellEutranNFreq.FrequencyCtrlSwitch** parameter.

- If this option is selected, the gNodeB cannot automatically adjust the **NRCellEutranNFreq.MeasActvFlag** parameter value for the corresponding neighboring frequency based on measurement reports.
- If this option is deselected, forbidding automatic neighboring frequency configuration does not take effect.

The change in this feature is supported only in low frequency bands.

Implementation

For details, see *ANR*. The specific sections are listed in the change history of the document.

Impact on the Network

- After intra-RAT forbidding automatic neighboring frequency configuration is enabled, the **NRCellFreqRelation.MeasActvFlag** parameter value will not be automatically adjusted for the corresponding neighboring frequency based on measurement reports. This may increase the number of unnecessary measurements on this frequency.
- After inter-RAT forbidding automatic neighboring frequency configuration is enabled, the **NRCellEutranNFreq.MeasActvFlag** parameter value will not be automatically adjusted for the corresponding neighboring frequency based on measurement reports. This may increase the number of unnecessary measurements on this frequency.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management

The change in this feature has no impact. It is controlled by the existing feature license.

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRCellFreqRelation.FrequencyCtrlSwitch	ADD NRCELLFREQUENCYRELATION MOD NRCELLFREQUENCYRELATION LST NRCELLFREQUENCYRELATION	gNodeB	The AUTO_NR_FREQ_REL_FORBID_SW option has been added to specify whether to forbid automatic neighboring frequency configuration for a specific neighboring NR frequency.
Modified	NRCellEutranFreqRelation.FrequencyCtrlSwitch	ADD NRCELLEUTRANFREQUENCYRELATION MOD NRCELLEUTRANFREQUENCYRELATION LST NRCELLEUTRANFREQUENCYRELATION	gNodeB	The AUTO_EUTRAN_FREQ_REL_FORBID_SW option has been added to specify whether to forbid automatic neighboring frequency configuration for a specific neighboring LTE frequency.

- Performance management

The change in this feature has no impact.

- Fault management

The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

The change in this feature has no impact.

- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details about related operations, see the corresponding "Operation and Maintenance" section in *ANR*.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0035

FeR-IR20250918000777-0010

RAT

NR FDD, NR TDD

5.3.3.14.6 Inter-RAT Neighboring Cell Class Label Management

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Inter-RAT neighboring cell class label management is now supported, which is controlled by the **EUTRAN_NCELL_CLASS_LABEL_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter.

During addition of missing neighboring cells, assume that cell A has two adjacent cells, cell B and cell C. Cell B and cell C work on the same frequency and are configured with the same PCI, and cell B is in the NRT of cell A. When subsequent event-triggered measurement reports contain the same PCI as cell B and cell C, the gNodeB queries the NRT and selects cell B in the NRT as the target cell for the handover. When the UE moves, the measured cell is either of the two adjacent cells corresponding to the PCI. Therefore, if the gNodeB selects an incorrect target cell, the UE handover fails. In this scenario, the gNodeB automatically sets **NRCeUtraNRelation.EutraNCellClassLabel** (this parameter cannot be set by running MML commands) to **FORMAL** or **EXTENDED**.

The **N2L_HO_SUCC_RATE_INTVL_CFG_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter specifies whether the measurement period is configurable for automatic neighboring cell class label adjustment.

- If this option is selected, the measurement period can be configured using the **gNodeBAlgo.N2LhoSuccRateStatInterval** parameter.
- If this option is deselected, the measurement period is fixed at 24 hours and the gNodeB performs a measurement at night every day.

When the number of handovers from cell A to cell B is less than the **NRCellAnr.LteNCellHoNumThld** parameter value within a measurement period, the gNodeB carries the number over the next measurement period until the number is greater than or equal to this parameter value. When the number of handovers from cell A to cell B is greater than or equal to the **NRCellAnr.LteNCellHoNumThld** parameter value, the gNodeB automatically sets the neighboring cell class label (specified by the **NRCellEutraNRelation.EutraNCellClassLabel** parameter) for cell B based on the neighboring cell handover success rate threshold (specified by the **NRCellAnr.EutraNCellHoSuccRateThld** parameter).

- When the **NRCellAnr.EutraNCellHoSuccRateThld** parameter is set to **0**, the gNodeB automatically sets the **NRCellEutraNRelation.EutraNCellClassLabel** parameter to **FORMAL** for cell B.
- When the **NRCellAnr.EutraNCellHoSuccRateThld** parameter is set to a value other than **0** and the handover success rate of cell B is less than this value, the gNodeB automatically sets the **NRCellEutraNRelation.EutraNCellClassLabel** parameter to **EXTENDED** for cell B. When the PCI of cell B and cell C is reported again in subsequent event-based measurements:
 - If the reported CGI of cell C is different from that of cell B corresponding to the PCI in the NRT, the gNodeB adds cell C to the NRT and sets the **NRCellEutraNRelation.EutraNCellClassLabel** parameter to **FORMAL** for cell B.
 - If the number of consecutive measurements with the report type of *reportCGI* performed by the gNodeB on cell B is greater than or equal to the value of the **NRCellAnr.EutraNCellRptCgiNumThld** parameter, the gNodeB sets the **NRCellEutraNRelation.EutraNCellClassLabel** parameter to **FORMAL** for cell B.

NOTE

After the gNodeB sets the **NRCellEutraNRelation.EutraNCellClassLabel** parameter to **FORMAL** for cell B through measurements of report type *reportCGI*, the handover success rate threshold is determined as follows in the next measurement:

- If the **NRCellAnr.EutraNCellHoSuccRateThld** parameter value remains unchanged in the next measurement, the handover success rate threshold in the next measurement is equal to that in the current measurement multiplied by **NRCellAnr.EutraNCellHoSuccRateThld**.
- If there is a change in the **NRCellAnr.EutraNCellHoSuccRateThld** parameter value in the next measurement, the handover success rate threshold in the next measurement is specified by this parameter.
- When the **NRCellAnr.EutraNCellHoSuccRateThld** parameter is set to a value other than **0** and the handover success rate of cell B is greater than or equal to this value, the value of the **NRCellEutraNRelation.EutraNCellClassLabel** parameter remains unchanged.

Specifically:

- The measurement of the number of handovers is controlled by the **SA_LTE_NCELL_SUCC_RT_STAT_SW** option of the **NRCellAnr.AnrAlgoSwitch** parameter. If this option is selected, the number of handover attempts is measured. If this option is deselected, the number of handover executions is measured.

- The handover success rate is equal to the result of the following: Number of successful handovers/Number of handovers x 100%.

The change in this feature is supported only in low frequency bands.

Implementation

For details, see *ANR*. The specific sections are listed in the change history of the document.

Impact on the Network

If neighboring cell class label management is enabled, neighboring LTE cells with the handover success rate lower than the threshold specified by the **NRCellAnr.EutraNCellHoSuccRateThld** parameter are configured as extended neighboring cells. The gNodeB can detect adjacent cells with PCI confusions and then add them as neighboring cells, preventing the handover success rate from decreasing due to PCI confusions. Specifically, the handover success rate is equal to the result of the following: **N.HO.InterRAT.N2E.ExecSuccOut/N.HO.InterRAT.N2E.PrepareAttOut** × 100%.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNodeBAlgo.SelfConfigSwitch	MOD GNODEBALGO LST GNODEBALGO	gNodeB	- The EUTRAN_NCELL_CLASS_LABEL_SW option has been added to control inter-RAT neighboring cell class label management. - The N2L_HO_SUCC_RATE_INTVL_CFG_SW option has been added to specify whether the interval at which the handover success rate is measured is configurable for the following functions: inter-RAT neighboring cell class label management, automatic optimization of neighbor relationship attributes based on the handover success rate, and automatic optimization of neighbor relationship attributes based on identification of over-distant neighboring cells.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRCeANr.AnrAlgoSwitch	MOD NRCELLANR LST NRCELLANR	gNodeB	The SA_LTE_NCELL_SUCC_RT_STAT_SW option has been added to control inter-RAT neighboring cell class label management.
Modified	NRCeANr.LteNCellHoNumThld	MOD NRCELLANR LST NRCELLANR	gNodeB	This parameter has been added to control inter-RAT neighboring cell class label management.
New	NRCeEutraNRelation.EutraNCellClassLabel	ADD NRCELLEUTRANRELATION MOD NRCELLEUTRANRELATION LST NRCELLEUTRANRELATION	gNodeB	This parameter has been added to specify the neighboring cell class label of a neighboring E-UTRAN cell.
New	NRCeANr.EutraNCellHoSuccRateThld	MOD NRCELLANR LST NRCELLANR	gNodeB	This parameter has been added to specify the threshold expressed as the success rate of handovers to a neighboring E-UTRAN cell, which is used in inter-RAT neighboring cell class label management.

Change Type	Parameter ID	MML Command	NE	Description
New	<i>NRCellAnr.EutraNCellRptCgiNumThld</i>	MOD NRCELLANR LST NRCELLANR	gNodeB	This parameter has been added to specify the threshold expressed as the number of reportCGI measurements on neighboring E-UTRAN cells.
New	<i>gNodeBAlgo.N2LhoSuccRateStatInterval</i>	MOD GNODEBALGO LST GNODEBALGO	gNodeB	This parameter has been added to specify the interval at which the success rate of NR-to-LTE handovers is measured for the following functions: inter-RAT neighboring cell class label management, automatic optimization of neighbor relationship attributes based on the handover success rate, and automatic optimization of neighbor relationship attributes based on identification of over-distant neighboring cells.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions

The change in this feature has no impact.

- Impacted features/functions

The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details about related operations, see the corresponding "Operation and Maintenance" section in *ANR*.

Requirement/Trouble Ticket Number

FeR-IR20250918000777-0010

RAT

NR FDD, NR TDD

5.3.3.14.7 Switch for Fast NR Networking Option Query#

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In the case of NCL update triggered by queries from the MAE-Access or external-cell PLMN list update triggered by queries from the MAE-Access, if the MAE-Access does not return any query result of the

NRExternalNCell.NrNetworkingOption parameter values of external NG-RAN cells or the query result is empty, the gNodeB cannot update the **NRExternalNCell.NrNetworkingOption** parameter value of the corresponding external NG-RAN cell. In this case, the gNodeB now supports fast NR networking option query. This function is controlled by the **NR_NTW_OPTN_FAST_QRY_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter.

- If this option is selected, the gNodeB queries the **NRExternalNCell.NrNetworkingOption** parameter values of external NG-RAN cells from the MAE-Access at night every day.
- If this option is deselected, the gNodeB queries the **NRExternalNCell.NrNetworkingOption** parameter values of external NG-RAN cells from the MAE-Access at night every seven days.

In the preceding scenario, if a version earlier than V100R022C10 is used, the gNodeB queries the NR networking options of external NG-RAN cells from the MAE-Access at night every day; after an upgrade to V100R022C10 or later, the **NR_NTW_OPTN_FAST_QRY_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter is deselected by default, and the gNodeB performs queries from the MAE-Access at night every seven days.

The change in this feature is supported only in low frequency bands.

Implementation

For details, see *ANR*. The specific sections are listed in the change history of the document.

Impact on the Network

After fast NR networking option query is enabled, the number of NR networking option queries increases. As a result, signaling resources are consumed due to repeated invalid requests.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNodeBAlgo.Sel fConfigSwitch	MOD GNODEBA LGO LST GNODEBA LGO	gNodeB	The NR_NTW_OPTN_FAS T_QRY_SW option has been added to specify whether to enable fast NR networking option query.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

- The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details about related operations, see the corresponding "Operation and Maintenance" section in *ANR*.

Requirement/Trouble Ticket Number

IR20250528001857

RAT

NR FDD, NR TDD

5.3.3.14.8 Added Support for Measurement of the Number of Times an External Cell is Used by Optimized Measurement for Automatic Removal of Neighboring Cells

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Optimized measurement for automatic removal of neighboring NR cells now supports the measurement on the number of times an external NR cell is used. This measurement is controlled by the **NR_NCELL_DEL_STAT_OPT_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter.

- If this option is selected, the measured number of times an external NR cell is used is the sum of the numbers of the following attempts performed by the gNodeB using this external NR cell: handover attempts, SCC addition/change attempts, and SUL carrier addition/change attempts.
- If this option is deselected, the measured number of times the external NR cell is used is the number of times the gNodeB detects this cell based on the PCI contained in measurement reports sent by UEs.

Optimized measurement for automatic removal of neighboring LTE cells now supports the measurement on the number of times an external LTE cell is used. This measurement is controlled by the **LTE_NCELL_DEL_STAT_OPT_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter.

- If this option is selected, the measured number of times an external LTE cell is used is the actual number of handover attempts the gNodeB performs using this external LTE cell.

- If this option is deselected, the measured number of times the external LTE cell is used is the number of times the gNodeB detects this cell based on the PCI contained in measurement reports sent by UEs.

The change in this feature is supported only in low frequency bands.

Implementation

For details, see *ANR*. The specific sections are listed in the change history of the document.

Impact on the Network

With automatic removal of neighboring cells enabled, if the **NR_NCELL_DEL_STAT_OPT_SW** or **LTE_NCELL_DEL_STAT_OPT_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter is selected, unused external cells are more likely to be removed.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNodeBAlgo.SelfConfigSwitch	MOD GNODEBA LGO LST GNODEBA LGO	gNodeB	- The NR_NCELL_DEL_STAT_OPT_SW option has been added to optimize the measurement on the number of times an external NR cell is used. - The LTE_NCELL_DEL_STAT_OPT_SW option has been added to optimize the measurement on the number of times an external LTE cell is used.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details about related operations, see the corresponding "Operation and Maintenance" section in *ANR*.

Requirement/Trouble Ticket Number

IR20241209002193

RAT

NR FDD, NR TDD

5.3.3.14.9 Support for the Configuration of the Measurement Period and the Threshold Expressed as the Number of reportCGI Measurements for Intra-RAT Neighboring Cell Class Label Management

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The measurement period and the threshold expressed as the number of reportCGI measurements are now configurable for intra-RAT neighboring cell class label management.

- The **NR_HO_SUCC_RATE_INTVL_CFG_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter specifies whether the measurement period is configurable for automatic neighboring cell class label adjustment.
 - If this option is selected, the measurement period can be configured using the **gNodeBAlgo.NrHoSuccRateStatInterval** parameter.
 - If this option is deselected, the measurement period is fixed at 24 hours and the gNodeB performs a measurement at night every day.
- The threshold expressed as the number of reportCGI measurements is specified by the **NRCeANr.NrNCellRptCgiNumThld** parameter.

The change in this feature is supported only in low frequency bands.

Implementation

For details, see *ANR*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNodeBAlgo.Sel fConfigSwitch	MOD GNODEBA LGO LST GNODEBA LGO	gNodeB	The NR_HO_SUCC_RATE_INTVL_CFG_SW option has been added to specify whether the interval at which the neighboring NR cell handover success rate is measured is configurable for the following functions: intra-RAT neighboring cell class label management, prohibition of handovers to neighboring cells based on the handover success rate, and automatic optimization of neighbor relationship attributes based on identification of over-distant neighboring cells.

Change Type	Parameter ID	MML Command	NE	Description
New	gNodeBAlgo.NrHoSuccRateStatInterval	MOD GNODEBALGO LST GNODEBALGO	gNodeB	This parameter has been added to specify the interval at which the neighboring NR cell handover success rate is measured for the following functions: intra-RAT neighboring cell class label management, prohibition of handovers to neighboring cells based on the handover success rate, and automatic optimization of neighbor relationship attributes based on identification of over-distant neighboring cells.
New	NRCellAnr.NrNCellRptCgiNumThld	MOD NRCELLANR LST NRCELLANR	gNodeB	This parameter has been added to specify the threshold expressed as the number of reportCGI measurements on neighboring NR cells.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.

- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details about related operations, see the corresponding "Operation and Maintenance" section in *ANR*.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0035

RAT

NR FDD, NR TDD

5.3.3.14.10 Support for the Configuration of the Measurement Periods for "Automatic Optimization of Neighbor Relationship Attributes Based on the Handover Success Rate" and "Automatic Optimization of Neighbor Relationship Attributes Based on Identification of Over-Distant Neighboring Cells"

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The measurement periods are now configurable for "automatic optimization of neighbor relationship attributes based on the handover success rate" and "automatic optimization of neighbor relationship attributes based on identification of over-distant neighboring cells".

After "prohibition of handovers to neighboring cells based on the handover success rate" (a function of "intra-RAT automatic optimization of neighbor relationship attributes based on the handover success rate") or "automatic optimization of neighbor relationship attributes based on identification of over-distant neighboring cells" is enabled, the gNodeB optimizes the setting of the **NRCellRelation.NoHoFlag** parameter for neighbor relationships based on the handover success rate or identification of over-distant neighboring cells within a measurement period. The **NR_HO_SUCC_RATE_INTVL_CFG_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter specifies whether the measurement period is configurable.

- If this option is selected, the measurement period can be configured using the **gNodeBAlgo.NrHoSuccRateStatInterval** parameter.
- If this option is deselected, the measurement period is fixed at 24 hours and the gNodeB performs a measurement at night every day.

After "inter-RAT automatic optimization of neighbor relationship attributes based on the handover success rate" or "automatic optimization of neighbor relationship attributes based on identification of over-distant neighboring cells" is enabled, the

gNodeB optimizes the setting of the **NRCellEutraNRelation.NoHoFlag** parameter for neighbor relationships based on the handover success rate or identification of over-distant neighboring cells within a measurement period. The **N2L_HO_SUCC_RATE_INTVL_CFG_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter specifies whether the measurement period is configurable.

- If this option is selected, the measurement period can be configured using the **gNodeBAlgo.N2LhoSuccRateStatInterval** parameter.
- If this option is deselected, the measurement period is fixed at 24 hours and the gNodeB performs a measurement at night every day.

The change in this feature is supported only in low frequency bands.

Implementation

For details, see *ANR*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNodeBAlgo.SelfConfigSwitch	MOD GNODEBALGO LST GNODEBALGO	gNodeB	- The NR_HO_SUCC_RATE_INTVL_CFG_SW option has been added to specify whether the interval at which the neighboring NR cell handover success rate is measured is configurable for the following functions: intra-RAT neighboring cell class label management, prohibition of handovers to neighboring cells based on the handover success rate, and automatic optimization of neighbor relationship attributes based on identification of over-distant neighboring cells. - The N2L_HO_SUCC_RATE_INTVL_CFG_SW option has been added to specify whether the interval at which the success rate of NR-to-LTE handovers is measured is configurable for the following functions: inter-RAT neighboring cell class label

Change Type	Parameter ID	MML Command	NE	Description
				management, automatic optimization of neighbor relationship attributes based on the handover success rate, and automatic optimization of neighbor relationship attributes based on identification of over-distant neighboring cells.
New	gNodeBAlgo.NrHoSuccRateStatInterval	MOD GNODEBALGO LST GNODEBALGO	gNodeB	This parameter has been added to specify the interval at which the neighboring NR cell handover success rate is measured for the following functions: intra-RAT neighboring cell class label management, prohibition of handovers to neighboring cells based on the handover success rate, and automatic optimization of neighbor relationship attributes based on identification of over-distant neighboring cells.

Change Type	Parameter ID	MML Command	NE	Description
New	gNodeBAlgo.N2LhoSuccRateStatInterval	MOD GNODEBALGO LST GNODEBALGO	gNodeB	This parameter has been added to specify the interval at which the success rate of NR-to-LTE handovers is measured for the following functions: inter-RAT neighboring cell class label management, automatic optimization of neighbor relationship attributes based on the handover success rate, and automatic optimization of neighbor relationship attributes based on identification of over-distant neighboring cells.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details about related operations, see the corresponding "Operation and Maintenance" section in *ANR*.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0035

FeR-IR20250918000777-0010

RAT

NR FDD, NR TDD

5.3.3.15 FOFD-021209 Inter-RAT Mobility from NG-RAN to E-UTRAN

5.3.3.15.1 Support for Downlink-SINR-based Mobility to E-UTRAN

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Downlink-SINR-based mobility to E-UTRAN is supported. With this function, the base station delivers SSB-SINR-based measurement configurations related to inter-RAT events A1 and A2 to a UE and, based on the downlink SINR measurement results, implements inter-RAT mobility from NG-RAN to E-UTRAN. Downlink-SINR-based mobility to E-UTRAN supports only handover and redirection.

This function is controlled by the **DL_SINR_MOBILITY_TO_EUTRAN_SW** option of the **NRCellMobilityConfig.InterRatHoExtSwitch** parameter. The thresholds for events A1 and A2 are specified by the **NRCellInterRHoMeaGrp.DlSinrHoToEutranA1Thld** and **NRCellInterRHoMeaGrp.DlSinrHoToEutranA2Thld** parameters, respectively. In addition, the value of **NRCellInterRHoMeaGrp.DlSinrHoToEutranA1Thld** must be greater than that of **NRCellInterRHoMeaGrp.DlSinrHoToEutranA2Thld**.

The change in this feature is supported only in SA networking.

Implementation

For details, see *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*. The specific sections are listed in the change history of the document.

Impact on the Network

When downlink-SINR-based mobility to E-UTRAN takes effect, the values of the following counters increase.

Counter ID	Counter Name
1911817063	N.HO.InterRAT.N2E.ExecSuccOut
1911817061	N.HO.InterRAT.N2E.ExecAttOut
1911817056	N.HO.InterRAT.N2E.PrepareAttOut

Counter ID	Counter Name
1911817057	N.HO.InterRAT.N2E.AMFAbnormRsp
1911817064	N.HO.InterRAT.N2E.Prep.FailOut.AMF
1911817065	N.HO.InterRAT.N2E.Prep.FailOut.PrepFailure
1911817059	N.HO.InterRAT.N2E.Prep.FailOut.NoReply
1911817054	N.HO.InterRAT.NCell.N2E.PrepAttOut
1911817053	N.HO.InterRAT.NCell.N2E.ExecAttOut
1911817055	N.HO.InterRAT.NCell.N2E.ExecSuccOut
1911817044	N.RRCRedirection.N2E.Coverage
1911822969	N.HO.InterRAT.N2E.Exec.FailOut.ReEst
1911822968	N.HO.InterRAT.N2E.Exec.FailOut.ReEst.Voice
1911851726	N.HO.InterRAT.N2E.DLSinr.PreAttOut
1911851727	N.HO.InterRAT.N2E.DLSinr.ExecAttOut
1911851728	N.HO.InterRAT.N2E.DLSinr.ExecSuccOut

Impact on NEs

This change is implemented on the gNodeB.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRCeIlMo bIlItyConfi g.InterRat HoExtSwit ch	MOD NRCELLMOB ILITYCONFIG LST NRCELLMOB ILITYCONFIG	gNodeB	This parameter has been added to specify whether to enable the extension of inter-RAT handover algorithms. The DL_SINR_MOBILITY_TO_EUTRAN_SW option specifies whether to enable downlink-SINR-based mobility to E-UTRAN.
New	NRCeIlInte rRHoMea Grp.DlSInr HoToEutra nA1Thld	ADD NRCELLINTE RRHOMEAG RP MOD NRCELLINTE RRHOMEAG RP LST NRCELLINTE RRHOMEAG RP	gNodeB	This parameter has been added to specify the threshold for event A1 related to downlink SSB-SINR-based inter-RAT handover from NG-RAN to E-UTRAN.
New	NRCeIlInte rRHoMea Grp.DlSInr HoToEutra nA2Thld	ADD NRCELLINTE RRHOMEAG RP MOD NRCELLINTE RRHOMEAG RP LST NRCELLINTE RRHOMEAG RP	gNodeB	This parameter has been added to specify the threshold for event A2 related to downlink SSB-SINR-based inter-RAT handover from NG-RAN to E-UTRAN.

- Performance management
The following counters have changed.

Change Type	Counter Name	Function Subset	NE	Description
New	N.HO.Inter RAT.N2E.DL Sinr.PreAtt Out	Outgoing Inter-RAT Handover Measurement	gNode B	This counter has been added to measure the number of downlink-SINR-based NR-to-EUTRAN outgoing handover preparation attempts.
New	N.HO.Inter RAT.N2E.DL Sinr.ExecAtt Out	Outgoing Inter-RAT Handover Measurement	gNode B	This counter has been added to measure the number of downlink-SINR-based NR-to-EUTRAN outgoing handover executions.
New	N.HO.Inter RAT.N2E.DL Sinr.ExecSu ccOut	Outgoing Inter-RAT Handover Measurement	gNode B	This counter has been added to measure the number of successful downlink-SINR-based NR-to-EUTRAN outgoing handovers.

- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low- frequ ency NR TDD	Downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN	MOBILITY_TO_EUTRAN_SW option of the NRCellAlgorithmSwitch.InterRatServiceMobilitySw parameter	<i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i>	Downlink-SINR-based mobility to E-UTRAN requires downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN.

- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*.

Requirement/Trouble Ticket Number

FeR-IR20240828001990-0001

RAT

NR FDD, NR TDD

5.3.3.15.2 Separate Priority Configuration for Voice Services in Blind Mode

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The "separate priority configuration for voice services in blind mode" function includes the following parts:

- During blind handovers or blind redirections for voice fallback or voice service mobility, the gNodeB selects the target LTE cells for blind handovers or the target LTE frequencies for blind redirections in descending order of priorities specified by the **NRCeUtranNFreq.VolteBlindPriority** parameter.
- Operator-specific frequency priorities can be configured for voice fallback or voice service mobility involved in blind handovers or blind redirections. These priorities are specified by the **gNBFreqPriorityGroup.VolteBlindPriority** parameter.

For an upgrade from a version earlier than V100R022C10SPC010 to V100R022C10SPC010 or later, the value of the **NRCeUtranNFreq.VolteBlindPriority** parameter after the upgrade inherits the value of the **NRCeUtranNFreq.VoltePriority** parameter before the upgrade.

For an upgrade from a version earlier than V100R022C10SPC010 to V100R022C10SPC010 or later, the value of the **gNBFreqPriorityGroup.VolteBlindPriority** parameter after the upgrade inherits the value of the **gNBFreqPriorityGroup.VoltePriority** parameter before the upgrade.

Implementation

For details, see *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature is implemented on the gNodeB.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRCellEutranNFreq. <i>VolteBlindPriority</i>	ADD NRCELLEUT RANNFREQ MOD NRCELLEUT RANNFREQ LST NRCELLEUT RANNFREQ	gNodeB	This parameter has been added to specify the frequency priority for blind handovers or blind redirections from NG-RAN to E-UTRAN for voice service UEs.
New	gNBFreqPriorityGroup. <i>VolteBlindPriority</i>	ADD GNBFREQPR IORITYGRO UP MOD GNBFREQPR IORITYGRO UP LST GNBFREQPR IORITYGRO UP	gNodeB	This parameter has been added to specify the operator-specific frequency priority for blind handovers or blind redirections from NG-RAN to E-UTRAN for voice service UEs.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change takes effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*.

Requirement/Trouble Ticket Number

FeR-IR20250117001886-0010

IR20250318001395

RAT

NR FDD, NR TDD

5.3.3.15.3 Optimized Downlink-Coverage-based Inter-RAT Mobility from NG-RAN to E-UTRAN

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Earlier, the **MOBILITY_TO_EUTRAN_SW** option of the **NRCellAlgoSwitch.InterRatServiceMobilitySw** parameter applies to both downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN and NSA/SA selection based on downlink traffic volume. Downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN cannot be disabled separately.

New options have been added to allow downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN to be disabled separately. These options are the **DL_COV_N2L_MOB_OFF_SWITCH** option of the **NRInterRatHoParam.HoModeSwitch** parameter and the **DL_COV_N2L_MOB_OFF_SWITCH** option of the **NRCellOpPolicy.InterRatHoPolicySwitch** parameter.

The change in this feature is supported only in SA networking.

Implementation

For details, see *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*. The specific sections are listed in the change history of the document.

Impact on the Network

If either the **DL_COV_N2L_MOB_OFF_SWITCH** option of the **NRInterRatHoParam.HoModeSwitch** parameter or the **DL_COV_N2L_MOB_OFF_SWITCH** option of the **NRCellOpPolicy.InterRatHoPolicySwitch** parameter is selected, downlink-coverage-based handovers and redirections from NG-RAN to E-UTRAN cannot be performed, which may cause service drops. If these options are both deselected, network performance is not affected.

Impact on NEs

This change is implemented on the gNodeB.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRInterRatHoParam.HoModeSwitch	MOD NRINTERRATHOPARAM LST NRINTERRATHOPARAM	gNodeB	The DL_COV_N2L_MOB_OFF_SWITCH option has been added to specify whether to disable downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN.
Modified	NRCellOpPolicy.InterRatHoPolicySwitch	ADD NRCELLOPPOLICY MOD NRCELLOPPOLICY LST NRCELLOPPOLICY	gNodeB	The DL_COV_N2L_MOB_OFF_SWITCH option has been added to specify whether to disable downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*.

Requirement/Trouble Ticket Number

FeR-IR20250213000671-0250

RAT

NR FDD, NR TDD

5.3.3.16 FOFD-021210 Voice Fallback

5.3.3.16.1 Separate Priority Configuration for Voice Services in Blind Mode

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The "separate priority configuration for voice services in blind mode" function includes the following parts:

- During blind handovers or blind redirections for voice fallback or voice service mobility, the gNodeB selects the target LTE cells for blind handovers or the target LTE frequencies for blind redirections in descending order of priorities specified by the **NRCeUtranNFreq.VolteBlindPriority** parameter.
- Operator-specific frequency priorities can be configured for voice fallback or voice service mobility involved in blind handovers or blind redirections. These priorities are specified by the **gNBFreqPriorityGroup.VolteBlindPriority** parameter.

For an upgrade from a version earlier than V100R022C10SPC010 to V100R022C10SPC010 or later, the value of the **NRCeUtranNFreq.VolteBlindPriority** parameter after the upgrade inherits the value of the **NRCeUtranNFreq.VoltePriority** parameter before the upgrade.

For an upgrade from a version earlier than V100R022C10SPC010 to V100R022C10SPC010 or later, the value of the **gNBFreqPriorityGroup.VolteBlindPriority** parameter after the upgrade inherits the value of the **gNBFreqPriorityGroup.VoltePriority** parameter before the upgrade.

Implementation

For details, see *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature is implemented on the gNodeB.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRCeUtranNFreq.VolteBlindPriority	ADD NRCeUTRANNFREQ MOD NRCeUTRANNFREQ LST NRCeUTRANNFREQ	gNodeB	This parameter has been added to specify the frequency priority for blind handovers or blind redirections from NG-RAN to E-UTRAN for voice service UEs.

Change Type	Parameter ID	MML Command	NE	Description
New	gNBFreqPriorityGroup.VolteBlindPriority	ADD GNBFREQPR ORITYGRO UP MOD GNBFREQPR ORITYGRO UP LST GNBFREQPR ORITYGRO UP	gNodeB	This parameter has been added to specify the operator-specific frequency priority for blind handovers or blind redirections from NG-RAN to E-UTRAN for voice service UEs.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change takes effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*.

Requirement/Trouble Ticket Number

FeR-IR20250117001886-0010

IR20250318001395

RAT

NR FDD, NR TDD

5.3.3.17 FOFD-021211 Multi-Operator Sharing

5.3.3.17.1 Prohibition of Multi-Operator Sharing During Xn-U Self-Setup

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In RAN sharing with common carrier, the "prohibition of multi-operator sharing during Xn-U self-setup" function is designed to enable separated Xn-U links for different operators to achieve service isolation. This function is controlled by the **XN_UP_SELF_SETUP_NO_MOCN_SW** option of the **gNBInterfaceParam.InterfaceSelfCfgOptSw** parameter. After this function is enabled, multiple operators each deploy their own self-setup Xn-U links, and these operators do not share user-plane transmission resources.

Implementation

For details, see *Multi-Operator Sharing*. The specific sections are listed in the change history of the document.

Impact on the Network

The "prohibition of multi-operator sharing during Xn-U self-setup" function allows multiple operators to deploy their own self-setup Xn-U links for the purpose of service isolation. However, if an Xn-U link transmission fault occurs for an operator, it may negatively impact that operator's services, as this function prohibits user-plane transmission resource sharing across operators.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameter has changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNBInterfaceParam.InterfaceSelfCfgOptSw	MOD GNBINTERFAC EPARAM LST GNBINTERFAC EPARAM	gNodeB	The XN_UP_SELF_SETUP_NO_MOCN_SW option has been added to specify whether to enable "prohibition of multi-operator sharing during Xn-U self-setup".

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationship has been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Basic networking for RAN sharing with common carrier	gNBSharingMode.gNBMultiOpSharingMode set to SHARED_FREQ	<i>Multi-Operator Sharing</i>	Prohibition of multi-operator sharing during Xn-U self-setup requires RAN sharing with common carrier to be deployed on the live network.

- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Multi-Operator Sharing*.

Requirement/Trouble Ticket Number

FeR-IR20251205000509-0010

RAT

NR FDD, NR TDD

5.3.3.18 FOFD-030201 mmWave 3D Coverage Pattern

5.3.3.18.1 Support for the Capacity Scenario with 2:3 Slot Configuration by the mmWave 3D Coverage Pattern (for High Frequency Bands Only)

NEs Involved

5900 series base stations

Feature/Function Description

The capacity scenario with 2:3 slot configuration (2_3_DSUUU) is now supported by the mmWave 3D coverage pattern. For details, see [Figure 5-4](#). Given that beams are discrete and discontinuous, there are dents between adjacent beams, causing signal strength attenuation. In this case, the capacity scenario can be selected to narrow down the dents between beams.

Figure 5-4 Capacity scenario with 2:3 slot configuration (2_3_DSUUU)



Specifically:

- The 2:3 slot configuration is configured by setting the **NRDUCell.SlotAssignment** parameter to **2_3_DSUUU**.
- The capacity scenario is configured by setting the **NRDUCellTrpMmwavBeam.CoverageScenario** parameter to **SCENARIO_101**.

The capacity scenario with 2:3 slot configuration has the following constraints:

- The PRACH configuration index (**NRDUCellPrach.PrachConfigurationIndex**) can only be set to **186**, **188**, or **65535** when all of the following conditions apply: (1) The cell slot configuration (**NRDUCell.SlotAssignment**) is set to **2_3_DSUUU**. (2) The subcarrier spacing (**NRDUCell.SubcarrierSpacing**) is set to **120KHZ**. (3) The cell is activated (by setting **NRCell.CellActiveState** to **CELL_ACTIVE**).

- If the PRACH configuration index (**NRDUCellPrach.PrachConfigurationIndex**) is set to **186**, the cell radius (**NRDUCell.CellRadius**) can only be set to a value in the range from **1** to **1160**.
- If the PRACH configuration index (**NRDUCellPrach.PrachConfigurationIndex**) is set to **188** or **65535**, the cell radius (**NRDUCell.CellRadius**) can only be set to a value in the range from **1** to **2410**.
- If the cell slot configuration (**NRDUCell.SlotAssignment**) is set to **2_3_DSUUU**, the SSB subcarrier spacing (**NRDUCell.SsbSubcarrierSpacing**) can only be set to **120KHZ**.

LampSite base stations do not support this function.

Implementation

For details, see *mmWave Beam Management (High-Frequency TDD)*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

HAAU5222, HAAU5323, HAAU5233w, HAAU5235w

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The change in this feature has no impact.
- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.

- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

The 2:3 slot configuration is set by running the **MOD NRDUCELL** command with the **NRDUCell.SlotAssignment** parameter set to **2_3_DSUUU**. The capacity scenario can be set by running the **MOD NRDUCELLTRPMMWAVBEAM** command with the **NRDUCellTrpMmwavBeam.CoverageScenario** parameter set to **SCENARIO_101**.

Requirement/Trouble Ticket Number

FeR-IR20250306002890-0003

RAT

NR TDD

5.3.3.19 FOFD-030204 COMP

5.3.3.19.1 Revised Intra-Base-Station DL CoMP Configurations for Distributed Massive MIMO Cells

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The intra-base-station DL CoMP configurations for distributed massive MIMO cells have been optimized, so that the intra-base-station DL CoMP switch (specified by the **INTRA_GNB_DL_JT_SW** option of the **NRDUCellAlgoSwitch.CompSwitch** parameter) can be turned on more flexibly. Specifically:

- Before the upgrade
The intra-base-station DL CoMP switch can be turned on only after the related parameters are set according to the following configuration constraints:
 - The corresponding **NRDUCell.LampSiteCellFlag** parameter is set to **YES**.
 - The corresponding **NRCCell.DuplexMode** parameter is set to **CELL_TDD**.
 - The corresponding **NRDUCellTrp.TrpType** parameter is set to **MASTER_DM_MIMO** or **MASTER_DM_MIMO_LIGHT**.
 - The corresponding **NRDUCellTrp.TxRxMode** parameter is set to **4T4R**, and the **NRDUCellTrp.TxRxMode** parameter is set to **4T4R** for the associated TRP with the **NRDUCellTrp.TrpType** parameter set to **SLAVE**.

- After the upgrade
The intra-base-station DL CoMP switch can be turned on directly. The following configuration constraints must be met only when the **NRCell.CellActiveState** parameter is set to **CELL_ACTIVE**.
 - The corresponding **NRDUCell.LampSiteCellFlag** parameter is set to **YES**.
 - The corresponding **NRCell.DuplexMode** parameter is set to **CELL_TDD**.
 - The corresponding **NRDUCellTrp.TrpType** parameter is set to **MASTER_DM_MIMO** or **MASTER_DM_MIMO_LIGHT**.
 - The corresponding **NRDUCellTrp.TxRxMode** parameter is set to **4T4R**, and the **NRDUCellTrp.TxRxMode** parameter is set to **4T4R** for the associated TRP with the **NRDUCellTrp.TrpType** parameter set to **SLAVE**.

Implementation

For details, see *CoMP*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This feature is implemented on the gNodeB.

Impact on Hardware Requirements

- Base station models
Intra-base-station DL CoMP between distributed massive MIMO cells requires DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.
- Boards
For details about the RF module requirements for a LampSite distributed massive MIMO cell, see *Distributed Antenna Solutions (Low-Frequency TDD)*.
- RF modules
For details about the RF module requirements for a LampSite distributed massive MIMO cell, see *Distributed Antenna Solutions (Low-Frequency TDD)*.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact.
- Configuration management
The change in this feature has no impact.
- Performance management

The change in this feature has no impact.

- Fault management

The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

The change in this feature has no impact.

- Mutually exclusive features/functions

The change in this feature has no impact.

- Impacted features/functions

The change in this feature has no impact.

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20250528001857

RAT

NR TDD

5.3.3.19.2 Intra-Base-Station UL CoMP for UEs with Specific QCIs

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Currently, after intra-base-station UL CoMP is enabled in a cell, the gNodeB delivers event A3 measurement configurations to all UEs in the cell, and intra-base-station UL CoMP takes effect for UEs that meet certain conditions. To enable intra-base-station UL CoMP to take effect for UEs with specific QCIs, intra-base-station UL CoMP for UEs with specific QCIs is introduced.

After this function is enabled (by selecting the **UL_2CELL_COMP_BEAR_POL_SW** option of the **NRDUCellComp.CompConfigPol** parameter), the gNodeB delivers measurement configurations to specific UEs. Specific UEs are those having at least one QCI for which the **COMP_SW** option of the **gNBQciBearer.ServiceDiffAlgoSwitch** parameter is selected.

Implementation

For details, see *CoMP*. The specific sections are listed in the change history of the document.

Impact on the Network

After the change in this feature takes effect, the number of UEs for which intra-base-station UL CoMP takes effect in a cell and the uplink PRB usage of the cell decrease.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellComp.CompConfigPool	MOD NRDUCELLCOMP LST NRDUCELLCOMP	gNodeB	This parameter has been added. The UL_2CELL_COMP_BEAR_POLICY_SW option of this parameter specifies the bearer policy for 2-cell UL CoMP.
Modified	gNBQCIBearer.ServiceDiffAlgorithm	ADD GNBQCIBEARER MOD GNBQCIBEARER LST GNBQCIBEARER	gNodeB	The COMP_SW option has been added to specify whether to allow CoMP to take effect for UEs with specified service bearers.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *CoMP*.

Requirement/Trouble Ticket Number

FeR-IR20241209002193-0032

RAT

NR FDD, NR TDD

5.3.3.20 FOFD-030205 Hyper Cell

5.3.3.20.1 FSD Transmission Mode of Hyper Cell

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

To improve the performance in overlapping areas, Hyper Cell now supports FSD transmission in addition to basic transmission.

In addition to the features of basic transmission mode, the FSD transmission mode considers user experience in overlapping areas. It dynamically identifies UEs in overlapping areas between TRPs and uplink and downlink interfering TRPs, and performs real-time interference avoidance on the identified interfering TRPs, thereby reducing inter-TRP interference on service and control channels. Compared with the basic transmission mode, the FSD transmission mode makes inter-TRP resource allocation more flexible and improves the performance of UEs in overlapping areas. The methods of identifying interfering TRPs are as follows:

- When the absolute value of the difference between the SRS RSRP of the serving TRP and that of another TRP is less than the value of **NRDUCellMultiTrp.UlIntrfTrpSrsRsrpDiffThld**, the gNodeB identifies the other TRP as an uplink interfering TRP for UEs in the overlapping area.
- When the absolute value of the difference between the SRS RSRP of the serving TRP and that of another TRP is less than the value of **NRDUCellMultiTrp.DlIntrfTrpSrsRsrpDiffThld**, the gNodeB identifies the other TRP as the downlink interfering TRP for UEs in the overlapping area.

The FDS transmission mode is enabled by setting
NRDUCellMultiTrp.DataTransMode to **FSD_MODE**.

Implementation

For details, see *Hyper Cell*. The specific sections are listed in the change history of the document.

Impact on the Network

- When the FSD transmission mode is used for hyper cells, the uplink and downlink scheduling and inter-TRP resource allocation of hyper cells become more flexible, and the performance of UEs in overlapping areas is further improved compared with those using the basic transmission mode. If inter-TRP interference, weak coverage, or overshoot coverage exists and coverage optimization cannot be performed in the short term, the FSD transmission mode is recommended.
- After a hyper cell is formed and the FSD transmission mode is used, inter-TRP PUSCH, PDSCH, and PDCCH resources are allocated more flexibly than when the basic transmission mode is used. Since scheduling of UEs changes, the following impacts may occur:
 - The CPU usage may increase.
 - The interference to UEs in the overlapping area decreases, the average MCS index of uplink and downlink UEs in the cell increases, the bit error rate decreases, the PDCCH CCE aggregation level decreases, and the service drop rate decreases.
 - In the overlapping area, UEs are scheduled separately in the frequency domain. In scenarios with the same traffic volume, the PRB usage of uplink and downlink traffic channels and the PDCCH CCE usage of the cell may decrease.
 - PDCCH resources are scheduled in frequency division mode for UEs in the overlapping area between TRPs. As a result, the available CCE resources may decrease.
 - If this mode is enabled in low inter-TRP interference and heavy-load scenarios, the number of RBs that can be scheduled may decrease due to frequency-domain staggering in the overlapping area. As a result, the cell-level uplink and downlink throughputs and the UE-level uplink and downlink throughputs decrease.
 - If the **NRDUCellMultiTrp.DlIntrfTrpSrsRsrpDiffThld** parameter is set to an excessively large or small value, interfering TRPs may not be accurately identified, and the overall cell performance may deteriorate. If this parameter is set to an excessively small value, interfering TRPs cannot be identified, and effective interference avoidance scheduling cannot be performed to improve the performance of UEs in the overlapping area. If this parameter is set to an excessively large value, the available resources for scheduling may be limited due to excessive avoidance in medium- and heavy-load scenarios. As a result, the risks of decreasing cell- and UE-level uplink and downlink throughputs increase. The default value is recommended.
 - If inter-BBP scenarios are involved and the number of SSB beams varies under different TRPs, the maximum number of SSB beams of each TRP in

the cell is used. As a result, the amount of PDSCH resources that can be scheduled may decrease.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameter has changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellMultiTrp.D ataTransMode	MOD NRDUCELL MULTITRP LST NRDUCELL MULTITRP	gNodeB	The value FSD_MODE has been added to specify FSD transmission mode.
New	NRDUCellMultiTrp.U lIntrfTrpSrsRsrpDiffThld	MOD NRDUCELL MULTITRP LST NRDUCELL MULTITRP	gNodeB	This parameter has been added to specify the uplink TRP SRS RSRP difference threshold.
New	NRDUCellMultiTrp.D lIntrfTrpSrsRsrpDiffThld	MOD NRDUCELL MULTITRP LST NRDUCELL MULTITRP	gNodeB	This parameter has been added to specify the downlink TRP SRS RSRP difference threshold.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

The change in this feature has no impact.

- Mutually exclusive features/functions

The following mutually exclusive relationships have been added.

RA T	Function Name	Function Switch	Reference	Description
Lo w- fre qu enc y TD D	NR DU cell resource management between network slices	RB_DYNAMIC_ CONTROL_SW option of the NRDUCellAlgo Switch.Network kSliceAlgoSwit ch parameter	<i>Network Slicing</i>	NR DU cell resource management between network slices cannot be enabled if the FSD transmission mode is used.
Lo w- fre qu enc y TD D	RFSP-group- based RB resource control	RB_DYNAMIC_ CONTROL_SW option of the NRDUCellCom monSw.RfspAl goSwitch parameter	<i>Resource Control in NSA Networking</i>	RFSP-group- based RB resource control cannot be enabled if the FSD transmission mode is used.
Lo w- fre qu enc y TD D	Dynamic RB control	RB_DYNAMIC_ CONTROL_SW option of the NRDUCellAlgo Switch.HighRe liabilitySwitch parameter	<i>URLLC</i>	Dynamic RB control cannot be enabled if the FSD transmission mode is used.
Lo w- fre qu enc y TD D	QCI-based resource control	RB_DYNAMIC_ CONTROL_SW option of the NRDUCellFeat ureSw.QciBase dDiffServExpS w parameter	<i>QCI-based Resource Control</i>	QCI-based resource control cannot be enabled if the FSD transmission mode is used.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Operator-specific guarantee on the minimum number of available RBs	MIN_RB_GUARANTEE_SW option of the NRDUCellAlgoSwitch.RanSharingAlgoSwitch parameter	<i>Multi-Operator Sharing</i>	Operator-specific guarantee on the minimum number of available RBs cannot be enabled if the FSD transmission mode is used.

- Impacted features/functions

The following impacted functions have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Smart frequency selective scheduling	SMART_FREQ_SEL_SWITCH_SW option of the NRDUCellPdsch.DIPdschAlgoSwitch parameter	None	If the FSD transmission mode is used for a hyper cell and this option is selected, the PDSCH throughput may significantly decrease due to staggering of overlapping areas. Therefore, enabling smart frequency selective scheduling is not recommended.
Low-frequency TDD	Spectral-efficiency-based adaptive switching between SU-MIMO and MU-MIMO	DL_SE_SU_MU_MIMO_ADAPT_SWITCH option of the NRDUCellDLIMimo.DLMuMimoSwitchSupplementSw parameter	<i>MIMO (TDD)</i>	Spectral-efficiency-based adaptive switching between SU-MIMO and MU-MIMO is not supported in hyper cells in the FSD transmission mode.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PUSCH MU-MIMO	UL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	Uplink MU-MIMO is not supported in hyper cells in the FSD mode.
Low-frequency TDD	PDSCH MU-MIMO	DL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	PDSCH spatial multiplexing is not supported in hyper cells in the FSD transmission mode.
Low-frequency TDD	Adaptive switching between hyper cell mode and combined cell mode	HYPERCELL_AND_COMBINECELL_ADPT option of the NRDUCellMultiTrp.MultiTrpAlgorithmSwitch parameter	None	Adaptive switching between hyper cell mode and combined cell mode is not supported in hyper cells in the FSD transmission mode.
Low-frequency TDD	Downlink correlation acceleration	DL_CORR_ACCELERATION_SW option of the NRDUCellDLIMimo.DLMuMimoSchSupplementSw parameter	None	Downlink correlation acceleration is not supported in hyper cells in the FSD mode.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink anti-interference scheduling	DL_ANTI_I NTRF_SCH_ SW option of the NRDUCellPdsch.DlPdsc hAlgoSwitch parameter	None	Downlink anti-interference scheduling is not supported in hyper cells in the FSD mode.
Low-frequency TDD	High Speed Intrf Random Sch PRB Thld	NRDUCellHighSpeedMob.HSIntrfRandomSchPrbThld	<i>High Speed Mobility</i>	It is not recommended that NRDUCellHighSpeed-Mob.HSIntrfRandomSchPrbThld be set to 100 for hyper cells in FSD mode.
Low-frequency TDD	RB Estimate Intrf Random PRB Thld	NRDUCellHighSpeedMob.UIRbEstIntrfRandomPrbThld NRDUCellHighSpeedMob.DIRbEstIntrfRandomPrbThld	<i>High Speed Mobility</i>	It is not recommended that NRDUCellHighSpeed-Mob.UIRbEstIntrfRandomPrbThld and NRDUCellHighSpeed-Mob.DIRbEstIntrfRandomPrbThld be set to 100 for hyper cells in FSD mode.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Hyper Cell*.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0076

FeR-IR20250102000700-0103

RAT

NR FDD, NR TDD

5.3.3.21 FOFD-031201 Intra-FR Inter-Band CA

5.3.3.21.1 Added the CSI-RS Beam Sweeping Optimization Function

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Without this function, CSI-RS beam sweeping is performed in both periodic and aperiodic modes. After the sweeping succeeds, the UE reports the optimal beam in an SCell. Resources must be reserved for such information reporting in both periodic and aperiodic sweeping scenarios (or only the supported sweeping scenario if the UE does not support both). The resources available for UEs to report information are limited. If resources are insufficient, other features that require UE information reporting for CA carriers may fail to take effect. Considering that periodic CSI-RS beam sweeping has a high success rate, CSI-RS beam sweeping optimization is introduced to reduce the resource consumption by UE information reporting. CSI-RS beam sweeping optimization is controlled by the **CSI_BEAM_SWEEPING_OPT_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoSwitch** parameter.

- If this option is selected:
 - If a UE supports periodic sweeping, such sweeping is performed and resources are reserved only for UE information reporting for periodic sweeping.
 - If a UE does not support periodic sweeping, aperiodic sweeping is performed and resources are reserved only for UE information reporting for aperiodic sweeping.
- If this option is deselected, CSI-RS beam sweeping optimization does not take effect.

The change in this feature is supported only in low frequency bands.

Implementation

For details, see *Carrier Aggregation*. The specific sections are listed in the change history of the document.

Impact on the Network

CSI-RS beam sweeping optimization optimizes the allocation of CA UE reporting resources among multiple carriers, improving CA UE performance.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management

The change in this feature has no impact. It is controlled by the existing feature license.

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellCarrMgmt.CaEnhancedAlgoSwitch	MOD NRDUCELLCAR RMGMT LST NRDUCELLCAR RMGMT	gNodeB	The CSI_BEAM_SWEEPING_OPTION_SW option has been added to specify whether to enable CSI-RS beam sweeping optimization.

- Performance management

The change in this feature has no impact.

- Fault management

The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

The change in this feature has no impact.

- Mutually exclusive features/functions

The change in this feature has no impact.

- Impacted features/functions

The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Carrier Aggregation*.

Requirement/Trouble Ticket Number

IR20241209002193

DTS2025021835017

RAT

NR FDD, NR TDD

5.3.3.21.2 Added a Method for Observing the Activation and Running State of Intra-FR Inter-Band CA

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The change in this feature involves two parts:

- The **N.User.CA.PCell.DL.Max** counter has been added for monitoring the activation and running state of downlink intra-FR inter-band CA. This counter measures the maximum number of downlink CA UEs that treat the local cell as their PCells.
- The **N.User.CA.PCell.UL.Max** counter has been added for monitoring the activation and running state of uplink intra-FR inter-band CA. This counter measures the maximum number of uplink CA UEs that treat the local cell as their PCells.

The change in this feature is supported only in low-frequency bands.

Implementation

For details, see *Carrier Aggregation*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The change in this feature has no impact.
- Performance management
The following counters have changed.

Change Type	Counter Name	Function Subset	NE	Description
New	N.User.CA.P Cell.DL.Max	Cell User Quantity Measurement	gNode B	This counter has been added to measure the maximum number of downlink CA UEs that treat the local cell as their PCells.
New	N.User.CA.P Cell.UL.Max	Cell User Quantity Measurement	gNode B	This counter has been added to measure the maximum number of uplink CA UEs that treat the local cell as their PCells.

- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0023

RAT

NR FDD, NR TDD

5.3.3.21.3 Added the SRS and PUSCH/PUCCH Concurrency Compatibility Function

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

As defined in 3GPP specifications, SRS and PUSCH/PUCCH concurrency is supported in intra-FR inter-band CA. In a CA scenario, SRS and PUSCH/PUCCH concurrency means that a PCC and an SCC work in the same symbols where one CC sends SRSs and the other CC sends PUSCH/PUCCH signals.

For intra-FR inter-band CA, SRS and PUSCH/PUCCH concurrency requires that UEs support this capability. Huawei gNodeBs comply with the protocol specifications to support SRS and PUSCH/PUCCH concurrency, while a UE supports this capability only if it reports the **parallelTxSRS-PUCCH-PUSCH** field.

Before the SRS and PUSCH/PUCCH concurrency compatibility function is introduced, the gNodeB by default considers that all UEs support the concurrency. If a UE does not support this capability, the gNodeB demodulates SRSs or PUSCH/PUCCH signals at incorrect time-domain positions, which compromises uplink performance of CA UEs. To address this issue, the SRS and PUSCH/PUCCH concurrency compatibility function is introduced. This function is controlled by the **SRS_PUSCH_CONCURRENCY_COMPT_SW** option of the **NRDUCellCarrMgmt.CaCompatibilitySwitch** parameter. After the SRS and PUSCH/PUCCH concurrency compatibility function is enabled, the gNodeB checks whether a UE supports this concurrency so that the processing can vary with the UE capability:

- If the UE supports SRS and PUSCH/PUCCH concurrency, the UE can concurrently send SRSs and PUSCH/PUCCH signals.
- If the UE does not support SRS and PUSCH/PUCCH concurrency, the UE does not send SRSs and PUSCH/PUCCH signals concurrently on the PCC and SCCs. Instead, in the corresponding symbols, the UE sends SRSs only but not PUSCH/PUCCH signals.

The change in this feature is supported only in low frequency bands.

Implementation

For details, see *Carrier Aggregation*. The specific sections are listed in the change history of the document.

Impact on the Network

After SRS and PUSCH/PUCCH concurrency compatibility takes effect, the average throughput of SCell-activated uplink CA UEs ($(N.CA.SccActiveUser.ThpVol.UL - N.CA.ThpVol.UL.SmallPkt)/N.CA.ThpTime.UL.RmvSmallPkt$) increases.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellCa rrMgmt.CaC ompatibility Switch	MOD NRDUCELLC ARRMGMT LST NRDUCELLC ARRMGMT	gNodeB	The SRS_PUSCH_CONCURREN CY_COMPT_SW option has been added to specify whether to enable the SRS and PUSCH/PUCCH concurrency compatibility function.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS interference coordination based on self-contained slots	SRS_INTRF_CO ORD_S_SLOT_S W option of the NRDUCellSrs.S rsAlgoExtSwitc h parameter	<i>Channel Management</i>	The SRS_INTRF_COORD_S_SLOT_SW option must be selected if the SRS and PUSCH/PUCCH concurrency compatibility function is enabled and the SRS remote interference avoidance switch is turned on.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS remote interference avoidance	SRS_RIM_INTRF_AVOID_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter	None	
Low-frequency TDD	PUCCH symbol length number increase	PUCCH_SYM_TYPE_NUM_INCR_SW option of the NRDUCellPucch.PucchAlgoSwitch parameter	<i>Channel Management</i>	When the SRS and PUSCH/PUCCH concurrency compatibility function is enabled, the PUCCH_SYM_TYPE_NUM_INCR_SW option must be selected.

- Mutually exclusive features/functions

The following mutually exclusive relationship has been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Light-load time sequence optimization	NRDUCellAlgoSwitch.LIGHT_LOAD_TIME_SEQ_OPT_SW option of the LowLatencySwitch parameter	None	When the SRS and PUSCH/PUCCH concurrency compatibility function is enabled, the light-load time sequence optimization function must be disabled.
Low-frequency TDD	Positioning-dedicated SRS transmission in uplink slots	SRS_FOR_POSITION_ON_USLOT_SW option of the NRDUCellPosAlgo.LcsMeasureAlgoSwitch parameter	<i>Positioning</i>	When the SRS and PUSCH/PUCCH concurrency compatibility function is enabled, positioning-dedicated SRS transmission in uplink slots cannot be enabled together with proactive avoidance of remote interference.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Proactive avoidance of remote interference	SRS_RIM_INTERF_AVOID_SW and SRS_INTERF_COORD_S_SLOT_SW options of the NRDUCellSrsResourcesAlgoExtSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	

- Impacted features/functions
The following impact relationships have been modified.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Proactive avoidance of remote interference	SRS_RIM_INTRF_AVOID_SW and SRS_INTRF_COORD_S_SLOT_SW options of the NRDUCellSrs.SrsAlgoExtSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	- When the uplink CA switch is on, the SRS_INTRF_AVOID_PUSCH_SCH_SW setting of the NRDUCellSrs.SrsAlgoExtSwitch parameter must be the same for the PCell and the SCell. Otherwise, the cells differ in their scheduling in self-contained slots in the uplink, affecting the reporting of the power headroom in the PCell and SCell. As a result, adjustments for TPC are performed frequently and the average uplink throughput of uplink CA UEs changes greatly. - It is good practice to keep the "proactive avoidance of remote interference" switch status consistent between the PCell and the SCell. Otherwise, within the same symbols, SRSs may be transmitted on one carrier but

RAT	Function Name	Function Switch	Reference	Description
				PUSCH data on the other. This increases bit errors of some UEs. In the event of inconsistent switch status between the two cells, it is good practice to enable the "SRS and PUSCH/PUCCH concurrency compatibility" function to avoid bit errors. For details about the "SRS and PUSCH/PUCCH concurrency compatibility" function, see <i>Carrier Aggregation</i> .

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Carrier Aggregation*.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0110

RAT

NR FDD, NR TDD

5.3.3.22 FOFD-031203 VoNR

5.3.3.22.1 Added CA Activation for VoNR UEs with a Specific 5QI

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

CA activation for VoNR UEs with a specific 5QI is introduced to provide differentiated user experience for UEs having multiple bearers and performing VoNR services.

This function allows differentiated configurations for VoNR UEs by activating CA for those with a specific 5QI while deactivating prohibition of measurement-based SCell configuration and CA prohibition for VoNR UEs.

This function is enabled when the **VONR_ALLOW_CA_SW** option of the **gNBop5qiAlgo.ServiceStrategySw** parameter is selected.

The change in this feature is supported only in low frequency bands.

The change in this feature is supported only in SA networking.

Implementation

For details, see *VoNR*. The specific sections are listed in the change history of the document.

Impact on the Network

The average uplink and downlink throughputs (**User Uplink Average Throughput (DU)** and **User Downlink Average Throughput (DU)**) of data services will increase and the voice quality will deteriorate when both of the following conditions are met: (1) Prohibition of measurement-based SCell configuration or CA prohibition for VoNR UEs has taken effect. (2) CA activation for VoNR UEs with a specific 5QI is enabled, and these UEs are performing both voice and data services.

Impact on NEs

The change in this feature is implemented on the gNodeB.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature is not under license control.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	gNBOP5qiAlgo.ServiceStrategySw	ADD GNBOP5QIALGO LST GNBOP5QIALGO MOD GNBOP5QIALGO	gNodeB	The VONR_ALLOW_CA_SW option has been added to specify whether to allow CA to take effect for VoNR UEs with a specific 5QI.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The following impact relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Prohibition of measurement-based SCell configuration	CA_INTERF_MEAS_VONR_COMPAT_SW option of the NRCellCaMgmtConfig.CaStrategySwitch parameter	<i>VoNR</i>	Prohibition of measurement-based SCell configuration does not take effect when CA takes effect for VoNR UEs with a specific 5QI.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CA prohibition for VoNR UEs	VONR_NOT_CONFIG_CA_SW option of the NRCellCaMgmtConfig.CaStrategySwitch parameter	<i>VoNR</i>	CA prohibition for VoNR UEs does not take effect when CA takes effect for VoNR UEs with a specific 5QI.

Related Operations

By default, the change does not take effect after an upgrade.
For details, see the corresponding "Operation and Maintenance" section in *VoNR*.

Requirement/Trouble Ticket Number

FeR-IR20241213000657-0001

RAT

NR FDD, NR TDD

5.3.3.22.2 Added CA Prohibition for UEs Initiating VoNR Calls

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

When a UE initiates a call in idle or inactive mode, SCell addition for CA may be triggered before a 5QI 1 bearer is set up. This can lead to reconfiguration or gap-assisted measurement, increasing the voice call delay.

To address this, CA prohibition for UEs initiating VoNR calls is now supported to prevent CA from taking effect immediately for UEs that initiate VoNR calls by sending the RRCSetupRequest message with the *EstablishmentCause* IE set to **mo-VoiceCall** or the RRCResumeRequest or RRCResumeRequest1 message with the *ResumeCause* IE set to **mo-VoiceCall**.

This function is enabled when the **VONR_CALL_INIT_NOT_CFG_CA_SW** option of the **NRCellCaMgmtConfig.CaStrategySwitch** parameter is selected.

The change in this feature is supported only in low frequency bands.

The change in this feature is supported only in SA networking.

Implementation

For details, see *VoNR*. The specific sections are listed in the change history of the document.

Impact on the Network

Enabling CA prohibition for UEs initiating VoNR calls may reduce the voice service delay for UEs that initiate VoNR calls. However, if UEs are initiating VoNR calls and performing data services simultaneously, their user-perceived rates of data services may decrease, and their average uplink and downlink throughputs (**User Uplink Average Throughput (DU)** and **User Downlink Average Throughput (DU)**) will reduce.

When CA prohibition for UEs initiating VoNR calls is in effect and CA activation for VoNR UEs with a specific 5QI is enabled, the average uplink and downlink throughputs (**User Uplink Average Throughput (DU)** and **User Downlink Average Throughput (DU)**) of data services increase for UEs that are initiating VoNR calls and performing data services simultaneously.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRCeIlCaMgmtConfig.CaStrategySwitch	MOD NRCeIlCAM GMTCONFIG LST NRCeIlCAM GMTCONFIG	gNodeB	The VONR_CALL_INIT_NOT_CFG_CA_SW option has been added to specify whether to enable CA prohibition for UEs initiating VoNR calls.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The following impact relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CA activation for VoNR UEs with a specific 5QI	VONR_ALL OW_CA_SW option of the gNB Op5qiA lgo. <i>ServiceStrategySw</i> parameter	<i>VoNR</i>	CA prohibition for UEs initiating VoNR calls does not take effect when CA takes effect for VoNR UEs with a specific 5QI.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *VoNR*.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0066

RAT

NR FDD, NR TDD

5.3.3.22.3 Added Support for Downlink Scheduling Optimization During Handovers for ViNR UEs

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

During handovers, UEs cannot be scheduled, causing jitter when ViNR UEs receive video packets after being handed over. This may result in asynchronous audio and video packets, degrading video call quality.

This function delays handovers for ViNR UEs and sends downlink video packets during the delay period to reduce jitter in ViNR services during handovers. When enabled together with both downlink buffer jitter elimination and downlink handover delay optimization for VoNR UEs, this function can also address the issue of asynchronous audio and video packets.

This function is enabled when the **VINR_DL_SCH_OPT_DUR_HO_SW** option of the **NRDUCellDLSch.VinrDLSchSwitch** parameter is selected.

The change in this feature is supported only in low-frequency SA networking.

Implementation

For details, see *ViNR*. The specific sections are listed in the change history of the document.

Impact on the Network

This function can reduce the jitter caused by handover interruption and the time difference in audio and video synchronization for ViNR UEs. However, service drops may occur due to delayed delivery of handover commands.

Impact on NEs

The change in this feature is implemented on the gNodeB.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellDL Sch. <i>VinrDLSchSwitch</i>	MOD NRDUCELLDLSCH LST NRDUCELLDLSCH	gNodeB	The VINR_DL_SCH_OPT_DURATION_SW option has been added to specify whether to enable downlink scheduling optimization during handovers for ViNR UEs.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	ViNR DL Jitter Eliminate Duration	NRCeLLServExp. <i>VinrDLJitterEliminateDur</i>	<i>ViNR</i>	When downlink scheduling optimization during handovers for ViNR UEs is enabled, the NRCeLLServExp. <i>VinrDLJitterEliminateDur</i> parameter must be set to a non-zero value.
Low-frequency TDD	Downlink handover delay optimization for VoNR UEs	VONR_DL_SCH_OPT_DURATION_SW option of the NRDUCellDL Sch. <i>VoiceDLSchSwitch</i> parameter	<i>VoNR</i>	Downlink scheduling optimization during handovers for ViNR UEs depends on downlink handover delay optimization for VoNR UEs.

- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *VoNR*.

Requirement/Trouble Ticket Number

FeR-IR20250103001850-0001

RAT

NR FDD, NR TDD

5.3.3.22.4 Added Aggregation Level Adjustment for Retransmissions upon DTXs

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Aggregation level adjustment for retransmissions upon DTXs is now supported and is controlled by the **DTX_RETRANS_AGG_LVL_ADJ_SW** option of the **gNBDUQciAlgoParamGrp.QciAlgoSwitch** parameter. When the gNodeB receives the HARQ feedback for retransmission indicating DTX from a voice service UE, it increases the PDCCH CCE aggregation level by an additional offset specified by the **gNBDUQciAlgoParamGrp.RetransCceAggLevelOfs** parameter.

The change in this feature is supported only in low frequency bands.

The change in this feature is supported only in SA networking.

Implementation

For details, see *VoNR*. The specific sections are listed in the change history of the document.

Impact on the Network

In addition, enabling aggregation level adjustment for retransmissions upon DTXs may decrease the CCE allocation success rate for data services and increase the uplink voice packet loss rate.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management

The change in this feature has no impact. It is controlled by the existing feature license.

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNB DUQciAl goParamGrp <i>.QciAlgoSwitch</i>	ADD GNBDUQCI LGOPARAMG RP MOD GNBDUQCI LGOPARAMG RP LST GNBDUQCI LGOPARAMG RP	gNodeB	The DTX_RETRANS_AGG_LVL_ADJ_SW option has been added to specify whether to enable aggregation level adjustment for retransmissions upon DTXs.

- Performance management

The change in this feature has no impact.

- Fault management

The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

The change in this feature has no impact.

- Mutually exclusive features/functions

The change in this feature has no impact.

- Impacted features/functions

The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *VoNR*.

Requirement/Trouble Ticket Number

FeR-IR20250102000700-0058

RAT

NR FDD, NR TDD

5.3.3.22.5 Added Downlink CCE Allocation Without Ratio Constraint

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Downlink CCE allocation without ratio constraint is now supported to remove the constraint of PDCCH uplink-to-downlink CCE ratio adaptation (controlled by the **UL_DL_CCE_RATIO_ADAPT_SW** option of the **NRDUCellPdcch.PdcchAlgoSwitch** parameter) on voice service UEs with an aggregation level of 16 that fail to be allocated downlink CCEs when all of the following conditions are met in a cell:

- There are voice service UEs with an aggregation level of 16 that fail to be allocated downlink CCEs.
- There are no voice service UEs with an aggregation level of 8 or 16 that fail to be allocated uplink CCEs.
- The total number of CCEs is at least 16.

This function maximizes the number of CCEs that can be allocated to voice service UEs during downlink scheduling.

It is controlled by the **DL_CCE_ALLO_NO_CONSTRAIN_SW** option of the **gNB DUQciAlgoParamGrp.QciAlgoSwitch** parameter.

The change in this feature is supported only in low frequency bands.

The change in this feature is supported only in SA networking.

Implementation

For details, see *VoNR*. The specific sections are listed in the change history of the document.

Impact on the Network

Enabling downlink CCE allocation without ratio constraint may decrease the downlink voice packet loss rate. However, it may increase the uplink and downlink CCE allocation failure rates of non-voice-service UEs and decrease the uplink and downlink user-perceived rates.

To achieve optimal gains, you are advised to enable this function when the value of **N.CCE.DL.AggLvl16Num** is greater than 0.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNBDUQciAlgoParamGrp.QciAlgoSwitch	ADD GNBDUQCIA LGOPARAMG RP MOD GNBDUQCIA LGOPARAMG RP LST GNBDUQCIA LGOPARAMG RP	gNodeB	The DL_CCE_ALLO_NO_CONSTRAIN_SW option has been added to specify whether to enable downlink CCE allocation without ratio constraint.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The following impact relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDCCH uplink-to-downlink CCE ratio adaptation	UL_DL_CCE_RATIO_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter	<i>Channel Management</i>	When downlink CCE allocation without ratio constraint takes effect, PDCCH uplink-to-downlink CCE ratio adaptation does not take effect for voice service UEs with an aggregation level of 16 that fail to be allocated downlink CCEs.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *VoNR*.

Requirement/Trouble Ticket Number

FeR-IR20250102000700-0058

RAT

NR FDD, NR TDD

5.3.3.22.6 Added Coverage-based Adaptation Between VoNR and EPS Fallback Prohibition for UEs in Idle or Inactive Mode

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

To reduce the EPS fallback failure rate in areas with weak LTE coverage, coverage-based adaptation between VoNR and EPS fallback prohibition for UEs in idle or inactive mode can be enabled. This prevents UEs of an operator from falling back to the LTE network when initiating voice services in idle or inactive mode. This function is controlled by the **VONR_EPS_FB_ADAPT_FORBID_SW** option of the **NRCellOpPolicy.VoiceStrategySwitch** parameter.

The change in this feature is supported only in low frequency bands.

The change in this feature is supported only in SA networking.

Implementation

For details, see *VoNR*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRCellOpPolicy.VoiceStrategySwitch	ADD NRCELLOPPOLICY MOD NRCELLOPPOLICY LST NRCELLOPPOLICY	gNodeB	The VONR_EPS_FB_ADAPT_FORBID_SW option has been added to specify whether to forbid coverage-based adaptation between VoNR and EPS fallback for UEs in idle or inactive mode.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *VoNR*.

Requirement/Trouble Ticket Number

FeR-IR20250103001850-0001

RAT

NR FDD, NR TDD

5.3.3.22.7 Added Coverage-based Adaptation Between VoNR and EPS Fallback Prohibition for UEs in Connected Mode

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

To reduce the EPS fallback failure rate in areas with weak LTE coverage, coverage-based adaptation between VoNR and EPS fallback prohibition for UEs in connected mode can be enabled. This prevents UEs of an operator from falling back to the LTE network when initiating voice services in connected mode. This function is controlled by the **CONN_EPS_FB_ADAPT_FORBID_SW** option of the **NRCellOpPolicy.VoiceStrategySwitch** parameter.

The change in this feature is supported only in low frequency bands.

The change in this feature is supported only in SA networking.

Implementation

For details, see *VoNR*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRCelOpPo licy.VoiceStr ategySwitch	ADD NRCELLOPP OLICY MOD NRCELLOPP OLICY LST NRCELLOPP OLICY	gNodeB	The CONN_EPS_FB_ADA PT_FORBID_SW option has been added to specify whether to forbid coverage-based adaptation between VoNR and EPS fallback for UEs in connected mode.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *VoNR*.

Requirement/Trouble Ticket Number

FeR-IR20250103001850-0001

RAT

NR FDD, NR TDD

5.3.3.22.8 Added a Parameter Specifying the ROHC Exit Threshold

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

A parameter specifying the ROHC exit threshold has been added. If ROHC (controlled by the **ROHC_SW** option of the **NRCelAlgoSwitch.RohcSwitch** parameter) is enabled in optimistic or reliable mode (with the **NRCelAlgoSwitch.RohcMode** parameter set to **O_MODE** or **R_MODE**) and the decompressor receives a number of consecutive CRC error packets greater than or equal to the proactive ROHC exit threshold, ROHC is disabled. The ROHC exit threshold is now specified by the **NRCelServExp.RohcExitThld** parameter, instead of being fixed at 6.

The change in this feature is supported only in low frequency bands.

The change in this feature is supported only in SA networking.

Implementation

For details, see *VoNR*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management

The change in this feature has no impact. It is controlled by the existing feature license.

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRCeLLSeRvE xp.RohcExitThld	MOD NRCELLSERV EXP LST NRCELLSERV EXP	gNodeB	This parameter has been added to specify the ROHC exit threshold.

- Performance management

The change in this feature has no impact.

- Fault management

The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

The change in this feature has no impact.

- Mutually exclusive features/functions

The change in this feature has no impact.

- Impacted features/functions

The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *VoNR*.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0135

RAT

NR FDD, NR TDD

5.3.3.22.9 Added ROHC IR State Optimization

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

ROHC IR state optimization is now supported.

During handovers, voice service UEs do not transfer context information to the target cell, necessitating them to resend initialization and refresh (IR) packets after migrating to the target cell. However, some UEs may experience ROHC decompression failures due to compatibility issues. To address this, ROHC IR state optimization is introduced.

With this function, when a voice service UE is handed over between cells that use profile 0x0001 (with the **PROFILE0X0001** option of the **NRCCellAlgoSwitch.RohcProfiles** parameter selected) in optimistic or reliable mode (with the **NRCCellAlgoSwitch.RohcMode** parameter set to **O_MODE** or **R_MODE**), the base station proactively sends a STATIC-NACK packet to the UE before the handover to expedite ROHC transition into the IR state. This function is enabled when the **ROHC_IR_STATE_OPT_SW** option of the **NRDUCellVoiceAlgo.VoiceUserAlgoSw** parameter is selected.

In addition, ROHC IR state optimization can be disabled for specified UEs by selecting the **ROHC_IR_STATE_OPT_SW_OFF** option of the **gNBUECompatSw.BlacklistCtrlExtSwitch** parameter.

For details about IR packets, IR state, and STATIC-NACK, see sections 5.10 "ROHC UNCOMPRESSED -- no compression (Profile 0x0000)", 4.3 "Compression and decompression states", and 5.2 "ROHC packets and packet types" in RFC 3095, respectively.

The change in this feature is supported only in low frequency bands.

The change in this feature is supported only in SA networking.

Implementation

For details, see *VoNR*. The specific sections are listed in the change history of the document.

Impact on the Network

Enabling ROHC IR state optimization will decrease the probability of consecutive ROHC decompression failures due to ROHC IR packet loss during handovers of voice service UEs and may reduce the value of **N.RLC.DL.Cell5QI.Delay**. However, this will reduce the ROHC compression ratio (**N.PDCP.UL.ROHC.VoNR.BeforeDecomp.Bytes/N.PDCP.UL.ROHC.VoNR.AfterDecomp.Bytes**).

Disabling ROHC IR state optimization for specified UEs (by selecting the **ROHC_IR_STATE_OPT_SW_OFF** option of the **gNBUECompatSw.BlacklistCtrlExtSwitch** parameter) may cause consecutive

decompression failures during handovers due to the absence of IR packets, resulting in consecutive packet loss.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellVoiceAlgo.VoiceUserAlgorithm	MOD NRDUCELLV OICEALGO LST NRDUCELLV OICEALGO	gNodeB	The ROHC_IR_STATE_OPT_SW option has been added to specify whether to enable ROHC IR state optimization.
Modified	gNBUEComPatSw.BlacklistCtrlExtSwitch	ADD GNBUECOM PATSW MOD GNBUECOM PATSW LST GNBUECOM PATSW	gNodeB	The ROHC_IR_STATE_OPT_SW_OFF option has been added to specify whether to disable ROHC IR state optimization.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	ROHC	ROHC_SW option of the NRCellAlgoSwitch.RohcSwitch parameter	<i>VoNR</i>	ROHC is required for ROHC IR state optimization to take effect.

- Mutually exclusive features/functions

The change in this feature has no impact.

- Impacted features/functions

The following impact relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink handover delay optimization	VONR_DL_SCH_OPT_DUR_HO_SW option of the NRDUCellDLISch.VoiceDLISchSwitch parameter	<i>VoNR</i>	When ROHC IR state optimization and downlink handover delay optimization are both enabled, downlink handover delay optimization may provide fewer benefits.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *VoNR*.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0135

RAT

NR FDD, NR TDD

5.3.3.23 FOFD-031205 RF Channel Intelligent Shutdown

5.3.3.23.1 Removed the Mutually Exclusive Relationship Between RF Channel Intelligent Shutdown and Cell Combination in Low-speed 8T8R Scenarios.

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

For macro base stations, RF channel intelligent shutdown can be enabled only when the **NRDUCellTrp.TxRxMode** parameter is set to **8T8R** for all TRPs serving a cell and all of the following conditions are met:

- **NRDUCell.NrDuCellNetworkingMode** is set to **HYPER_CELL_COMBINE_MODE**.
- **NRDUCellMultiTrp.DataTransMode** is set to **BASIC_MODE** or **DPS_MODE**.
- **NRDUCell.HighSpeedFlag** is set to **LOW_SPEED**.

The change in this feature is supported only in low frequency bands.

LampSite base stations do not support the change in this feature.

Implementation

For details, see *Energy Conservation and Emission Reduction*. The specific sections are listed in the change history of the document.

Impact on the Network

RF channel intelligent shutdown can be enabled in the scenarios described in the "Feature/Function Description" section. For details about the benefits and impacts of RF channel intelligent shutdown on the network, see *Energy Conservation and Emission Reduction*.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.

- Configuration management
The change in this feature has no impact.
- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The descriptions of the following mutually exclusive functions have been revised.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell Combination	- NRDUCell.NrDualNetworkMode set to HYPER_CELL_COMBINE_MODE - NRDUCellMultiTrp.DataTransmissionMode	Cell Combination	- When a combined cell served by a macro base station uses the basic or DPS transmission mode (with the NRDUCellMultiTrp.DataTransmissionMode parameter set to BASIC_MODE or DPS_MODE), RF channel intelligent shutdown can be enabled in high-speed scenarios only if the HIGH_SPEED_CELL_MUTING_SW option of the NRDUCellAlgorithmSwitch.PowerSavingSwitch parameter is selected. - When a combined cell served by a macro base station uses the basic or DPS transmission mode (with the NRDUCellMultiTrp.DataTransmissionMode parameter set to

RAT	Function Name	Function Switch	Reference	Description
				BASIC_MODE or DPS_MODE), RF channel intelligent shutdown can be enabled in non-high-speed scenarios only if NRDUCellTrp.TxRxMode is set to 8T8R for all TRPs serving the cell. • When a combined cell served by a macro base station uses the joint transmission mode (with the NRDUCellMultiTrp.DataTransMode parameter set to JOINT_MODE), RF channel intelligent shutdown can be enabled in high-speed scenarios only if the HIGH_SPEED_CELL_MUTING_SW option of the NRDUCellAlgorithmSwitch.PowerSavingSwitch parameter is selected.

RAT	Function Name	Function Switch	Reference	Description
				- When a combined cell served by a macro base station uses the joint transmission mode (with the NRDUCellMultiTrp.DataTr ansMode parameter set to JOINT_MODE), RF channel intelligent shutdown can be enabled in non-high-speed scenarios. After half of channels are shut down, the remaining TX/RX configuration must be compatible with the Cell Combination feature. - For a combined cell served by a LampSite base station, RF channel intelligent shutdown can be enabled.

- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Energy Conservation and Emission Reduction*.

Requirement/Trouble Ticket Number

FeR-IR20251027000417-0002

RAT

NR FDD, NR TDD

5.3.3.23.2 Added Support for RF Channel Intelligent Shutdown in Pseudo-Multimode Scenarios

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

RF channel intelligent shutdown can now take effect in pseudo-multimode scenarios by selecting the **PSE_MULTI_MODE_RF_SHUTDOWN_SW** option of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter. (In pseudo-multimode scenarios, an RRU is configured using **RRU.RS** to work in LTE and NR multimode, but only NR cells are activated and LTE cells are not configured or activated.)

The change in this feature is supported only in low frequency bands.

Implementation

For details, see *Energy Conservation and Emission Reduction*. The specific sections are listed in the change history of the document.

Impact on the Network

RF channel intelligent shutdown can take effect in pseudo-multimode scenarios. For details about the benefits and impacts of RF channel intelligent shutdown on the network, see *Energy Conservation and Emission Reduction*.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellPwrSave Algo.PowerSaving AlgoSwitch	MOD NRDUCELLPW RSVEALGO LST NRDUCELLPW RSVEALGO	gNodeB	The PSE_MULTI_MODE_RF_SHUTDOWN_SW option has been added to specify whether to enable RF channel intelligent shutdown in pseudo-multimode scenarios.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LNR coordinated carrier-channel shutdown	LNR_CARRIER_CHN_CO_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	For NR TDD, if the PSE_MULTI_MODE_RF_SHUTDOWN_SW option of the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter is selected, the LNR_CARRIER_CHN_CO_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter must be selected.

- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Energy Conservation and Emission Reduction*.

Requirement/Trouble Ticket Number

FeR-IR20250102000700-0068

RAT

NR FDD, NR TDD

5.3.3.24 FOFD-050201 Network Slicing Introduction

5.3.3.24.1 User-Rate-based Dynamic Network Slice Resource Management

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The "minimum rate guarantee for non-GBR services" and "minimum rate guarantee for GBR services" functions can be enabled respectively to enhance the service experience of UEs with non-GBR and GBR bearers, as detailed in *QoS Management*. However, the service experience of these UEs cannot be guaranteed if the minimum rate for non-GBR or GBR services is not met due to inappropriate settings of the **NRDUCellRes.UlRbAverageRatio** or **NRDUCellRes.DlRbAverageRatio** parameter. To address this, the "user-rate-based dynamic network slice resource management" function is introduced, which is controlled by the **DYNAMIC_NETWORKSLICE_SW** option of the **NRDUCellCommonSw.ServiceDiffAlgoExtSwitch** parameter. This function dynamically adjusts the average percentages of uplink and downlink RBs in NR DU cell resources. This function requires the **NRDUCellRes.ResStrategyType** parameter to be set to **MEAN_ONLY**. This function is supported only in low frequency bands.

When the aforesaid option is selected and the **NRDUCellRes.UlRbAverageRatio** or **NRDUCellRes.DlRbAverageRatio** parameter is configured for a resource group, the gNodeB dynamically adjusts RB allocation based on the minimum rate of non-GBR or GBR services of UEs in a network slice group. In this situation, RBs are not preferentially allocated based on the configured **NRDUCellRes.UlRbAverageRatio** or **NRDUCellRes.DlRbAverageRatio** parameter.

The change in this feature is supported only in SA networking.

LampSite base stations do not support the change in this feature.

The change in this feature is supported only in low frequency bands.

Implementation

For details, see *Network Slicing*. The specific sections are listed in the change history of the document.

Impact on the Network

If the RBs required by non-GBR or GBR services of UEs in a network slice group outnumber the RBs indicated by the configured average percentages of uplink or downlink RBs, enabling user-rate-based dynamic network slice resource management yields the following network impacts:

- If the target rate satisfaction level of the network slice group is low, user experience in the group improves.
- If the cell load is medium or heavy (PRB usage exceeding 30%) and there are a large number of cell edge UEs in the group, the uplink and downlink cell throughputs (**Cell Downlink Average Throughput (DU)** and **Cell Uplink Average Throughput (DU)**) decrease.

The rate satisfaction level is calculated using the following formulas:

- The formula below assumes that the counters specific to an uplink minimum GBR of 3 Mbit/s are used:

$$\text{Uplink rate satisfaction level} = (\text{N.Thp.UL.Flex.Index3} + \dots + \text{N.Thp.UL.Flex.Index14}) / (\text{N.Thp.UL.Flex.Index0} + \dots + \text{N.Thp.UL.Flex.Index14})$$

- The formula below assumes that the counters specific to a downlink minimum GBR of 5 Mbit/s are used:

$$\text{Downlink rate satisfaction level} = (\text{N.Thp.DL.Flex.Index2} + \dots + \text{N.Thp.DL.Flex.Index14}) / (\text{N.Thp.DL.Flex.Index0} + \dots + \text{N.Thp.DL.Flex.Index14})$$

 NOTE

Flexible performance counters (such as **N.Thp.UL.Flex.Index0**) can be configured using the **gNBFlexCounterObjCfg** MO. For details, see "Custom Object Type" in the "Object Type" section in *Performance Management*.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

Base station models: 3900 and 5900 series base stations (macro base stations).
3900 series base stations must be configured with the BBU3910.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameter has changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellCommonSw.ServiceDiffAlgorithmExtSwitch	MOD NRDUCELLC OMMONSW LST NRDUCELLC OMMONSW	gNodeB	The DYNAMIC_NETWORKSLICE_SW option has been added to this parameter to specify whether to enable user-rate-based dynamic network slice resource management.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

The following dependency relationship has been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR DU cell resource management between network slices	RB_DYNAMIC_C_CONTROL_SW option of the NRDUCellAlgoSwitch.NetworkSliceAlgoSwitch parameter	<i>Network Slicing</i>	User-rate-based dynamic network slice resource management (controlled by the DYNAMIC_NETWORKSLICE_SW option of the NRDUCellCommonSw.ServiceDiffAlgoExtSwitch parameter) can be enabled only after "NR DU cell resource management between network slices" is enabled.

- Mutually exclusive features/functions

The following mutually exclusive relationships have been added.

RAT	Function Name	Function Switch	Reference
Low-frequency TDD	QCI-based resource control	RB_DYNAMIC_C_CONTROL_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter	<i>QCI-based Resource Control</i>
Low-frequency TDD	Bi-Carrier	BI_CARRIER_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter	None
Low-frequency TDD	Slice-based QoS scheduling	SLICE_BASED_QOS_SCH_SW option of the NRDUCellAlgoSwitch.NetworkSliceAlgoSwitch parameter	None

RAT	Function Name	Function Switch	Reference
Low-frequency TDD	Service admission	QOS_FLOW_ADMISSION_SWITCH.RacAlgoSwitch parameter	<i>Admission Control</i>
Low-frequency TDD	Operator-specific resource sharing enhancement	NRDUCellNsMgmtConfig.NetSliceRfspGrpSharePol set to SAME_OPERATOR_PREFERRED	<i>Network Slicing</i>

- Impacted features/functions

The following impact relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Resource sharing between network slice groups and RFSP groups	NRDUCellRes.ResourceUsageObject set to NS_RFSP_GROUP	<i>Network Slicing</i>	When both "user-rate-based dynamic network slice resource management" and "resource sharing between network slice groups and RFSP groups" take effect, the average percentage of RBs estimated by the gNodeB is inaccurate.
Low-frequency TDD	Single-UE multi-slice RB management policy	SINGLEUE_MULTI_SLICE_RB_CTRL_SWITCH option of the NRDUCellAlgoSwitch.NetworkSliceAlgoSwitch parameter	<i>Network Slicing</i>	When both "user-rate-based dynamic network slice resource management" and "single-UE multi-slice RB management policy" take effect, the average percentage of RBs estimated by the gNodeB for UEs in a network slice group may be inaccurate.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UL and DL Decoupling	NRDUCellAlgoSwitch.UlDlDecouplingSwitch set to ON	<i>UL and DL Decoupling</i>	When user-rate-based dynamic network slice resource management is enabled for an SUL cell enabled with UL and DL Decoupling, you are advised to also enable user-rate-based dynamic network slice resource management for the corresponding TDD cell. Otherwise, the gNodeB may perform inaccurate dynamic resource adjustments.
Low-frequency TDD	Independent SUL deployment	SUPER_UL_TDM_SCH_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter	<i>Super Uplink</i>	When user-rate-based dynamic network slice resource management is enabled for an SUL cell enabled with independent SUL deployment, you are advised to also enable user-rate-based dynamic network slice resource management for the corresponding TDD cell. Otherwise, the gNodeB may perform inaccurate dynamic resource adjustments.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Network Slicing*.

Requirement/Trouble Ticket Number

FeR-IR20251216001872-0010

RAT

NR FDD, NR TDD

5.3.3.24.2 Optimized Method for Observing Network Slicing

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The method for observing the network slicing function has been optimized. Specifically, the **N.RLC.DL.TrfSDU.RxPackets.Ns** and **N.RLC.DL.TrfSDU.RxPackets.DuCell5QI.Ns** counters have been introduced to measure the number of downlink SDUs received at the RLC layer for a specific network slice and the number of downlink SDUs for 5QI-specific services received at the RLC layer for a specific network slice, respectively.

The change in this feature is supported only in SA networking.

Implementation

For details, see *Network Slicing*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The change in this feature has no impact.

- Performance management

The following counters have changed.

Change Type	Counter Name	Function Subset	NE	Description
New	N.RLC.DL.Tr fSDU.RxPac kets.Ns	Intra-DU Cell Network Slice RLC Measurement	gNode B	This counter has been added to measure the number of downlink SDUs received at the RLC layer for a specific network slice.
New	N.RLC.DL.Tr fSDU.RxPac kets.DuCell 5QI.Ns	DU Cell Network Slice 5QI-specific Service Quality Measurement	gNode B	This counter has been added to measure the number of downlink SDUs for 5QI-specific services received at the RLC layer for a specific network slice in a DU cell.

- Fault management

The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

The change in this feature has no impact.

- Mutually exclusive features/functions

The change in this feature has no impact.

- Impacted features/functions

The change in this feature has no impact.

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250528001856-0033

RAT

NR FDD, NR TDD

5.3.3.25 FOFD-050208 Service-Differentiated Low Latency

5.3.3.25.1 Policies for Controlling the Number of UEs for Preallocation

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The policies for controlling the number of UEs for preallocation are introduced to allow the maximum numbers of UEs (1) for which QCI-based uplink preallocation can take effect and (2) for which cell-level uplink preallocation can take effect to be controlled by different parameters. This function is controlled by the **PRESCH_UE_NUM_CTRL_POL_SW** option of the **NRDUCellUlsch.UlPreallocPolEnhSwitch** parameter.

- Policy 1: If this option is selected, the maximum numbers of UEs (1) for which QCI-based uplink preallocation can take effect and (2) for which cell-level uplink preallocation can take effect are separately configured.
 - The maximum number of UEs for which QCI-based uplink preallocation can take effect is specified by the **NRDUCellUlsch.QciPreschUeNumLimit** parameter.
 - The maximum number of UEs for which cell-level uplink preallocation can take effect is specified by the **NRDUCellUlsch.CellPreschUeNumLimit** parameter.
- Policy 2: If this option is deselected, the maximum number of UEs for which uplink preallocation can take effect is specified by the **NRDUCellPusch.PreschUeNumUpperLimit** parameter, irrespective of UE types.

Implementation

For details, see *Service-differentiated Experience Guarantee*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellUlsch.UlPreallocPolEnhSwitch	LST NRDUCELLDLS CH MOD NRDUCELLDLS CH	gNodeB	The PRESCH_UE_NUM_CTRL_POLICY_SW option has been added to specify the policy for controlling the number of UEs for preallocation.
New	NRDUCellUlsch.QciPreschUeNumLimit	LST NRDUCELLULS CH MOD NRDUCELLULS CH	gNodeB	This parameter has been added to specify the maximum number of UEs for which QCI-based uplink preallocation can take effect.
New	NRDUCellUlsch.CellPreschUeNumLimit	LST NRDUCELLULS CH MOD NRDUCELLULS CH	gNodeB	This parameter has been added to specify the maximum number of UEs for which cell-level uplink preallocation can take effect.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Service-differentiated Experience Guarantee*.

Requirement/Trouble Ticket Number

FeR-IR20250829003061-0001

RAT

NR FDD, NR TDD

5.3.3.25.2 QCI-specific Service Optimization for Specific UEs

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The "QCI-specific service optimization for specific UEs" function is now supported. This function is controlled by the **SPEC_USR_SERVICE_OPT_QCI_SW** option of the **gNBDUSchParamGroup.ScheduleOptSwitch** parameter.

- If the **SPEC_USR_SERVICE_OPT_QCI_SW** option is selected:
 - "QCI-specific scheduling based on the uplink target IBLER" takes effect only for UEs with a QCI-specific bearer which are not uplink full-buffer UEs.
 - "QCI-specific scheduling based on the downlink target IBLER" takes effect only for UEs with a QCI-specific bearer which are not downlink full-buffer UEs.
- If the **SPEC_USR_SERVICE_OPT_QCI_SW** option is deselected, both "QCI-specific scheduling based on the uplink target IBLER" and "QCI-specific scheduling based on the downlink target IBLER" can take effect for UEs with the QCI-specific bearer.

Specifically:

- "QCI-specific scheduling based on the uplink target IBLER" is controlled by the **UL_IBLER_QCI_SW** option of the **gNBDUSchParamGroup.ScheduleOptSwitch** parameter.

- "QCI-specific scheduling based on the downlink target IBLER" is controlled by the **DL_IBLER_QCI_SW** option of the **gNBDUSchParamGroup.ScheduleOptSwitch** parameter.

Implementation

For details, see *Service-differentiated Experience Guarantee*. The specific sections are listed in the change history of the document.

Impact on the Network

If "QCI-specific scheduling based on the uplink target IBLER" is enabled, enabling "QCI-specific service optimization for specific UEs" may increase the uplink data rates of uplink full-buffer UEs.

If "QCI-specific scheduling based on the downlink target IBLER" is enabled, enabling "QCI-specific service optimization for specific UEs" may increase the downlink data rates of downlink full-buffer UEs.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameter has changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNBDUSchParamGroup.ScheduleOptSwitch	ADD GNBDUSCHPARAM GROUP MOD GNBDUSCHPARAM GROUP LST GNBDUSCHPARAM GROUP	gNodeB	The SPEC_USR_SERVICE_OPT_QCI_SW option has been added to specify whether to enable "QCI-specific service optimization for specific UEs".

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Service-differentiated Experience Guarantee*.

Requirement/Trouble Ticket Number

IR20241209002193

DTS2025082913043

RAT

NR FDD, NR TDD

5.3.3.25.3 Frequency Selection Policy for QCI-based Preallocation

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The **QCI_PRESCH_FREQ_SEL_POL_SW** option of the **NRDUCellUISch.UlPreallocPolEnhSwitch** specifies whether uplink frequency selective scheduling takes effect for UEs during QCI-based uplink preallocation. For details about uplink frequency selective scheduling, see *Uplink Scheduling*.

- Policy 1: If this option is selected, the base station checks whether the estimated RBs for UEs in QCI-based uplink preallocation are small packets. If yes, uplink frequency selective scheduling does not take effect for these UEs. This reduces the probability for small-packet UEs to interrupt the scheduling of large-packet UEs, thereby increasing the average uplink cell throughput.
- Policy 2: If this option is deselected, uplink frequency selective scheduling can take effect for UEs in QCI-based uplink preallocation. Specifically, these UEs

can select the RBs with the optimal performance for scheduling, thereby reducing the uplink BER of these UEs.

 NOTE

This change was first incorporated into V100R022C10SPC010 but is first described in the network impact report of V100R022C10SPC050.

Implementation

For details, see *Service-differentiated Experience Guarantee*. The specific sections are listed in the change history of the document.

Impact on the Network

When the **QCI_PRESCH_FREQ_SEL_POL_SW** option of the **NRDUCellUISch.UlPreallocPolEnhSwitch** parameter is selected, the uplink cell throughput and uplink user-perceived rate can increase. However, this function may increase the uplink BLER of UEs with QCI-based uplink preallocation in effect, thereby increasing the E2E delay.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameter has changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellUISch.Ul PreallocPolEnhS- witch	LST NRDUCELLDLS CH MOD NRDUCELLDLS CH	gNodeB	The QCI_PRESCH_FREQ_SEL_POLICY_SW option has been added to specify whether to enable frequency selection policy for QCI-based uplink preallocation.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Service-differentiated Experience Guarantee*.

Requirement/Trouble Ticket Number

IR20250528001857

DTS2025120942141

RAT

NR FDD, NR TDD

5.3.3.26 FOFD-051201 Extended Cell Range

5.3.3.26.1 Support for Dual-Period 8:2 Slot Configuration by Extended Cell Range (TDD) and Extended Cell Range of 90 km (TDD)

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Extended Cell Range (TDD) and Extended Cell Range of 90 km (TDD) now support dual-period 8:2 slot configuration (with the **NRDUCell.SlotAssignment** parameter set to **8_2_DDDSUUDDDD**). Specific requirements for the PRACH configuration index and slot structure are as follows:

- Extended Cell Range (TDD)
 - PRACH configuration index: The **NRDUCellPrach.PrachConfigurationIndex** parameter must be set to **9, 14, 19, or 65535**.
 - Slot structure: The **NRDUCell.SlotStructure** parameter must be set to **SS86 or SS818**.
- Extended Cell Range of 90 km (TDD)
 - PRACH configuration index: The **NRDUCellPrach.PrachConfigurationIndex** parameter must be set to **49, 54, 59, or 65535**.
 - Slot structure: The **NRDUCell.SlotStructure** parameter must be set to **SS818**.

LampSite base stations do not support this feature.

The change in this feature is supported only in low frequency bands.

Implementation

For details, see *Extended Cell Range*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

RF modules:

- Extended Cell Range (TDD)
 - When the **NRDUCell.SlotAssignment** parameter is set to **8_2_DDDSUUDDDD**, all NR TDD-capable massive MIMO RF modules that work in low frequency bands support this function. For details, see

the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

- When the **NRDUCell.SlotAssignment** parameter is set to a value other than **8_2_DDDSUUDDDD**, all NR TDD-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.
- Extended Cell Range of 90 km (TDD)
All NR TDD-capable massive MIMO AAUs support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The change in this feature has no impact.
- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The descriptions of the following dependency relationships have been revised.
 - Extended Cell Range (TDD)

Function Name	Function Switch	Reference	Description
Slot configuration	NRDUCell.SlotAssignment	<i>Standards Compliance</i>	Extended Cell Range (TDD) requires that the NRDUCell.SlotAssignment parameter be set to 8_2_DDDDDDDSUU or 8_2_DDDSUUDDDD .
Slot structure	NRDUCell.SlotsStructure	<i>Standards Compliance</i>	Extended Cell Range (TDD) supports only SS56, SS518, SS86, and SS818.

Function Name	Function Switch	Reference	Description
3D coverage pattern	NRDUCellTrpBeam.CoverageScenario	<i>Beam Management</i>	When the NRDUCell.SlotStructure parameter is set to SS518 or SS818 , the NRDUCellTrpBeam.CoverageScenario parameter must be set to DEFAULT , EXPAND_SCENARIO_7 , EXPAND_SCENARIO_8 , SCENARIO_27 , SCENARIO_28 , SCENARIO_31 , SCENARIO_33 , SCENARIO_34 , SCENARIO_35 , SCENARIO_36 , CUSTOMIZED_BEAM_SCENARIO , or a value ranging from SCENARIO_1 to SCENARIO_16 .

– Extended Cell Range of 90 km (TDD)

Function Name	Function Switch	Reference	Description
Slot configuration	NRDUCell.SlotAssignment	<i>Standards Compliance</i>	Extended Cell Range of 90 km (TDD) requires that the NRDUCell.SlotAssignment parameter be set to 8_2_DDDDDDD SUU or 8_2_DDDSUUDDDD .

Function Name	Function Switch	Reference	Description
Slot structure	NRDUCell.Slot Structure	<i>Standards Compliance</i>	Extended Cell Range of 90 km (TDD) requires that the NRDUCell.SlotS tructure parameter be set to SS518 or SS818 .
PRACH Configuration Index	NRDUCellPrac h.PrachConfigu rationIndex	None	Extended Cell Range of 90 km (TDD) requires that the NRDUCellPrach .PrachConfigur ationIndex parameter be set to 49, 52, 54, 57, 59, 65535 , or a value ranging from 43 to 47 .

Function Name	Function Switch	Reference	Description
3D coverage pattern	NRDUCellTrpBeam.CoverageScenario	<i>Beam Management</i>	When Extended Cell Range of 90 km (TDD) is enabled and the NRDUCell.SlotsStructure parameter is set to SS518 or SS818 , then the NRDUCellTrpBeam.CoverageScenario parameter must be set to DEFAULT , EXPAND_SCENARIO_7 , EXPAND_SCENARIO_8 , SCENARIO_27 , SCENARIO_28 , SCENARIO_31 , SCENARIO_33 , SCENARIO_34 , SCENARIO_35 , SCENARIO_36 , CUSTOMIZED_BEAM_SCENARIO , or a value ranging from SCENARIO_1 to SCENARIO_16 .

- Mutually exclusive features/functions
The descriptions of the following mutually exclusive functions have been revised.
 - Extended Cell Range (TDD)

Function Name	Function Switch	Reference	Description
Co-MM	INTRA_GNB_SU_COMM_SW option of the NRDUCellColla bServ.MultiCell MimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	Co-MM cannot be enabled when the NRDUCell.Slot Structure parameter is set to SS518 or SS818 .

- Extended Cell Range of 90 km (TDD)

Function Name	Function Switch	Reference	Description
Co-MM	INTRA_GNB_SU_COMM_SW option of the NRDUCellColla bServ.MultiCell MimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	Co-MM cannot be enabled when the NRDUCell.Slot Structure parameter is set to SS518 or SS818 . Therefore, Extended Cell Range of 90 km (TDD) cannot be enabled with Co-MM.

- Impacted features/functions
The descriptions of the following impact relationships have been revised.
 - Extended Cell Range (TDD)

Function Name	Function Switch	Reference	Description
Fusion Cell	NRDUCell.NrDuCellNetworkingMode set to FUSION_CELL	<i>Fusion Cell (Low-Frequency TDD)</i>	If Fusion Cell is required to be enabled in a cell with Extended Cell Range enabled and the NRDUCell.SlotStructure parameter set to SS518 or SS818 , then the SS518_4GP_TIME_INTRF_AVOID_SW or SS818_4GP_TIME_INTRF_AVOID_SW option of the gNodeBAlgo.TimelntfAvoidSw parameter corresponding to the cell must be selected.

- Extended Cell Range of 90 km (TDD)

Function Name	Function Switch	Reference	Description
Fusion Cell	NRDUCell.NrDuCellNetworkingMode set to FUSION_CELL	<i>Fusion Cell (Low-Frequency TDD)</i>	If Fusion Cell is required to be enabled in a cell with Extended Cell Range of 90 km (TDD) enabled, then the SS518_4GP_TIME_INTRF_AVOID_SW or SS818_4GP_TIME_INTRF_AVOID_SW option of the gNodeBAlgo.TimelntfrfAvoidSw parameter corresponding to the cell must be selected.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Extended Cell Range*.

Requirement/Trouble Ticket Number

FeR-IR20251210001664-0001

RAT

NR TDD

5.3.3.27 FOFD-051202 Inter-gNodeB CA

5.3.3.27.1 Added a Method for Observing the Activation and Running State of Inter-gNodeB CA

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The **N.User.IntergNB.CA.PCell.DL.Max** counter has been added for monitoring the activation and running state of inter-gNodeB CA. This counter measures the maximum number of inter-gNodeB CA UEs that use the local cell as their PCells in the downlink.

Implementation

For details, see *CA Experience Improvement*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The change in this feature has no impact.
- Performance management
The following counters have changed.

Change Type	Counter Name	Function Subset	NE	Description
New	N.User.IntergNB.CA.PCell.DL.Max	Cell User Quantity Measurement	gNode B	This counter has been added to measure the maximum number of inter-gNodeB CA UEs that use the local cell as their PCells in the downlink.

- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0023

RAT

NR FDD, NR TDD

5.3.3.27.2 Added Adaptive HARQ Resource Limitation for CA

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

HARQ resource limitation for CA (controlled by the **CA_HARQ_RES_LIMIT_SW** option of **NRDUCellCarrMgmt.CaCompatibilitySwitch**) is used to limit the number of HARQ resources that can be used by UEs in the downlink inter-gNodeB CA state. This reduces the impact of these UEs on the HARQ resource request success rate of other UEs. However, the maximum number of HARQ resources that can be used by UEs in the downlink inter-gNodeB CA state is fixed and cannot adapt to dynamic network changes. For example, when there are a large number of UEs in the downlink inter-gNodeB CA state, these UEs cannot use resources beyond this fixed limit, leading to resource wastes and affecting their data rates. As such, adaptive HARQ resource limitation for CA is introduced in low frequency bands.

Adaptive HARQ resource limitation for CA is controlled by the **CA_F3_AND_F4_HARQ_ADAPT_ADJ_SW** option of the **NRDUCellPucch.PucchAlgoExtSwitch** parameter. After adaptive HARQ resource limitation for CA takes effect, the gNodeB monitors the HARQ resource requests from UEs in the downlink inter-gNodeB CA state over a fixed period of time. It then adaptively increases or decreases the upper limit of HARQ resources available for these UEs as needed. This dynamic adjustment fully utilizes HARQ resources and guarantees the benefits that can be delivered to the data rates of these UEs.

In earlier versions, adaptive HARQ resource limitation for CA is controlled by the **RSVDSWPARAM1_BIT18** option of **NRDUCellRsvdExt01.RsvdSwParam1**. In

V100R022C10SPC010, **RSVDSWPARAM1_BIT18** of **NRDUCellRsvdExt01.RsvdSwParam1** has been added to the disuse list and replaced with **CA_F3_AND_F4_HARQ_ADAPT_ADJ_SW** of **NRDUCellPucch.PucchAlgoExtSwitch**.

The relevant upgrade rules are as follows:

Table 5-33 Parameter value replacement rules

Upgrade Path	Switch On Before the Upgrade	Switch Off Before the Upgrade
For an upgrade to V100R022C10SPC010 or later from a version later than V100R021C10SPC160 but earlier than V100R022C00SPC010	Switch on after the upgrade	Switch off after the upgrade
For an upgrade from V100R022C00SPC050 or later to V100R022C10SPC010 or later	Switch on after the upgrade	Switch off after the upgrade
Other upgrade paths	Switch off after the upgrade	Switch off after the upgrade

Implementation

For details, see *CA Experience Improvement*. The specific sections are listed in the change history of the document.

Impact on the Network

After adaptive HARQ resource limitation for CA is in effect:

- The average downlink throughput of SCell-activated inter-gNodeB CA UEs ($(N_{\text{Inter}gNB.CA.SccActiveUser.ThpVol.DL} - N_{\text{Inter}gNB.CA.ThpVol.DL.LastSlot}) / N_{\text{Inter}gNB.CA.ThpTime.DL.RmvLastSlot}$) and **User Downlink Average Throughput (DU)** increase.
- There may be fewer HARQ resources available to non-inter-gNodeB CA UEs (including intra-gNodeB CA UEs and non-CA UEs), which could lead to decreased throughput for these UEs in extremely heavy-load scenarios.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Disused	NRDUCellRsvdExt01.RsvdSwParam1	MOD NRDUCELLRSV DEXT01 LST NRDUCELLRSV DEXT01	gNodeB	Adaptive HARQ resource limitation for CA, originally controlled by RSVDSWPARAM1_BIT18 of this reserved parameter, is now conveyed by the CA_F3_AND_F4_HARQ_ADAPT_ADJ_SW option of the NRDUCellPucch.PucchAlgoExtSwitch parameter, and this bit has been added to the disuse list. Although the configuration interface in this version supports synchronization and delivery of the setting of this bit, the system does not use this bit.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellPucch.PucchAlgoExtSwitch	MOD NRDUCELLPU CCH LST NRDUCELLPU CCH	gNodeB	The CA_F3_AND_F4_HARQ_ADAPT_ADJ_SW option has been added to specify whether to enable adaptive HARQ resource limitation for CA.

NOTE

NRDUCellRsvdExt01.RsvdSwParam1 is a reserved parameter.

For details about reserved parameters, see 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationship has been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	HARQ resource limitation for CA	CA_HARQ_RESOURCE_LIMIT_SW option of the NRDUCellCarrierMgmt.CaCompatibilitySwitch parameter	<i>CA Experience Improvement</i>	Adaptive HARQ resource limitation for CA can take effect only when HARQ resource limitation for CA is enabled.

- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

Whether the change takes effect by default after an upgrade varies as follows:

- With the **RSVDSWPARAM1_BIT18** option of **NRDUCellRsvdExt01.RsvdSwParam1** selected before an upgrade: Yes
- With **RSVDSWPARAM1_BIT18** of **NRDUCellRsvdExt01.RsvdSwParam1** deselected before an upgrade: No

For details, see the corresponding "Operation and Maintenance" section in *CA Experience Improvement*.

Requirement/Trouble Ticket Number

FeR-IR20250528001856-0001

RAT

NR FDD, NR TDD

5.3.3.28 FOFD-051211 3CC CA

5.3.3.28.1 Added Reselection Policy for 3CC and Massive CA UEs in Idle Mode

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

If a UE reselects a cell that does not support the 3CC or massive CA function, the function cannot take effect for the UE. To increase the proportion of UEs with 3CC or massive CA in effect, the "reselection policy for 3CC and massive CA UEs in idle mode" function is introduced in SA networking. This function is controlled by the **MASS_CA_IDLE_RESEL_POL_SW** option of the **NRCellOpQciBearer.ServiceDiffAlgoSwitch** parameter. With this function enabled, the gNodeB gives top dedicated reselection priority to the target PCC frequency supporting 3CC and massive CA during cell reselection for capable UEs with specific operator and QCI details. This increases the probability that these UEs can be reselected to cells capable of 3CC and massive CA, thereby increasing the proportion of UEs with 3CC and massive CA in effect.

There are multiple types of dedicated reselection priorities. If they are configured concurrently, they are delivered in different sequences. For details about the sequence in which dedicated reselection priorities are delivered and the basic information about cell reselection, see *SA Mobility Management in Idle Mode*.

The change in this feature is supported only in low-frequency SA networking.

NOTE

This change was first incorporated into V100R022C00SPC010 but is first described in the network impact report of V100R022C00SPC110.

Implementation

For details, see *CA Experience Improvement*. The specific sections are listed in the change history of the document.

Impact on the Network

After the "reselection policy for 3CC and massive CA UEs in idle mode" function is enabled, the cell may have an increased proportion of 3CC and massive CA UEs with specific operator and QCI details, and there may be an increase in the average throughput of SCell-configured CA UEs ((**N.CA.SccConfigUser.ThpVol.DL** – **N.CA.SccConfigUser.ThpVol.DL.LastSlot**)/**N.CA.SccConfigUser.ThpTime.DL.RmvLastSlot**).

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRCellOpQciBearer.ServiceDiffAlgoSwitch	ADD NRCELLOPQCI BEARER MOD NRCELLOPQCI BEARER LST NRCELLOPQCI BEARER	gNodeB	The NRCellOpQciBearer.ServiceDiffAlgoSwitch parameter has been added. The MASS_CA_IDLE_RESEL_POLICY_SW option of this parameter has been used to control the "reselection policy for 3CC and massive CA UEs in idle mode" function.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *CA Experience Improvement*.

Requirement/Trouble Ticket Number

FeR-IR20240807002404-0010

RAT

NR FDD, NR TDD

5.3.3.28.2 Added a Method for Observing the Activation and Running State of 3CC Aggregation

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The **N.User.CA.PCell.DL.3CC.Max** counter has been added for monitoring the activation and running state of 3CC aggregation. This counter measures the maximum number of 3CC CA UEs that use the local cell as their PCells in the downlink.

The change in this feature is supported only in low-frequency bands.

Implementation

For details, see *CA Experience Improvement*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The change in this feature has no impact.
- Performance management
The following counters have changed.

Change Type	Counter Name	Function Subset	NE	Description
New	N.User.CA.P Cell.DL.3CC. Max	Cell User Quantity Measurement	gNode B	This counter has been added to measure the maximum number of 3CC CA UEs that use the local cell as their PCells in the downlink.

- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0023

RAT

NR FDD, NR TDD

5.3.3.29 FOFD-060201 Virtual Grid-based Multi-Frequency Coordination

5.3.3.29.1 Added an Activation Verification Method for Virtual Grid-based Multi-Frequency Coordination

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In low-frequency scenarios, the following counters have been added for the observation of RSRP prediction model building and downlink intra-frequency spectral efficiency prediction model building:

- **N.NrVirtualGrid.Model.Max**, which measures the maximum number of available NR grid models in a cell
If the value of this counter is not 0 in the case of RSRP prediction models, one or more RSRP prediction models have been successfully built.
- **N.NrDISEGrid.Model.Avg**, which measures the average number of available NR downlink spectral efficiency grid models in a cell
If the value of this counter is not 0 in the case of downlink intra-frequency spectral efficiency prediction models, one or more downlink intra-frequency spectral efficiency prediction models have been successfully built.

Implementation

For details, see *Virtual Grid-based Multi-Frequency Coordination*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The change in this feature has no impact.
- Performance management
The following counters have changed.

Change Type	Counter Name	Function Subset	NE	Description
New	N.NrVirtualGrid.Model.Max	AI Grid Model Measurement	gNodeB	This counter has been added to measure the maximum number of available NR grid models in a cell.
New	N.NrDlSEGrid.Model.Avg	AI Grid Model Measurement	gNodeB	This counter has been added to measure the average number of available NR downlink spectral efficiency grid models in a cell.

- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20241209002193-0090

RAT

NR FDD, NR TDD

5.3.3.30 FOFD-060301 VoNR Performance Enhancement

5.3.3.30.1 Added Support for the 4:1 Slot Configuration by 3GPP R16-compliant Uplink Slot Aggregation

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The 4:1 slot configuration (with **NRDUCell.SlotAssignment** set to **4_1_DDDSU**) is now supported by 3GPP R16-compliant uplink slot aggregation. When **NRDUCell.SlotAssignment** is set to **4_1_DDDSU**, the gNodeB sets the *numberOfRepetitions-r16* IE in the RRC Reconfiguration message to **n8**, allowing DCI to dynamically instruct the UE to transmit the same voice data block in two consecutive uplink slots.

The change in this feature is supported only in low frequency bands.

The change in this feature is supported only in SA networking.

Implementation

For details, see *VoNR*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature is implemented on the gNodeB.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature is not under license control.
- Configuration management
The change in this feature has no impact.
- Performance management
The change in this feature has no impact.

- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Slot configuration	NRDUCell.SlotAssignment	<i>Standards Compliance</i>	3GPP R16-compliant uplink slot aggregation requires that the NRDUCell.SlotAssignment parameter be set to 8_2_DDDDDDDSUU , 8_2_DDDSUDDDD , 7_3_DDDSUDDSUU , or 4_1_DDDSU .

- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20241209002193-0062

RAT

NR TDD

5.3.3.31 FOFD-061203 Intelligent Channel Dynamic Shutdown

5.3.3.31.1 Added Support for Dynamic RF Channel Shutdown in SSB Slots with UE Scheduling

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

For TDD, dynamic RF channel shutdown can now be supported in SSB slots with UE scheduling. After common-channel dynamic shutdown in dynamic RF channel shutdown (controlled by **NRDUCellPowerSaving.SsbDynShutdownSwitch**) is enabled, the shutdown can take effect in SSB slots with UE scheduling by selecting the **RF_CHN_DYN_SHUTDOWN_SSB_SW** option of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter.

The change in this feature is supported only in low frequency bands.

LampSite base stations do not support this feature.

Implementation

For details, see *Energy Conservation and Emission Reduction*. The specific sections are listed in the change history of the document.

Impact on the Network

For TDD, when the **RF_CHN_DYN_SHUTDOWN_SSB_SW** option of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter is selected, dynamic RF channel shutdown can be performed in SSB slots with UE scheduling to deliver energy saving gains in this scenario. In addition, the impact of common-channel dynamic shutdown increases, and the downlink user-perceived rate and average downlink cell throughput decrease.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellPwrSave Algo.PowerSaving AlgoSwitch	MOD NRDUCELLPW RSVEALGO LST NRDUCELLPW RSVEALGO	gNodeB	The RF_CHN_DYN_SHUTDOWN_SSB_SW option has been added to specify whether to enable dynamic RF channel shutdown in SSB slots with UE scheduling.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Energy Conservation and Emission Reduction*.

Requirement/Trouble Ticket Number

FeR-IR20241217003450-0005

RAT

NR TDD

5.3.3.31.2 Added Support for Dynamic RF Channel Shutdown in Single-Carrier Scenarios

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

For TDD, dynamic RF channel shutdown can be supported in single-carrier scenarios. After dynamic RF channel shutdown (controlled by the **RF_CHN_DYN_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter) is enabled, dynamic RF channel shutdown can take effect in single-carrier scenarios by selecting the **RF_CHN_DYN_SHDN_SGL_CARR_SW** option of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter.

The change in this feature is supported only in low frequency bands.

LampSite base stations do not support this feature.

Implementation

For details, see *Energy Conservation and Emission Reduction*. The specific sections are listed in the change history of the document.

Impact on the Network

For TDD, when the **RF_CHN_DYN_SHDN_SGL_CARR_SW** option of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter is selected, dynamic RF channel shutdown can be performed in single-carrier scenarios to deliver energy saving gains in this scenario. The impacts of simultaneous shutdown are the same as those in multi-carrier scenarios. For details, see *Energy Conservation and Emission Reduction*.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management

The change in this feature has no impact. It is controlled by the existing feature license.

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellPwrSave Algo.PowerSaving AlgoSwitch	MOD NRDUCCELLPW RSVEALGO LST NRDUCCELLPW RSVEALGO	gNodeB	The RF_CHN_DYN_SHDN_SGL_CARR_SW option has been added to specify whether to enable dynamic RF channel shutdown in single-carrier scenarios.

- Performance management

The change in this feature has no impact.

- Fault management

The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

The change in this feature has no impact.

- Mutually exclusive features/functions

The change in this feature has no impact.

- Impacted features/functions

The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Energy Conservation and Emission Reduction*.

Requirement/Trouble Ticket Number

FeR-IR20241217003450-0005

RAT

NR TDD

5.3.3.32 FOFD-061223 Experience Boosting based on Multi-Band Coordination

5.3.3.32.1 Added a Measurement Timer for Protection Against Weak Uplink Coverage

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

To flexibly control the waiting time for inter-frequency measurements triggered due to weak uplink coverage, a measurement timer for protection against weak uplink coverage is added. The timer is specified by the **NRCellMobilityConfig.UlWeakCovPrtMeasTimer** parameter.

The inter-frequency measurement waiting timer for uplink-coverage-based inter-frequency handovers has been changed from a fixed value of 2s to the value of the **NRCellMobilityConfig.UlWeakCovPrtMeasTimer** parameter.

This change is supported only in low frequency bands.

Implementation

For details, see *Multi-Frequency Smart Selection*. The specific sections are listed in the change history of the document.

Impact on the Network

A larger value of the measurement timer for protection against weak uplink coverage results in a higher probability that a UE is handed over to an inter-frequency neighboring cell meeting the handover requirements. A smaller value of the timer produces the opposite effect.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRCellMobilityConfig.UlWeakCovPrtMeasTimer	MOD NRCELLMOBILITY CONFIG LST NRCELLMOBILITY CONFIG	gNode B	This parameter has been added to specify the waiting time for inter-frequency measurements triggered due to weak uplink coverage.

- Performance management

The change in this feature has no impact.

- Fault management

The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

The change in this feature has no impact.

- Mutually exclusive features/functions

The change in this feature has no impact.

- Impacted features/functions

The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Multi-Frequency Smart Selection*.

Requirement/Trouble Ticket Number

IR20250318001398

FeR-IR20250221001146-0010

RAT

NR FDD, NR TDD

5.3.3.32.2 Added LTE-NR Joint Evaluation of Large-packet Service Status for the "Protection of UEs Under Weak Uplink Coverage" Function

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

LTE-NR joint evaluation of large-packet service status is now supported to allow uplink-coverage-based inter-frequency handover for UEs with uplink fallback to LTE in effect.

Assume that a UE is identified as being under weak uplink coverage, the gNodeB-side traffic volume identification switch is on, and the uplink traffic volume to be transmitted by the UE is less than or equal to the threshold for identifying common large-packet UEs in the uplink. The gNodeB determines whether to initiate an uplink-coverage-based inter-frequency handover for the UE as follows:

- If LTE-NR joint evaluation of large-packet service status is enabled:
 - The gNodeB can initiate an uplink-coverage-based inter-frequency handover for the UE when all of the following conditions are met:
 - Uplink fallback to LTE is in effect for the UE.
 - The gNodeB can obtain the service status (large- or small-packet service) of the UE from the eNodeB.
 - The uplink traffic volume to be transmitted by the UE on the eNodeB side is greater than the uplink large-packet traffic volume threshold.
 - Otherwise, the uplink-coverage-based inter-frequency handover evaluation procedure ends and the next identification period starts.
- If LTE-NR joint evaluation of large-packet service status is disabled, the uplink-coverage-based inter-frequency handover evaluation procedure ends and the next identification period starts.

The following is related detailed information:

- LTE-NR joint evaluation of large-packet service status is controlled by the **LNR_LRG_PKT_UL_WEAK_COV_SW** option of the **NRCellMobilityConfig.InterFreqHoExtSwitch** parameter.
- The traffic volume identification switch is controlled by the **VOLUME_IDENTIFY_SW** option of the **NRCellAlgoSwitch.ServiceFunctionSwitch** parameter.
- The threshold for identifying common large-packet UEs in the uplink is specified by the **NRDUCellServExp.CommonULLargePktIdentThld** parameter.
- The uplink large-packet traffic volume threshold is specified by the **CellServiceDiffCfg.ULLargePacketDataVoThd** parameter.
- For details about uplink fallback to LTE and whether the gNodeB can obtain the service status (large- or small-packet service) of UEs with uplink fallback to LTE in effect from the eNodeB, see *EPC-based NSA Basics*.

This change is supported only in low-frequency NSA networking.

Implementation

For details, see *Multi-Frequency Smart Selection*. The specific sections are listed in the change history of the document.

Impact on the Network

In NSA networking, after LTE-NR joint evaluation of large-packet service status is enabled, the values of the following counters may increase:

- **N.NsaDc.InterFreq.UlCoverage.PSCell.Change.Att**
- **N.NsaDc.InterFreq.UlCoverage.PSCell.Change.Succ**

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRCellMobilityConfig.InterFreqHoExtSwitch	MOD NRCELLMOBILITY CONFIG LST NRCELLMOBILITY CONFIG	gNode B	The LNR_LRG_PKT_UL_WEAK_COV_SW option has been added to specify whether to evaluate the large-packet service status on both the LTE and NR sides for triggering protection of UEs under weak uplink coverage in NSA networking.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Protection of UEs under weak uplink coverage	UL_WEAK_COV_UE_PROTECT_SW option of the NRCellAlgoSwitch.MultiFreqAlgoSwitch parameter	<i>Multi-Frequency Smart Selection</i>	LTE-NR joint evaluation of large-packet service status (controlled by the LNR_LRG_PKT_UL_WEAK_COV_SW option of the NRCellMobilityConfig.InterFreqHoExtSwitch parameter) can be enabled only when protection of UEs under weak uplink coverage has been enabled.
Low-frequency TDD	Function of requesting LTE-side large-packet status of ULFB UEs	ULFB_LTE_UL_LRG_PKT_REQ_SW option of the gNodeBParam.NsaDcOptSwitch parameter	<i>EPC-based NSA Basics</i>	LTE-NR joint evaluation of large-packet service status (controlled by the LNR_LRG_PKT_UL_WEAK_COV_SW option of the NRCellMobilityConfig.InterFreqHoExtSwitch parameter) can be enabled only when the function of requesting LTE-side large-packet status of ULFB UEs has been enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Traffic volume identification	VOLUME_IDENTIFY_SW option of the NRCellAlgoSwitch.ServiceFunctionSwitch parameter	<i>Multi-Frequency Smart Selection</i>	LTE-NR joint evaluation of large-packet service status (controlled by the LNR_LRG_PKT_UL_WEAK_COV_SW option of the NRCellMobilityConfig.InterFreqHoExtSwitch parameter) can take effect only when the traffic volume identification switch is turned on.

- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Multi-Frequency Smart Selection*.

Requirement/Trouble Ticket Number

IR20250318001398

FeR-IR20250221001146-0010

RAT

NR FDD, NR TDD

5.3.3.33 FOFD-080011 Stable Video Surveillance Experience

5.3.3.33.1 Support for Enhanced Avoidance of RedCap Uplink Scheduling in Low-Frequency NR TDD Scenarios

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In low-frequency NR TDD scenarios, the avoidance of RedCap uplink scheduling function has been optimized as follows:

- The enumerated value **LEVEL3** has been added to the **NRDUCellRedCap.UIAvoidPol** parameter. In this case, non-RedCap UEs with medium, low, and relatively high priorities avoid the BWP frequency-domain positions corresponding to RedCap UEs adaptively based on the current frequency-domain SINR and frequency-domain positions of BWP1s activated for RedCap UEs during uplink scheduling.
- The coefficient can be configured for non-RedCap UEs to avoid the BWP frequency-domain positions corresponding to RedCap based on SINRs during uplink scheduling. Whether the SINR of non-RedCap UEs on the current frequency domain meets the avoidance conditions depends on the configuration of the **NRDUCellRedCap.UIAvoidSinrCoeff** parameter. A larger value of this parameter indicates a higher probability to trigger resource avoidance.

Implementation

For details, see *Stable Video Surveillance Experience*. The specific sections are listed in the change history of the document.

Impact on the Network

The average uplink throughput of non-RedCap UEs slightly decreases because non-RedCap UEs avoid the frequency-domain resources of all BWP1s activated for RedCap UEs and the scheduling positions change. The higher the value of the **NRDUCellRedCap.UIAvoidPol** parameter, the more non-RedCap UEs are affected.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellRedCap.UIAvoidSinrCoeff	MOD NRDUCELLR EDCAP LST NRDUCELLR EDCAP	gNodeB	This parameter has been added to specify the coefficient for non-RedCap UEs to avoid the BWP frequency-domain positions corresponding to RedCap based on SINRs during uplink scheduling.
Modified	NRDUCellRedCap.UIAvoidPol	MOD NRDUCELLR EDCAP LST NRDUCELLR EDCAP	gNodeB	The enumerated value LEVEL3 has been added.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Stable Video Surveillance Experience*.

Requirement/Trouble Ticket Number

IR20241209002193

DTS2025052022393

RAT

NR TDD

5.3.3.34 FOFD-080012 Massive CA

5.3.3.34.1 Added Reselection Policy for 3CC and Massive CA UEs in Idle Mode

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

If a UE reselects a cell that does not support the 3CC or massive CA function, the function cannot take effect for the UE. To increase the proportion of UEs with 3CC or massive CA in effect, the "reselection policy for 3CC and massive CA UEs in idle mode" function is introduced in SA networking. This function is controlled by the **MASS_CA_IDLE_RESEL_POL_SW** option of the **NRCCellOpQciBearer.ServiceDiffAlgoSwitch** parameter. With this function enabled, the gNodeB gives top dedicated reselection priority to the target PCC frequency supporting 3CC and massive CA during cell reselection for capable UEs with specific operator and QCI details. This increases the probability that these UEs can be reselected to cells capable of 3CC and massive CA, thereby increasing the proportion of UEs with 3CC and massive CA in effect.

There are multiple types of dedicated reselection priorities. If they are configured concurrently, they are delivered in different sequences. For details about the sequence in which dedicated reselection priorities are delivered and the basic information about cell reselection, see *SA Mobility Management in Idle Mode*.

The change in this feature is supported only in low-frequency SA networking.

NOTE

This change was first incorporated into V100R022C00SPC010 but is first described in the network impact report of V100R022C00SPC110.

Implementation

For details, see *CA Experience Improvement*. The specific sections are listed in the change history of the document.

Impact on the Network

After the "reselection policy for 3CC and massive CA UEs in idle mode" function is enabled, the cell may have an increased proportion of 3CC and massive CA UEs with specific operator and QCI details, and there may be an increase in the average throughput of SCell-configured CA UEs ((**N.CA.SccConfigUser.ThpVol.DL** – **N.CA.SccConfigUser.ThpVol.DL.LastSlot**)/**N.CA.SccConfigUser.ThpTime.DL.RmvLastSlot**).

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRCelOpQciBearer.ServiceDiffAlgoSwitch	ADD NRCELLOPQCI BEARER MOD NRCELLOPQCI BEARER LST NRCELLOPQCI BEARER	gNodeB	The NRCelOpQciBearer.ServiceDiffAlgoSwitch parameter has been added. The MASS_CA_IDLE_RESEL_POLICY option of this parameter has been used to control the "reselection policy for 3CC and massive CA UEs in idle mode" function.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

- The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *CA Experience Improvement*.

Requirement/Trouble Ticket Number

FeR-IR20240807002404-0010

RAT

NR FDD, NR TDD

5.3.3.35 FOFD-081202 Turbo Uplink 2.0

5.3.3.35.1 QCI-specific Uplink Data Compression

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Before this change, uplink data compression can be configured only by using the cell-level parameter (the **UDC_SW** or **UPLINK_SIG_COMPRESS_SW** option of the **NRDUCellChnCovAlgo.UlCoverageAlgoSwitch** parameter). This change enables QCI-specific uplink data compression, which is controlled by the **QCI_UDC_SW** option of the **gNBQciBearer.PdcpAlgoSwitch** parameter. After this option is selected, uplink data compression can take effect for bearers with the specified QCI.

- For such bearers carrying data services, if the **UDC_SW** option of the cell-level parameter **NRDUCellChnCovAlgo.UlCoverageAlgoSwitch** is selected, this cell-level parameter setting takes effect; otherwise, the QCI-level parameter setting takes effect.
- For such bearers carrying voice services (with QCI 5 or 69): If the **UPLINK_SIG_COMPRESS_SW** option of the cell-level parameter **NRDUCellChnCovAlgo.UlCoverageAlgoSwitch** is selected, this cell-level parameter setting takes effect; otherwise, the QCI-level parameter setting takes effect.

LampSite base stations do not support this function.

The change in this feature is supported only in SA networking.

The change in this feature is supported only in low frequency bands.

Implementation

For details, see *Turbo Uplink (Low-Frequency TDD)*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNBQciBearer.Pdc pAlgoSwitch	ADD GNBQCIBEAR ER MOD GNBQCIBEAR ER LST GNBQCIBEAR ER	gNo deB	The QCI_UDC_SW option has been added to specify whether to enable QCI-specific uplink data compression.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

The descriptions of the following dependency relationships have been revised.

Function Name	Function Switch	Reference	Description
Uplink data compression	UPLINK_SIG_CO MPRESS_SW option of the NRDUCellChnCo vAlgo.UlCoverag eAlgoSwitch parameter or QCI_UDC_SW option of the gNBQciBearer.P dcpAlgoSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	Uplink data compression based on the SIP standard dictionary takes effect only if the UPLINK_SIG_CO MPRESS_SW option of the NRDUCellChnCo vAlgo.UlCoverag eAlgoSwitch parameter or the QCI_UDC_SW option of the gNBQciBearer.P dcpAlgoSwitch parameter is selected.

- Mutually exclusive features/functions

The following mutually exclusive relationship has been added.

Function Name	Function Switch	Reference	Description
QoS Class Identifier	gNBQciBearer.Q ci	<i>QoS Management</i>	When the gNBQciBearer.Q ci parameter is set to 1 , 65 , or 66 , the QCI_UDC_SW option of the gNBQciBearer.P dcpAlgoSwitch parameter cannot be selected.

- Impacted features/functions

The descriptions of the following impact relationships have been revised.

Function Name	Function Switch	Reference	Description
Basic VoNR functions	VONR_SW option of the NRCellAlgoSwitch.VonrSwitch parameter	<i>VoNR</i>	If uplink data compression (controlled by the UPLINK_SIG_COMPRESS_SW option of the NRDUCellChnCoVAlgo.UlCoverageAlgoSwitch parameter or the QCI_UDC_SW option of the gNBQciBearer.PdcpAlgoSwitch parameter) and basic VoNR functions are enabled, voice quality may deteriorate (in terms of extra delay or jitter) or service drops may occur.
MCX switch	MCX_SW option of the NRCellAlgoSwitch.ServiceFunctionSwitch parameter	None	If uplink data compression (controlled by the UPLINK_SIG_COMPRESS_SW option of the NRDUCellChnCoVAlgo.UlCoverageAlgoSwitch parameter or the QCI_UDC_SW option of the gNBQciBearer.PdcpAlgoSwitch parameter) is enabled and the MCX switch is turned on, voice quality may deteriorate (in terms of extra delay or jitter) or service drops may occur.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Turbo Uplink (Low-Frequency TDD)*.

Requirement/Trouble Ticket Number

FeR-IR20251211000292-0001

RAT

NR TDD

5.3.3.36 FOFD-081203 MetaAAU Deep Energy Saving

5.3.3.36.1 Added Support for Uplink Symbol Power Saving in Multi-carrier Scenarios.

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

When the **UL_SYM_SHUTDOWN_MULTI_CARR_SW** option of the **gNBPwrSavingAlgo.PwrSavingAlgoSwitch** parameter is selected and multi-band RF modules are used, uplink symbol power saving can take effect independently in NR cells in multi-carrier scenarios.

The change in this feature is supported only in low frequency bands.

LampSite base stations do not support this feature.

Implementation

For details, see *Energy Conservation and Emission Reduction*. The specific sections are listed in the change history of the document.

Impact on the Network

Uplink symbol power saving can take effect in multi-carrier scenarios. For details about the benefits and impacts of uplink symbol power saving on the network, see *Energy Conservation and Emission Reduction*.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

All multi-band RF modules of MetaAAUs support this function. A multi-band RF module can work in multiple frequency bands. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNB PwrSavingAlgo o.PwrSavingAlgoSwitch	MOD GNBPWRS AVI NGALGO LST GNBPWRS AVI NGALGO	gNodeB	The UL_SYM_SHUTDOWN_MULTICARRIER_SWO option has been added to specify whether uplink symbol power saving takes effect in multi-carrier scenarios.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Energy Conservation and Emission Reduction*.

Requirement/Trouble Ticket Number

FeR-IR20241217003450-0003

RAT

NR TDD

5.3.3.37 FOFD-081231 8 Layers for a Single FWA User (Trial)

5.3.3.37.1 Added "Resource Allocation Adjustment for Downlink 8-Layer Transmission" and "MU Pairing Adjustment for Downlink 8-Layer Transmission"

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

To allocate SRS resources and data channel resources more properly in a cell with downlink 8-layer transmission for a single FWA user enabled, the following functions have been introduced:

- Resource allocation adjustment for downlink 8-layer transmission: This function enables the base station to properly allocate SRS resources to 8R UEs based on the overall cell performance. This function is controlled by the **DL_EIGHT_LAYER_RES_ALLO_ADJ_SW** option of the **NRDUCellDLRank.DIRankSelAlgoExtSw** parameter.
- MU pairing adjustment for downlink 8-layer transmission: This function enables 8R UEs to be paired when their channel conditions are good and there is multipath propagation, so that data channel resources are properly allocated in a cell. This function is controlled by the **DL_EIGHT_LAYER_MU_ADJ_SW** option of the **NRDUCellDLRank.DIRankSelAlgoExtSw** parameter and applies only to low-frequency TDD massive MIMO cells.

LampSite base stations do not support this feature.

The change in this feature is supported only in low frequency bands.

Implementation

Resource allocation adjustment for downlink 8-layer transmission: This function enables the base station to properly allocate SRS resources to 8R UEs based on the overall cell performance. This function is controlled by the **DL_EIGHT_LAYER_RES_ALLO_ADJ_SW** option of the **NRDUCellDLRank.DIRankSelAlgoExtSw** parameter.

MU pairing adjustment for downlink 8-layer transmission: This function enables 8R UEs to be paired when their channel conditions are good and there is multipath propagation, so that data channel resources are properly allocated in a cell. This function is controlled by the **DL_EIGHT_LAYER_MU_ADJ_SW** option of the **NRDUCellDLRank.DIRankSelAlgoExtSw** parameter and applies only to low-frequency TDD massive MIMO cells.

The upgrade policy is as follows:

- For an upgrade from V100R020C10SPC010 or later to V100R022C10SPC010 or later: (1) If the **DL_SU_EIGHT_LAYER_SW** option of **NRDUCellDLRank.DIRankSelAlgoSw** is selected before the upgrade, both the **DL_EIGHT_LAYER_RES_ALLO_ADJ_SW** and **DL_EIGHT_LAYER_MU_ADJ_SW** options of **NRDUCellDLRank.DIRankSelAlgoExtSw** will be deselected after the upgrade. (2) In other situations, both the **DL_EIGHT_LAYER_RES_ALLO_ADJ_SW** and **DL_EIGHT_LAYER_MU_ADJ_SW** options of **NRDUCellDLRank.DIRankSelAlgoExtSw** will be selected after the upgrade.
- In other upgrade paths, both the **DL_EIGHT_LAYER_RES_ALLO_ADJ_SW** and **DL_EIGHT_LAYER_MU_ADJ_SW** options of **NRDUCellDLRank.DIRankSelAlgoExtSw** will be selected after the upgrade.

Impact on the Network

Assume that downlink 8-layer transmission for a single FWA user is enabled. The impacts are as follows:

- If resource allocation adjustment for downlink 8-layer transmission is enabled, the following impacts will occur: (1) In the case of cell congestion, SRS resource consumption is reduced, and the downlink user-perceived rate may increase. (2) In the case that the cell serves a small number of UEs, the downlink user-perceived rate may fluctuate. (3) The number of RRC reconfiguration messages will increase, and the service drop rate may increase in extreme scenarios.
- If MU pairing adjustment for downlink 8-layer transmission is enabled, the downlink pairing-related indicators may change and the downlink user-perceived rate may fluctuate in medium- and heavy-load scenarios.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management

The change in this feature has no impact. It is controlled by the existing feature license.

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCell DLRank.DI RankSelAl goExtSw	MOD NRDUCELLD LRANK LST NRDUCELLD LRANK	gNodeB	This parameter has been added to specify whether to enable functions related to downlink rank adaptation. • The DL_EIGHT_LAYER_RES_ALLO_ADJ_SW option has been added to specify whether to enable resource allocation adjustment for downlink 8-layer transmission. • The DL_EIGHT_LAYER_MU_ADJ_SW option has been added to specify whether to enable MU pairing adjustment for downlink 8-layer transmission.

- Performance management

The change in this feature has no impact.

- Fault management

The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

The change in this feature has no impact.

- Mutually exclusive features/functions

The change in this feature has no impact.

- Impacted features/functions

The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *FWA*.

Requirement/Trouble Ticket Number

IR20250528001857

DTS2025103058445

RAT

NR TDD

5.3.3.37.2 Added Support for "Downlink 8-Layer Transmission for a Single FWA User" in NSA Networking and in CA Scenarios with Three or More CCs

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

"Downlink 8-layer transmission for a single FWA user" is now supported in NSA networking and in CA scenarios with three or more CCs. Enabling this function in these scenarios requires selecting both of the following options:

- **DL_SU_EIGHT_LAYER_SW** option of the **NRDUCellDlRank.DlRankSelAlgoSw** parameter
- **DL_EIGHT_LAYER_SCENARIO_EXT_SW** option of the **NRDUCellDlRank.DlRankSelAlgoSw** parameter

In CA scenarios with three or more CCs, the **NRDUCellCarrMgmt.DlEightLayerMaxCCNum** parameter specifies the maximum number of CCs where downlink 8-layer transmission can take effect. The details are as follows:

- When this parameter is set to **1CC**, downlink 8-layer transmission can take effect on at most one CC for a UE with three or more CCs aggregated.
- When this parameter is set to **2CC**, downlink 8-layer transmission can take effect on at most two CCs for a UE with three or more CCs aggregated.
- When this parameter is set to **3CC**, downlink 8-layer transmission can take effect on at most three CCs for a UE with three or more CCs aggregated.
- When this parameter is set to **NOT_CONFIG**, the maximum number of CCs where downlink 8-layer transmission can take effect equals the maximum number of CCs that can be aggregated.

LampSite base stations do not support this feature.

The change in this feature is supported only in low frequency bands.

Implementation

For details, see *FWA*. The specific sections are listed in the change history of the document.

Impact on the Network

- In NSA networking or in CA scenarios with three or more CCs, if the **DL_EIGHT_LAYER_SCENARIO_EXT_SW** option of the **NRDUCellDLRank.DIRankSelAlgoSw** parameter is selected, the **User Downlink Average Throughput (DU)** increases and the PUCCH consumes more resources.
- If downlink 8-layer transmission is in effect on a single carrier of a UE with three or more CCs aggregated and the cell load is light, there is an extremely low probability that the UE's downlink throughput decreases after the **DL_EIGHT_LAYER_SCENARIO_EXT_SW** option of the **NRDUCellDLRank.DIRankSelAlgoSw** parameter is selected.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellDLRank.DIRankSelAlgoSw	MOD NRDUCELLD LRANK LST NRDUCELLD LRANK	gNodeB	The DL_EIGHT_LAYER_SCENARIO_EXT_SW option has been added to specify whether to enable downlink 8-layer transmission for a single FWA user in NSA networking and in CA scenarios three or more CCs in SA networking.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellCarrMgmt.DlEightLayerMaxCCNum	MOD NRDUCELLC ARRMGMT LST NRDUCELLC ARRMGMT	gNodeB	This parameter has been added to specify the maximum number of CCs where downlink 8-layer transmission can take effect.

- Performance management

The following counters have changed.

Change Type	Counter Name	Function Subset	NE	Description
New	N.User.RRCConn.8R.Max	Cell User Quantity Measurement (User.Cell)	BTS3900, BTS3900 5G, BTS5900, BTS5900 5G	This counter has been added to measure the maximum number of 8R-capable UEs in a cell.
New	N.User.RRCConn.8R.Avg	Cell User Quantity Measurement (User.Cell)	BTS3900, BTS3900 5G, BTS5900, BTS5900 5G	This counter has been added to measure the average number of 8R-capable UEs in a cell.

- Fault management

The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

The change in this feature has no impact.

- Mutually exclusive features/functions

The change in this feature has no impact.

- Impacted features/functions

The descriptions of the following impact relationships have been revised.

Function Name	Function Switch	Reference	Description
3CC aggregation	NRDUCellCarrMgmt.CaDlMaxCNum set to DL3CC	<i>CA Experience Improvement</i>	- When the DL_EIGHT_LAYER_SCENARIO_EXT_SW option of the NRDUCellDirank.DlRankselAlgoSw parameter is deselected, downlink 8-layer transmission for a single FWA user cannot take effect together with 3CC aggregation for a UE. If 3CC aggregation is enabled, enabling downlink 8-layer transmission for a single FWA user may decrease the success rate of SCell configuration for 3CC aggregation. - When the DL_EIGHT_LAYER_SCENARIO_EXT_SW option of the NRDUCellDirank.DlRankselAlgoSw parameter is selected, downlink 8-layer transmission for a single

Function Name	Function Switch	Reference	Description
			FWA user can take effect for CA UEs with three or more CCs aggregated.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *FWA*.

Requirement/Trouble Ticket Number

FeR-IR20250213001423-0001

RAT

NR TDD

5.3.3.37.3 Optimized Memory Resource Management for 8R UEs

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

8R UEs and non-8R UEs can now share a memory resource pool. This raises the upper limit on memory resources allocated to 8R UEs and allows more 8R UEs to access a cell.

LampSite base stations do not support this feature.

The change in this feature is supported only in low frequency bands.

Implementation

One 8R UE now occupies two portions of memory resources, with each portion corresponding to four ports. This allows 8R UEs and non-8R UEs to share a memory resource pool.

Impact on the Network

When the average number of UEs per cell exceeds 455 on a base station, the downlink user-perceived rate may slightly change.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The change in this feature has no impact.
- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20241209002193-0038

RAT

NR TDD

5.3.3.37.4 Modified the Impact Relationship Between "Downlink 8-Layer Transmission for a Single FWA User" and "Interference Rejection Precoding for Downlink MU-MIMO in NLOS Scenarios"

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

When the **NRDUCellPdschPrecode.DIMuIRPrecodePol** parameter is set to **MU_ALLUSER_POWER_ENH**, "downlink 8-layer transmission for a single FWA user" no longer has an impact relationship with "interference rejection precoding for downlink MU-MIMO in NLOS scenarios". That is, these two functions can simultaneously take effect for a UE.

LampSite base stations do not support this feature.

The change in this feature is supported only in low frequency bands.

Implementation

For details, see *FWA*. The specific sections are listed in the change history of the document.

Impact on the Network

If "downlink 8-layer transmission for a single FWA user" and "interference rejection precoding for downlink MU-MIMO in NLOS scenarios" are both enabled and **NRDUCellPdschPrecode.DIMuIRPrecodePol** is set to **MU_ALLUSER_POWER_ENH**, these two functions can simultaneously take effect for a UE after an upgrade. This will increase the **Cell Downlink Average Throughput (DU)** and **User Downlink Average Throughput (DU)**.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The change in this feature has no impact.
- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The descriptions of the following impact relationships have been revised.

Function Name	Function Switch	Reference	Description
Interference rejection precoding for downlink MU-MIMO in NLOS scenarios	DL_MU_NLOS_IR_PRECODING_SWITCH option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>Massive MIMO AHR (Low-Frequency TDD)</i> <i>Coordinated Interference Management (Low-Frequency TDD)</i>	- When the NRDUCellPdschPrecode.DLMuIRPrecodePol parameter is set to MU_ALLUSER_LOWLATENCY or MU_ALLUSER and downlink 8-layer transmission for a single FWA user has taken effect, "interference rejection precoding for downlink MU-MIMO in NLOS scenarios" does not take effect for the user. - When the NRDUCellPdschPrecode.DLMuIRPrecodePol parameter is set to MU_ALLUSER_POWER_ENH and downlink 8-layer transmission for a single FWA user has taken effect, "interference rejection precoding for downlink MU-MIMO in NLOS scenarios" can take effect for the user.
DL MU Intrf Reject Precoding Policy	NRDUCellPdschPrecode.DLMuIRPrecodePol set to MU_ALLUSER_LOWLATENCY or MU_ALLUSER	<i>Massive MIMO AHR (Low-Frequency TDD)</i>	

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0021

RAT

NR TDD

5.3.3.38 FOFD-091201 Uplink Boosting

5.3.3.38.1 Added Uplink MU Grouping and Pairing Avoidance for RedCap UEs

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Uplink MU grouping and pairing avoidance for RedCap UEs is now supported. When this function takes effect, non-RedCap UEs with higher priorities avoid RedCap UEs. This increases the average uplink throughput of RedCap UEs.

This function is controlled by the **REDCAP_UL_MU_GRP_PAIR_AVOID_SW** option of the **NRDUCellRedCap.RedCapSchAlgoSwitch** parameter and takes effect only when the **REDCAP_UL_MU_GRP_PAIR_OPT_SW** option of the **NRDUCellRedCap.RedCapSchAlgoSwitch** parameter is selected.

LampSite base stations do not support the change in this feature.

Implementation

For details, see *Massive MIMO Uplink Boosting (Low-Frequency TDD)*. The specific sections are listed in the change history of the document.

Impact on the Network

In medium- and heavy-load scenarios, the average uplink throughput of RedCap UEs increases, and the average uplink cell throughput changes.

Impact on NEs

The change in this feature is implemented on the gNodeB.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellRedCap.RedCapSchAlgoSwitch	MOD NRDUCELLREDCAP LST NRDUCELLREDCAP	gNodeB	The REDCAP_UL_MU_GRP_PAIR_AVOID_SW option has been added to specify whether to enable uplink MU grouping and pairing avoidance for RedCap UEs.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.

Function Name	Function Switch	Reference	Description
Uplink MU grouping and pairing optimization for RedCap UEs	REDCAP_UL_MU_GRP_PAIR_OPT_SW option of the NRDUCellRedCap.RedCapSchAlgoSwitch parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	Uplink MU grouping and pairing avoidance for RedCap UEs can take effect only when uplink MU grouping and pairing optimization for RedCap UEs is enabled.

- Mutually exclusive features/functions
The change in this feature has no impact.

- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Massive MIMO Uplink Boosting (Low-Frequency TDD)*.

Requirement/Trouble Ticket Number

IR20241209002193

DTS2025041524846

RAT

NR TDD

5.3.3.38.2 Added Data-Volume-based Intelligent Power Control Optimization

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In certain cases with data-volume-based intelligent power control (controlled by the **DATA_VOL_INTEL_PWR_CTRL_SW** option of the **NRDUCellUIPcConfig.UlPwrCtrlAlgoExtSwitch** parameter) enabled, a large reduction in UE power can degrade user experience. Data-volume-based intelligent power control optimization has therefore been introduced to address this issue. This function enables the gNodeB to adaptively optimize the power adjustment amount based on UEs' channel quality to increase the average uplink UE throughput.

This function is controlled by the **DATA_VOL_INTEL_PWR_CTRL_OPT_SW** option of the **NRDUCellUIPcConfig.UlPwrCtrlAlgoExtSwitch** parameter and takes effect only when the **DATA_VOL_INTEL_PWR_CTRL_SW** option of the **NRDUCellUIPcConfig.UlPwrCtrlAlgoExtSwitch** parameter is selected.

Implementation

For details, see *Massive MIMO Uplink Boosting (Low-Frequency TDD)*. The specific sections are listed in the change history of the document.

Impact on the Network

Uplink interference to neighboring cells decreases, and the average uplink UE throughput increases.

Impact on NEs

The change in this feature is implemented on the gNodeB.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellUIPConfig.UlPwrCtrlAlgoExtSwitch	MOD NRDUCELLU LPCCONFIG LST NRDUCELLU LPCCONFIG	gNodeB	The DATA_VOL_INTEL_PWR_CTL_OPT_SW option has been added to specify whether to enable data-volume-based intelligent power control optimization.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Massive MIMO Uplink Boosting (Low-Frequency TDD)*.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0046

RAT

NR TDD

5.3.3.39 FOFD-091203 Layered QoS

5.3.3.39.1 Support for Layered QoS by the Scenario Where the Core Network Transmits Only the Downlink PSIHI Set to False

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Layered QoS is supported a scenario where the SMF indicates that the *PDU Set QoS Parameters* IE contains only the downlink PSIHI (value: **False**). In such a scenario, if PDU Set identification is enabled by setting **gNBQciBearer.IdentificationMode** to **R18_PDU_SET_MODE**, the **PSIHI_FALSE_ONLY_PROC_SW** option of the **gNBQciBearer.DLLayeredQosSwitch** parameter also needs to be selected.

The change in this feature is supported only in low-frequency bands.

The change in this feature is supported only in SA networking.

Implementation

For details, see *Layered QoS*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameter has changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNBQciBearer.DLayeredQoSSwitch	ADD GNBQCIBEARER MOD GNBQCIBEARER LST GNBQCIBEARER	gNodeB	The PSIHI_FALSE_ONLY_PROC_SW option has been added to specify whether to allow algorithms related to Layered QoS to take effect for bearers that carry only the PSIHI (value: False).

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Layered QoS*.

Requirement/Trouble Ticket Number

FeR-IR20240626000421-0004

RAT

NR FDD, NR TDD

5.3.3.40 FOFD-091206 Premium RedCap

5.3.3.40.1 Added Uplink-Experience-based Inter-frequency Handover Collaboration

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The "uplink-experience-based inter-frequency handover collaboration" function is now supported. After this function is enabled, RedCap/eRedCap UEs performing large-packet services each can continuously camp on the target inter-frequency cell after an inter-frequency handover.

This function is controlled by the **UL_EXP_INTER_FREQ_HO_COLLAB_SW** option of the **NRCellRedCap.RedCapAlgoSwitch** parameter.

The change in this feature is supported only in low-frequency SA networking.

Implementation

For details, see *RedCap*. The specific sections are listed in the change history of the document.

Impact on the Network

After this function takes effect, RedCap/eRedCap UEs performing uplink large-packet services at the cell edge or a medium distance from the cell center each can continuously camp on the target frequency of the corresponding RAT after being handed over to an inter-frequency cell that provides better user experience, improving uplink user experience of these UEs.

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management

The change in this feature has no impact. It is controlled by the existing feature license.

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRCellRedCap.RedCapAlgoSwitch	MOD NRCELLRED CAP LST NRCELLRED CAP	gNodeB	The UL_EXP_INTER_FREQ_HO_COLLAB_SW option has been added to specify whether to enable uplink-experience-based inter-frequency handover collaboration.

- Performance management

The change in this feature has no impact.

- Fault management

The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions

The change in this feature has no impact.

- Mutually exclusive features/functions

The change in this feature has no impact.

- Impacted features/functions

The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *RedCap*.

Requirement/Trouble Ticket Number

FeR-IR20250102000700-0136

RAT

NR FDD, NR TDD

5.3.3.41 MRFD-171262 NSA/SA Selection Based on User Experience (NR)

5.3.3.41.1 Added Cell-level Control on SA-to-NSA Switching Based on Weak Uplink Coverage

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The **NSA_SA_UL_WEAK_FRBD_SW** option has been added to the **NRInterRatHoParam.MobilityHOCControlSwitch** parameter to specify whether to disable SA-to-NSA switching based on weak uplink coverage for a specific cell. If this option is selected, SA-to-NSA switching based on weak uplink coverage does not take effect for the cell.

Implementation

For details, see *NSA-SA Selection Based on User Experience*. The specific sections are listed in the change history of the document.

Impact on the Network

After the function of disabling SA-to-NSA switching based on weak uplink coverage takes effect, the values of the following counters decrease:

- N.HO.InterRAT.N2E.ResourceOpt.PrepAttOut
- N.HO.InterRAT.N2E.ResourceOpt.ExecAttOut
- N.HO.InterRAT.N2E.ResourceOpt.ExecSuccOut
- N.HO.InterRAT.N2E.ResourceOpt.SA2NSA.PrepAttOut.UL
- N.HO.InterRAT.N2E.ResourceOpt.SA2NSA.ExecAttOut.UL
- N.HO.InterRAT.N2E.ResourceOpt.SA2NSA.ExecSuccOut.UL
- L.IRATHO.N2E.Direct2EnDc.PrepAttIn
- L.IRATHO.N2E.Direct2EnDc.ExecSuccIn
- L.IRATHO.N2E.Direct2EnDc.PrepInSucc
- L.IRATHO.N2E.PrepAttIn
- L.IRATHO.N2E.PrepInSucc
- L.IRATHO.N2E.PrepInSucc

Impact on NEs

The change in this feature is implemented on the gNodeB.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRInterRatHoParam.MobilityHOControlSwitch	MOD NRINTERRAT HOPARAM LST NRINTERRAT HOPARAM	gNodeB	The NSA_SA_UL_WEAK_FRBD_SW option has been added to specify whether to disable SA-to-NSA switching based on weak uplink coverage.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *NSA-SA Selection Based on User Experience*.

Requirement/Trouble Ticket Number

FeR-IR20250221001146-0010

RAT

NR TDD

5.3.3.41.2 Added Cell-level Control on the "NSA/SA Selection Based on Downlink Traffic Volume" Function

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Once NSA/SA selection based on downlink traffic volume is enabled, it applies to all NR cells under a gNodeB. To disable this function for a specific NR cell, the **NSA_SA_DL_OPT_FRBD_SW** option of the **NRInterRatHoParam.MobilityHOControlSwitch** parameter can be selected for the cell accordingly. Other NR cells remain unaffected.

Implementation

For details, see *NSA-SA Selection Based on User Experience*. The specific sections are listed in the change history of the document.

Impact on the Network

After this function takes effect, the values of the following counters decrease:

- N.HO.InterRAT.N2E.ResourceOpt.PrepAttOut
- N.HO.InterRAT.N2E.ResourceOpt.ExecAttOut
- N.HO.InterRAT.N2E.ResourceOpt.ExecSuccOut
- N.HO.InterRAT.N2E.ResourceOpt.SA2NSA.PrepAttOut.DL
- N.HO.InterRAT.N2E.ResourceOpt.SA2NSA.ExecAttOut.DL
- N.HO.InterRAT.N2E.ResourceOpt.SA2NSA.ExecSuccOut.DL
- L.IRATHO.N2E.Direct2EnDc.PrepAttIn
- L.IRATHO.N2E.Direct2EnDc.ExecSuccIn
- L.IRATHO.N2E.Direct2EnDc.PrepInSucc
- L.IRATHO.N2E.PrepAttIn
- L.IRATHO.N2E.PrepInSucc
- L.IRATHO.N2E.PrepInSucc

Impact on NEs

The change in this feature is implemented on the gNodeB.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRInterRatHoParam.MobilityHOControlSwitch	MOD NRINTERRAT HOPARAM LST NRINTERRAT HOPARAM	gNodeB	The NSA_SA_DL_OPT_FRBD_S W option has been added to specify whether to disable NSA/SA selection based on downlink traffic volume for a specific cell.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *NSA-SA Selection Based on User Experience*.

Requirement/Trouble Ticket Number

FeR-IR20250530001992-0020

RAT

NR FDD, NR TDD

5.3.3.41.3 Added Counters to Monitor the Number of Times UEs with High Downlink Traffic Volume Are Identified in a Cell and Experience-based NSA-SA Selection

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The following counters have been added:

- **N.DLHighTraffic.UE.Count**: used to observe the number of times UEs with high traffic volume are identified in a cell.
- **N.HO.InterRAT.N2E.ResourceOpt.SA2NSA.PrepAttOut.DL**, **N.HO.InterRAT.N2E.ResourceOpt.SA2NSA.ExecAttOut.DL**, and **N.HO.InterRAT.N2E.ResourceOpt.SA2NSA.ExecSuccOut.DL**: used to observe whether NSA/SA selection based on downlink traffic volume has taken effect.
- **N.HO.InterRAT.N2E.ResourceOpt.SA2NSA.PrepAttOut.UL**, **N.HO.InterRAT.N2E.ResourceOpt.SA2NSA.ExecAttOut.UL**, and **N.HO.InterRAT.N2E.ResourceOpt.SA2NSA.ExecSuccOut.UL**: used to observe whether SA-to-NSA switching based on weak uplink coverage has taken effect.

Implementation

For details, see *NSA-SA Selection Based on User Experience*. The specific sections are listed in the change history of the document.

Impact on the Network

The change in this feature has no impact.

Impact on NEs

The change in this feature is implemented on the gNodeB.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The change in this feature has no impact.

- Performance management

The following counters have changed.

Change Type	Counter Name	Function Subset	NE	Description
New	N.DLHighTraffic.UE.Count	Cell Algorithm Measurement	gNode B	This counter has been added to measure the number of times UEs whose traffic volume changes from low to high are identified in a cell.
New	N.HO.InterRAT.N2E.ResourceOpt.SA2NSA.PrepareAttOut.DL	Outgoing Inter-RAT Handover Measurement	gNode B	This counter has been added to measure the number of downlink-based SA-to-NSA outgoing handover preparation attempts.
New	N.HO.InterRAT.N2E.ResourceOpt.SA2NSA.ExecAttOut.DL	Outgoing Inter-RAT Handover Measurement	gNode B	This counter has been added to measure the number of downlink-based SA-to-NSA outgoing handover execution attempts.
New	N.HO.InterRAT.N2E.ResourceOpt.SA2NSA.ExecSuccOut.DL	Outgoing Inter-RAT Handover Measurement	gNode B	This counter has been added to measure the number of successful downlink-based SA-to-NSA outgoing handover executions.
New	N.HO.InterRAT.N2E.ResourceOpt.SA2NSA.PrepareAttOut.UL	Outgoing Inter-RAT Handover Measurement	gNode B	This counter has been added to measure the number of uplink-based SA-to-NSA outgoing handover preparation attempts.
New	N.HO.InterRAT.N2E.ResourceOpt.SA2NSA.ExecAttOut.UL	Outgoing Inter-RAT Handover Measurement	gNode B	This counter has been added to measure the number of uplink-based SA-to-NSA outgoing handover execution attempts.
New	N.HO.InterRAT.N2E.ResourceOpt.SA2NSA.ExecSuccOut.UL	Outgoing Inter-RAT Handover Measurement	gNode B	This counter has been added to measure the number of successful uplink-based SA-to-NSA outgoing handover executions.

- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250530001992-0040

FeR-IR20250530001992-0010

RAT

NR FDD, NR TDD

5.3.3.41.4 Added NSA-to-SA Switching Based on Downlink User Experience

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

NSA-to-SA switching based on downlink user experience is triggered for UEs whose downlink experience in NR cells is considered better than that in LTE cells. This function provides a networking scheme with better service experience for UEs and improves the camping probability of UEs in SA networking. Currently, NSA-to-SA switching is supported in the following scenarios:

- A UE accesses an LTE sub-1 GHz cell and is at the cell edge. The function is controlled by the **DL_NSA_TO_SA_SWITCH_SW** option of the **ENodeBAlgoExtSwitch.NsaDcAlgoExtSwitch** parameter in this scenario.
- A UE performing large-packet downlink services is handed over to the eNodeB due to SA-to-NSA switching based on downlink traffic volume and then transitions from the large-packet state to the small-packet state. The function is controlled by the following options in this scenario:
 - **DL_NSA_TO_SA_SWITCH_SW** option of the **ENodeBAlgoExtSwitch.NsaDcAlgoExtSwitch** parameter

- **LARGE_PKT_UE_IDENT_SW** option of the **ENodeBAlgoExtSwitch**.*NetworkPrfmOptExtSwitch* parameter
- **DL_NSA_TO_SA_SWITCH_SW** option of the **gNodeBParam**.*NetworkingOptionOptSw* parameter
- **NR_TO_LTE_BIG_PKT_STA_TRANS_SW** option of the **gNodeBParam**.*NetworkingOptionOptSw* parameter

Implementation

For details, see *NSA-SA Selection Based on User Experience*. The specific sections are listed in the change history of the document.

Impact on the Network

Benefits: After NSA-to-SA switching based on downlink user experience takes effect, the average downlink UE throughput (indicated by **User Downlink Average Throughput (DU)**) increases.

Impacts:

NR side:

- The downlink RB usage in the NR cell, indicated by **Downlink Resource Block Utilizing Rate (DU)**, increases.
- The downlink traffic volume in the NR cell, indicated by **Downlink Traffic Volume (DU)**, increases.
- During NSA-to-SA switching, the average downlink UE throughput, indicated by **User Downlink Average Throughput (DU)**, temporarily decreases.
- After a UE switches from NSA to SA, its transmit power may increase, which can affect the average uplink UE throughput, indicated by **User Uplink Average Throughput (DU)**, when the UE is located at the cell edge.
- The proportion of UEs camping in SA networking increases.
- The average uplink or downlink UE throughput decreases when NR cell channel quality deteriorates or the number of available RBs decreases, for example, when a large number of UEs are handed over to the NR cell simultaneously.

LTE side:

- The downlink RB usage in the LTE cell, indicated by **Downlink Resource Block Utilizing Rate**, decreases.
- The downlink traffic volume in the LTE cell, indicated by **Downlink Traffic Volume**, decreases.
- The downlink RB usage of NSA DC UEs, indicated by **Downlink Resource Block Utilizing Rate (NSA DC)**, decreases.

Impact on NEs

The change in this feature is implemented on the gNodeB.

Impact on Hardware Requirements

Main control boards: Only UMPTe, UMPTg/ga, and UMPTha series boards support this function.

Baseband processing units: Only UBBPe, UBBPf/fw, UBBPg, and UBBPh series boards support this function.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNodeBParam.NetworkOptionOptSw	MOD GNODEBPARAM LST GNODEBPARAM	gNodeB	- The DL_NSA_TO_SA_SWITCH_SW option has been added to specify whether to enable downlink NSA-to-SA switching on the NR side for NSA/SA Selection Based on User Experience. - The NR_TO_LTE_BIG_PKT_STA_TRANS_SW option has been added to specify whether to enable NR-to-LTE large-packet status transfer.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference
NR FDD Low-frequency NR TDD LTE FDD LTE TDD	NSA/SA selection based on downlink traffic volume	• NSA_SA_DL_SEL_OPT_SW option of the gNodeBParam.NetworkingOptionOptSw parameter • LTE_FDD_NSA_SA_DL_SEL_OPT_SW or LTE_TDD_NSA_SA_DL_SEL_OPT_SW option of the EnodebAlgoExtSwitch.MultiNetworkingOptionOptSw parameter	<i>NSA-SA Selection Based on User Experience</i>

- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *NSA-SA Selection Based on User Experience*.

Requirement/Trouble Ticket Number

FeR-IR20250102000700-0089

RAT

NR FDD, NR TDD

5.3.3.41.5 ANR for Missing Neighboring LTE Cells

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

To support handovers from SA to the UE-supported but missed neighboring LTE cell with the highest anchor priority (specified by **NRCellEutranNFreq.NsaPccFreqPriority**) and highest RSRP after the gNodeB has received all LTE B1 measurement reports, the **NSA_SA_SEL_MISS_CELL_ANR_SW** option of the **NRCellAnr.AnrAlgoSwitch** parameter can be selected to enable delivery of CGI measurement configurations. The gNodeB then determines whether to add missed neighbor relationships based on the CGI measurement results according to the configured ANR mode. Specifically:

- In ANR controlled mode (with the **NR_EUTRAN_ANR_CTRL_MODE_SW** option of the **NRCellAnr.AnrModeSwitch** parameter selected):
 - If handovers are allowed (with the **NR_EUTRAN_ANR_CTRL_MODE_HO_SW** option of the **NRCellAnr.AnrModeSwitch** parameter selected): CGI measurement configuration can be delivered. The neighboring LTE cell is added as a to-be-confirmed neighboring cell based on the CGI measurement result, and SA-to-NSA switching can be triggered normally after the experience-based evaluation result is obtained.
 - If handovers are not allowed (with the **NR_EUTRAN_ANR_CTRL_MODE_HO_SW** option of the **NRCellAnr.AnrModeSwitch** parameter deselected): Handovers to the to-be-confirmed neighboring cell is not allowed, but CGI measurement configurations are still delivered once. After this neighboring cell is confirmed as a formal one on the OSS, handovers to this neighboring cell can be performed.
- In ANR free mode (with the **NR_EUTRAN_ANR_CTRL_MODE_SW** option of the **NRCellAnr.AnrModeSwitch** parameter deselected): CGI measurement configurations can be delivered. After receiving a CGI measurement report, the gNodeB adds the neighboring cell as a formal one based on the CGI measurement result. SA-to-NSA switching can be triggered normally after the experience-based evaluation result is obtained.

Implementation

For details, see *NSA-SA Selection Based on User Experience*. The specific sections are listed in the change history of the document.

Impact on the Network

After ANR for missing neighboring LTE cells takes effect, the following counters related to the number of SA-to-NSA switching times increase.

- **N.HO.InterRAT.N2E.ResourceOpt.PrepAttOut**
- **N.HO.InterRAT.N2E.ResourceOpt.ExecAttOut**
- **N.HO.InterRAT.N2E.ResourceOpt.ExecSuccOut**

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRCCellAnr <i>.AnrAlgoSwitch</i>	MOD NRCELLANR LST NRCELLANR	gNodeB	The NSA_SA_SEL_MISS_CELL_ANR_SW option has been added to specify whether to enable ANR for missing neighboring LTE cells.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.

Function Name	Function Switch	Reference	Description
Inter-RAT ANR	NR_EUTRAN_ANR_SW option of the NRCCellAlgoSwitch.AnrSwitch parameter	<i>ANR in 5G RAN Feature Documentation</i>	The NSA_SA_SEL_MISS_CELL_ANR_SW option can be selected only when the NR_EUTRAN_ANR_SW option is selected.

- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions

The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *NSA-SA Selection Based on User Experience*.

Requirement/Trouble Ticket Number

FeR-IR20250708001473-0290

RAT

NR FDD, NR TDD

5.3.3.41.6 Support for NSA/SA Downlink Selection Handover Optimization

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

NSA/SA downlink selection handover optimization enables the NSA/SA downlink selection function to still take effect when there are no available SA bandwidth resources for UEs due to a small value of the **NRCellMobilityConfig.NrCellDLExperienceCoeff** parameter or other reasons. NSA/SA downlink selection handover optimization is controlled by the **NSA_SA_DL_SEL_HO_OPT_SW** option of the **NRCellMobilityConfig.InterRatHoExtSwitch** parameter.

NOTE

This change was first incorporated into V100R022C00SPC010 but is first described in the network impact report of V100R022C00SPC050.

Implementation

For details, see *NSA-SA Selection Based on User Experience*. The specific sections are listed in the change history of the document.

Impact on the Network

After NSA/SA downlink selection handover optimization takes effect, the number of SA-to-NSA switching times increases.

- **N.HO.InterRAT.N2E.ResourceOpt.PrepAttOut**
- **N.HO.InterRAT.N2E.ResourceOpt.ExecAttOut**
- **N.HO.InterRAT.N2E.ResourceOpt.ExecSuccOut**

Impact on NEs

The change in this feature has no impact.

Impact on Hardware Requirements

The change in this feature has no impact.

Impact on Inter-NE Interfaces

The change in this feature has no impact.

Impact on Operation and Maintenance

- License management
The change in this feature has no impact. It is controlled by the existing feature license.
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRCeIlMo bIlItyConfi g.InterRat HoExtSwit ch	MOD NRCELLMOB ILITYCONFIG LST NRCELLMOB ILITYCONFIG	gNodeB	The NSA_SA_DL_SEL_HO_OPT _SW option has been added to specify whether to enable NSA/SA downlink selection handover optimization.

- Performance management
The change in this feature has no impact.
- Fault management
The change in this feature has no impact.

Impact on Features/Functions

- Prerequisite features/functions
The change in this feature has no impact.
- Mutually exclusive features/functions
The change in this feature has no impact.
- Impacted features/functions
The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *NSA-SA Selection Based on User Experience*.

Requirement/Trouble Ticket Number

IR20250528001008

RAT

NR TDD

5.3.4 Function Changes

5.3.4.1 Radio

5.3.4.1.1 Improved Efficiency of Processing Control-Plane Signaling (V100R022C00)#

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

To address the low efficiency in processing control-plane signaling and in calculating and allocating radio resources when gNodeBs face heavy traffic, the following two functions are introduced for better performance:

- Control-plane flow control optimization: When the CPU usage of the main control board is high, flow control is imposed on low-priority RRC reconfiguration procedures. In this way, the system can handle more high-priority services and procedures such as UE access and handover.
- Control-plane signaling processing capability improvement: The gNodeB capability of processing signaling has been optimized, so that control-plane signaling is processed more efficiently in procedures such as UE access and handover.

Implementation

Control-plane flow control optimization: Flow control is imposed on low-priority RRC reconfiguration procedures to improve the capability of processing signaling for procedures such as UE access and handover.

Control-plane signaling processing capability improvement: A lightweight communication mechanism is used and internal message processing is optimized to improve the signaling processing efficiency.

Impact on the Network

Control-plane flow control optimization: When the CPU usage is high, the uplink and downlink traffic volumes and the average uplink and downlink cell or UE throughputs may decrease. Specifically, the values of the following KPIs may decrease:

- **Downlink Traffic Volume (DU)**
- **Uplink Traffic Volume (DU)**
- **Cell Downlink Average Throughput (DU)**
- **Cell Uplink Average Throughput (DU)**
- **User Downlink Average Throughput (DU)**
- **User Uplink Average Throughput (DU)**

Control-plane signaling processing capability improvement:

- Given the same traffic model, equipment parameter settings, and number of UEs, the CPU usage of the main control board and BBPs may be lowered. Specifically, the values of the following counters may decrease:
 - **VS.BBUBoard.CPULoad.Max**
 - **VS.BBUBoard.CPULoad.Mean**
 - **VS.BBUBoard.CPULoad.Over**
 - **VS.BBUBoard.CPULoad.CumulativeHighloadCount**
- The UE handover delay may be shortened, with the values of the following counters decreasing:
 - **N.HO.Xn.IntergNB.Duration.Avg**
 - **N.HO.Ng.IntergNB.Duration.Avg**
 - **N.HO.Xn.IntergNB.Duration.Avg.PLMN**
 - **N.HO.Ng.IntergNB.Duration.Avg.PLMN**

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20240806000570-0001

RAT

NR FDD, NR TDD

5.3.4.1.2 BBP Performance Optimization on the User Plane in V100R022C00

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

This function has optimized software to improve the processing capability of the CPU serving the user plane of a gNodeB BBP and thereby improve the BBP performance on the user plane.

Implementation

This function has optimized software to improve the processing capability of the CPU serving the user plane of a gNodeB BBP and thereby improve the BBP performance on the user plane.

Impact on the Network

Given the same traffic model, equipment parameter settings, and number of UEs, the user-plane CPU usage of a BBP may be lower than that in earlier versions. This change may also have the following impacts on the network:

- The values of the following counters decrease:
 - **VS.NRBoard.UPlane.CPULoad.Avg**
 - **VS.NRBoard.UPlane.CPULoad.Max**
 - **VS.NRBoard.UPlane.CumulativeHighloadCount**
- The values of the following counters change:

- N.Traffic.DL.RlcFirstPktDelay.Time
- N.QoS.DL.PktDelaygNBDU.Time
- N.QoS.DL.PktDelayAirInterface.Time
- N.ThpVol.UE.UL.SmallPkt
- N.ThpVol.DL.LastSlot
- N.MAC.UL.Cell5QI.Delay and N.MAC.DL.Cell5QI.Delay
- N.RLC.UL.Cell5QI.Delay and N.RLC.DL.Cell5QI.Delay
- N.Thp.DL.Samp.Index0 to N.Thp.DL.Samp.Index14
- N.ChMeas.PDSCH.MCS.0 to N.ChMeas.PDSCH.MCS.31

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20240806000581-0002

RAT

NR FDD, NR TDD

5.3.4.1.3 Improved Efficiency of Processing Control-Plane Signaling (V100R022C10)#

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

To address the low efficiency in processing control-plane signaling and in calculating and allocating radio resources when gNodeBs face heavy traffic, the following two functions are introduced for better performance:

- Control-plane flow control optimization: When the CPU usage of the main control board is high, flow control is imposed on low-priority RRC reconfiguration procedures. In this way, the system can handle more high-priority services and procedures such as UE access and handover.
- Control-plane signaling processing capability improvement: The gNodeB capability of processing signaling has been optimized, so that control-plane signaling is processed more efficiently in procedures such as UE access and handover.

Implementation

Control-plane flow control optimization: Flow control is imposed on low-priority RRC reconfiguration procedures to improve the capability of processing signaling for procedures such as UE access and handover.

Control-plane signaling processing capability improvement: A lightweight communication mechanism is used and internal message processing is optimized to improve the signaling processing efficiency.

Impact on the Network

Control-plane flow control optimization: When the CPU usage is high, the uplink and downlink traffic volumes and the average uplink and downlink cell or UE throughputs may decrease. Specifically, the values of the following KPIs may decrease:

- Downlink Traffic Volume (DU)
- Uplink Traffic Volume (DU)
- Cell Downlink Average Throughput (DU)
- Cell Uplink Average Throughput (DU)
- User Downlink Average Throughput (DU)

- **User Uplink Average Throughput (DU)**

Control-plane signaling processing capability improvement:

- Given the same traffic model, equipment parameter settings, and number of UEs, the CPU usage of the main control board and BBPs may be lowered. Specifically, the values of the following counters may decrease:
 - **VS.BBUBoard.CPULoad.Max**
 - **VS.BBUBoard.CPULoad.Mean**
 - **VS.BBUBoard.CPULoad.Over**
 - **VS.BBUBoard.CPULoad.CumulativeHighloadCount**
- The UE handover delay may be shortened, with the values of the following counters decreasing:
 - **N.HO.Xn.IntergNB.Duration.Avg**
 - **N.HO.Ng.IntergNB.Duration.Avg**
 - **N.HO.Xn.IntergNB.Duration.Avg.PLMN**
 - **N.HO.Ng.IntergNB.Duration.Avg.PLMN**

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions

No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250208000111-0005

RAT

NR FDD, NR TDD

5.3.4.1.4 BBP Performance Optimization on the User Plane in V100R022C10

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

This function has optimized software to improve the processing capability of the CPU serving the user plane of a gNodeB BBP and thereby improve the BBP performance on the user plane.

Implementation

This function has optimized software to improve the processing capability of the CPU serving the user plane of a gNodeB BBP and thereby improve the BBP performance on the user plane.

Impact on the Network

Given the same traffic model, equipment parameter settings, and number of UEs, the user-plane CPU usage of a BBP may be lower than that in earlier versions. This change may also have the following impacts on the network:

- The values of the following counters decrease:
 - **VS.NRBoard.UPlane.CPULoad.Avg**
 - **VS.NRBoard.UPlane.CPULoad.Max**
 - **VS.NRBoard.UPlane.CumulativeHighloadCount**
- The values of the following counters change:
 - **N.ThpVol.UL.Cell** and **N.ThpVol.DL.Cell**
 - **N.ThpVol.UL** and **N.ThpVol.DL**
 - **N.Traffic.DL.RlcFirstPktDelay.Time**
 - **N.QoS.DL.PktDelaygNBDU.Time**
 - **N.QoS.DL.PktDelayAirInterface.Time**
 - **N.ThpVol.UE.UL.SmallPkt**

- **N.MAC.UL.Cell5QI.Delay** and **N.MAC.DL.Cell5QI.Delay**
- **N.Thp.DL.Samp.Index0** to **N.Thp.DL.Samp.Index14**
- **N.ChMeas.PDSCH.MCS.0** to **N.ChMeas.PDSCH.MCS.31**
- **N.RA.Contention.Msg3**
- **N.RA.Contention.Resolution.Succ**
- **DL IBLER**
- **DL RBLER**
- **Average DownLink MCS**
- **Average DownLink RANK**
- **Average Downlink Packet Delay in MAC**

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250208000111-0004

FeR-IR20250409002396-0011

RAT

NR FDD, NR TDD

5.3.4.1.5 Changes in Some Parameters in V100R022C10SPC010 and Related Impacts#

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

This change describes only some parameter changes in V100R022C10SPC010 and the impacts of these changes.

Implementation

Parameter ID	Parameter Change Type	Solution After Parameter Change	Impact of Parameter Change
NRDUCellRsvdExt03.RsvdSwParam0	Reserved-parameter bit addition to the disuse list	RSVDSWPARAM0_BIT9 of this reserved parameter has been added to the disuse list, and the solution with this bit set to 0 takes effect by default. That is, downlink MU-MIMO pairing enhancement is enabled by default.	If this bit is set to 1 before an upgrade, the values of the following indicators may change after the upgrade, depending on the traffic model: • N.User.Schedule.DL.Sum • Downlink Average MuPairing Layer • Average DownLink RANK • Average DownLink MCS • Downlink MuPairing RB Ratio • Cell Downlink Average Throughput (DU) • User Downlink Average Throughput (DU) • Downlink Resource Block Utilizing Rate (DU) • DL IBLER • DL DCI Allocate Succ Rate • Average DL CCE AGG Level In other scenarios, the upgrade has no impact on network performance.

Parameter ID	Parameter Change Type	Solution After Parameter Change	Impact of Parameter Change
NRDUCellRsvdExt03.RsvdSwParam0	Reserved-parameter bit addition to the disuse list	RSVDSWPARAM0_BIT10 of this reserved parameter has been added to the disuse list, and the solution with this bit set to 0 takes effect by default. That is, resynchronization optimization is enabled by default.	If this bit is set to 1 before an upgrade, the impacts after the upgrade are as follows: - The values of the following indicators may increase: N.User.Schedule.DL.Sum VS.NRBoard.UPlane.CPULoad.Max VS.NRBoard.UPlane.CPULoad.Avg QoS Flow Setup Success Rate (CU) - The values of the following indicators may change, depending on the traffic model: DL IBLER DL RBLER Cell Downlink Average Throughput (DU) Cell Uplink Average Throughput (DU) User Downlink Average Throughput (DU) User Uplink Average Throughput (DU) Downlink Resource Block Utilizing Rate (DU) Average DownLink RANK Average DownLink MCS DL DCI Allocate Succ Rate PDCCH CCE Utilizing Rate

Parameter ID	Parameter Change Type	Solution After Parameter Change	Impact of Parameter Change
			– Average DL CCE AGG Level In other scenarios, the upgrade has no impact on network performance.

NOTE

For details about reserved parameters, see 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List.

Impact on the Network

For details, see the **Impact of Parameter Change** column in **Implementation**.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact

- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250528001856-0001

RAT

NR FDD, NR TDD

5.3.4.1.6 Added the Function of Requesting LTE-side Large-Packet Status of ULFB UEs

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

If the gNodeB is required to obtain the packet status (large or small) of UEs that have undergone uplink fallback to LTE (ULFB UEs), the function of requesting LTE-side large-packet status of ULFB UEs needs to be enabled. Once this function is enabled, the gNodeB reports the function switch status to the eNodeB. After learning that the switch is on, and both the **UL_LARGE_PKT_SERV_DECT_SW** option of the **EnodebAlgoExtSwitch.MultiNetworkingOptionOptSw** parameter and the **ULFB_UL_LRG_PKT_FLAG_TO_NR_SW** option of the **EnodebAlgoExtSwitch.NsaDcAlgoSwitch** parameter are selected, the eNodeB provides the uplink large- and small-packet status of ULFB UEs to the gNodeB. The function of requesting LTE-side large-packet status of ULFB UEs is controlled by the **ULFB_LTE_UL_LRG_PKT_REQ_SW** option of the **gNodeBParam.NsaDcOptSwitch** parameter.

This function is supported only in NSA networking.

The BTS5929R does not support this change.

Implementation

For details, see *EPC-based NSA Basics*. The specific sections are listed in the change history of the document.

Impact on the Network

Enabling the function of requesting LTE-side large-packet status of ULFB UEs increases the X2 signaling overhead.

Impact on NEs

This function is implemented on the gNodeB and eNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNodeBParameter.NsaDcOptSwitch	MOD GNODEBPA RAM LST GNODEBPA RAM	gNodeB	The ULFB_LTE_UL_LRG_PKT_REQ_SW option has been added to specify whether the gNodeB requests the eNodeB to report the large-packet status of UEs that have undergone uplink fallback to LTE.

- Performance management

No impact

- Fault management

No impact

Impact on Features/Functions

- Prerequisite features/functions

No impact

- Mutually exclusive features/functions

No impact

- Impacted features/functions

No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *EPC-based NSA Basics*.

Requirement/Trouble Ticket Number

FeR-IR20250221001146-0010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD

5.3.4.1.7 Added Support for NR Local-Cell-Load-based RF Module Dormancy

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The following functions have been added:

- Optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module deep dormancy
- Optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module super dormancy

In addition to the entry conditions of scheduled RF module dormancy, the local cell load is now also considered as an energy saving entry condition. The details are as follows:

If NR is included in the RATs indicated by the **RRU.RS** parameter for the RF module and optimized evaluation of NR RF module dormancy is enabled (by selecting the **NR_DORMANCY_JUDGE_THLD_OPT_SW** option of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter), the base station periodically evaluates whether the local cell can enter scheduled RF module dormancy based on the cell load.

- An energy saving policy is configured for the local cell. The **NRDUCellPowerSaving.PowerSavingPolicyIndex** parameter specifies an energy saving policy index. It uniquely identifies an energy saving policy within the local cell. The **NRDUCellPowerSaving.PowerSavingType** parameter must be set to **NR_DORMANCY_JUDGE_THLD_OPT**. The **NRDUCellPowerSaving.EvalPeriodGranularity** parameter must be set to **MINUTE_LEVEL**.
- An NR cell cannot enter dormancy unless both of the following conditions are met:
 - The current time falls within the effective energy saving period determined by the **NRDUCellPowerSaving.StartTime** and **NRDUCellPowerSaving.StopTime** parameters.
 - The NR cell is not in an energy saving penalty state.
- Regarding cell load, an NR cell cannot enter dormancy unless all of the following conditions are met:

- Downlink PRB usage of the cell $\leq$ **NRDUCellPowerSaving.DlPrbThld**.
Downlink PRB usage of the cell = $(\text{N.PRB.DL.DrbUsed.Avg} - \text{N.PRB.DL.ExtendUsed.Avg}) / \text{N.PRB.DL.Actual.Avail.Avg} \times 100\%$
- Uplink PRB usage of the cell $\leq$ **NRDUCellPowerSaving.UlPrbThld**. Uplink PRB usage of the cell = $\text{N.PRB.PUSCH.Used.Avg} / \text{N.PRB.UL.PUSCH.Actual.Avail.Avg} \times 100\%$
- Number of RRC_CONNECTED UEs in the cell $\leq$ **NRDUCellPowerSaving.UserNumThld**
 - The number of RRC_CONNECTED UEs in an SUL cell with co-deployment is the sum of the **N.User.RRCConn.Avg** and **N.User.SUL.UL.Avg** values.
 - The number of RRC_CONNECTED UEs in any other cell is indicated by **N.User.RRCConn.Avg**.
- In SUL co-deployment scenarios, the total number of SUL UEs in the local cell is less than or equal to the value of the **NRDUCellPowerSaving.SulUserNumThld** parameter.
- For NR TDD, when the **ELIM_INCR_PWR_SAVING_JUDGE_SW** option of the **gNBpwrSavingAlgo.PwrSavingAlgoSwitch** parameter is selected, the number of UEs in the cell excludes the number of iNCR modules. For details about iNCR, see *LampSite iNCR Product Documentation*.

LampSite base stations do not support this function.

Implementation

For details, see *Multi-RAT Coordinated Energy Saving*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellPwrSaveAlgo. <i>PowerSavingAlgoSwitch</i>	MOD NRDUCEL LPWRS AV EALGO LST NRDUCEL LPWRS AV EALGO	gNodeB eNodeB	The NR_DORMANCY_JUDGE_T HLD_OPT_SW option has been added to specify whether to optimize the evaluation of NR RF module dormancy.
Modified	NRDUCellPowerSaving. <i>PowerSavingType</i>	ADD NRDUCEL LPOWERS AVING MOD NRDUCEL LPOWERS AVING DSP NRDUCEL LPOWERS AVING LST NRDUCEL LPOWERS AVING	gNodeB eNodeB	The enumerated value NR_DORMANCY_JUDGE_T HLD_OPT has been added to specify the energy saving policy for optimized evaluation of NR RF module dormancy.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
The following mutually exclusive relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD High-frequency NR TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch . <i>PowerSavingSwitch</i> parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	The time period configured for timing carrier shutdown cannot overlap with the period configured for optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module dormancy. You are advised to preferentially plan the periods during which timing carrier shutdown takes effect.
Low-frequency NR TDD	Deployment Position	NRDUCell.DeploymentPosition set to DEPLOY_IAB	None	If this parameter is set to DEPLOY_IAB , optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module dormancy cannot be enabled.
NR FDD	MBS broadcast	MBS_BROADCAST_SWITCH option of the NRCellFeatureSwitch . <i>MbsSwitch</i> parameter	None	When the MBS broadcast switch is on, optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module dormancy cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	<i>URLLC in 5G RAN Feature Documentation</i>	In low latency and high reliability scenarios, optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module dormancy cannot be enabled.
Low-frequency NR TDD	Dynamic MIMO layer number decrease	DYNAMIC_MIMO_LAYER_DECSW option of the gNodeBAlgo.DedicatedNetworkFuncSw parameter	None	Dynamic MIMO layer number decrease does not work with optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module dormancy.
Low-frequency NR TDD	Cell deployment in IAB mode	NRDUCell.DeploymentPosition set to DEPLOY_IAB	None	Cells deployed in IAB mode do not support optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module dormancy.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Supplementary Cell Indicator	NRDUCell.SupplementaryCellInd	None	If the NRDUCell.DuplexMode parameter is set to CELL_TDD and the value of supplementary cell indicator is INTRA_FREQ_ASSISTANT_CELL , optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module dormancy cannot be enabled.
NR FDD Low-frequency NR TDD High-frequency NR TDD	NR load-based RF module dormancy	INTRA_NR_DORMANCY_BY_LOAD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	NR load-based RF module dormancy does not work with optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module dormancy.
NR FDD Low-frequency NR TDD	LNR load-based RF module dormancy	LNR_DORMANCY_BY_LOAD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	LNR load-based RF module dormancy does not work with optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module dormancy.

- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Multi-RAT Coordinated Energy Saving*.

Requirement/Trouble Ticket Number

FeR-IR20250526000353-0010

RAT

NR FDD, NR TDD

5.3.4.1.8 Support for QCI-specific Uplink MU Pairing Differentiation

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Uplink UE pairing increases the inter-UE interference and BER and reduces the link reliability of UEs. To address this issue, QCI-specific uplink MU pairing differentiation is introduced. This function enables differentiated configurations of whether UEs with QCI-specific bearers participate in PUSCH MU-MIMO pairing.

- When the **UL_MU_MIMO_SW** option of the **NRDUCellQciBearer.ServiceDiffAlgoSwitch** parameter is deselected, UEs with QCI-specific bearers are not allowed to participate in PUSCH MU-MIMO pairing, reducing inter-layer interference of pairing and improving the link reliability of these UEs.
- When the **UL_MU_MIMO_SW** option of the **NRDUCellQciBearer.ServiceDiffAlgoSwitch** parameter is selected, UEs with QCI-specific bearers are allowed to participate in PUSCH MU-MIMO pairing.

Implementation

For details, see *Service-differentiated Experience Guarantee*. The specific sections are listed in the change history of the document.

Impact on the Network

Benefits: With this function, the uplink MU-MIMO pairing ratio of UEs with QCI-specific bearers and the uplink IBLER decrease.

Impacts: With this function, the network impact is related to the uplink traffic volume of UEs with QCI-specific bearers. A larger number of UEs with QCI-specific bearers results in higher uplink RB usage and lower uplink MU-MIMO pairing ratio and uplink user-perceived rate in a cell. In high-load scenarios, the average uplink cell throughput decreases.

Impact on NEs

No impact

Impact on Hardware Requirements

- Base station models
3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- Boards
All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- RF modules
All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PUSCH MU-MIMO	UL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	QCI-specific uplink MU pairing differentiation can take effect only when the UL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter is selected.

- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Service-differentiated Experience Guarantee*.

Requirement/Trouble Ticket Number

IR20241209002193

DTS2025082033228

RAT

NR FDD, NR TDD

5.3.4.1.9 Prohibition of Multi-Frequency Smart Selection for NSA UEs Whose RRC Connections Are Reestablished

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In NSA networking, the SCG service drop rate may increase due to carrier combination changes or handover-triggered NR cell changes for UEs whose RRC connections are reestablished on the LTE side. To reduce the SCG service drop rate, prohibition of multi-frequency smart selection for NSA UEs whose RRC connections are reestablished is introduced. After this function is enabled, experience-based smart carrier selection does not take effect for NSA UEs whose RRC connections are reestablished on the LTE side, and there will be decreases in the number of handovers triggered by experience-based smart carrier selection and the number of SCC additions.

A UE that has sent an RRCConnectionReestablishmentRequest message to the eNodeB is determined as a UE whose RRC connection is reestablished on the LTE side.

This change is supported only in low frequency bands.

Implementation

After "prohibition of multi-frequency smart selection for NSA UEs whose RRC connections are reestablished" is enabled, experience-based smart carrier selection does not take effect for NSA UEs whose RRC connections are reestablished on the LTE side, and there will be decreases in the number of handovers triggered by experience-based smart carrier selection and the number of SCC additions.

The function is controlled by the following options:

- NR side: **NSA_REEST_FORB_MUL_FREQ_SEL_SW** option of the **gNodeBParam.NetworkingOptionOptSw** parameter
- LTE side: **NSA_REEST_NOTIFY_TO_NR_SW** option of the **EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch** parameter

Impact on the Network

After prohibition of multi-frequency smart selection for NSA UEs whose RRC connections are reestablished takes effect, the SCG service drop rate decreases on NSA networks with UEs whose RRC connections are reestablished on the LTE side. The SCG service drop rate is calculated as follows:

- On the LTE side: $(L.NsaDc.SgNB.Rmv.Att - L.NsaDc.MenbTrig.SgNB.NormRel - L.NsaDc.SgnbTrig.SgNB.NormRel) / L.NsaDc.SgNB.Rmv.Att \times 100\%$
- On the NR side: $(N.NsaDc.SgNB.Rel - N.NsaDc.SgNB.Rel.SgNBTrigger - N.NsaDc.SgNB.Rel.MeNBTrigger.NormalRel + N.NsaDc.SgNB.AbnormRel - N.NsaDc.SgNB.Rel.InitialAdd.RachFail) / N.NsaDc.SgNB.Rel \times 100\%$

After prohibition of multi-frequency smart selection for NSA UEs whose RRC connections are reestablished takes effect, the number of handovers triggered by experience-based smart carrier selection and the number of SCC additions decrease in the scenario where NSA networking is used and there are RRC connection reestablishments on the LTE side. The related counters are as follows:

- Counters related to the number of handovers triggered by experience-based smart carrier selection: **N.MultiFreqSmartSel.Exec**, **N.MultiFreqSmartSel.Exec.CA**, **N.MultiFreqSmartSel.DlIntraFreqSEModel.Exec**, **N.MultiFreqSmartSel.DlIntraFreqSEModel.Exec.CA**, **N.MultiFreqSmartSel.Exec.DL**, and **N.MultiFreqSmartSel.Exec.DL.CA**
- Counters related to the number of SCC additions: **N.CA.SCell.Add.Att** and **N.CA.SCell.Add.Succ**

Impact on NEs

This function is implemented on the gNodeB and eNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameter has changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNodeBParam.NsaWorkingOptionOptSw	MOD GNODEBPARAM LST GNODEBPARAM	gNodeB	The NSA_REEST_FORB_MUL_FREQ_SEL_SW option has been added to specify whether to prohibit multi-frequency smart selection for NSA UEs whose RRC connections are reestablished.
Modified	EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch	MOD ENODEBALGOEXT SWITCH LST ENODEBALGOEXT SWITCH	eNodeB	The NSA_REEST_NOTIFY_TO_NR_SW option has been added to specify whether the eNodeB notifies NR of NSA UEs whose RRC connections are reestablished.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

To enable "prohibition of multi-frequency smart selection for NSA UEs whose RRC connections are reestablished", run the **MOD GNODEBPARAM** command on the NR side with the **NSA_REEST_FORB_MUL_FREQ_SEL_SW** option of the **gNodeBParam.NsaWorkingOptionOptSw** parameter selected and run the **MOD ENODEBALGOEXTSWITCH** command on the LTE side with the

NSA_REEST_NOTIFY_TO_NR_SW option of the **EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch** parameter selected.

Requirement/Trouble Ticket Number

FeR-IR20250708001473-0260

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD

5.3.4.1.10 Added UE Selection Guarantee for QCI Groups

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

When the RB-usage-based UE selection policy takes effect (that is, the **NRDUCellRes.QciGrpRbCtrlUeSelPol** parameter is set to **AUTO**, **ARP_BASED**, or **MCS_BASED**), a smaller number of UEs in a QCI group increases the likelihood of a single UE repeatedly gaining and losing its RB priority, while a higher traffic volume of UEs in a QCI group increases the risk of service degradation for UEs outside the group. In such cases, you are advised to enable UE selection guarantee for QCI groups, which is controlled by the **QCI_GRP_UE_SEL_GURA_POL_SW** option of the **NRDUCellRes.SchedulingPolicySwitch** parameter.

- If this option is deselected, the function does not take effect. The gNodeB can remove RB priority from one UE and grant it to another UE within each UE selection period.
- If this option is selected, the function takes effect. The gNodeB removes or grants RB priority for only one UE within each UE selection period.

UE selection guarantee for QCI groups takes effect only when the RB-usage-based UE selection policy is in effect.

Implementation

For details, see *QCI-based Resource Control*. The specific sections are listed in the change history of the document.

Impact on the Network

After an upgrade, when the RB-usage-based UE selection policy takes effect and UE selection guarantee for QCI groups is enabled, fewer UEs are selected within each UE selection period, resulting in slower uplink and downlink rate changes within a QCI group. In other scenarios, there is no impact.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellRes.SchedulingPolicySwitch	ADD NRDUCELLRES MOD NRDUCELLRES LST NRDUCELLRES	gNodeB	The QCI_GRP_UE_SEL_GURA_POLICY_SW option has been added to specify whether to enable UE selection guarantee for QCI groups.

- Performance management
No impact

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RB-usage-based UE selection policy	NRDUCellRes.QciGrpRbCtrlUeSelPol	<i>QCI-based Resource Control</i>	UE selection guarantee for QCI groups takes effect only when the RB-usage-based UE selection policy (specified by setting the NRDUCellRes.QciGrpRbCtrlUeSelPol parameter to AUTO , ARP_BASED , or MCS_BASED) is in effect.

- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *QCI-based Resource Control*.

Requirement/Trouble Ticket Number

IR20241209002193

DTS2025072212257

RAT

NR FDD, NR TDD

5.3.4.1.11 Added SCC Removal due to Consecutive Non-Split in Uplink CA

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The total power of a UE may decrease after SCC configuration. If there is consecutive non-split due to unqualified carrier channel quality, the UE cannot

benefit from CA. However, the UE rate on a single carrier (PCC) may decrease due to the decrease in the total power.

To mitigate this, SCC removal due to consecutive non-split in uplink CA is introduced. This function is controlled by the **NRDUCellCarrMgmt.UlCaContNonSplitSccDelThld** parameter. If the count of consecutive non-split events due to unqualified carrier channel quality exceeds the value of this parameter, the gNodeB removes the corresponding uplink SCC to ensure that the UE rate on the PCC is not affected.

This change is supported only in low frequency bands.

Implementation

For details, see *Carrier Aggregation*. The specific sections are listed in the change history of the document.

Impact on the Network

After this function is enabled, **User Uplink Average Throughput (DU)** of cell-edge UEs may increase.

After this function takes effect, an excessively small value of **NRDUCellCarrMgmt.UlCaContNonSplitSccDelThld** may cause unnecessary SCC removal, decreasing the average throughput of SCell-activated uplink CA UEs (calculated as follows: $(N.CA.SccActiveUser.ThpVol.UL - N.CA.ThpVol.UL.SmallPkt) / N.CA.ThpTime.UL.RmvSmallPkt$).

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellCarrierMgmt.UplinkSCCRemovalThld	MOD NRDUCELLC ARRMGMT LST NRDUCELLC ARRMGMT	gNodeB	This parameter has been added to specify whether to enable SCC removal due to consecutive non-split in uplink CA.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Carrier Aggregation*.

Requirement/Trouble Ticket Number

IR20241209002193

DTS2025062410265

RAT

NR FDD, NR TDD

5.3.4.1.12 Added Support for Flexible Configuration of the SINR Threshold for Uplink Data Split to SCells in CA

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Before the upgrade, the SINR threshold for uplink data split to SCells in CA is fixed at -1 dB.

After the upgrade, this threshold can be configured using the **NRDUCellCarrMgmt.UlCaSccSplitSinrThld** parameter to flexibly adapt to network requirements.

This change is supported only in low frequency bands.

Implementation

For details, see *Carrier Aggregation*. The specific sections are listed in the change history of the document.

Impact on the Network

After this function takes effect, the SINR threshold for uplink data split to SCells in CA can be configured more flexibly.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellCarrMgmt.UlCaSccSplitSinrThld	MOD NRDUCELLC ARRMGMT LST NRDUCELLC ARRMGMT	gNodeB	This parameter has been added to specify the SINR threshold for uplink data split to SCells in CA.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Carrier Aggregation*.

Requirement/Trouble Ticket Number

IR20241209002193

DTS2025062410265

RAT

NR FDD, NR TDD

5.3.4.1.13 Support for QCI-specific MAC- and RLC-Layer Delay Measurement in NSA Scenarios

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In this version, the base station supports measurement of QCI-level delay at the MAC and RLC layers for delay-sensitive services in NSA scenarios. The following counters have been added:

- Uplink: See [Table 5-34](#).

Table 5-34 Counters related to uplink

Counter Name	Counter Description
N.MAC.UL.QCI.Delay	Total delay of uplink services with a specific QCI at the MAC layer in a cell
N.MAC.UL.Packets.QCI	Total number of uplink packets with a specific QCI at the MAC layer in a cell
N.RLC.UL.QCI.Delay	Total delay of uplink services with a specific QCI at the RLC layer in a cell
N.RLC.UL.Packets.QCI	Total number of uplink packets with a specific QCI at the RLC layer in a cell
N.RLC.UL.TrfSDU.BfrDelay.QCI	Number of uplink RLC SDUs whose buffer delay is less than the threshold for services with a specific QCI in a DU cell

For **N.RLC.UL.TrfSDU.BfrDelay.QCI**, the buffer delay threshold is specified by the **gNBQciBearer.RlcUlDelayStatThld** parameter.

- Downlink: See [Table 5-35](#).

Table 5-35 Counters related to downlink

Counter Name	Counter Description
N.MAC.DL.QCI.Delay	Total transmission delay of downlink services with a specific QCI at the MAC layer in a cell
N.MAC.DL.Packets.QCI	Total number of downlink packets with a specific QCI at the MAC layer in a cell
N.RLC.DL.QCI.Delay	Total transmission delay of downlink services with a specific QCI at the RLC layer in a cell
N.RLC.DL.TrfSDU.RxPackets.QCI	Number of downlink SDUs for services with a specific QCI received at the RLC layer in a DU cell
N.RLC.DL.TrfSDU.BfrDelay.QCI	Number of downlink RLC SDUs whose buffer delay is less than the threshold for services with a specific QCI in a DU cell

For **N.RLC.DL.TrfSDU.BfrDelay.QCI**, the buffer delay threshold is specified by the **gNBQciBearer.RlcDlDelayStatThld** parameter.

This change is supported only in NSA networking.

Implementation

To measure the QCI-specific delay at the MAC and RLC layers for delay-sensitive services, the following counters have been added:

- Uplink
 - **N.MAC.UL.QCI.Delay**
 - **N.MAC.UL.Packets.QCI**
 - **N.RLC.UL.QCI.Delay**
 - **N.RLC.UL.Packets.QCI**
 - **N.RLC.UL.TrfSDU.BfrDelay.QCI**
- Downlink
 - **N.MAC.DL.QCI.Delay**
 - **N.MAC.DL.Packets.QCI**
 - **N.RLC.DL.QCI.Delay**
 - **N.RLC.DL.TrfSDU.RxPackets.QCI**
 - **N.RLC.DL.TrfSDU.BfrDelay.QCI**

Impact on the Network

No impact

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
The following function subsets have changed.

Change Type	Function Subset	NE	Description
New	DU Cell QCI Traffic Measurement	gNode B	This function subset has been added to measure QCI-related service performance of a DU cell, including delay and the number of packets.

 NOTE

- The counters in the new function subsets are not listed separately.
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20241209002193-0059

RAT

NR FDD, NR TDD

5.3.4.1.14 Optimized Cooperating Cell Selection

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In scenarios with dense deployment of multiple cells, cooperation-based functions such as intra-base-station UL CoMP, intra-base-station DL CoMP, intra-base-station Co-MM, and CS/CBF may not take effect due to failure to select a suitable cooperating cell. To address this issue, optimized cooperating cell selection

(controlled by the **COOP_CELL_ALLOC_OPT_SW** option of the **NRDUCellComp.CompOptSwitch** parameter) has been introduced to expand the range of cells that can be selected as cooperating cells in overlapping areas, thereby increasing the probability that the coordination-based functions take effect and increasing the average uplink and downlink throughputs of UEs in the overlapping areas.

This change is supported only in low-frequency bands.

Implementation

For details, see *CoMP, Co-MM (Low-Frequency TDD)*, and *Coordinated Interference Management (Low-Frequency TDD)*. The specific sections are listed in the change history of the documents.

Impact on the Network

After this function takes effect, the probability that coordination-based functions take effect increases in scenarios with dense deployment of multiple cells, increasing the average uplink and downlink throughputs of UEs in overlapping areas.

- Average uplink throughput of UEs in overlapping areas (Mbit/s) = $(N.ThpVol.UL.CellOverlap - N.ThpVol.UL.SmallPkt.CellOverlap) / N.ThpTime.UL.RmvSmallPkt.CellOverlap \times 1000$
- Average downlink throughput of UEs in overlapping areas (Mbit/s) = $(N.ThpVol.DL.CellOverlap - N.ThpVol.DL.LastSlot.CellOverlap) / N.ThpTime.DL.RmvLastSlot.CellOverlap \times 1000$

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellComp.CompOptSwitch	MOD NRDUCELLCOMP LST NRDUCELLCOMP	gNodeB	The COOP_CELL_ALLOC_OPT_SW option has been added to specify whether to enable optimized cooperating cell selection.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS interference coordination based on self-contained slots	SRS_INTRF_COORD_S_SLOT_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter	<i>Channel Management</i>	In low-frequency NR TDD scenarios, optimized cooperating cell selection (controlled by the COOP_CELL_ALLOC_OPT_SW option of the NRDUCellComp.CompOptSwitch parameter) requires that SRS interference coordination based on self-contained slots be enabled first.

- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *CoMP*, *Co-MM (Low-Frequency TDD)*, and *Coordinated Interference Management (Low-Frequency TDD)*.

Requirement/Trouble Ticket Number

FeR-IR20241209002193-0009

RAT

NR FDD, NR TDD

5.3.4.1.15 Support for Inter-Frequency MLB for mmWave PCells (for High Frequency Bands Only)

NEs Involved

5900 series base stations

Feature/Function Description

Inter-frequency MLB for mmWave PCells is now supported. This function is controlled by the **MMW_PCELL_INTER_FREQ_MLB_SW** option of the **NRCellMlb.MlbAlgoEnhSwitch** parameter. If this option is selected:

- Inter-frequency MLB for mmWave PCells is supported. The gNodeB performs inter-cell MLB among normal cells in the same sector based on the number of UEs in connected mode in each of these cells. A normal cell is one with the **NRCellMlb.SupplementaryCellInd** parameter set to **NORMAL_CELL**.
- The uplink UE number threshold for MLB is specified by the **NRCellMlb.InterFreqMlbUeNumThld** parameter, and the offset to the uplink UE number threshold for MLB is specified by the **NRCellMlb.MlbUeNumOffset** parameter. At the end of a configured MLB evaluation period, each cell checks whether the conditions for starting the MLB procedure are met. If so, the MLB procedure is started in the cell. If not, the cell will check again at the end of the next MLB evaluation period. The conditions for starting the MLB procedure are as follows:
 - New UEs access the cell in the MLB evaluation period.
 - The number of UEs in connected mode in the sector meets the specific threshold. That is, the number of UEs in connected mode in the sector is greater than or equal to the value of MLB uplink UE number threshold and MLB uplink UE number offset throughout the period.
- In load redistribution, if there are more UEs than the target UE number for MLB in a cell, the gNodeB regards the cell as a high-load cell. In all other cases, the gNodeB regards the cell as a low-load cell. The target UE number for MLB is the rounded-up average number of UEs in connected mode in the sector when the cell enters the MLB state.
- The number of selected UEs must be less than or equal to the upper limit on the total number of UEs allowed for load redistribution. If the number of UEs that can be selected is greater than this upper limit, the gNodeB randomly selects some of them. This upper limit takes the minimum value among the following three:

- Total number of UEs that have newly accessed the cell and have camped on the cell for a period of time within the MLB evaluation period
- Total number of UEs in the local cell minus the target UE number for MLB
- UE number upper limit for connected mode MLB, which is specified by the **NRCellMlb.ConnMlbUeNumThld** parameter

Intra-sector UE-number-based connected mode MLB does not take effect for cells in the energy saving state. You can run the **DSP NRDUCELLPOWERSAVING** command to check whether a cell is in the energy saving state. If the value of **Current State** for any **Power Saving Type** is **Enabled**, the cell is in the energy saving state.

Implementation

For details, see *Mobility Load Balancing in Connected Mode*. The specific sections are listed in the change history of the document.

Impact on the Network

If inter-frequency MLB for mmWave PCells takes effect for a sector for which intra-sector UE-number-based connected mode MLB does not take effect, the number of handovers increases, the load of cells in the same sector decreases, and the resource usage and capacity of the cells involved in MLB increases. These gains are directly proportional to the load of cells in the sector and the inter-cell load difference.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRCeLLMLb .MlbAlgoEnhSwitch	MOD NRCELLMLB LST NRCELLMLB	gNodeB	The MMW_PCELL_INTER_FREQ_MLB_SW option has been added to specify whether to enable inter-frequency MLB for mmWave PCells.
Modified	NRCeLLMLb .InterFreqMlbULUeNumThld	MOD NRCELLMLB LST NRCELLMLB	gNodeB	This parameter now also applies to high-frequency TDD.
Modified	NRCeLLMLb .MlbULUeNumOffset	MOD NRCELLMLB LST NRCELLMLB	gNodeB	This parameter now also applies to high-frequency TDD.
Modified	NRCeLLMLb .ConnMlbUeNumThld	MOD NRCELLMLB LST NRCELLMLB	gNodeB	This parameter now also applies to high-frequency TDD.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
High - frequency TDD	Intra-sector UE-number-based connected mode MLB	NRCellMlb.MlbTriggerMode set to INTRA_SECTOR_CELL_USER_NUM	<i>Mobility Load Balancing in Connected Mode</i>	If the MMW_PCELL_INTER_FREQ_MLB_SW option of the NRCellMlb.MlbAlgoEnhSwitch is selected, the INTER_FREQ_CONNECTED_MLB_SW option of the NRCellAlgoSwitch.MlbAlgoSwitch parameter must be selected and the NRCellMlb.MlbTriggerMode parameter must be set to INTRA_SECTOR_CELL_USER_NUM .

- Mutually exclusive features/functions

No impact

- Impacted features/functions

The following impact relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
High - frequency TDD	Multi-operator sharing	gNBSharingMode.gNBMultiOpSharingMode set to SHARED_FREQ or SEPARATED_FREQ	<i>Multi-Operator Sharing</i>	If a UE only supports certain operators in multi-operator sharing scenarios, the UE may fail to be handed over blindly during load redistribution.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Mobility Load Balancing in Connected Mode*.

Requirement/Trouble Ticket Number

FeR-IR20241209002208-0002

RAT

NR TDD

5.3.4.1.16 DRX-based Data Split Optimization for the SCG in NSA DC Scenarios

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In NSA DC Option 3x, if downlink data is distributed to the SCG when the SCG is in the DRX sleep time, the data can be transmitted only after the SCG enters the DRX active time. This increases the downlink data packet delay. To address this issue, DRX-based data split optimization for the SCG in NSA DC scenarios is introduced. With this function, downlink data is distributed to the MCG when the SCG is in the DRX sleep time, thereby reducing the downlink data packet delay.

This function is controlled by the **NSA_SCG_DRX_SPLIT_OPT_SW** option of the **NRCellNsaDcConfig.NsaDcAlgoSwitch** parameter.

This change is supported only in NSA networking.

This change is supported only in low-frequency bands.

This change is irrelevant to LampSite base stations.

Implementation

For UEs whose primary path for uplink data split is the SCG in NSA DC Option 3x, DRX-based data split optimization for the SCG in NSA DC scenarios distributes downlink data packets originally intended for the SCG to the MCG when the SCG is in the DRX sleep time, thereby shortening the downlink data packet delay.

Impact on the Network

When DRX-based data split optimization for the SCG in NSA DC scenarios is enabled, the data transmission delay is shortened for UEs whose primary path for uplink data split is the SCG in NSA DC Option 3x. If this function is disabled, network performance is not affected.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact

- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRCellNsaDcConfig.NsaDcAlgorithmSwitch	MOD NRCELLNSA DCCONFIG LST NRCELLNSA DCCONFIG	gNodeB	The NSA_SCG_DRX_SPLIT_OPT_SW option has been added to specify whether to enable DRX-based data split optimization for the SCG in NSA DC scenarios.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Uplink preallocation	NSA_DC_PREALLOCAT ION_SW option of the NRCellNsaDcConfig.NsaDcAlgorithmSwitch parameter	<i>EPC-based NSA Basics</i>	DRX-based data split optimization for the SCG in NSA DC scenarios requires that uplink preallocation in NSA DC scenarios be enabled first.
NR FDD Low-frequency NR TDD	NSA downlink data split optimization in DRX scenarios	DRX_NSA_SPLIT_OPT_SW option of the gNodeBParam.NsaDcOptSwitch parameter	<i>EPC-based NSA Basics</i>	DRX-based data split optimization for the SCG in NSA DC scenarios requires that NSA downlink data split optimization in DRX scenarios be enabled first.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Fast retransmission for downlink data split	NSA_DC_FAST_RETRANS_SWITCH option of the gNodeBParameter.NsaDcOptSwitch parameter	<i>EPC-based NSA Basics</i>	DRX-based data split optimization for the SCG in NSA DC scenarios requires that fast retransmission for downlink data split be enabled first.

- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

Run the **MOD NRCELLNSADCCONFIG** command with the **NSA_SCG_DRX_SPLIT_OPT_SW** option of the **NRCellNsaDcConfig.NsaDcAlgoSwitch** parameter selected to enable DRX-based data split optimization for the SCG in NSA DC scenarios.

Requirement/Trouble Ticket Number

FeR-IR20241209002193-0027

RAT

NR FDD, NR TDD

5.3.4.1.17 RLC-layer Fast Retransmission for Sparse-Packet Services

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In NSA DC Option 3x, if RLC-layer retransmission on the SCG side repeatedly fails for sparse-packet services, the downlink data delay will increase. To address this issue, RLC-layer fast retransmission for sparse-packet services is introduced (sparse-packet services currently refer to the ping service). This function migrates the downlink-retransmitted data packets of sparse-packet services from the SCG to the MCG after RLC-layer retransmission on the SCG side fails for the sparse-packet services, thereby reducing the downlink data packet delay.

If the threshold of the number of downlink retransmissions for RLC-layer fast retransmission (indicated by the **NRDUCellServExp.FastDLRLcRetransTimesThld**

parameter) is set to a value other than **0**, RLC-layer fast retransmission for sparse-packet services is enabled.

This change is supported only in NSA networking.

This change is supported only in low-frequency bands.

This change is irrelevant to LampSite base stations.

Implementation

In NSA DC Option 3x, after RLC-layer fast retransmission for sparse-packet services is enabled, the downlink-retransmitted data packets of sparse-packet services will be migrated to the MCG for transmission when the number of RLC-layer retransmissions for the downlink sparse-packet services is greater than or equal to the threshold of the number of downlink retransmissions for RLC-layer fast retransmission.

Impact on the Network

A smaller value of the **NRDUCellServExp.FastDLRLcRetransTimesThld** parameter results in a higher probability that RLC-layer fast retransmission for sparse-packet services takes effect and a shorter data transmission delay. A larger value of this parameter results in the opposite effects.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellServExp.FastDLRLcRetransTimesThld	MOD NRDUCELLS ERVEXP LST NRDUCELLS ERVEXP	gNodeB	This parameter has been added to specify the threshold expressed as the number of downlink retransmissions for RLC-layer fast retransmission for sparse-packet services.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

Run the **MOD NRDUCELLSERVEEXP** command with the **NRDUCellServExp.FastDLRLcRetransTimesThld** parameter set to a value other than **0** to enable RLC-layer fast retransmission for sparse-packet services.

Requirement/Trouble Ticket Number

FeR-IR20241209002193-0027

RAT

NR FDD, NR TDD

5.3.4.1.18 Support for SUL CoMP to Take Effect for UEs with Specific QCLs

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

With SUL CoMP enabled, the gNodeB only delivers measurement configurations to all UEs in a cell, and SUL CoMP can take effect for UEs that meet certain conditions. SUL CoMP can now take effect for UEs with specific QCI.

After the **UL_2CELL_COMP_BEAR_POL_SW** option of **NRDUCellComp.CompConfigPol** is selected to enable SUL CoMP to take effect for UEs with specific QCIs, the gNodeB regards specific UEs as candidate UEs for SUL CoMP. Specific UEs are those having at least one QCI for which the **COMP_SW** option of the **gNBQciBearer.ServiceDiffAlgoSwitch** parameter is selected.

Implementation

For details, see *Super Uplink* and *UL and DL Decoupling*. The specific sections are listed in the change history of each document.

Impact on the Network

After this function takes effect, the number of UEs for which SUL CoMP is enabled and the uplink PRB usage decrease in the cell.

Impact on NEs

This change is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellComp.CompConfigPol	MOD NRDUCELLC OMP LST NRDUCELLC OMP	gNodeB	This parameter has been added, and the UL_2CELL_COMP_BEAR_POL_SW option specifies the bearer policy for 2-cell UL CoMP.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNBQciBearer.ServiceDiffAlgorithm	ADD GNBQCIBEARER MOD GNBQCIBEARER LST GNBQCIBEARER	gNodeB	The COMP_SW option has been added to specify whether to allow CoMP to take effect for UEs with specified service bearers.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Super Uplink* and *UL and DL Decoupling*.

Requirement/Trouble Ticket Number

FeR-IR20241209002193-0032

RAT

NR TDD

5.3.4.1.19 Added CA-related Counters for Bandwidth-based Measurement of CA UEs

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Before this change, CA-related counters are measured without distinguishing the bandwidths of CA UEs. In some scenarios, however, bandwidth-specific measurement is required for CA-related counters to count in only CA UEs meeting the specified bandwidth limit. Given this, the following CA-related counters are added to distinguish between the bandwidths of CA UEs:

- **N.User.CA.PCell.DL.Avg.CBW**: average number of downlink CA UEs using the local cell as their PCell and meeting the specified bandwidth limit
- **N.User.CA.PCell.DL.SCellAct.Avg.CBW**: average number of downlink CA UEs using the local cell as their PCell, having activated SCells, and meeting the specified bandwidth limit
- **N.CA.SccActiveUser.ThpVol.DL.CBW**: total volume of downlink data sent at the RLC layer to CA UEs having activated SCells and meeting the specified bandwidth limit in a cell
- **N.CA.ThpVol.DL.LastSlot.CBW**: total volume of downlink data sent at the RLC layer to CA UEs meeting the specified bandwidth limit in the last slots during which the buffer becomes empty in a cell
- **N.CA.ThpTime.DL.RmvLastSlot.CBW**: data transmission duration excluding the last slots during which the downlink buffer becomes empty for CA UEs meeting the specified bandwidth limit in a cell

A CA UE meets the specified bandwidth limit when the total bandwidth of its low-frequency PCell and SCells is greater than or equal to that specified by **gNodeBParam.LfCaDLBandwidthThld**.

Implementation

Before this change, CA-related counters are measured without distinguishing the bandwidths of CA UEs. In some scenarios, however, bandwidth-specific measurement is required for CA-related counters to count in only CA UEs meeting the specified bandwidth limit. Given this, the following CA-related counters are added to distinguish between the bandwidths of CA UEs:

- **N.User.CA.PCell.DL.Avg.CBW**: average number of downlink CA UEs using the local cell as their PCell and meeting the specified bandwidth limit
- **N.User.CA.PCell.DL.SCellAct.Avg.CBW**: average number of downlink CA UEs using the local cell as their PCell, having activated SCells, and meeting the specified bandwidth limit
- **N.CA.SccActiveUser.ThpVol.DL.CBW**: total volume of downlink data sent at the RLC layer to CA UEs having activated SCells and meeting the specified bandwidth limit in a cell
- **N.CA.ThpVol.DL.LastSlot.CBW**: total volume of downlink data sent at the RLC layer to CA UEs meeting the specified bandwidth limit in the last slots during which the buffer becomes empty in a cell
- **N.CA.ThpTime.DL.RmvLastSlot.CBW**: data transmission duration excluding the last slots during which the downlink buffer becomes empty for CA UEs meeting the specified bandwidth limit in a cell

A CA UE meets the specified bandwidth limit when the total bandwidth of its low-frequency PCell and SCells is greater than or equal to that specified by **gNodeBParam.LfCaDLBandwidthThld**.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	gNodeBParameter.LfCaDlBandwidthThreshold	MOD GNODEBPARAM LST GNODEBPARAM	gNodeB	This parameter has been added to specify the bandwidth requirements of CA UEs for bandwidth-based measurement of CA-related counters.

- Performance management

The following counters have changed.

Change Type	Counter Name	Function Subset	NE	Description
New	N.User.CA.PCell.DL.Avg.CBW	Cell User Quantity Measurement	gNodeB	This counter has been added to measure the average number of downlink CA UEs using the local cell as their PCell and meeting the specified bandwidth limit.

Change Type	Counter Name	Function Subset	NE	Description
New	N.User.CA.PCell.DL.SCellAct.Avg.CBW	NR DU Cell Scheduled User Quantity Measurement	gNode B	This counter has been added to measure the average number of downlink CA UEs using the local cell as their PCell, having activated SCells, and meeting the specified bandwidth limit.
New	N.CA.SccActiveUser.ThpVol.DL.CBW	Throughput and Data Volume Measurement	gNode B	This counter has been added to measure the total volume of downlink data sent at the RLC layer to CA UEs having activated SCells and meeting the specified bandwidth limit in a cell.
New	N.CA.ThpVol.DL.LastSlot.CBW	Throughput and Data Volume Measurement	gNode B	This counter has been added to measure the total volume of downlink data sent at the RLC layer to CA UEs meeting the specified bandwidth limit in the last slots during which the buffer becomes empty in a cell.
New	N.CA.ThpTime.DL.RmvLastSlot.CBW	Throughput and Data Volume Measurement	gNode B	This counter has been added to measure the data transmission duration excluding the last slots during which the downlink buffer becomes empty for CA UEs meeting the specified bandwidth limit in a cell

- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact

- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20241209002193-0098

RAT

NR FDD, NR TDD

5.3.4.1.20 Added the Uplink CA Data Split Threshold Optimization Function

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In uplink intra-gNodeB CA scenarios, the SCell uplink data split threshold is scenario-specific and internally calculated by the gNodeB. The threshold accuracy and stability have a significant impact on CA performance. Uplink CA data split threshold optimization optimizes the threshold calculation for improved accuracy and stability. This function is controlled by the **UL_CA_SPLIT_THLD_OPT_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoSwitch** parameter.

This change is supported only in low frequency bands.

NOTE

This change was first incorporated into V100R022C00SPC010 but is first described in the network impact report of V100R022C00SPC050.

Implementation

For details, see *Carrier Aggregation*. The specific sections are listed in the change history of the document.

Impact on the Network

This function has the following impacts:

- The total volume of uplink data received at the MAC layer from uplink CA UEs that treat the local cell as their SCell (**N.CA.SCell.ThpVol.Ul.Cell**) increases.
- The total volume of uplink data received at the MAC layer from uplink CA UEs that treat the local cell as their PCell (**N.CA.PCell.ThpVol.Ul.Cell**) decreases.

- The average throughput of SCell-activated uplink CA UEs (calculated as follows: $(N.CA.SccActiveUser.ThpVol.UL - N.CA.ThpVol.UL.SmallPkt) / N.CA.ThpTime.UL.RmvSmallPkt$) varies, influenced by the SCell load. Under low SCell load, this throughput increases.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellCarrierMgmt.CaEnhancedAlgorithmSwitch	MOD NRDUCELLC ARRMGMT LST NRDUCELLC ARRMGMT	gNodeB	The UL_CA_SPLIT_THLD_OPT_SW option has been added to specify whether to enable uplink CA data split threshold optimization.

- Performance management

No impact

- Fault management

No impact

Impact on Features/Functions

- Prerequisite features/functions

No impact

- Mutually exclusive features/functions

No impact

- Impacted features/functions

No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Carrier Aggregation*.

Requirement/Trouble Ticket Number

IR20241209002193

DTS2025062316721

RAT

NR FDD, NR TDD

5.3.4.1.21 Blacklist Control for Non-Gap-Assisted Coverage-based Inter-Frequency PSCell Change

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Non-gap-assisted coverage-based inter-frequency PSCell change can be disabled for specific UEs. This function is controlled by the **INTERF_HO_NO_GAP_SW_OFF** option of the **gNBUECompatSw.BlacklistCtrlExtSwitch** parameter on the NR side.

This change is supported only in low-frequency bands.

This change is supported only in NSA networking.

Implementation

For details, see *NSA Mobility Management*. The specific sections are listed in the change history of the document.

Impact on the Network

When the "blacklist control for non-gap-assisted coverage-based inter-frequency PSCell change" function is enabled, non-gap-assisted inter-frequency handover does not take effect for specific UEs, and the user-perceived rates of these UEs may decrease. When this function is disabled, network performance is not affected.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNBUEComPatSw.Bla acklistCtrl ExtSwitch	ADD GNBUECOM PATSW MOD GNBUECOM PATSW LST GNBUECOM PATSW	gNodeB	The INTERF_HO_NO_GAP_SW_OFF option has been added to specify whether to disable non-gap-assisted inter-frequency PSCell change for specific UEs.

- Performance management

No impact

- Fault management

No impact

Impact on Features/Functions

- Prerequisite features/functions

No impact

- Mutually exclusive features/functions

No impact

- Impacted features/functions

No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *NSA Mobility Management*.

Requirement/Trouble Ticket Number

FeR-IR20241205002359-0010

IR20250115001577

RAT

NR FDD, NR TDD

5.3.4.1.22 Added Uplink Enhancement for Smart Carrier Selection

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Service-based carrier selection (controlled by the **SERVICE_AWARE_CARR_SEL_SW** option of the **NRDUCellCollabServ.MultiCarrMgmtSw** parameter) can change the carrier selection mode based on different service requirements of UEs, so that UEs can obtain the optimal user-perceived rate. When the traffic volume model changes for a UE capable of 3CC or massive CA, however, preference based on the uplink peak rate cannot take effect in the CA capability combination selection policy in scenarios where the optimal uplink CA combination differs from the optimal downlink CA combination. As a result, uplink CA performance cannot be ensured. To address this restriction, uplink enhancement for smart carrier selection is introduced. This function is controlled by the **SMART_CARR_SEL_UL_ENH_SW** option of the **NRDUCellCollabServ.MultiCarrMgmtSw** parameter. After this enhancement function is enabled and the traffic volume model changes for a UE capable of 3CC or massive CA, preference based on the uplink peak rate can take effect in the CA capability combination selection policy in scenarios where the optimal uplink CA combination differs from the optimal downlink CA combination. As a result, uplink CA performance can be ensured.

This change is supported only in low frequency bands.

Implementation

For details, see *Carrier Aggregation*. The specific sections are listed in the change history of the document.

Impact on the Network

If uplink enhancement for smart carrier selection takes effect in suitable scenarios:

- The average number of uplink CA UEs that treat the local cell as their PCell (**N.User.CA.PCell.UL.Avg**) increases.
- The number of UEs for which 3CC CA or massive CA takes effect decreases. As a result, the average number of downlink CA UEs that treat the local cell as their PCell (**N.User.CA.PCell.DL.Avg**) decreases.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellCollabServ.MultiCarrMgmtSw	MOD NRDUCELLC OLLABSERV LST NRDUCELLC OLLABSERV	gNodeB	The SMART_CARR_SEL_UL_ENH_SW option has been added to specify whether to enable uplink enhancement for smart carrier selection.

- Performance management

No impact

- Fault management

No impact

Impact on Features/Functions

- Prerequisite features/functions

The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Smart carrier selection	SMART_CARR_SEL_SW option of the NRDUCellCollabServ.MultiCarrMgmtSw parameter	<i>Carrier Aggregation</i>	Uplink enhancement for smart carrier selection takes effect only when both smart carrier selection and service-based optimal carrier selection are enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Service-based optimal carrier selection	SERVICE_AWARE_CARRIER_SEL_SW option of the NRDUCellCollabServ.MultiCarrierMgmtSw parameter	<i>Carrier Aggregation</i>	

- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Carrier Aggregation*.

Requirement/Trouble Ticket Number

FeR-IR20250806001609-0320

RAT

NR FDD, NR TDD

5.3.4.1.23 Optimized the Maintenance and Test for RedCap Resources and the Number of UEs in a Specific Network Slice

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Counters related to CCE resources, operator-level PRB resources, and the number of RedCap UEs in the RRC_CONNECTED state in a specific network slice have been added, to improve maintenance and test capabilities in RedCap scenarios.

- **N.CCE.RedCap.Used.Avg**: measures the average number of PDCCH CCEs used by RedCap UEs.
- **N.CCE.RedCap.Avail.Avg**: measures the average number of PDCCH CCEs available for RedCap UEs.

- **N.PRBUUsed.UL.RedCap.PLMN**: measures the number of uplink PRBs used by RedCap UEs of a specific operator.
- **N.PRBUUsed.DL.RedCap.PLMN**: measures the number of downlink PRBs used by RedCap/ UEs of a specific operator.
- **N.User.RedCap.RRCConn.Avg.Ns**: measures the average number of RedCap UEs in RRC_CONNECTED mode in a specific network slice.
- **N.User.RedCap.RRCConn.Max.Ns**: measures the maximum number of RedCap UEs in RRC_CONNECTED mode in a specific network slice

Implementation

For details, see *RedCap*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
The following counters have changed.

Change Type	Counter Name	Function Subset	NE	Description
New	N.CCE.RedCap.Used.Avg	PRB Measurement	gNodeB	This counter has been added to measure the average number of PDCCH CCEs used by RedCap UEs.

Change Type	Counter Name	Function Subset	NE	Description
New	N.CCE.RedCap.Avail.Avg	PRB Measurement	gNodeB	This counter has been added to measure the average number of PDCCH CCEs available for RedCap UEs.
New	N.PRBUse.d.UL.RedCap.PLMN	DU Cell Operator PRB Measurement	gNodeB	This counter has been added to measure the number of uplink PRBs used by RedCap UEs of a specific operator.
New	N.PRBUse.d.DL.RedCap.PLMN	DU Cell Operator PRB Measurement	gNodeB	This counter has been added to measure the number of downlink PRBs used by RedCap/UEs of a specific operator.
New	N.User.RedCap.RRC Conn.Avg.Ns	Intra-Cell Network Slice User Number Measurement	gNodeB	This counter has been added to measure the average number of RedCap UEs in RRC_CONNECTED mode in a specific network slice.
New	N.User.RedCap.RRC Conn.Max.Ns	Intra-Cell Network Slice User Number Measurement	gNodeB	This counter has been added to measure the maximum number of RedCap UEs in RRC_CONNECTED mode in a specific network slice.

- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250911002729-0001

FeR-IR20250528001857-0126

RAT

NR FDD, NR TDD

5.3.4.1.24 Dual-Plane Function in Intra-Frequency Dual-Network Redundancy Networking in ToB Scenarios

NEs Involved

5900 series base stations

Feature/Function Description

In 5GtoB scenarios, if a board (baseband processing unit or main control board) in a BBU or an RF module (AAU or RRU) is faulty, services in the coverage area will be abnormal or interrupted. To solve this problem, "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" is introduced.

The transmission-link-fault-triggered switchover can be sped up by configuring the **SCTEMPLATE** MO and setting **GUPTRANSPARA.UPSTATFASW** to **ENABLE**.

To ensure the success rate of the active/standby switchover and no impact on the performance during the switchover, a check can be performed on the consistency of the configurations or licenses between the active and standby BBUs. Such a check can be performed only if **SYSTEMOM.DualBBUConsChkSw** is set to **ON** on both the primary and backup gNodeBs. If this parameter is set to **ON**, the NE will check the consistency of the configurations and licenses between the active and standby BBUs. In the case of an inconsistency, ALM-26405 Dual Network Active/Standby Inconsistency is reported, which can be cleared by referring to the alarm reference.

Implementation

For details, see *Base Station Equipment Reliability*. The specific sections are listed in the change history of the document.

Impact on the Network

With "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" enabled, if services are affected because a board (baseband processing unit or main control board) in a BBU or an RF module is removed, powered off, or faulty, a fast switchover to the standby BBU is performed to restore network services.

Impact on NEs

This change is implemented on the gNodeB.

Impact on Hardware Requirements

- Boards
 - Main control boards: Only the UMPTg, UMPTha, and UMPThb series boards support this function.
 - Baseband processing units: Only the NR TDD-capable UBBPg and UBBPh series boards support this function.

- RF Modules

The RF module requirements are the same as those for RF channel intelligent shutdown when this function is enabled on a macro base station working in low-frequency TDD. For details, see *Energy Conservation and Emission Reduction* in *5G RAN Feature Documentation*.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNBRedundancy.gNBReliabilitySwitch	ADD GNBREDUNDANCY MOD GNBREDUNDANCY LST GNBREDUNDANCY	gNodeB	The DUAL_PLANE_TOB_SW option has been added to control "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios".

- Performance management

No impact

- Fault management

No impact

Impact on Features/Functions

- Prerequisite features/functions

The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low - frequency NR TDD	Intra-mode Dual-site Ring Topology Mode	EQUIPMENT.IMDSRTMODE	None	"Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" requires this parameter to be set to NONE .

- Mutually exclusive features/functions

The following mutually exclusive relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low - frequency NR TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	"Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "timing carrier shutdown" cannot be both enabled.

RAT	Function Name	Function Switch	Reference	Description
Low - frequency NR TDD	Power saving scheduling	POWER_SAVING_SCHEDULE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	"Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "power saving scheduling" cannot be both enabled.
Low - frequency NR TDD	Deep symbol power saving	DEEP_SYMBOL_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	"Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "deep symbol power saving" cannot be both enabled.
Low - frequency NR TDD	Energy saving based on flexible frequency-domain scheduling	FLEX_FREQ_SCH_ENERGY_SAVING_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	"Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "energy saving based on flexible frequency-domain scheduling" cannot be both enabled.

RAT	Function Name	Function Switch	Reference	Description
Low - frequency NR TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	"Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "RF channel intelligent shutdown" cannot be both enabled.
Low - frequency NR TDD	Intelligent carrier shutdown	INTRA_GNB_MULTI_CARRER_SD_SW or INTER_GNB_MULTI_CARRER_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	"Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "intelligent carrier shutdown" cannot be both enabled.

RAT	Function Name	Function Switch	Reference	Description
Low - frequency NR TDD	Fast carrier shutdown	- For capacity-layer cells: INTRA_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter selected (intra-base-station multi-carrier scenarios) or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter selected (inter-base-station multi-carrier scenarios) FAST_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter selected - For basic-layer cells: FAST_CARR_SHUT_SERV_ASSUR_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter selected	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	"Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "fast carrier shutdown" cannot be both enabled.
Low - frequency NR TDD	LTE and NR intelligent carrier shutdown	LNR_SMART_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	"Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "LTE and NR intelligent carrier shutdown" cannot be both enabled.

RAT	Function Name	Function Switch	Reference	Description
Low - frequency NR TDD	LTE and NR coordinated symbol power saving	MULTI_RAT_SYMBOL_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	"Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "LTE and NR coordinated symbol power saving" cannot be both enabled.
Low - frequency NR TDD	LTE and NR bundled scheduling	LNR_BUNDLED_SCHEDULING_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	"Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "LTE and NR bundled scheduling" cannot be both enabled.
Low - frequency NR TDD	LNR coordinated channel shutdown	MULTI_RAT_RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	"Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "LNR coordinated channel shutdown" cannot be both enabled.
Low - frequency NR TDD	LTE FDD and NR Uplink Spectrum Sharing	- NR: LTE_NR_UL_SPECTRUM_SHARING_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter - LTE: SpectrumCloud.SpectrumCloudSwitch set to LTE_NR_UPLINK_SPECTRUM_SHR	<i>LTE FDD and NR SUL Dynamic Spectrum Sharing</i>	"Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "LTE FDD and NR Uplink Spectrum Sharing" cannot be both enabled.

RAT	Function Name	Function Switch	Reference	Description
Low - frequency NR TDD	Baseband shutdown for energy saving	BB_SHUTDOWN_POWERSAVING_SW option of the gNodeBParam.BaseStationDeepDormancySw parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	"Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "baseband shutdown for energy saving" cannot be both enabled.
Low - frequency NR TDD	TTI-level channel shutdown	TTI_RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	"Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "TTI-level channel shutdown" cannot be both enabled.
Low - frequency NR TDD	RF channel dynamic muting	RF_CHN_DYN_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	"Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "RF channel dynamic muting" cannot be both enabled.

RAT	Function Name	Function Switch	Reference	Description
Low - frequency NR TDD	Intra-frequency multi-carrier shutdown	INTRA_FREQ_MULTI_CARR_SD_SW option of the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter	None	"Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "intra-frequency multi-carrier shutdown" cannot be both enabled.
Low - frequency NR TDD	Adaptive BBU deep dormancy	BBP_DEEP_DORMANCY_SW option of the gNB PwrSavingAlgo.DormancySwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	If "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" is configured for a gNodeB, adaptive BBU deep dormancy cannot be enabled.
Low - frequency NR TDD	Backup power saving policy	NRDUCellPowerSaving.BackupPwrSavingPolicy	None	When "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" is enabled, NRDUCellPowerSaving.BackupPwrSavingPolicy cannot be set to CUSTOMIZED_BACKUP_POLICY .
Low - frequency NR TDD	Dual-plane function in intra-frequency dual-network redundancy networking	DUAL_PLANE_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	"Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "dual-plane function in intra-frequency dual-network redundancy networking" cannot be both enabled.

RAT	Function Name	Function Switch	Reference	Description
Low - frequency NR TDD	Primary/Backup function in intra-frequency dual-network redundancy networking	PRIMARY_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	"Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "primary/backup function in intra-frequency dual-network redundancy networking" cannot be both enabled.
Low - frequency NR TDD	Dual-BBU backup in dual-network redundancy networking	DUAL_BBU_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	"Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "dual-BBU backup in dual-network redundancy networking" cannot be both enabled.

RAT	Function Name	Function Switch	Reference	Description
Low - frequency NR TDD	RF module fast deep dormancy	- NR load-based RF module fast deep dormancy: FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUD OROPTFEATSW parameter and NR_FAST_DORMANCY_BY_LOAD_SW option of the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter - LNR load-based RF module fast deep dormancy: FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUD OROPTFEATSW parameter and LNR_FAST_DORMANCY_BY_LOAD_SW option of the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	"Dual-BBU backup in dual-network redundancy networking" and "NR or LNR load-based RF module fast deep dormancy" cannot be both enabled.
Low - frequency NR TDD	Intra-frequency multi-carrier shutdown	INTRA_FREQ_MULTICARRIER_SD_SW option of the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter	None	"Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "intra-frequency multi-carrier shutdown" cannot be both enabled.

- Impacted features/functions
The following impact relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency Network	Adaptive baseband shutdown	BBP.CELLDEPLOY set to ADAPT_BBP_SHUTDOWN	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Adaptive baseband shutdown does not take effect when "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" is in effect.
Low-frequency Network	Power saving first	BBP.CELLDEPLOY set to POWERSAVING	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	Power saving first does not take effect when "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" is in effect.
Low-frequency Network	RF module deep dormancy	RRU.DORMANCYSW set to ON or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	When RF module deep dormancy is enabled, RF modules enter the deep dormancy state, which causes an active/standby BBU switchover failure. Therefore, RF module deep dormancy and "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" cannot be both enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	RF module super dormancy	RRU.DORMANCYSW set to ON_SUPER or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	When RF module super dormancy is enabled, RF modules enter the super dormancy state, which causes an active/standby BBU switchover failure. Therefore, RF module super dormancy and "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" cannot be both enabled.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Base Station Equipment Reliability*.

Requirement/Trouble Ticket Number

FeR-IR20250205001396-0001

RAT

NR TDD

5.3.4.1.25 Added Support for Multi-TRS Optimization in Multi-TRP Scenarios

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Added support for multi-TRS optimization in multi-TRP scenarios. This function allocates multiple sets of TRS resources based on the actual TRP deployment in a multi-TRP cell (a combined cell or hyper cell) to reduce the TRS measurement capability overhead of UEs.

This function is controlled by the **MULTI_TRP_MULTI_TRS_OPT_SW** option of the **NRDUCellCsirs.CsirsExtSwitch** parameter.

Implementation

For details, see *Cell Combination* and *Hyper Cell*. The specific sections are listed in the change history of the documents.

Network Impacts

After multi-TRS optimization in multi-TRP scenarios takes effect in combined cells and hyper cells, the capability overhead of UEs for TRS measurement is reduced, and more features that depend on CSI-RS measurement can take effect.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellCsirs.CsirsExtSwitch	MOD NRDUCELLC SIRS LST NRDUCELLC SIRS	gNodeB	The MULTI_TRP_MULTI_TRS_OPT_SW option has been added to specify whether to enable multi-TRS optimization in multi-TRP scenarios.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.

RA T	Function Name	Function Switch	Reference	Description
Lo w- fre qu enc y TD D	Cell Combination	NRDUCell.NrD uCellNetworki ngMode set to HYPER_CELL_C OMBINE_MOD E and NRDUCellMulti Trp. DataTrans Mode set to BASIC_MODE or DPS_MODE	<i>Cell Combination</i>	Multi-TRS optimization in TRP scenarios can be enabled only in combined cells and hyper cells when the TRS resource allocation policy is set to MULTI_TRS .
Lo w- fre qu enc y TD D	Hyper Cell	NRDUCell.NrD uCellNetworki ngMode set to HYPER_CELL	<i>Hyper Cell</i>	Multi-TRS optimization in TRP scenarios can be enabled only in combined cells and hyper cells when the TRS resource allocation policy is set to MULTI_TRS .
Lo w- fre qu enc y TD D	TRS Allocation Strategy	NRDUCellCsirs. TrsAllocationS trategy set to MULTI_TRS	<i>Cell Combination Hyper Cell</i>	Multi-TRS optimization in TRP scenarios can be enabled only in combined cells and hyper cells when the TRS resource allocation policy is set to MULTI_TRS .

- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Cell Combination* and *Hyper Cell*.

Requirement/Trouble Ticket Number

FeR-IR20250528001856-0025

RAT

NR FDD, NR TDD

5.3.4.1.26 Support for Configurable T320 Timer

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The T320 timer specifies the validity period of dedicated cell reselection priorities. This change supports the configurable T320 timer. Specifically:

- In RRC connection release triggered by load balancing, the T320 timer is configured by **NRCCellReselConfig.T320ForMlb**.
- In RRC connection release not triggered by load balancing, the T320 timer is configured by **NRCCellReselConfig.T320ForOther**.

This change is supported only in low frequency bands.

This change is supported only in SA networking.

Implementation

For details, see *SA Mobility Management in Idle Mode* and *Mobility Load Balancing in Idle Mode*. The specific sections are listed in the change history of the documents.

Impact on the Network

A smaller value of the timer results in a shorter duration before the cell-reselection parameters contained in the RRCRelease message become invalid. A larger value of the timer results in a longer duration, which may lead to delayed update of cell-reselection parameters on the UE side.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRCellReselConfig <i>.T320ForOther</i>	LST NRCELLRESEL CONFIG MOD NRCELLRESEL CONFIG	gNodeB	This parameter has been added to specify the length of the T320 timer used in RRC connection release triggered by an operation other than load balancing.
New	NRCellReselConfig <i>.T320ForMlb</i>	LST NRCELLRESEL CONFIG MOD NRCELLRESEL CONFIG	gNodeB	This parameter has been added to specify the length of the T320 timer used in RRC connection release triggered by load balancing.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
The following impact relationship has been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Reselection policy for 3CC and massive CA UEs in idle mode	MASS_CA_IDLE_RESEL_POLICY_SW option of the NRCellOpQciBearer.ServiceDiffAlgoSwitch parameter	<i>CA Experience Improvement</i>	If both dedicated-priority-based idle mode MLB and the reselection policy for 3CC and massive CA UEs in idle mode are enabled, the T320 timer takes the value of NRCellReselConfig.T320ForOther for UEs in the 3CC aggregation state and massive CA UEs.
Low-frequency TDD	Dedicated-priority-based idle mode MLB	All of the following settings apply: • NRCellMLB.IdleMLBPolicy set to DEDICATED_PRIORITY • INTER_FREQ_IDLE_MLB_SW option of the NRCellAlgoSwitch.MlbAlgoSwitch parameter selected	<i>Mobility Load Balancing in Idle Mode</i>	

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *SA Mobility Management in Idle Mode* and *Mobility Load Balancing in Idle Mode*.

Requirement/Trouble Ticket Number

FeR-IR20250528001856-0032

RAT

NR FDD, NR TDD

5.3.4.1.27 Added Uplink Symbol Power Saving Optimization in Remote Interference Avoidance Scenarios

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

When remote interference adaptive avoidance (controlled by the **RIM_ADAPT_AVOID_SW** option of **NRDUCellAlgoSwitch.RimAlgoSwitch**) or enhanced remote interference adaptive avoidance (controlled by the **RIM_ADAPT_AVOID_ENH_SW** option of **NRDUCellAlgoSwitch.RimAlgoSwitch**) takes effect, the benefits of uplink symbol power saving decrease due to the reduced power saving duration. In this case, uplink symbol power saving optimization in remote interference avoidance scenarios can be enabled so that uplink symbol power saving can take effect in a higher proportion of cases, increasing energy saving gains. Uplink symbol power saving optimization in remote interference avoidance scenarios is controlled by the **RIM_UL_SYMBOL_SHUTDOWN_OPT_SW** option of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter.

This function is supported only in low frequency bands.

LampSite base stations do not support this function.

Implementation

For details, see *Energy Conservation and Emission Reduction*. The specific sections are listed in the change history of the document.

Impact on the Network

When remote interference adaptive avoidance (controlled by the **RIM_ADAPT_AVOID_SW** option of **NRDUCellAlgoSwitch.RimAlgoSwitch**) or enhanced remote interference adaptive avoidance (controlled by the **RIM_ADAPT_AVOID_ENH_SW** option of **NRDUCellAlgoSwitch.RimAlgoSwitch**) takes effect, enabling uplink symbol power saving optimization in remote interference avoidance scenarios increases energy saving gains because uplink symbol power saving can take effect in a higher proportion of cases.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch	MOD NRDUCELLPW RSVEALGO LST NRDUCELLPW RSVEALGO	gNodeB	The RIM_UL_SYMBOL_SHUTDOWN_OPTION_SWITCH option has been added to specify whether to enable uplink symbol power saving optimization in remote interference avoidance scenarios.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink symbol power saving	UL_SYMBOL_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Uplink symbol power saving optimization in remote interference avoidance scenarios requires that the UL_SYMBOL_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter be selected and the RIM_ADAPT_AVOID_SW or RIM_ADAPT_AVOID_ENH_SW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter be selected.
Low-frequency TDD	Remote interference adaptive avoidance	RIM_ADAPT_AVOID_SW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	
Low-frequency TDD	Enhanced remote interference adaptive avoidance	RIM_ADAPT_AVOID_ENH_SW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	

- Mutually exclusive features/functions

No impact

- Impacted features/functions

The descriptions of the following impact relationships have been revised.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Remote interference adaptive avoidance	RIM_ADAPT_AVOID_SW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	When remote interference adaptive avoidance is in effect, the benefits of uplink symbol power saving decrease due to the reduced power saving duration. In this scenario, the RIM_UL_SYMBOL_SHUTDOWN_OPT_SW option of the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter can be selected so that uplink symbol power saving can take effect in a higher proportion of cases, increasing energy saving gains.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Enhanced remote interference adaptive avoidance	RIM_ADAPT_AVOID_ENH_SWITCH.RimAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	When enhanced remote interference adaptive avoidance is in effect, the benefits of uplink symbol power saving decrease due to the reduced power saving duration. In this scenario, the RIM_UL_SYMBOL_SHUTDOWN_OPT_SW option of the NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch parameter can be selected so that uplink symbol power saving can take effect in a higher proportion of cases, increasing energy saving gains.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Energy Conservation and Emission Reduction*.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0001

RAT

NR TDD

5.3.4.1.28 Added Non-CQI-based SCell Deactivation for CA UEs Performing Downlink Large-packet Services

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

With SCell deactivation based on downlink CQI in effect (**NRDUCellCarrMgmt.SccDeactivCqiThld** set to a value other than **0**), the gNodeB delivers the MAC CE to deactivate an SCell whose spectral efficiency indicated by the CQI reported by a CA UE falls behind that configured using **NRDUCellCarrMgmt.SccDeactivCqiThld**.

An SCell with low spectral efficiency (namely poor channel quality) as indicated by its CQI, can still deliver certain throughput for a CA UE performing downlink large-packet services. In this case, if the SCell is deactivated based on channel quality, the throughput of the CA UE will be affected. As such, non-CQI-based SCell deactivation for CA UEs performing downlink large-packet services has been introduced. This function is controlled by the **NOT_DEACT_SCC_BASED_CQI_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoSwitch** parameter. If this option is selected, channel-quality-triggered SCell deactivation does not take effect for CA UEs performing downlink large-packet services, regardless of whether **NRDUCellCarrMgmt.SccDeactivCqiThld** is set to **0**. Non-CQI-based SCell deactivation for CA UEs performing downlink large-packet services is supported only in low frequency bands.

Implementation

For details, see *Carrier Aggregation*. The specific sections are listed in the change history of the document.

Impact on the Network

Non-CQI-based SCell deactivation for CA UEs performing downlink large-packet services increases the average throughput of SCell-activated downlink CA UEs, which is calculated using the following formula: $(N.CA.SccActiveUser.ThpVol.DL - N.CA.ThpVol.DL.LastSlot)/N.CA.ThpTime.DL.RmvLastSlot$.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellCarrierMgmt.CaEnhancedAlgorithmSwitch	MOD NRDUCELLC ARRMGMT LST NRDUCELLC ARRMGMT	gNodeB	The NOT_DEACT_SCC_BASED_CQI_SW option has been added to specify whether to enable non-CQI-based SCell deactivation for CA UEs performing downlink large-packet services.

- Performance management

No impact

- Fault management

No impact

Impact on Features/Functions

- Prerequisite features/functions

No impact

- Mutually exclusive features/functions

No impact

- Impacted features/functions

No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Carrier Aggregation*.

Requirement/Trouble Ticket Number

FeR-IR20250528001856-0040

RAT

NR FDD, NR TDD

5.3.4.1.29 Added a Parameter to Control Paging Signaling Shutdown in Common-Channel Dynamic Shutdown

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

For common-channel dynamic shutdown in TTI-level channel shutdown and dynamic RF channel shutdown, the **PAGING_DYN_SHUTDOWN_SW** option of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter has been added to specify whether shutdown takes effect for paging signaling.

- In versions earlier than V100R022C10SPC010, when **NRDUCellPowerSaving.ComChnDynShutdownSwitch** is set to **ON**, shutdown takes effect for RMSI, OSI, and paging signaling.
- In V100R022C10SPC010 and later versions, when **NRDUCellPowerSaving.ComChnDynShutdownSwitch** is set to **ON**, shutdown takes effect for RMSI and OSI signaling; to support shutdown of paging signaling, the **PAGING_DYN_SHUTDOWN_SW** option of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter must also be selected.

For an upgrade from a version earlier than V100R022C10SPC010 to V100R022C10SPC010 or later: If (1) **NRDUCellPowerSaving.PowerSavingType** is set to **TTI_RF_SHUTDOWN** or **RF_CHN_DYN_SHUTDOWN** and (2) any **NRDUCellPowerSaving.ComChnDynShutdownSwitch** in the corresponding energy saving policy is set to **ON** before the upgrade, the **PAGING_DYN_SHUTDOWN_SW** option of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter is selected after the upgrade; otherwise, the **PAGING_DYN_SHUTDOWN_SW** option of the **NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch** parameter is deselected after the upgrade.

This function is supported only in low frequency bands.

LampSite base stations do not support this function.

Implementation

For details, see *Energy Conservation and Emission Reduction*. The specific sections are listed in the change history of the document.

Impact on the Network

When common-channel dynamic shutdown takes effect for paging signaling, the base station energy saving gains increase, the coverage of the paging channel slightly deteriorates, and the paging success rate decreases.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellPwrSave Algo.PowerSaving AlgoSwitch	MOD NRDUCELLPW RSAVEALGO LST NRDUCELLPW RSAVEALGO	gNodeB	The PAGING_DYN_SHUTDOWN_SW option has been added to specify whether to enable common-channel dynamic shutdown for paging signaling.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Energy Conservation and Emission Reduction*.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0060

RAT

NR FDD, NR TDD

5.3.4.1.30 New Methods for Observing the QoS-specific Service Running Status

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The counters **N.User.RRCCConn.Active.DuCell5QI.PLMN.Max** (maximum number of 5QI-specific active UEs of a specific operator in a DU cell) and **N.User.RRCCConn.Active.DuCell5QI.PLMN.Avg** (average number of 5QI-specific active UEs of a specific operator in a DU cell) are added for observing the QoS-specific service running status.

Implementation

The counters **N.User.RRCCConn.Active.DuCell5QI.PLMN.Max** (maximum number of 5QI-specific active UEs of a specific operator in a DU cell) and **N.User.RRCCConn.Active.DuCell5QI.PLMN.Avg** (average number of 5QI-specific active UEs of a specific operator in a DU cell) are added for observing the QoS-specific service running status.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact

- Configuration management
No impact
- Performance management
The following counters have changed.

Change Type	Counter Name	Function Subset	NE	Description
New	N.User.RRC Conn.Active .DuCell5QI. PLMN.Max	DU Cell Operator 5QI Traffic Measurement	gNode B	This counter has been added to measure the maximum number of 5QI-specific active UEs of a specific operator in a DU cell.
New	N.User.RRC Conn.Active .DuCell5QI. PLMN.Avg	DU Cell Operator 5QI Traffic Measurement	gNode B	This counter has been added to measure the average number of 5QI-specific active UEs of a specific operator in a DU cell.

- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250911002729-0003

RAT

NR FDD, NR TDD

5.3.4.1.31 Added Support for Inter-TRP F4 HARQ Resource Sharing

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In a multi-TRP cell, UEs that transmit HARQ feedback exceeding two bits can use more format-4 resources for HARQ feedback, which increases the average downlink throughput of these UEs.

This function is controlled by the **INTER_TRP_F4_HARQ_SHARE_SW** option of the **NRDUCellPucch.PucchAlgoExtSwitch** parameter.

The upgrade rules for inter-TRP format-4 HARQ resource sharing are as follows:

- After an upgrade from V100R021C10SPC160 or an earlier version, or from a version between V100R022C00SPC010 (included) and V100R022C00SPC050 (excluded) to V100R022C10SPC010 or later, inter-TRP F4 HARQ resource sharing is disabled.
- For an upgrade from a version between V100R021C10SPC160 (excluded) and V100R022C00SPC010 (included) or from V100R022C00SPC050 or a later version to V100R022C10SPC010 or later, if the **RSVDSWPARAM1_BIT17** option of the **NRDUCellRsvdExt01.RsvdSwParam1** parameter is set to **1** before the upgrade, inter-TRP F4 HARQ resource sharing will be enabled after the upgrade. Otherwise, inter-TRP F4 HARQ resource sharing will be disabled after the upgrade.

Implementation

For details, see *Cell Combination* and *Hyper Cell*. The specific sections are listed in the change history of the documents.

Network Impacts

After a combined cell or a hyper cell is set up, if inter-TRP F4 HARQ resource sharing is enabled, the cell can support more UEs with HARQ feedback of more than two bits. The proportion of PUCCH allocation failures decreases, downlink throughput increases, but the probability of inter-TRP F4 HARQ resource conflicts may increase, increasing the downlink bit error rate.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellPucch.PucchAlgoExtSwitch	MOD NRDUCCELLPUCCH LST NRDUCCELLPUCCH	gNodeB	The INTER_TRP_F4_HARQ_SHARE_SW option has been added to specify whether to enable inter-TRP F4 HARQ resource sharing.
Modified	NRDUCellRsvdExt01.RsvdSwParam1	MOD NRDUCCELLR SVDEXT01 LST NRDUCCELLR SVDEXT01	gNodeB	RSVDSWPARAM1_BIT17 of the reserved parameter NRDUCellRsvdExt01.RsvdSwParam1 has been added to the parameter disuse list and replaced with the INTER_TRP_F4_HARQ_SHARE_SW option of the NRDUCellPucch.PucchAlgoExtSwitch parameter.

NOTE

NRDUCellRsvdExt01.RsvdSwParam1 is a reserved parameter.

For details about reserved parameters, see 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
The following impact relationship has been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Format-3 and Format-4 ACK RB Resource Adaptation	F3_AND_F4_ACK_RB_RESOURCE_ADAPT_SWITCH option of the NRDUCellPucchAlgorithm parameter	<i>Channel Management</i>	When inter-TRP format-4 resource sharing for HARQ feedback is enabled together with format-3 and format-4 ACK RB resource adaptation, only format-3 and format-4 ACK RB resource adaptation takes effect.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Cell Combination* and *Hyper Cell*.

Requirement/Trouble Ticket Number

FeR-IR20250528001856-0001

RAT

NR FDD, NR TDD

5.3.4.1.32 Added UE-Number-based SCell Configuration Control

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

CA UEs count towards the RRC_CONNECTED UE number limits for both the PCell and SCells. In scenarios where these limits are likely to be reached (for example, during major sports events), non-CA UEs may struggle to connect to the network. As such, UE-number-based SCell configuration control is introduced in low frequency bands. This function is controlled by the **UE_NUM_BASED_SCELL_CONFIG_SW** option of the **gNodeBCarrMgmt.CaEnhancedAlgoSwitch** parameter.

This function includes two subfunctions: UE-number-based 2CC aggregation prohibition and UE-number-based 3CC aggregation prohibition. After the **UE_NUM_BASED_SCELL_CONFIG_SW** option of the

gNodeBCarrMgmt.CaEnhancedAlgoSwitch parameter is selected, the related UE proportion thresholds must be configured for the subfunctions to take effect. UE-number-based SCell configuration control works when either of the subfunctions takes effect.

- UE-number-based 2CC aggregation prohibition: takes effect when the **NRCellCaMgmtConfig.Forbid2ccUeRatioThld** parameter is set to a value other than **100**. After this function takes effect in a cell, the gNodeB periodically calculates the proportion of UEs in the cell. If this proportion is greater than the value of this parameter, CA with two or more CCs aggregated is disallowed for UEs that treat this cell as their PCell. Also, this cell is prohibited from being configured as an SCell for UEs in other cells.
- UE-number-based 3CC aggregation prohibition: takes effect when the **NRCellCaMgmtConfig.Forbid3ccUeRatioThld** parameter is set to a value other than **100**. After this function takes effect in a cell, the gNodeB periodically calculates the proportion of UEs in the cell. If this proportion is greater than the value of this parameter, CA with three or more CCs aggregated is disallowed for UEs that treat this cell as their PCell.

Note the following: Proportion of UEs in a cell = (Number of UEs in a cell / Maximum number of RRC_CONNECTED UEs in a cell) x 100%, where the number of UEs in a cell includes the number of RRC_CONNECTED UEs in a cell and the number of UEs that treat the local cell as their SCell (including SUL UEs). For details about the maximum number of RRC_CONNECTED UEs in a cell, see the UE number specifications of the corresponding RAT in "NR Technical Specifications of Baseband Processing Units" in "BBU Technical Specifications" in *3900 & 5900 Series Base Station Product Documentation*.

NOTE

When the value of **NRCellCaMgmtConfig.Forbid2ccUeRatioThld** or **NRCellCaMgmtConfig.Forbid3ccUeRatioThld** is changed (for example, from **100** to a value other than **100** to enable UE-number-based SCell configuration control), the latest setting does not take effect immediately for CA UEs in the RRC_CONNECTED state. Instead, the latest setting takes effect only when an SCell addition or change procedure is triggered.

This change is supported only in low frequency bands.

Implementation

For details, see *Carrier Aggregation*. The specific sections are listed in the change history of the document.

Impact on the Network

After the UE-number-based SCell configuration control function takes effect:

- Fewer RRC_CONNECTED UEs use CA in scenarios where the number of RRC_CONNECTED UEs may reach the upper limit (for example, during major sports events). This ensures that more UEs can access the network.
- The proportion of UEs for which CA takes effect and the network-wide UE throughput (including the uplink and downlink throughputs) change, depending on the configured thresholds expressed as the proportion of UEs for prohibiting 2CC or 3CC aggregation. Higher thresholds indicate higher proportions of CA UEs and higher network-wide UE throughput. Lower

thresholds indicate lower proportions of CA UEs and lower network-wide UE throughput. In addition, after the **UE_NUM_BASED_SCELL_CONFIG_SW** option of the **gNodeBCarrMgmt.CaEnhancedAlgoSwitch** parameter is selected, regardless of whether UE-number-based 2CC aggregation prohibition and UE-number-based 3CC aggregation prohibition take effect, more cell load information will be exchanged over the eXn interface, leading to increased length of a single message and possible more message fragments in some scenarios. As a result, the values of counters related to receive and transmit traffic in the SCTP Link Measurement subset may increase.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRCeIlCaMg mtConfig.Fo rbid2ccUeRatioThld	MOD NRCeIlCaM GMTCONFIG LST NRCeIlCaM GMTCONFIG	gNodeB	This parameter has been added to specify the threshold expressed as the proportion of UEs for prohibiting CA with two CCs aggregated from taking effect.
New	NRCeIlCaMg mtConfig.Fo rbid3ccUeRatioThld	MOD NRCeIlCaM GMTCONFIG LST NRCeIlCaM GMTCONFIG	gNodeB	This parameter has been added to specify the threshold expressed as the proportion of UEs for prohibiting CA with three CCs aggregated from taking effect.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNodeBCarrMgmt.CaEnhancedAlgoSwitch	MOD GNODEBCAR RMGMT LST GNODEBCAR RMGMT	gNodeB	The UE_NUM_BASED_SCELL_CONFIG_SW option has been added to specify whether to enable UE-number-based SCell configuration control.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Carrier Aggregation*.

Requirement/Trouble Ticket Number

FeR-IR20250211000074-0001

RAT

NR FDD, NR TDD

5.3.4.1.33 Added C-RNTI Conflict Detection and Conflict-triggered Reconfiguration

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

A cell radio network temporary identifier (C-RNTI) is a temporary ID of a UE in connected mode in a cell. The gNodeB uses different C-RNTIs for the UEs in uplink and downlink scheduling. The C-RNTIs of UEs in a cell must be different to avoid inter-UE interference.

Without C-RNTI conflict detection, the C-RNTI of a UE in an SCell to be configured may have been used by another UE in this SCell. In this case, a C-RNTI conflict occurs, which intensifies inter-UE interference and affects UE performance. As such, the C-RNTI conflict detection function is introduced. This function is controlled by the **INTRA_GNB_CRNTI_CON_DETECT_SW** option of the **gNodeBCarrMgmt.CaEnhancedAlgoSwitch** parameter. Note that the preceding parameter-controlled C-RNTI conflict detection function is an intra-gNodeB function, because inter-gNodeB C-RNTI conflict detection takes effect by default and does not need to be configured.

NOTE

Inter-gNodeB C-RNTI conflict detection has been supported before this change.

After C-RNTI conflict detection takes effect, the gNodeB checks whether the C-RNTI of a UE is unique within an SCell to be configured.

- If so, the SCell is configured.
- If not, the gNodeB does not configure the SCell.

When intra- or inter-gNodeB C-RNTI conflict detection takes effect and the gNodeB determines that the C-RNTI of a UE in an SCell to be configured has been used by another UE within the SCell, the C-RNTI conflict-triggered reconfiguration function is introduced to ensure normal SCell configuration for the UE and guarantee UE performance. This function is controlled by the **CRNTI_CONFLICT_RECONFIG_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoSwitch** parameter. With this function, the gNodeB re-allocates C-RNTIs to UEs for which C-RNTI conflicts are detected, and attempts to reconfigure the SCell. This reduces the risk of SCell configuration failures caused by C-RNTI conflicts.

The preceding option controls both intra-gNodeB and inter-gNodeB C-RNTI conflict-triggered reconfiguration. In intra-gNodeB scenarios, C-RNTI conflict-triggered reconfiguration takes effect only when C-RNTI conflict detection is enabled.

C-RNTI conflict detection and conflict-triggered reconfiguration take effect only in measurement-based SCell configuration, but not blind SCell configuration.

NOTE

SCell change also involves SCell configuration. Therefore, C-RNTI conflict detection and conflict-triggered reconfiguration are also triggered during SCell change.

Consider an inter-FR CA scenario where (1) two low-frequency 2CCs and at least one high-frequency SCC have been configured; (2) During low-frequency SCC change, a C-RNTI conflict is detected in the SCC to be configured. The gNodeB will delete all configured high-frequency SCCs during C-RNTI reconfiguration. The gNodeB will subsequently reconfigure high-frequency SCCs based on the high-frequency SCC configuration procedure.

Implementation

For details, see *Carrier Aggregation*. The specific sections are listed in the change history of the document.

Impact on the Network

- C-RNTI conflict detection prevents interference caused by C-RNTI conflicts, thereby increasing the average throughput of SCell-activated CA UEs in scenarios (such as heavy-load scenarios) where C-RNTI conflicts are likely to occur.
 - Average throughput of SCell-activated downlink CA UEs =
$$\frac{(N.CA.SccActiveUser.ThpVol.DL - N.CA.ThpVol.DL.LastSlot)}{N.CA.ThpTime.DL.RmvLastSlot}$$
 - Average throughput of SCell-activated uplink CA UEs =
$$\frac{(N.CA.SccActiveUser.ThpVol.UL - N.CA.ThpVol.UL.SmallPkt)}{N.CA.ThpTime.UL.RmvSmallPkt}$$
- C-RNTI conflict-triggered reconfiguration reduces the risk of SCell configuration failures due to C-RNTI conflicts, thereby increasing the average throughput of SCell-activated CA UEs in scenarios (such as heavy-load scenarios) where C-RNTI conflicts are likely to occur.
 - Average throughput of SCell-activated downlink CA UEs =
$$\frac{(N.CA.SccActiveUser.ThpVol.DL - N.CA.ThpVol.DL.LastSlot)}{N.CA.ThpTime.DL.RmvLastSlot}$$
 - Average throughput of SCell-activated uplink CA UEs =
$$\frac{(N.CA.SccActiveUser.ThpVol.UL - N.CA.ThpVol.UL.SmallPkt)}{N.CA.ThpTime.UL.RmvSmallPkt}$$
- C-RNTI conflict-triggered reconfiguration: The RRC reconfiguration times and service drop rate vary, depending on the number of CA capability combinations selected during SCell configuration. Selecting more CA capability combinations increases the likelihood of fewer RRC reconfiguration times and a lower service drop rate. Selecting fewer CA capability combinations increases the likelihood of more RRC reconfiguration times and a higher service drop rate. In addition, this function increases the value of the following counters:
 - **N.CA.SCell.Add.Att**
 - **N.CA.SCell.Add.Succ**
 - **N.CA.SCell.Mod.Att**
 - **N.CA.SCell.Mod.Succ**

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNodeBCarrMgmt.CaEnhancedAlgorithmSwitch	MOD GNODEBCAR RMGMT LST GNODEBCAR RMGMT	gNodeB	The INTRA_GNB_CRNTI_CON_DETECT_SW option has been added to specify whether to enable intra-gNodeB C-RNTI conflict detection.
Modified	NRDUCellCarrMgmt.CaEnhancedAlgorithmSwitch	MOD NRDUCELLC ARRMGMT LST NRDUCELLC ARRMGMT	gNodeB	The CRNTI_CONFLICT_RECONFIG_SW option has been added to specify whether to enable C-RNTI conflict-triggered reconfiguration.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Carrier Aggregation*.

Requirement/Trouble Ticket Number

FeR-IR20250211000074-0003

RAT

NR FDD, NR TDD

5.3.4.1.34 Added Time-Domain Power Saving BWP

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The time-domain power saving BWP function is introduced. With this function, a base station can adaptively switch the UE to a BWP with a low PDCCH monitoring density based on the UE traffic volume to save UE power.

Implementation

For details, see *UE Power Saving*. The specific sections are listed in the change history of the document.

Impact on the Network

Benefits:

This function reduces the power consumption of UEs. Because different UEs have varying chips and service types, it is recommended that UE vendors observe and evaluate the power consumption gains. This function is most effective when UEs are performing services with a small amount of data. In other cases, the UE power saving effect is not as pronounced.

Network impacts:

- The average uplink and downlink cell throughputs and the average uplink and downlink UE throughputs decrease.
- Uplink and downlink cell edge throughputs increase or decrease.
- The uplink and downlink IBLERs and RBLERs increase or decrease.
- The delay increases.
- The CCE usage decreases.
- The uplink and downlink CCE allocation success rates decrease.
- The QoS flow drop rate of all services increases.
- The values of the counters related to the uplink and downlink RB usage increase or decrease.
- The RRC connection setup success rate fluctuates.
- The average uplink and downlink ranks may increase or decrease.
- The total number of CQI reporting times increases, and the measured CQI value increases or decreases.

Impact on NEs

This change is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellUePwrSaving.BwpPwrSavingSw	MOD NRDUCELLUEPWR SAVING LST NRDUCELLUEPWR SAVING	gNodeB	The TDM_BWP2_SWITCH option has been added to specify whether to enable time-domain power saving BWP.
New	NRDUCellUePwrSaving.Bwp1ToTdmBwp2DlThptThld	MOD NRDUCELLUEPWR SAVING LST NRDUCELLUEPWR SAVING	gNodeB	This parameter has been added to specify the downlink RLC throughput threshold for UEs to switch from BWP1 to TDM BWP2.
New	NRDUCellUePwrSaving.Bwp1ToTdmBwp2UlThptThld	MOD NRDUCELLUEPWR SAVING LST NRDUCELLUEPWR SAVING	gNodeB	This parameter has been added to specify the uplink RLC throughput threshold for UEs to switch from BWP1 to TDM BWP2.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellUePwrSaving.TdmBwp2SwitchFailRateThld	MOD NRDUCELLUEPWR SAVING LST NRDUCELLUEPWR SAVING	gNodeB	This parameter has been added to specify the threshold expressed as the UE scheduling failure probability in time-domain power saving BWP2 in the cell.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions

Function Name	Function Switch	Reference	Description
Power saving BWP	UE_PWR_SAVING_ENHANCE_SW option of the NRDUCellUePwrSaving.UePowerSavingSwitch parameter and the BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	When the TDM_BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter is selected, the BWP2_SWITCH option must be deselected.

Function Name	Function Switch	Reference	Description
Uplink DCI extension-based scheduling	DCI_EXTEND_UL_CONGEST_SW option of the NRDUCellPusch.UlPuschAlgoSwitch parameter	<i>Uplink Scheduling</i>	If the TDM_BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter is selected, the DCI_EXTEND_UL_CONGEST_SW option of the NRDUCellPusch.UlPuschAlgoSwitch parameter must be deselected.

- Impacted features/functions
Impact relationships with functions related to scheduling:

Function Name	Function Switch	Reference	Description
Rate-matching-pattern-configuration-based PDCCH rate matching	PDCCH_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	When both rate-matching-pattern-configuration-based PDCCH rate matching and time-domain power saving BWP are enabled, newly admitted UEs are not allowed to enter BWP2 as long as rate-matching-pattern-configuration-based PDCCH rate matching is in effect. The air interface resources of the UEs that have already switched to BWP2 are deleted through RRC reconfiguration messages, and such UEs are switched back to BWP1 from BWP2.

Impact relationships with functions related to carrier management

Function Name	Function Switch	Reference	Description
Single DCI	DL_SINGLE_DCI_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	When both time-domain power saving BWP and single DCI are enabled on the PCC, whether DCI format 1_3 can be configured on BWP2 of the PCC depends on whether the "single DCI configuration on BWP2" function (controlled by the BWP2_CONFIG_SINGLE_DCI_SW option of the NRDUCellCarrMgmt.CaCompatibilitySwitch parameter) is enabled. • If single DCI configuration on BWP2 is enabled, DCI format 1_3 can be configured on the BWP2 of the PCC. • If single DCI configuration on BWP2 is disabled, DCI format 1_3 cannot be configured on the BWP2 of the PCC. If UEs have compatibility issues when both single DCI and time-domain power saving BWP take effect, it is recommended that "single DCI

Function Name	Function Switch	Reference	Description
			configuration on BWP2" be enabled to reduce the downlink IBLER and RBLER of these UEs.

Function Name	Function Switch	Reference	Description
Downlink intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAlgoSwitch . CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	BWP2 is not configured for CA UEs in their SCells.
Downlink intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAlgoSwitch . CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	When CA and time-domain power saving BWP are both enabled, there are the following impacts: • A larger value of NRDUCellCarrierMgmt.CaDisCellActThldSlotNum results in a higher traffic volume threshold for UE switching from BWP2 to BWP1, a lower probability of such switching, and lower UE power consumption. • A larger value of NRDUCellCarrierMgmt.CaDisCellDeactThldSlotNum results in a higher traffic volume threshold for UE switching from BWP1 to BWP2, a higher probability of such switching, and lower UE power consumption. • A larger value of NRDUCellCarrierMgmt.DlStateTransitBufVolT

Function Name	Function Switch	Reference	Description
			<i>hld</i> results in a lower probability of UE switching from BWP2 to BWP1 and a higher probability of UE switching from BWP1 to BWP2. As a result, UEs stay in BWP2 for a longer time and their power consumption decreases. • A smaller value of NRDUCellCarr Mgmt.DlStateTransitBufVolT <i>hld</i> results in a higher probability of UE switching from BWP2 to BWP1 and a lower probability of UE switching from BWP1 to BWP2. As a result, UEs stay in BWP2 for a shorter time and their power consumption increases. BWP switching and RRC reconfiguration may occur more frequently. For cell-edge UEs under weak coverage, the service drop

Function Name	Function Switch	Reference	Description
			rate may increase. • SCells are more likely to be deactivated if the NRDUCellCarrMgmt.RlcDlThroughputThld parameter value is less than the NRDUCellUePwrSaving.Bwp1ToBwp2DlThptThld parameter value. If NRDUCellCarrMgmt.RlcDlThroughputThld is set to 0, SCells will not be deactivated and UEs will not switch to BWP2.

Function Name	Function Switch	Reference	Description
Uplink intra-band CA (TDD)	INTRA_BAND_UL_CA_SW option of the NRDUCellAlgoSwitch parameter	<i>Carrier Aggregation</i>	BWP2 is not configured for CA UEs in their SCells.
Uplink intra-FR inter-band CA (low-frequency TDD)	INTRA_FR_INTER_BAND_UL_CA_SW option of the NRDUCellAlgoSwitch parameter	<i>Carrier Aggregation</i>	When CA and time-domain power saving BWP are both enabled, there are the following impacts: • A larger value of NRDUCellCarrierMgmt.CaUISCellActThldSlotNum results in a higher traffic volume threshold for UE switching from BWP2 to BWP1, a lower probability of such switching, and lower UE power consumption. • A larger value of NRDUCellCarrierMgmt.CaUISCellDeactThldSlotNum results in a higher traffic volume threshold for UE switching from BWP1 to BWP2, a higher probability of such switching, and lower UE power consumption. • SCells are more likely to be deactivated if the NRDUCellCarrier

Function Name	Function Switch	Reference	Description
			Mgmt.RlcUlThroughtputThld parameter value is less than the NRDUCellUePwrSaving.Bwp1ToBwp2UlThptThld parameter value. If NRDUCellCarrMgmt.RlcUlThroughtputThld is set to 0 , SCells will not be deactivated and UEs will not switch to BWP2.
Inter-FR CA	INTER_FR_FR1TOFR2_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	Time-domain power saving BWP does not take effect on inter-FR CA UEs.

Impact relationships with MIMO-related functions

Function Name	Function Switch	Reference	Description
Fusion Cell	NRDUCell.NrDuCellNetworkingMode set to FUSION_CELL	<i>Fusion Cell (Low-Frequency TDD)</i>	When there is a large proportion of BWP2 UEs, the gains on the downlink user-perceived rate brought by Fusion Cell are affected.

Function Name	Function Switch	Reference	Description
PDSCH MU-MIMO	DL_MU_MIMO_SW option of the NRDUCellAlgoSwitch h.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	Downlink MU-MIMO is not performed on UEs that work on BWP2 for power saving. If time-domain power saving BWP is enabled after MIMO, the following may occur:⁽¹⁾ • The number of MIMO layers, BLER, and average number of downlink RBs for MU-MIMO UE pairing in the downlink fluctuate. • The number of MIMO layers, BLER, and average number of RBs for MU-MIMO UE pairing in the uplink fluctuate. • The numbers of uplink and downlink scheduling times decrease. • The average cell throughput and average UE throughput fluctuate.
PUSCH MU-MIMO	UL_MU_MIMO_SW option of the NRDUCellAlgoSwitch h.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	
Downlink SU-MIMO	DL_SU_MULTI_LAYER_SW option of the NRDUCellAlgoSwitch h.SuMimoMultipleLayerSw parameter	<i>MIMO (TDD)</i>	
Uplink SU-MIMO	UL_SU_MULTI_LAYER_SW option of the NRDUCellAlgoSwitch h.SuMimoMultipleLayerSw parameter	<i>MIMO (TDD)</i>	

Function Name	Function Switch	Reference	Description
			(1) For details about the formulas for calculating the uplink and downlink BLERs, average cell throughput, and average UE throughput, see "Impacts" of the time-domain power saving BWP function in <i>UE Power Saving</i>. The following are indicators related to the number of layers, number of scheduling times, and average number of RBs for MU pairing: Average number of downlink MU-MIMO layers per PRB in a cell = N.ChMeas.MIMO.DL.Pair.Layer/N.ChMeas.MIMO.DL.Pair.PRB Average number of downlink layers per PRB in a cell = N.ChMeas.MIMO.DL.Transmission.Layer/[(N.PRB.DL.DrbUsed.Avg + N.PRB.DL.SrbUsed.Avg) × N.DL.PDSCH.Tti.Num] Maximum number of downlink layers per PRB in a cell = N.ChMeas.MIMO.DL.Transmission.Layer.Max Average number of RBs for downlink MU pairing = (1 × N.ChMeas.MIMO.DL.MuPairing.1Layer.RB + 2 × N.ChMeas.MIMO.DL.MuPairing.2Layers.RB + ... + 16 × N.ChMeas.MIMO.DL.MuPairing.16Layers.RB)/ (N.ChMeas.MIMO.DL.MuPairing.1Layer.RB + N.ChMeas.MIMO.DL.MuPairing.2Layers.RB + ... + N.ChMeas.MIMO.DL.MuPairing.16Layers.RB) Average number of uplink MU-MIMO layers per PRB in a cell = N.ChMeas.MIMO.UL.Pair.Layer/N.ChMeas.MIMO.UL.Pair.PRB Average number of uplink layers per PRB in a cell = N.ChMeas.MIMO.UL.Trans.Layer/(N.PRB.PUSCH.Used.Avg × N.UL.PUSCH.Tti.Num) Average number of RBs for uplink MU pairing = (1 × N.ChMeas.MIMO.UL.MuPairing.1Layer.RB + 2 × N.ChMeas.MIMO.UL.MuPairing.2Layers.RB + ... + 8 × N.ChMeas.MIMO.UL.MuPairing.8Layers.RB)/ (N.ChMeas.MIMO.UL.MuPairing.1Layer.RB + N.ChMeas.MIMO.UL.MuPairing.2Layers.RB + ... + N.ChMeas.MIMO.UL.MuPairing.8Layers.RB) Number of uplink scheduling times = N.User.Schedule.UL.Sum Number of downlink scheduling times = N.User.Schedule.DL.Sum

Impact relationships with UCN-related functions

Function Name	Function Switch	Reference	Description
Cell Combination	NRDUCell.NrDuCell NetworkingMode set to HYPER_CELL_COMBI NE_MODE , and NRDUCellMultiTrp. DataTransMode set to BASIC_MODE or DPS_MODE	<i>Cell Combination</i>	With time-domain power saving BWP enabled, UEs are more likely to use BWP2 after the value of the NRDUCellMultiTrp. DataTransMode parameter is changed from BASIC_MODE to DPS_MODE for Cell Combination. Related performance indicators on the base station side deteriorate. For details, see "Impacts" of the time-domain power saving BWP function in <i>UE Power Saving..</i>

Function Name	Function Switch	Reference	Description
Co-MM	INTRA_GNB_SU_CO_MM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	If the COOR_UE_PWR_SAV E_PRI_ADJ_SW option of the NRDUCellCollabServ.CommAlgoSwitch parameter is selected, the downlink throughput of Co-MM UEs increases. However, the proportion of time in which UEs work on BWP2 may decrease, the value of the N.Bwp2.Dur.Total counter decreases, and the UE power saving effect may deteriorate. The SRS SINR measurement value may increase (indicated by a decrease in the proportion of SINR measurement values that fall into small-index ranges).
Intra-base-station UL CoMP	INTRA_GNB_UL_COMMP_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	UEs with CoMP in effect will not be switched to BWP2. When a UE is working on BWP2 and the conditions for CoMP to take effect are met, the UE is first switched to BWP1 and then CoMP takes effect.

Function Name	Function Switch	Reference	Description
Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	UEs with CoMP in effect will not be switched to BWP2. When a UE is working on BWP2 and the conditions for CoMP to take effect are met, the UE is first switched to BWP1 and then CoMP takes effect.

Impact relationships with functions related to radio services

Function Name	Function Switch	Reference	Description
RedCap Introduction	REDCAP_SW option of the NRDUCellFeatureSwitch.UeCapBasedFunction parameter	<i>RedCap</i>	- Time-domain power saving BWP does not take effect on RedCap UEs. - When time-domain power saving BWP takes effect, the scheduling of RedCap UEs on BWP1 is affected, and RedCap user experience fluctuates.

Function Name	Function Switch	Reference	Description
Downlink resource isolation	DL_EMBB_PREFERENTIAL_SCH_SW option of the NRDUCellAlgoSwitch.FwaAlgoSwitch parameter	<i>FWA</i>	After the downlink resource isolation function is enabled, if the time-domain power saving BWP function takes effect on FWA UEs, the downlink experience of FWA UEs deteriorates more severely. In this case, it is good practice to select the FWA_BWP2_DISABLE_SW option of the NRDUCellAlgoSwitch.FwaAlgoSwitch parameter so that BWP2 will not be configured for FWA UEs.
Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliability-Switch parameter	<i>URLLC</i>	When "low latency and high reliability" and time-domain power saving BWP are both enabled, the UEs using the "low latency and high reliability" function will not be switched to BWP2.
3GPP R15-compliant uplink slot aggregation	VONR_UL_SLOT_AGGREGATION_SW option of the NRDUCellUISch.VoiceUISchSwitch parameter	<i>VoNR</i>	VoNR UEs with uplink slot aggregation in effect will not be switched to BWP2.

Function Name	Function Switch	Reference	Description
Optimization of collaboration between VoNR and UE power saving	VONR_UE_PWR_SAVING_COOP_OPT_SW option of the NRDUCellAlgoSwitch.ServiceDiffSwitch parameter	<i>VoNR</i>	When "optimization of collaboration between VoNR and UE power saving" is enabled, the duration in which BWP2 is used by UEs reduces.
SRS field strength-based positioning	SRS_RSRP_LOCATION_SW option of the NRCellAlgoSwitch.LcsSwitch parameter	<i>Positioning</i>	Time-domain power saving BWP can take effect for UEs with SRS field strength-based positioning in effect if the UEs can report srs-PosResources-r15 .
Live streaming UE identification	gNBQciBearer.ServiceType set to EMBB_LIVE_VIDEO	<i>Ultimate Video Experience</i>	Live streaming UEs will not be switched to BWP2.
Cloud X UE identification	gNBQciBearer.ServiceType set to EMBB_CLOUD_VR or EMBB_CLOUD_GAMING	<i>Ultimate Cloud X Experience</i>	UEs with Cloud X services in effect will not be switched to BWP2.

Impact relationships with functions related to power saving

Function Name	Function Switch	Reference	Description
SSSG switching	UE_PWR_SAVING_ENHANCE_SW and SSSG_SWITCHING_SW options of the NRDUCellUePwrSaving.UePowerSavingSwitch parameter	<i>UE Power Saving</i>	If the SSSG_SWITCHING_SW option of the NRDUCellUePwrSaving.UePowerSavingSwitch is selected for an SSSG-switching-capable UE, only SSSG takes effect for the UE. The time-domain power saving BWP function takes effect for UEs only when the SSSG_SWITCHING_SW option of the NRDUCellUePwrSaving.UePowerSavingSwitch is deselected and the TDM_BWP2_SWITCH option of the NRDUCellUePwrSaving.UePowerSavingSwitch parameter is selected.

Function Name	Function Switch	Reference	Description
DRX	BASIC_DRX_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgorithmSwitch parameter	<i>DRX</i>	When both time-domain power saving BWP and DRX are enabled: - If the duration configured using the gNBDUDrxParamGroup.DrxInactivityTimer parameter is less than 20 ms, UEs may remain working on BWP2 and cannot be switched to BWP1. - If the BWP2_APER_CSI_UE_DRX_ENH_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgorithmSwitch parameter is selected, UEs in BWP2 do not aperiodically report CSI-RS for CM measurement results even though configured to do so when there is no uplink data transmission in non-On Duration periods. This way, these

Function Name	Function Switch	Reference	Description
			UEs can enter the DRX sleep time and the number of CSI-RS for CM measurement reports decreases, affecting network performance. For details about the impact analysis, see <i>DRX</i>. • If the base station triggers a BWP2-to-BWP1 switching during the DRX sleep time, the corresponding DCI may fail to be delivered. As a result, the base station stops downlink scheduling for the UE during the BWP switching, leading to downlink scheduling failures. After the DRX_BWP2_OPTION_SW option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter

Function Name	Function Switch	Reference	Description
			is selected, the base station can trigger UE switching from BWP2 to BWP1 only when the DRX inactivity timer is running.
PDCCH skipping	PDCCH_SKIP_SW option of the NRDUCellUePwrSaving.SchPwrSavingSw parameter	<i>UE Power Saving</i>	Time-domain power saving BWP does not take effect on UEs with PDCCH skipping in effect.

Function Name	Function Switch	Reference	Description
Data convergence in slots in power saving scheduling	POWER_SAVING_SCHEDULE_SW option of the NRDUCellAlgorithm.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	A larger value of the NRDUCellPowerSaving.RlcFirstPktDelayThld parameter results in a higher degree of data convergence with a longer time. As a result, the number of scheduling failures of UEs on BWP2 in a cell increases. When the scheduling failure rate of BWP2 UEs in a cell exceeds the value of the NRDUCellUePwrSaving.Bwp2SchFailRateThld parameter, UEs are more likely to switch from BWP2 back to BWP1.

Other impacts:

Function Name	Function Switch	Reference	Description
Super uplink	SUPER_UL_TDM_SCH_SW, FLEX_SUPER_UPLINK_SW, or INTER_GNB_SUPER_UPLINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter	<i>Super Uplink</i>	Only UEs on the NUL carrier can be switched to BWP2. UEs on the SUL carrier cannot be switched to BWP2. The scheduling delay for UEs on the NUL carrier increases when the SUL carrier is activated.
UL and DL decoupling	NRDUCellAlgoSwitch.UlDlDecouplingSwitch	<i>UL and DL Decoupling</i>	Only UEs on the NUL carrier can be switched to BWP2. UEs on the SUL carrier cannot be switched to BWP2. The scheduling delay for UEs on the NUL carrier increases when the SUL carrier is activated.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *UE Power Saving*.

Requirement/Trouble Ticket Number

FeR-IR20241217003452-0003

RAT

NR TDD

5.3.4.1.35 Added BWP2 PUCCH Frequency-Domain Position Adjustment

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Added the BWP2 PUCCH frequency-domain position adjustment function. This function adjusts the PUCCH frequency-domain position of BWP2 to ensure contiguous PUSCH in the frequency domain, thereby increasing the number of available PUSCH resources.

Implementation

For details, see *UE Power Saving*. The specific sections are listed in the change history of the document.

Impact on the Network

- Benefits:
BWP2 PUCCH frequency-domain position adjustment increases the average uplink UE throughput (**User Uplink Average Throughput (DU)**).
- Network impacts:
 - The number of available PUSCH RBs may increase, and the interference may change. As a result, the values of the following indicators may fluctuate:
 - Average uplink UE throughput: **User Uplink Average Throughput (DU)**
 - Average uplink cell throughput: **Cell Uplink Average Throughput (DU)**
 - Average uplink RB usage in a cell: **Uplink Resource Block Utilizing Rate (DU)**
 - Total number of PRBs that are paired for uplink MIMO: **N.ChMeas.MIMO.UL.Pair.PRB**
 - Due to RB adjustment, the PUCCH interference differs, and therefore the PUCCH SINR fluctuates. The values of the following indicators may fluctuate:
 - Average MCS indexes for uplink and downlink scheduling
Average MCS index for uplink scheduling = $(0 \times \text{N.ChMeas.PUSCH.MCS.0} + 1 \times \text{N.ChMeas.PUSCH.MCS.1} + 2 \times \text{N.ChMeas.PUSCH.MCS.2} + 3 \times \text{N.ChMeas.PUSCH.MCS.3} + \dots + 26 \times \text{N.ChMeas.PUSCH.MCS.26} + 27 \times \text{N.ChMeas.PUSCH.MCS.27} + 28 \times \text{N.ChMeas.PUSCH.MCS.28}) / (\text{N.ChMeas.PUSCH.MCS.0} + \text{N.ChMeas.PUSCH.MCS.1} + \text{N.ChMeas.PUSCH.MCS.2} + \text{N.ChMeas.PUSCH.MCS.3} + \dots + \text{N.ChMeas.PUSCH.MCS.26} + \text{N.ChMeas.PUSCH.MCS.27} + \text{N.ChMeas.PUSCH.MCS.28})$
Average MCS index for downlink scheduling = $(0 \times \text{N.ChMeas.PDSCH.MCS.0} + 1 \times \text{N.ChMeas.PDSCH.MCS.1} + 2 \times$

$$\frac{N.ChMeas.PDSCH.MCS.2 + 3 \times N.ChMeas.PDSCH.MCS.3 + \dots + 26 \times N.ChMeas.PDSCH.MCS.26 + 27 \times N.ChMeas.PDSCH.MCS.27 + 28 \times N.ChMeas.PDSCH.MCS.28}{(N.ChMeas.PDSCH.MCS.0 + N.ChMeas.PDSCH.MCS.1 + N.ChMeas.PDSCH.MCS.2 + N.ChMeas.PDSCH.MCS.3 + \dots + N.ChMeas.PDSCH.MCS.26 + N.ChMeas.PDSCH.MCS.27 + N.ChMeas.PDSCH.MCS.28)}$$

■ Uplink/Downlink IBLER

$$\text{Uplink IBLER} = \frac{(N.UL.SCH.HalfPiBPSK.ErrTB.Ibler + N.UL.SCH.QPSK.ErrTB.Ibler + N.UL.SCH.16QAM.ErrTB.Ibler + N.UL.SCH.64QAM.ErrTB.Ibler)}{(N.UL.SCH.HalfPiBPSK.TB + N.UL.SCH.QPSK.TB + N.UL.SCH.16QAM.TB + N.UL.SCH.64QAM.TB + N.UL.SCH.256QAM.TB)} \times 100\%$$

$$\text{Downlink IBLER} = \frac{(N.DL.SCH.QPSK.ErrTB.Ibler + N.DL.SCH.16QAM.ErrTB.Ibler + N.DL.SCH.64QAM.ErrTB.Ibler)}{(N.DL.SCH.QPSK.TB + N.DL.SCH.16QAM.TB + N.DL.SCH.64QAM.TB + N.DL.SCH.256QAM.TB)} \times 100\%$$

■ Uplink/Downlink RBLER

$$\text{Uplink RBLER} = \frac{(N.UL.SCH.HalfPiBPSK.ErrTB.Rbler + N.UL.SCH.QPSK.ErrTB.Rbler + N.UL.SCH.16QAM.ErrTB.Rbler + N.UL.SCH.64QAM.ErrTB.Rbler)}{(N.UL.SCH.HalfPiBPSK.TB + N.UL.SCH.QPSK.TB + N.UL.SCH.16QAM.TB + N.UL.SCH.64QAM.TB + N.UL.SCH.256QAM.TB)} \times 100\%$$

$$\text{Downlink RBLER} = \frac{(N.DL.SCH.QPSK.ErrTB.Rbler + N.DL.SCH.16QAM.ErrTB.Rbler + N.DL.SCH.64QAM.ErrTB.Rbler)}{(N.DL.SCH.QPSK.TB + N.DL.SCH.16QAM.TB + N.DL.SCH.64QAM.TB + N.DL.SCH.256QAM.TB)} \times 100\%$$

■ Average downlink UE throughput = **User Downlink Average Throughput (DU)**

■ Average downlink cell throughput = **Cell Downlink Average Throughput (DU)**

- Due to RB adjustment, the PRACH frequency-domain position changes, and therefore the contention-based and non-contention-based random access success rates may fluctuate.

$$\text{Contention-based random access success rate} = \frac{N.RA.Contention.Resolution.Succ}{N.RA.Contention.Att} \times 100\%$$

$$\text{Non-contention-based random access success rate} = \frac{N.RA.Dedicated.Msg3}{N.RA.Dedicated.Att} \times 100\%$$

Impact on NEs

This change is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellPucch.PucchAlgoExtSwitch	MOD NRDUCCELLPUCCH LST NRDUCCELLPUCCH	gNodeB	The BWP2_PUCCH_FREQ_POS_ADJ option has been added to specify whether to enable BWP2 PUCCH frequency-domain position adjustment.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions

Function Name	Function Switch	Reference	Description
Power saving BWP	UE_PWR_SAVING_ENHANCE_SW option of the NRDUCellUePwrSaving.UePowerSavingSwitch parameter and the BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	BWP2 PUCCH frequency-domain position adjustment can be enabled only after power saving BWP is enabled.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *UE Power Saving*.

Requirement/Trouble Ticket Number

FeR-IR20250528001856-0010

RAT

NR TDD

5.3.4.1.36 Added Allocation of Aperiodic SRS Resources to RedCap/eRedCap UEs Without Periodic SRS Resources

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

This function allows the base station to allocate aperiodic SRS resources to RedCap/eRedCap UEs that do not have periodic SRS resources.

Implementation

In heavy-load scenarios (with a large number of RedCap/eRedCap UEs), the performance of some RedCap/eRedCap UEs deteriorates because they cannot be allocated periodic SRS resources. In extreme scenarios, service drops may occur. This function allows the base station to allocate aperiodic SRS resources to RedCap/eRedCap UEs, thereby reducing the risk of service drops in extreme scenarios.

Impact on the Network

The service drop rate in a cell may decrease.

Impact on NEs

This change is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0136

RAT

NR FDD, NR TDD

5.3.4.1.37 Added Uplink Voice Jitter Elimination Optimization

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Uplink voice jitter elimination optimization is now supported.

After enabling uplink buffer jitter elimination, you are advised to also enable uplink voice jitter elimination optimization by selecting the **UL_VOICE_JIT_ELIMINATE_OPT_SW** option of the **NRCellServExp.VoiceAlgoSwitch** parameter. This excludes RTCP packets from jitter elimination, thereby reducing RTCP packet delay.

The change in this feature is supported only in low frequency bands.

The change in this feature is supported only in SA networking.

Implementation

For details, see *VoNR*. The specific sections are listed in the change history of the document.

Impact on the Network

Reduces RTCP packet delay after uplink voice jitter elimination optimization is enabled.

Impact on NEs

This change is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRCe11SeryE xp.Vo1ceAlgo Switch	MOD NRCe11SERV EXP LST NRCe11SERV EXP	gNodeB	The UL_VOICE_JIT_ELIMINATE_OPT_SW option has been added to specify whether to enable uplink voice jitter elimination optimization.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions

The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink buffer jitter elimination	NRCe11OpPolicy.UlVoic eJitterEliminateTime	<i>VoNR</i>	Uplink voice jitter elimination optimization requires uplink buffer jitter elimination to be enabled.

- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *VoNR*.

Requirement/Trouble Ticket Number

FeR-IR20250102000700-0132

RAT

NR FDD, NR TDD

5.3.4.1.38 Added an Operator-level Method for Observing the Number of CA UEs and SCell Configuration

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The change in this feature involves two parts:

- The following counters have been added to measure the maximum number of CA UEs for a specific operator:
 - **N.User.CA.PCell.UL.Max.PLMN** (maximum number of uplink CA UEs that treat the local cell as their PCell for a specific operator)
 - **N.User.CA.PCell.DL.Max.PLMN** (maximum number of downlink CA UEs that treat the local cell as their PCell for a specific operator)
 - **N.User.CA.PCell.DL.3CC.Max.PLMN** (maximum number of downlink 3CC CA UEs that treat the local cell as their PCell for a specific operator)
- The "CA Measurement per Operator" subset has been added and the following new counters in this subset are used to observe SCell addition for CA UEs:
 - **N.CA.SCell.Add.Att.PLMN** (number of SCell addition attempts for CA UEs of a specific operator)
 - **N.CA.SCell.Add.Succ.PLMN** (number of successful SCell additions for CA UEs of a specific operator)
 - **N.UL.CA.SCell.Add.Att.PLMN** (number of uplink SCell addition attempts for CA UEs of a specific operator)
 - **N.UL.CA.SCell.Add.Succ.PLMN** (number of successful uplink SCell additions for CA UEs of a specific operator)

Implementation

- The following counters have been added to measure the maximum number of CA UEs for a specific operator:
 - **N.User.CA.PCell.UL.Max.PLMN** (maximum number of uplink CA UEs that treat the local cell as their PCell for a specific operator)
 - **N.User.CA.PCell.DL.Max.PLMN** (maximum number of downlink CA UEs that treat the local cell as their PCell for a specific operator)
 - **N.User.CA.PCell.DL.3CC.Max.PLMN** (maximum number of downlink 3CC CA UEs that treat the local cell as their PCell for a specific operator)
- The "CA Measurement per Operator" subset has been added and the following new counters in this subset are used to observe SCell addition for CA UEs:
 - **N.CA.SCell.Add.Att.PLMN** (number of SCell addition attempts for CA UEs of a specific operator)

- **N.CA.SCell.Add.Succ.PLMN** (number of successful SCell additions for CA UEs of a specific operator)
- **N.UL.CA.SCell.Add.Att.PLMN** (number of uplink SCell addition attempts for CA UEs of a specific operator)
- **N.UL.CA.SCell.Add.Succ.PLMN** (number of successful uplink SCell additions for CA UEs of a specific operator)

Impact on the Network

No impact

Impact on NEs

No impact

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management

The following function subsets have changed.

Change Type	Function Subset	NE	Description
New	CA Measurement per Operator	gNodeB	This function subset has been added to measure CA procedures for a specific operator in a cell.

NOTE

The counters in these new or deleted function subsets are no longer listed separately.
The following counters have changed.

Change Type	Counter Name	Function Subset	NE	Description
New	N.User.CA.P Cell.UL.Max .PLMN	Cell User Quantity Measurement per Operator	gNode B	This counter has been added to measure the maximum number of uplink CA UEs that treat the local cell as their PCells for a specific operator.
New	N.User.CA.P Cell.DL.Max .PLMN	Cell User Quantity Measurement per Operator	gNode B	This counter has been added to measure the maximum number of downlink CA UEs that treat the local cell as their PCell for a specific operator.
New	N.User.CA.P Cell.DL.3CC. Max.PLMN	Cell User Quantity Measurement per Operator	gNode B	This counter has been added to measure the maximum number of downlink 3CC CA UEs that treat the local cell as their PCell for a specific operator.

- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0170

FeR-IR20250528001857-0146

RAT

NR FDD, NR TDD

5.3.4.1.39 Added a Cell-level Method for Observing the Number of CA UEs and SCell Configuration

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The change in this feature involves two parts:

- The **N.User.UL.CA.Capability.Avg** counter has been added to measure the average number of uplink CA-capable UEs in a cell.
- The following counters have been added to observe uplink SCell addition for CA UEs in a cell:
 - **N.UL.CA.SCell.Add.Att** (number of uplink SCell addition attempts for CA UEs)
 - **N.UL.CA.SCell.Add.Succ** (number of successful uplink SCell additions for CA UEs)

Implementation

- The **N.User.UL.CA.Capability.Avg** counter has been added to measure the average number of uplink CA-capable UEs in a cell.
- The following counters have been added to observe uplink SCell addition for CA UEs in a cell:
 - **N.UL.CA.SCell.Add.Att** (number of uplink SCell addition attempts for CA UEs)
 - **N.UL.CA.SCell.Add.Succ** (number of successful uplink SCell additions for CA UEs)

The measurement rule of the **N.User.UL.CA.Capability.Avg** counter is subject to optimized measurement of the numbers of CA-capable UEs in cells (controlled by the **CA_CAP_USER_STATS_POL_SW** option of the **gNodeBAlgo.NrSmartMultiCarrSw** parameter). For details about optimized measurement of the numbers of CA-capable UEs in cells, see *Carrier Aggregation*.

Impact on the Network

No impact

Impact on NEs

No impact

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
The following counters have changed.

Change Type	Counter Name	Function Subset	NE	Description
New	N.User.UL.CA.Capability.Avg	Cell User Quantity Measurement	gNode B	This counter has been added to measure the average number of UEs capable of uplink CA in a cell.
New	N.UL.CA.SCell.Add.Att	Cell CA Measurement	gNode B	This counter has been added to measure the number of SCell addition attempts in the uplink for CA UEs.
New	N.UL.CA.SCell.Add.Succ	Cell CA Measurement	gNode B	This counter has been added to measure the number of successful SCell additions in the uplink for CA UEs.

- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0170
FeR-IR20250528001857-0146

RAT

NR FDD, NR TDD

5.3.4.1.40 Added Support for TRP Overshoot-coverage Handover Optimization

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Added Support for TRP overshoot-coverage handover optimization function, which specifies whether to optimize handovers between overshoot-coverage TRPs in the hyper cell or combined cell. When the **TRP_OVER_COV_HO_OPT_SW** option of the **NRDUCellMultiTrp.MultiTrpAlgoSwitch** parameter is selected, the TA of the source TRP is considered during handovers between overshoot-coverage TRPs. Specifically, the setting of the **NRDUCellMultiTrp.SrcTrpOverCovHoRsrpThldOfs** parameter takes effect only if this TA is valid.

Implementation

For details, see *Cell Combination* and *Hyper Cell*. The specific sections are listed in the change history of the documents.

Network Impacts

After TRP overshoot-coverage handover optimization takes effect in combined cells and hyper cells, TRP handovers are more accurate and the UE throughput is higher in overshoot coverage scenarios.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellMultiTrp. <i>MultiTrpAlgoSwitch</i>	MOD NRDUCELL MULTITRP LST NRDUCELL MULTITRP	gNodeB	The TRP_OVER_COV_HO_OPT_SW option has been added to specify whether to enable TRP overshoot-coverage handover optimization.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Cell Combination* and *Hyper Cell*.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0172

RAT

NR FDD, NR TDD

5.3.4.1.41 User-Plane Peer Address Fault Status Evaluation Optimization

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

This function optimizes the criteria for evaluating user-plane path fault status. In NSA Option 3x networking, it can increase the number of neighboring NR cells that remain in the available state.

This function is supported only in NSA networking.

The BTS5929R does not support this change.

Implementation

In NSA Option 3x networking, when determining whether a neighboring NR cell is available, the eNodeB checks the status of the user-plane peer address reported by the gNodeB. The gNodeB determines whether a user-plane peer address is available based on the following rule: If any user-plane path associated with that peer address in the **EPGROUP** MO for the S1-U interface is faulty, the gNodeB marks the user-plane peer address as faulty and reports the fault status to the eNodeB. Because the determination criteria are strict, the eNodeB cannot fully utilize neighboring NR cells for services. To address this issue, the "user-plane peer address fault status evaluation optimization" function can be enabled to increase the number of neighboring NR cells in the available state. After this function is enabled on the gNodeB, the gNodeB determines that a user-plane peer address is faulty only when all user-plane paths associated with that peer address in the **EPGROUP** MO are faulty.

The "user-plane peer address fault status evaluation optimization" function is controlled by the **UPPEER_FLT_RPT_CTRL_SW** option of the **TRANSALGEXTPARA.TNLBEARERMNGALGSW** parameter.

Impact on the Network

After this function takes effect, in NSA Option 3x networking, the number of neighboring NR cells that the eNodeB can select from during SCG addition increases. For example, if only some user-plane paths associated with a user-plane peer address on the gNodeB are faulty, the peer address is considered normal after this function is enabled, allowing the NR cells of the gNodeB to be used as candidate SCG cells.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	TRANSALGEXTPARA.TNLBEARERMNGALGSW	SET TRANSALGEXTPARA LST TRANSALGEXTPARA	gNodeB	The UPPEER_FLT_RPT_CTRL_SW option has been added to specify whether to enable the "user-plane peer address fault status evaluation optimization" function.

- Performance management

No impact

- Fault management

No impact

Impact on Features/Functions

- Prerequisite features/functions

No impact

- Mutually exclusive features/functions

No impact

- Impacted features/functions

No impact

Related Operations

By default, the change does not take effect after an upgrade.

Run the **SET TRANSALGEXTPARA** command with the **UPPEER_FLT_RPT_CTRL_SW** option of the **TRANSALGEXTPARA.TNLBEARERMNGALGSW** parameter selected to enable the "user-plane peer address fault status evaluation optimization" function.

Requirement/Trouble Ticket Number

IR20250908001644

RAT

NR FDD, NR TDD

5.3.4.1.42 Added User-Rate-based Dynamic QCI Resource Management

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

To improve user experience on non-GBR and GBR bearers, minimum rate guarantee can be enabled for both non-GBR or GBR services, as detailed in *QoS Management*. However, if the **NRDUCellRes.UlRbAverageRatio** or **NRDUCellRes.DlRbAverageRatio** parameter is set to an inappropriate value, the minimum rate cannot be reached, compromising service quality for UEs with these bearers. To address this, user-rate-based dynamic QCI resource management is introduced to dynamically adjust the **Uplink/Downlink RB Average Ratio** parameter value for QCI-group-based resource groups. This function is controlled by the **DYNAMIC_QCI_GROUP_SW** option of the **NRDUCellFeatureSw.QciBasedDiffServExpSw** parameter.

This change is supported only in SA networking.

LampSite base stations do not support this change.

This change is supported only in low frequency bands.

Implementation

For details, see *QCI-based Resource Control*. The specific sections are listed in the change history of the document.

Impact on the Network

When user-rate-based dynamic QCI resource management is enabled and the resources required by non-GBR or GBR services of UEs in a QCI group exceed the resources represented by **Uplink/Downlink RB Average Ratio**, the network impacts are as follows:

- If the target rate satisfaction level for the group is low, the user experience of UEs in this group improves.
- If the cell load is medium or heavy (PRB usage exceeding 30%) and there are a large number of cell edge UEs in the group, the uplink and downlink cell throughputs (**Cell Uplink Average Throughput (DU)** and **Cell Downlink Average Throughput (DU)**) decrease.

The rate satisfaction level is calculated using the following formulas:

- For the uplink, assuming counters specific to a minimum GBR of 3 Mbit/s:
$$\text{Uplink rate satisfaction level} = (\text{N.Thp.UL.Flex.Index3} + \dots + \text{N.Thp.UL.Flex.Index14}) / (\text{N.Thp.UL.Flex.Index0} + \dots + \text{N.Thp.UL.Flex.Index14})$$
- For the downlink, assuming counters specific to a minimum GBR of 5 Mbit/s:
$$\text{Downlink rate satisfaction level} = (\text{N.Thp.DL.Flex.Index2} + \dots + \text{N.Thp.DL.Flex.Index14}) / (\text{N.Thp.DL.Flex.Index0} + \dots + \text{N.Thp.DL.Flex.Index14})$$

NOTE

Flexible performance counters (such as **N.Thp.UL.Flex.Index0**) can be configured using the **gNBFlexCounterObjCfg** MO. For details, see "Custom Object Type" in the "Object Type" section in *Performance Management*.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

Base station models: 3900 and 5900 series base stations (macro base stations).
3900 series base stations must be configured with the BBU3910.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellFeatureSw.QciBasedDiffServExpSw	MOD NRDUCELLF EATURESW LST NRDUCELLF EATURESW	gNodeB	The DYNAMIC_QCI_GROUP_SW option has been added to specify whether to enable user-rate-based dynamic QCI resource management.

- Performance management

No impact

- Fault management

No impact

Impact on Features/Functions

- Prerequisite features/functions

The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	QCI-based resource control	RB_DYNAMIC_IC_CONTROL_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter	<i>QCI-based Resource Control</i>	User-rate-based dynamic QCI resource management (controlled by the DYNAMIC_QCI_GROUP_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw) parameter can be enabled only when the switch controlling QCI-based resource control is turned on.

- Mutually exclusive features/functions

The following mutually exclusive relationships have been added.

RAT	Function Name	Function Switch	Reference
Low-frequency TDD	NR DU cell resource management between network slices	RB_DYNAMIC_CONTROL_SW option of the NRDUCellAlgoSwitch.NetworkSliceAlgoSwitch parameter	<i>Network Slicing</i>
Low-frequency TDD	Service admission	QOS_FLOW_ADMISSION_SW option of the NRDUCellAlgoSwitch.RacAlgoSwitch parameter	<i>Admission Control</i>
Low-frequency TDD	RB-usage-based UE selection policy	NRDUCellRes.QciGrpRbCtrlUeSelPol set to AUTO , ARP_BASED , or MCS_BASED	<i>QCI-based Resource Control</i>

- Impacted features/functions

The change in this feature has no impact.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *QCI-based Resource Control*.

Requirement/Trouble Ticket Number

FeR-IR20251216001872-0010

RAT

NR FDD, NR TDD

5.3.4.1.43 Added Functions for Improving the SCell Activation and Deactivation Success Rates

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The SCell activation and deactivation success rates are subject to multiple factors. The following functions improve the SCell activation and deactivation success rates from the perspectives of DRX status consistency and coverage.

- SCell activation and deactivation suspension during DRX wakeup protection: When DRX is in effect, there may be false scheduling request (SR) detection. That is, the gNodeB determines that a UE is in the DRX active time, but the UE is actually in the DRX sleep time. When false SR detection occurs in the PCell, the SCell activation success rate is affected. As such, the "SCell activation and deactivation suspension during DRX wakeup protection" function is introduced. With this function in effect, the gNodeB suspends SCell activation and deactivation for a UE during the DRX wakeup protection period (a period where false SR detection may occur); and resumes SCell activation and deactivation when this period elapses. This function is controlled by the **RSVDSWPARAM3_BIT32** option of the **NRDUCellRsvdExt01.RsvdSwParam3** parameter. For details about DRX, see *DRX*.
- SCell activation prohibition under weak coverage: The SCell activation success rate is low for a UE under weak coverage in the PCell. As such, the SCell activation prohibition under weak coverage function is introduced. This function is controlled by **NRDUCellRsvdExt01.RsvdParam15**.
 - When this parameter ranges from **1** to **126**, this function takes effect. If the SRS RSRP of a UE in the PCell is less than or equal to the parameter value, the gNodeB considers the UE to be under weak coverage and does not activate any SCell for the UE.
 - If this parameter is set to **0** or a value greater than **126**, this function does not take effect.

The actual value of **NRDUCellRsvdExt01.RsvdParam15** is the configured value minus 157. For example, if this parameter is set to **1**, the actual value is -156 dBm.

This change is supported only in low frequency bands.

Implementation

For details, see *Carrier Aggregation*. The specific sections are listed in the change history of the document.

Impact on the Network

- SCell activation and deactivation suspension during DRX wakeup protection: After this function takes effect, the number of UEs for which SCells can be activated and deactivated may decrease in the PCell, the SCell activation success rate (**N.CA.SCell.Act.Succ/N.CA.SCell.Act.Att**) and SCell deactivation success rate (**N.CA.SCell.Deact.Succ/N.CA.SCell.Deact.Att**) may increase, but the number of SCell activation times (**N.CA.SCell.Act.Att**) and the number of SCell deactivation times (**N.CA.SCell.Deact.Att**) may decrease.
- SCell activation prohibition under weak coverage: When this function is in effect, the following impacts may occur depending on the value of **NRDUCellRsvdExt01.RsvdParam15**:
 - A smaller value of **NRDUCellRsvdExt01.RsvdParam15** results in more UEs for which SCells can be activated in the PCell, a lower SCell activation success rate (**N.CA.SCell.Act.Succ/N.CA.SCell.Act.Att**), more SCell activation times (**N.CA.SCell.Act.Att**), and a higher average throughput of SCell-configured CA UEs ($(\text{N.CA.SccConfigUser.ThpVol.DL} - \text{N.CA.SccConfigUser.ThpVol.DL.LastSlot}) / \text{N.CA.SccConfigUser.ThpTime.DL.RmvLastSlot}$).
 - A larger value of **NRDUCellRsvdExt01.RsvdParam15** results in fewer UEs for which SCells can be activated in the PCell, a higher SCell activation success rate (**N.CA.SCell.Act.Succ/N.CA.SCell.Act.Att**), fewer SCell activation times (**N.CA.SCell.Act.Att**), and a lower average throughput of SCell-configured CA UEs ($(\text{N.CA.SccConfigUser.ThpVol.DL} - \text{N.CA.SccConfigUser.ThpVol.DL.LastSlot}) / \text{N.CA.SccConfigUser.ThpTime.DL.RmvLastSlot}$).

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellRsvdExt01.RsvdSwParam3	MOD NRDUCELLRS VDEXT01 LST NRDUCELLRS VDEXT01	gNodeB	RSVDSWPARAM3_BIT32 has been activated to specify whether to suspend SCell activation and deactivation suspension during DRX wakeup protection
Modified	NRDUCellRsvdExt01.RsvdParam15	MOD NRDUCELLRS VDEXT01 LST NRDUCELLRS VDEXT01	gNodeB	This parameter has been activated to specify whether to enable SCell activation prohibition under weak coverage.

NOTE

NRDUCellRsvdExt01.RsvdSwParam3 and **NRDUCellRsvdExt01.RsvdParam15** are reserved parameters.

For details about reserved parameters, see 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Carrier Aggregation*.

Requirement/Trouble Ticket Number

FeR-IR20250329000527-0810

RAT

NR FDD, NR TDD

5.3.4.1.44 QCI-specific CCE Aggregation Level Adaptation

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

However, the insufficiency of CCE resources will be exacerbated if "QCI-specific PDCCH aggregation level optimization" takes effect in cells with a high CCE allocation failure rate. As a result, the scheduling opportunities of UEs with QCI-specific bearers and other UEs in the cell are reduced, thereby affecting user experience. To address this issue, QCI-specific CCE aggregation level adaptation is introduced. This function is controlled by the **CCE_AGG_LEVEL_ADAPT_SW** option of the **gNBDSchParamGroup.ScheduleOptSwitch** parameter.

After "QCI-specific CCE aggregation level adaptation" is enabled together with "QCI-specific PDCCH aggregation level optimization", the following rules apply to UEs with QCI-specific bearers:

- When the uplink or downlink CCE allocation failure rate of a cell exceeds the upper threshold specified by the **gNBDSchParamGroup.CceAllocFailRateUpperThld** parameter, "QCI-specific PDCCH aggregation level optimization" does not take effect for UEs with QCI-specific bearers. This prevents the scheduling opportunities of UEs with QCI-specific bearers and other UEs in the cell from being affected, thereby improving their experience.
- When both the uplink and downlink CCE allocation failure rates in the cell fall below the lower threshold specified by the **gNBDSchParamGroup.CceAllocFailRateLowerThld** parameter, "QCI-specific PDCCH aggregation level optimization" can take effect for UEs with QCI-specific bearers. This reduces the probability of DCI miss-detection for these UEs, thereby improving their experience.

Implementation

For details, see *Service-differentiated Experience Guarantee*.

Impact on the Network

Benefits:

Assume that QCI-specific CCE aggregation level adaptation is enabled together with any optimization of "QCI-specific PDCCH aggregation level optimization" in scenarios with a high CCE allocation failure rate in a cell. In this situation, the cell's uplink and downlink CCE allocation success rates increase, thereby increasing

the values of **Cell Uplink Average Throughput (DU)** and **Cell Downlink Average Throughput (DU)**.

Impacts:

Assume that QCI-specific CCE aggregation level adaptation is enabled together with any optimization of "QCI-specific PDCCH aggregation level optimization" in scenarios with a high CCE allocation failure rate in a cell. In this situation, the average uplink and downlink PDCCH CCE aggregation levels of UEs with QCI-specific bearers decrease, which increases the uplink and downlink CCE allocation success rates of the cell and possibly increases the DCI miss-detection rate of such UEs. As a result, the uplink and downlink rates of a specific QCI or 5QI, and 5QI-specific average uplink and downlink delays at the MAC layer fluctuate.

Impact on NEs

No impact

Impact on Hardware Requirements

- Boards
All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- RF modules
All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNBDUSchParamGroup.ScheduleOptSwitch	ADD GNBDUSCHP ARAMGROUP MOD GNBDUSCHP ARAMGROUP LST GNBDUSCHP ARAMGROUP	gNo deB	The CCE_AGG_LEVEL_ADAPT_SW option has been added to specify whether to enable QCI-specific CCE aggregation level adaptation.
New	gNBDUSchParamGroup.CceAllocFailureRateUpperThld	ADD GNBDUSCHP ARAMGROUP MOD GNBDUSCHP ARAMGROUP LST GNBDUSCHP ARAMGROUP	gNo deB	This parameter has been added to specify the upper threshold of the CCE allocation failure rate.
New	gNBDUSchParamGroup.CceAllocFailureRateLowerThld	ADD GNBDUSCHP ARAMGROUP MOD GNBDUSCHP ARAMGROUP LST GNBDUSCHP ARAMGROUP	gNo deB	This parameter has been added to specify the lower threshold of the CCE allocation failure rate.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	QCI-specific PDCCH aggregation level optimization	PUSCH_DTX_BASED_PDCCH_ADJ_SW option of the gNBDUSchParamGroup.ScheduleOptSwitch parameter, DL_BLER_TAR_QCI_SW option of the gNBDUSchParamGroup.ScheduleOptSwitch parameter, or the gNBDUSchParamGroup.PdcchSinrOffset parameter	<i>Service-differentiated Experience Guarantee</i>	QCI-specific CCE aggregation level adaptation takes effect only when any of the following conditions is met: • The PUSCH_DTX_BASED_PDCCH_ADJ_SW option of the gNBDUSchParamGroup.ScheduleOptSwitch parameter is selected. • The DL_BLER_TAR_QCI_SW option of the gNBDUSchParamGroup.ScheduleOptSwitch parameter is selected. • The gNBDUSchParamGroup.PdcchSinrOffset parameter is set to a value other than 0.

- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Service-differentiated Experience Guarantee*.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0143

RAT

NR FDD, NR TDD

5.3.4.1.45 PUSCH Scheduling Prohibition in Self-Contained Slots

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In high-load scenarios, the downlink user-perceived rate may decrease after the function controlled by the **SRS_INTRF_AVOID_PUSCH_SCH_SW** option of the **NRDUCellSrs.SrsAlgoExtSwitch** parameter takes effect. To address this issue, the PUSCH scheduling prohibition in self-contained slots function is introduced. This function is controlled by the **S_SLOT_PUSCH_SCH_PROHIB_SW** option of the **NRDUCellSrs.SrsAlgoExtSwitch** parameter. When this option is selected, the function controlled by the **SRS_INTRF_AVOID_PUSCH_SCH_SW** option of the **NRDUCellSrs.SrsAlgoExtSwitch** parameter can take effect only when the uplink or downlink PRB usage is less than or equal to the value of the **NRDUCellRimConfig.IntrfAvoidPuschSchPrbThld** parameter.

LampSite base stations do not support this function.

NOTE

This change was first incorporated into V100R022C10SPC010 but is first described in the network impact report of V100R022C10SPC050.

Implementation

In high-load scenarios, the downlink user-perceived rate may decrease after the function controlled by the **SRS_INTRF_AVOID_PUSCH_SCH_SW** option of the **NRDUCellSrs.SrsAlgoExtSwitch** parameter takes effect. To address this issue, the PUSCH scheduling prohibition in self-contained slots function is introduced. This function is controlled by the **S_SLOT_PUSCH_SCH_PROHIB_SW** option of the **NRDUCellSrs.SrsAlgoExtSwitch** parameter. When this option is selected, the function controlled by the **SRS_INTRF_AVOID_PUSCH_SCH_SW** option of the **NRDUCellSrs.SrsAlgoExtSwitch** parameter can take effect only when the uplink or downlink PRB usage is less than or equal to the value of the **NRDUCellRimConfig.IntrfAvoidPuschSchPrbThld** parameter.

Impact on the Network

If the **SRS_INTRF_AVOID_PUSCH_SCH_SW** option of the **NRDUCellSrs.SrsAlgoExtSwitch** parameter is selected, selecting the **S_SLOT_PUSCH_SCH_PROHIB_SW** option helps the network avoid degradation of the downlink user-perceived rate. However, the number of uplink scheduling opportunities and the number of TTIs may decrease, which in turn reduces the uplink cell traffic volume.

Impact on NEs

This change is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellSr s.SrsAlgoExt Switch	MOD NRDUCELLSR S LST NRDUCELLSR S	gNodeB	The S_SLOT_PUSCH_SCH_PROHIB_SW option has been added to control whether to enable the PUSCH scheduling prohibition in self-contained slots function.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Remote Interference Management (Low-Frequency TDD)*.

Requirement/Trouble Ticket Number

IR20250528001857

RAT

NR TDD

5.3.4.1.46 Added the CA UE Data Split Stop Optimization Function

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In some scenarios, the gNodeB does not distribute data of specific CA UEs to their SCCs. Without the CA UE data split stop optimization function, the PCC data split weight (proportion of the data volume distributed to the PCC to the total data volume) is not the optimal value when data split to SCCs is stopped for a CA UE. This increases the data scheduling duration and affects user experience. As such, the CA UE data split stop optimization function is introduced. This function is controlled by the **SPLIT_STOP_OPT_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoSwitch** parameter. After this function takes effect, the gNodeB optimizes the PCC data split weight when data split to SCCs is stopped for a CA UE. This shortens the data scheduling duration and guarantees user experience.

The change in this function is supported only in low frequency bands. The change in this function is supported only in downlink CA scenarios.

NOTE

This change was first incorporated into V100R022C10SPC010 but is first described in the network impact report of V100R022C10SPC050.

Implementation

For details, see *Carrier Aggregation*. The specific sections are listed in the change history of the document.

Impact on the Network

After CA UE data split stop optimization takes effect and data split is stopped for some CA UEs:

- The data scheduling delay of CA UEs may decrease.
- Due to the optimized PCC data split weight, more data is scheduled on the PCC. This may decrease the user-perceived rate of non-CA UEs on the PCC.

Impact on NEs

No impact

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellCarrierMgmt.CaEnhancedAlgorithmSwitch	MOD NRDUCELLC ARRMGMT LST NRDUCELLC ARRMGMT	gNodeB	The SPLIT_STOP_OPT_SW option has been added to specify whether to enable CA UE data split stop optimization.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Carrier Aggregation*.

Requirement/Trouble Ticket Number

IR20250528001857

DTS2025121029142

RAT

NR FDD, NR TDD

5.3.4.1.47 Support for UEs' Resource Set Capability Verification During SCC Configuration

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

When a UE reports its capability to the gNodeB, the *maxConfiguredResourceSetsAllCC* IE indicates the maximum number of resource sets supported by the UE on all carriers.

Without the change in this function, the gNodeB does not check whether the number of PCC and SCC resource sets exceeds the UE-supported upper limit after SCC configuration. For some UEs, if the number of resource sets exceeds the upper limit, service drops may occur. As such, the change in this function adds verification of UEs' resource set capability during SCC configuration. If the number of PCC and SCC resource sets exceeds the UE-supported upper limit, the SCC configuration stops, preventing issues such as service drops of UEs.

Implementation

The change in this function adds verification of UEs' resource set capability during SCC configuration. If the number of PCC and SCC resource sets exceeds the UE-supported upper limit, the SCC configuration stops, preventing issues such as service drops of UEs.

Impact on the Network

After the change in this function takes effect:

- The following indicators may decrease: (1) **Service Call Drop Rate (CU, Inactive)** of UEs; (2) RRC connection reestablishment rate ($\frac{\text{N.RRC.ReEst.Att}}{\text{N.RRC.SetupReq.Succ} + \text{N.RRC.ResumeReq.Succ} + \text{N.HO.InterRAT.E2N.ExecSucIn}} \times 100\%$).
- The values of the **N.CA.SCell.Add.Att** and **N.CA.SCell.Add.Succ** counters may decrease.

Impact on NEs

No impact

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250427003315-0010

RAT

NR FDD, NR TDD

5.3.4.1.48 Added Conditions for Checking the Number of UEs in an Energy Saving Policy Group for Entering or Exiting Intelligent Carrier Shutdown/Fast Carrier Shutdown

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Checking the number of UEs in an energy saving policy group has been added to the conditions for entering or exiting intelligent carrier shutdown/fast carrier shutdown. The number of UEs in an energy saving policy group is checked when all of the following conditions are met:

- The **SERV_TYPE_DYN_EXIT_PWR_SAVE_SW** option of the **NRDUCellAlgoSwitch.PowerSavingExtSwitch** parameter is selected.
- For the capacity-layer cell, the **NRCeLLPsPolicyGrp** MO is configured, and the **USER_NUM_CSD_EXIT_SW** option of the **NRCeLLPsPolicyGrp.ServTypeDynExitPwrSavingSw** parameter corresponding to the **NRCeLLPsPolicyGrp.PowerSavingPolicyGrpIndex** parameter is selected. The **NRCeLLPsPolicyGrp.UserIdentMethod** parameter is set to **SERVICE_TYPE**.

The specific check conditions are as follows:

- Entering condition: Total number of UEs in the energy saving policy groups of capacity-layer and basic-layer cells $\leq$ Threshold specified by **NRCeLLPsPolicyGrp.GrpUserNumThld**
- Exiting condition: Number of UEs in a power saving policy group of a basic-layer cell $>$ Threshold specified by **NRCeLLPsPolicyGrp.GrpUserNumThld** + Offset specified by **NRCeLLPsPolicyGrp.GrpUserNumOffset**

The change in this function is supported only in low frequency bands.

Implementation

For details, see *Energy Conservation and Emission Reduction*. The specific sections are listed in the change history of the document.

Impact on the Network

This function adds the number of UEs in an energy saving policy group as a check condition for entering and exiting intelligent carrier shutdown/fast carrier shutdown. This affects the effectiveness of intelligent carrier shutdown/fast carrier shutdown, and the energy saving gains may decrease. For details about the impact of intelligent carrier shutdown/fast carrier shutdown on the network, see *Energy Conservation and Emission Reduction*.

Impact on NEs

This change is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

The *GrpIndexQciUeNumInfo* IE is added to *Vendor Xn NR Resource Status* in a RESOURCE STATUS UPDATE message over the Xn interface. This IE indicates the number of UEs in an energy saving policy group.

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRCeLLPsPolicyGrp.ServTypeDynExitPwrSavingSw	ADD NRCELLPSPOLICYGRP MOD NRCELLPSPOLICYGRP LST NRCELLPSPOLICYGRP	gNodeB	The USER_NUM_CSD_EXIT_SW option has been added to specify whether to check the number of UEs in an energy saving policy group for entering or exiting intelligent carrier shutdown/fast carrier shutdown.
New	NRCeLLPsPolicyGrp.GrpUserNumThld	ADD NRCELLPSPOLICYGRP MOD NRCELLPSPOLICYGRP LST NRCELLPSPOLICYGRP	gNodeB	This parameter has been added to specify the threshold of the number of UEs in an energy saving policy group.
New	NRCeLLPsPolicyGrp.GrpUserNumOffset	ADD NRCELLPSPOLICYGRP MOD NRCELLPSPOLICYGRP LST NRCELLPSPOLICYGRP	gNodeB	This parameter has been added to specify the offset to the number of UEs in an energy saving policy group.

- Performance management

No impact

- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Energy Conservation and Emission Reduction*.

Requirement/Trouble Ticket Number

FeR-IR20251204000006-0210

RAT

NR FDD, NR TDD

5.3.4.1.49 Added n1 and n65 as Additional Frequency Bands for an External NR Cell and the Frequency Relationship of an NR Cell

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

n1 and n65 can now be added as additional frequency bands during configuration of an external NR cell and during frequency relationship configuration of an NR cell on the gNodeB.

This change is supported only in low-frequency bands.

Implementation

- Frequency relationship configuration of an NR cell: The **NRCellFreqRelation.AdditionalFrequencyBand** parameter can now be set to **N1** or **N65**.
- Configuration of an external NR cell: The **NRExternalNCell.AdditionalFrequencyBand** parameter can now be set to **N1** or **N65**.

The following constraints have been added:

- When **NRCellFreqRelation.AdditionalFrequencyBand** is set to **N1**, **NRCellFreqRelation.FrequencyBand** must be set to **N65**.
- When **NRCellFreqRelation.AdditionalFrequencyBand** is set to **N65**, **NRCellFreqRelation.FrequencyBand** must be set to **N1**.
- When the **NRCellFreqRelation.SsbDescMethod** parameter is set to **SSB_DESC_TYPE_GSCN**:
 - When **NRCellFreqRelation.AdditionalFrequencyBand** is set to **N1**, the value of **NRCellFreqRelation.SsbFreqPos** must range from **5279** to **5419**.
 - When **NRExternalNCell.AdditionalFrequencyBand** is set to **N1**, the value of **NRExternalNCell.SsbFreqPos** must range from **5279** to **5419**.
 - When **NRCellFreqRelation.AdditionalFrequencyBand** is set to **N65**, the value of **NRCellFreqRelation.SsbFreqPos** must range from **5279** to **5494**.
 - When **NRExternalNCell.AdditionalFrequencyBand** is set to **N65**, the value of **NRExternalNCell.SsbFreqPos** must range from **5279** to **5494**.
- When **NRCellFreqRelation.SsbDescMethod** is set to **SSB_DESC_TYPE_NARFCN**:
 - When **NRCellFreqRelation.AdditionalFrequencyBand** is set to **N1**, the value of **NRCellFreqRelation.SsbFreqPos** must range from **422000** to **434000**.
 - When **NRExternalNCell.AdditionalFrequencyBand** is set to **N1**, the value of **NRExternalNCell.SsbFreqPos** must range from **422000** to **434000**.
 - When **NRCellFreqRelation.AdditionalFrequencyBand** is set to **N65**, the value of **NRCellFreqRelation.SsbFreqPos** must range from **422000** to **440000**.
 - When **NRExternalNCell.AdditionalFrequencyBand** is set to **N65**, the value of **NRExternalNCell.SsbFreqPos** must range from **422000** to **440000**.
- When **NRExternalNCell.AdditionalFrequencyBand** is set to **N1**, **NRExternalNCell.FrequencyBand** must be set to **N65**.
- When **NRExternalNCell.AdditionalFrequencyBand** is set to **N65**, **NRExternalNCell.FrequencyBand** must be set to **N1**.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRCellFreqRelation.AdditionalFrequencyBand	ADD NRCELLFREQLRELATION MOD NRCELLFREQLRELATION LST NRCELLFREQLRELATION	gNodeB	This parameter specifies the additional frequency band of an intra-RAT neighboring frequency. The values N1 and N65 have been added.
Modified	NRExternalNCell.AdditionalFrequencyBand	ADD NREXTERNALNCELL MOD NREXTERNALNCELL LST NREXTERNALNCELL	gNodeB	This parameter specifies the additional frequency band in which the external cell operates. The values N1 and N65 have been added.

- Performance management

No impact

- Fault management

No impact

Impact on Features/Functions

- Prerequisite features/functions

No impact

- Mutually exclusive features/functions

No impact

- Impacted features/functions

The following impact relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-RAT ANR	NR_NR_ANR_SW option of the NRCellAlgoSwitch.AnrSwitch parameter	<i>ANR</i>	The system can automatically detect and add missing neighboring cells in the additional frequency band n1 or n65, and automatically update the NCL of this frequency band.

Related Operations

By default, the change does not take effect after an upgrade.

The **NRCellFreqRelation.AdditionalFrequencyBand** and **NRExternalNCell.AdditionalFrequencyBand** parameters can now be set to **N1** or **N65** to configure n1 or n65 as an additional frequency band during frequency relationship configuration of an NR cell and during configuration of an external NR cell.

Requirement/Trouble Ticket Number

FeR-IR20250102000700-0057

RAT

NR FDD, NR TDD

5.3.4.1.50 Added PDCCH NEBF Based on DFT-based Weights in NR TDD

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

PDCCH normalized eigenvector beamforming (NEBF) based on discrete Fourier transform (DFT)-based weights is now supported in TDD cells with at least 8T8R. With this function, PDCCH beamforming is performed using both DFT-based weights and normalized eigenvectors, preventing PDCCH power from exceeding the upper limit for UEs.

This function is controlled by the **PDCCH_DFT_WEIGHT_NEBF_SW** option of the **NRDUCellPdcch.PdcchWeightAlgoSwitch** parameter.

LampSite base stations do not support this function.

Implementation

In cells with at least 8T8R, when the **PDCCH_DFT_WEIGHT_NEBF_SW** option of the **NRDUCellPdcch.PdcchWeightAlgoSwitch** is selected, PDCCH beamforming is performed using both DFT-based weights and normalized eigenvectors, preventing PDCCH power from exceeding the upper limit for UEs.

Impact on the Network

When this function is enabled, for both the uplink and downlink, the average CCE aggregation level may increase and the average UE throughput may decrease in the following scenarios:

- Scenario 1: The **NRDUCellPdcchAlgo.PdcchPrecodingPolicy** parameter is set to **ENH_PMI** for a TDD cell in NSA networking.
- Scenario 2: The **NRDUCellPowerSaving.RfShutdownNum** parameter is set to **THREE_QUARTERS** and the **NRDUCellPdcchAlgo.PdcchPrecodingPolicy** parameter is set to **ENH_PMI** for a TDD 32T32R cell.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

- Boards
 - Main control boards: UMPTe series, UMPTg series, UMPTga series, and UMPTha series
 - Baseband processing units: All baseband processing units that support NR TDD 8T8R or higher
- RF modules

All NR TDD-capable RF modules with at least 8T8R support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact
- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellPdcch.PdcchWeightAlgorithmSwitch	MOD NRDUCCELLP DCCH LST NRDUCCELLP DCCH	gNodeB	This parameter has been added to specify whether to enable PDCCH weight algorithms.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0142

RAT

NR TDD

5.3.4.1.51 Support for LampSite mmWave Scenario-based Beam in mmWave Beam Management (for High Frequency Bands Only)#

NEs Involved

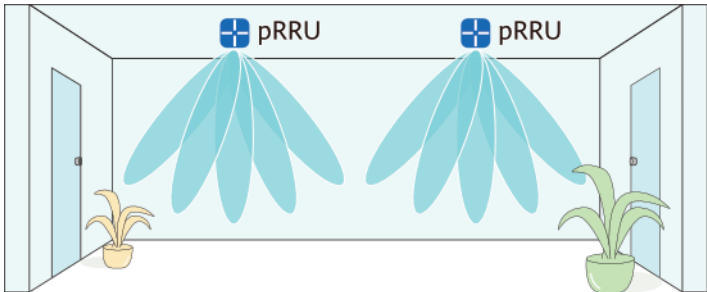
5900 series base stations

Feature/Function Description

LampSite mmWave scenario-based beam is now supported in mmWave beam management.

LampSite mmWave scenario-based beam supports the default scenario, as shown in [Figure 5-5](#). The default scenario can be configured by setting **NRDUCellTrpMmwavBeam.CoverageScenario** to **DEFAULT**.

Figure 5-5 Default scenario



"LampSite mmWave scenario-based beam" has certain requirements for RF modules. [Table 5-36](#) lists the related specifications of RF modules supporting "LampSite mmWave scenario-based beam".

Table 5-36 Related specifications of RF modules supporting "LampSite mmWave scenario-based beam"

Scenario ID	RF Module	(Horizontal 3 dB Beamwidth, Vertical 3 dB Beamwidth)
DEFAULT (default scenario)	pRRU5551JR 2T2R (26 GHz)	(360°, ±80°)

 NOTE

- For pRRUs of DBS5900 LampSite:
- The tilt (**NRDUCellTrpMmwavBeam.Tilt**) must be set to **255** and cannot be adjusted. (In this case, the default downtilt supported by the RF modules is used.)
 - The azimuth (**NRDUCellTrpMmwavBeam.Azimuth**) must be set to **0** and cannot be adjusted. (In this case, the default azimuth 0° is used.)

This function is only applicable to LampSite base stations.

Implementation

For details, see *mmWave Beam Management (High-Frequency TDD)*. The specific sections are listed in the change history of the document.

Impact on the Network

LampSite mmWave scenario-based beam supports the default scenario, thereby ensuring user experience in LampSite base station scenarios.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

- Base station models
DBS5900 LampSite, which must be configured with the BBU5900 or BBU5910
- Boards
All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- RF modules
Only the pRRU5551JR (RHUB5873) supports this function.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *mmWave Beam Management (High-Frequency TDD)*.

Requirement/Trouble Ticket Number

IR20250703005716

IR20241010000514

FeR-IR20240812001197-001

FeR-IR20240812001208-002

RAT

NR TDD

5.3.4.1.52 SUL UE Quantity Considered for Entering or Exiting LTE and NR Intelligent Carrier Shutdown and RF Module Dormancy

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The number of SUL UEs is now considered when determining whether to enter or exit NR load-based RF module deep dormancy or NR load-based RF module super dormancy.

- Dormancy entry condition: In SUL co-deployment scenarios, the total number of SUL UEs in the capacity-layer and basic-layer cells is less than or equal to the value of the **NRDUCellPowerSaving.SulUserNumThld** parameter. If the capacity-layer cell is associated with multiple basic-layer cells, each pair of the basic- and capacity-layer cells must meet this condition.
- Dormancy exit condition: In SUL co-deployment scenarios, the number of SUL UEs in a basic-layer cell is greater than the sum of the **NRDUCellPowerSaving.SulUserNumThld** and **NRDUCellPowerSaving.SulUserNumOffset** parameter values. If the capacity-layer cell is associated with multiple basic-layer cells, the number of SUL UEs in one of the basic-layer cells must meet the condition.

NOTE

This change requires that the base stations serving the capacity-layer and corresponding basic-layer cells be upgraded to V100R022C10SPC010 or later.

For LTE and NR intelligent carrier shutdown, optimized evaluation of NR RF module dormancy in NR local-cell-load-based RF module dormancy, LNR load-based RF module deep dormancy, and LNR load-based RF module super dormancy, the number of SUL UEs is considered as a dormancy entry condition. Specifically, in SUL co-deployment scenarios, the total number of SUL UEs in the NR cell is less than or equal to the value of the **NRDUCellPowerSaving.SulUserNumThld** parameter.

Implementation

For details, see *Multi-RAT Coordinated Energy Saving*. The specific sections are listed in the change history of the document.

Impact on the Network

In SUL co-deployment scenarios, setting **NRDUCellPowerSaving.SulUserNumThld** to a value other than **65535** may reduce the chance of entering the energy saving state and decrease the probability that SUL scheduling cannot be performed for SUL UEs.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellPowerSaving.SulUserNumThld	ADD NRDUCELLPOWERSAVING MOD NRDUCELLPOWERSAVING LST NRDUCELLPOWERSAVING	gNodeB	This parameter has been added to specify the SUL UE number threshold for RF module dormancy.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellPowerSaving.SulUserNumOffset	ADD NRDUCELLPOWERSAVING MOD NRDUCELLPOWERSAVING LST NRDUCELLPOWERSAVING	gNodeB	This parameter has been added to specify the SUL UE number offset for RF module dormancy.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Multi-RAT Coordinated Energy Saving*.

Requirement/Trouble Ticket Number

FeR-IR20250605000478-0010

IR20250701003015

RAT

NR FDD, NR TDD

5.3.4.1.53 Added SUL UE Number Check to the Conditions for Entering or Exiting Intelligent Carrier Shutdown/Fast Carrier Shutdown/Timing Carrier Shutdown

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In SUL co-deployment scenarios, if the **NRDUCellPowerSaving.SulUserNumThld** parameter is set to a value other than **65535**, the SUL UE number check has been added to the conditions for entering and exiting intelligent carrier shutdown and fast carrier shutdown as well as the conditions for entering timing carrier shutdown if load-based evaluation applies.

The related thresholds are defined by the **NRDUCellPowerSaving.SulUserNumThld** and **NRDUCellPowerSaving.SulUserNumOffset** parameters.

For intelligent carrier shutdown and fast carrier shutdown, this function requires that the base stations serving the capacity-layer and corresponding basic-layer cells be upgraded to V100R022C10SPC010 or later.

This function is supported only in low frequency bands.

Implementation

For details, see *Energy Conservation and Emission Reduction*. The specific sections are listed in the change history of the document.

Impact on the Network

In SUL co-deployment scenarios, setting **NRDUCellPowerSaving.SulUserNumThld** to a value other than **65535** may reduce the chance of entering the energy saving state and decrease the probability that SUL scheduling cannot be performed for SUL UEs.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

The *sulUserNum* IE has been added to the **Power Saving Information** structure under **Vendor Xn NR Resource Status** in a RESOURCE STATUS UPDATE message over the Xn interface.

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	NRDUCellPowerSaving.SulUserNumThld	ADD NRDUCELLPO WERSAVING MOD NRDUCELLPO WERSAVING LST NRDUCELLPO WERSAVING	gNodeB	This parameter has been added to specify the SUL UE number threshold for intelligent carrier shutdown/fast carrier shutdown/timing carrier shutdown.
New	NRDUCellPowerSaving.SulUserNumOffset	ADD NRDUCELLPO WERSAVING MOD NRDUCELLPO WERSAVING LST NRDUCELLPO WERSAVING	gNodeB	This parameter has been added to specify the SUL UE number offset for intelligent carrier shutdown/fast carrier shutdown.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Energy Conservation and Emission Reduction*.

Requirement/Trouble Ticket Number

FeR-IR20250605000478-0010

IR20250701003015

RAT

NR FDD, NR TDD

5.3.4.1.54 Added the UeType parameter to the STR NRCELLSERVICEPARAMEFFECT Command

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The **UeType** parameter has been added to the **STR NRCELLSERVICEPARAMEFFECT** command to distinguish between RedCap UEs and eMBB UEs so that differentiated control can be implemented for different types of UEs.

Implementation

For details, see *RedCap*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following MML commands have changed.

Change Type	MML Command	NE	Description
Modified	STR NRCELLSER VICEPARAM EFFECT	gNodeB	The UeType parameter has been added to the STR NRCELLSERVICEPARAMEFFECT command.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0043

RAT

NR FDD, NR TDD

5.3.4.1.55 Added SCell Deactivation Cause Values to an External CHR Event

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

To enhance the efficiency of locating SCell activation failures, SCell deactivation cause values have been added to the existing external CHR event PERIOD_PRIVATE_UE_MEASUREMENT_EXT.

 NOTE

External CHRs refer to SIG logs.

Implementation

The following SCell deactivation cause values have been added to the existing external CHR event PERIOD_PRIVATE_UE_MEASUREMENT_EXT.

Scenario	SCell Deactivation Cause Value
No deactivation (default)	NO_SCELL_DEACT
Deactivation triggered by the traffic volume of a CA UE	SCELL_DEACT_FOR_TRAFFIC_VOLUME
	SCELL_DEACT_FOR_DP_CHIP_OVERLOAD
	SCELL_DEACT_FOR_ANR_DRX
Channel quality-triggered SCell deactivation	SCELL_DEACT_FOR_CHANNEL_QUALITY(CQI)
Consecutive residual block error-triggered SCell deactivation	SCELL_DEACT_FOR_CONTINUOUS_ERROR
Deactivation triggered by the volume of data distributed from the PCell to an SCell	SCELL_DEACT_FOR_NO_TRAFFIC_DIVERSION
Deactivation protection	SCELL_DEACT_FOR_ACT_FAILURE
	SCELL_DEACT_FOR_HYPERCELL_CSIRS_BEAMSCAN_INVALID
	SCELL_DEACT_FOR_L3_NOTIFY
	SCELL_DEACT_FOR_UE_OVERHEATING
Delayed deactivation	SCELL_DEACT_FOR_ALIGN_UE_STATUS

Impact on the Network

No impact

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20241209002193-0035

RAT

NR FDD, NR TDD

5.3.4.1.56 Support for Simulated Loading When Joint Transmission Cell Combination Is Enabled

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Simulated loading is now supported when Joint Transmission Cell Combination is enabled. For a combined cell in joint transmission mode, different beams are selected based on the specified simulated loading rate and with actual services considered to conduct simulated loading for neighboring cell interference tests.

Implementation

Simulated loading in wide beam mode is now supported with **WIDE_BEAM_MODE** added to the **NRDUCellSimulateLoad.SimulatedLoadWorkMode** parameter. This mode takes effect only in a TDD combined cell in joint transmission mode. In addition, when this mode is used in such a cell, the **NRDUCellSimulateLoad.TestMode** parameter can be set to **TDD_SIMULATED_LOAD_MODE** for the cell.

Impact on the Network

When simulated loading is in effect in a combined cell in joint transmission mode, the throughput of UEs in neighboring cells decreases.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellSimulateLoad.SimulatedLoadWorkMode	STR NRDUCELLSIMULATELOAD DSP NRDUCELLSIMULATELOAD	gNodeB	The value WIDE_BEAM_MODE has been added, indicating wide beam mode. In this mode, only one wide transmit beam can be always used.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions

The descriptions of the following impact relationships have been revised.

RA T	Function Name	Function Switch	Reference	Description
Lo w- fre qu enc y TD D	TDD simulated loading mode	NRDUCellSimulateLoad.Test Mode set to TDD_SIMULATED_LOAD_MODE	None	TDD simulated loading mode is supported only for combined cells in joint transmission mode with the NRDUCellSimulateLoad.SimulatedLoadWork Mode parameter set to WIDE_BEAM_ MODE . Activating TDD simulated loading mode in a cell will decrease the UE throughput in its neighboring cells.

Related Operations

By default, the change does not take effect after an upgrade.

Run the **STR NRDUCELLSIMULATELOAD** command with the **NRDUCellSimulateLoad.SimulatedLoadWorkMode** parameter set to **WIDE_BEAM_MODE** to enable simulated loading.

Requirement/Trouble Ticket Number

FeR-IR20241209002193-0013

RAT

NR TDD

5.3.4.1.57 Specification Expansion for QCI-specific Bearers

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

To increase the proportion of QCI-specific bearers that can be configured, the instance specifications and value ranges for 5QI/QCI-specific bearers have been expanded, considering a maximum of 256 5QI/QCI values defined in 3GPP specifications.

Implementation

[Table 5-37](#) describes the expanded instance specifications and value ranges for 5QI/QCI-specific bearers.

Table 5-37 Changes in the instance specifications and value ranges for 5QI/QCI-specific bearers

MO	Before Optimization	After Optimization	Description
gNBQciBeare r	gNBQciBearer MO instance specifications: [15, 45]	gNBQciBearer MO instance specifications: [15, 256]	N/A
gNB5qiConfi g	gNB5qiConfig MO instance specifications: [15, 45]	gNB5qiConfig MO instance specifications: [15, 256]	N/A
NRCellQciBe arer	NRCellQciBeare r MO instance specifications per cell: [15, 50] NRCellQciBeare r MO instance specifications per base station: [15, 6912]	NRCellQciBearer MO instance specifications per cell: [15, 256] NRCellQciBearer MO instance specifications per base station: [15, 6912]	N/A

MO	Before Optimization	After Optimization	Description
gNBDUSchParamGroup	gNBDUSchParamGroup MO instance specifications: [0, 60]	gNBDUSchParamGroup MO instance specifications: [0, 155]	As the gNBDUSchParamGroup MO instance specifications expand up to 155, the value range of the gNBDUSchParamGroup.ScheduleParamGroupId parameter has been changed from [1, 60] to [1, 155].
NRDuCellQciBearer	NRDuCellQciBearer MO instance specifications per cell: [15, 50] NRDuCellQciBearer MO instance specifications per base station: [15, 6912]	NRDuCellQciBearer MO instance specifications per cell: [15, 256] NRDuCellQciBearer MO instance specifications per base station: [15, 6912]	As the NRDuCellQciBearer.ScheduleParamGroupId parameter is associated with the gNBDUSchParamGroup.ScheduleParamGroupId parameter, the value range of the NRDuCellQciBearer.ScheduleParamGroupId parameter has been changed from [1, 60] and 255 to [1, 155] and 255.
gNBOP5qiConfig	N/A	N/A	The value range of both the gNBOP5qiConfig.Qci and gNBOP5qiConfig.RedCapQci parameters has been changed from [201, 255] and 65535 to [101, 255] and 65535.

MO	Before Optimization	After Optimization	Description
gNBFlexCounterObjCfg	gNBFlexCounterObjCfg MO instance specifications: [0, 30]	gNBFlexCounterObjCfg MO instance specifications: [0, 64]	As the gNBFlexCounterObjCfg MO instance specifications expand up to 64, the value range of the gNBFlexCounterObjCfg.FlexCounterObjId parameter has been changed from [0, 29] to [0, 63].

Impact on the Network

After the expansion of instance specifications and value ranges for 5QI/QCI-specific bearers, creating instances that exceed the pre-expansion specifications will occupy additional system memory.

Impact on NEs

This change is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250806001609-0390

RAT

NR FDD, NR TDD

5.3.4.1.58 Support for Querying the Channel Maintenance and Test Details of an NR DU Cell TRP

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The **DSP NRDUCELLTRPCHNMNT** command has been added to query the channel maintenance and test details of an NR DU cell TRP.

Implementation

A cell may fail to be activated due to incorrect RF module power configuration. In such cases, the cell can be reactivated only after the power of each RF module channel is reconfigured. During power reconfiguration, the most recent transmit power of each RF module channel for the cell may need to be referenced. This information can be obtained using the newly added **DSP NRDUCELLTRPCHNMNT** command, which is used to query the channel maintenance and test details of a cell. In the command output, if the value of **TX Channel No** is **65535**, the **TX Channel Pwr(mW)** parameter indicates the total transmit power of all RF module channels for the cell. If the value of **TX Channel No** is not **65535**, the **TX Channel Pwr(mW)** parameter indicates the transmit power of the corresponding channel for the cell.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following MML commands have changed.

Change Type	MML Command	NE	Description
New	DSP NRDUCELLTRPCHNMNT	gNodeB	This MML command has been added to query the channel maintenance and test details of an NR DU cell TRP.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

Run the **DSP NRDUCELLTRPCHNMNT** command to query the channel maintenance and test details of an NR DU cell TRP.

Requirement/Trouble Ticket Number

FeR-IR20250102000700-0064

RAT

NR FDD, NR TDD

5.3.4.1.59 Change in the Value Range of the Uplink Scheduling Priority Weight Factor Parameter in the gNBDUMacParamGroup MO

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The **gNBDUMacParamGroup.UlSchPriWeightFactor** parameter specifies the weighting factor in the formula for calculating the uplink scheduling priority of a radio bearer. A larger value of this parameter results in more scheduling opportunities for services on the radio bearer of the corresponding QCI and a higher user-perceived data rate when multiple services are scheduled concurrently in the cell. Before the change in this function is introduced, the maximum GUI value of the **gNBDUMacParamGroup.UlSchPriWeightFactor** parameter is **1000**. In some scenarios, more scheduling opportunities need to be provided for services on QCI-specific radio bearers. Therefore, the value range of the **gNBDUMacParamGroup.UlSchPriWeightFactor** parameter has been expanded in this function.

Implementation

The value range of the **gNBDUMacParamGroup.UlSchPriWeightFactor** parameter has been changed from range "**1-1000**" to range "**1-10000**". Considering the step of 0.001, the actual value range has been changed from range "**0.001-1**" to range "**0.001-10**".

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNBDUMacParamGroup .UISchPriWeightFactor	ADD GNBDUMAC PARAMGROUP MOD GNBDUMAC PARAMGROUP LST GNBDUMAC PARAMGROUP	gNodeB	This parameter now supports the values ranging from 1001 to 10000 . The value range has been changed from range " 1-1000 " to range " 1-10000 ". Considering the step of 0.001, the actual value range has been changed from range " 0.001-1 " to range " 0.001-10 ".

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250724000484-0001

FeR-IR20250915000390-0001

RAT

NR FDD, NR TDD

5.3.4.1.60 Changed the Value Range of the CarrSelMinInterval Parameter

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The value range of the **NRCellCaMgmtConfig.CarrSelMinInterval** parameter has been changed.

This change is supported only in low frequency bands.

Implementation

The value range of the **NRCellCaMgmtConfig.CarrSelMinInterval** parameter has been changed from "30-600, 65535" to "10-600, 65535".

Impact on the Network

No impact

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameter has changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRCellCaMgmtConfig.CarrSelMinInterval	MOD NRCELLCAMG MTCONFIG LST NRCELLCAMG MTCONFIG	gNodeB	The value range of this parameter has been extended to include values 10 to 29 . Due to the extension, the GUI value range has been changed from " 30–600 , and 65535 " to " 10–600 , and 65535 ".

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250418001250-0010

RAT

NR FDD, NR TDD

5.3.4.1.61 Changed the Mutually Exclusive Relationships of Uplink DMRS Adaptation with SRS Period Shortening for Downlink 1T4R UEs and SRS Period Shortening for Downlink 2T4R UEs to Impact Relationships

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The mutually exclusive relationships of uplink DMRS adaptation with SRS period shortening for downlink 1T4R UEs and SRS period shortening for downlink 2T4R UEs have been changed to impact relationships.

This change is supported only in low frequency bands.

This change is supported only in SA networking.

Implementation

For details, see *VoNR*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions

No impact

- Mutually exclusive features/functions

The following mutually exclusive relationships have been deleted.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS period shortening for downlink 1T4R UEs	DL_1T4R_SRS_PERIOD_SHORTEN_SW option of the NRDUCellPdschPrecode.SuMimoSrsPeriodShortSw parameter	None	Uplink DMRS adaptation cannot be enabled together with SRS period shortening for downlink 1T4R UEs.
Low-frequency TDD	SRS period shortening for downlink 2T4R UEs	DL_2T4R_SRS_PERIOD_SHORTEN_SW option of the NRDUCellPdschPrecode.SuMimoSrsPeriodShortSw parameter	None	Uplink DMRS adaptation cannot be enabled together with SRS period shortening for downlink 2T4R UEs.

- Impacted features/functions

The following impact relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS period shortening for downlink 1T4R UEs	DL_1T4R_SRS_PERIOD_SHORTEN_SW option of the NRDUCellPdschPrecode.SuMimoSrsPeriodShortSw parameter	None	When SRS period shortening for downlink 1T4R UEs takes effect for UEs with VoNR enabled, uplink DMRS adaptation does not take effect in an uplink slot in which the PUSCH occupies at most 12 symbols.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS period shortening for downlink 2T4R UEs	DL_2T4R_SRS_PERIOD_SHORTEN_SW option of the NRDUCellPdschPrecodeSchemeMimoSrsPeriodShortSw parameter	None	When SRS period shortening for downlink 2T4R UEs takes effect for UEs with VoNR enabled, uplink DMRS adaptation does not take effect in an uplink slot in which the PUSCH occupies at most 12 symbols.

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250528001856-0035

RAT

NR FDD, NR TDD

5.3.4.1.62 More Frequencies Allowed in a gNodeB Frequency Priority Group

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

More frequencies can now be configured in a gNodeB frequency priority group. This is achieved by raising the maximum number of **gNBFreqPriorityGroup** MOs that can have the same **gNBFreqPriorityGroup.gNBFreqPriorityGroupId** parameter value from 8 to 32. With more frequencies allowed to be configured, frequency priorities can now be configured more flexibly.

Implementation

This change uses the following methods to increase the maximum number of frequencies that can be configured in a gNodeB frequency priority group:

- The following restriction has been deleted: A maximum of eight **gNBFreqPriorityGroup** MOs can have the same combination of

gNBFreqPriorityGroup.gNBFreqPriorityGroupId and
gNBFreqPriorityGroup.RatType values.

- The value range of the **gNBFreqPriorityGroup.FreqIndex** parameter has been extended from [0, 15] to [0, 31].

Impact on the Network

No impact

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNBFreqPriorityGroup.FreqIndex	ADD GNBFREQPRIO RITYGROUP LST GNBFREQPRIO RITYGROUP MOD GNBFREQPRIO RITYGROUP RMV GNBFREQPRIO RITYGROUP	gNodeB	Values 16 to 31 have been added to this parameter, whose GUI value range has therefore changed from [0, 15] to [0, 31].

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20241209002193-0014

RAT

NR FDD, NR TDD

5.3.4.1.63 Improved Integrity of External CHR Data

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

This change has improved the integrity of reported external CHR data. This is achieved by reducing the probability of flow control on external CHR data and shortening the period for selecting UEs involved in reporting. Reported external CHR data is typically incomplete in the following scenarios:

- Flow control is imposed on the management plane.
- There are UEs involved in ping-pong handovers and high-speed UEs.

The external CHRs affected by this change are SIG logs and TRC logs.

Implementation

This change adopts the following optimizations to improve the integrity of reported external CHR data:

- Optimization 1: This optimization is controlled by the **CHR_E2E_PACKAGED_REPORT_SW** option of the **gNBChrOutputCtrl.ChrOutputCtrlSwitch** parameter. If this option is selected, data processing is optimized. After external CHR messages are generated, the base station groups multiple messages for packaged reporting. This reduces the number of messages forwarded within the base station.
- Optimization 2: This optimization is controlled by the **CELL_LEVEL_SELECT_UE_TIMER_SW** option of the

gNBChrOutputCtrl.ChrOutputCtrlSwitch parameter. The following approaches are taken if this option is selected:

- The base station staggers the reporting time of periodic measurement events across different cells, ensuring that reporting occurs at distinct seconds. This avoids simultaneous reporting.
- A shorter period is used for selecting UEs involved in reporting. Specifically, the default period has been changed from 5s to 2s. This speeds up the selection of newly admitted UEs for periodic event reporting.

Impact on the Network

The following impacts arise after an upgrade:

- Optimization 1:
 - The volume of data reported over the OMCH may increase, the values of **VS.BBUBoard.CPULoad.Max** and **VS.BBUBoard.CPULoad.Mean** may decrease, and the values of **VS.NRBoard.UPlane.CPULoad.Avg** and **VS.NRBoard.UPlane.CPULoad.Max** may increase.
 - The packaging mechanism prevents the alignment among timers (at the level of seconds) for reporting different types of external CHR events (such as control-plane events and user-plane events). As a result, the event reporting sequence may change.
- Optimization 2: Due to staggered data reporting, the values of **VS.BBUBoard.CPULoad.Max** and **VS.BBUBoard.CPULoad.Mean** may decrease. UEs involved in ping-pong handovers and high-speed UEs are more likely to be selected for periodic event reporting, and therefore the sample integrity of these UEs improves.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNBChrOutputCtrl <i>.ChrOutputCtrlSwitch</i>	MOD GNBCHROUT PUTCTRL LST GNBCHROUT PUTCTRL	gNodeB	The CHR_E2E_PACKAGED_REPORT_SW option has been added to this parameter to specify whether to enable external CHR packaging.
Modified	gNBChrOutputCtrl <i>.ChrOutputCtrlSwitch</i>	MOD GNBCHROUT PUTCTRL LST GNBCHROUT PUTCTRL	gNodeB	The CELL_LEVEL_SELECT_UE_TIMER_SW option has been added to this parameter to specify whether to enable staggered data reporting by second and cell-specific UE selection timers.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

To enable external CHR packaging, staggered data reporting, and cell-specific UE selection timers, run the **MOD GNBCHROUTPUTCTRL** command to select the **CHR_E2E_PACKAGED_REPORT_SW** and **CELL_LEVEL_SELECT_UE_TIMER_SW** options of the **gNBChrOutputCtrl.ChrOutputCtrlSwitch** parameter.

Requirement/Trouble Ticket Number

FeR-IR20240815002763-0005

FeR-IR20240815002763-0007

RAT

NR FDD, NR TDD

5.3.4.1.64 Optimized Uplink Performance of 4T UEs#

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The uplink performance of 4T UEs has been optimized through the following mechanisms:

- MU-MIMO compatibility with uplink high-rank UEs: Uplink high-rank UEs can now participate in pairing after their ranks are lowered.
- Channel estimation enhancement: Channel estimation for 4T UEs has been further enhanced based on the protocol.

Implementation

When the **UL_SU_FOUR_LAYER_SW** option of the **NRDUCellUIRank.UlRankAlgoSw** parameter is selected:

- Before an upgrade, UEs with rank 3 or 4 do not participate in pairing.
- After an upgrade, UEs with rank 3 or 4 can participate in pairing after their ranks are lowered.

When the **UL_DENSER_CODEBOOK_SW** option of the **NRDUCellUeCoop.SpecUeCooperationSwitch** parameter is selected, channel estimation for 4T UEs is further enhanced after an upgrade.

Impact on the Network

- With the **UL_SU_FOUR_LAYER_SW** option of the **NRDUCellUIRank.UlRankAlgoSw** parameter selected, the uplink MU pairing rate, average number of paired layers for uplink MU-MIMO, uplink PRB usage, average uplink cell throughput, and average uplink UE throughput may change after an upgrade.
- With the **UL_DENSER_CODEBOOK_SW** option of the **NRDUCellUeCoop.SpecUeCooperationSwitch** parameter selected, the

average uplink interference and average uplink UE throughput may change after an upgrade.

 NOTE

- Uplink MU pairing rate = **N.ChMeas.MIMO.UL.Pair.Ratio**
- Average number of paired layers for uplink MU-MIMO = **N.ChMeas.MIMO.UL.Pair.Layer/N.ChMeas.MIMO.UL.Pair.PRB**
- Uplink PRB usage = **Uplink Resource Block Utilizing Rate (DU)**
- Average uplink cell throughput = **Cell Uplink Average Throughput (DU)**
- Average uplink UE throughput = **User Uplink Average Throughput (DU)**
- Average uplink interference = **N.UL.NI.Avg**

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0081

RAT

NR TDD

5.3.4.1.65 Optimized Threshold Configuration in RFSP-specific Inter-Frequency Handover Scenarios

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

To adapt to low-altitude UAV scenarios and allow UAVs to use specific inter-frequency handover thresholds, the configuration of thresholds for RFSP-specific inter-frequency handover incorporates the following optimizations in this version:

- Optimized configuration range of RFSP-specific inter-frequency handover threshold
 - **gNBRRspConfig.InterFreqA1A2A5RsrpThd1Ofs**. The value range has been changed from **-20-20** to **-100-100**.
 - **gNBRRspConfig.InterFreqA5RsrpThd2Ofs**. The value range has been changed from **-20-20** to **-100-100**.
 - **gNBRRspConfig.FreqPriHoA1A2RsrpThldOfs**. The value range has been changed from **-40-0** to **-100-100**.
- Flexible configuration of the threshold for event A4 and threshold 2 for event A5 in frequency-priority-based inter-frequency handover scenarios
 - Threshold for event A4
 - If the RFSP-specific frequency-priority-based inter-frequency handover function is disabled, the threshold for event A4 cannot be customized for UEs. In this case, the threshold for event A4 equals the value of **NRCeIlInterFHoMeaGrp.FreqPriInterFA4RsrpThld**.
 - If the RFSP-specific frequency-priority-based inter-frequency handover function is enabled, the threshold for event A4 can be customized for UEs. In this case, the threshold for event A4 equals the sum of the values of **NRCeIlInterFHoMeaGrp.FreqPriInterFA4RsrpThld** and **gNBRRspConfig.FreqPriFHoA4RsrpThldOfs**.
 - Threshold 2 for event A5
 - If the RFSP-specific frequency-priority-based inter-frequency handover function is disabled, threshold 2 for event A5 cannot be customized for UEs. In this case, threshold 2 for event A5 equals the value of **NRCeIlInterFHoMeaGrp.FreqPriInterFA4RsrpThld**.
 - If the RFSP-specific frequency-priority-based inter-frequency handover function is enabled, threshold 2 for event A5 can be

customized for UEs. In this case, threshold 2 for event A5 equals the sum of the values of **NRCellInterFHoMeaGrp.FreqPriInterFA4RsrpThld** and **gNBRfspConfig.FreqPriFHoA4RsrpThldOfs**.

 NOTE

The RFSP-specific frequency-priority-based inter-frequency handover function is controlled by the **FREQ_PRIORITY_BASED_HO_SW** option of the **NRCellAlgoSwitch.RfspAlgoSwitch** parameter.

Implementation

For details, see *Flexible User Steering, SA Mobility Management in Connected Mode, NSA Mobility Management*. The specific sections are listed in the change history of the documents.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNBRefspConfig. / nterFreqA1A2A5 RsrpThd1Ofs	ADD GNBRFSPC ONFIG MOD GNBRFSPC ONFIG LST GNBRFSPC ONFIG	gNodeB	The value range of this parameter has been changed from -20-20 to -100-100 .
Modified	gNBRefspConfig. / nterFreqA5Rsrp Thd2Ofs	ADD GNBRFSPC ONFIG MOD GNBRFSPC ONFIG LST GNBRFSPC ONFIG	gNodeB	The value range of this parameter has been changed from -20-20 to -100-100 .
Modified	gNBRefspConfig. FreqPriHoA1A2 RsrpThldOfs	ADD GNBRFSPC ONFIG MOD GNBRFSPC ONFIG LST GNBRFSPC ONFIG	gNodeB	The value range of this parameter has been changed from -40-0 to -100-100 .
New	gNBRefspConfig. FreqPriIFHoA4R srpThldOfs	ADD GNBRFSPC ONFIG MOD GNBRFSPC ONFIG LST GNBRFSPC ONFIG	gNodeB	This parameter has been added to specify the RFSP-specific user-level offset to the following thresholds: • Threshold for event A4 related to frequency-priority-based inter-frequency handover • Threshold 2 for event A5 related to frequency-priority-based inter-frequency handover

- Performance management
No impact
- Fault management

No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

Whether the change takes effect by default after an upgrade varies as follows:

- **gNBRfspConfig.InterFreqA1A2A5RsrpThd1Ofs**, **gNBRfspConfig.InterFreqA5RsrpThd2Ofs**, and **gNBRfspConfig.FreqPriHoA1A2RsrpThldOfs**: yes
- **gNBRfspConfig.FreqPriFHoA4RsrpThldOfs**: no. You can run the **MOD GNBRFSPCONFIG** command to configure the **gNBRfspConfig.FreqPriFHoA4RsrpThldOfs** parameter.

Requirement/Trouble Ticket Number

FeR-IR20240627000892-0001

RAT

NR FDD, NR TDD

5.3.4.1.66 Optimized Mechanism for Reporting CODR Status

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

After detecting that a cell enters or recovers from sleeping, the base station reports the latest status of the cell to the MAE CODR module, which did not provide any feedback in previous versions. Afterwards, the base station automatically triggers reporting every six hours to ensure the reporting reliability. To improve the reliability of sleeping cell status reporting, two mechanisms have been added: CODR status reporting with feedback on the base station side and proactive synchronization on the MAE side.

NOTE

A sleeping cell does not trigger any alarm but experiences UE access issues and has no traffic within a long period, due to potential gNodeB software or hardware faults. CODR stands for cell outage detection and recovery. For details about sleeping cells and CODR, see *Cell Outage Detection and Recovery*.

Implementation

On the base station side, CODR status reporting with feedback works as follows: After detecting that a cell enters or recovers from sleeping, the base station reports the latest status of the cell to the MAE CODR module. If the base station does not receive an ACK from the MAE for the reporting within 5 minutes, it retransmits the status message up to three times.

- If the base station still does not receive an ACK after three retransmissions, it turns back to automatic reporting every six hours.
- If automatic reporting succeeds in each early morning of the next three days, it is determined that this latest status has been successfully reported to the MAE. The base station then stops reporting this status.

On the MAE side, proactive synchronization works as follows: The MAE sends a status synchronization request to a base station in scenarios such as when the MAE detects OMCH recovery, the base station is newly subscribed to, a rule is newly selected, and the SON service is restarted. After receiving the request, the base station starts CODR status reporting with feedback.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the gNodeB and MAE.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact

- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20240815002936-0001

RAT

NR FDD, NR TDD

5.3.4.1.67 Optimized Baseband Processor Shutdown

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The baseband processor shutdown function is optimized as follows:

- Allowed baseband processor shutdown to be performed at the baseband processing unit (BBP) level.
 - Adaptive baseband shutdown: When the **BBP.CELLDEPLOY** parameter is set to **ADAPT_BB_P_SHUTDOWN**, BBP-level baseband processor shutdown is supported. On a BBP, baseband processors carrying no services can enter the low power consumption state. It is recommended that this function be enabled together with RF module deep dormancy or RF module super dormancy.
 - Power saving first: When the **BBP.CELLDEPLOY** parameter is set to **POWERSAVING**, base-station-level baseband processor shutdown is supported. Compared to adaptive baseband shutdown, this function prioritizes power saving by deploying services centrally on fewer BBPs. This further reduces power consumption, but affects cell performance.
- Deleted the mutually exclusive relationship between base-station-level baseband shutdown for energy saving (controlled by the **BB_SHUTDOWN_POWERSAVING_SW** option of the **gNodeBParam.BaseStationDeepDormancySw** parameter) and adaptive BBU deep dormancy.

This function is supported only in low frequency bands.

Implementation

For details, see *Energy Conservation and Emission Reduction*. The specific sections are listed in the change history of the document.

Impact on the Network

After baseband processor shutdown takes effect, the energy consumption of the base station decreases.

During baseband processor shutdown, activation or establishment of cells causes some baseband processors to exit the low power consumption state, and therefore the activation or establishment duration is prolonged. The duration typically lasts for 2 to 5 minutes.

After power saving first is enabled, resources are deployed based on power saving optimization, leading to changes in user-plane performance indicators such as the number of UEs, user-perceived rate, and traffic volume, as well as slight increase of the CPU usage of BBPs.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

All NR-capable main control boards and all NR-capable baseband processing units except the UBBPg1 support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

Only resources of the UBBPg3/UBBPh3a/UBBPh3b can be deployed for the power saving first function.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	BBP.CELLDEPLOY	ADD BRD MOD BBP LST BBP	gNodeB	The value ADAPT_BBP_SHUTDOWN has been added to specify whether to enable adaptive baseband shutdown.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
The following mutually exclusive relationships have been deleted.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Adaptive BBU deep dormancy	• BBP_DEEP_DORMANCY_SW option of the gNBpwrSavingAlgo.DormancySwitch parameter selected • BBP.PWR_SVGSW (5G gNodeB, LTE eNodeB) set to BBP_DEEP_DORMANCY_ADAPT	<i>Multi-RAT Coordinated Energy Saving</i>	Adaptive BBU deep dormancy cannot be enabled together with baseband shutdown for energy saving. If the BB_SHUTDOWN_POWER_SAVING_SW option of the gNodeBParam.BaseStationDeepDormancySw parameter is selected, the BBP_DEEP_DORMANCY_SW option of the gNBpwrSavingAlgo.DormancySwitch parameter must be deselected.

- Impacted features/functions
The following impact relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Adaptive BBU deep dormancy	• BBP_DEEP_DORMANCY_SW option of the gNBpwrSavingAlgo.DormancySwitch parameter selected • BBP.PWRSVG SW (5G gNodeB, LTE eNodeB) set to BBP_DEEP_DORMANCY_ADAPT	<i>Multi-RAT Coordinated Energy Saving</i>	When adaptive BBU deep dormancy is in effect, adaptive baseband shutdown, power saving first, and baseband shutdown for energy saving do not take effect.
Low-frequency TDD	Primary/Backup function in intra-frequency dual-network redundancy networking	PRIMARY_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	When the primary/backup function in intra-frequency dual-network redundancy networking is in effect, adaptive baseband shutdown and power saving first do not take effect.
Low-frequency TDD	Dual-BBU backup in dual-network redundancy networking	DUAL_BBU_BACKUP_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	None	When dual-BBU backup in dual-network redundancy networking is in effect, adaptive baseband shutdown and power saving first do not take effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios	DUAL_PLANE_TOB_SW option of the gNBRedundancyReliabilitySwitch parameter	<i>Base Station Equipment Reliability</i>	When the dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios is in effect, adaptive baseband shutdown and power saving first do not take effect.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Energy Conservation and Emission Reduction*.

Requirement/Trouble Ticket Number

FeR-IR20250206002228-0010
IR20250318001006

RAT

NR FDD, NR TDD

5.3.4.1.68 Optimized Uplink SRB Flow Control

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Uplink SRB flow control has been optimized to reduce air interface resource consumption by abnormal UEs.

Implementation

If a UE's uplink SRB traffic in 1s exceeds a specified threshold, the UE is considered abnormal. The gNodeB then releases the UE to reduce air interface resource consumption by abnormal UEs. In addition, when the number of measurement reports received by the gNodeB per second reaches a specified threshold, flow

control is imposed on measurement reports. This prevents system overload and consequent UE service exceptions caused by excessive measurement reports.

Impact on the Network

- If there are abnormal UEs on the network, the gNodeB will release these UEs after an upgrade. The following impacts on performance indicators may occur:
 - The **Service Call Drop Rate (CU)** increases slightly.
 - The number of RRC_CONNECTED UEs and the number of active UEs decrease slightly.
 - The **Cell Uplink Average Throughput** and **User Uplink Average Throughput** increase slightly.
- If the gNodeB receives a large number of measurement reports, the gNodeB will impose flow control on measurement reports after an upgrade. The following impacts on performance indicators will occur:
 - The **RRC Setup Success Rate (CU)** increases slightly.
 - The **Service Call Drop Rate (CU)** decreases slightly.
 - The number of RRC_CONNECTED UEs and the number of active UEs increase slightly.
 - The numbers of intra-frequency, inter-frequency, and inter-RAT handovers decrease slightly.
 - The **Downlink Traffic Volume (DU)** and **Uplink Traffic Volume (DU)** increase slightly.
 - The values of **VS.BBUBoard.CPULoad.Max**, **VS.BBUBoard.CPULoad.Mean**, **VS.BBUBoard.CPULoad.Over**, and **VS.BBUBoard.CPULoad.CumulativeHighloadCount** may decrease.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact

- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250208000127-0034

FeR-IR20250528001857-0031

RAT

NR FDD, NR TDD

5.3.4.1.69 Optimized Possible Causes of RF Module Dormancy Not Taking Effect

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

This change involves the following functions: LTE load-based RF module deep dormancy, LNR load-based RF module deep dormancy, NR load-based RF module deep dormancy, multi-frequency flexible dormancy, LTE load-based RF module super dormancy, LNR load-based RF module super dormancy, and NR load-based RF module super dormancy.

The possible causes why RF module dormancy does not take effect refer to those displayed in the **DSP NRDUCELLPOWERSAVING** and **DSP CELLPWRSAVINGPOLICY** command outputs.

The possible causes why RF module dormancy does not take effect are optimized as follows:

- Added the cause **Reliability Requirements Not Met**. Displayed the cause **Reliability Requirements Not Met** instead of **Others** when dormancy is ineffective due to the existence of dormancy-affecting alarms.
- Optimized the reporting mechanism of the possible causes.

LampSite base stations do not support this function.

Implementation

For details, see *Multi-RAT Coordinated Energy Saving*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the gNodeB and eNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250708001473-0040

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD

5.3.4.1.70 Optimized Out-of-Order Packet Processing Performance

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

This change has optimized the out-of-order packet processing performance, improving user experience in scenarios with out-of-order packets.

Implementation

This change has optimized the out-of-order packet processing performance, improving user experience in scenarios with out-of-order packets.

Impact on the Network

After an upgrade, as the out-of-order packet processing performance has been optimized, the following network impacts will occur:

- The values of the following indicators may increase:
 - **N.RLC.UL.AM.TrfPDU.Retrans.RxPackets**
 - **N.RLC.UL.AM.TrfPDU.Retrans.RxPackets.DuCell5QI**
 - **N.PDCP.UL.TrfSDU.RxPacket.Loss**
 - **N.PDCP.UL.TrfSDU.RxPacket.Loss.Cell5QI**
 - **N.RLC.UL.Cell5QI.Delay**
 - **N.PDCP.UL.Cell5QI.Delay**
 - **N.PDCP.UL.ROHC.VoNR.DecompFail.Pkts**
 - **N.PDCP.UL.ROHC.VoNR.BeforeDecomp.Bytes**
 - **N.VoNR.VQI.UL.Poor.Times**
 - **N.VoNR.VQI.UL.Poor.Times**
 - **N.VoNR.VQI.UL.Accept.Times**
 - **N.VoNR.E2EVQI.Bad.Times**
 - **N.VoNR.E2EVQI.Poor.Times**
 - **N.VoNR.E2EVQI.Accept.Times**
- The values of the following indicators may decrease:
 - **N.ThpVol.UE.UL.SmallPkt**
 - **N.PDCP.UL.TrfPDU.RxPackets**
 - **N.PDCP.DL.TrfSDU.RxPackets**
 - **N.PDCP.DL.TrfSDU.TxPacket.Discard**
 - **N.MAC.SDU.ThpVol.UL.Cell**

- N.MAC.SDU.ThpVol.DL.Cell
- N.PDCP.UL.ROHC.VoNR.DecompTotal.Pkts
- N.VoNR.VQI.UL.Good.Times
- N.VoNR.VQI.UL.Excellent.Times
- N.VoNR.E2EVQI.Good.Times
- N.VoNR.E2EVQI.Excellent.Times
- Uplink Traffic Volume (DU)
- Downlink Traffic Volume (DU)

In other scenarios, there is no impact.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

Only the UBBPg series boards support this change.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0167

RAT

NR FDD, NR TDD

5.3.4.1.71 Refined Logic for Enforcing Configuration Constraints Between Some Switches and the NRDUCellTrp MO

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

This change refines the logic for enforcing configuration constraints between some switches and the **NRDUCellTrp** MO.

Implementation

To turn on some switches, specific parameters must be configured in a certain way in the **NRDUCellTrp** MO. The timing of enforcing these configuration constraints may affect O&M efficiency.

Take the following configuration constraint as an example: Option *Y* of parameter *B* can be selected only if the **NRDUCellTrp.A** parameter is set to *X*. In earlier versions, enforcing this constraint in the following scenarios affects O&M efficiency:

- Scenario 1: When the **RMV NRDUCELLTRP** command is to be delivered, the system checks whether option *Y* is deselected. If it is selected, the command cannot be successfully delivered. You must deselect option *Y* before this command can be successfully delivered.
- Scenario 2: When option *Y* is to be selected, the system checks whether the corresponding **NRDUCellTrp** MO has been added. If it has not, you must run the **ADD NRDUCELLTRP** command to add the corresponding **NRDUCellTrp** MO before selecting option *Y*.

To improve O&M efficiency, the logic for enforcing configuration constraints related to the **NRDUCellTrp** MO has been refined. The refined logic is as follows (using the preceding constraint as an example):

- Scenario 1: When the **RMV NRDUCELLTRP** command is to be delivered, the system does not check whether option *Y* is deselected.
- Scenario 2: Option *Y* can be selected even when the corresponding **NRDUCellTrp** MO has not been added. The system checks against this constraint only when the **ADD NRDUCELLTRP** command is executed; this constraint must be met before this command can be successfully delivered.

If a configuration constraint places requirements not only on the **NRDUCellTrp** MO but also on the **NRDUCellCoverage** MO, the same refined logic applies for enforcing the constraint when the **RMV NRDUCELLCOVERAGE** and **ADD NRDUCELLCOVERAGE** commands are executed.

 NOTE

This change only refines the logic for enforcing certain configuration constraints. The related configuration constraints still apply, which means that if a constraint is not met, the related functions cannot take effect.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0164

RAT

NR FDD, NR TDD

5.3.4.1.72 Optimized Tail Packet Data Conversion for Optimized Measurement of Downlink CA Throughput

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Optimized measurement of downlink CA throughput (controlled by the **DL_CA_THPT_L_SLOT_STATS_POL_SW** option of **NRDUCellOamParam.StatisticsStrategySwitch**) optimizes the method for measuring the average throughput of SCell-activated UEs by converting the tail packet data, so that this average throughput can more accurately reflect the air interface capabilities of CA UEs. For details about optimized measurement of downlink CA throughput, see the corresponding sections in *Carrier Aggregation*.

The change in this function optimizes the tail packet data conversion method when the "optimized measurement of downlink CA throughput" function is enabled, further allowing the average throughput of SCell-activated UEs to more accurately reflect the air interface capabilities of CA UEs in medium- and heavy-load scenarios. The average throughput of SCell-activated UEs includes the following:

- Average throughput of SCell-activated CA UEs, which is equal to $(N.CA.SccActiveUser.ThpVol.DL - N.CA.ThpVol.DL.LastSlot) / N.CA.ThpTime.DL.RmvLastSlot$
- Average throughput of SCell-activated inter-gNodeB CA UEs, which is equal to $(N.IntergNB.CA.SccActiveUser.ThpVol.DL - N.IntergNB.CA.ThpVol.DL.LastSlot) / N.IntergNB.CA.ThpTime.DL.RmvLastSlot$
- Average throughput of SCell-activated CA UEs in the 3CC aggregation state, which is equal to $(N.3CC.CA.ThpVol.DL - N.3CC.CA.ThpVol.DL.LastSlot) / N.3CC.CA.ThpTime.DL.RmvLastSlot$
- Average throughput of SCell-activated massive CA UEs, which is equal to $(N.Massive.CA.ThpVol.DL - N.Massive.CA.ThpVol.DL.LastSlot) / N.Massive.CA.ThpTime.DL.RmvLastSlot$

Implementation

The change in this function optimizes the tail packet data conversion method when the "optimized measurement of downlink CA throughput" function is enabled, further allowing the average throughput of SCell-activated UEs to more accurately reflect the air interface capabilities of CA UEs in medium- and heavy-load scenarios. The average throughput of SCell-activated UEs includes the following:

- Average throughput of SCell-activated CA UEs, which is equal to $(N.CA.SccActiveUser.ThpVol.DL - N.CA.ThpVol.DL.LastSlot) / N.CA.ThpTime.DL.RmvLastSlot$

- Average throughput of SCell-activated inter-gNodeB CA UEs, which is equal to $(N.\text{IntergNB.CA.SccActiveUser.ThpVol.DL} - N.\text{IntergNB.CA.ThpVol.DL.LastSlot}) / N.\text{IntergNB.CA.ThpTime.DL.RmvLastSlot}$
- Average throughput of SCell-activated CA UEs in the 3CC aggregation state, which is equal to $(N.\text{3CC.CA.ThpVol.DL} - N.\text{3CC.CA.ThpVol.DL.LastSlot}) / N.\text{3CC.CA.ThpTime.DL.RmvLastSlot}$
- Average throughput of SCell-activated massive CA UEs, which is equal to $(N.\text{Massive.CA.ThpVol.DL} - N.\text{Massive.CA.ThpVol.DL.LastSlot}) / N.\text{Massive.CA.ThpTime.DL.RmvLastSlot}$

Impact on the Network

After the change in this function takes effect, the average throughput of SCell-activated UEs may increase in medium- and heavy-load scenarios, including:

- Average throughput of SCell-activated CA UEs
- Average throughput of SCell-activated inter-gNodeB CA UEs
- Average throughput of SCell-activated CA UEs in the 3CC aggregation state
- Average throughput of SCell-activated massive CA UEs

Impact on NEs

No impact

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions

- No impact
- Impacted features/functions
- No impact

Related Operations

Whether the change takes effect by default after an upgrade varies as follows:

- Optimized measurement of downlink CA throughput enabled before the upgrade: Yes
- Optimized measurement of downlink CA throughput disabled before the upgrade: No

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0160

RAT

NR FDD, NR TDD

5.3.4.1.73 Optimized Activation Verification of RSRP Prediction based on Virtual Grids

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The activation verification of RSRP prediction based on virtual grids is optimized by adding the **N.LnrVirtualGrid.Model.Max** counter, which measures the maximum number of LTE-NR inter-RAT grid models available for a cell.

This change is supported only in SA networking.

Implementation

For details, see *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
The following counters have changed.

Change Type	Counter Name	Function Subset	NE	Description
New	N.LnrVirtualGrid.Model.Max	AI Grid Model Measurement	gNode B	This counter has been added to measure the maximum number of LTE-NR inter-RAT grid models available for a cell.

- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20241209002193-0090

RAT

NR FDD, NR TDD

5.3.4.1.74 Optimized the Policy of Collaboration Between Virtual Grid-based Multi-Frequency Coordination and RF Channel Intelligent Shutdown

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The policy of collaboration between Virtual Grid-based Multi-Frequency Coordination and RF channel intelligent shutdown has been optimized.

- Before the optimization: If RF channel intelligent shutdown is not in effect in a local cell and is in effect in its neighboring cells, building of virtual grid models in the local cell may be suspended, and existing virtual grid models of the local cell may go offline and become unavailable.
- After the optimization: Virtual Grid-based Multi-Frequency Coordination and RF channel intelligent shutdown can collaborate with each other when optimization of collaboration between virtual grid models and energy-saving-oriented channel shutdown functions is enabled on both the local and neighboring base stations by selecting the **VIRTUAL_GRID_CH_SD_COOP_OPT_SW** option of the **gNodeBAlgo.NrSmartMultiCarrSw** parameter. In the scenario where RF channel intelligent shutdown is not in effect in a local cell and is in effect in its neighboring cells, the optimization increases the proportion of online virtual grid models and reduces the probability that building of virtual grid models in the local cell is suspended and the probability that existing virtual grid models of the local cell go offline and become unavailable.

This change is supported only in low frequency bands.

Implementation

For details, see *Virtual Grid-based Multi-Frequency Coordination*. The specific sections are listed in the change history of the document.

Impact on the Network

Optimization of collaboration between virtual grid models and energy-saving-oriented channel shutdown functions reduces the probability that building of virtual grid models is suspended and the probability that virtual grid models go offline and become unavailable due to RF channel intelligent shutdown.

The impacts of changes in the setting of the option that controls the optimization are as follows:

- When the setting is changed from off to on, models incapable of collaboration optimization go offline and models capable of collaboration optimization come online. Building of the models capable of collaboration optimization requires certain data from UEs, and the UEs involved in data collection will experience an increase in their signaling overheads. In addition, after the models incapable of collaboration optimization go offline, features that used the models can no longer use them.

- When the setting is changed from on to off, models capable of collaboration optimization go offline and models incapable of collaboration optimization come online. Building of the models incapable of collaboration optimization requires certain data from UEs, and the UEs involved in data collection will experience an increase in their signaling overheads. In addition, after the models capable of collaboration optimization go offline, features that used the models can no longer use them.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact.

Impact on Inter-NE Interfaces

No impact.

Impact on Operation and Maintenance

- License management
No impact.
- Configuration management
The following parameter has changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNodeBALgo.NrSmartMultiCarrierSw	MOD GNODEBAL GO LST GNODEBAL GO	gNodeB	The VIRTUAL_GRID_CH_SD_C_OOP_OPT_SW option has been added to specify whether to enable the optimization of collaboration between virtual grid models and energy-saving-oriented channel shutdown functions.

- Performance management
No impact.
- Fault management
No impact.

Impact on Features/Functions

- Prerequisite features/functions
No impact.

- Mutually exclusive features/functions
No impact.
- Impacted features/functions
The descriptions of the following impact relationships have been revised.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When RF channel intelligent shutdown and Virtual Grid-based Multi-Frequency Coordination are both enabled: • If RF channel intelligent shutdown is not in effect in a local cell and is in effect in its neighboring cells, building of virtual grid models in the local cell may be suspended, and existing virtual grid models of the local cell may go offline and become unavailable. To increase the proportion of online virtual grid models in this scenario, optimization of collaboration between virtual grid models and energy-saving-oriented channel shutdown functions can be enabled on the local and neighboring base stations by selecting the VIRTUAL_GRID_CH_SD_COOP_OPT_SW option of the gNodeBAlgo.NrSmartMultiCarrier parameter. The optimization

RAT	Function Name	Function Switch	Reference	Description
				reduces the probability that building of virtual grid models in the local cell is suspended and the probability that existing virtual grid models of the local cell go offline and become unavailable. • If RF channel intelligent shutdown is in effect in a local cell, building of virtual grid models in the local cell will be suspended and existing virtual grid models of the local cell will go offline and become unavailable.

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Virtual Grid-based Multi-Frequency Coordination*.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0089

RAT

NR FDD, NR TDD

5.3.4.1.75 Replaced the Reserved Option That Controls "Optimization of the Policy of Collaboration Between Virtual Grid-based Multi-Frequency Coordination and TTI-Level Channel Shutdown as Well as Dynamic RF Channel Shutdown" with an Option of a Formal Parameter

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The **RSVDSWPARAM4_BIT16** option of the **gNBRsvd.RsvdSwParam4** parameter, which controls optimization of the policy of collaboration between Virtual Grid-based Multi-Frequency Coordination and TTI-level channel shutdown as well as dynamic RF channel shutdown, has been replaced by the **VIRTUAL_GRID_CH_SD_COEXIST_SW** option of the **gNBPwrSavingAlgo.PwrSavingAlgoSwitch** parameter and added to the disuse list.

This change is supported only in low frequency bands.

This change is irrelevant to LampSite base stations.

Implementation

For details, see *Virtual Grid-based Multi-Frequency Coordination*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact.

Impact on Inter-NE Interfaces

No impact.

Impact on Operation and Maintenance

- License management
No impact.
- Configuration management
The following parameter has changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNBPrSavingAlgo.PwrSavingAlgoSwitch	MOD GNBPW RSA VINGALGO LST GNBPW RSA VINGALGO	gNodeB	The VIRTUAL_GRID_CH_SD_C OEXIST_SW option of this parameter has been added to specify whether the virtual grid feature can take effect together with TTI-level channel shutdown or RF channel dynamic shutdown.
Modified	gNBRsvd.RsvdSwParam4	MOD GNBR SVD LST GNBR SVD	gNodeB	The function controlled by the RSVDSWPARAM4_BIT16 option of the gNBRsvd.RsvdSwParam4 parameter is now controlled by the VIRTUAL_GRID_CH_SD_C OEXIST_SW option of the gNBPrSavingAlgo.PwrSavingAlgoSwitch parameter, and the RSVDSWPARAM4_BIT16 option of the gNBRsvd.RsvdSwParam4 parameter has been added to the disuse list. Although the configuration interface in this version supports synchronization and delivery of the setting of this option, the system does not use this option.

NOTE

RSVDSWPARAM4_BIT16 of the **gNBRsvd.RsvdSwParam4** parameter is a reserved option.

For details about reserved parameters, see 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List.

- Performance management
No impact.
- Fault management
No impact.

Impact on Features/Functions

- Prerequisite features/functions
No impact.
- Mutually exclusive features/functions
No impact.
- Impacted features/functions
The descriptions of the following impact relationships have been revised.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	TTI-level channel shutdown	TTI_RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	After TTI-level channel shutdown takes effect, the building of virtual grid models will be suspended and existing virtual grid models will be brought offline and cannot be used. If virtual grid models need to be used while TTI-level channel shutdown is in effect, the VIRTUAL_GRID_CH_SD_COEXIST_SW option of the gNBPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected for both the local and neighboring base stations.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Dynamic RF channel shutdown	RF_CHN_DYN_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	After dynamic RF channel shutdown takes effect, the building of virtual grid models will be suspended and existing virtual grid models will be brought offline and cannot be used. If virtual grid models need to be used while dynamic RF channel shutdown is in effect, the VIRTUAL_GRID_CH_SD_COEXIST_SW option of the gNBPwrSavingAlgo.PwrSavingAlgoSwitch parameter must be selected for both the local and neighboring base stations.

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

DTS2025041439607

IR20241209002193

RAT

NR FDD, NR TDD

5.3.4.1.76 Added a Formal Option to Replace the Reserved Bit That Used to Control "Optimized Rate-Matching-Pattern-Configuration-Free PDCCH Rate Matching in Joint Transmission Scenarios"

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

"Optimized rate-matching-pattern-configuration-free PDCCH rate matching in joint transmission scenarios" was introduced in V100R021C10 to increase the proportion of occasions where rate-matching-pattern-configuration-free PDCCH rate matching takes effect in joint transmission scenarios. A formal option has now been added to replace the reserved bit that used to control this optimization function. LampSite base stations do not support this function.

Implementation

"Optimized rate-matching-pattern-configuration-free PDCCH rate matching in joint transmission scenarios" used to be controlled by **RSVDSWPARAM0_BIT2** of the **NRDUCellRsvdExt03.RsvdSwParam0** parameter. This bit has now been added to the disuse list and replaced with the new **JT_PDCCH_NO_PATTERN_RM_OPT_SW** option of the **NRDUCellComp.DlCompAlgoExtSwitch** parameter.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellRsvdExt03.RsvdSwParam0	MOD NRDUCELLRSV DEXT03 LST NRDUCELLRSV DEXT03	gNodeB	RSVDSWPARAM0_BIT2 of this reserved parameter has been added to the disuse list. The controlled function is now controlled by the JT_PDCCH_NO_PATTERN_RM_OPT_SW option of the NRDUCellComp.DlCompAlgoExtSwitch parameter. Although the configuration interface in this version supports synchronization and delivery of the setting of this bit, the system does not use this bit.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellComp.DI CompAlgoExtSwitch	MOD NRDUCELLCO MP LST NRDUCELLCO MP	gNodeB	The JT_PDCCH_N O_PATTERN_R M_OPT_SW option has been added to this parameter to specify whether to enable "optimized rate-matching-pattern-configuration-free PDCCH rate matching in joint transmission scenarios".

NOTE

NRDUCellRsvdExt03.RsvdSwParam0 is a reserved parameter.

For details about reserved parameters, see 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20241209002193

DTS2025032400796

RAT

NR FDD, NR TDD

5.3.4.1.77 Replaced the Reserved Parameters for "Optimized Experience Evaluation for NSA Carrier Combination Selection Based on User Experience" with Formal Parameters

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The **RSVD_SW_PARAM0_BIT20** option of the LTE reserved parameter **EnodebRsvdParamExt.RsvdSwParam0** and the **RSVDSWPARAM4_BIT19** option of the NR reserved parameter **gNBRsvd.RsvdSwParam4**, which jointly control "optimized experience evaluation for NSA carrier combination selection based on user experience", have been replaced by the **NSA_NR_EVAL_REQ_NR_MEAS_IND_SW** option of the **EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch** parameter and the **EXP_BASED_NSA_PCC_SEL_OPT_SW** option of the **gNodeBAlgo.NrSmartMultiCarrSw** parameter, respectively.

Implementation

For details, see *EPC-based NSA Experience Improvement*. The specific sections are listed in the change history of the document.

Impact on the Network

None

Impact on NEs

This function is implemented on the gNodeB and eNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	EnodebRsvdParamExt.RsvdSwParam0	MOD ENBCELLRSVDPARAMEXT LST ENBCELLRSVDPARAMEXT	eNodeB	The RSVD_SW_PARAM0_BIT20 option of the EnodebRsvdParamExt.RsvdSwParam0 parameter has been added to the disuse list and replaced by the NSA_NR_EVAL_REQ_NR_MEAS_IND_SW option of the EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch parameter.
Modified	EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch	MOD ENODEBALGOEXTSWITCH LST ENODEBALGOEXTSWITCH	eNodeB	The NSA_NR_EVAL_REQ_NR_MEAS_IND_SW option has been added to specify whether to enable "optimized experience evaluation for NSA carrier combination selection based on user experience".
Modified	gNBRsvd.RsvdSwParam4	MOD GNBRSDV LST GNBRSDV	gNodeB	The RSVDSWPARAM4_BIT19 option of the gNBRsvd.RsvdSwParam4 parameter has been added to the disuse list and replaced by the EXP_BASED_NSA_PCC_SEL_OPT_SW option of the gNodeBAlgo.NrSmartMultiCarrSw parameter.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNodeBAlgo.NrSmartMultiCarrierSw	MOD GNODEBAL GO LST GNODEBAL GO	gNodeB	The EXP_BASED_NSA_PCC_SEL_OPT_SW option has been added to specify whether to enable "optimized experience evaluation for NSA carrier combination selection based on user experience".

NOTE

EnodebRsvdParamExt.RsvdSwParam0 and **gNBRsrd.RsrdSwParam4** are reserved parameters.

For details about reserved parameters, see 3900 & 5900 Series Base Station eNodeBFunction Used Reserved Parameter List and 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact.
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *EPC-based NSA Experience Improvement*.

Requirement/Trouble Ticket Number

IR20250528001008

DTS2025060918621

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD

5.3.4.1.78 Replaced the Reserved Parameter Controlling the "Multi-Carrier Combination Selection Based on Multiple Bandwidth Levels" function with a Formal Parameter

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The **RSVDSWPARAM0_BIT31** option of the **NRCeRsvd.RsvdSwParam0** parameter, originally used to control the "multi-carrier combination selection based on multiple bandwidth levels" function, has been added to the disuse list and replaced by the **MULTI_CARR_MULTI_BWP_OPT_SW** option of the **NRCeCaMgmtConfig.CaStrategySwitch** parameter.

The relevant upgrade rules are as follows:

- After an upgrade from V100R021C10SPC210 or a later V100R021C10 version, or from V100R022C00SPC050 or a later V100R022C00 version, to V100R022C10SPC010 or later, the pre-upgrade setting of the **RSVDSWPARAM0_BIT31** option of **NRCeRsvd.RsvdSwParam0** will be inherited.
- In other upgrade paths, this option will be deselected after the upgrade.

This change is supported only in low frequency bands.

Implementation

For details, see *Carrier Aggregation*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

No impact

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Disused	NRCeIIrsvd.RsvdSwParam0	MOD NRCELLRSVD LST NRCELLRSVD	gNodeB	The RSVDSWPARAM0_BIT31 option of this reserved parameter has been added to the disuse list, and the multi-carrier combination selection based on multiple bandwidth levels function, originally conveyed by this option, is now controlled by the MULTI_CARR_MULTI_BWP_OPT_SW option of the NRCeIIcaMgmtConfig.CaStrategySwitch parameter. Although the configuration interface in this version supports synchronization and delivery of the setting of this bit, the system does not use this bit.
Modified	NRCeIIcaMgmtConfig.CaStrategySwitch	MOD NRCELLCAMGMTCONFIG LST NRCELLCAMGMTCONFIG	gNodeB	The MULTI_CARR_MULTI_BWP_OPT_SW option has been added to specify whether to enable multi-carrier combination selection based on multiple bandwidth levels.

NOTE

NRCeIIrsvd.RsvdSwParam0 is a reserved parameter.

For details about reserved parameters, see 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250726001541-0010

RAT

NR FDD, NR TDD

5.3.4.1.79 Changed the Reserve Parameter for Adaptive CSI-RS Beam Sweeping for CA UEs to a Formal Parameter

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Before the upgrade, adaptive CSI-RS beam sweeping for CA UEs is controlled by **RSVDSWPARAM1_BIT18** of the **NRDUCellRsvdExt03.RsvdSwParam1** parameter. After the upgrade, this function is controlled by the **CA_CSIRS_BM_SWEEP_ADPT_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoSwitch** parameter.

Implementation

For details, see *Cell Combination* and *Hyper Cell*. The specific sections are listed in the change history of the documents.

Network Impacts

No impact

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellCarrMgmt.CaEnhancedAlgoSwitch	MOD NRDUCELLC ARRMGMT LST NRDUCELLC ARRMGMT	gNodeB	The CA_CSIRS_BM_SWEEP_ADPT_SW option has been added to specify whether to enable adaptive CSI-RS beam sweeping for CA UEs.
Modified	NRDUCellRsvdExt03.RsvdSwParam1	MOD NRDUCELLR SVDEXT03 LST NRDUCELLR SVDEXT03	gNodeB	RSVDSWPARAM1_BIT18 of the NRDUCellRsvdExt03.RsvdSwParam1 parameter has been added to the disuse list. The function of this bit is now controlled by the CA_CSIRS_BM_SWEEP_ADPT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter. Although the configuration interface in this version supports synchronization and delivery of the setting of this bit, the system does not use this bit.

NOTE

NRDUCellRsvdExt03.RsvdSwParam1 is a reserved parameter.

For details about reserved parameters, see 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List.

- Performance management

No impact

- Fault management

No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250806001609-0250

RAT

NR FDD, NR TDD

5.3.4.1.80 Added a Formal Parameter to Replace the Reserved Parameter that Controls Transmit Power Optimization When PMI-based Weights Are Used for the PDCCH

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The transmit power in PDCCH symbols occasionally does not meet the expectation, causing an inconsistency between the actual transmit power in PDCCH symbols with the value of the **N.DL.Pwr.Avg** counter. This may occur when both of the following conditions are met:

- The **NRDUCellPdcchAlgo.PdcchPrecodeEnhPolicy** parameter is set to **ENH_PMI**.
- One of the following conditions is met:
 - The **DM_MIMO_SERVICE_SWITCH** option of the **NRDUCellAlgoSwitch.DmMimoSwitch** parameter is selected.
 - The **D_MIMO_SERVICE_SWITCH** option of the **NRDUCellAlgoSwitch.DmMimoSwitch** parameter is selected.
 - The **NRDUCellAlgoSwitch.VmimoSwitch** parameter is set to **ON**.
 - The **SRS_DIFF_BASED_RANK_SELECT_SW** option of the **NRDUCellAlgoSwitch.RankSelectAlgoSw** parameter is selected.

In response, "transmit power optimization when PMI-based weights are used for the PDCCH" was introduced in V100R021C10SPC160. This function was described as a resolved issue in the release notes of this version. The trouble ticket number was DTS2025041810831. This function was controlled by **RSVDSWPARAM0_BIT6**

of the reserved parameter **NRDUCellRsvdExt05.RsvdSwParam0**. When this bit is set to **1**, transmit power optimization takes effect when PMI-based weights are used for the PDCCH, and the transmit power of the PDCCH is adjusted based on the actual number of transmit antennas. By default, this bit is set to **1**, indicating that this function is enabled. In upgrade scenarios, this bit is set to **0** by default if the **NRDUCell.LampSiteCellFlag** parameter is set to **NO**, and is set to **1** by default if the **NRDUCell.LampSiteCellFlag** parameter is set to **YES**.

This reserved bit has been added to the disuse list in V100R022C10SPC010. The function of this reserved bit is now controlled by the **PDCCH_PMI_WEIGHT_PWR_OPT_SW** option of the **NRDUCellPdcch.PdcchAlgoEnhSwitch** parameter.

Implementation

RSVDSWPARAM0_BIT6 of the reserved parameter **NRDUCellRsvdExt05.RsvdSwParam0** has been added to the disuse list, and is replaced with the **PDCCH_PMI_WEIGHT_PWR_OPT_SW** option of the **NRDUCellPdcch.PdcchAlgoEnhSwitch** parameter.

- For an upgrade from a version earlier than V100R021C10SPC160 to V100R022C10SPC010 or later:
 - For a cell with **NRDUCell.LampSiteCellFlag** set to **NO** before the upgrade, the switch will be on after the upgrade if either of the following conditions is met and will be off in other cases: (1) **NRDUCellPdcchAlgo.PdcchPrecodeEnhPolicy** is set to **ENH_OFF**. (2) **NRDUCellAlgoSwitch.VmimoSwitch** is set to **OFF**, the **DM_MIMO_SERVICE_SWITCH** and **D_MIMO_SERVICE_SWITCH** options of **NRDUCellAlgoSwitch.DmMimoSwitch** are deselected, and the **SRS_DIFF_BASED_RANK_SELECT_SW** option of **NRDUCellAlgoSwitch.RankSelectAlgoSw** is deselected.
 - For a cell with **NRDUCell.LampSiteCellFlag** set to **YES** before the upgrade, the switch will be on after the upgrade.
- For an upgrade from a version between V100R021C10SPC160 (inclusive) and V100R022C00SPC100 (not inclusive) to V100R022C10SPC010 or later:
 - For a cell with **NRDUCell.LampSiteCellFlag** set to **NO** before the upgrade: (1) The switch will be on if **NRDUCellPdcchAlgo.PdcchPrecodeEnhPolicy** is set to **ENH_OFF**, or if **NRDUCellAlgoSwitch.VmimoSwitch** is set to **OFF**, the **DM_MIMO_SERVICE_SWITCH** and **D_MIMO_SERVICE_SWITCH** options of **NRDUCellAlgoSwitch.DmMimoSwitch** are deselected, and the **SRS_DIFF_BASED_RANK_SELECT_SW** option of **NRDUCellAlgoSwitch.RankSelectAlgoSw** is deselected. (2) In other cases, the switch will be off if **RSVDSWPARAM0_BIT6** of **NRDUCellRsvdExt05.RsvdSwParam0** is set to **0** before the upgrade and will be on otherwise.
 - For a cell with **NRDUCell.LampSiteCellFlag** set to **YES** before the upgrade, the switch will be off in the case that **RSVDSWPARAM0_BIT6** of **NRDUCellRsvdExt05.RsvdSwParam0** is set to **0** before the upgrade and will be on otherwise.
- For an upgrade from V100R022C00SPC100 or later to V100R022C10SPC010 or later, if **RSVDSWPARAM0_BIT6** of **NRDUCellRsvdExt05.RsvdSwParam0** is set

to **0** before the upgrade, the switch will be off after the upgrade; otherwise, the switch will be on after the upgrade.

Impact on the Network

After an upgrade from a version earlier than V100R021C10SPC160 to V100R022C10SPC010 or later, if the **NRDUCell.LampSiteCellFlag** parameter is set to **YES** before the upgrade, the impacts are as follows:

- The values of the following indicators may increase:
 - Average uplink PDCCH aggregation level
 - Average downlink PDCCH aggregation level
 - Downlink IBLER
 - Uplink IBLER
 - Uplink RBLER
 - Downlink RBLER
 - Uplink CCE allocation failure rate
 - Downlink CCE allocation failure rate
- The values of the following indicators may decrease:
 - Average uplink UE throughput
 - Average downlink UE throughput
- The average downlink cell power may change depending on UE models.

The involved indicators are defined as follows.

Indicator	Definition
Average uplink PDCCH aggregation level	$\frac{(N.CCE.UL.AggLvl16Num \times 16 + N.CCE.UL.AggLvl8Num \times 8 + N.CCE.UL.AggLvl4Num \times 4 + N.CCE.UL.AggLvl2Num \times 2 + N.CCE.UL.AggLvl1Num)}{(N.CCE.UL.AggLvl16Num + N.CCE.UL.AggLvl8Num + N.CCE.UL.AggLvl4Num + N.CCE.UL.AggLvl2Num + N.CCE.UL.AggLvl1Num)}$
Average downlink PDCCH aggregation level	$\frac{(N.CCE.DL.AggLvl16Num \times 16 + N.CCE.DL.AggLvl8Num \times 8 + N.CCE.DL.AggLvl4Num \times 4 + N.CCE.DL.AggLvl2Num \times 2 + N.CCE.DL.AggLvl1Num)}{(N.CCE.DL.AggLvl16Num + N.CCE.DL.AggLvl8Num + N.CCE.DL.AggLvl4Num + N.CCE.DL.AggLvl2Num + N.CCE.DL.AggLvl1Num)}$
Uplink IBLER	$\frac{(N.UL.SCH.HalfPiBPSK.ErrTB.Ibler + N.UL.SCH.QPSK.ErrTB.Ibler + N.UL.SCH.16QAM.ErrTB.Ibler + N.UL.SCH.64QAM.ErrTB.Ibler + N.UL.SCH.256QAM.ErrTB.Ibler)}{(N.UL.SCH.HalfPiBPSK.TB + N.UL.SCH.QPSK.TB + N.UL.SCH.16QAM.TB + N.UL.SCH.64QAM.TB + N.UL.SCH.256QAM.TB)} \times 100\%$

Indicator	Definition
Downlink IBLER	$\frac{(N.DL.SCH.QPSK.ErrTB.Ibler + N.DL.SCH.16QAM.ErrTB.Ibler + N.DL.SCH.64QAM.ErrTB.Ibler + N.DL.SCH.256QAM.ErrTB.Ibler)}{(N.DL.SCH.QPSK.TB + N.DL.SCH.16QAM.TB + N.DL.SCH.64QAM.TB + N.DL.SCH.256QAM.TB)} \times 100\%$
Uplink RBLER	$\frac{(N.UL.SCH.HalfPiBPSK.ErrTB.Rbler + N.UL.SCH.QPSK.ErrTB.Rbler + N.UL.SCH.16QAM.ErrTB.Rbler + N.UL.SCH.64QAM.ErrTB.Rbler + N.UL.SCH.256QAM.ErrTB.Rbler)}{(N.UL.SCH.HalfPiBPSK.TB + N.UL.SCH.QPSK.TB + N.UL.SCH.16QAM.TB + N.UL.SCH.64QAM.TB + N.UL.SCH.256QAM.TB)} \times 100\%$
Downlink RBLER	$\frac{(N.DL.SCH.QPSK.ErrTB.Rbler + N.DL.SCH.16QAM.ErrTB.Rbler + N.DL.SCH.64QAM.ErrTB.Rbler + N.DL.SCH.256QAM.ErrTB.Rbler)}{(N.DL.SCH.QPSK.TB + N.DL.SCH.16QAM.TB + N.DL.SCH.64QAM.TB + N.DL.SCH.256QAM.TB)} \times 100\%$
Uplink CCE allocation failure rate	$\frac{(N.CCE.UL.AllocReq.Num - N.CCE.UL.AggLvl1Num - N.CCE.UL.AggLvl2Num - N.CCE.UL.AggLvl4Num - N.CCE.UL.AggLvl8Num - N.CCE.UL.AggLvl16Num)}{N.CCE.UL.AllocReq.Num} \times 100\%$
Downlink CCE allocation failure rate	$\frac{N.RadioRes.AllocFail.PDCCH}{(N.RadioRes.AllocFail.PDCCH + N.CCE.DL.AggLvl1Num + N.CCE.DL.AggLvl2Num + N.CCE.DL.AggLvl4Num + N.CCE.DL.AggLvl8Num + N.CCE.DL.AggLvl16Num)} \times 100\%$
Average uplink UE throughput	User Uplink Average Throughput (DU)
Average downlink UE throughput	User Downlink Average Throughput (DU)
Average downlink cell power	N.DL.Pwr.Avg

Impact on NEs

This change is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellRsvdExt05.RsvdSwParam0	MOD NRDUCELLRSV DEXT05 LST NRDUCELLRSV DEXT05	gNodeB	RSVDSWPARAM0_BIT6 of this reserved parameter has been replaced with the PDCCH_PMI_WEIGHT_PWR_OPT_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter, therefore has been added to the disuse list. Although the configuration interface in this version supports synchronization and delivery of the setting of this bit, the system does not use this bit.

 NOTE

NRDUCellRsvdExt05.RsvdSwParam0 is a reserved parameter.
For details about reserved parameters, see 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List.

- Performance management
No impact

- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250528001856-0001

RAT

NR TDD

5.3.4.1.81 Replaced the Reserved Parameter for "Optimized Measurement of RRC Connection Setup Success Rate Based on Cell Radius" with a Formal Parameter

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Replaced **RSVDSWPARAM0_BIT15** of the reserved parameter **NRCellRsvd.RsvdSwParam0** with the **RRC_SETUP_SUCC_RADIUS_OPT_SW** option of the **NRCellFeatureSwitch.CounterConfigSwitch** parameter for the function of "optimized measurement of RRC connection setup success rate based on cell radius".

This function is supported only by macro base stations in SA networking.

Implementation

For details, see *Random Access Control*. The specific sections are listed in the change history of the document.

Impact on the Network

After "optimized measurement of RRC connection setup success rate based on cell radius" is enabled, UEs whose cell radius exceeds the limit are excluded in the measurement of the RRC connection setup success rate. This prevents a potential

decrease in the RRC connection setup success rate that could arise from enabling "UE random access control based on cell radius."

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRCellRsvd.RsvdSwParam0	MOD NRDUCELLRsvd LST NRDUCELLRsvd	gNodeB	Replaced RSVDSWPARAM0_BIT15 of the reserved parameter NRCellRsvd.RsvdSwParam0 with the RRC_SETUP_SUCC_RADIUS_OPT_SW option of the NRCellFeatureSwitch.CounterConfigSwitch parameter.

Change Type	Parameter ID	MML Command	NE	Description
New	NRCellFeatureSwitch.CounterConfigSwitch	MOD NRCELLFEATURSWITCH LST NRCELLFEATURSWITCH	gNodeB	Added the RRC_SETUP_SUCCESS_RADIUS_OPT_SW option to the NRCellFeatureSwitch.CounterConfigSwitch parameter to control the function of "optimized measurement of RRC connection setup success rate based on cell radius".

 NOTE

NRCellRsvd.RsvdSwParam0 is a reserved parameter.
For details about reserved parameters, see 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE random access control based on cell radius	ACCESS_BY ND_CELL_R AD_PROHIB _SW option of the NRDUCellPrach.PrachAlgoExtSwitch parameter	<i>Random Access Control</i>	"Optimized measurement of RRC connection setup success rate based on cell radius" takes effect only after "UE random access control based on cell radius" is enabled.

- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *Random Access Control*.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0079

RAT

NR FDD, NR TDD

5.3.4.1.82 Changed the Value Range of the DlyCarrSelDlPrbUsageThld Parameter

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The value range of the **gNBCloudXParamGroup.DlyCarrSelDlPrbUsageThld** parameter has been changed.

This change is supported only in low frequency bands.

Implementation

The value range of the **gNBCloudXParamGroup.DlyCarrSelDlPrbUsageThld** parameter has been changed from **10–45** to **10–60**.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNBCloudXParamGroup.DlyCarrSelDlPrbUsageThld	ADD GNBCLOUDXP ARAMGROUP MOD GNBCLOUDXP ARAMGROUP LST GNBCLOUDXP ARAMGROUP	gNodeB	This parameter can now be set to a value in the range of 46 to 60 , and therefore the GUI value range of this parameter has therefore changed from 10–45 to 10–60 .

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20250528001008

DTS2025052904985

RAT

NR FDD, NR TDD

5.3.4.1.83 Modified Policy for Measuring the Counters in Number of Active Users with a Certain 5QI and Number of QCI-specific Active Users

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Before the upgrade, only the number of RRC_CONNECTED UEs that have (1) bearers with a specific 5QI or QCI and (2) data in the downlink buffer at the sampling time is counted in the counters in **Number of Active Users with a Certain 5QI** and **Number of QCI-specific active users**.

After the upgrade, the policy for measuring the counters in **Number of Active Users with a Certain 5QI** and **Number of QCI-specific active users** is related to the setting of the **NR5QI_QCI_ACT_USER_STAT_POL_SW** option of the **NRDUCellOamParam.StatisticsStrategySwitch** parameter.

- If this option is deselected, the number of RRC_CONNECTED UEs that have (1) bearers with a specific 5QI or QCI and (2) data in the downlink buffer at the sampling time is counted.
- If this option is selected, the number of RRC_CONNECTED UEs that have (1) bearers with a specific 5QI or QCI and (2) data in the uplink or downlink buffer at the sampling time is counted.

Number of Active Users with a Certain 5QI includes the following counters:

- **N.User.RRConn.Active.DuCell5QI.Avg** (Average number of active users with a certain 5QI in a DU cell)

- **N.User.RRCCConn.Active.DuCell5QI.Max** (Maximum number of active users with a certain 5QI in a DU cell)

Number of QCI-specific active users includes the following counters:

- **N.User.RRCCConn.Active.QCI.Avg** (Average number of QCI-specific active users in NSA networking)
- **N.User.RRCCConn.Active.QCI.Max** (Maximum number of QCI-specific active users in NSA networking)

Implementation

Before the upgrade, only the number of RRC_CONNECTED UEs that have (1) bearers with a specific 5QI or QCI and (2) data in the downlink buffer at the sampling time is counted in the counters in **Number of Active Users with a Certain 5QI** and **Number of QCI-specific active users**.

After the upgrade, the policy for measuring the counters in **Number of Active Users with a Certain 5QI** and **Number of QCI-specific active users** is related to the setting of the **NR5QI_QCI_ACT_USER_STAT_POL_SW** option of the **NRDUCellOamParam.StatisticsStrategySwitch** parameter.

- If this option is deselected, the number of RRC_CONNECTED UEs that have (1) bearers with a specific 5QI or QCI and (2) data in the downlink buffer at the sampling time is counted.
- If this option is selected, the number of RRC_CONNECTED UEs that have (1) bearers with a specific 5QI or QCI and (2) data in the uplink or downlink buffer at the sampling time is counted.

Number of Active Users with a Certain 5QI includes the following counters:

- **N.User.RRCCConn.Active.DuCell5QI.Avg** (Average number of active users with a certain 5QI in a DU cell)
- **N.User.RRCCConn.Active.DuCell5QI.Max** (Maximum number of active users with a certain 5QI in a DU cell)

Number of QCI-specific active users includes the following counters:

- **N.User.RRCCConn.Active.QCI.Avg** (Average number of QCI-specific active users in NSA networking)
- **N.User.RRCCConn.Active.QCI.Max** (Maximum number of QCI-specific active users in NSA networking)

Impact on the Network

After the **NR5QI_QCI_ACT_USER_STAT_POL_SW** option of the **NRDUCellOamParam.StatisticsStrategySwitch** parameter is selected, the values of the following counters increase:

- **N.User.RRCCConn.Active.DuCell5QI.Avg** (Average number of active users with a certain 5QI in a DU cell)
- **N.User.RRCCConn.Active.DuCell5QI.Max** (Maximum number of active users with a certain 5QI in a DU cell)
- **N.User.RRCCConn.Active.QCI.Avg** (Average number of QCI-specific active users in NSA networking)

- **N.User.RRCCConn.Active.QCI.Max** (Maximum number of QCI-specific active users in NSA networking)

Impact on NEs

This change is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameter has changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCelloamParam.StatisticsStrategySwitch	MOD NRDUCELLO AMPARAM LST NRDUCELLO AMPARAM	gNodeB	The NR5QI_QCI_ACT_USER_STAT_POL_SW option has been added to specify the policy for measuring the number of active UEs with the specific 5QI or QCI.

- Performance management
The following counters have changed.

Change Type	Counter Name	Function Subset	NE	Description
Modified	N.User.RRC Conn.Active .DuCell5QI.Avg	DU Cell 5QI Traffic Measurement	gNode B	The policy for measuring the counters has been modified. - If the NR5QI_QCI_ACT_USER_STAT_POL_SW option of the NRDUCellOamParam.StatisticsStrategySwitch parameter is selected, the number of RRC_CONNECTED UEs that have (1) bearers with a specific 5QI or QCI and (2) data in the uplink or downlink buffer at the sampling time is counted. - If the NR5QI_QCI_ACT_USER_STAT_POL_SW option of the NRDUCellOamParam.StatisticsStrategySwitch parameter is deselected, the number of RRC_CONNECTED UEs that have (1) bearers with a specific 5QI or QCI and (2) data in the downlink buffer at the sampling time is counted.
Modified	N.User.RRC Conn.Active .DuCell5QI.Max	DU Cell 5QI Traffic Measurement	gNode B	
Modified	N.User.RRC Conn.Active .QCI.Avg	DU Cell QCI User Number Measurement	gNode B	
Modified	N.User.RRC Conn.Active .QCI.Max	DU Cell QCI User Number Measurement	gNode B	

- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact

- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

The policy for measuring certain counters can be modified by running the **MOD NRDUCELLOAMPARAM** command with the status of the **NR5QI_QCI_ACT_USER_STAT_POL_SW** option of the **NRDUCellOamParam.StatisticsStrategySwitch** parameter configured.

Requirement/Trouble Ticket Number

FeR-IR20250911002729-0003

RAT

NR FDD, NR TDD

5.3.4.1.84 Modified the Dependency of "Data Split of Sparse Packets in DRX Scenarios in NSA Networking" on NSA Downlink Data Split Optimization

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

This change removes the restriction that data split of sparse packets in DRX scenarios in NSA networking can be enabled only after NSA downlink data split optimization on the NR side is enabled. Now, data split of sparse packets in DRX scenarios in NSA networking will take effect only after NSA downlink data split optimization on the NR side is enabled.

This change is supported only in NSA networking.

NOTE

This change was first incorporated into V100R022C00SPC010 but is first described in the network impact report of V100R022C00SPC050.

Implementation

Before the change, data split of sparse packets in DRX scenarios in NSA networking can be enabled only after NSA downlink data split optimization on the NR side is enabled.

After the change, data split of sparse packets in DRX scenarios in NSA networking can be enabled independently of NSA downlink data split optimization on the NR side; however, it will only take effect once the latter is enabled.

Data split of sparse packets in DRX scenarios in NSA networking is controlled by the **DRX_NSA_SPARSE_PKT_SPLIT_SW** option of the **gNodeBParam.NsaDcOptSwitch** parameter.

NSA downlink data split optimization on the NR side is controlled by the **DRX_NSA_SPLIT_OPT_SW** option of the **gNodeBParam.NsaDcOptSwitch** parameter.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
The following dependency relationship has been changed.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency NR TDD	NSA downlink data split optimization	DRX_NSA_SPLIT_OPT_SW option of the gNodeBParameter.NsaDcOptSwitch parameter	<i>EPC-based NSA Basics</i>	Data split of sparse packets in DRX scenarios in NSA networking can be enabled independently of NSA downlink data split optimization on the NR side; however, it will only take effect once the latter is enabled.

- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20241209002193

RAT

NR FDD, NR TDD

5.3.4.1.85 Modified Way in Which the gNodeB Measures Random-Access-related Counters of NR TDD Cells in SUL Scenarios

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In SUL scenarios, random access-related counters are now measured in NR TDD cells instead of SUL cells.

- When only UL and DL Decoupling is in effect, the random access-related counters listed in [Table 5-38](#) need to be measured in SUL cells instead of NR TDD cells to ensure that the values are accurate. Such measurements do not need to be performed in independently deployed SUL cells or SUL cells co-deployed with NR FDD cells.
- When only super uplink is in effect:
 - In independent SUL deployment scenarios, the random access-related counters listed in [Table 5-38](#) (except for **N.RA.Contention.Msg3**, **N.RA.Dedicated.HO.Msg3**, **N.RA.Dedicated.Msg3**, and **N.RA.SDT.Contention.Msg3**) need to be measured in SUL cells instead of NR TDD cells to ensure that the values are accurate. Such measurements do not need to be performed in independently deployed SUL cells. If false preamble detection occurs in SUL cells, the values of the access-related counters of NR TDD cells need to be measured. In the case of false preamble detection, the **SUPER_UL_TDM_SCH_SW** option of **NRDUCellAlgoSwitch.SuperUplinkSwitch** needs to be selected, **NRDUCellAlgoSwitch.UlDlDecouplingSwitch** needs to be set to **OFF**, and the **SUL_CHANNEL_DECOUPLE_SW** option of **NRDUCellCollabServ.SulAlgoSwitch** needs to be deselected.
 - In SUL and NR FDD co-deployment scenarios, random access to SUL cells cannot be initiated.
- When UL and DL Decoupling and super uplink are both in effect, the random access-related counters listed in [Table 5-38](#) need to be measured in SUL cells instead of NR TDD cells to ensure that the values are accurate. Such measurements do not need to be performed for independently deployed SUL cells or SUL cells co-deployed with NR FDD cells.

Table 5-38 Random access-related counters measured in a modified way

Counters	Description
N.RA.Contention.Att	Number of random preamble receptions
N.RA.Contention.Msg3	Number of Msg3 receptions during contention-based random access
N.RA.Dedicated.Att	Number of dedicated preamble receptions
N.RA.Dedicated.HO.Att	Number of dedicated preamble receptions in case of handovers in a cell
N.RA.Dedicated.HO.Msg3	Number of Msg3 receptions during non-contention-based random access triggered by handover

Counters	Description
N.RA.Dedicated.Msg3	Number of Msg3 receptions during non-contention based random access
N.RA.RedCap.Contention.Att	Number of random preamble receptions for RedCap UEs
N.RA.SDT.Contention.Att	Number of random preamble receptions for RA-SDT
N.RA.SDT.Contention.Msg3	Number of Msg3 receptions during RA-SDT
N.RA.TA.Total	Total TA value in random access for UEs in a cell
N.RA.TA.UE.Index0	Number of random access times when the TA values are in the range indicated by index 0
N.RA.TA.UE.Index1	Number of random access times when the TA values are in the range indicated by index 1
N.RA.TA.UE.Index2	Number of random access times when the TA values are in the range indicated by index 2
N.RA.TA.UE.Index3	Number of random access times when the TA values are in the range indicated by index 3
N.RA.TA.UE.Index4	Number of random access times when the TA values are in the range indicated by index 4
N.RA.TA.UE.Index5	Number of random access times when the TA values are in the range indicated by index 5
N.RA.TA.UE.Index6	Number of random access times when the TA values are in the range indicated by index 6
N.RA.TA.UE.Index7	Number of random access times when the TA values are in the range indicated by index 7
N.RA.TA.UE.Index8	Number of random access times when the TA values are in the range indicated by index 8
N.RA.TA.UE.Index9	Number of random access times when the TA values are in the range indicated by index 9

Counters	Description
N.RA.TA.UE.Index10	Number of random access times when the TA values are in the range indicated by index 10
N.RA.TA.UE.Index11	Number of random access times when the TA values are in the range indicated by index 11
N.RA.TA.UE.Index12	Number of random access times when the TA values are in the range indicated by index 12
N.RA.TA.UE.Index13	Number of random access times when the TA values are in the range indicated by index 13
N.RA.TA.UE.Index14	Number of random access times when the TA values are in the range indicated by index 14
N.RA.TA.UE.Index15	Number of random access times when the TA values are in the range indicated by index 15

Implementation

For details, see *UL and DL Decoupling* and *Super Uplink*. The specific sections are listed in the change history of the documents.

Impact on the Network

- When only UL and DL Decoupling is in effect, the random access-related counters need to be measured in SUL cells instead of NR TDD cells to ensure that the values are accurate. Such measurements do not need to be performed in independently deployed SUL cells or SUL cells co-deployed with NR FDD cells.
 - In independent SUL deployment mode, UEs randomly access an SUL cell. Random access-related counters of the NR TDD cell are measured, which may cause the values to fluctuate. The values of random access-related counters of the SUL cell are all 0 (except for **N.RA.Contention.Att.Max**, **N.RA.Dedicated.PreambleAssign.Num**, and **N.RA.Dedicated.PreambleAssign.Num**).
 - In SUL and NR FDD co-deployment mode, UEs randomly access an SUL cell. Random access-related counters of the NR TDD cell are measured, which may cause the values to fluctuate. The values of random access-related counters of the SUL cell co-deployed with the NR FDD cell are not measured (except for **N.RA.Contention.Att.Max**, **N.RA.Dedicated.PreambleAssign.Num**, and **N.RA.Dedicated.PreambleAssign.Num**).

- When only super uplink is in effect:
 - In independent SUL deployment mode, false preamble detection may occur in SUL cells and random access-related counters are measured in NR TDD cells, causing the measured Msg1-related counter value may be greater than the measured Msg2-related counter value in NR TDD cells. That is, the value of **N.RA.Contention.Att** may be greater than that of **N.RA.Contention.Resp** in contention-based random access scenarios, and the value of **N.RA.Dedicated.Att** may be greater than that of **N.RA.Dedicated.Resp** in non-contention-based random access scenarios, causing a probability of a decrease in the success rate of access to NR TDD cells. If the **SUL_CHANNEL_DECOUPLE_SW** option of the **NRDUCellCollabServ.SulAlgoSwitch** parameter is selected, there is no such impact.
 - In SUL and NR FDD co-deployment mode, there are no impacts.
- When super uplink and UL and DL Decoupling are both in effect, the impacts are the same as those when only UL and DL Decoupling is enabled.

Impact on NEs

This change is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
The change in this function has no impact. It is controlled by the existing license.
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

DTS2025111316832

DTS2025121105950

IR20250528001857

RAT

NR TDD

5.3.4.1.86 Changed the Default Value of Cell Medium Load UE Ratio Threshold Parameter

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The default value of the **NRDUCellRedCap.CellMediumLoadUeRatioThld** parameter has been changed from **25** to **100**.

Implementation

The default value of the **NRDUCellRedCap.CellMediumLoadUeRatioThld** parameter has been changed from **25** to **100**.

After an upgrade from a version earlier than V100R022C00SPC050 to V100R022C00SPC050 or later, the value of the **NRDUCellRedCap.CellMediumLoadUeRatioThld** parameter is **100**.

Impact on the Network

As the number of activated RedCap BWP1s in a cell increases, more RedCap UEs can access the cell. The number of RedCap UEs camping on each RedCap BWP1 in the cell changes. The impact of the increase in the number of RedCap UEs in the cell on the network is the same as that of more common NR UEs accessing the cell, the values of cell-related indicators such as those related to random access and uplink/downlink scheduling, fluctuate.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellRedCap.CellMediumLoadUeRatioThld	MOD NRDUCELLREDCAP LST NRDUCELLREDCAP	gNodeB	The default value of this parameter is changed from 25 to 100 .

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20241209002193

RAT

NR FDD, NR TDD

5.3.4.1.87 Outdated Antenna-Selection-based Resource Sharing for Downlink Beamforming#

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Antenna-selection-based resource sharing for downlink beamforming no longer takes effect.

Implementation

In low-frequency scenarios, antenna-selection-based resource sharing for downlink beamforming no longer takes effect. Accordingly, the **DL_BF_RES_SHR_SW** option of the **NRDUCellPdschPrecode.PrecodingAlgoSwitch** parameter, which controls this function, has been added to the disuse list. This change is supported only by macro base stations.

Impact on the Network

If antenna-selection-based resource sharing for downlink beamforming was enabled before an upgrade, the values of the following indicators may increase after the upgrade for a cell with the **Downlink Resource Block Utilizing Rate (DU)** greater than 50%:

- **Cell Downlink Average Throughput (DU)**
- **User Downlink Average Throughput (DU)**
- **Average DownLink RANK**

In other scenarios, there is no impact.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

- Baseband processing units: Only the UBBPg series and later baseband processing units support this function.
- RF modules: All RF modules capable of at least 32T32R support this function.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Disused	NRDUCellPdschPre code.PrecodingAlgoSwitch	MOD NRDUCELLPD SCHPRECODE LST NRDUCELLPD SCHPRECODE	gNodeB	The DL_BF_RES_SHR_SW option of this parameter has been added to the parameter disuse list, and the controlled function is now outdated. Although the configuration interface in this version supports synchronization and delivery of the setting of this option, the system does not use this option.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20241209002193-0012

RAT

NR TDD

5.3.4.1.88 Added the Parameter Controlling Low-Complexity SRS Measurement to the Parameter Disuse List#

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Before an upgrade, low-complexity SRS measurement is controlled by the **SRS_MEAS_LOW_CPLX_SW** option of the **NRDUCellSrsMeas.SrsMeasOptSwitch** parameter. If this option is selected, low-complexity SRS measurement takes effect, reducing SRS measurement complexity. If this option is deselected, this function does not take effect.

After an upgrade, the **SRS_MEAS_LOW_CPLX_SW** option of the **NRDUCellSrsMeas.SrsMeasOptSwitch** parameter is added to the parameter disuse list, and low-complexity SRS measurement takes effect by default.

Implementation

Low-complexity SRS measurement now takes effect by default, and the **SRS_MEAS_LOW_CPLX_SW** option of the **NRDUCellSrsMeas.SrsMeasOptSwitch** parameter, which previously controlled this function, has been added to the parameter disuse list. This option will be deleted from later versions. Although the configuration interface in this version supports synchronization and delivery of the setting of this option, the system does not use this option. Therefore, avoid using this option.

Impact on the Network

If the **SRS_MEAS_LOW_CPLX_SW** option of the **NRDUCellSrsMeas.SrsMeasOptSwitch** parameter is deselected before an upgrade, the following impact is observed after the upgrade: With low-complexity SRS measurement taking effect by default, when no interference is present on SRS and more than 20% of UEs in the cell have an SRS PreSINR less than -10 dB, the **User Downlink Average Throughput (DU)** fluctuates slightly.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellSrsMeas.SrsMeasOptSwitch	MOD NRDUCELLS RSMEAS LST NRDUCELLS RSMEAS	gNodeB	The SRS_MEAS_LOW_CPLX_SW option has been added to the parameter disuse list. The controlled function, low-complexity SRS measurement, now takes effect by default. This option will be deleted from later versions. Although the configuration interface in this version supports synchronization and delivery of the setting of this option, the system does not use this option. Therefore, avoid using this option.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250528001856-0001

RAT

NR TDD

5.3.4.1.89 Enabled "Complexity Reduction for SRS Narrowband Interference Avoidance" to Take Effect by Default and Added the Corresponding Function Switch to the Disuse List#

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

"Complexity reduction for SRS narrowband interference avoidance" was introduced in V100R019C10 for low frequency bands. This function reduces the complexity of narrowband interference avoidance (controlled by the **UL_NARROWBAND_INTRF_AVOID** or **UL_UE_LEVEL_INTRF_AVOID** option of the **NRDUCellIntrfIdent.IntrfOptimizationMethod** parameter).

The switch for "complexity reduction for SRS narrowband interference avoidance" has been added to the disuse list, and this function now takes effect by default.

Implementation

The **NRDUCellSrsMeas.SrsMeasExtSw** parameter and the **SRS_NB_INTRF_AVOID_CPLX_LOWER** option of this parameter have been added to the disuse list. "Complexity reduction for SRS narrowband interference avoidance" now takes effect by default.

Impact on the Network

Assume that before an upgrade, "complexity reduction for SRS narrowband interference avoidance" was disabled in either of the following scenarios:

- The **UL_NARROWBAND_INTRF_AVOID** option of the **NRDUCellIntrfIdent.IntrfOptimizationMethod** parameter is selected, the interference identification method is set to **UL_MANUAL**, and an interference range is configured.
- The **UL_UE_LEVEL_INTRF_AVOID** option of the **NRDUCellIntrfIdent.IntrfOptimizationMethod** parameter is selected, and the interference identification method is set to **UL_AUTO**.

After the upgrade, due to the impact of SRS PreSINRs, indicator values may change for UEs at the cell edge as follows:

- The average downlink rank and average downlink MCS index slightly increase.
- The values of indicators related to the downlink BLER slightly decrease.
- The **Cell Downlink Average Throughput (DU)** and **User Downlink Average Throughput (DU)** slightly increase.

In other scenarios, there is no impact.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Disused	NRDUCellSrsMeas. <i>SrsMeasExtSw</i>	MOD NRDUCELLSRS MEAS LST NRDUCELLSRS MEAS	gNodeB	This parameter and the SRS_NB_INTERFACE_AVOID_CPLX_LOWER option of this parameter have been added to the disuse list, and the solution with this option selected takes effect by default. This parameter will be deleted in later versions. Although the configuration interface in this version supports synchronization and delivery of the setting of this parameter, the system does not use this parameter. Therefore, avoid using this parameter.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact

- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20241209002193-0045

RAT

NR TDD

5.3.4.1.90 Deleted Certain Descriptions of the Configuration Constraints on the Frequency Band and Duplex Mode Parameters

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Certain descriptions of the configuration constraints on some parameters have been deleted.

Implementation

There are many configuration constraints on the **NRDUCell.FrequencyBand** and **NRDUCell.DuplexMode** parameters, and the descriptions in the "Parameter Relationship" field are excessively long, affecting reading experience. To improve reading experience, certain descriptions in the "Parameter Relationship" field have been deleted since V100R022C00SPC010.

The descriptions are deleted as follows: Assume that parameter *A* is described as a condition in a configuration constraint, for example, "if parameter *A* meets certain conditions, parameters *B*, *C*, and *D* must satisfy certain constraints". In this case, this configuration constraint is deleted from the "Parameter Relationship" field of parameter *A* but retained in the "Parameter Relationship" fields of parameters *B*, *C*, and *D*.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCell. Frequency Band	ADD NRDUCELL MOD NRDUCELL LST NRDUCELL	gNodeB	Certain descriptions of the configuration constraints have been deleted from the "Parameter Relationship" field of this parameter. For details, see Implementation .
Modified	NRDUCell. DuplexMode	ADD NRDUCELL MOD NRDUCELL LST NRDUCELL	gNodeB	Certain descriptions of the configuration constraints have been deleted from the "Parameter Relationship" field of this parameter. For details, see Implementation .

- Performance management

No impact

- Fault management

No impact

Impact on Features/Functions

- Prerequisite features/functions

No impact

- Mutually exclusive features/functions

No impact

- Impacted features/functions

No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20241209002193

DTS2025061118082

RAT

NR FDD, NR TDD

5.3.4.1.91 Deleted Some Configuration Constraints on the Power Saving BWP Function

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The following configuration constraints on the power saving BWP function have been deleted.

- If the **BWP2_SWITCH** option of the **NRDUCellUePwrSaving.BwpPwrSavingSw** parameter is selected and the **NRDUCell.NrDuCellNetworkingMode** parameter is set to **HYPER_CELL_COMBINE_MODE** or **HYPER_CELL**, the **NRDUCell.HighSpeedFlag** parameter must be set to **LOW_SPEED**.
- If the **BWP2_SWITCH** option of the **NRDUCellUePwrSaving.BwpPwrSavingSw** parameter is selected and the **NRDUCell.NrDuCellNetworkingMode** parameter is set to **HYPER_CELL_COMBINE_MODE** or **HYPER_CELL**, the **INTER_TRP_SSB_STAGGER_SWITCH** option of the **NRDUCellMultiTrp.MultiTrpAlgoSwitch** parameter must be deselected.

Implementation

The following configuration constraints on the power saving BWP function have been deleted.

- If the **BWP2_SWITCH** option of the **NRDUCellUePwrSaving.BwpPwrSavingSw** parameter is selected and the **NRDUCell.NrDuCellNetworkingMode** parameter is set to **HYPER_CELL_COMBINE_MODE** or **HYPER_CELL**, the **NRDUCell.HighSpeedFlag** parameter must be set to **LOW_SPEED**.
- If the **BWP2_SWITCH** option of the **NRDUCellUePwrSaving.BwpPwrSavingSw** parameter is selected and the **NRDUCell.NrDuCellNetworkingMode** parameter is set to **HYPER_CELL_COMBINE_MODE** or **HYPER_CELL**, the **INTER_TRP_SSB_STAGGER_SWITCH** option of the **NRDUCellMultiTrp.MultiTrpAlgoSwitch** parameter must be deselected.

Although the preceding constraints are deleted, when power saving BWP is enabled, the **NRDUCell.HighSpeedFlag** must be set to **LOW_SPEED** and the **INTER_TRP_SSB_STAGGER_SWITCH** option of the **NRDUCellMultiTrp.MultiTrpAlgoSwitch** parameter must be deselected. This is because power saving BWP depends on SRS period adaptation (controlled by the **SRS_PERIOD_ADAPT_SW** option of the **NRDUCellSrs.SrsAlgoSwitch** parameter)

and UE-characteristic-based SRS period adaptation (controlled by the **USER_CHARACTER_SRS_ADAPT_SW** option of the **NRDUCellSrs.SrsAlgoSwitch** parameter). Both High-speed mobility and inter-TRP SSB staggering are mutually exclusive with the preceding two functions.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
The following mutually exclusive relationships have been deleted.

Function Name	Function Switch	Reference	Description
High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	If the BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter is selected and the NRDUCell.NrDuCellNetworkingMode parameter is set to HYPER_CELL_COMBINE_MODE or HYPER_CELL , the NRDUCell.HighSpeedFlag parameter must be set to LOW_SPEED .

Function Name	Function Switch	Reference	Description
Inter-TRP SSB staggering	INTER_TRP_SSB_STAGGER_SWITCH option of the NRDUCellMultiTrp.MultiTrpAlgoSwitch parameter	<i>Hyper Cell Cell Combination</i>	If the BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter is selected and the NRDUCell.NrDuCellNetworkingMode parameter is set to HYPER_CELL_COMBINE_MODE or HYPER_CELL , the INTER_TRP_SSB_STAGGER_SWITCH option of the NRDUCellMultiTrp.MultiTrpAlgoSwitch parameter must be deselected.

- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

For details, see the corresponding "Operation and Maintenance" section in *UE Power Saving*.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0103

RAT

NR TDD

5.3.4.2 Transport

5.3.4.2.1 Support for Backup Power Saving Data Transmission from the Base Station to the Microwave Device

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The **BKP_PSAV_EVALUATE_DATA_SW** option of the **LLDPGLOBAL.BTSMWCOSW (LTE eNodeB, 5G gNodeB)** parameter has been added to support the transmission of backup power saving data from the base station to the microwave device. When this option is selected and the microwave device sends a backup power saving data subscription request, the base station is enabled to periodically send backup power saving data to the microwave device.

Implementation

For details, see *IPv4 Transmission* or *IPv6 Transmission*. The specific sections are listed in the change history of the document.

Impact on the Network

After the upgrade, backup power saving data is now included in the data transmitted from the base station to the microwave device through LLDP. As a result, exchanging LLDP data between the base station and the microwave device over Ethernet ports will consume a small amount of transmission bandwidth.

Impact on NEs

This change is implemented on the gNodeB and eNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	LLDPGLOB AL.BTSM WCOSW (LTE eNodeB, 5G gNodeB)	SET LLDPGLOBA LINFO LST LLDPGLOBA LINFO	eNodeB gNodeB	The BKP_PSAV_EVALUATE_DATA_SW option has been added to specify whether the base station transmits its backup power saving data to the microwave device.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

Run the **SET LLDPGLOBALINFO** command with the **BKP_PSAV_EVALUATE_DATA_SW** option of the **LLDPGLOBAL.BTSMWCOSW (LTE eNodeB, 5G gNodeB)** parameter selected to enable this change. If this option needs to be selected, ensure that the peer end is a directly connected Huawei microwave device and the corresponding switch on this peer device is also turned on. Otherwise, this function does not take effect.

Requirement/Trouble Ticket Number

FeR-IR20250220002033-0010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD

5.3.4.2 Enhanced Capability of Information Exchange over the eXn Interface

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The capability of information exchange over the eXn interface has been enhanced. More cell configuration information can now be exchanged over the eXn interface during inter-base-station services.

Implementation

The capability of information exchange over the eXn interface has been enhanced. More cell configuration information can now be exchanged over the eXn interface during inter-base-station services.

Impact on the Network

After an upgrade, more cell configuration information is exchanged over the eXn interface between base stations. As a result, a single message becomes longer and, in certain scenarios, may be broken into more fragments. This may increase the values of counters related to the received and transmitted traffic volumes in the **SCTP Link Measurement** function subset.

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions

No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250408001167-0005

IR20241209002193

DTS2025040125607

DTS2025040125339

RAT

NR FDD, NR TDD

5.3.4.2.3 Support for Performance Counter Data Transmission from the Base Station to the Microwave Device

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Performance counter data transmission from the base station to the microwave device is now supported. After successfully parsing a request packet from the microwave device to subscribe to performance counters, the base station can periodically send performance counter data to the microwave device.

Implementation

When the microwave device requests to subscribe to performance counters, the **NETWORK_EVALUATE_DATA_SW** option of the **LLDPGLOBAL.BTSMWCOSW (LTE eNodeB, 5G gNodeB)** parameter is selected, and the corresponding switch on the microwave device is also turned on, the base station will periodically send the collected performance counter data to the microwave device through LLDP packets. If this option is deselected, the base station refuses to send its performance counter data to the microwave device.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB and eNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following MML commands have changed.

Change Type	MML Command	NE	Description
Modified	LST LLDPGLOBA LINFO	eGBTS NodeB eNodeB gNodeB	Querying the status of Network Evaluate Data Switch of the Base Station Microwave Collaboration Switch parameter is now supported.

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	LLDPGLOBA L.BTSM WCOSW (LTE eNodeB, 5G gNodeB)	LST LLDPGLOBA LINFO SET LLDPGLOBA LINFO	eGBTS NodeB eNodeB gNodeB	The NETWORK_EVALUATE_DATA_SW option of the LLDPGLOBAL.BTSMWCOSW (LTE eNodeB, 5G gNodeB) parameter has been added to specify whether the base station sends its performance counter data to the microwave device.

- Performance management

No impact

- Fault management

No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

Run the **SET LLDPGLOBALINFO** command with the **NETWORK_EVALUATE_DATA_SW** option of the **LLDPGLOBAL.BTSMWCOSW (LTE eNodeB, 5G gNodeB)** parameter selected to enable this change. If this option needs to be selected, ensure that the peer end is a directly connected Huawei microwave device and the corresponding switch on this peer device is also turned on. Otherwise, this function does not take effect.

Requirement/Trouble Ticket Number

FeR-IR20240822003052-0010

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD

5.3.4.2.4 Optimized NSA SgNB Addition Success Rate in EN-DC X2-U Fault Scenarios

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

This function improves the NSA SgNB addition success rate in EN-DC X2-U fault scenarios. When an SCG is to be added, the eNodeB does not initiate the SCG addition procedure if it determines that the X2-U link between itself and the peer gNodeB is faulty. The eNodeB and gNodeB both require that only one X2-U local IP address be configured for an operator. If this requirement is not met, this function does not take effect.

Implementation

This function takes effect when the function switches are turned on for both the gNodeB and eNodeB. The details are as follows:

- The **X2U_IPADDR_SEND_SW** option of the **gNodeBParam.NsaDcOptSwitch** parameter has been added for the gNodeB to specify whether to send the X2-U IP address. When this option is selected, the gNodeB sends the X2-U IP address to the eNodeB.

- The **X2U_STATUS_BASED_SCG_ADD_SW** option of the **ENodebAlgoExtSwitch.NsaDcAlgoExtSwitch** parameter has been added for the eNodeB to specify whether to enable SCG addition based on the X2-U status. When this option is selected, the eNodeB checks the X2-U link status after receiving the X2-U IP address from the gNodeB. The eNodeB does not initiate an SCG addition procedure if it determines that the X2-U link between itself and the peer gNodeB is faulty.

Impact on the Network

The impacts on the LTE side are as follows:

If the EN-DC X2-C is available but the X2-U is unavailable, the value of **SgNB Addition Success Rate (L.NsaDc.SgNB.Add.Succ/L.NsaDc.SgNB.Add.Att × 100%)** increases, the total number of successful SCG changes for UEs that treat the local cell as their PCell in the LTE-NR NSA DC state (**L.NsaDc.SCG.Change.Succ**) may decrease, and the number of SCG change failures for NSA DC UEs in a cell (indicated by **L.NsaDc.SCG.Change.Failure** and **L.NsaDc.SCG.Change.Failure.TNL**) may increase.

The impacts on the NR side are as follows:

If the EN-DC X2-C is available but the X2-U is unavailable, the value of **SgNB Addition Success Rate (CU) (N.NsaDc.SgNB.Add.Succ/N.NsaDc.SgNB.Add.Att × 100%)** increases, the number of successful inter-SgNB PSCell changes in LTE-NR NSA DC scenarios (indicated by **N.NsaDc.InterSgNB.PSCell.Change.Succ**) may decrease, and the number of inter-SgNB PSCell change failures caused by transmission faults in LTE-NR NSA DC scenarios (indicated by **N.NsaDc.InterSgNB.PSCell.Change.Fail.Trans**) may increase.

Impact on NEs

No impact

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

An IE for transferring the gNodeB X2-U IP address and a field indicating whether the eNodeB supports SCG addition based on the X2-U status have been added.

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch	MOD ENODEBALG OEXTSWITCH LST ENODEBALG OEXTSWITCH	eNodeB	The X2U_STATUS_BASED_SCG_ADD_SW option has been added to specify whether the eNodeB checks the status of the X2-U interface between itself and the gNodeB when adding an SCG in Option 3x.
Modified	gNodeBParam.NsaDcOptSwitch	MOD GNODEBPARAM LST GNODEBPARAM	gNodeB	The X2U_IPADDR_SEND_SW option has been added to specify whether to send or update the X2-U IP address to the eNodeB.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions

Function Name	Function Switch	Reference	Description
GTP-U static detection	GTPU.STATICCHK (LTE eNodeB, 5G gNodeB) set to ENABLE	None	This function change takes effect only when GTP-U static detection is enabled.

- Mutually exclusive features/functions

Function Name	Function Switch	Reference	Description
Operator-specific NCI configuration	gNBSharingMode.MultipleNCISwitch set to OFF	<i>Multi-Operator Sharing</i>	Operator-specific NCI configuration cannot be enabled when the X2-U IP address sending switch is turned on.

- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

The **MOD ENODEBALGOEXTSWITCH** command can be executed to select the **X2U_STATUS_BASED_SCG_ADD_SW** option of the **EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch** parameter.

The **MOD GNODEBPARAM** command can be executed to select the **X2U_IPADDR_SEND_SW** option of the **gNodeBParam.NsaDcOptSwitch** parameter.

Requirement/Trouble Ticket Number

FeR-IR20250102000700-0020

FeR-IR20250102000700-0021

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD

5.3.4.3 Equipment

5.3.4.3.1 Extended Fields in Uplink Transmission Packets at the Physical Layer of the eCPRI Interface

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

This function extends the fields in uplink transmission packets at the physical layer in eCPRI scenarios, increasing the data traffic over the eCPRI interface.

This change is irrelevant to LampSite base stations.

Implementation

This function extends the fields in uplink transmission packets at the physical layer in eCPRI scenarios.

Impact on the Network

As the traffic over the eCPRI interface increases, the values of the following counters may increase.

Counter Name	Counter Description
VS.ECPRIPORT.TxBytes	Number of Bytes Transmitted from an eCPRI Port
VS.ECPRIPORT.RxBytes	Number of Bytes Received by an eCPRI Port
VS.ECPRIPORT.TxPackets	Number of Packets Transmitted from an eCPRI Port
VS.ECPRIPORT.RxPackets	Number of Packets Received by an eCPRI Port
VS.ECPRIPORT.TxMaxSpeed	Maximum Transmit Rate of an eCPRI Port
VS.ECPRIPORT.RxMaxSpeed	Maximum Receive Rate of an eCPRI Port
VS.ECPRIPORT.TxMeanSpeed	Average Transmit Rate of an eCPRI Port
VS.ECPRIPORT.RxMeanSpeed	Average Receive Rate of an eCPRI Port
VS.ECPRIPORT.TxMinSpeed	Minimum Transmit Rate of an eCPRI Port
VS.ECPRIPORT.RxMinSpeed	Minimum Receive Rate of an eCPRI Port
VS.ECPRIPORT.RxErrPackets	Number of Error Packets Received by an eCPRI Port
VS.ECPRIPORT.PCS.ErrBlocks	Number of PCS error blocks over the eCPRI port
VS.ECPRIPORT.PCS.ES	Number of PCS ESs over the eCPRI port
VS.ECPRIPORT.PCS.SES	Number of PCS SESs over the eCPRI port
VS.ECPRIPORT.PCS.UAS	Number of PCS UASs over the eCPRI port
VS.FronthaulLink.eCPRILink.UplinkDropMeans	VS.eCPRILink.UplinkDropMeans
VS.FronthaulLink.eCPRILink.DownlinkDropMeans	VS.eCPRILink.DownlinkDropMeans
VS.FronthaulLink.eCPRILink.UplinkDelayMeans	VS.eCPRILink.UplinkDelayMeans
VS.FronthaulLink.eCPRILink.DownlinkDelayMeans	VS.eCPRILink.DownlinkDelayMeans

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20240816001195-0002

RAT

NR FDD, NR TDD

5.3.5 Operation and Maintenance

5.3.5.1 General Management

There are no changes in general management.

5.3.5.2 Configuration Management

This section describes the changes in configuration management.

For details about changes in MML commands, MOs, and parameters, see *MML Command Changes* and *MO and Parameter Changes*, which are provided with the software version. You can obtain *MML Command Changes* and *MO and Parameter Changes* from the following directory: **BTS3900&BTS5900 VXXXRXXXCXXSPCXXX Release Documents-EN\01-Release Notes\Delta(MMI)**.

5.3.5.2.1 Added Auto Configuration Flag Fields to the Command Output of the X2 and Xn Interfaces

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

This function adds the auto configuration flag fields to the output of the **DSP GNBCUX2INTERFACE** and **DSP GNBCUXNINTERFACE** commands, respectively. The flag fields are used to indicate whether the gNodeB CU X2 and gNodeB CU Xn interfaces are created automatically using SON or manually.

Implementation

If **SCTPPEER.REMOTEID** is set to **gNBXn:MCC_xxx MNC_xxx GlobalgNBId_xxx-xxx-xxxxxxx** or **eNB:MCC_xxx MNC_xxx GlobaleNBId_xxx-xxx-xxxxxxx**, and the auto configuration flag field is set to **AUTO_CREATED**, the gNodeB CU X2 or gNodeB CU Xn interface is created automatically using SON. Otherwise, the interface is created manually.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following MML commands have changed.

Change Type	MML Command	NE	Description
Modified	DSP GNBCUX2INTERF ACE	gNode B	The auto configuration flag field has been added to the command output to indicate whether the gNodeB CU X2 interface is created automatically using SON or manually.
Modified	DSP GNBCUXNINTERF ACE	gNode B	The auto configuration flag field has been added to the command output to indicate whether the gNodeB CU Xn interface is created automatically using SON or manually.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250321003120-0010

RAT

NR FDD, NR TDD

5.3.5.2.2 MEHC Used in Neutral Host Sharing and Hybrid Neutral Host Sharing Scenarios

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In neutral host sharing and hybrid neutral host sharing scenarios, the MEHC is introduced on the DCU to support RF input cells and BBU access cells.

Implementation

This function enables the basic operation and maintenance functions of LampSite Sharing base stations to be adapted to the networking with the MEHC. The MEHC integrates the functions of the MERC and MEDC, supports RF input cells and BBU access cells, and enhances networking and combination capabilities.

This function requires that the BBU and DCU versions be V100R022C10SPC010 or later.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB and eNodeB.

Impact on Hardware Requirements

Only the MEHC supports this function.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following MML commands have changed.

Change Type	MML Command	NE	Description
New	DSP EXTCONNPORT	eNodeB gNodeB	This command has been added to query the dynamic information about the port connected to an external NE.

Change Type	MML Command	NE	Description
Modified	DSP EXTRRURES	eNodeB gNodeB	The following output fields have been added: RCS Server No. , Ext Conn Port ID List , RRU Type , and Server RRU Type .

 NOTE

The parameters in these new or deleted MML commands are no longer listed separately.

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	RCSSERVER.CONNPORTLST (LTE eNodeB, 5G gNodeB)	-	eNodeB gNodeB	The value range has been changed from 0 to 4 to 0 to 6 .
Modified	EXTCONNPORT.PN (LTE eNodeB, 5G gNodeB)	ADD EXTCON NPORT MOD EXTCON NPORT DSP EXTCON NPORT LST EXTCON NPORT	eNodeB gNodeB	The value range of the EXTCONNPORT.PN (LTE eNodeB, 5G gNodeB) parameter has been changed from 0 to 1 to 0 to 2 when the EXTCONNPORT.PT (LTE eNodeB, 5G gNodeB) parameter is set to HEI .
Modified	RCSSERVER.CONNPORTNUM (LTE eNodeB, 5G gNodeB)	ADD RCSSERV ER MOD RCSSERV ER	eNodeB gNodeB	The value range has been changed from 1 to 4 to 1 to 6 .
New	RCSSERVER.CONNPORT5ID (LTE eNodeB, 5G gNodeB)	ADD RCSSERV ER MOD RCSSERV ER	eNodeB gNodeB	This parameter has been added to specify the ID of the fifth local NE port that is connected to an RCS server.

Change Type	Parameter ID	MML Command	NE	Description
New	RCSSERVER.CON NPORT6ID (LTE eNodeB, 5G gNodeB)	ADD RCSSERV ER MOD RCSSERV ER	eNodeB gNodeB	This parameter has been added to specify the ID of the sixth local NE port that is connected to an RCS server.

- Performance management

No impact

- Fault management

The following alarms have changed.

Change Type	Name	NE	Description
Modified	ALM-26230 BBU CPRI Optical Module Fault	eNodeB gNodeB	The following reporting scenario has been added: This alarm is triggered when the optical module located on the BBU side and connected to the DCU is faulty in a BBU-DCU interconnection network.
Modified	ALM-26231 BBU CPRI Optical Module or Electrical Port Not Ready	eNodeB gNodeB	The following reporting scenario has been added: This alarm is triggered when the optical module located on the BBU side and connected to the DCU is not inserted in a BBU-DCU interconnection network.
Modified	ALM-26234 BBU CPRI Interface Error	eNodeB gNodeB	The following reporting scenario has been added: This alarm is triggered when the optical link between the BBU and DCU is abnormal in a BBU-DCU interconnection network.
New	ALM-26318 BBU-DCU Interconnection Configuration Failure	eNodeB gNodeB	This alarm is reported when a failure occurs in the BBU-DCU interconnection configuration or the configuration is inconsistent with the actual situation.

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20250123000983

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD

5.3.5.2.3 Support for Querying Messages Related to Xn Self-Setup

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The **DSP GNBINTERFACEINFO** command has been enhanced to support query of messages related to Xn self-setup.

Implementation

Users can query messages related to Xn self-setup by setting the **gNBInterfaceParam.InterfaceType** parameter to **XN**.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

To query messages related to Xn self-setup, run the **DSP GNBINTERFACEINFO** command with the **gNBInterfaceParam.InterfaceType** parameter set to **XN**.

Requirement/Trouble Ticket Number

FeR-IR20240930001479-0001

RAT

NR FDD, NR TDD

5.3.5.2.4 Support for the DSP Command to Query the Subboard Type of the Serving BBP for LTE and NR

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The following changes have been introduced:

- LTE: The subboard type of the serving BBP of a cell, instead of the serving BBP type, is displayed in the **DSP CELLPHYTOPO** command output.
- NR: The subboard type of the serving BBP of a cell can now be queried in the **DSP NRDUCELLANTENNA** command output.

Implementation

This function is implemented as follows:

- LTE: The meaning of the **Cell.ServingBbpType** field in the output of **DSP CELLPHYTOPO** has been updated: it now indicates the subboard type of the serving BBP, instead of the serving BBP type.
- NR: The serving BBP type field has been added to the output of **DSP NRDUCELLANTENNA** to support querying the subboard type of the serving BBP.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the eNodeB and gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management

The following MML commands have changed.

Change Type	MML Command	NE	Description
Modified	DSP CELLPHYTOPO	eNodeB	The meaning of the Cell.ServingBbpType field in the command output has been updated: it now indicates the subboard type of the BBP, instead of the BBP type.

Change Type	MML Command	NE	Description
Modified	DSP NRDUCELLANTENNA	gNodeB	The serving BBP type field has been added to the command output to support querying the subboard type of the serving BBP.

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	Cell.ServingBbpType	DSP CELLPHYTOP O	eNodeB	The output has been changed from the BBP type to the BBP subboard type.

- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250102000700-0009

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD

5.3.5.3 Performance Management

This section describes the changes in performance management.

For details about changes in performance counters, see *Performance Counter Changes* delivered with the software version. You can obtain *Performance Counter Changes* from the following directory: **BTS3900&BTS5900 VXXXRXXXCXXSPCXXX Release Documents-EN\01-Release Notes\Delta(MMI)**.

5.3.5.3.1 Support for Operator-Specific Measurement of Uplink/Downlink PDCP SDUs and the Average Number of Active UEs in Downlink

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In RAN sharing with common carrier, the following counters have been added to support operator-specific performance evaluation:

- **N.User.RRCCConn.Active.DL.Avg.PLMN**, which measures the average number of active UEs in the downlink for a specific operator in a DU cell
- **N.PDCP.Vol.DL.TrfSDU.Rx.PLMN** and **N.PDCP.Vol.UL.TrfSDU.Tx.PLMN**, which measure the data volume of downlink SDUs received at the PDCP layer and data volume of uplink SDUs sent from the PDCP layer for a specified operator in a cell, respectively

Implementation

In RAN sharing with common carrier, the following counters have been added to support operator-specific performance evaluation:

- **N.User.RRCCConn.Active.DL.Avg.PLMN**, which measures the average number of active UEs in the downlink for a specific operator in a DU cell
- **N.PDCP.Vol.DL.TrfSDU.Rx.PLMN** and **N.PDCP.Vol.UL.TrfSDU.Tx.PLMN**, which measure the data volume of downlink SDUs received at the PDCP layer and data volume of uplink SDUs sent from the PDCP layer for a specified operator in a cell, respectively

Impact on the Network

No impact

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
The following counters have changed.

Change Type	Counter Name	Function Subset	NE	Description
New	N.User.RRCCConn.Active.DL.Avg.PLMN	NR DU Cell Scheduled User Quantity Measurement per Operator	gNodeB	This counter has been added to measure the average number of active UEs in the downlink for a specific operator in a DU cell.
New	N.PDCP.Vol.DL.TrfS DU.Rx.PLMN	PDCP Measurement per Operator	gNodeB	This counter has been added to measure the data volume of downlink SDUs received at the PDCP layer for a specific operator in a cell.
New	N.PDCP.Vol.UL.TrfS DU.Tx.PLMN	PDCP Measurement per Operator	gNodeB	This counter has been added to measure the data volume of uplink SDUs sent from the PDCP layer for a specified operator in a cell.

- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250528001856-0013

RAT

NR FDD, NR TDD

5.3.5.3.2 New Way to Observe Uplink Interference and Noise

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Uplink interference and noise can now be observed using the following counters, whose reporting is controlled by the **HIGH_PRE_LINEAR_INTER_RPT_SW** option of the **NRDUCellOamParam.StatisticsStrategySwitch** parameter:

- **N.UL.NI.LinearAvg**
- **N.TRP.UL.NI.LinearAvg**
- **N.UL.NI.LinearAvg.PRBO** to **N.UL.NI.LinearAvg.PR272**

Implementation

When the preceding option is selected, the following counters can be reported: **N.UL.NI.LinearAvg**, **N.TRP.UL.NI.LinearAvg**, and **N.UL.NI.LinearAvg.PRBO** to **N.UL.NI.LinearAvg.PR272**.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	NRDUCellOamParam.StatisticsStrategySwitch	MOD NRDUCELLOAMPARAM LST NRDUCELLOAMPARAM	gNodeB	The HIGH_PRE_LI NEAR_INTER_RPT_SW option has been added to specify whether to report the following counters: N.UL.NI.LinearAvg , N.TRP.UL.NI.LinearAvg , and N.UL.NI.LinearAvg.PRBO to N.UL.NI.LinearAvg.PRB272 .

- Performance management
The following function subsets have changed.

Change Type	Function Subset	NE	Description
New	Uplink Linear Interference Measurement on PRBs 0 to 99	gNodeB	This function subset has been added to measure the average interference and noise on uplink PRBs 0 to 99.
New	Uplink Linear Interference Measurement on PRBs 100 to 199	gNodeB	This function subset has been added to measure the average interference and noise on uplink PRBs 100 to 199.
New	Uplink Linear Interference Measurement on PRBs 200 to 272	gNodeB	This function subset has been added to measure the average interference and noise on uplink PRBs 200 to 272.

 NOTE

The counters in these new or deleted function subsets are no longer listed separately. The following counters have changed.

Change Type	Counter Name	Function Subset	NE	Description
New	N.UL.NI.LinearAvg	Channel Quality Measurement	gNodeB	This counter measures the linear average interference and noise received on each uplink PRB.
New	N.TRP.UL.NI.Linear Avg	DU Cell TRP Channel Quality Measurement	gNodeB	This counter measures the linear average interference and noise received on each uplink PRB for a TRP.

- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change does not take effect after an upgrade.

You can run the **MOD NRDUCELLOAMPARAM** command with the **HIGH_PRE_LINEAR_INTER_RPT_SW** option of the **NRDUCellOamParam.StatisticsStrategySwitch** parameter selected.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0158

RAT

NR FDD, NR TDD

5.3.5.3.3 Added Counter-based Activation Verification for Fast Carrier Shutdown

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

The **N.PowerSaving.FastCarrierShutdown.Dur** counter can now be used to determine whether fast carrier shutdown takes effect. The **N.PowerSaving.FastCarrierShutdown.Dur** counter measures the energy saving duration of fast carrier shutdown. If the value of the **N.PowerSaving.FastCarrierShutdown.Dur** counter is not 0, fast carrier shutdown has taken effect in the cell.

This function is supported only in low frequency bands.

Implementation

For details, see *Energy Conservation and Emission Reduction*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
The following counters have changed.

Change Type	Counter Name	Function Subset	NE	Description
New	N.PowerSaving.FastCarrierShutdown.Dur	DU Cell Algorithm Measurement	gNode B	This counter has been added to measure the fast carrier shutdown duration of a cell.

- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20251031001797-0020

RAT

NR FDD, NR TDD

5.3.5.3.4 Extended Effective Scenarios of the External CHR Event to Measure the Uplink Data Packet Delay

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

To better measure the delay of uplink data packets of services such as Cloud X, the effective scenarios of the external CHR event PERIOD_PRIVATE_DRB_DELAY_SENSITIVE_SERVICE are extended.

LampSite base stations do not support this function.

This change is supported only in low frequency bands.

Implementation

The effective scenarios of the external CHR event PERIOD_PRIVATE_DRB_DELAY_SENSITIVE_SERVICE have been extended. The event can be measured and reported when either of the following conditions is met:

- The **CLOUD_X_FRAME_IDENT_SW** option of the **NRCellAlgoSwitch.CloudVrSwitch** parameter is selected, and the **gNBQciBearer.ServiceType** parameter of the corresponding bearer is set to **EMBB_CLOUD_VR**, **EMBB_CLOUD_GAMING**, or **EMBB_CLOUD_PHONE**.
- The **DL_LAYERED_QOS_SW** option of the **NRCellFeatureSwitch.IntelligentExpGuarSwitch** parameter is selected, and the **gNBQciBearer.IdentificationMode** parameter of the corresponding bearer is set to **R18_PDU_SET_MODE**.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

Main control boards: All NR-capable main control boards except the UMPTe series can be used.

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

Whether the change takes effect by default after an upgrade varies as follows:

- Before the upgrade, if the **CLOUD_X_FRAME_IDENT_SW** option of the **NRCCellAlgoSwitch.CloudVrSwitch** parameter is selected and the **gNBQciBearer.ServiceType** parameter of the corresponding bearer is set to **EMBB_CLOUD_VR**, **EMBB_CLOUD_GAMING**, or **EMBB_CLOUD_PHONE**, or the **DL_LAYERED_QOS_SW** option of the **NRCCellFeatureSwitch.IntelligentExpGuarSwitch** parameter is selected and the **gNBQciBearer.IdentificationMode** of the corresponding bearer is set to **R18_PDU_SET_MODE**: Yes
- Other scenarios: No

You can run the **MOD NRCELLALGOSWITCH**, **MOD NRCELLFEATURESWITCH**, or **MOD GNBQCIBEARER** command to configure the **CLOUD_X_FRAME_IDENT_SW** option of the **NRCCellAlgoSwitch.CloudVrSwitch** parameter, the **DL_LAYERED_QOS_SW** option of the **NRCCellFeatureSwitch.IntelligentExpGuarSwitch** parameter, the **gNBQciBearer.ServiceType** parameter, or the **gNBQciBearer.IdentificationMode** parameter based on network conditions to use the external CHR event **PERIOD_PRIVATE_DRB_DELAY_SENSITIVE_SERVICE** for monitoring.

Requirement/Trouble Ticket Number

FeR-IR20250102000700-0011

RAT

NR FDD, NR TDD

5.3.5.3.5 Configuration-based Activation and Deactivation of Some NR Cell Performance Measurement Objects

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Before an upgrade, the activation conditions for different performance measurement objects are inconsistent. Some are activated based on configuration objects, whereas some are activated based on the cell running status. Some cells on the live network may be deactivated either manually or due to faults. Such deactivation events can cause inconsistencies in the values of semantically related performance counters collected across different performance measurement objects during the related measurement period.

After an upgrade, **gNBOamParam.PmObjActBasedOnCfg**, a new parameter, is available for specifying configuration-based activation of performance measurement objects, with each object controlled by an individual option.

Implementation

The **gNBOamParam.PmObjActBasedOnCfg** parameter is added. It is used to specify configuration-based activation of performance measurement objects, with each object controlled by an individual option. The options are as follows:

- **NRCELL_BASED_PFM**: This option specifies whether performance measurement objects related to NR cells are activated based on configuration.
- **NRCELL_OPERATOR_BASED_PFM**: This option specifies whether performance measurement objects related to NR cell operators are activated based on configuration.
- **NRCELL_5QI_BASED_PFM**: This option specifies whether performance measurement objects related to NR cell 5QIs are activated based on configuration.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
New	gNBOamParam. <i>PmObjActBasedOnCfg</i>	LST GNBOAMPAR AM MOD GNBOAMPAR AM	gNodeB	This parameter has been added. It is used to specify configuration-based activation of performance measurement objects, with each object controlled by an individual option.

- Performance management

No impact

- Fault management

No impact

Impact on Features/Functions

- Prerequisite features/functions

No impact

- Mutually exclusive features/functions

No impact

- Impacted features/functions

No impact

Related Operations

By default, the change does not take effect after an upgrade.

You need to run the **MOD GNBOAMPARAM** command with the **NRCELL_BASED_PFM**, **NRCELL_OPERATOR_BASED_PFM**, or **NRCELL_5QI_BASED_PFM** option of the **gNBOamParam.*PmObjActBasedOnCfg*** parameter selected, depending on your requirements.

Requirement/Trouble Ticket Number

FeR-IR20250528001857-0033

RAT

NR FDD, NR TDD

5.3.5.4 Software Management

There are no changes in software management.

5.3.5.5 Fault Management

This section describes the changes in fault management.

5.3.5.5.1 Support for NR Cell Unavailable Alarm Optimization

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

NR Cell Unavailable alarm optimization is introduced in NR to improve the accuracy of reported root cause for ALM-29841 NR Cell Unavailable caused by a faulty NR DU cell bound to the NR cell. This optimization improves O&M efficiency and is implemented as follows:

- If the **Specific Problem** of ALM-29844 NR DU Cell Unavailable is not **Other fault cause**, it is used as the **Specific Problem** of ALM-29841 NR Cell Unavailable.
- Otherwise, the **Specific Problem** of ALM-29870 NR DU Cell TRP Unavailable is used as the **Specific Problem** of ALM-29841 NR Cell Unavailable.

NR Cell Unavailable alarm optimization is controlled by the **NR_CELL_UNAVAILABLE_ALM_OPT_SW** option of the **gNBOamParam.AlarmSwitch** parameter. It is recommended that this optimization be disabled for multi-TRP cells, such as combined cells and hyper cells, to avoid potential inaccuracies in the displayed **Specific Problem** of ALM-29841 NR Cell Unavailable for these cells.

Implementation

For details, see *Fault Management*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management

No impact

- Configuration management

The following parameters have changed.

Change Type	Parameter ID	MML Command	NE	Description
Modified	gNBOamParam.AlarmSwitch	MOD GNBOAMPA RAM LST GNBOAMPA RAM	gNodeB	The NR_CELL_UNAVAILABLE_ALM_OPT_SW option has been added to specify whether to enable NR Cell Unavailable alarm optimization.

- Performance management

No impact

- Fault management

The following alarms have changed.

Change Type	Name	NE	Description
Modified	ALM-29841 NR Cell Unavailable	gNodeB	The descriptions of the Specific Problem field are revised for this alarm when NR Cell Unavailable alarm optimization is enabled.

Impact on Features/Functions

- Prerequisite features/functions

No impact

- Mutually exclusive features/functions

No impact

- Impacted features/functions

No impact

Related Operations

By default, the change does not take effect after an upgrade.

Run the **MOD GNBOAMPARAM** command with the **NR_CELL_UNAVAILABLE_ALM_OPT_SW** option of the **gNBOamParam.AlarmSwitch** parameter selected to enable NR Cell Unavailable alarm optimization.

Requirement/Trouble Ticket Number

FeR-IR20241209002193-0110

RAT

NR FDD, NR TDD

5.3.5.5.2 Added Support for the MOCN Scenario by All Scenario KPI Overview

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

When an accident or emergency fault occurs on the live network, it usually affects only the services of a single operator. The original indicators do not support operator-specific analysis. When the service volume differs greatly between operators, the fault occurrence time and impact degree for the operator with a smaller traffic volume cannot be identified in the overall performance indicators. This affects fault identification and quick recovery. To improve the fault locating efficiency in the MOCN scenario, operator-specific indicator analysis in the MOCN scenario is added to the all scenario KPI overview function.

The following LTE indicators can be analyzed in the MOCN scenario:

- E-RAB Setup Success Rate PLMN
- VoLTE E-RAB Setup Success Rate PLMN
- Call Drop Rate PLMN
- Call Drop Rate PLMN QCI1
- DL Traffic PLMN(MB)
- UL Traffic PLMN(MB)

The following NR indicators can be analyzed in the MOCN scenario:

- SgNB Addition Success Rate PLMN
- SgNB-triggered Abnormal SgNB Released Rate PLMN
- Abnormal SgNB Released Rate(Include MeNB triggered) PLMN
- SA NgSig Setup Success Rate PLMN
- QoS Flow Setup Success Ratio PLMN
- VONR SA QoSFlow Drop Rate PLMN

- SA QoSFlow Drop Rate PLMN
- VONR QoS Flow Setup Success Ratio PLMN
- DL RLC Traffic PLMN
- UL RLC Traffic PLMN
- N.User.RRCCConn.Avg.Operator

Implementation

For details, see *FMA*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the gNodeB and eNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

IR20240830001035

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD

5.3.5.5.3 Added Support for Analysis of RedCap-related Performance Counters by All Scenario KPI Overview in NR

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Currently, the 5G online FMA supports only cell-level KPI check and does not support RedCap-level KPI check. To quickly determine whether RedCap KPI deterioration is related only to RedCap UEs during online accident analysis, RedCap-related indicators can now be analyzed, improving accident recovery efficiency.

- RedCap RRC Setup Success Rate
- RedCap NgSig Setup Success Rate
- RedCap QosFlow Setup Success Rate
- RedCap Service Drop Rate
- RedCap Intra Freq HO Out Success Rate
- RedCap Inter Freq HO Out Success Rate
- RedCap Average User Number
- RedCap NR2L HO Out Success Rate
- Downlink Traffic Volume of PDCP Services of RedCap Ues
- Uplink Traffic Volume of PDCP Services of RedCap Ues
- RedCap User DL Experienced Rate(Excluding Last Slot Packets)
- RedCap User UL Experienced Rate(Excluding Small Packets)
- RedCap BWP PRB Usage
- RedCap UE PRB Usage

Implementation

For details, see *FMA*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20240618001009-0001

RAT

NR FDD, NR TDD

5.3.5.5.4 Optimization of CHR Log Parsing Specifications on the LTE/NR Online FMA

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

Currently, the wireless FMA can parse a maximum of 24 CHR files within 5 minutes. As the number and frequency of CHR-recorded key events are unlimited, a large number of abnormal events may be reported within a short period of time during peak hours, which exceeds the processing and analysis capability of the FMA. As a result, a network error is displayed on the GUI. The processing logic is optimized in the service process to improve the efficiency of fault analysis, thereby improving service reliability.

Implementation

The CHR file processing method is optimized to reduce the statistical time of a single CHR file and improve the efficiency of fault analysis.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the gNodeB and eNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20240729003544-0004

RAT

NR FDD, NR TDD, LTE FDD, LTE TDD

5.3.5.5.5 Optimization for Remote Distance of Certain LampSite RHUBs Exceeding Maximum Supported Distance

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

In LampSite RF combination scenarios involving multiple levels of RHUBs, if the distance between certain RHUBs and the BBU exceeds the maximum supported distance, these RHUBs can be removed and cells can be set up with reduced specifications.

Implementation

In LampSite RF combination scenarios involving multiple levels of RHUBs, optimization is performed for common cells, hyper cells, and combined cells for best-effort activation of devices within maximum supported distances, preventing TRP setup failures. If the RHUB is installed at a distance longer than 40 km away from the BBU, no change is required. After the optimization, the number of ALM-29870 NR DU Cell TRP Unavailable may decrease. Alternatively, the system may report ALM-29871 NR DU Cell TRP Capability Degraded instead of ALM-29870 NR DU Cell TRP Unavailable.

Impact on the Network

No impact

Impact on NEs

This change is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
No impact
- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250303001254-009

RAT

NR FDD, NR TDD

5.3.5.6 Inventory Management

There are no changes in inventory management.

5.3.5.7 License Management

This section describes the changes in license management.

For details about changes in licenses, see *License Changes*, which is provided with the software version. You can obtain *License Changes* from the following directory: **BTS3900&BTS5900 VXXXRXXCXXSPCXXX Release Documents-EN\01-Release Notes\Delta(MMI)**.

5.3.5.7.1 Added License Sales Number Precheck for Some License Control Items

NEs Involved

3900 series base stations, 5900 series base stations

Feature/Function Description

If a feature is not enabled for a cell and no license-related alarm is reported for the cell, it is possible that the cell cannot be activated due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified. To prevent service interruption caused by the preceding issue when operators enable features, license sales number precheck is supported. This function is controlled by the **LIC_SALE_NUM_PRE_CHECK_SW** option of the **gNBLicenseAlmThld.LicenseChkSw** parameter. This function may not take effect for cells in the deactivated or abnormal state.

- If this function is enabled and the licensed value of a license control item for a feature that affects cell activation is insufficient, the parameters for enabling the feature cannot be modified. This prevents cell activation failures caused by license insufficiency. This function will only take effect after a delay because it takes the base station seconds to obtain the licensed value.
- If this function is disabled and the configuration of a feature that affects cell activation changes, the base station does not check the licensed value of the corresponding license control item.

The change allows the license sales number precheck function to check the license control items listed in [Table 5-39](#).

Table 5-39 License control items related to license sales number precheck

Model	Name
NR0SULADLD00	UL and DL Decoupling (per Cell)
NR0S00HPCL00	Hyper Cell (per Cell)
NR0S00SUPC00	Super Uplink (per Cell)
NR0S0NRSFN00	Cell Combination (per Cell)
NR0S00VMMM00	Distributed Massive MIMO (NR TDD) (per Cell)

Implementation

For details, see *License Management*. The specific sections are listed in the change history of the document.

Impact on the Network

No impact

Impact on NEs

This function is implemented on the gNodeB.

Impact on Hardware Requirements

No impact

Impact on Inter-NE Interfaces

No impact

Impact on Operation and Maintenance

- License management
No impact
- Configuration management
No impact
- Performance management
No impact
- Fault management
No impact

Impact on Features/Functions

- Prerequisite features/functions
No impact
- Mutually exclusive features/functions
The following mutually exclusive relationships have been added.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UL and DL decoupling	NRDUCellAlgoSwitch.UIDIDecouplingSwitch set to ON	<i>UL and DL Decoupling</i>	When license sales number precheck is enabled, the base station checks the licensed value of the license control item NROSULADLD00. If the licensed value is insufficient, the NRDUCellAlgoSwitch.UIDIDecouplingSwitch parameter is not allowed to be set to ON .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	When license sales number precheck is enabled, the base station checks the licensed value of the license control item NROS00HPCL00. If the licensed value is insufficient, the NRDUCell.NrDuCellNetworkingMode parameter is not allowed to be set to HYPER_CELL .
Low-frequency TDD	Super uplink	SUPER_UL_TDM_SCH_SW or FLEX_SUPER_UPLINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter	<i>Super Uplink</i>	When license sales number precheck is enabled, the base station checks the licensed value of the license control item NROS00SUPC00. If the licensed value is insufficient, the SUPER_UL_TDM_SCH_SW and FLEX_SUPER_UPLINK_SW options of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter are not allowed to be selected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	When license sales number precheck is enabled, the base station checks the licensed value of the license control item NROS0NRSFN00. If the licensed value is insufficient, the NRDUCell.NrDuCellNetworkingMode parameter is not allowed to be set to HYPER_CELL_COMBINE_MODE .
Low-frequency TDD	Distributed Massive MIMO (NR TDD)	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter NRDUCellTrp.TrpType set to MASTER_DM_MIMO	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	When license sales number precheck is enabled, the base station checks the licensed value of the license control item NROS00VMMM00. If the licensed value is insufficient, the following setting combination is not allowed: The DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter is selected, and the NRDUCellTrp.TrpType parameter is set to MASTER_DM_MIMO .

- Impacted features/functions
No impact

Related Operations

By default, the change takes effect after an upgrade.

Requirement/Trouble Ticket Number

FeR-IR20250528001856-0022

FeR-IR20250102000700-0041

RAT

NR FDD, NR TDD

6 Reference Documents

This chapter lists the reference documents related to NIR.

1. 5G RAN10.0 Network Impact Report
2. 5G RAN9.1 Network Impact Report
3. MAE Release Notes
4. 5G RAN10.1 Feature List
5. 5G RAN10.1 feature description documents